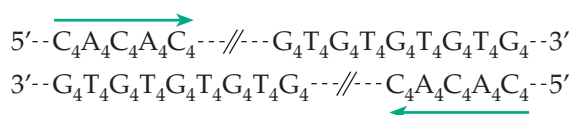


for adaptation to conditions of stress,⁶⁵⁰ and in eukaryotes.⁶⁵¹ In oocytes of amphibia, such as *Xenopus*, excess DNA accumulates around the nucleoli and later breaks up to form 1000 or more separate nucleoli. As many as 3000 copies of the rDNA (which forms a distinct satellite band upon centrifugation) may be present. Much of this DNA exists as extrachromosomal rings containing 1–20 rDNA units. Using these genes as many as 10^{12} ribosomes per oocyte are synthesized.

Tetrahymena contains only one set of rRNA genes per haploid genome in its diploid **micronucleus**. Following sexual conjugation the chromosomes of the micronucleus undergo multiple replications to form polytene chromosomes containing thousands of copies. However, about 5% of the resulting DNA is excised as linear pieces with characteristic inverted repeat sequences and 3' single-stranded tails:



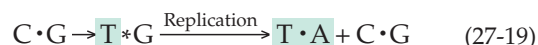
These tails are joined by a protein to form the circles, which are segregated in the **macronucleus**. The other 95% of the amplified DNA is degraded. At the next meiosis the macronucleus is discarded entirely, and a new one is formed at the next diploid stage.

Toxic drugs often cause cells to amplify genes that help resist the drug.⁶⁴⁷ This can be a major problem in the chemotherapy of cancer. For example, a culture of human leukemia cells grown in the presence of increasing concentrations of methotrexate increased its level of dihydrofolate reductase 240-fold.⁶⁵² The cause is an increase in the number of copies of a chromosomal region containing the gene.⁶⁵³ Cancer cells tend to amplify oncogenes such as *c-myc*.⁶⁵⁴

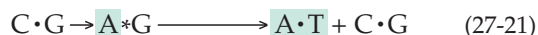
There are several mechanisms for gene amplification. The formation and breakup of polytene chromosomes in *Tetrahymena* is one. The circular copies of rDNA in *Xenopus* and many other species are generated by a rolling circle mechanism similar to that in Eq. 27-7. If each circle excised from the original chromosomes contains an origin of replication, many copies can be formed. Another possible mechanism is replication of a local region of DNA several times followed by excision of pieces of the DNA. A mechanism that may give rise to homogeneously staining regions is unequal crossing-over (Section 2) repeated several times within a gene cluster. Transposition can cause excision of DNA in a circular form, which can be amplified by a rolling-circle mechanism.

E. Damage and Repair of DNA

A characteristic of living things is their high degree of mutability. Harmful mutations take a toll of human life at an early age, and the very high incidence of cancer in older persons is largely a result of the accumulation of somatic mutations. Mutations are a major factor in aging and are continuously introducing new genetic defects into the population. Mutations can be described as **base substitutions**, **deletions**, or **additions**. Base substitution mutations are classified as **transitions**, in which a pyrimidine in one strand is replaced by a different pyrimidine. In the complementary strand a purine is replaced by the other purine, e.g.,



In a *transversion* a purine in one chain is replaced by a pyrimidine, while the pyrimidine in the complementary chain is replaced by a purine:



Here the central asterisks designate mismatches and the green shade marks mismatched bases and also the resulting mutant base pair formed in the next replication cycle.

1. Causes of Mutations

DNA can be damaged in many ways.⁶⁵⁵ Spontaneous hydrolysis of the glycosidic bonds between nucleic acid bases and the deoxyribose to which it is connected cause the loss of $\sim 10^5$ purines and pyrimidine rings per day from the DNA in a mammalian cell.⁶⁵⁶ About 100 residues per day of cytosine are deaminated by such agents as nitrite or bisulfite (Chapter 5, Section H,3) to form uracil. Like thymine, uracil will pair with adenine causing a $\text{C} \cdot \text{G} \rightarrow \text{T} \cdot \text{G}$ transition mutation as in Eq. 27-19. A few adenines per day are also deaminated to form hypoxanthine. Oxidized bases are formed by attack of $\text{HO}^\bullet$ radicals and other species of reduced oxygen.^{657–659} Alkylating agents from the external environment as well as S-adenosylmethionine carry out slow, nonspecific alkylation of purine and pyrimidine bases. Polycyclic aromatic hydrocarbons and other carcinogens are converted to metabolites that alkylate DNA, and alkylated bases often mispair during replication.^{660,661} Ultraviolet light induces formation of photohydrates (Eq. 23-25),⁶⁶² pyrimidine dimers (Eq. 23-26), and other photochemical products.⁶⁶³ X-rays and gamma rays cleave nucleic acid bases and break chromosomes.^{663a}

Natural radioactivity has a measurable effect.^{663b} Mistakes caused by mispairing⁶⁶⁴ or misalignment⁶⁶⁵ are made during replication and recombination and even during repair of DNA. Errors in replication are especially numerous in highly repetitive DNA sequences. Some errors probably arise as a result of tautomerization and others from incorporation of uracil in place of thymine. Thus, to keep its DNA in repair a cell must continuously deal with missing bases, wrong bases, altered bases or sugars, pyrimidine dimers and other crosslinkages, deletions, and insertions.

2. Fidelity of Replication

During DNA replication in *E. coli* only one mistake is made on the average during polymerization of 10^9 – 10^{10} nucleotides.^{666–668} The rate varies among different sites in the genome.^{668a} In eukaryotes the error rate may be only 1 in 10^{10} base pair or less per generation.^{669,669a} Early workers often attributed the specificity in base pairing and the resultant high precision of replication entirely to the strength of the two or three hydrogen bonds formed together with the stabilization provided by the adjacent helix. However, the Gibbs energy of formation of the base pairs is small (Chapter 5), and the additional energy of binding to the end of an existing helix is insufficient to account for the specificity of pairing.^{670,671} Thus, according to Eq. 6-30 a difference in ΔG of binding between the correct nucleotide and an incorrect one of 11 kJ/mol would give an error rate of 1 in 100.

The role of polymerases. The polymerase enzymes play a major role in ensuring correct pairing during replication, transcription, and protein synthesis. RNA and DNA polymerases are large molecules. The binding site on the enzyme can completely surround the double helix. Water will be excluded as the enzyme folds around the base pairs. This may have the effect of greatly increasing differences in the Gibbs energies of binding and thereby enhancing selectivity.⁶⁷¹ It has traditionally been assumed that formation of proper hydrogen bonds is essential for the base selection. A hypothetical active site is portrayed in Fig. 27-33. A guanine ring of the template strand of DNA is portrayed at the point where the complementary DNA strand is growing from the 3' end. The proper nucleoside triphosphate must be fitted in to form the correct GC base pair before the displacement reaction takes place to link the new nucleotide unit to the growing chain. Let us suppose that the enzyme possesses binding sites for the deoxyribose unit of the template nucleotide and for the sugar unit of the incoming nucleoside triphosphate and that the two binding sites are held at a fixed distance one from the

other. As indicated in Fig. 27-33, some group H-Y might also be present at each binding site to hydrogen bond to the nitrogen or oxygen indicated by the heavy green arrows. All four of the bases could form hydrogen bonds of this type in the same position. Hydrophobic interactions could provide additional stabilization. With such an arrangement the correct nucleoside triphosphate could be selected no matter which one of the four bases occupied the binding site on the left side of the figure. (The outlines of the thymine and adenine rings have been drawn in with dotted and dashed lines, respectively.) If a purine is present on the left side, as is shown in the drawing, there is room on the right side only for a pyrimidine ring. Thus, A and G are excluded, and the choice is only between C and U (or T). However, U will be excluded because the dipoles needed to form the hydrogen bonds point in the wrong direction. These dipolar groups are hydrated in solution. They are unlikely to give up their associated water molecules unless hydrogen bonds can be formed within the base pair. Not only would a molecule of U (or T) be unable to form the stabilizing hydrogen bonds within the vacant site but also the electrostatic repulsion of the like ends of the dipoles would tend to prevent association. This would lower the affinity of the polymerase for mispaired bases. Verification that the proper base pair has been formed could be accomplished by using its tautomeric properties to sense an electronic

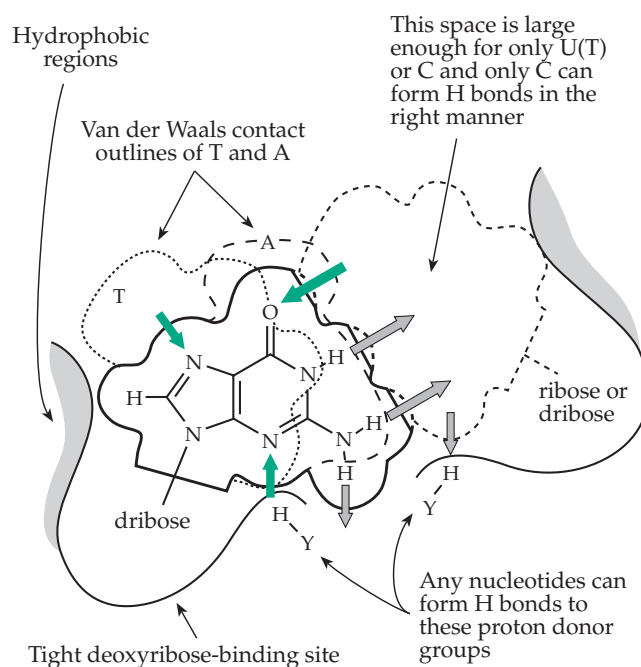
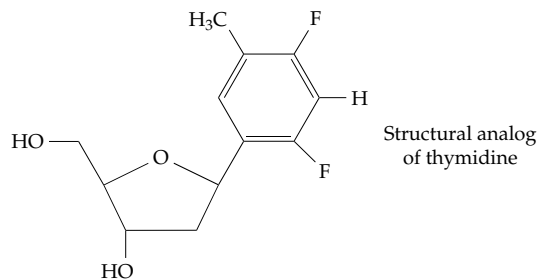


Figure 27-33 Selecting the right nucleotide for the next unit in a growing RNA or DNA chain. A deoxyguanosine unit of the template chain is shown bound to a hypothetical site of a DNA polymerase.

displacement through the hydrogen-bonded network.^{672,673} Considerable experimental evidence suggests, however, that hydrogen bonding may not be important to the selectivity of DNA polymerases.^{666,674–676} For example, the triphosphate of the following analog of thymidine is efficiently incorporated by *E. coli* DNA polymerase I into DNA opposite the thymine base-pair partner adenine.⁶⁷⁵



Similar results have been obtained for a variety of other analogs with poor hydrogen bonding characteristics. Apparently *shape* is most important.^{666,674,676,676a} Binding strengths in transition state structures must also be considered.^{676b}

Editing (proofreading). As is discussed in Section C.2, base-pairing is checked after a new monomer is added at the 3' end of a growing polynucleotide chain, as well as before polymerization occurs. If the wrong base pair has been formed, the newly created linkage is hydrolyzed, and the incorrect nucleotide is released. It has been estimated⁶⁷⁷ that in *E. coli* the error rate for DNA polymerase III holoenzyme is $\sim 1 \times 10^{-7}$ per base pair of which proofreading by the ϵ subunit may provide $\sim 10^{-2}$. Additional mismatch repair reduces the error rate 200- to 300-fold to give the overall error rate of $<10^{-9}$. Fidelity of replication, which seems to be higher on the lagging strand than on the leading strand,⁶⁷⁸ may also be improved by operation of other proteins, such as the sliding clamp, that are part of the replication apparatus.⁶⁷⁹ Replicative proteins also appear to be designed to minimize **frameshift mutations** that could arise by slippage of the template versus the replicated strand in long runs of dA•dT pairs.⁶⁸⁰

3. Repair of Damaged DNA

A final check of the fidelity of replication is made after a new strand has been formed. Mismatched base pairs are identified, and the incorrect nucleotides are cut out and replaced by correct ones.^{655,670,681–683} Some of the thymine dimers created by the action of light are also repaired photochemically by photolyases (see Chapter 23). **Photoreactivation** was the first DNA repair process recognized.⁶⁸⁴ However, most thymine

dimers in human DNA must be excised and replaced. Loops, gaps, and double-stranded breaks are also sensed, and appropriate corrections are made. If necessary the sequence of a badly damaged segment of DNA may be copied from the DNA in a sister chromatid before mitosis is completed or from the homologous chromosome.⁶⁸⁵

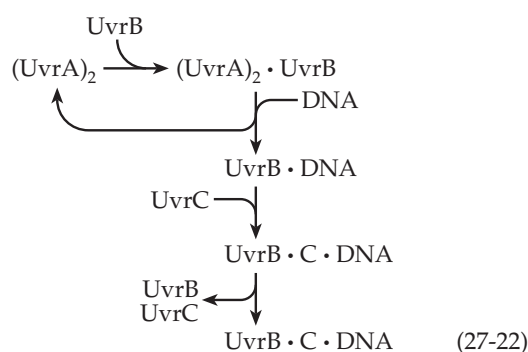
Much of our understanding of DNA repair comes from investigations of mutant strains of *E. coli*, which display an elevated incidence of mutations. On this basis, the “mutator” genes *dam*, *mutD*, *mutH*, *mutL*, *mutS*, *mutU* (also called *uvrD*), and *mutY* were implicated in DNA repair.^{677,686} An additional series of genes, *uvrA*, *B*, *C*, and *D*, were identified by their participation in resistance to ultraviolet radiation damage and were also recognized as being involved in DNA repair. Many corresponding genes were found later in the yeast, *Drosophila*, and human genomes. The study of human DNA repair has also been facilitated by the recognition of a group of human inherited defects (Box 27-A). There are 130 known human DNA repair genes.^{686a}

Methyl-directed and other mismatch repair. In *E. coli* the methylase encoded by gene *dam* (Fig. 29-4), within ~ 7 min of replication, methylates adenine at N-6 in the sequence GATC, which occurs at many locations.^{670,686,687} This methylation provides a label for the template strand in a newly replicated duplex. The GATC sites in the newly synthesized strand will not yet be methylated, and repair enzymes recognize it as the strand on which to carry out excision repair. Another system, which may function during recombination, acts on fully *dam* methylated DNA.⁶⁵⁵ Since eukaryotes do not use methylation of GATC sites to distinguish the old template and newly replicated DNA strands, other mechanisms must operate.⁶⁶⁹

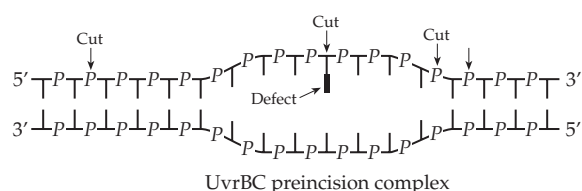
Ten proteins are required for methyl-directed mismatch correction. Of the required *mut* genes, *mutD* (also called *dnaQ*) was found to be the structural gene for the proofreading subunit ϵ of DNA polymerase III (Table 27-2).⁶⁸⁸ *MutH*, *L*, and *S* as well as ATP are also needed, as is a helicase, an exonuclease, DNA pol III, and a ligase. Homodimers of proteins MutS and MutL preferentially bind to DNA with mismatched base pairs such as Pur-Pur (e.g., A•G) or Pyr•Pyr (less well recognized). Some of these mismatched pairs may contain modified bases. After **recognition** by the MutS•MutL complex (see banner, p. 1527)^{688a–c} the endonuclease MutH is activated and cuts the DNA chain at the nearest unmethylated GATC sequence.⁶⁸⁹ An exonuclease then degrades the chain past the mismatched base leaving a gap that must be filled by the action of DNA polymerase and DNA ligase. This **long-patch mismatch repair (MMR)** is utilized also by eukaryotes,^{688a,b,690–692a} using proteins homologous to *E. coli* proteins MutS and MutL. However, methyl-

tion is not involved. In both yeast and human beings there are at least six MutS homologs, some of which have functions other than repair. For example, in yeast MutS- and MutL-related proteins are essential for normal levels of meiotic crossing-over.⁶⁹² Defects in human mismatch proteins may cause hereditary non-polyposis colorectal cancer (Box 27-A).^{692b}

Excision repair. The *E. coli* mismatch repair is a type of excision repair. However, a different **nucleotide excision repair** system (**NER**) is utilized by all organisms from bacteria to human to remove a variety of defects. These include thymine dimers, photohydrates, oxidized bases, adducts of cisplatin (Box 5-B), mutagens derived from polycyclic aromatic compounds,⁶⁸³ and poorly recognized C•C mismatched pairs.⁶⁹² In *E. coli* this excision repair process depends upon proteins encoded by genes **UvrA**, **B**, **C**, and **D** and also DNA polymerase I and DNA ligase.^{693–695a} A dimer of protein UvrA forms a complex with helicase UvrB (Eq. 27-22).^{696,696a}



The helicase, driven by ATP hydrolysis, may move along the DNA chain with its associated UvrA protein in search of defects.⁶⁹⁶ When one is located UvrB binds tightly.⁶⁹⁷ UvrA then dissociates, and the nuclease UvrC binds and cleaves the DNA chain in two places as in the following scheme. One is at the fourth or fifth phosphodiester linkage in the 3' direction from the defect. The other is at the eighth phosphodiester on the 5' side.⁶⁹⁸ The resulting gap is filled by a DNA polymerase and DNA ligase. Another helicase, encoded by *UvrD*, is also required.⁶⁹⁹



Both yeast and human cells have similar but more complex systems of nucleotide excision repair.^{700–703}

The dual incision steps require at least six components. Several of these (XPA, C, F, G) have been found defective in various forms of the inherited disease **xeroderma pigmentosum** (Box 27-A), in which there is a high incidence of UV-induced skin cancer.^{700,703a} Replication protein A (Section C,10), a three-subunit ssDNA binding protein, is also essential.⁷⁰⁴ In eukaryotic excision repair the DNA is cleaved at the same position as in the bacterial preincision complex (preceding scheme), on the 3' side of the defect but further out on the 5' side, the excised polynucleotide being ~24–32 nucleotides in length. Repair of the resulting gap in the DNA duplex is accomplished by synthesis mediated by DNA polymerase δ or ϵ and action of a DNA ligase.

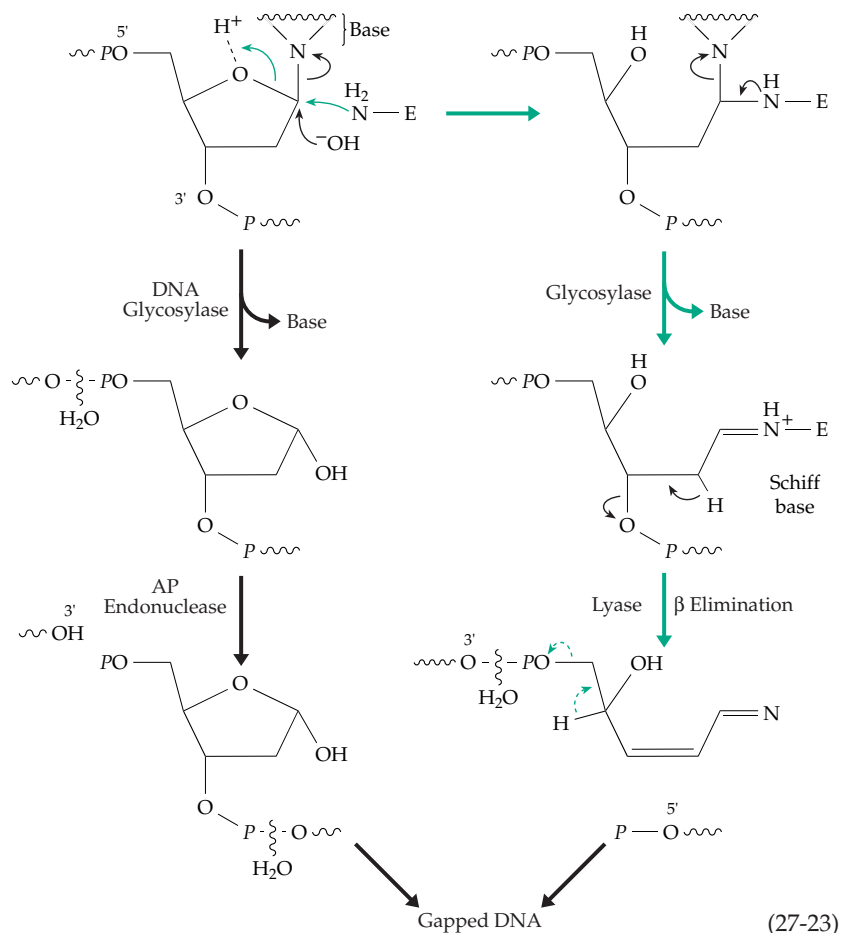
One of the lesions removed by nucleotide excision is the thymine photodimer.⁷⁰⁵ In the fission yeast *S. pombe* an **alternative excision repair** system, specialized for removal of thymine dimers and 6–4 photo-products (Chapter 23), produces two-nucleotide gaps with 3'-OH and 5'-phospho-group ends.^{702,705a} Alternative NER pathways are also employed by bacteria.^{705b}

Base excision repair (BER). The N-glycosyl linkages of the purine and pyrimidine bases to the deoxyribose residues of the sugar–phosphate backbone of DNA are subject to spontaneous hydrolysis, one important source of damage to DNA. Similar hydrolytic reactions are catalyzed by **DNA glycosylases**, which remove many mismatched or damaged bases.^{706–708a} At least seven enzymes of this type are present in cells of *E. coli*. One of them, a **uracil-DNA glycosylase**, hydrolyzes the glycosyl linkage wherever uracil has been incorporated accidentally in place of thymine or has been produced by deamination of cytosine.^{706,709–711d} Acting via a nucleophilic displacement mechanism (Eq. 27-23, black arrows), it removes the uracil, leaving the DNA backbone intact but with an **apyrimidinic site** in one chain. The resulting apyrimidinic (apurinic sites) are recognized and cleaved by **apurinic/apyrimidinic DNA endonucleases** (AP nucleases).^{708,712–714}

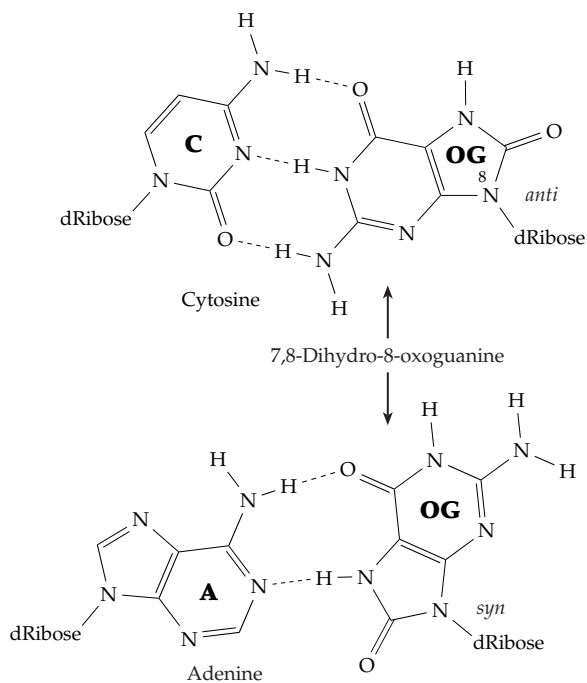
These enzymes cut the DNA backbone, some leaving a 3'-phosphate group and a 5'-OH terminus and others a 3'-OH and 5'-phosphate. The resulting gap can be filled, for example by the action of DNA polymerase I and a ligase. To prevent the eliminated uracil from being converted to dUTP and reincorporated into DNA a **deoxyUTPase**, essential to both *E. coli* and yeast, hydrolyzes dUTP to dUMP and inorganic phosphate, decreasing the concentration of dUTP.⁷¹⁵ Other enzymes, known as **DNA glycosylase/AP lyases**,^{706,716–718} use an amino group of an enzyme side chain as a nucleophile in a ring-opening reaction that is followed by β elimination of the nucleotide base with formation of a Schiff base intermediate (green arrow, Eq. 27-23). An example is the bacteriophage T4

enzyme **T4 endonuclease V**.^{706,719} Others include the *E. coli* **endonuclease III**, encoded by the **Nth** gene, its yeast homolog **Nth-Spo**,⁷¹⁷⁻⁷¹⁸ and human DNA pol ι (iota).^{718a} After the Schiff base is formed, the DNA backbone is cut on the 3' side of the abasic site by a second β elimination reaction. A δ elimination (green dashed arrows in Eq. 27-23) may also occur, cleaving the DNA chain on the 5' side of the lesion. Alternatively, a hydrolytic cleavage is possible as is also indicated in Eq. 27-23. Either type of DNA glycosylase forms a single-nucleotide gap or a gap missing just a few nucleotides. This gap can also be filled by polymerase and ligase action.

Both NER and BER forms of excision repair remove a great variety of defects, many of which are a result of oxidative damage.^{657,720} Most prominent among these is **7,8-dihydro-8-oxoguanine** (8-OG), which is able to base pair with either cytosine (with normal Watson–Crick hydrogen bonding) or with adenine, which will yield a purine–purine mismatch and a C•G $\rightarrow$ A•T transversion mutation (Eq. 27-24), a frequent mutation in human cancers.^{721,722}



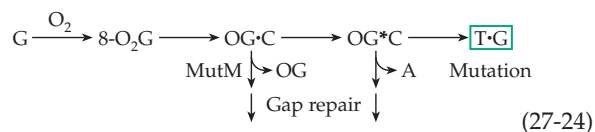
(27-23)



In *E. coli* three enzymes protect against 8-OG DNA mismatches. A glycosyltransferase encoded by the

mutM (or *fpg*) gene removes 8-OG from DNA.^{723,724}

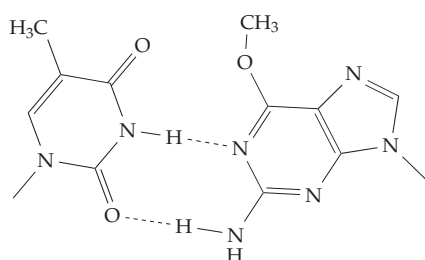
Because some A•OG base pairs may remain, a second glycosyltransferase (**MutY**) removes adenine from them.^{723,725-725b} Interestingly, MutY contains an Fe_4S_4 cluster, essential to its function.^{725,726} Both yeast and human cells have a corresponding enzyme.^{727,728} The third *E. coli* enzyme (**MutT**) is a nucleoside triphosphate pyrophosphohydrolase, similar to the previously mentioned dUTPase. It preferentially hydrolyzes free 8-oxo-dGTP, preventing its incorporation into DNA.⁷²⁹



(27-24)

Other DNA glycosylases remove thymine rings that have been converted to saturated or fragmented forms by oxidizing agents or ionizing radiation.^{720,730,731} Among these are 5,6-hydrated thymine and urea, which are still attached in ribosyl linkage. Purines such as hypoxanthine and 3-methyladenine can all be removed by glycosylases.

Reactions of alkylated bases. The O⁶ alkylation of guanine rings in DNA is highly mutagenic, presumably because mispairing with thymine will cause C•G → T•A mutations.



Thymine–O⁶-methylguanine pair

Both bacteria and higher eukaryotes have an enzyme-like **O⁶-methylguanine–DNA methyltransferase**, whose synthesis may be induced by culture of cells in the presence of alkylating agents. This 354-residue Zn²⁺-containing **Ada protein** acts as both acceptor and catalyst for transfer of the methyl group off from the O⁶-methylguanine onto the sulfur atom of a cysteine side chain in the protein.^{732–734b} The presence of alkylating agents also induces synthesis of DNA glycosylases that remove 3-methyladenine and other alkylated bases.^{735,736} 5-Methylcytosine, present in CpG islands in human DNA, will occasionally be hydrolyzed to thymine, giving T•G mismatches. The mismatched thymines are removed in cells of *E. coli* by a specialized endonuclease (**vsr** gene),⁷³⁷ which hydrolyzes the phosphodiester linkage preceding the mismatch. In eukaryotic cells a **thymine–DNA glycosylase**^{738,739} accomplishes the removal.

Repair of double-strand breaks. Following replication both in bacteria and in eukaryotes, recombinational exchanges may occur between sister chromatid duplexes or between homologous pairs of chromosomes.⁷⁴⁰ A newly replicated chain may have gaps because of defects in the template strand from which it was copied. Copying from a sister duplex may permit assembly of a correct chromosome sequence and survival for the cell.^{551a} Recombinational repair is also a major mechanism for preventing loss of genetic information from **double-strand breaks**. Such breaks are induced by ionizing radiation or chemical damage. They are often created at replication forks, stalling DNA synthesis with potentially disastrous consequences to the cell. Repair of these breaks, which is required for completion of replication, depends upon a large complex of replication and recombination proteins.^{741–741b} In *E. coli* this includes protein RecF.^{741,742} Repair of double-strand breaks by homologous recombination may be the most frequently employed type of DNA repair in bacteria,

yeasts such as *S. cerevisiae*^{530d} and also in mammals and other organisms.^{742a–d}

A second repair pathway, called **nonhomologous end-joining (NHEJ)**, is also utilized and may be relatively more important in higher organisms.^{743–746b} The NHEJ mechanism repairs double-strand breaks caused by ionizing radiation and is also employed in specialized genomic rearrangements of developing B and T cells of the immune system. This V(D)J recombination (Chapter 31) is used to create the huge array of antibodies and antigen receptors required for immunological recognition and protection.^{745,746a,747,748} NHEJ doesn't depend upon templates but simply rejoins duplex ends. Errors can be made if the ends have been damaged or if the wrong ends are joined. Several special proteins are required. One is a DNA-activated protein kinase (**DNA-PK**)⁷⁴⁹ and a heterodimeric **end-binding protein** with subunits **KU70** and **KU80**.^{745,750,750a} The serine / threonine protein kinase ATM (Box 27-A) is also activated and phosphorylates protein p53 in response to γ -irradiation of cells.⁷⁵⁰ This is thought to be part of a signaling pathway for control of the cell cycle following DNA damage. Mitosis may be delayed while repair is completed. Alternatively, the damaged cell may be killed by apoptosis.

Subjects of current interest are the mechanisms by which completion of DNA synthesis is signaled at the various checkpoints in the cell cycle.^{750b,c} These include points in addition to those marked in Fig. 11-15. At every point signals must accumulate to indicate that replication must be delayed to allow repair or that the cell must be allowed to die by apoptosis. Failure to accomplish repair may lead to cancer.^{750d} Even one double-strand break will prevent the completion of mitosis.^{750b} Among other problems faced by a dividing cell are slow replication and stalling at replication forks.^{750c} A range of chemical alterations in chromatin and other proteins are associated with DNA damage and repair.^{750b,e–h} Among these are conjugation of PCNA with ubiquitin and SUMO (Fig. 27-12D).⁷⁵⁰ⁱ

The SOS response and translesion repair. If cells of bacteria or eukaryotes are heavily damaged by UV, X-ray irradiation, or mutagenic chemicals, an emergency or **SOS response** is initiated.^{711a,741,751–753a} In *E. coli* the two proteins specified by genes *lexA* and *recA* initiate the response. The *lexA* protein is a repressor that prevents transcription of a group of SOS genes (see Chapter 28). It is thought that some product from damaged DNA activates the RecA protease activity. The activated RecA protein then cleaves the *lexA* protein allowing transcription of the SOS genes. The SOS response is transient but complex. It includes increased recombinational activity, alterations in replication initiation, inhibition of nucleases, and induction of an **error-prone DNA synthesis**. The cell now

replicates DNA more rapidly than normal but with an increased frequency of errors.^{754,754a} For example, it will bypass thymine dimers and other defects, and it will continue the DNA synthesis even though incorrect bases have been put into the chain opposite the thymine dimer. Later the errors may be corrected by recombinational repair or by photoreactivation. In *E. coli* this translesional repair depends upon DNA polymerase III, the RecA protein, and two proteins encoded by genes *umuC* and *umuD*. These genes were recognized as providing resistance to mutations induced by UV radiation.^{753–754b} Study of similar genes in yeast led to the discovery of two new and unusual DNA polymerases essential for translesional repair. DNA polymerase ζ (zeta), encoded by gene *REV3* and *REV7* subunits, bypasses abasic sites by inserting a dCMP residue into the growing DNA chain. Because C may be the wrong base the process is “error-prone.” Polymerase η (eta) bypasses thymine photodimers by placing two consecutive dAMP residues in the growing strand. This is an error-free process.^{755,756} Genes for human polymerases ζ and η have also been identified,^{757–759b} and additional polymerases have been found in yeast and other organisms.^{760–760e} The human XP-V gene for polymerase η is defective in xeroderma pigmentosa variant type (Box 27-A). Polymerase ι may act sequentially with pol η to bypass highly distorting lesions.^{760b} The very imprecise pol θ may also be used to bypass hard-to-remove lesions.^{760c} Polymerase λ is thought to play a role in DNA repair during meiosis,^{760d} while pol κ is needed in some way to provide cohesion between sister chromatids.^{760e} Some of these DNA polymerases are very inaccurate.^{760a} It has been suggested that human pol ι may participate in hypermutation of immunoglobulin variable genes.^{760f} Polymerase ϕ of yeast may be required for synthesis of ribosomal RNA.^{760g}

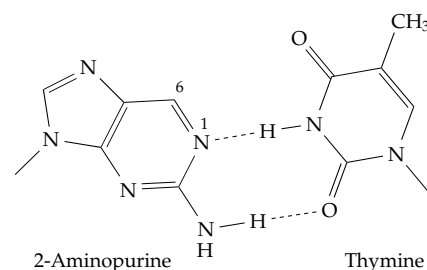
Poly (ADP-ribose). A eukaryotic peculiarity, which is not well understood, is the synthesis of poly(ADP-ribose) chains attached to many sites in nuclear proteins (see also Eq. 15-16). Increased synthesis is observed following damage to DNA.^{761–763a} The poly (ADP-ribose) polymerase binds to DNA near strand breaks or nicks and, using NAD⁺ as a substrate, synthesizes the highly branched polymer attached to a small number of nuclear target proteins.

F. Mutagens in the Environment

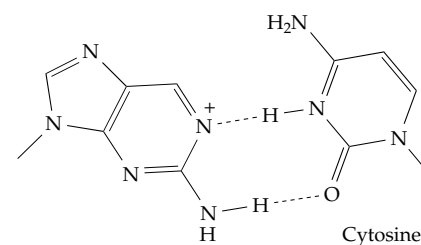
More than 500 new chemicals are introduced into the environment industrially each year. Some widely used drugs, e.g., **hycanthone** (Fig. 5-22), are mutagenic. Powerful mutagens are present naturally in some foods.^{764–766} Others have been added through ignorance. Although many of these have now been re-

moved, the problem cannot be ignored.

One way in which chemical compounds can induce base substitution mutation is through their incorporation into the structure of DNA itself. Thus, 5-bromodeoxyuridine (or bromouracil) can replace thymidine in DNA, where it serves as an efficient mutagenic agent.⁷⁶⁷ 2-Aminopurine, an analog of adenine, pairs with thymine, just as does adenine when incorporated into DNA.



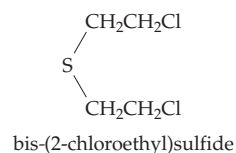
However, if it protonates on N-1, it can pair with cytosine, causing mutation.^{768,769}



Another mutagenic base is *N*⁶-hydroxylaminopurine.⁷⁷⁰

Many alkylating agents are powerful mutagens. Alkylation can occur at many places in DNA, but the N-7 position of guanine is especially susceptible (Eq. 5-18). The resultant positive charge on the imidazole ring portion of the alkylated guanine causes hydrolysis of the *N*-glycosyl linkage and depurination. However, this may be a lethal rather than a mutagenic event. The previously mentioned methylation on O-6 of guanine is probably more important in inducing mutations.⁷⁷¹

Among the most biologically reactive alkylating agents are the nitrogen and sulfur “mustards” such as bis-(2-chloroethyl)sulfide. These toxic bifunctional compounds cause lethal crosslinking of DNA chains



(see Eq. 5-20). The monofunctional half-mustards are mutagenic but are less acutely toxic. Another group of alkylating agents, the **nitrosamines** (Chapter 5), are

BOX 27-A DEFICIENCIES IN HUMAN DNA REPAIR

Problems with human repair systems account for several serious diseases. One is the rare skin condition **xeroderma pigmentosum (XP)**, an autosomal recessive hereditary defect. Homozygous individuals are extremely sensitive to ultraviolet radiation and have a high incidence of multiple carcinomas.^{a-e} Defects in at least seven human genes, listed in the accompanying table, can cause XP. Most of the genes have been cloned and found to be homologs of genes for nucleotide excision repair (NER). In rodents the *ERCC* genes have the same functions as do genes for resistance to ultraviolet light in yeast (*RAD* genes). This information has elucidated the functions of the XP genes and the basis for this group of human diseases.^{a,b,f}

The XP-A protein (see table) is a zinc-finger-containing DNA-binding protein, which interacts with human replication protein A (Section C,10). It may also interact with proteins corresponding to the rodent ERCC1 and ERCC4 proteins to form a complex that recognizes DNA defects.^{g-i} Proteins XPC and XPE may be specialized DNA binding proteins for cyclobutane dimers and other photoproducts.^{j,k} The XPG protein is an endonuclease that probably cuts on the 3' side of the DNA defect.^{a,l} In human cell lines with a variety of XP mutations the DNA problems have been corrected by transfer of the corresponding normal genes into the cells.^{m,n}

While defects in protein XPD often cause typical XP symptoms, some defects in the same protein lead to **trichothiodystrophy (TTD)**, brittle hair disease). The hair is sulfur deficient, and scaly skin (ichthyosis, Box 8-F), mental retardation, and other symptoms are observed.^o Like their yeast counterparts (proteins RAD3 and RAD25), XPB and XPD are both DNA helicases.^o They also constitute distinct subunits of the human transcription factor TFIIHP, which is discussed in Chapter 28. It seems likely that XPD is involved in **transcription-coupled repair (TCR)** of DNA.^{o,q-s} This is a subpathway of the nucleotide excision repair (NER) pathway, which allows for rapid repair of the transcribed strand of DNA. This is important in tissues such as skin, where the global NER process may be too slow to keep up with the need for rapid protein synthesis. Transcription-coupled repair also appears to depend upon proteins CSA and CSB, defects which may result in the rare **cockayne syndrome**.^{b,o,t,u} Patients are not only photosensitive but have severe mental and physical retardation including skeletal defects and a wizened appearance.

In a variant form of XP, designated XP-V, nucleotide excision repair is normal, but DNA replication is very slow. Postreplicational translesional repair

(bypass repair) is also slow, and patients are cancer-prone.^{v,w} The recently discovered DNA polymerase η may be defective.^x

Children with **ataxia telangiectasia (AT)** have progressive neurological problems, a weak immune system, premature aging, and a high incidence of cancer.^y Their skin fibroblasts are deficient in the ability to repair X-ray damage, which causes many double-strand breaks. Apparently in this disease cells do not wait until DNA repair has been carried out after exposure to ionizing radiation but attempt to replicate the damaged DNA.^y The defective protein **ATM** (ataxia telangiectasia, mutated) has been identified as a large 370-kDa Ser / Thr protein kinase with a carboxyl terminal domain similar to phosphatidylinositol 3-kinase.^{z,aa} It appears to play a crucial role in the cell cycle DNA damage checkpoints (Fig. 11-15) by participating in the detection of double-strand breaks and in delay of replication, while they are repaired by homologous recombination.^{y,bb,cc} Although AT is an autosomal recessive disease, women heterozygous for the ATM gene have an increased susceptibility to breast cancer. This observation led to the discovery that the proteins encoded by the well known breast cancer genes *Brca1* and *Brca2* form a complex with ATM. Phosphorylation of BRCA1 by ATM may initiate a signaling pathway through the p53, c-Abl, and Chk2 proteins that cause cell cycle arrest (Fig. 11-15). BRCA1 and BRCA2 also form a complex with protein RAD51, a RecA homolog necessary for homologous recombination. BRCA1 also may be essential to transcription-coupled repair.^{dd,de}

There are two major forms of hereditary susceptibility to colon cancer.^{ee} Familial adenomatous polyposis is caused by defects in the *APC* gene (see Chapter 32). The more common **hereditary non-polyposis colorectal cancer (HNPCC)**, which includes many endometrial, stomach, and urinary tract tumors, results from defects in DNA mismatch repair.^{ff-ji} The proteins hMSH2 and hMSL1 are homologs of the *E.coli* MutS and MutL (main text).

Cells of patients with **Bloom syndrome (BS)** have many chromosome breaks and a high frequency of sister chromatid exchanges, perhaps in an effort to correct these breaks. The body is small but well-proportioned.^{kk} A somewhat similar disease, the **Werner syndrome (WS)**, is associated with premature aging.^{ll} The Bloom's protein **BLM** and the WS gene product **WRN** are both helicases related to *E.coli* RecQ. Protein BLM colocalizes with replication protein A as discrete foci in the meiotic synaptonemal complex.^{mmm} Protein WRN also seems to be associated with DNA replication. Defects

BOX 27-A DEFICIENCIES IN HUMAN DNA REPAIR (continued)

appear to increase homologous and illegitimate recombination.ⁿⁿ Both proteins may also function in transcription.^{oo}

Many other diseases leading to a high incidence of cancer are known. Among them is the Nijmegen breakage syndrome, in which chromosomes are

hypersensitive to breakage by ionizing radiation. The gene has been identified by positional cloning, and its protein is apparently involved in repair of double strand breaks.^{pp} **Fanconi anemia**, **Gardner syndrome**, and hereditary **retinoblastoma** (Box 11-D) may also involve defects in DNA repair.

Some Human Hereditary Defects of DNA Repair

Human disease	Human gene involved		Yeast gene	Function
Xeroderma pigmentosum (XP)				
XP-A	XPA		RAD14	DNA-binding, damage recognition
XP-B	XPB	ERCC3	RAD25 (SSL2)	DNA helicase
XP-C	XPC		RAD4	DNA-binding, thymine dimers
XP-D	XPD	ERCC2	RAD3	DNA helicase
XP-E	XPE		RAD16	
XP-F	XPF		RAD10	DNA nuclease complex
XP-G	XPG		RAD2	DNA nuclease
XP-V	XPV		RAD30	DNA polymerase η
Cockayne syndrome (CS)				
CSA	CSA			Interaction with helicase CSB and with TFIIH
CSB	CSB			DNA helicase
Trichothiodystrophy (TID, brittle hair disease)				
	XPDE			
Ataxia telangiectasia (AT)				
	ATM			Cell cycle delay for repair of ds breaks
Human nonpolyposis colorectal cancer (HNPCC)				
	hMSH2		MSH2	Mismatch repair
	hMSL1		MSL1	
	GTBP			

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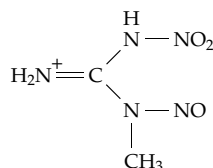
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BOX 27-A (continued)

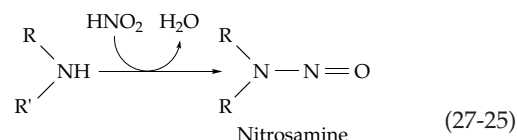
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highly mutagenic.^{772,773} Much used in the laboratory as a mutagen is **N-methyl-N'-nitro-N-nitroso-guanidine**:



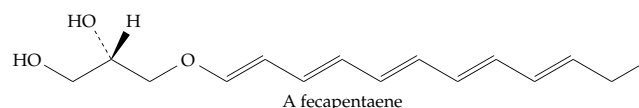
N-Ethyl-N-nitrosourea is one of the most potent carcinogens known.^{773,774} These compounds, as well as diazomethane and other related substances, probably act via a common intermediate: CH_3N_2^+ or $\text{C}_2\text{H}_5\text{N}_2^+$.⁷⁷⁵

Any secondary amine will react with nitrous acid to form a nitrosamine (Eq. 27-25). Tertiary amines can also react with loss of one alkyl group. This can occur in the stomach, and the nitrosamines may be absorbed into the system. All plants contain some nitrate and some, such as spinach and beets, have large amounts. Bacon and other cured R' R meats contain both nitrites and nitrates, and many drugs and natural food



constituents are secondary amines. Cigarette smoke also contains a nitrosamine.⁷⁷⁶ There is a possibility that these substances may induce human cancer.^{772,777}

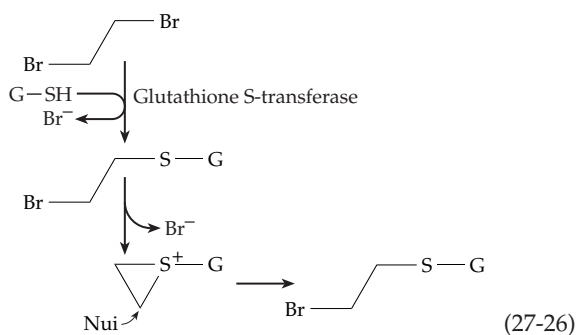
About 3% of residents of North America excrete feces that contain mutagenic unsaturated glyceryl ethers, which have been named **fecapentaenes**.^{778,779}



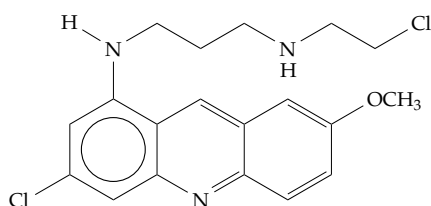
They could be among the causative agents of colorectal cancer. Protonation of a double bond will produce a reactive carbocation, which may be an active alkylating agent.⁷⁷⁹

Many halogenated compounds are carcinogenic. Among these is **1,2-dibromoethane**, which has been

produced in the United States in quantities as great as 10^8 kg / year. It has been widely used as a fumigant for foods, as an industrial solvent, and as an additive to gasoline. It is a **procarcinogen**, whose carcinogenic properties are expressed only after **metabolic activation**. One pathway of activation is reaction with glutathione to form a thioether, which can cyclize to a sulfonium compound.⁷⁸⁰ The latter alkylates nucleophilic groups in DNA as in Eq. 27-26. Other bifunctional electrophiles such as acrolein, malondialdehyde, vinyl chloride, and urethane are procarcinogens.⁷⁸¹⁻⁷⁸⁴ Styrene, another procarcinogen, is converted to the carcinogenic styrene oxide by action of a cytochrome P450.⁷⁸⁵ Metabolites of glucose such as **pyruvaldehyde**, **methylglyoxal**, and other reactive aldehydes can also attack DNA.^{786,787}

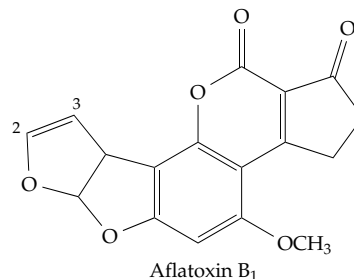


Less common than base pair exchanges are frame-shift mutations. A characteristic of such mutations is that they do not revert (back mutate) as readily as do base substitution mutations, and reversion is not induced by chemicals known to cause base substitutions. However, reversion of frame-shift mutations is induced by acridines and other flat molecules that are known to act as intercalating agents in DNA helices and which promote frame-shift mutations. They are especially effective in causing mutations of regions in which long repeated sequences of a single base such as $(A)_n$ or $(G)_n$ occur.^{770,788} For example, deletion of two base pairs from a “hot spot” (site of frequent mutation) in the *Salmonella* histidine⁷⁸⁹ operon with the following sequence is induced by 2-nitrosofluorene and causes reversion of a (-1) histidine-requiring mutant: 5'-CGCGCGCG. Whereas simple intercalating agents are often not very mutagenic, compounds that are both an intercalating agent and an alkylating agent are especially potent. An example is the following compound, which contains a half-mustard side chain:

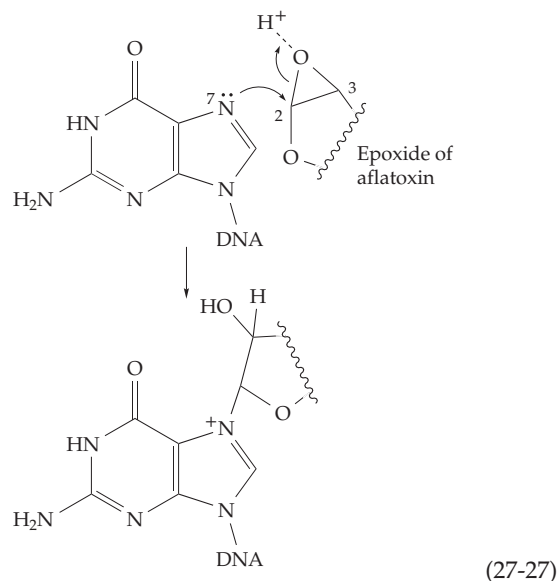


When the CH_2Cl group of the side chain is replaced by CH_2OH , the compound is 100 times less mutagenic.

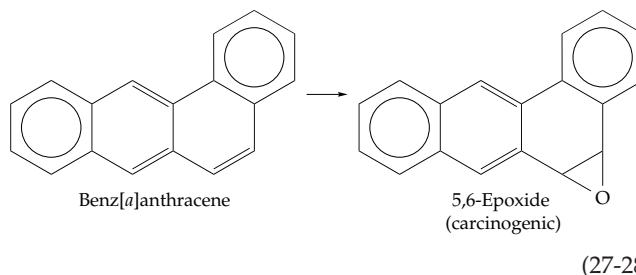
The carcinogenic **aflatoxins**, which are produced by *Aspergillus flavus*, may be present in infected peanuts and other foodstuffs.⁷⁹⁰ Like many other compounds that are carcinogenic or mutagenic, the aflatoxins are not unusually reactive chemically.

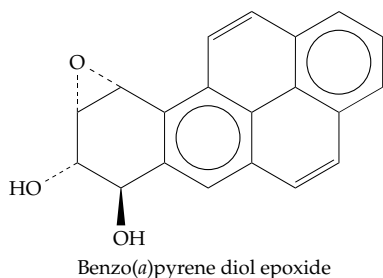


However, they are activated in the animal body by hydroxylation and formation of 2,3-epoxides. The latter may react with N-7 of guanyl residues in DNA (Eq. 27-27).^{791,792}



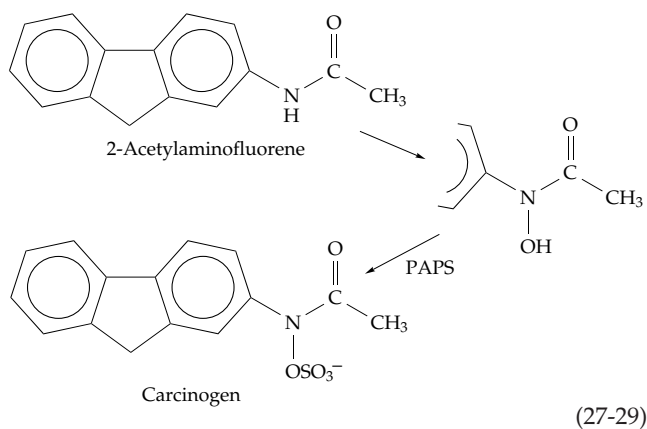
Benz(a)anthracene is also converted in the body to a carcinogenic epoxide (Eq. 27-28).⁷⁹³ Benzo(a)pyrene was isolated from coal tar in 1929 and in 1930 provided the first demonstration of the carcinogenicity of a pure chemical compound.⁷⁹⁴ It can also be activated by conversion in the ER to the 7,8-dihydrodiol 9,10-





epoxide.⁷⁹⁵⁻⁷⁹⁷ Hydroxylation activates 2-acetylaminofluorene to a carcinogenic *N*-sulfate (Eq. 27-29),^{788,798} which reacts with guanine rings.^{788,799}

Although it seems incongruous, many of the most potent antitumor drugs are powerful mutagens.



Among these are intercalating agents such as daunomycin (Figs. 5-22, 5-23), neocarzinostatin, and bleomycin (Box 5-B). These are alkylating reagents⁸⁰⁰ or attack DNA in other ways. The fact that such compounds are in use for chemotherapy emphasizes the need for new approaches to cancer treatment.

How can compounds be recognized as mutagenic? This is an important question that is difficult to answer and which has led to many controversies. For example, how dangerous is formaldehyde in the environment?⁸⁰¹ Butadiene?⁸⁰² Dioxin?^{790,803} Is fluoride a carcinogen?^{804,805}

A quick test for mutagenic activity makes use of **tester strains** of bacteria developed by Ames and associates. These are *Salmonella* mutants that are unable to synthesize their own histidine, but which

can grow when a mutagenic agent produces a back mutation.^{806,807} One of the strains can be mutated by agents causing base exchanges, while the other three, which contain different types of frame-shift mutations, are affected differently by various mutagens. About 10^9 bacteria are spread on a Petri plate, and a small amount of the mutagenic chemical is introduced in the center of the plate. Where back mutation has occurred, a colony of the bacteria appears. The strains all carry a mutation in the main DNA excision-repair system so that most mutations are not repaired, and the test is very sensitive. Addition of a liver homogenate plus an NADPH-generating system to the sample tested allows activation of many aromatic chemicals by hydroxylation.⁸⁰⁸ Feeding of mutagens to *Drosophila* eye color mutants permits testing for back mutation in a eukaryote.⁸⁰⁹

The bacterial tests have been widely used and have been of great value. For example, they revealed that certain chemicals that were being used as flame retardants in children's clothing are mutagens,⁸¹⁰ and that mutagens can be generated during cooking of meat and other foods.⁸¹¹ However, there are good reasons for using other methods of identifying mutagens together with the bacterial tests.⁸¹² Government regulatory agencies in the United States have relied largely on tests with rodents. Compounds are tested at very high doses with these short-lived animals. Extrapolation to human exposures at very low levels has been criticized.^{764,765,813,814} However, rodent tests have identified many true carcinogens.⁸¹⁵ A broader range of tests are now in use.⁸¹⁶ Direct monitoring of the accumulation of defects in animal and human cells is now possible. For example, in the **³²P-postlabeling technique** DNA is enzymatically digested to the nucleotide level, and the adducts with mutagens are labeled with ³²P, separated, and their quantities measured.⁸¹⁷⁻⁸¹⁹ Laser-excited fluorescence from such adducts, GC / mass spectrometry, and immunological methods can also be used to identify DNA adducts.⁸¹⁹ Careful vigilance is needed to keep our mutation rate at as low a level as possible.

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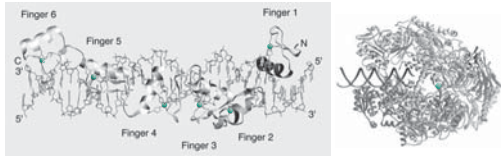
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Study Questions

1. Demethylation of 5-methylcytosine in DNA during early embryonic development has been proposed (see Chapter 32). Can you suggest one or more mechanisms by which such demethylation could occur?
2. Describe the structures and functions of histones and of nucleosomes in eukaryotic cells. Are there comparable proteins and structures in bacteria? Do you think that our knowledge of histones and nucleosomes is nearly complete?
3. List the major molecular components required for replication of DNA in *E. coli*. Describe briefly the functions of each protein or other component.
4. Compare replication in bacteria and in eukaryotes.
5. Compare synthesis of the leading and lagging strands in the elongation phase of DNA replication. Explain why DNA polymerases may have difficulty in replicating the 3'-end of the *lagging* strand of **linear** DNA. How has this problem been solved in many bacterial and viral systems? In eukaryotic cells?
6. The circular chromosome of an *E. coli* cell contains 4.6×10^6 base pairs. If a replication fork moves at a rate of ~1000 nucleotides per second, how much time will be required for replication of the DNA? Cells of *E. coli* can divide every 20 minutes under favorable conditions. How can you explain this rapid rate of growth?
7. DNA ligase, whose reaction is reversible, is able to relax supercoiled circular DNA in the presence of AMP but not in its absence. Outline the chemical mechanism of the ligase reaction. Why is it dependent on AMP? What other DNA ligase mechanism is known?
8. DNA polymerases involved in replication require a primer. Why? What is the nature of the primer?
9. Why is it essential for a cell to have several different mechanisms of DNA repair? Describe some of these mechanisms.
10. Why do cells use error-prone DNA polymerases under some circumstances?
11. Do you see a relationship of some types of DNA repair to the chemical events during meiosis?
12. Is O⁶-methyltransferase an enzyme?
13. Why are high rates of mutation observed in regions of DNA that contain 5-methylcytosine?
14. Why is uracil-DNA glycosylase important in DNA repair? Is it important for DNA replication?
15. Why do cells exposed to visible light following irradiation by ultraviolet light have a greater survival rate than cells kept in the dark after UV irradiation?
16. Can exposure of *E. coli* to nitrous acid (HNO₂) lead to mutation of a tRNA^{GLY} to an amber suppressor? The Gly codons are GGX (where X = any nucleotide) and the amber codon is UAG.



Left. The N-terminal 190-residue fragment of transcription factor TFIIIA of *Xenopus laevis* bound to a 31 bp DNA segment of the promoter region for 5S ribosomal RNA. Six zinc finger motifs (zinc atoms are green) bind in several ways into the major groove of DNA and across the minor groove. From Nolte *et al*, *Proc. Natl. Acad. Sci. USA* 95, 2938–2943, 1998. Courtesy of Raymond S. Brown. Right. Ribbon drawing of the three-dimensional structure of a 10-subunit form of yeast RNA polymerase II, which transcribes genes to form messenger RNA. A 20 base pair segment of B-DNA has been modeled but the transcription bubble is not shown. The active site Mg^{2+} is green. From Cramer *et al*, *Science* 288, 640–649, 2000. Courtesy of Roger D. Kornberg.

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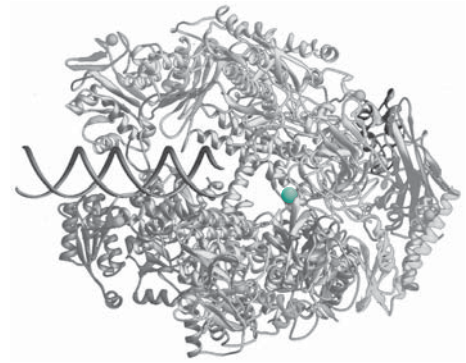
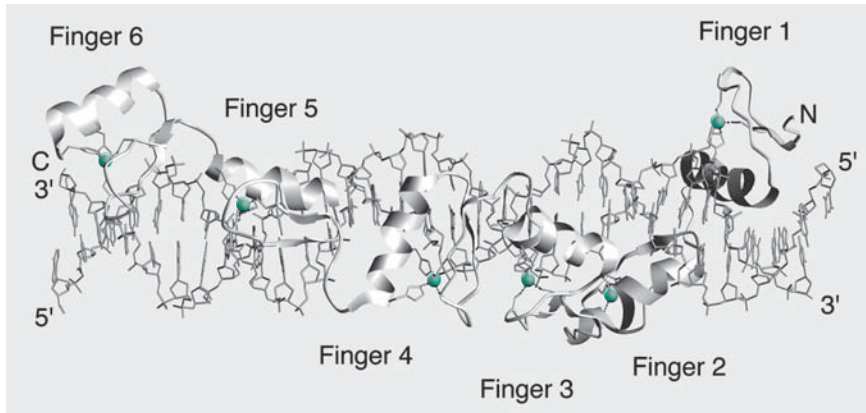
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The Transcription of Genes

28



The copying of genetic information from DNA into messenger RNA is the initial step in the chain of reactions leading to synthesis of the multitude of proteins and specialized RNA molecules needed by cells. The requirement for these macromolecules varies with conditions, and in eukaryotic cells, with the stage of differentiation. Therefore, it is not surprising that transcription is highly controlled.

Cells make four principal kinds of RNA: ribosomal (rRNA), transfer (tRNA), messenger (mRNA), and a variety of small RNAs. The last, which range in length from a few up to several hundred nucleotide units,^{1-1b} are designated variously as sRNAs, ncRNAs, miRNAs, siRNAs, snRNAs, and snoRNAs. The abbreviation *s*, *nc*, *mi*, *si*, *sn*, and *sno* stand for small, non-coding, micro, silencing, small nuclear, and small nucleolar, respectively. All of these RNAs are synthesized as larger transcripts, which undergo cleavage and other modifications within the cell. Therefore, a second major topic in this chapter is the processing of RNA precursors. We will also consider the fact that cells may be hosts for RNA viruses, may occasionally harbor RNA plasmids, and must sometimes transcribe viral DNA.

The absence of a nuclear membrane is a characteristic of bacteria that has a profound effect on transcription. Bacterial transcripts are processed rapidly, and their 5' ends often enter ribosomes and are directing protein synthesis, while the 3' ends of the genes are still being transcribed. In contrast, most eukaryotic RNA transcripts must be processed and transported out of the nucleus before they can function. As consequence, many aspects of the control of transcription differ between prokaryotes and eukaryotes.

A. Transcription and Processing of RNA in Bacteria

Even after the existence of mRNA had been recognized, it was not obvious how formation of single-stranded (ss) RNA would be accomplished, using a double-stranded (ds) template. The fact that purified RNA polymerases can synthesize RNA from the four ribonucleoside triphosphates using ssDNA as the template suggested that transcription, like DNA replication, involves base pairing. In line with this conclusion was the fact that the ssDNA obtained from bacteriophage ϕ X174 was converted by RNA polymerases into a dsRNA–DNA hybrid. However, when dsDNA served as the template, free ssRNA was formed. Thus, it appeared likely that at the site of the polymerase action the dsDNA was momentarily pulled apart into single strands and that one of these was copied by the polymerase. More recent experiments have confirmed this view.

1. The *lac* Operon

Much of the terminology used to describe the control of transcription originated with Jacob and Monod. Based on studies of the induction of enzymes in bacteria they proposed the **operon model**.^{1c-3} An operon is a regulated cluster of genes, one of which is shown diagrammatically in Fig. 28-1. This is the *lac* operon of *E. coli*. Found at position 8 min on the genetic map of Fig. 26-4, it is probably the most intensively studied group of *E. coli* genes. Three structural genes encode the amino acid sequences of β -galactosidase (*lacZ*),^{3a}

permease (*lacY*),⁴ and a transacetylase (acetyltransferase, *lacA*), which transfers acetyl groups from acetyl-CoA to β galactosides. To account for the apparently synchronous control of these three genes, Jacob and Monod proposed that they function as a **transcriptional unit** the operon, which encodes a single molecule of mRNA. They proposed that each operon is controlled by a segment of the DNA molecule located at the beginning of the operon, i.e., at the 5' end of the coding chain or 3' end of the template chain. The first part of this **control region** they called the **promoter** (*P*). The promoter is the site of the initial binding of the RNA polymerase to the DNA, the binding constants for the association being very high. The rates of association and of initiation may be influenced strongly by various other proteins. One of these, the **catabolite gene activator protein (CAP**; also called cAMP receptor protein, CRP), is important to the *lac* operon. It also binds in the promoter region (Fig. 28-1) and stimulates transcription.

Repression and induction. Immediately adjacent to the promoter is the **operator** (*O*), which is a binding site for a **repressor** (*R*). When the operator is free, transcription is initiated and proceeds through the operator region and on to the genes coding for the three proteins. On the other hand, if the repressor is bound to the operator, transcription is blocked. When the operon model was first proposed, the chemical nature of the repressor was unknown. However,

many repressors have been identified as oligomeric proteins able to undergo allosteric alteration. The *lac* repressor is made up of four identical 360-residue subunits. Each subunit has a helix–turn–helix binding domain that is specific for the DNA sequence of the operator and an allosteric binding site for an effector.^{5,6} The drawing in Fig. 28-1 is simplified to show only two of the four subunits (see also Chapter 5, Section F,1).

The *lac* operon is ordinarily subject to repression and is activated by the presence of an **inducer**, now known to be **allolactose**, $D\text{-Galp-}\beta 1 \rightarrow 6\text{-D-Glc}$. However, in experimental work artificial inducers such as **isopropyl- β -D-thiogalactoside (IPTG)** are most often used. Jacob and Monod postulated that the free repressor protein binds to the operator. In the presence of the inducer a conformational change takes place, destroying the affinity of the repressor protein for the operator site. Thus, in the presence of inducer the operator is not blocked, and the transcription takes place. Such an operon is said to be **negatively controlled** and **inducible**.

Important to the control of the operon is the **regulatory gene**, which codes for the synthesis of the repressor protein. In the case of the *lac* operon, the regulatory gene (the *I* gene) is located immediately preceding the operon itself (Fig. 28-1). However, for some operons the regulatory gene is located a considerable distance away. For example, the *gal* operon of *E. coli*, which codes for enzymes of galactose metabolism, is found at map position 17 min, while the regulatory gene is at 61 min.⁷

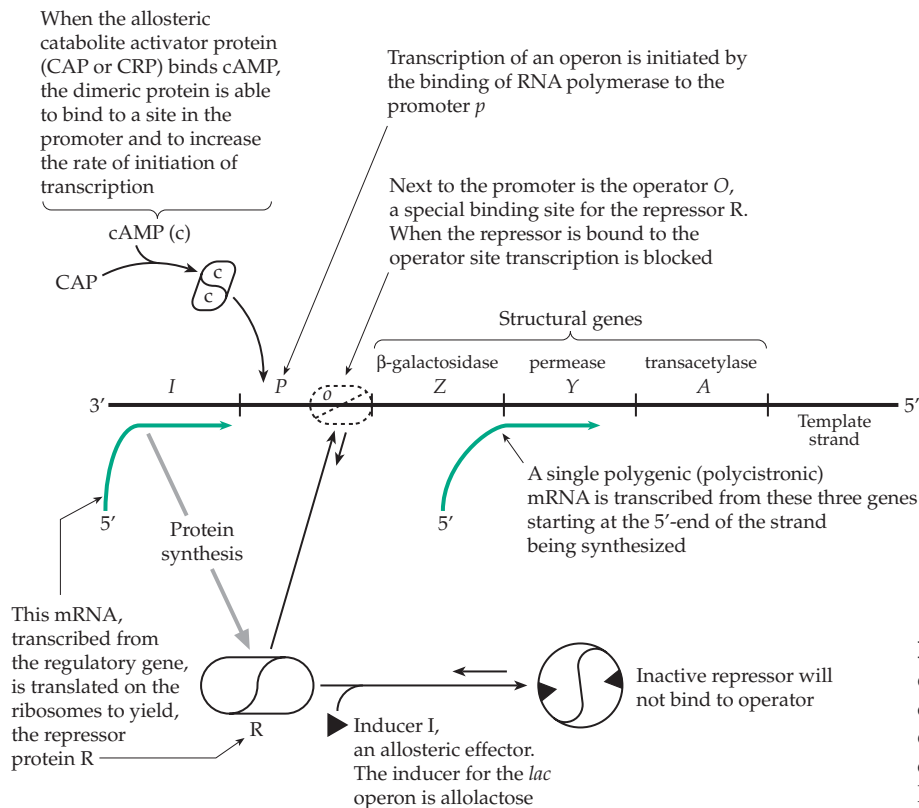


Figure 28-1 Schematic representation of the *lac* operon of *E. coli* and of its control. Here only the template strand of the DNA is shown. However, the coding (nontranscribed) strand is usually the one labeled, as in Fig. 28-2.

Regulatory genes are normally transcribed at a slow but steady rate, presumably because RNA polymerase initiates RNA chains infrequently at the promoter sites of regulatory genes. Thus, each cell of *E. coli* normally contains only about ten molecules of the *lac* repressor protein. A mutation in a regulatory gene may lead to a defective repressor, which no longer binds at the operator. Then, transcription of the operon is uncontrolled, and mRNA is produced in greater amounts. In such a mutant strain (designated *I*⁻ in contrast to the normal *I*⁺ strain) production of the enzyme representing the gene product becomes constitutive, just as is the formation of the enzymes of the central pathways of metabolism. The latter enzymes also appear to be produced regularly in large amounts without control by a repressor, establishing that *transcription* rather than translation regulates the expression of these genes. The operon can also become unregulated, i.e., **constitutively expressed**, if a mutation occurs in the operator site and the repressor can no longer bind because of the altered DNA sequence.

Nucleotide sequence of the *lac* control region.

The sequence of the *E. coli* DNA representing the promoter-operator region of the *lac* operon is shown in Fig. 28-2. It includes the end of the *I* gene, at the left, and the beginning of the *Z* gene, at the right.² The series of codons representing the peptide sequence Glu-Ser-Gly-Gln-Stop at the left-hand end corresponds

to the known C-terminal sequence of the repressor, while the three codons at the right are those of formyl-Met-Thr-Met, the known N-terminal sequence of the *Z* gene product β -galactosidase. Detailed genetic mapping of the region in 1973 made it possible to assign operator and promoter regions with confidence as indicated. The mRNA transcript begins in the operator region as shown in the figure. The initiation codon for the *Z* gene is 39 bases from the end of the transcript. In this figure the original numbering of the nucleotides is printed in black. Now it is customary to number from the transcription initiation position (+1). Green numbers are used in this way in Fig. 28-2. Positions to the left and toward the 5' end of the nontranscribed coding strand precede the initiation position. They are referred to as **upstream** and are numbered with negative integers. Positions to the right are **downstream** and numbered with positive integers.

The operator region was located by digesting the DNA with deoxyribonuclease in the presence of the repressor protein.⁸ The bound repressor protected a region of 27 base pairs as indicated in the figure. The operator is centered on a region of local twofold rotational symmetry (Chapter 5; Fig. 5-34). The symmetry is not perfect, the sequence being **quasipalindromic**. The following precisely symmetric synthetic sequence, which contains an 11-bp inverted repeat of the left half of the *lac* operator sequence, binds *lac* repressor 8-fold more tightly than does the natural *E. coli* operator.⁹

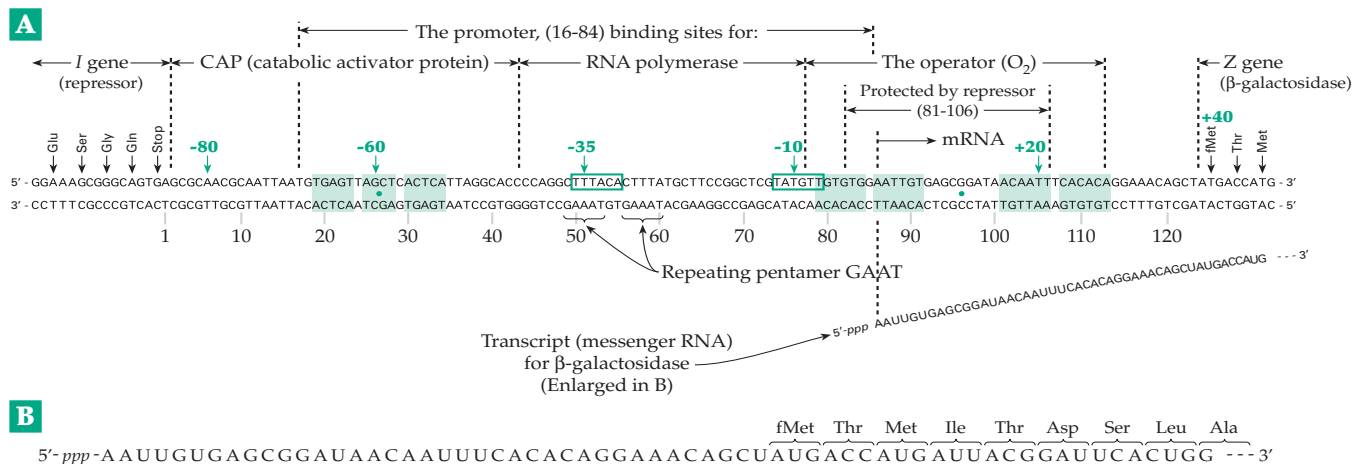
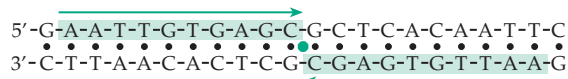


Figure 28-2 (A) Nucleotide sequence of the *lac* promoter-operator region of the *E. coli* chromosome.⁷ The proposed locations of the *I* gene, the promoter (which contains CAP and RNA polymerase binding sites), the operator, and the beginning of the *Z* gene (β -galactosidase) are shown. Note the two regions of local 2-fold rotational symmetry, which are marked by bars and central dots and the repeating pentamer. Positions upstream (–) or downstream (+) from the +1 start position for transcription are marked in green. The –10 (Pribnow) and –35 promoter elements are boxed on the coding strand (top). Labels are usually applied to the coding strand. The mRNA is copied from the complementary template strand (bottom). (B) The sequence of an mRNA molecule initiated in the *lac* promoter-operator region of a mutant strain of *E. coli* with an altered promoter.¹⁰ The peptide initiation amino acid is identified by the symbol fMet, and the successive amino acids from the known N-terminal sequence of β -galactosidase have been matched with the codons.



The dyad symmetry of the operator sequence is probably important in providing tight binding to two subunits of the symmetric tetrameric protein.^{11–13} It is also possible that repressor molecules move along DNA chains in a one-dimensional diffusion process, and that the symmetry of the operator site facilitates recognition by a protein moving from either direction.^{14,15}

The repressor structure. The *lac* repressor protein is a member of the **helix–turn–helix** family of DNA binding proteins (Fig. 28-3; see also Fig. 5-35). The first helix and the turn of this motif fit into the major groove of the DNA, the side chains from the helix interacting with specificity-determining groups in the major groove. The protein consists of three domains; the N-terminal DNA binding “head” (residues ~1–~50), a core domain (residues 62–340), and a leucine heptad repeat domain (residues 340–360) that forms the dimer–dimer interface. The dimeric form of the repressor binds to the palindromic operator sequence. The tetramer can bind to two operator sequences.^{16–20} The inducer IPTG binds to the core domain near the ONPF site shown in Fig. 28-3. The sequence, structure, and binding site of the core domain resemble those of sugar transport proteins such as the galactose-binding chemoreceptor protein (Fig. 4-18A). Binding of the inducer causes the conformational change that appears to disrupt the interactions of the “hinge helices,” seen in the center of Fig. 28-3A, with the DNA. This causes the repressor to dissociate from the operator (Fig. 28-1) and allows RNA polymerase to bind and to initiate transcription.

The *E. coli lac* repressor is one of the most investigated of all proteins. For example, 4000 single-amino-acid mutants have been prepared and studied.^{6,17,21} Suppressor mutations were used to determine the function of various portions of the protein.^{22,23} Many of the mutant proteins were created using *amber* mutations that were induced in the gene at many positions. The mutated genes were transferred into plasmids for cloning. Each plasmid was used to infect five different strains of bacteria, each carrying a suppressor mutation that would introduce a different amino acid when the (termination) codon UAG was encountered (see Chapter 29, Section C.3). From these infected bacteria large quantities of the mutant forms of the *lac* repressor were isolated. It was found that many mutations near the N-terminal end interfered with binding of the repressor to DNA, whereas mutations near the center interfere with binding to the inducer.

In addition to the main *lac* operator O_1 , which is marked in Fig. 28-2, there are two weaker auxiliary operator sequences designated O_2 and O_3 located 401

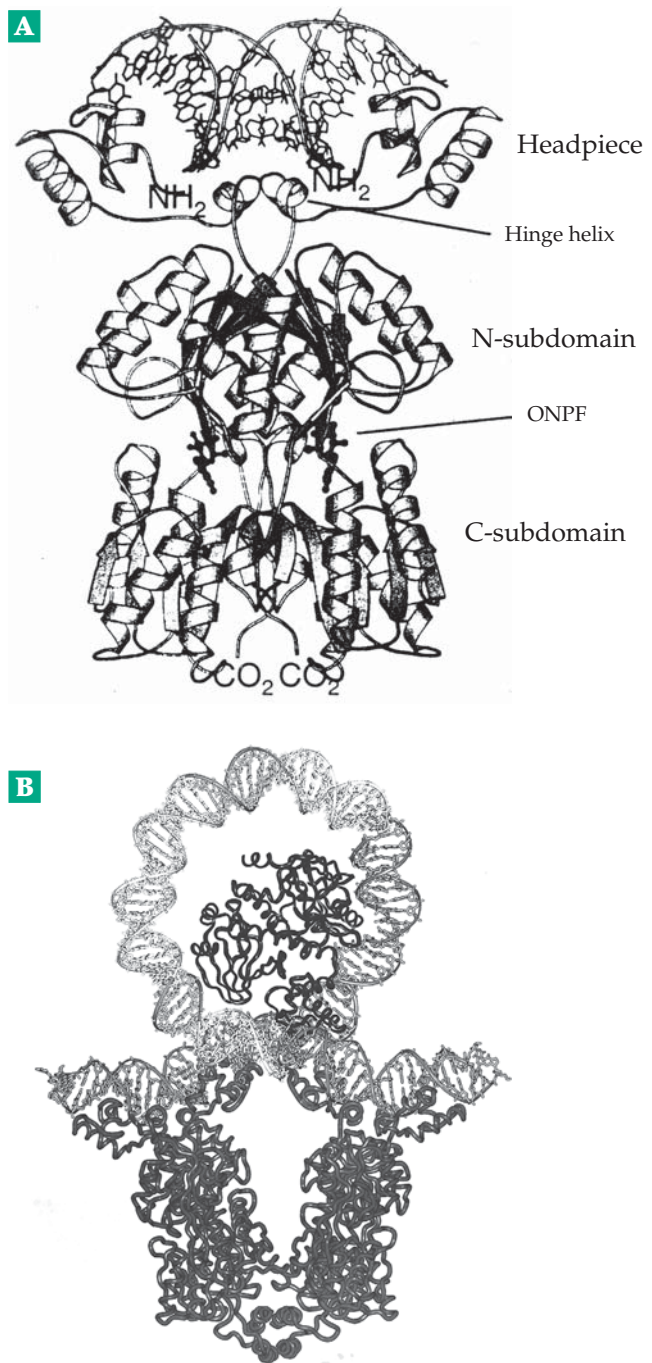


Figure 28-3 (A) Ribbon view of the dimeric *lac* repressor bound to a natural operator and to the anti-inducer *o*-nitrophenylfucoside (ONPF). The headpiece (residues 2–46) and the hinge helix (residues 50–58) form the DNA-binding domains. The core (residues 62–330), which is divided into N- and C-terminal subdomains, forms the binding site for ONPF. The C-terminal residues 334–360, which form a tetramerization domain, are absent from this MolScript drawing. Notice that the hinge helices bind to and widen the minor groove at the center of the operator. From Lewis *et al.*^{5a} (B) Model of a 93-bp DNA loop corresponding to residues –82 to +11 of the *lac* operon (Fig. 28-2) bound to the tetrameric *lac* repressor. The active sites of the repressor are bound to the major operator O_1 and to the secondary operator O_3 . From Lewis *et al.*⁵

bp downstream of O_1 in the lacZ coding region, and 92 bp upstream of O_1 , respectively.²⁴ The DNA can bind to both O_1 and either O_2 or O_3 with a loop between them as in Fig. 28-3B. Binding of the 10–20 copies of the lac repressor present in a cell of *E. coli* to the operator sequence is so tight that expression of the genes controlled is reduced 1000-fold. However, when placed in front of other operators or in different positions relative to the Pribnow sequence (Fig. 28-2), its effectiveness varied greatly. The extent of repression appears to be affected both by thermodynamic factors^{12,13} and by relative rates of repressor binding and of RNA polymerase movement (see also Eq. 28-1).²⁵

2. Initiation of Transcription

The rate of RNA synthesis varies from one operon to another. Sequences of promoters, operators, and other control sequences as well as the state of repressors and activator proteins all affect these rates.^{10,26,27} However, in every instance the first steps in transcription involve the binding of RNA polymerase to DNA.

Bacterial RNA polymerase. Most RNA polymerases (**RNAPs**) are large multisubunit proteins. However, bacterial viruses sometimes induce their own RNA polymerases, and these may be monomeric. For example, the 99-kDa (883-residue) phage T7-encoded polymerase is a single peptide chain with a structure and two-metal-ion active site resembling those of *E. coli* DNA polymerase I.^{28–30a} It is able to carry out all of the steps of the transcription cycle of the virus. In contrast, the most studied bacterial RNAP, that from *E. coli*, consists of five kinds of subunits^{31–33d} with the composition $\alpha_2\beta\beta'\sigma\omega$. A similar composition has been found for RNAPs of other bacteria.^{33e} Functions of the five subunits can be correlated directly with components of archaeal and eukaryotic RNAPs.^{33f} However, the latter contain additional subunits. The two α subunits in the *E. coli* enzyme have identical sequences, but their locations and interactions are different.^{33a–c}

Gene symbol	Subunit	Molecular mass (kDa)	Number of amino acid residues
α	<i>rpoA</i>	36.5	329
β	<i>rpoB</i>	150.6	1342
β'	<i>rpoC</i>	155.2	1407
σ	<i>rpoD</i>	70.2	
ω	<i>rpoZ</i>	6	

The three-dimensional structure of the *E. coli* RNAP bound to DNA in an initiation complex shows

that the enzyme forms a groove into which the DNA can fit. It can then close to form a tunnel in which the template and nontemplate strands are separated (Fig. 28-4). The polymerase is present in large amounts, ~3000 molecules per cell in *E. coli*.²⁶ The β and β' subunits of the *E. coli* enzyme each contain an essential zinc ion. The Zn^{2+} in the β' subunit is present in a zinc finger motif near the N terminus.³⁴ It is thought to interact with the template strand of the DNA. The Zn^{2+} in the β strand is more loosely bound.^{33,35} The active site is largely in the β subunit. In the assembly of the RNAP complex a dimer of the small α subunit binds to β , after which β' is added.³⁶ The sequences of the β and β' subunits have several highly conserved regions with homologous sequences from bacteria and from functionally equivalent regions in eukaryotic RNAPs. Three aspartate residues in the sequence NADFDGD may chelate two Mg^{2+} ions as in the active site of DNA polymerases (Fig. 27-13).³⁷ The basic chemistry of all of the polymerases may be similar, but the modular structure of the bacterial RNAP differs markedly from that of the DNA Pol I family.³⁸

Of the RNA polymerase subunits σ (sigma) plays a unique role in initiation of transcription. It is required for the recognition of promoter sites.³⁹ However, it is not needed for elongation of an RNA chain and dissociates from the $\alpha_2\beta\beta'$ core complex soon after transcription is initiated. In a given bacterial species there is one predominant σ factor, but there are often smaller amounts of other σ factors with homologous sequences.^{26,40,40a} In *E. coli* σ^{70} (where the superscript number is the molecular mass in kDa) is predominant, but other specialized σ subunits recognize different groups of promoters. For example, $\sigma^N(\sigma^{54})$ binds to promoters that allow transcription of genes involved in assimilation of nitrogen^{41,41a} as well as in aromatic catabolism.⁴² Synthesis of protein σ^S of *E. coli* is induced by stress such as carbon starvation.⁴³ At high temperatures (e.g., 40–49°C) σ^{32} is synthesized and permits transcription of genes for “heat shock proteins.”⁴⁴ Actively growing cells of *Bacillus subtilis* contain at least five different sigma factors. An additional four control gene expression during spore formation.⁴⁵

Promoter sequences. In 1975, Pribnow pointed out⁴⁶ that a series of six known promoters had a conserved 7-base sequence beginning six nucleotides upstream from the initiation site for transcription. Although this sequence varies somewhat from one promoter to another, it has been found in hundreds of *E. coli* promoters. This is called the **–10 region**, the **Pribnow sequence**, or Pribnow box (the last in recognition of the fact that people like to draw boxes around these special sequences). A typical 6-base consensus Pribnow sequence is 5'-TATAAT as written for the coding strand, whose sequence corresponds to that of the mRNA. Only three of these bases are highly

conserved: 5'-TA__T. For example, in the *lac* promoter (Fig. 28-2) the sequence is TATGTT. The nucleotides of the -10 consensus sequence are present with the following frequencies (as percentages): T(80)A(95)T(45)A(60)A(50)T(96).⁴⁷ The position of the -10 sequence is not exactly the same in all promoters but usually starts 5–9 bp upstream of the start position for transcription. About 16–19 bp upstream from the Pribnow sequence is another conserved sequence, the **-35 region**. A consensus sequence is TTGACA, the TTG sequence to the left being the most highly conserved. Both -10 and -35 regions are needed for efficient promoter activity. Lewin suggested the following “optimal” promoter sequence.⁴⁷ The location of these two regions in transcription initiation complex is shown in Fig. 28-4B.



Naturally occurring promoters usually do not have the exact -10 and -35 consensus sequences, but artificially constructed promoters containing them are highly effective *in vivo*.²⁶ The fact that most promoters depart from these “ideal” sequences is not surprising because cells need varying amounts of different proteins. Promoter strengths vary over a range of at least 10^4 . Much of this variation comes from variations in the specific -10 and -35 sequences, which appear to be specifically recognized by RNA polymerase. This variation includes an upward extension of the -10 region⁴⁸ for some promoters. Activator-binding sites are also often present in an **upstream activating region (UAR)**; see following diagram). This may extend from the -35 sequence through the remainder of the promoter region and as far upstream as -200 to -500 bp. For example, the CAP-binding site in the *lac* operon DNA is centered at -60 (Fig. 28-2). In spite of the variation it was possible to locate 2584 operators (of which only 392 were previously known) and to predict the location of 2405 promoters, when the complete *E. coli* genome sequence became known.⁴⁹

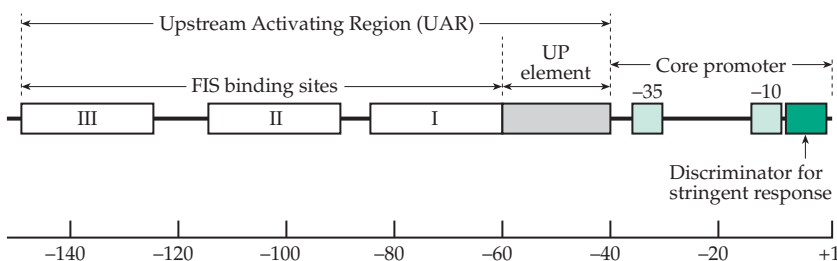
The binding of the various RNAP subunits, repressors, and activators has been studied using mutant promoter sequences,^{31,37} antibiotic-binding sites,^{50,51}

chemical crosslinking reagents,^{34,37} and a cysteine-tethered Fe-dependent DNA-cutting reagent.^{32,52} Other “footprinting” techniques, e.g., observing cleavage of the DNA by hydroxyl radicals generated by reduction of H_2O_2 by Fe(II) (Fig. 5-50), have also been employed. It was shown that RNA polymerase binds to both the -10 and -35 sequences and also to sequences further upstream. The σ^{70} subunit associates with the DNA, principally the transcribed strand, along a region from about the -25 to the +12 position relative to the transcription start site.^{32–32b} The α subunits bind to an UP element from ~ -40 to -60 via their C-terminal domains (CTDs). See Fig. 28-4B.^{53–54d}

Control of stable RNA synthesis. Whereas most mRNA has a relatively short lifetime, the stable ribosomal RNAs and transfer RNAs have much longer lives. Furthermore, in *E. coli* their transcription is coordinately controlled by seven *rrnB* P1 promoters.⁵⁵ The genes for stable RNA have promoters with the usual -10 and -35 sequences, but they contain a complex upstream activating sequence that includes the UP element and three binding sites for protein FIS (**the factor for inversion stimulation**).⁵⁶ This name reflects a second function, that of promoting inversion of a DNA segment in the *Hin* recombinase system (Chapter 27, Section D,3). FIS is a dsDNA-bending protein.^{55,57} A dimer of 11.2-kDa subunits, it is an abundant protein. Like HU, IHF, H-NS, and Dps, it coats a significant fraction of the DNA in the *E. coli* chromosome.⁵⁸ It binds to the FIS sites using a helix–turn–helix motif. In addition, there is a GC-rich region at positions -7 to -1 with the consensus sequence 5'-GCGCC_C. It has been suggested that this **discriminator** is involved in the **stringent response**, the diminished rate of stable RNA synthesis observed during amino acid starvation.^{59,59a}

The very complex stringent response, which involves ribosomes in the synthesis of guanosine 5'-triphosphate-3'-diphosphate (pppGpp) under some conditions,⁶⁰ is dealt with further in Chapter 29 (Section C,8). The regulator pppGpp, whose concentration may rise from $\sim 50 \mu M$ to $\sim 1 mM$ within minutes after deprivation of amino acids, may react directly with RNA polymerase at an allosteric site to inhibit transcription.^{61–62a} Another possibility is that pppGpp acts

on another protein that binds to the discriminator sequence. However, the fact that a phage T7 gene, which is also under stringent control, lacks the discriminator sequence argues against this.⁶³ pppGpp, whose concentration is usually high under conditions of slow growth, may be a major growth-controlling effector, which acts by inhibiting the replication of the rRNA needed for

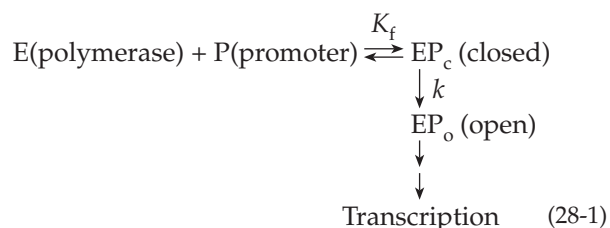


ribosome formation and protein synthesis when nutrients are scarce. However, some operons are activated by the same effector.⁶²

The initiation reaction. A promoter not only locates the site of initiation but also determines the direction of transcription and, therefore, the strand of the DNA duplex that is to serve as the template. The requirement for two specific recognition sequences ensures this directionality. The RNA polymerase may bind randomly to DNA, then move rapidly along the double helix until it locates a strong binding site^{64–66} where it binds to the recognition sequences of the promoter through specific interactions in the major groove of the DNA helix (see Fig. 5-3).

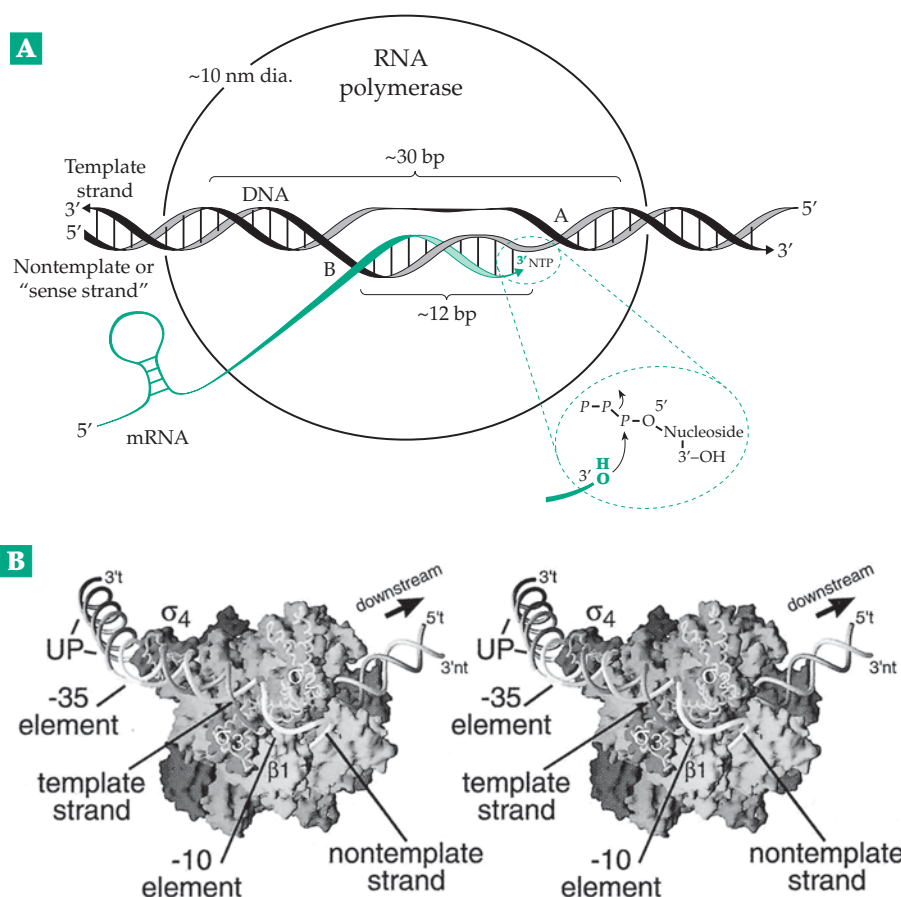
A satisfactory mathematical model for initiation of transcription supposes that the polymerase and DNA bind reversibly to form a complex with formation constant K_f . This initial specific polymerase–promoter complex is referred to as a **closed complex** because it is thought that the bases in the DNA chain are all still paired. It is postulated that in a rate-determining step the closed complex is converted into an **open complex**, which is ready to initiate mRNA synthesis (Eq. 28-1).^{26,67} In the open complex the hydrogen bonds

holding together the base pairs have been broken, and the bases of the template chain are available for pairing with incoming ribonucleotide triphosphates.



It is clear from Eq. 28-1 that the efficiency of initiation depends upon both the affinity K_f and the rate constant k for opening of the double helix. Notice that the Pribnow sequence is AT-rich; therefore, opening of the helix at this point would be easier than in a GC-rich region. Thus, the Pribnow sequence may represent a point of entry of RNA polymerase to form the open complex.⁶⁷ Other upstream A•T tracts are often present frequently at about the –43 position in the UP element. They also seem to strengthen promoter activity.⁶⁸ The open complex is thought to undergo some kind of isomerization to form an **initial transcribing**

Figure 28-4 (A) Hypothetical structure of a “transcription bubble” formed by an RNA polymerase. Shown is a double-stranded length of DNA with the unwound bubble in the center. This contains a short DNA–RNA hybrid helix formed by the growing mRNA. The DNA double helix is undergoing separation at point A as the hybrid helix at point B. NTP is the ribonucleotide triphosphate substrate. See Yager and von Hippel.⁷¹ (B) Stereoscopic view of the structure of RNA polymerase from *Thermus aquaticus* in a complex with a promoter DNA. Included are the α I, α II, ω , β , β' , and σ subunits. However, the α C-terminal domains have been omitted. The template (t) strand passes through a tunnel, which is formed by the β and β' subunits and two of the structural domains of the σ subunit. The nontemplate (nt) strand follows a different path. The position of the –10, –35, and UP elements of the DNA are marked. From Murakami *et al.*^{33d} Courtesy of Seth A. Darst.



complex in which the first chemical steps in RNA formation occur.^{69,70} These initial steps may involve rearrangements of subunit interactions and untwisting of DNA by torsional movements between subunits.⁵⁷

Initiation of an RNA chain begins by reaction within the transcription bubble of either ATP or GTP with a second ribonucleotide triphosphate (Eq. 28-2) to form a dinucleotide still bearing a triphosphate at the 5' end. Further addition of nucleotide units at the 3' end by the same type



of reaction leads to rapid transcription at a rate of ~ 50 nucleotides s^{-1} at 25°C . This is about one-thirtieth the rate of replication. The action of the RNA polymerase is apparently processive, a single molecule of the enzyme synthesizing the entire mRNA transcript.

3. Elongation of RNA Transcripts

After the newly initiated transcript has grown to 8–9 nucleotides, the sigma factor is lost from the RNA polymerase complex, the complex becomes very stable toward increased salt concentrations, and transcription proceeds processively in a **stably elongating mode**. As it does, the strands of the DNA duplex are pulled apart ahead of the polymerase and close up again behind the polymerase, the polymerase itself moving in a “transcription bubble” (Fig. 28-4). This is thought to lie within the ~ 10 nm diameter RNA polymerase complex and to encompass ~ 30 bp of DNA.⁷¹ At the leading edge (point A in Fig. 28-4) a “separator” opens the DNA, which then closes behind the bubble. Behind the polymerase active site (next to substrate NTP in Fig. 28-4) the transcribed RNA forms a short hybrid helix with the DNA, presumably with an A type structure (Chapter 5). A severe topological problem is avoided if the transcribed RNA is separated at point B (Fig. 28-4) as the polymerase moves along the double helix. The driving force for the polymerization lies largely in the hydrolysis of the inorganic PP_i formed in the polymerization (Eq. 17-57). Stabilization of the RNA transcript by formation of loops and other secondary structure may also be a factor.

Although this picture seems clear and simple, many uncertainties remain. Transcription does not proceed evenly but by pauses and spurts. This has suggested the possibility of an “inchworm” type of movement of RNA polymerase.^{72–75} However, the observations may also be explained by variations in the sequence. There are both pausing or stalling sites⁷⁶ and terminator sequences. The concentrations of the needed ribonucleotide triphosphate precursors will also affect the kinetics. In addition, defects in the

DNA will be met. Transcriptionally linked repair (Box 27-A)^{76a} may have to be called into play before transcription can continue.^{77,78} Mismatching of bases can also occur in the growing RNA chain necessitating a pause during which the mismatch is recognized. Specialized proteins **GreA** and **GreB** participate in an editing step during which the RNAP backtracks for a few nucleotides, while a piece is hydrolyzed off from the 3' end before transcription can continue.^{69,79,80}

If the transcription complex moves straight along a DNA double helix, separation of the strands will create positive supercoils (overwound DNA) in front of it and negative supercoils behind the bubble.⁸¹ Experimental data support this prediction.⁸² In *E. coli* the transcription of a plasmid generates positively supercoiled plasmid DNA when DNA gyrase (Chapter 27) is inhibited selectively.⁸³ A similar result was observed in yeast.⁸⁴ These and other data suggest that DNA gyrase may act to remove these positive supercoils, and that topoisomerase I may function in removing the negative supercoils generated behind the transcription bubble.

The electron micrograph in Fig. 28-5 shows RNA polymerase complexes apparently moving along a DNA strand with ribosomes assembled on the RNA and

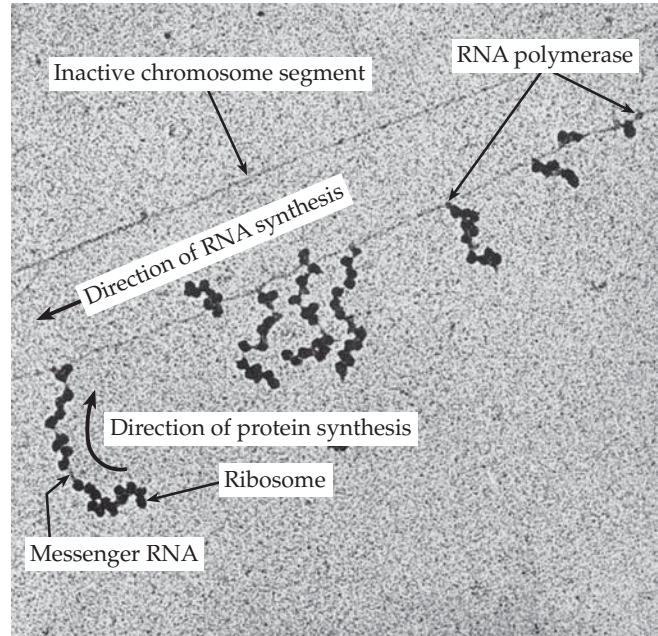


Figure 28-5 Electron micrograph showing transcription from an unidentified operon in *E. coli*. Note the DNA duplexes (horizontal) and the mRNA chains with ribosomes attached. The mRNA chains are shorter at the right side where transcription begins and larger to the left where transcription has proceeded for a longer time. From O. L. Miller, Jr.⁸⁵

presumably synthesizing proteins. Actually, the DNA may be moving through the polymerase complexes.

4. Control of Various Bacterial Operons

With more than 2400 promoters an *E. coli* cell can be expected to utilize a great variety of control mechanisms. The best known of these involve the basic biosynthetic pathways and energy-yielding reactions. Here is a small sample.

Many repressors. Bacteria tend to keep most operons relatively inactive by use of repressors, most of which are proteins. Repressor proteins come in a variety of sizes and three-dimensional structures. Most are oligomers, and all have a DNA-binding motif, often at the N terminus but sometimes at the C terminus or elsewhere. The most studied and perhaps most abundant family⁸⁶ have the helix–turn–helix (HTH) motif. Within this family three-dimensional structures have been established for the *lac* repressors, the *trp* repressor (Fig. 5-35), the 66-residue *cro* repressor,^{87–89} and a 92-residue DNA-binding “head piece” of the 236-residue lambda (cI) repressor.^{90–91} The latter two proteins, both of which occur as symmetric dimers, are involved in preserving the lysogenic state of the λ prophage (Section B,1). The related 71-residue *cro* repressor from phage 434⁹² and the head piece from the phage 434 repressor have similar structures.^{93,94} All of these proteins have the N-terminal HTH structural motif, which binds to DNA and recognizes the appropriate sequence by interactions in the major groove of the DNA. Experiments in which the recognition helix of the 434 repressor was replaced with the corresponding helix from a phage P22 repressor supported the concept of an N-terminal HTH DNA-binding domain.^{95,96} The hybrid repressor bound to the P22 operator rather than to that of phage 434. A similar HTH domain is present near the C terminus in the catabolite-activator protein (Fig. 28-6). The fit of the helix–turn–helix recognition corner varies from one protein to another. The small 53-residue **Arc** repressor of bacteriophage P22 of *Salmonella* has a very different interaction with DNA: the β sheet of an arc dimer fits into the major groove of the operator DNA making specific contacts.^{97,98} A similar interaction characterizes the dimeric methionine repressor.⁹⁹

Basic to the functions both of repressors and of activator proteins are allosterically induced conformational changes caused by the binding of inducers or corepressors. The changes begin at the binding sites of these small effector molecules but are transmitted to the DNA-binding heads. In several cases the conformational changes have been observed by X-ray crystallography and are seen to involve a movement of the recognition helices. This is also true for the *lacI*

repressor (Fig. 28-3) and for the *TrpR* repressor (Fig. 5-35B). Notice that the tryptophan binds to the aporepressor immediately adjacent to the DNA-binding site, where it may control not only the shape but perhaps also the charge distribution within the recognition motif.

Another important factor in determining the strength of a repressor–operator interaction is the twist of the DNA or any other distortion of its regular helical structure. For example, in the center of the *cro* 434-operator complex the DNA is wound, while at the ends it is unwound.⁹² Conformational changes in either the repressor or in the DNA or in both may be needed to provide optimal binding. As is seen in Fig. 28-3 the *lac* repressor causes a distinct bend in the DNA.

A theoretical possibility would be for the DNA in an operator site to be extruded as a cruciform structure like that in Fig. 5-34A. Such structures bind well to certain oligomeric proteins (e.g., see Fig. 27-26B), and they do appear to form in some promoters.¹⁰⁰ However, crystallographic structure determinations have ruled out cruciform structures for many repressors. A change from a linear helical duplex to cruciform structure would require a substantial unwinding of the helix and would mean that negatively superhelical DNA molecules would bind repressor much more tightly than does DNA without superhelical turns. Negative supercoiling does facilitate the *lac* operator–repressor interaction. However, there is only a 50–90° unwinding of the DNA.¹⁰¹ Some AT-rich palindromes are readily converted to cruciform structures when negative supercoiling is increased. Placement of a promoter sequence within such a structure represses transcription unless the supercoiling is relaxed.¹⁰²

Binding of repressors and activator proteins to DNA control sequences is being studied in many ways. Among them are ¹H and ¹⁹F NMR measurements on *lac* and *cro* repressors with specific tyrosine side chains replaced by deuterated tyrosine or 3-fluorotyrosine¹⁰³ or with 5-F-uracil replacing specific thymines in an operator sequence.¹⁰⁴ Footprinting shows that λ and *cro* repressors bind to only one side of the double helix, as is depicted in Fig. 5-50. Addition of dimethyl sulfate to growing *E. coli* causes methylation at many sites in the DNA and has been used to obtain footprints *in vivo*.¹⁰⁵ Not only are certain guanosines in a promoter protected from methylation when active transcription is occurring but also guanosines on the opposite side of the double helix become unusually reactive. This suggested that the DNA helix is bent in the transcriptional initiation complex and that the resulting distortion makes the bases on the outside of the bend more reactive.¹⁰⁵ There is also evidence that the repressor binds to the opposite face of the DNA that binds RNA polymerase in the –35 region of the promoter.¹⁰⁶

Repressors may have similar recognition domains but may vary greatly both in size and in the functioning of their other domains, which may react both with small allosteric effectors and with other proteins. The repressor **BirA** of the *E. coli* biotin synthesis operon is an enzyme. The 321-residue protein activates biotin to form biotiny 5'-adenylate and transfers the biotiny group to proteins such as acetyl-CoA carboxylase^{107–109a} and also represses transcription.

Inducible operons. The operon model as presented in Fig. 28-1 describes the negatively controlled inducible *lac* operon. There are many other examples of this type of control. Among them are control of the utilization of *N*-acetylglucosamine and xylose.¹¹⁰ Ten genes for catabolism of inositol are encoded in two negatively controlled operons in *Bacillus subtilis*.¹¹¹ Similar controls are used by bacteria to protect against the antibiotic tetracycline. The **TetR** repressor controls a membrane associated protein TetA, which acts as an energy-dependent pump to remove tetracycline from the bacterial cell. Synthesis of TetA is normally repressed, but tetracycline, if present, acts as an inducer to cause TetR to dissociate from its operator site, allowing transcription of the transporter gene.^{112–113a} The gene order, gene content, and regulatory mechanisms in an operon are often poorly conserved among related species of bacteria. However, functional and regulatory relationships may be maintained.^{113b}

Feedback repression. A simple modification of the operon model accounts for **feedback repression** by end products of biosynthetic sequences (Fig. 11-1). The product, e.g., an amino acid, vitamin, purine, or pyrimidine base, acts as a corepressor binding to the aporepressor and causing an allosteric modification that inhibits transcription. It is not the aporepressor but the effector–repressor complex that binds to the operon and blocks transcription. An example is the tryptophan–repressor complex, whose 3-dimensional structure is shown in Fig. 5-35. If free tryptophan accumulates within a bacterial cell, it binds to the allosteric site in the *TrpR* repressor, inducing a conformational change that permits the tryptophan–repressor complex to bind to at least three 21-bp operator sites. Binding of the repressor–corepressor complex at the *aroH* operator locus represses the genes for the first three steps in the aromatic biosynthetic pathway (Fig. 25-1, step *a*). Binding at the *trpEDBCA* operator represses the genes for conversion of chorismate to tryptophan (Fig. 25-2), and binding at the *trpR* locus lowers the rate of synthesis of the *trp* repressor, providing a counteracting effect that may be important in controlling the growth rate. The *trp* operon is also well known for another type of control called attenuation (see Fig. 28-9 and associated text).

Feedback repression controls methionine synthesis

in *E. coli*. The corepressor *S*-adenosylmethionine binds to an aporepressor that recognizes the following palindromic sequence in the DNA. This sequence occurs,



with minor variation, in front of four different operons encoding enzymes of methionine biosynthesis.^{99,114} Together they form the methionine **regulon**. Other *E. coli* operons that are negatively regulated by operator–repressor interactions include those involved in biosynthesis of phenylalanine, tyrosine, arginine, threonine, and isoleucine. The *tyr* repressor modulates gene expression in at least eight operons, largely by repression.¹¹⁵ The *ile* repressor acts on both the *ile* and *thr* operons.¹¹⁶ The purine repressor, 341-residue PurR, belongs to the LacI family of repressors. Its C-terminal domain also has some sequence similarity to that of periplasmic sugar-binding proteins.¹¹⁷ PurR represses several steps in the biosynthesis of IMP and in its conversion to GMP and AMP (Fig. 25-15).^{118,119} In *E. coli* twelve biosynthetic genes, organized as nine transcriptional units, provide for the synthesis of arginine in eight enzymatic steps (Fig. 24-10).^{120,121} As with the *trp* operons, synthesis of the repressor ArgR is autoregulated. The hexameric repressor has an N-terminal winged HTH DNA-binding domain.^{121–123}

As is pointed out in Chapter 16, the acquisition of iron and control of its concentration is of crucial importance to bacteria. In *E. coli* the Fe²⁺-binding protein **Fur** (ferric uptake regulator) represses promoters controlling siderophore biosynthesis as well as other responses. It is a **global regulator** that controls ~40 transcriptional units.¹²⁴ Similar proteins repress synthesis of the diphtheria toxin by *Corynebacterium diphtheriae*,^{125,126} uptake of iron in these bacteria and in *Mycobacterium tuberculosis*,¹²⁷ and uptake of molybdate.^{127a}

Positive control by activator proteins. Cyclic AMP in bacterial cells mediates the phenomenon of **catabolite repression**. This is the inhibition of the transcription of genes for enzymes needed in catabolism of lactose and other energy-yielding substrates, when the more efficient energy source glucose is present. Glucose causes a decrease in the concentration of cAMP by a complex mechanism^{59,128} that may also cause a decrease in the concentration of inducer. When the glucose concentration decreases, the concentration of cAMP rises and stimulates the initiation of transcription in many operons. This is accomplished through the mediation of the 209-residue catabolite gene activator protein, CAP (also known as cyclic AMP receptor protein, CRP). This protein is a “global”

regulator of gene expression that activates transcription at over 100 promoters in *E. coli*.¹²⁹ The CAP–cAMP complex binds to the *lac* promoter adjacent to the RNA polymerase site at the palindromic sequence in the DNA as is indicated in Fig. 28-2. The CAP molecule is a 45-kDa dimer of identical subunits, which resembles the repressors in having the HTH reading head that binds to the DNA.^{130–132} However, the HTH motif is at the C terminus. The cAMP binds to two sites in each monomer (Fig. 28-6). Tightly bound cAMP molecules in an *anti* conformation bind in the center of the large N-terminal domains in the major allosteric sites. At higher cAMP concentration the second sites in the C-terminal DNA-binding domains are occupied.¹³¹

The CAP binds to DNA with the consensus sequence 5'-AAATGTGATCT/5'-AGATCACATTT, which may be located at variable distances from the promoter.¹³³ How does binding of the CAP–cAMP complex increase the rate of initiation of mRNA transcription? The answer evidently lies in direct interaction between CAP and the N-terminal domain of the RNAP α subunit.^{54d,129} Binding of CAP induces a 90° bend in the DNA, which may facilitate the protein–protein interaction and may lead to looping.^{130,134}

The galactose (*gal*) operon of *E. coli* is negatively controlled and inducible by D-galactose or D-fucose, which bind to the *gal* repressor. There are two overlapping promoter sites, one of which is stimulated

by adjacent binding of a CAP molecule.^{135,136} A surprise came from the discovery that the operator was *upstream* from the promoter, that is, it comes before both the promoter and the structural genes to be transcribed. Later, a second operator sequence was found 90 bp away from the first and within the first structural gene. This suggested that the dimeric *gal* repressor binds the two operators to form a loop that blocks transcription.

Another example of positive regulation by CAP is provided by the seven proteins required for uptake of maltose and its catabolism by *E. coli*. These are encoded in two operons that are controlled as a single regulon. An apo-activator protein becomes an activator when it binds maltose.^{137,138}

An *E. coli* protein known as **FRN** (for fumarate nitrate reduction) is a global transcription regulator homologous to CAP. It is active only under anaerobic conditions in which it controls more than 100 genes.^{139–141} FRN contains an $[\text{Fe}_4\text{S}_4]^{2+}$ cluster, which is required for dimerization and binding to DNA. Exposure to O_2 converts the cluster into an $[\text{Fe}_2\text{S}_2]^{2+}$ cluster with loss of activator activity. The photosynthetic *Rhodospirillum rubrum* is able to adapt to growth on carbon monoxide as a carbon source. A CAP-like transcriptional activator **CooA** contains heme. It acts as a sensor for CO, which activates transcription, as does cyclic AMP with CAP.^{142,142a} Sequence homologies suggest that several other bacterial activator proteins also have the HTH DNA-binding motif near their C termini.

There are other types of transcriptional activators in bacteria. One is transcription factor 1 (TF1) encoded by a *Bacillus subtilis* phage. It is a member of the protein HU family (Chapter 27). However, unlike the nonspecific HU it binds to some sites specifically and activates transcription.¹⁴³ The *E. coli* Ada protein is the acceptor protein in removal of methyl groups from DNA (Chapter 27). The same protein is an inducer of transcription of DNA repair enzymes in the large *ada* regulon. Methylation of Cys 69 of the Ada protein itself converts it into a gene activator.¹⁴⁴

Control by looping. The arabinose utilization operon of *E. coli*, *araBAD*, encodes proteins needed for uptake of arabinose and conversion to D-xylulose 5-P. The repressor AraC in the absence of arabinose binds at operator 1 (O_1) to prevent further synthesis of repressor (autorepression) and also at the *araI* region to repress transcription of operon *araBAD*. The operator 2 (O_2) site, which is 211 bp upstream from *araI*, is also needed for this repression.^{145–147b} A loop is apparently formed by repressor binding (Fig. 28-7). Binding of arabinose to the repressor converts it into an activator, which stimulates initiation of transcription at the *BAD* promoter. Further stimulation is provided by the CAP–cAMP complex, which binds next at *araI*.



Figure 28-6 MolScript ribbon drawing of the CAP dimer bound to DNA with two molecules of the coactivator cAMP bound per monomer. A *syn*-cAMP molecule is bound to the HTH domain and a loop from the N-terminal domain, while the second *anti*-cAMP is bound more tightly in the center of the larger N-terminal domain. The DNA sequence for each half site is 5'-ATGTCACATTAATTGCGTTGCGC-3'. From Passner and Steitz.¹³¹ Courtesy of Thomas A. Steitz.

Looping is a recognized control mechanism for a number of other operons as well.¹⁴⁸

Bacterial enhancers. Positive regulatory DNA sequences that are distant from the genes controlled are often called enhancers.^{149,150} Their function is usually independent of position over a range of hundreds or more base pairs either upstream or downstream of the transcription initiation site. Quite common in eukaryotes (Section C.4), enhancers are less often found in bacteria. However, the binding sites for the **nitrogen regulatory protein C (NtrC or Nr_I)** of *E. coli* has the characteristics of an enhancer.^{151,152} It functions with the rather complex glutamine synthetase (*glnALG*) operon in a major control point for nitrogen metabolism. The enzymology is illustrated in Fig. 24-7. When the supply of nitrogen from NH₃ is low the NtrC protein, a product of gene *glnG*, binds to the enhancer, which is located over 100 bp upstream and is thought to contact the σ^N (σ^{54}) subunit of the RNAP by formation of a loop.^{153,153a} The process has been visualized by scanning force microscopy.¹⁵² Another interesting aspect of this control system is the activation of Ntrc by phosphorylation of a specific aspartate side chain (Asp 54). The NtrC-P form is the active enhancer-binding protein. NtrC is a member of the family of two-component sensor-response regulator pairs, which frequently control bacterial metabolism and behavior (Chapter 11, Section C, 2; Fig. 19-5). The sensor protein is **NtrB (Nr_{II})**, which is an auto-phosphorylating histidine kinase similar to the CheY protein of bacterial chemotaxis (Fig. 19-5).

NtrC-P dimerizes and binds to the enhancer sequence, where it appears to catalyze an ATP-dependent isomerization of the closed to open forms of the transcription initiation complex (Eq. 28-1).^{153,154} The isomerization may depend upon looping.¹⁵² Other operons that utilize the σ^N subunit of RNAP often also have upstream or downstream enhancers.^{155,156}

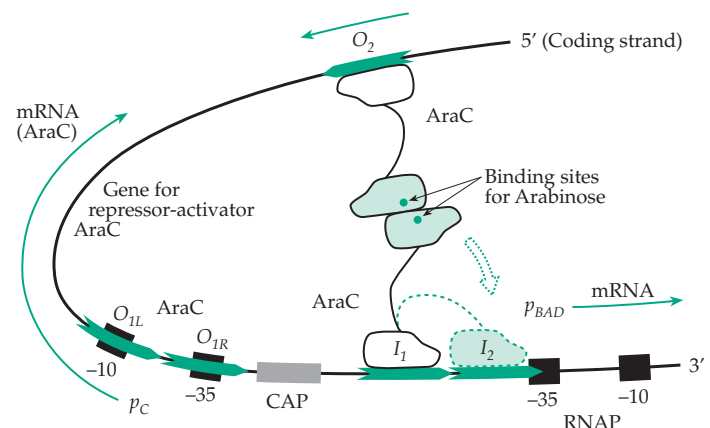
Other two-component control systems. More than 60 different sensor-response regulator pairs have

been discovered in bacteria. Many are associated with nutrition. For example, more than 30 genes of the phosphate (*pho*) regular are controlled by the sensor kinase **phoR**, which detects low phosphate ion concentrations and phosphorylates the response regulator **phoB**.^{157–159} Protein **ArB** senses changes in environmental O₂ levels, and response regulator **ArcA** regulates ~30 operons in response.^{159a} One effect is to activate ~30 genes needed for the conjugative transfer of DNA (Fig. 26-3).¹⁶⁰ Transcription of rhizobial *nif* and *fix* genes (Fig. 24-4) is controlled by the O₂ sensor **FixL** and its response regulator **FixJ**.¹⁶¹ FixL is a heme protein whose kinase is active only when the heme is deoxygenated. The *E. coli* proteins discussed on p. 1075 mediate transcriptional responses to accumulation of superoxide anions or hydrogen peroxide.^{161a}

In the absence of O₂ the *E. coli* FNR protein induces proteins of the anaerobic respiration pathways. Nitrate also has its own two-component system that senses nitrate availability and activates transcription of enzymes catalyzing nitrate respiration.¹⁶² An expanded two-component system induces sporulation in *Bacillus subtilis* in response to poor growth conditions.^{163,164} The crystal structure of one of two response regulators (**Spo0F**) has a structure closely related to that of CheY and the nitrate response regulator NarL.

Antisense RNA. Another mechanism of control of either transcription or of plasmid replication involves small molecules of RNA that are transcribed from the opposite strand than the template strand used for mRNA synthesis.^{1,1a,165–166b} These **antisense** RNA molecules have at least some part of their sequence complementary to that of the mRNA and to the corresponding sequence in DNA. A well-studied example is control of the copy number of the colicin E1 and other plasmids of *E. coli*.^{167–169} Two transcripts, RNAI and RNAII, are initiated upstream from the origin of replication (Fig. 28-8). RNA II is a 555-nucleotide primer of replication. It is synthesized as a longer transcript that is cut by RNase H at *ori*. This

Figure 28-7 Scheme showing regulatory region of the *araCBAD* operon of *E. coli*. In the absence of arabinose the protein AraC acts as a repressor, which binds to upstream region I of the promoter and also to operator O₂, forming a loop of ~210 bp of DNA. Binding of arabinose to the N-terminal domains of the dimeric repressor allows the dimer to dissociate and the N-terminal domain of one subunit to bind to region I₂ of the promoter for genes *B, A, D* (dashed outline) activating their transcription. The AraC protein also binds to promoter P_C, repressing its own synthesis. After Zhang *et al.*¹⁴⁶



cleavage is inhibited if the 108 nucleotide RNA I forms a duplex with the 5' end of the RNA II, which has a complementary base sequence. The process is more complex than this because both RNA II and RNA I have complex secondary structures and are brought together with the help of the small protein product of gene *rop*, which permits them to recognize each other prior to duplex formation.

The major outer membrane porins of *E. coli* (Fig. 8-20) are encoded by genes *ompC* and *ompF*. A small 174-nucleotide RNA called **mRNA-interfering complementary RNA** (mic RNA), whose gene is upstream of the *ompC* promoter, is transcribed in the opposite direction from that of the porin gene.¹⁶⁷ It is homologous with the 5' end of *ompF* mRNA, and its function is evidently to inhibit translation of the *ompF* mRNA. Since *ompC* and the *mic*RNA gene are apparently regulated coordinately, synthesis of the *ompF* product is inhibited if the *ompC* product is being synthesized, as happens when the bacteria are growing in a medium of high osmotic strength.^{166b,170,171}

Many other examples of regulation by antisense RNA are being discovered. Small noncoding RNA molecules (**ncRNAs**) often serve as templates or **guides**^{1,171a} in processes ranging from synthesis of telomere ends (Fig. 27-21) to editing,¹⁷² modifying,¹⁷³ and splicing mRNA¹⁷⁴ (Section D). Both in bacteria^{174a} and in eukaryotes (p. 1640) dsRNAs are often formed and subsequently cleaved to single-stranded antisense RNAs that act as guides to initiate hydrolytic destruction of defective, toxic, or unwanted mRNAs.^{171a} The recognition of these natural regulatory mechanisms has led to keen interest in artificial regulation by antisense RNA. Synthetic antisense RNA injected into

cells will inhibit expression of selected genes. There is the possibility of effective therapy against viruses, cancer, and inflammation if suitable antisense RNA could be generated within eukaryotic cells or introduced as drugs.^{175–176b} Such drugs, which are typically ~15 nucleotides in length, are most satisfactory if they are stable, enter cells, and interact specifically with complementary sequences of DNA or RNA.¹⁷⁷ Stability can be improved by use of linkages other than the natural phosphodiester.^{178–181} Phosphorothioates, in which sulfur replaces one nonbridging oxygen atom on phosphorus, is a favorite. Synthetic antisense oligonucleotide mimics may cause adverse reactions with proteins within cells.^{178,182} Nevertheless, future successes seem likely.^{182a} See also RNA interference (p. 1640).

5. Termination of Transcription in Bacteria

Encoded in DNA are not only the initiation signals for transcription but also termination signals or **terminators**.¹⁸³ Some of these are sensed by the bacterial RNA polymerase itself while the “reading” of others requires auxiliary proteins. Terminators can be either constitutive or regulatable. The simplest terminators result from GC-rich regions of dyad symmetry in the DNA. The RNA transcript is able to form a stable hairpin loop, possibly within the transcription bubble. If such a loop is followed closely by a series of uracils, the RNA and the polymerase will dissociate from the DNA template terminating transcription. The low stability of AU base pairs may facilitate dissociation, but RNA polymerase may also recognize the terminator loop. Sometimes a terminator will have a series of adenines preceding the loop. This is often a bidirectional terminator; the transcript from the other strand of DNA will have a loop followed by a series of U's. There are many more complex termination mechanisms.

Attenuation. A major mechanism of feedback repression, known as attenuation, depends not upon a repressor protein but upon control of premature termination. It was first worked out in detail by Yanofsky *et al.* for the *trp* operon of *E. coli* and related bacteria.^{184–186} Accumulation of tryptophan in the cell represses the *trp* biosynthetic operon by the action of accumulating tryptophanyl-tRNA^{Trp}, which specifically induces termination in the *trp* operon. Other specific “charged” aminoacyl-tRNA molecules induce termination at other amino acid synthesis operons. The first structural gene in the *trp* operon,

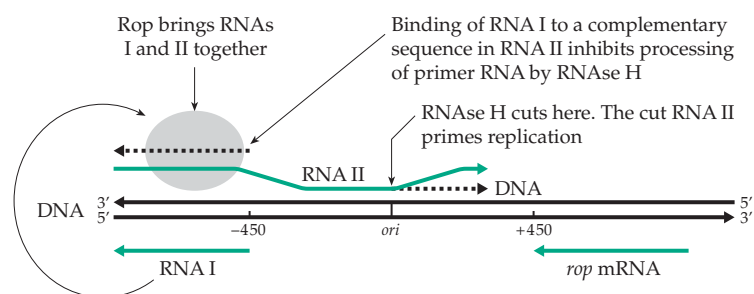


Figure 28-8 Simplified scheme for control of replication of the ColE1 type plasmid by antisense RNA. The primer for DNA synthesis is RNA II whose initial transcript extends past the replication *ori*. It is cut by RNase H at *ori* and then primes replication of the upper strand as shown in the figure. The antisense RNA is RNA I. It binds to protein Rop whose gene location is also indicated in the figure. Rop assists RNA I and RNA II in undergoing a complementary interaction. However, both RNAs apparently maintain a folded tertiary structure, and only some segments interact. The interaction with the Rop protein evidently in some way prevents initiation of replication until the Rop concentration falls because of replication of the host cell.^{167,168}

trpE, is preceded at the 5' end by a 162-bp **leader sequence**, which is transcribed into mRNA. Within this RNA are two adjacent hairpin loops. The second of these loops (labeled **3:4** in Fig. 28-9) has a GC-rich stem and is followed by eight consecutive U's. It is a typical efficient terminator. An RNA polymerase, having just passed this sequence, will interact with the looped RNA formed behind it and will dissociate from the DNA to terminate transcription. However, if the terminator loop is prevented from forming, transcription will continue, and the structural genes of the operon will be expressed.

The *trp* operon contains a short gene for a **leader peptide** preceding the terminator. Its RNA transcript is shaded, and its initiation codon AUG and termination codon AAU are boxed in Fig. 28-9. In bacteria translation of mRNA begins while transcription is still in progress (Fig. 28-5). The 5' end of the mRNA may enter a ribosome before the RNA polymerase reaches the terminator loop, after which the leader peptide will be synthesized. The ninth and tenth codons of the leader peptide gene are of special importance. They lie at the beginning of the **1:2** or "protector" loop (Fig. 28-9) and code for tryptophan. If the level of tryptophan is high, tryptophanyl-tRNA^{Trp} will be formed and the mRNA will move rapidly through the ribosome, these tryptophan codons will pass through, and tryptophan will be incorporated into the peptide. The 1:2 loop will be opened, but the terminator loop will remain intact. The result will be termination of transcription. However, if the tryptophan concentration is low, there will be a shortage of charged tRNA^{Trp}, and peptide synthesis will be "stalled" with these tryptophan codons in the active sites of the ribosome. This will allow time for the attenuator region to assume the alternative secondary structure shown in Fig. 28-9B. Here the **1** limb of the protector loop is stalled in the ribosome allowing the **2** limb to form the alternative **2:3** loop. The terminator has been destroyed, and transcription continues through the rest of the operon. Thus a low level of tryptophan (and of tryptophanyl-tRNA) favors transcription of the *trp* synthetic operon.

Cells of *Bacillus subtilis* also synthesize a *trp* operon transcript that can form either an antiterminator or a terminator loop (Fig. 28-9C).^{187,187a-c} Tryptophan, when present in a high enough concentration, binds to a *trp* RNA-binding attenuation protein (TRAP). This is an 11-subunit protein, which has 11 tryptophan-binding pockets and also 11 binding sites for GAG or UAG RNA triplets. When tryptophan accumulates within the cell it binds to TRAP, which then wraps ~53 residues of RNA transcript containing 11 GAG or UAG triplets around its perimeter (Fig. 28-9D). This prevents formation of the antiterminator loop but allows the terminator loop to form. At low tryptophan concentrations the antiterminator loop is formed, preventing formation of the terminator loop.¹⁸⁸

Attenuation is also an important mechanism of control of transcription of biosynthetic operons for histidine, phenylalanine, leucine, isoleucine, and threonine.¹⁸⁹ Like the *trp* attenuator region, attenuators for these operons contain codons for the amino acid whose synthesis is being regulated: seven Phe codons in the *phe* attenuator, seven His codons in the *his* attenuator, four Leu codons in the *leu* attenuator. The *thr* operon, which is sensitive to both threonine and isoleucine, has eight Thr and four Ile codons, while the *ilv* attenuator has four Leu, five Ile, and six Val codons permitting feedback repressor by three kinds of charged tRNA.¹⁹⁰ The pyrimidine synthesis operon *pyr* has three attenuator sequences, one right after the promoter and two others, one just before each of the two genes in the operon. This permits partially independent control of the two genes.¹⁹¹

Rho and other termination factors. Termination proteins can also react with specific regions of DNA or of an RNA transcript to terminate transcription.¹⁸³ The best known termination factor is the rho protein; a hexamer of 45-kDa subunits. It interacts with transcripts at specific termination sequences, which are often C-rich, and in a process accompanied by hydrolysis of ATP causes release of both RNA and the polymerase from the DNA.^{192,193} Additional *E. coli* proteins, products of genes *nus A* and *nus G*, cooperate with the rho factor at some termination sequences.^{194-196c} The rho hexamer is a helicase that moves along the RNA transcript in the 5' → 3' direction driven by ATP hydrolysis. If it locates an appropriate termination signal, it may utilize its helicase activity to uncoil the DNA-RNA hybrid segment within the transcription bubble (Fig. 28-4).^{197-198b}

Cells also contain **antitermination proteins**, which prevent termination of transcription of rRNA or tRNA genes at the many loops of secondary structure that are possible with these transcripts.^{59,199-200b} These antitermination factors are also important in regulating transcription during the lytic phase of growth of phage λ (see Section B,1). Also important are rates of hydrolytic degradation of mRNA molecules.²⁰¹

6. Effects of Antibiotics

The antibiotic **rifamycin** (Box 28-A) appears to interfere with initiation by competing for the binding of the initial purine nucleoside 5'-triphosphate. The same bacterial RNAP that synthesizes mRNA also transcribes both rRNA and the tRNAs. Thus, the synthesis of all forms of RNA is inhibited by rifamycin. When a population of bacteria is subjected to this antibiotic, a few individuals survive. These rifamycin-resistant mutants are no longer sensitive to the antibiotic. Among them are some mutants that produce an

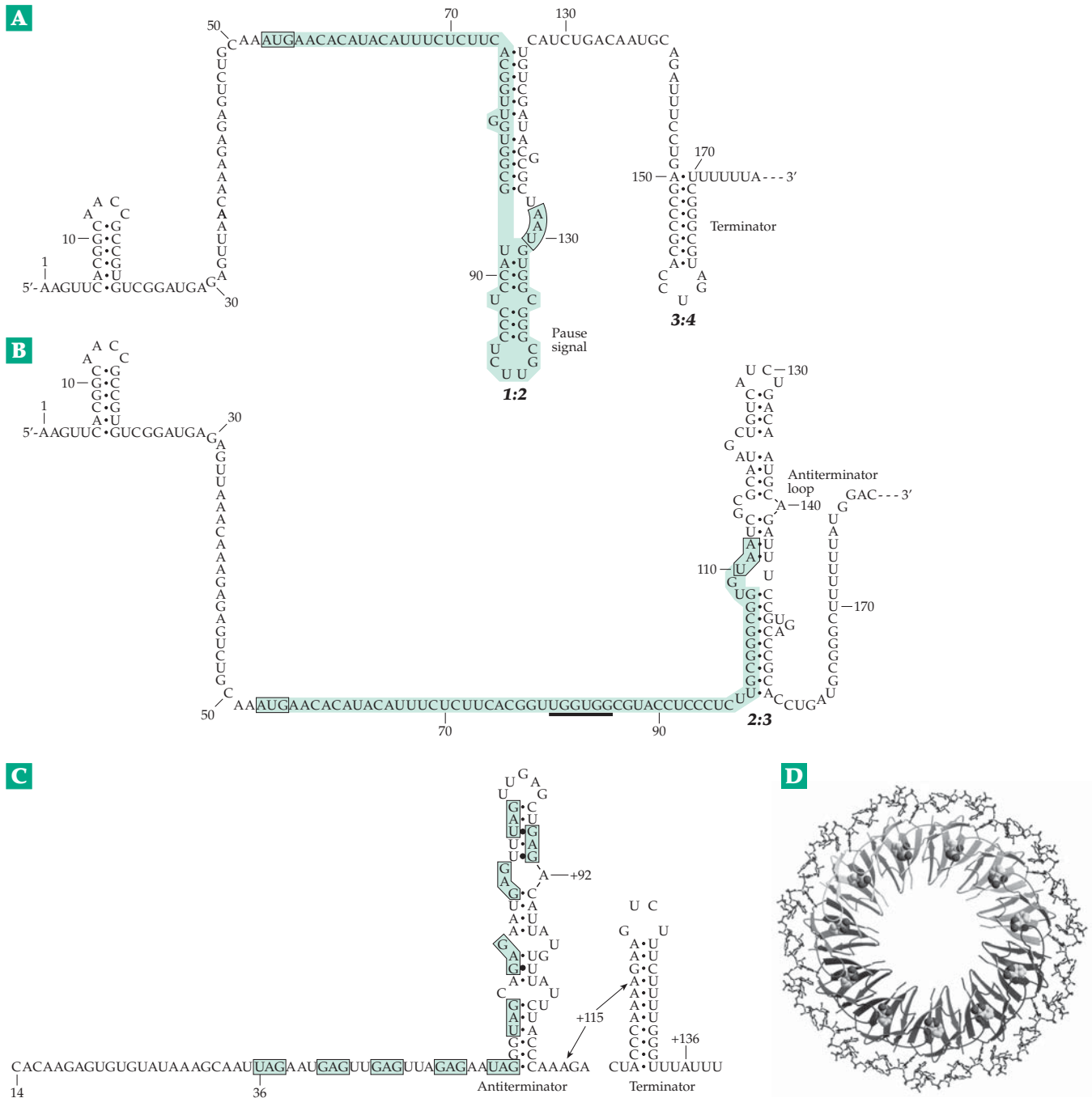


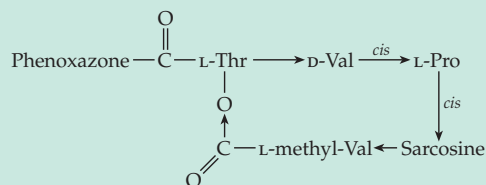
Figure 28-9 (A,B) Alternative leader RNA structures that mediate control of attenuation in the *trp* operon of *Serratia marcescens*. The leader peptide initiation and termination codons are boxed. Tryptophan codons are indicated by underlining in B. Formation of the 5' hairpin structure is predicted by computer analysis but is not implicated in control by attenuation. A structure **1:2** is thought to serve as a transcription pause signal. Structure **3:4** is predicted to form when there is an adequate supply of charged tRNA^{Trp} and is thought to function as the transcription termination signal recognized by RNA polymerase. The **2:3** structure, or **antiterminator loop**, is predicted to form when charged tRNA^{Trp} is unavailable. Its formation presumably precludes formation of structure **3:4**, thereby allowing RNA polymerase to continue transcription into the structural genes of the operon. From Kuroda and Yanofsky.¹⁸⁶ (C) Antiterminator and terminator loops, one of which may form from the leader sequence of the *Bacillus subtilis trp* operon mRNA. Numbering refers to the start point of transcription. The triplet repeats involved in attenuation are shaded. From Baumann *et al.*¹⁸⁷ (D) Structure of the 11-subunit tryptophan RNA-binding attenuation protein (TRAP) as a ribbon diagram with 11 molecules of L-tryptophan shown as van der Waals spheres. The apparently circular RNA structure reflects the fact that the gap between the beginning and end of the bound RNA segment is averaged (randomized) over eleven binding sites in the crystal structure. The 53-residue RNA containing 11 triplet repeats of GAG or UAG is bound around the perimeter and is shown as a ball-and-stick model. From Antson *et al.*¹⁸⁸

BOX 28-A THE ANTIBIOTICS RIFAMYCIN, RIFAMPICIN, AND ACTINOMYCIN D

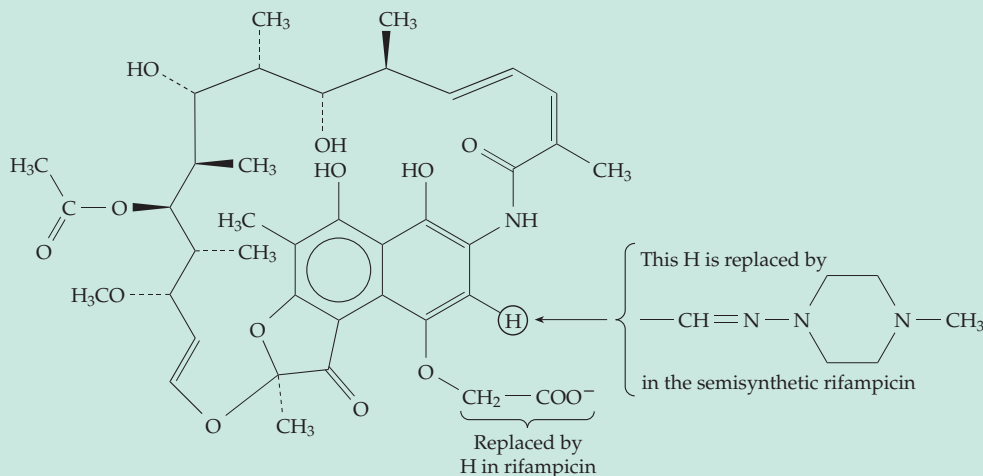
Rifamycin, produced by *Streptomyces mediterranei*, is of medical value because it affects **acid-fast** as well as gram-positive bacteria. The semisynthetic rifampicin has been especially useful in the treatment of tuberculosis. The ether linkage at the bottom of the ring at the right in the structural formula is cleaved, and the resulting hydroquinone is oxidized to a quinone within the bacteria.^a At a concentration of 2×10^{-8} M rifampicin inhibits bacterial RNA polymerase 50%. It does not prevent the binding of polymerase to DNA but inhibits initiation of transcription. Mutants of *E. coli* resistant to rifampicin produce RNA polymerase whose β subunit has been altered, sometimes with a change in electrophoretic mobility. The related antibiotic **streptolydigin** also binds to the subunit of RNA polymerase and blocks elongation, resistant mutants mapping close to *rif* mutants.

The actinomycins, which are also produced by *Streptomyces*, not only kill bacteria but also are among the most potent antitumor agents known.^b However, because of their extreme toxicity they are of little medicinal value. Actinomycin D, which is a specific inhibitor of RNA polymerase, contains a planar phenoxazone chromophore bearing two carboxyl groups, each one linked to an identical cyclic peptide made up of L-threonine, D-valine, L-proline, sarcosine (*N*-methylglycine), and L-methylvaline.

An ester linkage joins the methylvaline residue of the peptide to the side-chain hydroxyl of threonine. Two *cis* peptide linkages are present. Ignoring the obvious asymmetry of the phenoxazine ring, actinomycin possesses approximate twofold symmetry.



The antibiotic binds tightly to double-stranded DNA in regions containing guanine. A 2:1 deoxyguanosine-actinomycin complex has been crystallized, and the structure has been determined by X-ray diffraction.^{c-e} The phenoxazine ring lies at the center of the complex, one peptide loop extending above it. The twofold symmetry is present in the dideoxyguanosine complex as well as in actinomycin itself. The phenoxazine ring lies between the two flat guanosine rings in van der Waals contact. The two amino groups of the guanine rings form strong hydrogen bonds with the carbonyl groups of the threonine residues. There are also weaker non-linear hydrogen bonds from the N-3 atoms of the guanines to the NH groups of the same threonines.



RNAP with an altered β subunit. Since the mutant polymerases do not bind rifamycin, it was concluded that rifamycin binds to the β subunit and that the rifamycin-resistance gene *rpoB* or *rif* (which maps at 89 min) is the gene for the β subunits of RNA polymerase.

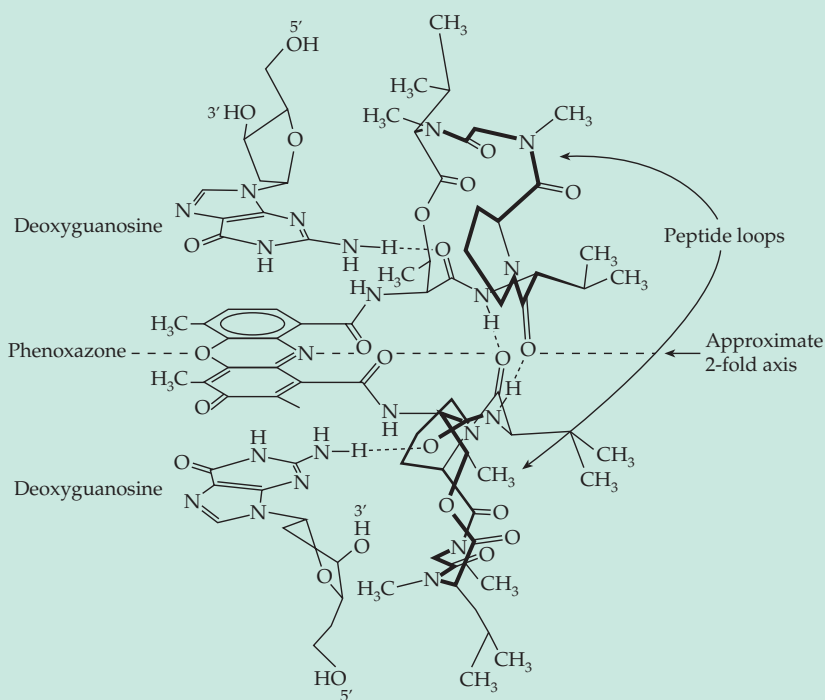
Streptolydigin inhibits both initiation and elongation. **Actinomycin D**, by binding to DNA, inhibits DNA polymerases as well as RNA polymerases, the

latter at a concentration of only 10^{-6} M (Box 28-A). The eukaryotic RNA polymerases are not inhibited by rifamycin, but RNA polymerases II and III are completely inhibited by the mushroom poison **α -amanitin** (see Box 28-B). Inhibitors of DNA gyrase (Chapter 27) also interfere with transcription as do chain terminators such as cordycepin (3'-deoxyadenosine) and related nucleosides.

BOX 28-A (continued)

A symmetric pair of hydrogen bonds join the two carbonyl and NH groups of the D-valine residues in the peptide loops.

Model building studies show that a similar complex can be formed with double-stranded DNA.^f While the amino groups of the guanine rings (see drawing) are hydrogen-bonded to the actinomycin, the other hydrogen atoms of the same amino groups as well as the N-1 hydrogen atoms and the carbonyl groups of the guanine ring are available for hydrogen bonding to form a GC base pair. Thus, the structure above can be modified readily into part of the double-stranded DNA molecule in which the phenoxazone ring of actinomycin is intercalated between two CG pairs. To do this the normal DNA structure has to be unwound by 18° at the point of insertion of the extra ring. Binding also occurs at other sites.^f Sobell suggested that actinomycin binds to a premelted region of the DNA helix within the transcription bubble and immobilizes it. This interferes with the elongation of growing RNA chains.^g



^a Goldberg, I. H., and Friedman, P. A. (1971) *Ann. Rev. Biochem.* **40**, 775–810

^b Perlman, D. (1970) in *Medicinal Chemistry*, 3rd ed. (Burger, A., ed), pp. 309–316, Wiley (Interscience), New York (Part 1)

^c Sobell, H. M. (1973) *Prog. Nucleic Acid Res. Mol. Biol.* **13**, 153–190

^d Sobell, H. M., Jain, S. C., Sakore, T. D., and Ponticello, G. (1971) *Cold Spring Harbor Symp. Quant. Biol.* **36**, 263–270

^e Sobell, H. M. (1974) *Sci. Am.* **231**(Aug), 82–91

^f Robinson, H., Gao, Y.-G., Yang, X.-I., Sanishvili, R., Joachimiak, A., and Wang, A. H.-J. (2001) *Biochemistry* **40**, 5587–5592

^g Sobell, H. M. (1985) *Proc. Natl. Acad. Sci. U.S.A.* **82**, 5328–5331

7. Processing of Bacterial RNA

Newly formed rRNA and tRNA molecules are usually not functional but must undergo chain cleavage, methylation, and other alterations before they are “mature.” However, most bacterial mRNA does not require processing. Bacteria produce a series of mRNA molecules of variable length, some corresponding to polycistronic (polygenic) and some to monocistronic operons. Most of the mRNA molecules produced are unstable with an average lifetime of about two min; however, some, such as those produced in bacteria about to undergo sporulation, survive much longer. Bacterial mRNA sometimes does undergo processing before it reaches the ribosomes. For example, following infection of *E. coli* cells by phage T7, ribonuclease RNase III cleaves the large 7000 nucleotide “early RNA” transcript from the virus DNA into five defined fragments.²⁰² Each fragment presumably carries the message for a single viral gene.

Genes containing introns have been identified in several archaeobacteria^{203–205} and in certain phage.²⁰⁶ The corresponding transcripts must be spliced, as are most eukaryotic transcripts.

Ribosomal RNA. Quantitatively the most important RNA, making up 90% of that present in cells, is ribosomal RNA. Synthesis of rRNA must be rapid, for an *E. coli* cell produces 5–10 new ribosomes per second, or 2×10^4 molecules of RNA per generation. Bacterial ribosomes contain three pieces of RNA. These are designated, according to their sedimentation constants, as 5S, 16S, and 23S and contain about 120, 1700, and 3300 nucleotides, respectively. All three pieces appear in cells as parts of larger **pre-rRNA** precursor molecules with extra nucleotide sequences at both the 3' and 5' ends.^{207,208}

There are seven rRNA regions in the *E. coli* chromosome.^{208a} Each region consists of a single transcriptional unit containing a gene each for 16S, 23S, and 5S

rRNA with interspersed tRNA genes as follows: 16S, tRNA, 23S, tRNA, 5S, tRNA. A single transcript (which can, in certain mutant strains, appear as a 30S molecule) is cut by the endonuclease RNase III into the smaller pre-rRNA molecules.^{209,210} Other nucleases trim these to their final sizes, and methylases act to modify 24 residues in *E. coli* rRNAs.^{208,211–213} Most RNAs of all organisms contain **pseudouridine** (ψ), which is formed by isomerization of specific uridines present in the RNAs (Eq. 28-3). In *E. coli* there is one pseudouridine in the 16S ribosome RNA and nine in the 23S RNA as well as one or more in most tRNAs.^{214–217} The isomerization depends upon a carboxylate group of the enzyme, which evidently adds to the 6-position of the uracil ring to form a pivot around which the ring can rotate after it is eliminated from its attachment to the RNA and before it is reattached with a C–N linkage (Eq. 28-3).²¹⁷ Both bacteria and eukaryotes contain several pseudouridine synthases, which act to isomerize specific uridine residues in the precursor RNAs.^{218,218a,b} In eukaryotes special **guide RNAs** direct the pseudouridine synthases to specific locations in their substrates.^{174,219} The same thing is true for 2'-O-methylases that modify selected ribose rings in precursor RNAs.^{219,220}

Transfer RNA. The genes for tRNA molecules in both bacteria and mammalian cells are grouped in clusters, which are transcribed as large precursors sometimes containing more than one kind of tRNA or containing tRNA fused to rRNA or mRNA sequences. At least three different nucleases are needed for cutting and trimming to form the mature tRNA. These enzymes may not always act in the same sequence. Thus, for some but not all tRNAs cleavage near the 3' end is needed before cleavage can take place at the 5' end.

The best known processing nuclease is **RNase P**, which cleaves bacterial tRNA precursors to create the 5' ends of the mature tRNAs. All of the 64 tRNA precursors present in *E. coli* are cleaved by this unusual enzyme,^{221–222c} which contains an essential piece of RNA (Chapter 12, Section D.6). Cleavage of polycistronic tRNA precursors by RNase P or of the previous-

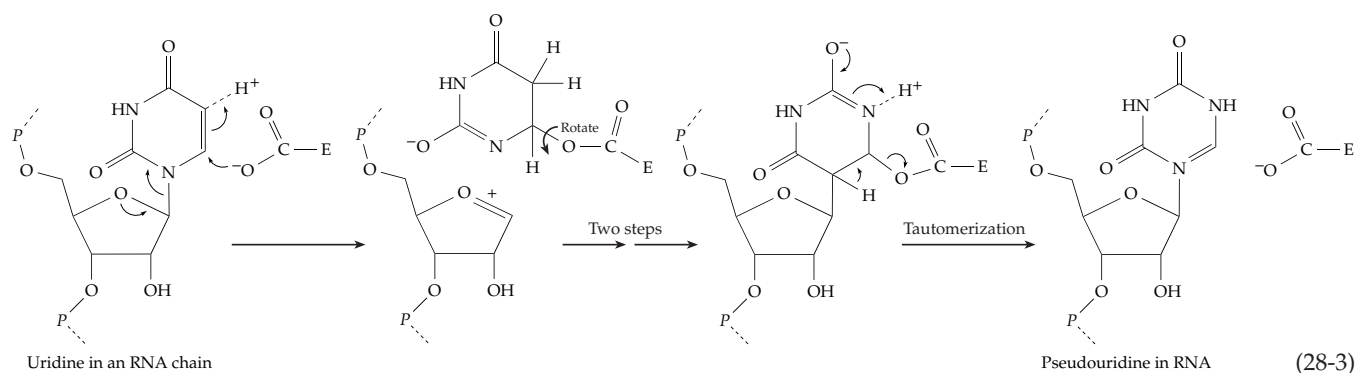
ly mentioned rRNA and tRNA precursor by RNases P and III releases mature tRNAs carrying extra nucleotides at the 3' ends are removed by exoribonucleases, a surprising number of which are present in cells. In *E. coli* they include RNases II, D, BN, and T,^{223,224} as well as polynucleotide phosphorylase.²²⁵

The structure of the precursor to the minor *E. coli* tyrosine tRNA₁ is shown in Fig. 28-10. This is encoded by the *amber* suppressor gene *SupF* (see Chapter 29). Transcription of its gene is initiated by GTP 43 bp upstream of the 5' end of the mature tRNA and usually terminates at a ρ -dependent signal 225 bp beyond the CCA terminus of the tRNA. An endonuclease cuts the transcript a few nucleotides beyond the CCA end. It is then trimmed to an ~130-nucleotide piece still containing 2–3 extra nucleotides at the 3' end. This intermediate is cut by RNase P at the 5' end after which final trimming is done at the 3' end (Fig. 28-10).^{226,227}

An important chemical achievement was the synthesis by H. G. Khorana and associates of the double-stranded DNA segment coding for this *E. coli* tyrosine tRNA²²⁸ and its precursor.²²⁹ This was one of the first synthetic genes (Chapter 5). The synthesis was extended to include the gene termination region, which lies beyond the CCA end of the tRNA. Two noteworthy features appeared. There is a local center of dyad symmetry (indicated by vertical bars and a central dot in Fig. 28-10), which may serve as a termination signal. The operator is located in the 29-nucleotide sequence preceding the tyrosine tRNA gene.^{230–232}

The 3'-terminal group of three nucleotides, CCA, is invariant among all tRNA molecules and is labile, undergoing active removal and resynthesis. The rate of this turnover is sufficient to involve about 20% of the tRNA molecules of a cell per generation, but it is very much slower than the rate of participation of the tRNA molecules in protein synthesis. The physiological significance of end turnover is unknown.²³³ While this CCA sequence is encoded in bacterial tRNA genes, it is added in a separate reaction in eukaryotes.²³⁴

In addition to the cutting and trimming of precursors by nucleases, extensive modification of purine and pyrimidine bases is required to generate mature tRNAs.²³⁵ Some of these modification reactions are



(28-3)

vitamin. However, germ-free mice reared on a queuine-deficient diet seemed normal. Therefore, the essentiality of queuine in the human diet is in doubt.^{242,243}

B. Viral RNA in Prokaryotes

Bacteria not only transcribe their own genes but sometimes transcribe, or assist in transcribing, genes of invading DNA viruses or of integrated proviruses. In addition, they assist in replication of RNA viruses, another process that requires RNA synthesis. Viruses sometimes make use of host RNA polymerases but often synthesize their own catalytic subunits. Bacteriophage T4 uses the *E. coli* RNA polymerase and σ factors but modifies their action through the binding of several phage-encoded proteins.²⁴⁸ In contrast, phage T7 encodes its own relatively simple RNAP whose initiation complex (Section A,2)²⁹ and elongation complexes have been studied.^{249–249b}

1. The Lysogenic State of Phage λ

The study of bacteriophage lambda has provided many insights into biological process.²⁵⁰ As we have seen (Chapter 27), the DNA of phage λ can become incorporated into the genomic DNA of *E. coli*. The resulting prophage contains many genes (Fig. 28-11), but they remain largely unexpressed until the SOS signal (Chapter 27) is generated. Certain prophage genes are then expressed with the result that the λ DNA is excised as a replicating virus.²⁵¹ How can the λ genes remain unexpressed in the prophage but be expressed rapidly at the time of excision? Part of the answer has been found^{252–254} in the *cI* and *Cro* repressors. The short L1 operon (Fig. 28-11) of the λ prophage is transcribed continuously by the *E. coli* RNA polymerase. This operon contains genes *cI* and *rex*, which are transcribed from the *l* strands of the prophage DNA as indicated in Fig. 28-11. The protein C_I (or CI) specified by gene *cI* is the **lambda repressor**, which binds to two operator sites in the prophage DNA. One operator (o_L) is to the left and the other (o_R) to the right of the *cI* gene. From a study of fragments of DNA protected by the repressor, it was concluded that each operator has three subsites, which are filled from left to right at o_L and from right to left at o_R successively by up to six repressor monomers. Each presumed subsite has a similar 17-bp quasipalindromic sequence to which a dimeric repressor can bind. The binding is cooperative, probably because the repressor molecules contact each other, apparently binding the DNA into a loop.

The right operator o_R controls not only the R1 operon but also the L1 operon, which encodes the λ repressor (*cI* gene). The first of the three subsites in

the operator is adjacent to the L1 promoter P_{RM} , and binding of λ repressor activates that promoter at the same time that binding to the adjacent subsites blocks transcription of the R1 operon. Thus, the λ repressor positively controls its own synthesis. At the same time, blocking of promoters P_R and P_L prevents synthesis of virally encoded enzymes that catalyze excision of the λ DNA and replication and transcription of the rest of the genes.

The SOS signal causes rapid hydrolytic cleavage of the λ repressor and transcription of the other λ operons. The matter is more complex than this. Gene products *cIII* and *cII*, from the **early left** and **early right operons**, respectively, stimulate the transcription of *cI* and are needed for establishing the lysogenic state initially.²⁵⁶ Once established these genes do not function since they are never transcribed. There are only a few molecules of the λ repressor present in a cell, but this is ordinarily sufficient to maintain the prophage state. On the other hand, irradiation of the bacterium with ultraviolet light activates the SOS response and results in rapid hydrolytic cleavage of the λ repressor and transcription of other phage operons.

Of special significance to the lytic cycle is the Cro repressor gene *cro*, found at the beginning of operon R1. Although it binds to the same operator sequences as does the λ repressor, the Cro repressor has opposite effects.²⁵⁴ It represses transcription of operon L2 and hence synthesis of λ repressor, but it positively activates P_R and P_L . The earliest proteins synthesized during lytic development are the Cro protein and the product of the first gene *N* in the left operon L2. The *N* protein, an antiterminator that permits transcription to continue on past points t_L and t_R , is an unstable, short-lived molecule of $t_{1/2} \sim 2$ min.²⁵⁷ Leftward transcription proceeds through genes *exo* and β , which are involved with recombination, and *xis*, which is required for excision. When the DNA is integrated into the *E. coli* chromosome, it is cut at points *aa'* (Fig. 28-11) and is inserted just to the right of the *gal* operon (Fig. 26-4). Prophage transcription can now continue past point *a'* and into the genes of the bacterium. Translation of the mRNA formed from this early left operon generates the enzymes needed to free the prophage and to permit reformation of the circular replicative form of the phage DNA. The excision is also made near point *a'*, and it is easy to see how the nearby *gal* genes can sometimes be included in the excised λ genome.

The product of gene *N* also permits rightward transcription through genes *O*, *P*, and *Q* and at a slower rate on along the rest of the chromosome to point *a*. Genes *O* and *P* code for proteins that permit the host replication system to initiate formation of new λ DNA molecules. Replication begins at the point *ori* and occurs in both directions. Gene *Q* codes for a

protein that activates transcription of the **late genes** beginning at promoter P_R .

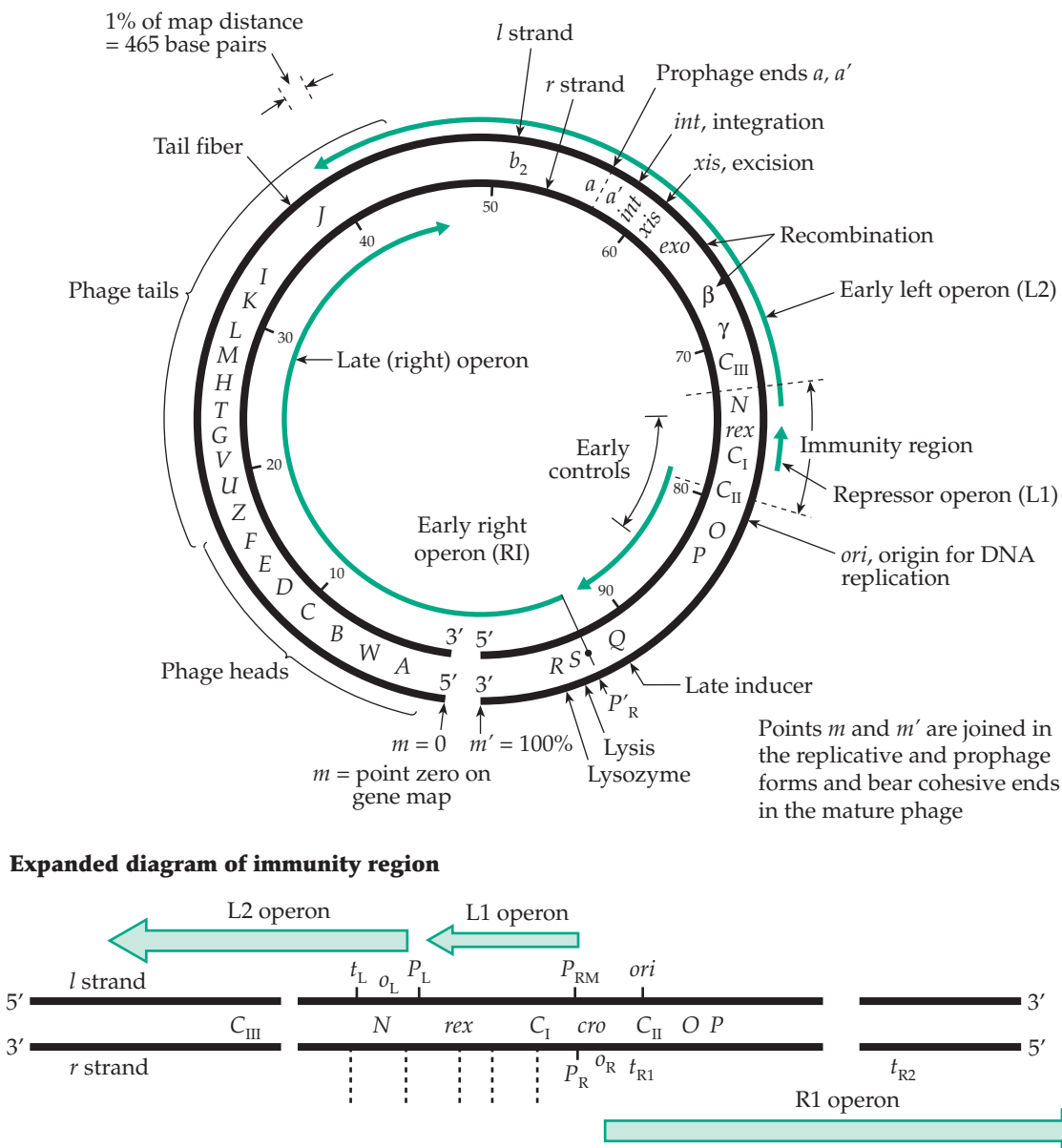
As indicated in Fig. 28-11, the chromosome can be divided into four major operons, the short one that produces repressor, and the early left, early right, and late operons. The early operons code largely for replication and recombination enzymes and control proteins. The late operon is concerned with production of proteins needed for assembly of the virus particles and must be transcribed at an even higher rate; hence the need for the product of gene Q. Within the late operon, genes A to F are involved in packaging of λ DNA and in formation of heads, while genes S to J are

concerned with the production and assembly of tails. Genes S and R produce proteins that lead to destruction of the host membrane and to lysis of the cell.

During the late stages of lytic growth the early genes are largely shut off by the Cro repressor. We can see that even in a virus the control of transcription can be a complex process.

2. Replication of RNA Bacteriophages

The small icosahedral RNA-containing bacteriophage are of interest because of the small number of



genes and the possibility of obtaining a detailed understanding of their replication. The four genes in the 3569-nucleotide MS2 RNA (Fig. 29-17) code for the A protein (maturation protein), the coat protein, a replicase (RNA polymerase) subunit, and a protein needed for lysis of the host cell. The last gene overlaps both the replicase gene and that of the coat protein.^{258,259} The somewhat larger phage Q β contains a 4.2-kb RNA genome. One subunit of the Q β replicase is encoded by the virus, but three bacterial proteins are needed to form the complete replicase.²⁶⁰ They are ribosomal protein S1 and elongation factors EF-Tu and EF-Ts, proteins that normally function within *E. coli* in translation of mRNA (Chapter 29). Their ability to associate with RNA has been exploited by the phage for a quite different purpose.

Replication of a single-stranded virus must take place in two steps. From the (+) strand present in the virus a complementary (–) strand is first formed. Initiation of this step requires another bacterial host factor Hfq^{260,261} and GTP. The (–) strands formed do not associate with the (+) strands. They are apparently released from the replicase in a single-stranded form and presumably fold into highly structured molecules with many hairpin loops (as for the (+) strand of MS2 RNA shown in Fig. 29-17). The (–) strands are then copied (Hfq is not needed for this) to make a large number of new (+) strands for incorporation into the finished virus particles. The Q β replicase is able to synthesize *in vitro* complete complementary strands to either (+) or (–) viral RNA molecules. However, the system is specific for the viral RNA and will not copy any arbitrary nucleotide sequence, certain sequences at the 3' end being essential for initiation of replication. During replication in the test tube mistakes are made including premature termination and mispairing of bases.^{262,263} Thus mutation takes place, and it is possible to select RNA molecules much smaller than the original viral RNA that will be replicated readily by the Q β replicase system. One such fragment contains only 114 nucleotides in a known sequence.²⁶⁴

C. Transcription in Eukaryotic Cells and in Archaea

There are three primary domains of life, represented by the bacteria, archaea, and eukaryotes. Some of the clearest evidence for the independent evolution of these three groups of organisms is found in the transcriptional apparatus. While the basic chemistry is the same, the details of initiation and control of transcription in bacteria and in eukaryotes are very different.^{264a} The archaea share characteristics of both bacteria and eukaryotes. Archaeal RNA polymerases have a complexity similar to that of eukaryotes and also share a similar mechanism of initiation of transcription.^{265–266b}

Several of the protein transcription factors of archaea also resemble those of eukaryotes.^{267,268} However, in a comparison of DNA sequences from the complete genomes of four archaeal species, it was found that of 280 predicted transcription factors or transcription-associated proteins 168 were homologous to bacterial proteins and only 51 to eukaryotic proteins.²⁶⁸ This tends to confirm the ancient divergence of the three primary domains of life.

In bacteria transcription and translation are closely linked. Polyribosomes may assemble on single DNA strands as shown in Fig. 28-5. It has often been assumed that RNA synthesis occurs on loops of DNA that extend out into the cytosol. However, recent studies indicate that most transcription occurs in the dense nucleoid and that assembly of ribosomes takes place in the cytosol.^{268a} In a similar way eukaryotic transcription occurs in the nucleus and protein synthesis in the cytosol. Nevertheless, some active ribosomes are present in the nucleus.^{268b}

1. Eukaryotic Nuclei and Transcription

In cells with true membrane-enclosed nuclei the messenger RNA molecules are relatively long lived. They must move out from the nucleus to the sites of protein synthesis in the cytoplasm. In addition to the need for eukaryotic mRNA to travel further and to last longer than that of bacteria, a number of other differences are evident. Eukaryotic mRNAs are usually transcribed from single genes. Polygenic operons are uncommon in most animals but are numerous in *C. elegans*.^{268c} Eukaryotic cells appear to rely less on negative control through specialized protein repressors than do bacteria but use a greater variety of positive control mechanisms. However, most genes are repressed by being held in a **silent state**.

Another characteristic of eukaryotes is the extensive processing of transcripts. Most primary transcripts that give rise to mRNA appear first in the nucleus as **heterogeneous nuclear RNA (hnRNA)**. Like mRNA it has a base composition resembling that of DNA. The molecular size varies from 1.5 to 30 kb or more. It turns over rapidly, most of it having a half-life of ~10 min. However, some may last as long as 20 h. Only about 5% of the hnRNA ever leaves the nucleus as mRNA, most being degraded without export to the cytoplasm.²⁶⁹ The processing consists of **capping** at the 5' ends, removal of introns (splicing), **cleavage** by nucleases, **polyadenylation** at the 3' ends, **methylation**, formation of pseudouridines, other covalent base modifications, and sometimes **editing**. Because of the complexity of eukaryotic transcription there are many points at which control can be exerted during initiation of transcription, termination of transcription, splicing and methylation, transport of mRNA out of

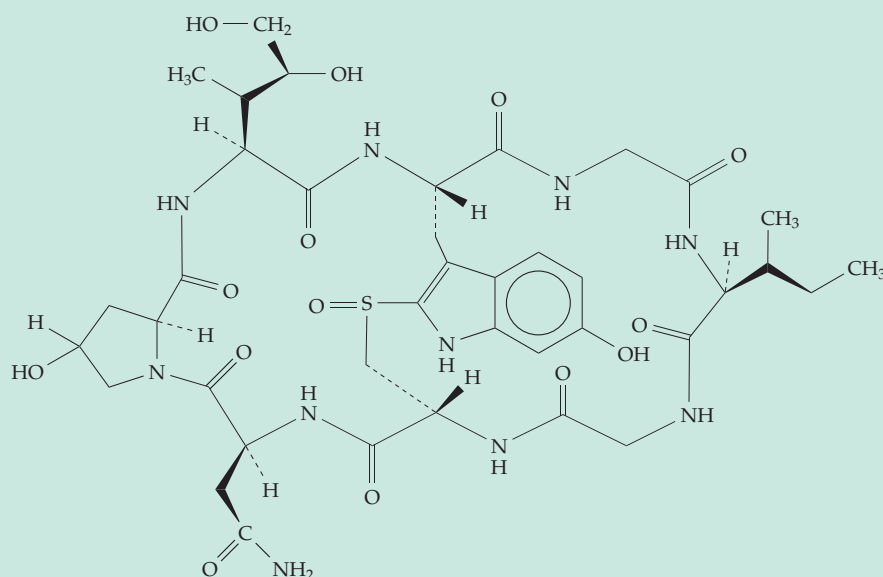
the nucleus, and the degradation of mRNA. However, the first points of control are found in the nucleosomes and in the structure of the chromatin.

It has been recognized for many years that genes in heterochromatin (Chapter 27) are usually not transcribed. In most chromatin the great majority of the genes are silent most of the time. Genes are repressed by the folding of the DNA into nucleosomes and by the further folding into higher order folds or coils (Fig. 27-5).^{270–272} Various activating transcription factors as well as RNA polymerases must bind to the DNA, displacing it from the histones around which it is wrapped in the nucleosomes. The processes by which

inactive nucleosomal DNA becomes active in response to external signals are beginning to be understood. Chemical alterations in the histones in the nuclear matrix and in other nuclear proteins and also in the DNA itself may be involved.^{273–275} As pointed out in Chapter 27, the CpG “islands” that lie upstream of many genes are heavily methylated in the silent heterochromatin. Repressor proteins may bind to methyl-CpG groups.^{276,276a}

In recent years attention has been focused on the N-terminal “tails” of histones H3 and H4 (Fig. 27-4) in which lysine side chains undergo reversible acetylation and which may also be phosphorylated and

BOX 28-B POWERFUL POISONS FROM MUSHROOMS



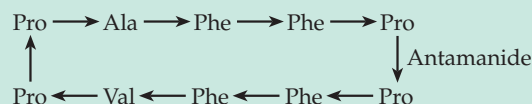
Several deadly species of the genus *Amanita* produce colorless toxic octapeptides, the **amanitins**.^{a,b} Two residues of glycine, one of L-isoleucine, one of the unusual L-dihydroxyisoleucine, one of L-asparagine, and one of L-hydroxyproline are present in α -amanitin. In the center a modified tryptophan residue has been combined oxidatively with an SH group of a cysteine residue. If the dihydroxyisoleucine residue of α -amanitin is replaced with unhydroxylated leucine, the resulting compound, known as amanullin, is nontoxic. The LD₅₀ for mice is 0.3 mg kg⁻¹ and 50 g of fresh *Amanita phalloides* may be sufficient to kill a person. Amanitins act slowly, and it is impossible to kill mice in less than 15 h, no matter how high the dose.

α -Amanitin completely blocks transcription by eukaryotic RNA polymerases II and III. Polymerase II is the major nuclear RNA polymerase, and its inhibition prevents almost all protein synthesis by

the cell. Note that the amanitin molecule is semisymmetric overall, much as is an actinomycin (Box 28-A), with an aromatic group protruding from behind in the center.

The same mushrooms contain several fast-acting toxic heptapeptides, the **phalloidins**, whose structures are similar to those of the amanitins. However, they contain a reduced sulfur atom (—S—) in the cross-bridge. They are specifically toxic to the liver.^c The same mushrooms also contain an antidote to the phalloidins, **antamanide**. This cyclic

decapeptide, like the toxins, is made up entirely of L-amino acids, and it apparently competes for the



binding site of the phalloidins. Unfortunately, it is of little value in treating cases of mushroom poisoning. Antamanide is a specific sodium-binding ionophore.

^a Wieland, T., and Wieland, O. (1972) *Microb. Toxins* **8**, 249–280

^b Wieland, T., and Faulstich, H. (1983) *Handbook of Natural Toxins*, Marcel Dekker, New York

^c Wieland, T., Nassal, M., Kramer, W., Fricker, G., Bickel, U., and Kurz, G. (1984) *Proc. Natl. Acad. Sci. U.S.A.* **81**, 5232–5236

methyated.^{33b,271,275a–c} The resulting modifications in shape and electrical charge can affect the ways in which the histones interact with the DNA, with each other, and with other proteins in transcription complexes.^{271a,275a} The histone tails may interact with adjacent molecules either to pack the chromatin more tightly or to loosen it and allow transcription to take place. Active chromatin has long been known to be highly acetylated, while silent chromatin has a low degree of acetylation, but a high degree of methylation of both histones and DNA (pp. 1541–1542). Methylation of histone H3 on lysine 9 (H3-Lys⁹ or H3-K9) is especially significant.^{275d,e} See Chapter 32, Section A,1 and C,1 for further discussion.

Several transcriptional activators form complexes with acetylating enzymes, the **histone acetyltransferases (HATs)**, while transcriptional repressor proteins often associate with **histone deacetylases**.^{277–279c} The deacetylases are often found in very large complexes. For example, the mammalian complex **Sin3** contains two histone deacetylases plus at least five other subunits, some of which evidently bind to histones.^{277–278a} Sodium butyrate in millimolar concentrations is a powerful inhibitor of these deacetylases.²⁸⁰ Special **silencer** sequences in the DNA are sometimes present and provide sites for binding of transcriptional repressors. Among these are **silent information repressors (Sir** proteins). They regulate regions of DNA that can be converted to a heterochromatin-like state. They have been located in silenced mating type loci in yeast (see p. 1880), in telomeres, and in DNA containing ribosomal RNA genes.^{280a,b} Sir proteins have an unusual histone deacetylase activity. The acetyl groups removed from histones are transferred by reaction with NAD⁺ to ADP ribose (see Eq. 15-16).^{280c,d} Regions of silenced DNA are often set apart by **insulator** or bounding regions.^{280e–g}

An important mechanism of silencing some genes is the use of an antisense RNA strand, as is illustrated for a bacterial system in Fig. 28-8. This **RNA interference** is also used in animals and plants, often as a way of blocking replication of viruses.^{1,1a,280h–j} The small 20–25 nucleotide siRNAs that function in this way are abundant in *C. elegans* and in *Drosophila*.

Histone acetyltransferases also form large complexes that acetylate not only histones but also other nuclear proteins.^{281–283} The **SAGA HAT** complex of *S. cerevisiae* has a molecular mass of ~2.0 MDa and contains at least 14 subunits.^{283a} Large acetylating complexes have also been identified in *Tetrahymena*, *Drosophila*, *Arabidopsis*, and mammalian species.²⁸³ Some subunits of these complexes have been identified as previously known transcription factors. Such multiprotein complexes are sometimes described as **cis-regulatory elements (CREs)**.^{283b,c}

Changes in the structural properties of chromatin observed during silencing of genes or during their

activation are often described as **chromatin remodeling**. Large multisubunit complexes are involved. Their action is characteristically dependent upon ATP hydrolysis.^{284–284b} Complexes **SWI/SNF** and **RSC**, first found in yeast but also present in human cells, appear to participate in the disruption of nucleosomes needed for initiation of transcription.^{284–287c} Among other distinctly different remodeling complexes are **ISWI** of *Drosophila*, its human homolog, and the human Williams syndrome transcription factor **WCRF**.²⁸⁴ The ATP-dependent component in these complexes has a conserved sequence that is shared with DNA helicases. RNA helicases are also required for all processes that form, modify, or utilize RNA.^{288,288a–c} However, the chromatin remodeling complexes appear not to unwind DNA but to open the nucleosomal DNA for initiation of transcription.^{272,284} They may act in a processive fashion and be coupled to transcription. **Peptidyl-prolyl isomerases**, such as the cyclophilins (Box 9-F) may also be essential components of chromatin-remodeling complexes.²⁸⁹

Just as they participate in driving the cell cycle, ubiquitin and proteasomes also function in the control of transcription. In many cases specific transcription factors are targeted for destruction after they are used to activate or repress a gene.^{289a–c} However, a 19S regulatory complex, which consists of a base and a lid of the proteasome (Box 7-A), may participate directly in control of transcription rather than in mediating proteolysis.^{289a–e} In addition to ubiquitin a 97-residue relative designated SUMO-1 is linked to proteins by enzymes resembling E₁ and E₂ of the ubiquitin system (Box 10-C). Conjugation with **SUOMO-1** regulates some transcription factors and has other functions,^{289f,g} e.g., participation in control of nuclear pores. The 81-residue protein **NEDD8**, which is 80% homologous with ubiquitin, controls some transcriptional processes in heart and skeletal muscle.^{289h,i} Furthermore, ubiquitin-like sequences (UBX domain) are present in the C-terminal ends of a variety of specific proteins.^{289j}

2. RNA Polymerases

Eukaryotic nuclei contain at least three RNA polymerases^{269,290–292} which have the following functions:

Polymerase I	Formation of large pre-rRNAs
Polymerase II	Transcription of most genes to give precursors to mRNA and most small nuclear RNAs (snRNAs and small nucleolar RNAs (snoRNAs)
Polymerase III	Formation of 5S rRNA, tRNAs, and small RNA U6

Polymerase I is localized in the nucleolus.^{293,294} Mitochondria contain a fourth RNA polymerase^{295,295a} and chloroplasts a fifth.²⁹⁶

Like the bacterial polymerase, eukaryotic RNA polymerases are large 500–600 kDa aggregates of 9–14 subunits each. Yeast and human RNA Pol IIs each contain twelve subunits (Fig. 28-12).²⁹⁰ There are two large nonidentical subunits, which in mammalian cells have masses of 214 kDa and 140 kDa and are homologous to the β' and β core subunits of the *E. coli* polymerase, respectively.^{291,297–299c} The active site contains one or two catalytic Mg^{2+} ions.²⁹⁰ The largest subunit has an unusual singly glycosylated C-terminal domain (known as the **CTD**). It contains the repeating sequence $(YSPTSPS)_n$. The number of repeats varies: $n = 18$ in plasmodia, 27 in yeast, 45 in *Drosophila*, and 52 in mammals.³⁰⁰ The numerous serine side chains in this tail domain undergo phosphorylation and

dephosphorylation to varying extents during each catalytic cycle.³⁰¹ This may be a way of easing the transcriptional complex through nucleosomes,³⁰² but its most important function appears to be the linking of transcription to pre-mRNA processing.^{303,304}

RNA polymerases I and III have properties similar to those of Pol II.^{304a,b} Polymerases II and III are very sensitive to inhibition by the lethal mushroom poison **amanitin** (Box 28-B). However, both RNA polymerase I and RNA polymerases of mitochondria resemble the bacterial enzyme in being resistant. Genes transcribed by polymerase I, II, and III are often referred to as genes of classes I, II, and III, respectively. While mRNA is transcribed from class II genes, rRNAs, tRNAs, and some small RNAs, which must undergo processing but are not polyadenylated, are transcribed from genes of classes I and III. Each type of nuclear RNA functions in its own sites as independent “factories.”^{305,306}

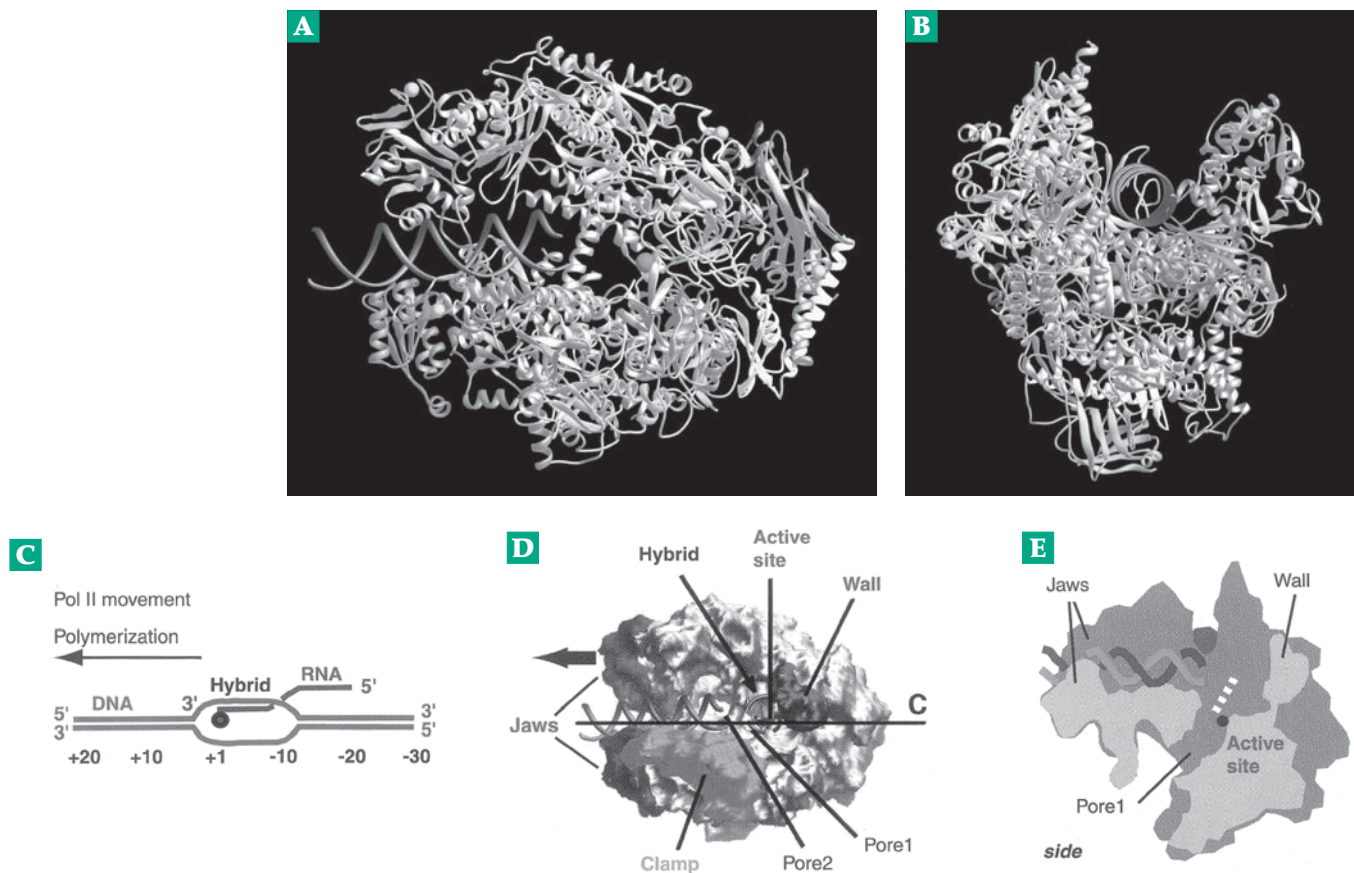
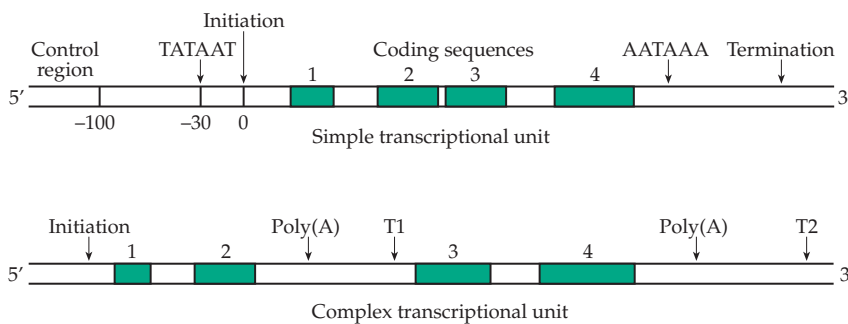


Figure 28-12 Three-dimensional structure of yeast RNA polymerase II. (A) View from the “top,” with backbones of ten subunits (of 12) in the 514-kDa protein shown as ribbon drawings. A 20-base-pair segment of B-DNA has been modeled in a location indicated by electron crystallography. (B) Side view looking toward the end of the DNA. Eight zinc atoms as well as an active-site magnesium (green) are visible. (C) Schematic drawing showing the transcription bubble as proposed for a transcribing polymerase. (D) Surface representation of the polymerase viewed as in (A). (E) Side view of a section cut along the line marked C in D. The dashed white line represents the axis of the DNA-RNA hybrid segment. The hybrid axis must lie at an angle with respect to the axis of the incoming DNA. Pore 1 may be a route for exit of RNA during “backtracking.” The nucleotide triphosphate substrates may also enter via pore 1. From Cramer *et al.*²⁹⁰ Courtesy of Roger D. Kornberg.

3. Transcriptional Units and Initiation of RNA Synthesis

Typical simple **transcriptional units** for class II eukaryotic genes contain the following elements: (a) site of initiation, (b) **TATA sequence** (Goldberg–Hogness sequence) at position ~ -30 bp, (c) **upstream regulatory elements**, (d) **enhancers**, (e) a series of **coding sequences** or exons separated by introns, an **AATAAA sequence** that in the RNA transcript may establish the 3' polyadenylation site, (f) a termination region.³⁰⁷



The preceding paragraph describes a **simple transcriptional unit**. There are also **complex transcriptional units**.³⁰⁸ For example, the terminator sequence T1 may be followed by additional exons and a second polyadenylation signal and second termination sequence T2. Termination may sometimes occur at T1 and sometimes at T2 resulting, after splicing, in two mRNAs, one containing only exons 1 and 2 and the other all four exons. In some cases two or more different modes of splicing may occur with one or more exons omitted from the final processed mRNA. Thus, a single transcriptional unit can give rise to two or more different proteins which share some common sequences.^{309–312} Multiple initiation sites sometimes exist as well.

Initiation of RNA synthesis is a complex process, which is summarized diagrammatically in Eq. 28-5. Some details are given in Sections 4–7. Elongation of the RNA being formed is also complex, often requiring splicing and other processing steps before synthesis can be terminated. These steps, which are discussed in Section 8 and in Section D, also depend upon large complexes of proteins, which are directly coupled to the RNA polymerase.^{312a–c}

4. Promoters, Transcription Factors, Enhancers, and Activators

Eukaryotic promoter sequences are less well defined than are those of bacteria, and the initiation

points for transcription are more variable. Most promoters contain the TATA sequence, which is usually required for binding of polymerase II. In addition, there are upstream regulatory elements (Table 28-1). Many promoters contain the sequence 5'-**CCAAT** at about -75 , and others have 5'-**GGGCGG** or similar sequences at close to -100 .^{273, 273a,b} A sequence found upstream from many yeast genes is 5'-**TGACTC** or the longer semipalindromic 5'-**ATGACTCAT**.^{313,314} Some sequences are unique to small sets of genes such as the “heat-shock” genes³¹⁵ considered in Section 6. Most upstream sequences are not polymerase-binding sites but attachment sites for additional protein transcription factors. Computer programs that help locate promoter sequences have been devised.^{316,317}

The SV40 early promoter and transcription factor Sp1. The study of transcription in eukaryotes has been difficult because purified polymerases do not initiate transcription at most promoter sites. As a consequence, much of the early work was done with viruses such as adenovirus and SV40.

Their genes are transcribed by RNA polymerase II and have unusually effective promoters. A protein known as **Sp1**, isolated from human cultured cells, protects an SV40 promoter from digestion by DNase. Sp1, which is now known as an **accessory factor** or **coactivator** for transcription,^{318,318a} protects a region that extends from about -45 to -104 bp and contains the hexanucleotide GGGCGG sequence repeated six times. The Sp1 protein was isolated as a mixture of related 95- and 105-kDa peptides³¹⁹ and was found to contain two DNA-binding zinc-finger domains (Figs. 5-37, 5-38).³²⁰ The Sp1 protein is synthesized in most cells and binds not only to SV40 promoters but also to many promoters of host cells³²¹ (an example is provided by the mouse mitochondrial aspartate aminotransferase gene whose sequence is shown in Fig. 5-4). In genes that lack a TATA sequence the binding of Sp1 or the related Sp2, 3, and 4 to GGGCGG or similar GC-rich sequences is essential to initiation of transcription.^{322,323} A possible role is to assist in nucleosome remodeling.³²² Sp1 also binds, together with other transcription factors, to certain enhancer sequences.^{324,324a,b} Its effects are modulated by posttranscriptional phosphorylation and glycosylation.³²¹

The TATA binding protein and general transcription initiation factors. A slow basal level of transcription can be observed when all but a small part of the control region at the 5' end of a gene is deleted.³²⁵ This minimum promoter, which includes the TATA sequence, is the binding site of both the RNA

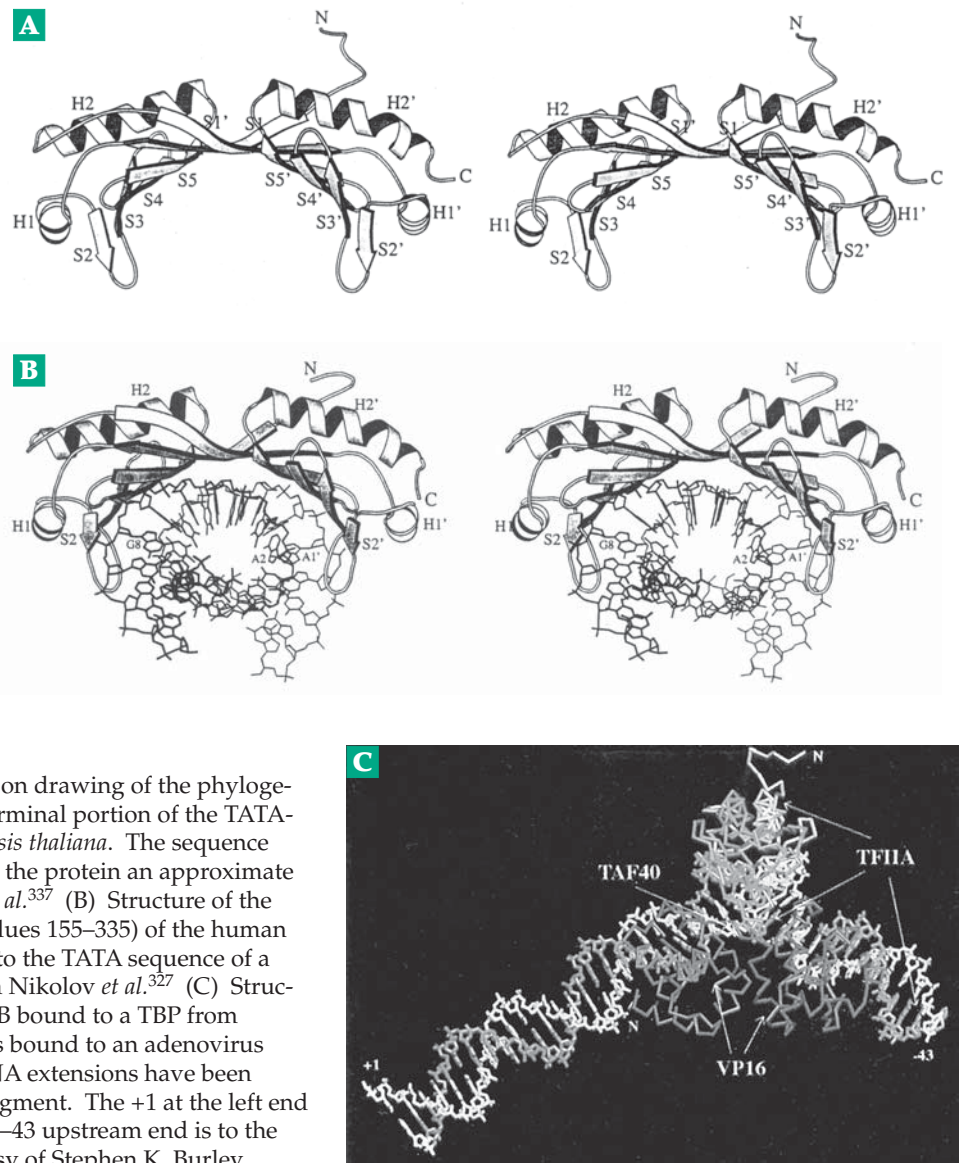
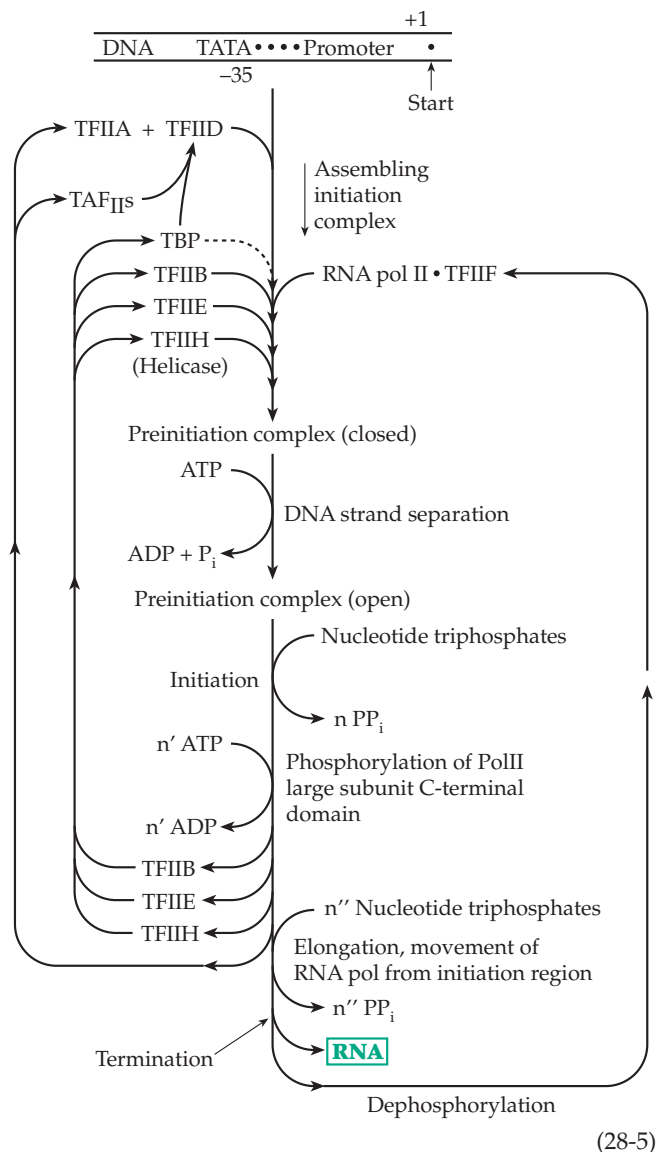


Figure 28-13 (A) Stereoscopic ribbon drawing of the phylogenetically conserved 180-residue C-terminal portion of the TATA-binding protein (TBP) from *Arabidopsis thaliana*. The sequence consists of two direct repeats, giving the protein an approximate twofold symmetry. From Nikolov *et al.*³³⁷ (B) Structure of the corresponding C-terminal core (residues 155–335) of the human TATA-binding protein (TBP) bound to the TATA sequence of a promoter in adenovirus DNA. From Nikolov *et al.*³²⁷ (C) Structure of human transcription factor IIB bound to a TBP from *Arabidopsis thaliana*, which, in turn, is bound to an adenovirus TATA sequence. Hypothetical B DNA extensions have been modeled at both ends of the DNA segment. The +1 at the left end is the transcription start site and the –43 upstream end is to the right. From Nikolov *et al.*³³⁸ Courtesy of Stephen K. Burley.

polymerase and of transcription initiation factors that are designated TFII-A, -B, -D, -E, -F, and H. Because they affect many genes these are called **general transcription factors**.^{326–329} TFIID is a large complex of a DNA-binding subunit known as the **TATA-binding protein (TBP)** together with 8–12 additional tightly bound subunits known as **TBP-associated factors (TAF_{II}s)**. Many of the TAF_{II}s have histone-fold structures. Some possess histone acetyltransferase and other enzymatic activities.^{329a} They may stabilize initiation complexes on specific gene promoters.^{330–332a} TBP binds specifically to the TATA sequence (Table 28-1),³³³ which is found in most promoters for RNA polymerases I, II, and III. Its three-dimensional structure resembles a saddle, which sits astride the TATA sequence (Fig. 28-13). The DNA is bent, untwisted by ~117°, and the minor groove broadened to allow a good fit.^{327,328,334} TFIIB is thought to bind first to

the DNA–TBP complex, after which the RNA polymerase II complex binds and becomes positioned on a promoter site. Other factors, including the ATP-dependent bidirectional helicase TFIIF,^{335,335a} also add (Eq. 28-5). ATP may be needed for more than one step in initiation.³³⁶ TFIID contains TAF_{II} subunits. They may bind along with TBP^{336a} as indicated in Eq. 28-5, or they may bind at a later point in an assembly pathway. It is often stated in current literature that the growing initiation complex “recruits” the next subunit. It is important to realize that this simply means that the next subunit that strikes the complex by diffusion sticks, perhaps with cooperativity in binding. The word recruit doesn’t mean that the complex advertises a vacancy for the next subunit. The very large complexes (**transcriptosomes**) that are formed vary in composition and in assembly pathways. Proteins related to TBP bind to different promoters.³¹⁰ It is also



important to recognize that some promoters bind transcription factors tightly while others do so only weakly. Some are constitutive, always functioning, but most are inducible, acting only upon appropriate stimulation. The many operators present in a genome and the many species of eukaryotes present on earth ensure a vast variety of detailed pathways and control mechanisms.

Transcriptional activators. Many proteins serve as activators of transcription, causing larger increases in rate over those observed with TBP alone. Some of these are listed in Table 28-2.^{338a} The table also lists two proteins (Sp1 and NF1), and the DNA sequence CCAAT, which control constitutive or continuously active genes. A large group of transcription factors are active in development. Receptors may be resident in cytoplasm, cell membrane or nucleus, as indicated in Table 28-2. Some cytoplasmic factors are *latent*, becoming

active only following stimulation by an external signal. All of the factors in Table 28-2 are positive-acting.^{338a} However, some negative-acting factors are known. One, designated **Ssn6-Tup1** in *S. cerevisiae* is a global repressor, affecting many genes.^{338b} For example, it opposes the activator GAL4. Proteins related to Ssn6-Tup1 are found in flies, worms, and mammals.

Genetic studies indicate that gene activator proteins often bind to TFIIB, TFIID, and TFIIF.²⁷² The coactivator **TafII130** (which binds to TFIID), and the bound transcriptional activator Sp1 apparently interact with the protein **huntingtin** in regulation of transcription in the brain.^{338c} A defect in huntingtin leads to the fatal neurodegenerative Huntington disease (pp. 1516, Chapter 30). Other activator proteins bind to upstream activator sequences, as in prokaryotes. Among the most studied of these is GAL4, an 881-residue yeast protein that binds to a specific 17-bp palindromic upstream site near the TATA sequence. It activates transcription of genes needed for galactose metabolism.^{339–343} The GAL4 protein contains a binuclear metal cluster composed of two Zn²⁺ ions and six cysteine side chains, two of which bridge the pair of metal ions.^{341,343} GAL4 is able to activate genes of *Drosophila* and of human cells. The specific GAL4-binding sequence has been introduced into 5' control regions of genes in various positions.³⁴⁴ It was found that neither an exact distance nor alignment between the GAL4 binding site and the TATA sequence is required, but activation is best when this distance is not too large. The explanation for the lack of a requirement for alignment seems to lie in the flexibility of the C-terminal segment of GAL4, which carries a large negative charge and may bind to the repeated C-terminal sequence of RNA polymerase II to activate it.³⁴⁵ Many other promoter-specific activators are known.

Mediators and coactivators. Transcriptional activators that act in a crude cell-free system often do not function with purified DNA, RNA polymerase, and the basal transcription factors as indicated in Eq. 28-5. Studies with yeast, *Drosophila*, and human cells revealed that additional large multisubunit complexes known as mediators are needed.^{272,346–348} A yeast mediator complex consists of 20 subunits.^{349–350b} Many activator proteins bind to the DNA sequences known as enhancers, discussed in the next section. Mediator complexes may also interact with enhancer-bound activators. Individual proteins, such as the TAF subunits, that bind to and cooperate with activator proteins are often called coactivators.³⁵¹

Enhancers. Complex DNA sequences called enhancers help to regulate transcription of many eukaryotic genes. The first of these was discovered in an upstream control region of the virus SV40 DNA

and consists of two repeats of a 72-bp sequence.^{47,352,353} The presence of an enhancer sequence may cause as much as 100- to 1000-fold increase in the rate of transcription as compared with the same transcriptional unit from which the enhancer has been deleted. A surprising fact is that enhancers as far as 1–2 kbp upstream or even far downstream of the promoter and in either of the two possible orientations are effective.

This finding suggested that enhancers induce long-range conformational alterations in DNA. Alternatively, they might contain points of entry for RNA polymerase or for an initiation factor that could move along the DNA to the promoter region. However, the synthetic DNA molecule shown in Fig. 28-14 contains two copies of an enhancer in opposite orientations in one strand but none in the other strand.³⁵⁴ The

TABLE 28-1
Nucleotide Sequences of Some Commonly Found Regulatory Elements in Promoters and Upstream Activator Sites

Sequence ^a transcription	Description. Positions are relative to the start site
RNA polymerase binding sites:	
5'- TATAAT ^b	Bacterial –10 or Pribnow promoter sequence
5'- TTGACA	Bacterial –35 region promoter sequence
5'-(C/T)TTA(A/T)Ann	Archaeal –30 region, TBP binding site ^c
5'-TATA(A/T)A(A/T) or 5'-TATA@A@n ^d	Eukaryotic –30 region; yeast –60 to –120 region
Upstream eukaryotic promoter sites:	
5'-GGGCGG	–100 region, Sp1
5'-CCAAT	–75 region, CTF
Small GC clusters ~5 bp apart	Binding site for TFIID
5'-GCGCC-C	~ –5; “discriminator” sequence: inhibition of gene expression by ppGpp
Enhancer elements and transcription factors:	
5'-ATGA(C/G)TCAT	AP-1, cJun, GCN4 (yeast)
5'-CCCCAGGC	AP-2
5'-CAC(G/T)	Myc / Max heterodimer
5'-ATGACGTCAT	CRE (cAMP responsive element)
5'-GGTCAnnnTGACC	Estrogen-responsive element
5'-GGGTGAnnnGGGTGA	Vitamin D-responsive element; direct repeats
5'-CC(A/T) ₆ GC	SRE (serum response element)
5'-ATGCAAAT	Homeotic genes; Oct-2
5'-GGTCAnnnTGACC	ERE (erythroid responsive element)

^a The sequences are all for the sense strand of the DNA, n = any base. See Keller, W., König, P., and Richmond, T. J. (1995) *J. Mol. Biol.* **254**, 657–667; Adams, R. L. P., Knowler, J. T., and Leader, D. P. (1992) *The Biochemistry of the Nucleic Acids*, 11th ed., Chapman & Hall, London; Lewin, B. (2000) *Genes VII*, Oxford Univ. Press, New York.

^b Consensus sequence. The bases in boldface are the most highly conserved.

^c From DeDecker, B. S., O'Brien, R., Fleming, P. J., Geiger, J. H., Jackson, S. P., and Sigler, P. B. (1996) *J. Mol. Biol.* **264**, 1072–1084.

^d The symbol @ refers to either A/T or T/A. See Juo, Z. S., Chiu, T. K., Leiberman, P. M., Baikalov, I., Berk, A. J., and Dickerson, R. E. (1996) *J. Mol. Biol.* **261**, 239–254.

Figure 28-14 A “tailed circle” consisting of an enhancer linked to, but topologically separated from, a gene. One of the DNA strands of this plasmid bears two copies of an SV40 enhancer sequence, one copy inverted with respect to the other. This extra region protrudes from the circle and self-pairs to form a functional enhancer. The main body of the circle contains the β -globin gene, transcription of which is increased by the enhancer. Twisting of the enhancer has no effect on the winding of the strands on the main body of the circle; nevertheless, the enhancer efficiently increases β -globin transcription. From Ptashne.³⁵⁵

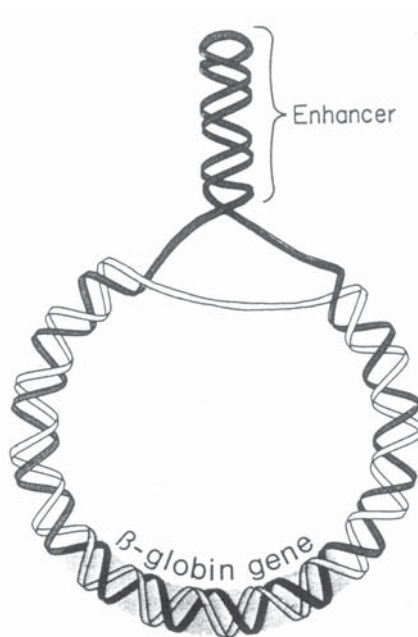
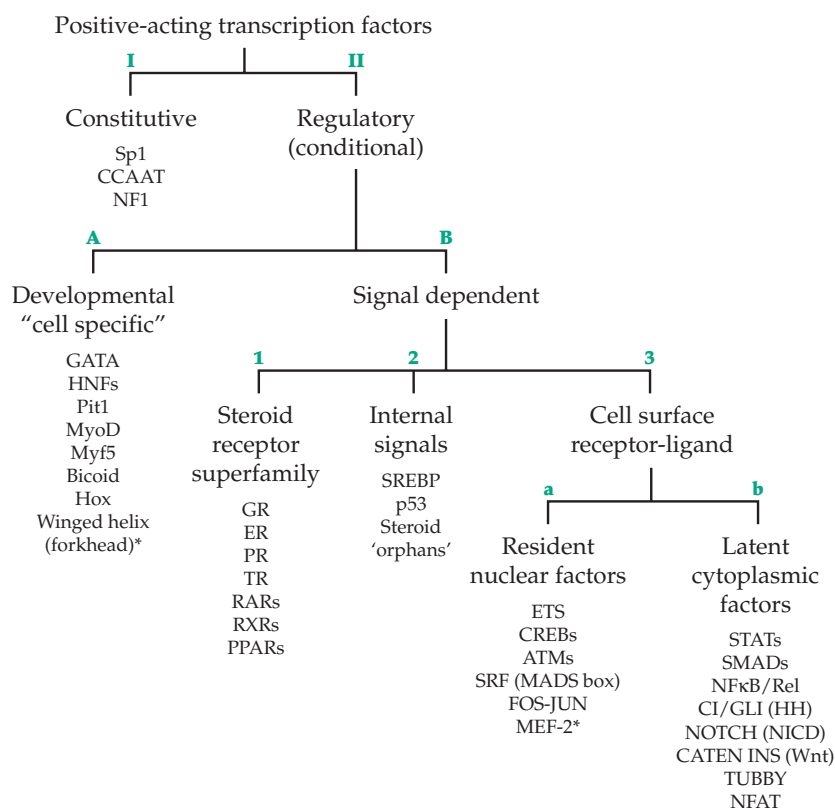


TABLE 28-2
Functional Classification of Positive-Acting Eukaryotic Transcription Factors^a



^a Major groups are labeled in large type and specific examples are listed below (green). Some bacterial proteins are included. Proteins designated by asterisks can be trapped in the cytoplasm by phosphorylation of serine site chains. From Brivanlou and Darnell.^{338a}

enhancer is thus topologically separated from the globin gene present in the DNA. Nevertheless, the enhancer functioned efficiently. This suggested that enhancers are protein-binding sequences and that the bound proteins may, perhaps, close a loop to affect the transcription initiation complex.^{149,355,356} An enhancer may control a whole loop of DNA in a chromosome affecting many genes.³⁵⁷ Enhancer DNA sequences are often complex or *modular*, consisting of several shorter elements. For example, the SV40 enhancer contains 8- to 10-bp sequences that are repeated with minor variations and are also present in other enhancers. These are sites for the binding of activator proteins, some of which carry messages from signaling cascades such as the MAP kinase pathway of Fig. 11-13. These pathways are central to the control of cell growth. Their activation is recognized as a mitogenic response. Some enhancers (*nonmodular*) are more compact and directly tied to the RNA pol complex. An example is the human interferon- β (IFN- β) enhancer.^{357a,b} It responds cooperatively to three separate activator proteins: NF- κ B, interferon regulatory factors, and ATF-2/c-Jun together with an architectural HMG protein. These form an **enhanceosome**, which interacts directly with the transcriptosome complex. The whole complex is sometimes referred to as a cis-regulatory module (CRM). The DNA domains affected by enhancers may be separated by insulator or boundary regions (p. 1626).

5. The Variety of DNA-Binding Proteins

A large number of proteins, often present in very small amounts, bind to DNA and may affect transcription. Some, such as the histones, are relatively nonselective. The more specific transcription factors are often capable of binding tightly, but may do so only upon

allosteric activation. Architectures of transcription factors vary. Some place an α helix in the major groove of B-DNA while for others β strands or peptide loops may interact with the DNA in the wide groove. Many transcription factors bind only to a bent or distorted DNA helix, often with a broadened groove. Intercalation of groups from other domains of a transcription factor within the minor groove may help to bring about necessary distortion.³⁵⁸ Transcription factors sometimes, perhaps often, have other roles, e.g., as enzymes. For example, a pterin dehydratase acts as a transcriptional cofactor in liver cells.³⁵⁹

While architectures vary greatly, the DNA-binding domain or domains are usually rich in positively charged side chains, which may interact directly by hydrogen bonding with the charged phosphate groups of the DNA backbone or indirectly via bound water molecules (see Fig. 5-36).

Leucine zipper transcription factors. A large family of DNA enhancer-binding proteins, which are involved in regulating cell growth in specialized cells of differentiated tissues and also in yeast cells, have related structure domains with a characteristic coiled coil that holds two subunits together.^{360,361} An example is the Max structure shown in Fig. 2-21. One of the first proteins of this type to be recognized was the mammalian phorbol ester-induced **cell activator protein AP-1**, which is a heterodimer of the protein **cJun**, encoded by cellular protooncogene *c-jun*, and **cFos**, encoded by protooncogene *c-fos*.³⁶² The heterodimer binds to the palindromic sequence 5'-TGACTCA. Each of the monomeric proteins *c-jun* and *c-fos*, as well as other members of the leucine zipper family, has an N-terminal DNA-binding domain rich in positively charged basic amino acid side chains, an **activation domain** that can interact with other proteins in the initiation complex, and the leucine-rich dimerization domain.³⁶³ The parallel coiled-coil structure (Fig. 2-21) allows for formation of either homodimers or heterodimers. However, cFos alone does not bind to DNA significantly and the cJun/cFos heterodimer binds much more tightly than does cJun alone.³⁶⁴ The yeast transcriptional activator protein GCN4 binds to the same 5'-TGACTCA sequence as does the mammalian AP-1 and also has a leucine zipper structure.^{360,364,365}

Several mammalian leucine zipper proteins bind to the CCAAT sequence (Table 28-1) and are, therefore, as a family designated C/EBP.^{361,366,367} A 30-residue segment of C/EBP contains four leucine residues at 7-residue intervals. When plotted as a helical wheel (Fig. 2-20) the four leucines are aligned on one side.³⁶⁸ Similar sequences are present in the proteins cMyc, cJun, and cFos and in GCN4. These observations suggested that if the peptide sequence forms an α helix, the leucine side chain from two identical subunits or closely related proteins might interdigitate in

a knobs-in-holes fashion to form the leucine zipper (Fig. 2-21).³⁶⁸ The structures of these leucine zipper proteins have now been thoroughly investigated and verified by X-ray and NMR methods.^{360,369-371} Mutational alterations in the zipper regions of these proteins decreases both activation³⁷² and dimerization.³⁷³ The CCAAT sequence is found in many enhancers^{366,374,375} and is present in ~25% of all eukaryotic promoters that function in differentiated tissues. A trimeric protein known as NF-Y binds specifically to this sequence.³⁶⁶ Its subunits contain glutamine-rich domains and histone-fold domains that suggest formation of nucleosome-like structures.^{366,376}

Another transcriptional activator of the leucine zipper class is **Myc**, a product of the *c-myc* oncogene (Chapter 11) and a key regulator of both cell growth and programmed cell death (apoptosis).^{377-379a} It binds to the sequence 5'-CACGTG, and its binding is greatly enhanced by formation of the heterodimer **Myc/Max** (Fig. 2-21). Max is not an activator and may dimerize with certain other proteins to become an inhibitor of transcription. However, it is ubiquitously present and ready to join with Myc to activate an appropriate series of genes. A related enhancer sequence 5'-TGACGTCA is a **cyclic AMP response element** (CRE) that functions within hormone-responsive tissues that use cAMP as a second messenger.^{364,380} Cyclic AMP activates protein kinase A (Fig. 11-4), whose catalytic subunit diffuses into the nucleus and phosphorylates the cyclic AMP-response element-binding protein **CREB**, a coactivator that binds to the CRE and which also contacts the general transcriptional factor complex.^{365,381-382b} Transcriptional responses to cAMP are quite complex. For example, activation of the phosphoenolpyruvate carboxykinase (PEPCK) gene is maximal only when CREB, C/EBP, and AP-1 are all bound at adjacent DNA sites.^{382c} CREB (also known as ATF) is a family of proteins that control the activities of hundreds of genes. Participating in this control is a coactivator, the CREB-binding protein (**CBP**),^{382d} which is, in turn, subject to control by methylation.³⁸³

Control of growth. A large variety of transcription factors control cell growth. Some of these are indicated in Figs. 11-14 and 11-15 and Table 28-2. Since growth in cell numbers requires completion of the cell cycle, the specialized transcription factors involved are necessary. As is indicated in Fig. 11-5, factor E2F is of central importance. In fact, there are at least six mammalian E2F proteins, five of which have both a conserved DNA-binding domain and conserved activation domains.^{383a,b} E2F1, 2, and 3 can all induce the S phase of the cycle. E2F6, in contrast, appears to bind to E2F-binding sites in DNA and also to Myc-binding sites to silence these genes and to help keep the cell in the G₀ state.^{383c,d} As shown in Fig. 11-5, the

retinoblastoma protein Rb also binds to EF2 and represses transcription. However, it allows transcription when phosphorylated.^{383e}

Among the many other proteins that influence growth are cFos and cJun, which may be activated by the MAP kinase pathway (Fig. 11-13).^{383f,g} Binding of cytokines (pp. 1571, 1845) activates signaling pathways from cell membrane receptors to two other families of transcription factors. These are the **STATS** (p. 1845)^{383h-j} and **SMADS**.^{383h,k,l} Upon activation STATS and SMADS move from the cytosol into the nucleus where they find their binding sites on DNA.

Response elements. DNA binding sites for activator proteins are often described as “response elements.”⁴⁷ Thus, the site for the cAMP-responsive protein CREB is the response element CRE. The binding site for AP-1 is **TRE**, named after the phorbol ester TPA. HSE is the heat shock response element, GRE the glucocorticoid response element (or the glucose response element), and **SRE** the sterol regulatory element.^{384,384a} Response elements tend to be present in many enhancers and cooperate with other enhancer-binding proteins to activate groups of genes. An especially large group of genes respond to the **serum-response element** (also SRE), which is found within the *c-fos* promoter region.^{384b,c} It is the DNA binding site for the ubiquitous serum protein SRF (serum response factor),³⁸⁵⁻³⁸⁷ which is involved in growth control, cell cycle progression, and wound repair.

Zinc-containing transcription factors. The zinc finger domain (Fig. 5-37), which is also designated Krüppel-like finger in reference to a *Drosophila* protein,^{388,388a} is a repeated motif present in many transcription factors (see Fig. 5-38).³⁸⁹⁻³⁹¹ In each finger a Zn^{2+} ion is coordinated by two -SH groups and two imidazole groups to form the $\text{Cys}_2\text{His}_2\text{Zn}$ domain. One of the first proteins in which zinc fingers were recognized is TFIIIA, one of the factors that controls transcription of 5S RNA genes. A large 30-kDa N-terminal domain contains nine ~30-residue repeats of the sequence $\text{XF/YXCX}_{2-4}\text{CX}_3\text{FX}_{4-5}\text{LX}_{2-3}\text{HX}_{3-4}\text{HX}_{2-6}$ where X can be any amino acid. The fact that TFIIIA binds 7–11 Zn^{2+} ions per polypeptide chain suggested that the repeated sequences might be Zn^{2+} -binding domains in which each Zn^{2+} is held by two cysteine and two histidine side chains. Each of the nine Zn^{2+} -binding domains might constitute a metal-binding “finger” able to interact with about five bases in the DNA.³⁹¹⁻³⁹⁴ In agreement with this idea was the observation of an ~5-bp repeat of guanine clusters in the DNA.³⁹⁵ The three-dimensional structures of a large fragment of the TFIIIA N-terminal domain (see title page banner for this chapter)³⁹⁶ and of numerous other zinc finger proteins are known.

Yeast proteins often contain a pair of zinc fingers,

but in the nematode *Caenorhabditis elegans* and in *Drosophila* there are more proteins with three or more zinc fingers.³⁹¹ The previously discussed Sp1 has three.³⁹⁰ In *C. elegans* there are more than 100 genes that encode proteins with the Cys_2His_2 zinc-binding motif.³⁹¹ In addition, there are many proteins with four-cysteine zinc-binding motifs. These include Cys_4Zn proteins of the **GATA family** of transcription factors, which are found in fungi, plants, and animals.^{397,397a} GATA-1 is a specific transcription factor for regulation of erythroid genes. It binds to the consensus sequence 5'(T/A)GATA(A/G) found in globin genes.³⁹⁸ In fungi members of the GATA family regulate nitrogen metabolism, biosynthesis of siderophores, and uptake of iron.^{397,399} Another family (LIM) has $\text{Cys}_2\text{HisCysZn}$ domains.³⁹¹ A widely distributed motif in transcriptional repressors is a $\text{Cys}_3\text{HisCys}_4\text{Zn}_2$ or RING finger domain.⁴⁰⁰

If zinc-containing domains lose their Zn^{2+} they do not bind tightly to DNA. Regulation of the flow of zinc ions from storage sites in metallothioneins (Box 6-E) into transcription factors as well as into more than 300 enzymes poses interesting mechanistic questions.⁴⁰¹

Winged helix transcription factors. Liver-specific expression of certain genes in rats depends upon **hepatocyte nuclear factor-3** (HNF-3). Related proteins are encoded by the **forkhead family** of genes in *Drosophila*. These proteins have characteristic C-terminal DNA-binding domains, each consisting of three helices, one of which fits into the major groove of DNA. Also present is a twisted three-strand β structure and two flexible loops or “wings.”^{402,403} The structure of the DNA-binding domain is similar to that of histone H5 and also resembles the HTH domains of prokaryotic repressors (Figs. 5-35, 28-3) and of CAP (Fig. 28-6).⁴⁰⁴ There are many members of the HNF-3/Forkhead family of proteins and of the related **Ets-domain** transcription factors.^{405-406a}

The NF- κ B/Rel proteins. Nuclear factor NF- κ B plays a crucial role in cellular immune responses and in inflammatory disease.^{407-408c} This transcription factor was first recognized for its function in regulating transcription of the κ light chains of immunoglobulins. It is a member of the larger NF- κ B/Rel family, which act in concert with a group of DNA-binding inhibitors of the I κ B family.⁴⁰⁹ The structure of an NF- κ B dimer bound to its DNA target, whose consensus sequence is 5'-GGGRNYYYCC, is shown in Fig. 5-40. Its architecture⁴¹⁰ is quite unlike that of other transcription factors discussed in this book.

HMG proteins as transcription factors. The abundant high mobility group (HMG) nuclear proteins (Chapter 27) bind to DNA, some of them to four-

way junctions. The latter may be present in cruciform structures thought to play a role in regulation of transcription.⁴¹¹ They are often regarded as modulators of chromatin structure. The ~80-residue HMG domain contains three helices and binds into a flattened, underwound, and bent DNA minor groove.^{412,413} HMG proteins also act as transcription factors, which may interact directly with TBP, p53, steroid hormone receptors, and enhancers.⁴¹⁴ Cooperative binding with other DNA-binding proteins is characteristic of the effects of HMG proteins.^{414a} Such interactions may be affected by acetylation. Members of the enhancer-binding HMG-14/-17 family undergo acetylation at seven specific sites.⁴¹⁵

6. The Variety of Transcriptional Responses

Every protein has specialized functions, and specific regulatory mechanisms often control transcription of its genes. A cell must respond to a large number of stimuli, and responses often include activation or repression of transcription.^{415a} In some cases an internal signal, such as a change in concentration of a nutrient or a key metabolite, provides the stimulus. In other cases an external stimulus such as heat or light is the inducer. A few examples follow. Others are mentioned throughout the book.

Many types of hormonal response, including those of insulin^{415b,416} (Chapter 11), are transcriptionally mediated. This is also true for plants (Chapter 30). Defensive responses of both animals and plants (Chapter 31) are mediated in part by transcriptional responses.

Nuclear hormone receptors. Among the best known transcription factors are a large family of hormone receptors, which not only bind specific hormones but also contain in a central domain a pair of Cys₄Zn fingers that interact with response elements (discussed in Chapter 22).^{416a-c} A subfamily of these binding proteins includes receptors for glucocorticoids, progesterone,⁴¹⁷ androgens, and mineralocorticoids. Another member binds the insect hormone 20-hydroxyecdysone (Fig. 22-12) and regulates the puffing seen in giant salivary gland chromosomes of *Drosophila*.^{418,418a} A larger subfamily binds estrogens, vitamin D₃, thyroid hormone,⁴¹⁹ and retinoic acid.⁴²⁰ The vitamin D receptor appears to serve also as a **bile acid sensor**.^{420a} The same subfamily also includes many “orphan receptors.”⁴²¹⁻⁴²⁴ The latter have been discovered by DNA sequence comparisons and have led through “endocrinology in reverse” to discovery of new hormonal signaling pathways.⁴²⁵ The binding of a nuclear receptor to its response element in DNA is well illustrated by the estrogen receptor,^{426-427b} which binds to a palindromic **estrogen response element**

with the consensus sequence 5'-GGTCAnnnTCACC. It is regulated both by hormone-binding and by phosphorylation. The latter is catalyzed by a cyclinA-CDK2 complex in response to cell cycle alterations.⁴²⁸ The glucocorticoid receptor protein forms a complex with a second protein, which has been identified as the 90-kDa chaperonin hsp90.⁴²⁹ A **sterol regulatory element-binding protein** functions in a more general way to activate over 20 different genes that encode enzymes needed for synthesis of cholesterol and unsaturated fatty acids by animal cells.^{429a,b} The steroid hormones often bind to their receptors in the cytosol, and the resulting complex is translocated into the nucleus (p. 1264).

Nutrient control. In addition to the sterols and fat-soluble vitamins, other dietary constituents are also recognized by transcriptional activators or repressors.⁴³⁰ These include glucose,^{431,432} amino acids,⁴³³ phosphate ions,^{434,435} and various metal ions. Best known among the latter is iron. The mammalian **iron response element** (IRE) is a hairpin loop RNA structure, which like the bacterial attenuator system (Fig. 28-9) functions posttranscriptionally.^{436-438a} Iron regulatory proteins (IRP1 and IRP2), which contain Fe₄S₄ clusters, bind to the IRE sequences and inhibit translation. IRP1 is identical to cytosolic aconitase. A high intracellular iron concentration promotes assembly of the Fe₄S₄ cluster and binding to the IRE (see also Chapter 16). In the green algae *Chlamydomonas* a **copper response element** (GTAC) in DNA induces expression of genes important to copper uptake.⁴³⁹ Many nutritional response systems have been recognized first in bacteria. For example, *E. coli* controls uptake of molybdate^{427a,438,440} and of phosphate^{437,441} as well as of sugars and ammonia (discussed in Section A,4).

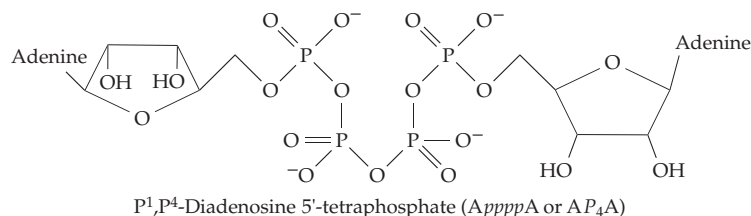
Stress proteins. Ritossa reported in 1962 that when *Drosophila* were suddenly warmed from 25°C to 36–37°C, a series of new puffs could be observed within the polytene chromosomes of the salivary glands.⁴⁴² These same puffs, which were also seen after other stresses, appeared within a few minutes and were associated with formation of new mRNAs. This **heat-shock** phenomenon was subsequently found to be universal. It is observed in all species of animals as well as in plants and bacteria.⁴⁴³⁻⁴⁴⁸

The principal **heat-shock proteins** (hsp), encoded by the new mRNAs, belong to five conserved classes: hsp100, hsp90, hsp70, hsp60, and small heat shock proteins. The function of some proteins as chaperonins has been discussed in Chapter 10. The *E. coli* chaperonin GroEL, a member of the hsp60 group, forms cylindrical aggregates with chambers in which proteins may fold (Box 7-A).⁴⁴⁹ In a similar way a 16.5-kDa small heat-shock protein of *Methanococcus*

jannaschii forms 24-subunit hollow spheres with octahedral geometry.⁴⁵⁰ The structure of the 68- to 70-kDa hsp 70 (dnaK) in *E. coli* has been conserved with high homology throughout evolution. Using DNA chip technology 77 heat-shock genes have been identified recently in *E. coli*.⁴⁵¹

Induction of heat-shock proteins depends upon a heat-shock promoter element (**HSE**) that binds an activating transcription factor **HSF**.^{452–455} An increase in temperature not only induces synthesis of heat-shock proteins but represses synthesis of most other proteins. Thus, in *E. coli* or *Salmonella* a shift from 30°C to 42°C causes the appearance of 13 heat-shock proteins. At 50°C synthesis of almost all other proteins stops. In *E. coli* transcription of heat-shock genes is controlled by alternative factors, σ^{32} and σ^E .^{456,456a}

In *S. typhimurium* a series of unusual nucleotides such as P^1 , P^4 -diadenosine 5'-tetrphosphate (ApppA or Ap_4A) accumulate. The related compounds ApppGpp (with a 3' pyrophosphate), ApppG, AppG, and ApppA also accumulate and appear not only in bacteria but in eukaryotes as well.⁴⁵⁷ They are formed as a side product in aminoacylation of transfer RNAs (Chapter 29).^{458,459} Lee *et al.*⁴⁶⁰ proposed that Ap_4A and related nucleotides are formed in response to oxidative stress and serve as **alarmones** that signal the need to reduce transcription of most genes and to increase transcription of genes for protective proteins. However, there is little correlation between the Ap_4A concentration and the heat shock response.⁴⁶¹ See also p. 1715.



Another type of **stress response protein** is related to glutathione *S*-transferase.⁴⁶² Cells of *E. coli* also accumulate a 15.8-kDa **universal stress protein** in response to most types of stress. It is an autophosphorylating phosphoprotein, a member of a little-studied group of phosphoproteins normally present in very small amounts.⁴⁶³ Cells also respond to various other types of stress such as deprivation of glucose, hypoxia,⁴⁶⁴ ultraviolet irradiation, presence of hydrogen peroxide,^{465,465a} or change in osmotic pressure.⁴⁶⁶ Salt-tolerant plants synthesize new proteins in response to increased salinity of water.⁴⁶⁷

Responses to low oxygen tension in tissues (**hypoxia**) are important to all aerobic organisms.^{464,467a–d} In mammals transcription of hypoxia-responsive genes is regulated by **hypoxia inducible factors** HIF-1 and

HIF-2. A subunit of HIF-1 undergoes 2-oxoglutarate-dependent hydroxylation on proline and asparagine residues. This may be a step in induction of ubiquitination and destruction of this component of the transcription factor complex. The human **von Hippel-Lindau** (VHL) tumor suppressor is a ubiquitin E3 ligase, which is also present in this transcription factor complex.^{467b,d–h} One subunit of HIF-1 also interacts with the tumor suppressor p53.⁴⁶⁷ⁱ Together with the VHL protein and the elongation factor elongin (p. 1637)^{467j} HIF participates in controlling both production of red blood cells and growth of new blood vessels (angiogenesis, Chapter 32, Section D).^{467k}

Light-induced transcription. Light has a strong effect on transcription, especially in plants and photosynthetic bacteria. The photosystem II subunits D1, D2, CP47, and CP43 (see Fig. 23-34) are encoded in the chloroplast genome. D1 and D2 are unstable in light, and their rate of synthesis is increased as a result of elevated levels of transcription that are induced by a blue light response.⁴⁶⁸ The light-induced conversion of phytochrome to its far-red absorbing form **Pfr** (Eq. 23-42) causes increased transcription of a variety of plant genes.⁴⁶⁹ See also Chapter 23.

Homeotic genes and homeodomain proteins.

Geneticists discovered in *D. melanogaster* and other species genes that establish the placement of antennae and legs on particular segments and in general to specify the body plan.^{470,471} These homeotic genes encode a series of proteins containing a 60-residue **homeodomain**, a DNA-binding domain of the helix–turn–helix class.^{470,472} Some homeodomain-containing proteins bind to DNA containing the octameric sequence shown in Table 28-1 and are known as **octamer-binding transcription factors** (Oct). One of these (Oct-2) is specifically needed for activation of immunoglobulin genes,^{473,474} while Oct-1 binds to promoters of various other genes including that of histone H2B, U1, and U2 snRNAs.^{475,475a} Another transcription factor, **Pit-1**, which activates genes for growth hormone and for prolactin in the pituitary, binds to the same octamer.^{476,476a} Homeotic genes are considered further in Chapter 32 and immunoglobulin genes in Chapter 31.

7. Transcription by RNA Polymerases I and III

Promoters for RNA Pol I, like those of Pol II, lie upstream of the initiation site for transcription. At least two transcription factors have been identified^{477,477–478a} and vary among species. The human factors bind to a G•C-rich DNA sequence in the –45 to

+20 region and to a related upstream control element **UCE** at -180 to -107.

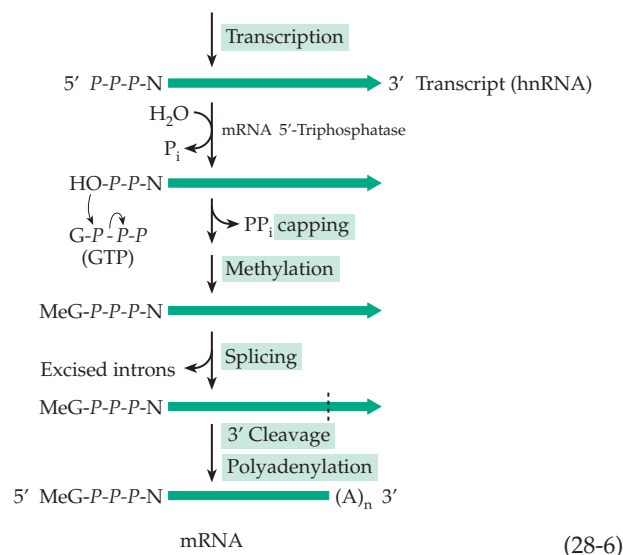
In vertebrate cells transcription by RNA pol III is controlled by three types of promoter: class 1 (5S RNA), class 2 (tRNAs), and class 3 (U6 snRNA).^{47,479,480} Initiation of transcription of class 1 genes requires the 40-kDa **transcription factor TF IIIA**.⁴⁷⁹ This was the first eukaryotic transcription factor to be purified and characterized,^{481,482} and one of the first of the zinc finger proteins to be recognized. It was a surprise to find that TFIIIA does not bind to the promoter region but to a DNA sequence in the center of the 5S RNA gene between positions +55 and +80.⁴⁷ The presence of the TFIIIA binding site in the middle of the gene it controls suggests that TFIIIA interacts with other proteins that bind in the promoter region to form a loop. TFIIIA is involved in initiation of transcription of the ~24,000 oocyte type 5S RNA genes of *Xenopus*, but the ~400 somatic type 5S RNA genes are not activated in the same way. TFIIIA is also unusual in that it binds not only to DNA but also to a specific CCUGG sequence in the transcribed 5S rRNA. This RNA is stored as a 7S ribonucleoprotein particle until it is needed after the oocyte is fertilized and begins rapid protein synthesis.⁴⁸³

After TFIIIA binds, proteins TFIIIC and then TFIIIB also bind. Although promoters of classes 2 and 3 do not require TFIIIA, all three classes depend upon TFIIIB and TFIIIC.⁴⁸⁴ The TATA-binding protein TBP is one of three components present in TFIIIB, which may be regarded as the true initiation factor.^{47,485} Both TFIIIA and TFIIIC can be described as **assembly factors**.⁴⁷ A silkworm RNA pol III has been reported to require a transcription factor consisting of RNA.⁴⁸⁶

8. Elongation, Modification, and Termination of Transcription

As in prokaryotic transcription (Fig. 28-4) elongation by RNA polymerase II occurs within a transcription bubble of ~20–30 nucleotides in length.⁴⁸⁷ Most transcriptionally active DNA is still in the form of nucleosomes, which must be unwound as the transcription bubble moves. Details are still uncertain.^{269,488} All of the major steps in processing of the pre-mRNA transcripts, which include capping, splicing, 3'-end cleavage, and polyadenylation (Eq. 28-6), are coupled to transcription. This is apparently accomplished, in part, by physical connections of the necessary proteins to the CTD domain of RNA polymerase II.^{304,312a,b} While pre-mRNA usually undergoes all of the steps of Eq. 28-6, rRNA and tRNAs are not capped or polyadenylated and often are not spliced.

Elongation of the RNA chain depends upon five **general elongation factors**, designated **P-TEFb**, **SII** (TFIIS), **TFIIF**, **Elongin** (SIII), and **ELL**.^{489–492} Many of



these subunits of the transcriptosome function in the suppression of pausing in the uneven movement of the template DNA through the complex. Some may have to deal with torsional strain induced in the transcribed DNA.⁴⁹³ Elongin is a heterotrimer of subunits A, B, and C.⁴⁹⁴ Most of the other elongation factors are heterodimers,⁴⁸⁹ but SII is a monomer that contains two conserved Zn²⁺-binding **zinc ribbon** motifs.⁴⁹² It promotes cleavage of the growing transcript within the transcription bubble at stalled sites, allowing transcription to be restarted from a fresh 3' end. Defects in elongation factors P-TEFb, ELL, and **CSB** (Cockayne syndrome complementation group) have been correlated with human diseases including cancer.⁴⁹⁵

Despite the complexity of the processes represented by Eq. 28-6, a yeast cell is able to transcribe genes at rates of about one in every 6–8 s.⁴⁹⁶ This can be compared with a rate of about once in 2–3 s for RNA polymerase of *E. coli*.

The least well defined step in eukaryotic transcription is termination, which follows the various steps of processing, discussed in Section D. The final 3' end of processed transcripts of RNA pol II action in mammals is marked by the sequence AAUAAA, which is found about 10–30 nucleotides upstream from the end.⁴⁹⁷ This is usually followed by a polyU or GU-rich sequence.⁴⁹⁸ In yeast the termination and polyadenylation signals are less clear.^{499,500} The initial transcript almost always continues beyond the AAUAAA signal, sometimes for hundreds of nucleotides. However, the excess RNA is rapidly degraded by a large complex of proteins.^{499–502} The precise 3' end cleavage is energy-dependent, requiring creatine phosphate rather than ATP or GTP.⁵⁰⁰ Transcription termination by RNA polymerases I and III is more like that of bacteria. Terminator sequences are present in the DNA, and terminator proteins interact with them.^{480,503–504b}

9. Conformational Properties of RNA

Newly formed RNA transcripts fold quickly into structures of complex shapes,^{505–508b} the folding being influenced by interactions with proteins and with other RNA molecules. RNA chains are flexible, with many sterically allowed conformations.⁵⁰⁹ As with proteins folding probably begins with a nucleation event, perhaps involving monovalent or divalent metal ions^{510–512} and continues rapidly.⁵¹³ Folding is affected by hydrogen bonding,⁵¹⁴ base stacking, and binding of ions, and by formation of pseudoknots (Fig. 5-29).⁵¹⁵

As is apparent from the structures of tRNAs (Figs. 5-30, 5-31, 28-20), the *Tetrahymena* self-splicing ribozyme (Fig. 12-26), and ribosomal RNA structures (Fig. 29-2), a large fraction of a folded RNA exists as hairpin or **stem-loop** structures. These are A-type structures with largely Watson-Crick base pairs. However, mismatched pairs, triples, and quadruples of bases are also formed. Recently discovered RNA structural elements include **base platforms**, formed by pairing of adjacent bases,⁵¹⁶ interdigitation of unpaired bases (also seen in DNA; Fig. 5-27), and wobble pairs (Chapter 5).⁵¹⁷ Hydrogen bonding between riboses of consecutive nucleotides in two strands may help to form a **ribose zipper**.^{437,507} Guanine-rich tetraplexes (pp. 208, 227), cytosine-rich i-motif structures (p. 228), and water-mediated U•C base pairs also arise in RNAs.^{517a–c}

The terminal loops, which usually contain the consensus sequence **GNRA**, may constitute up to one-third of the entire molecule.⁵¹⁸ These loops interact with many binding proteins, such as those in the snRNA-protein particles.⁵¹⁹ GNRA loops may also dock into the shallow groove of RNA helices.⁵¹⁸ Adenosines that are not paired in double helices, e.g., those in GNRA loops, are able to interact in a variety of ways with other parts of an RNA molecule or with other molecules. They are involved in helix packing interactions in virtually every RNA studied.^{517d} Although examples are still rare, specific mRNA molecules may provide binding pockets for small regulatory molecules, e.g., amino acids and thiamin.^{517e}

D. Processing of Eukaryotic RNAs

All RNA found in eukaryotes undergoes major alterations prior to functioning. The cutting out of rRNA and tRNA molecules from larger precursors resembles that in bacteria, but subsequent processing is much more complex, as is that of mRNA.

1. Ribosomal RNA

Eukaryotic ribosomes contain four pieces of RNA (Tables 5-4 and 29-1), which are usually designated by their sedimentation coefficients. The 18S, 5.8S, and 28S RNAs are encoded as single transcriptional units with spacers separating the sequences that encode the mature RNAs. A typical animal cell contains several hundred copies of this transcriptional unit, all located in the DNA in the nucleolus (Fig. 28-15), and each having its own set of promoter sequences, enhancers, and transcription factors.^{47,520–522} The promoter sequences vary substantially among different species.⁵²³ The primary transcripts from these units are the sole product of RNA polymerase I.

Electron micrographs of portions of unwound cores of nucleoli have revealed fibrils of RNA coated with protein growing from the DNA strands of the pre-rRNA genes (Fig. 28-16), ~80–100 RNA chains of different length being transcribed concurrently from a single gene. The overall gene length in the electron microscope is 2.3 μm , only a little less than the calculated length for a fully extended DNA molecule in

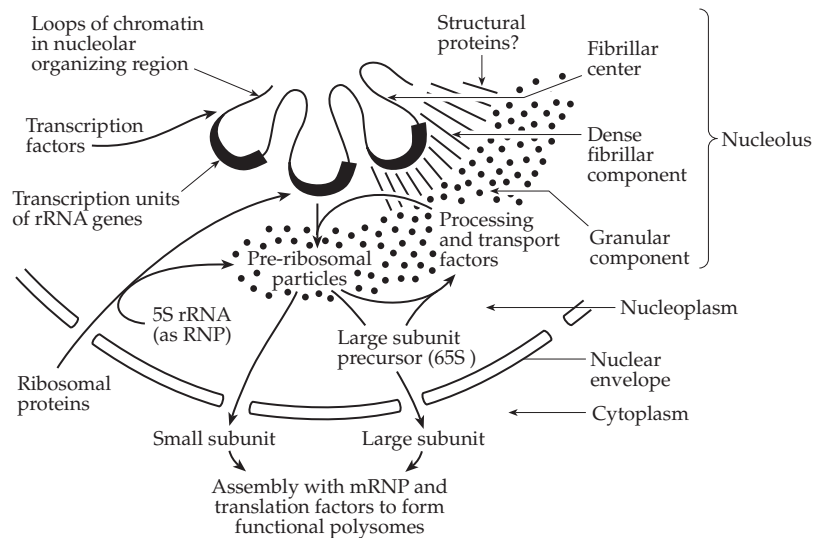


Figure 28-15 Steps in ribosome formation in the nucleolus. From Sommerville.⁵²⁴ Abbreviations are NOR, nucleolar organizing regions; RNP, ribonucleoproteins.

the B form. However, judging by the lengths of the transcripts formed, the pre-rRNA chains are folded extensively.

The primary eukaryotic rRNA transcripts extend several hundred nucleotides past the 3' termini of the

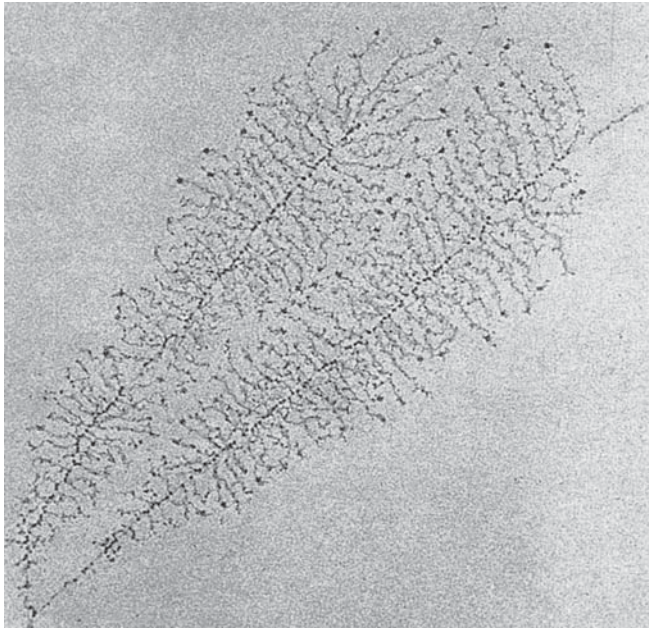
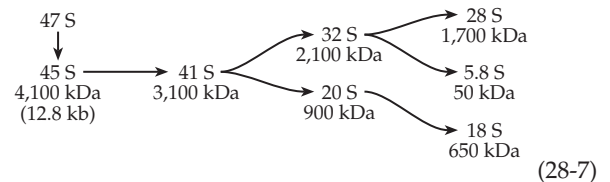


Figure 28-16 Ribosomal RNA genes from an embryo of *Drosophila melanogaster* in the process of transcription. The densely packed ribonucleoprotein strands are shorter where transcription begins and contain increasing lengths of rRNA where transcription has proceeded for a longer time. Also note the characteristic granular knobs at the tips of the strands. From Miller.⁸⁵

mature 28S RNA molecules. As the primary transcripts, which are formed in the core region of the nucleolus, move away into the outer cortex of the nucleolus, cleavage occurs in a number of steps (Eq. 28-7).^{525,526} Electron microscopy provided the first direct confirmation of the relationship of one pre-rRNA molecule to another (Fig. 28-17).^{527,528} The 18S portion of the 45S RNA seen in Fig. 28-17 lies nearest to the 5'-end just as does the 16S rRNA in the large transcript of the prokaryotic rRNA genes (Section A,7).



As is indicated in Fig. 28-15, transcription is thought to occur from the loops of DNA that form the nucleolar organizing region. The 100-kDa **nucleolin**, the major protein of the nucleolus, binds to the non-transcribed spacer sequences in the DNA.⁵²⁹⁻⁵³⁰ It also binds to the newly formed transcripts, as do various proteins that enter the nucleus from the cytoplasm.^{524,531} More than 270 proteins, many of which participate in synthesis of ribosomes, have been detected in the nucleolus.^{531a} Some of these proteins, acting together with the snoRNAs discussed in the next section, catalyze hydrolytic cleavage of the pre-rRNA molecules. For completion of pre-ribosomal particles additional protein molecules enter the nucleolus and associate with the pre-rRNA particles, then diffuse out of the nucleus.

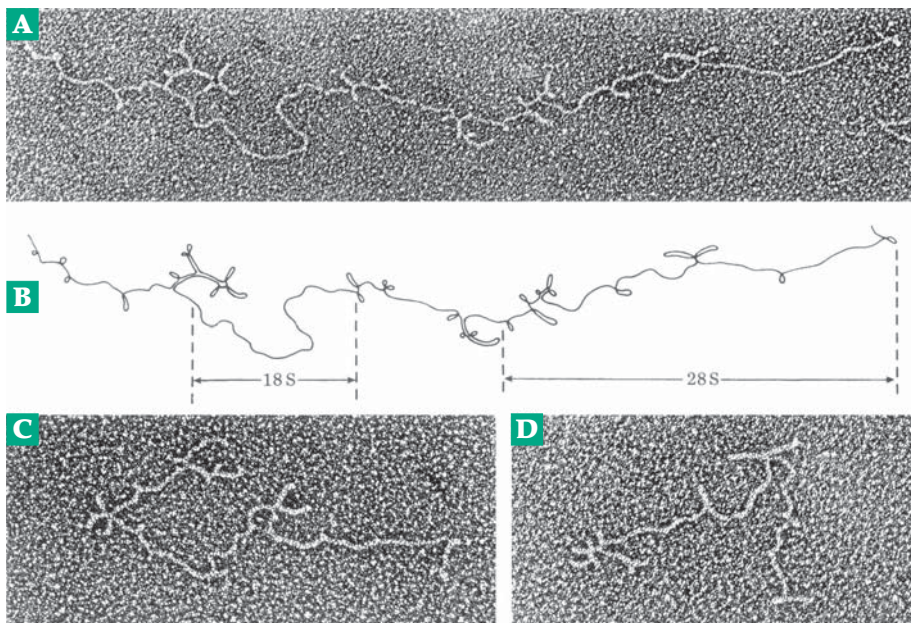


Figure 28-17 (A) Electron micrograph of the 45S precursor of rRNA from HeLa cells after spreading from 80% formamide and 4 M urea. The molecule is shown in reverse contrast. (B) Tracing of molecule in (A) showing several regions of secondary structure as hairpin loops. The 28S and 18S rRNA regions are indicated. (C) 32S rRNA. (D) 28S rRNA. Notice that the same secondary structure can be seen in the 28S RNA as in its 32S and 45S precursors. From Wellauer and Dawid.⁵²⁷

2. Small Nuclear, and Nucleolar, and Cytoplasmic RNAs

The nuclei of all eukaryotic cells contain a group of small nuclear RNAs (**snRNAs**), some of which (**snoRNAs**) are localized to the nucleolus and function there.^{532–535e} At least ten of these are always present (Table 28-3), and yeast appears to contain more than 100. Some are present in small amounts, and it may be that a large number of snRNAs will also be found in other eukaryotes. The uridine-rich or U series of snRNA are especially abundant (10^4 – 10^6 molecules per nucleus). Several of them (U1,2,4–6,11,12,16,18) function in RNA splicing. Species U1–U10 contain from ~60 to ~215 nucleotides but some snRNAs from yeast have over 300 nucleotides and one over 1000.⁵³² The most abundant snRNAs, the metabolically stable U1, U2, U4, U5, and U6, are exported from the nucleus. In the cytoplasm each snRNA associates with a complex of several proteins to form ribonucleoprotein particles (**snRNP particles**). The proteins, known as **Sm proteins**, range in mass from 11 to 70 kDa⁵³⁶ and are designated B, B', D1, D2, D3, E, F, and G.^{535,537,537a} The proteins may associate to a complex B₂D₂EFG, which then binds the snRNAs. Whereas many proteins are found in most of the snRNP particles, some are associated with specific RNAs. Thus, mature U1 snRNP may have a partial stoichiometry U1A₂B₂C₂D₄E.^{536,538} The snRNA particle, known as U4/U6snRNP, contains two snRNAs. After returning to the nucleus the four snRNP complexes U1, U2, U5, and U4/U6, together with the pre-mRNA molecules, associate via an ordered assembly pathway that gives rise to the large **spliceosomes** in which the removal of introns takes place.⁵³⁹ Some of the snRNP complexes, e.g. the spliceosomal U6 snRNP and snRNPs involved in mRNA degradation, contain Sm-like (Lsm) proteins.^{537a}

More recently small 20–25 nucleotide antisense micro RNAs (**miRNAs**) have become recognized as important in control of the breakdown of mRNA, in formation of heterochromatin, and in control of specific stages in development (Chapter 32), and in cellular defense mechanisms (Chapter 31, Section G).^{1a,537b} In animals the 21–23 nt miRNAs are formed by enzymatic cleavage of double-stranded dsRNA by a ribonuclease III-type enzyme called **Dicer**.^{171a,537c} In the resulting fragments, which have 5'-phosphate termini, one strand is antisense to a “target” sequence in mRNA. **Small temporal RNAs** (stRNAs) that guide development are cut by Dicer from RNA stem-loop structures. In other cases, e.g., in formation of the siRNAs that silence individual genes, an RNA-dependent RNA polymerase acts on an RNA transcript to form a long dsRNA that can be cleaved by Dicer. This often happens with foreign RNAs, e.g., from viruses or trans-genes. Unwinding of the small

fragments formed by Dicer, in a process that may require ATP, provides the single-stranded antisense siRNA molecules. Such antisense molecules bind to their target RNAs together with a group of proteins to form **RNA-induced silencing complexes** (RISCs). These complexes contain an RNase (different from Dicer) that cuts the dsRNA of an siRNA-target complex, initiating destruction of the mRNA and silencing of its gene.^{1a,171a,539a} This **RNA-interference** is widely utilized by plants and animals. It has also provided the basis for development of many practical tools for understanding gene sequences, for genetic engineering, and for design of new drugs.

Processing of ribosomal RNA. Transcripts of rRNA genes vary in size from ~35 to 47S (6–15 kb) and often contain spacer regions at both ends as well as between the 18S, 5.8S, and 28S sequences. For example, human 47S transcripts have 414 extra nucleotides at the 5' ends.⁵⁴⁰ One group of snoRNAs participates in the hydrolytic cleavage of pre-rRNA. RNAs U3, U14, snR10, snR30, as well as MRP RNA (Table 28-3) are always required.^{541–545} Also needed is U22, an intron-encoded RNA.⁵³¹ The reactions represented by Eq. 28-7 are best known for *Saccharomyces cerevisiae*.^{525,541–541c} Four spacers, the **5'-external transcribed spacer** (5'-ETS), the first and second **internal transcribed spacers** (ITS1 and ITS2), and the 3'-external transcribed sequence (3'-ETS), must be removed.^{525,541,546} Removal of the 5'-ETS depends upon the snoRNA U3, which contains two highly conserved sequences able to form base-paired structures with the 5' end of the 5.8S rRNA region of the pre-rRNA gene.^{542–544,546a} Both U8 and U22 are also needed for cleavage of the pre-rRNA.^{546b} Although the exact functions of snoRNAs and their associated proteins in the cleavage of pre-rRNA are still uncertain, these RNAs probably act as guide molecules for the cleavage reactions. They may have ribozyme activity and may perhaps be chaperones.



Cleavage at the 5' end of the 5.8S region requires RNase MRP, a relative of the RNase P that cleaves at the 5' ends of tRNAs (Fig. 28-10).^{525,547} MRP (**mitochondrial processing protein**) also cleaves primers for mitochondrial DNA replication. The importance of the enzyme is emphasized by the existence of a hereditary defect in the MRP RNA (Table 28-3) that causes abnormalities in bone, cartilage, hair, and the immune system.^{547a} Most bacterial rRNA genes have a tRNA gene in the position corresponding to that of 5.8S RNA

TABLE 28-3
Some Eukaryotic Small Nuclear and Nucleolar RNA Molecules

Designation	Number of nucleotides	
	Vertebrate	Yeast
U1 ^a	164	568
U2 ^a	188	1175
U3 ^b	206 – 228	333
U4 ^a	142 – 146	160
U4 _{atac} ^c	131	
U5 ^a	116 – 118	183 or 196
U6 ^{a,d}	116	214
U6 _{atac} ^c	125	
U7 ^e	57 – 58	
U8 ^b	136 – 140	
U9	130	
U10	60	
U11 ^c	131 – 135	
U12 ^c	150	
U13 ^{b,f}	105	
U14 ^{b,f}	87 – 96	125 – 128
U18 ^{b,f}	67 – 70	102
X ^b	150	
U20 ^{b,f}	80	
U21 ^{b,f}	93	
U22(Y) ^b	125	
U24 ^{b,f,g}	77	
U32 – U40 ^{b,f,h}		
SnR10 ^b		245
SnR30 ^b		605
SnR38 ^f		93
SnR39 ^f		85
SnR40 ^f		96
SnR41 ^f		
MRP RNA (RNA 7–2)	260 – 280	339

^a Major spliceosomal RNAs.

^b Fibrillarin-associated SnoRNAs that function in pre-ribosomal RNA processing. See Morrissey, J. P., and Tollervy, D. (1995) *Trends Biochem. Sci.* **20**, 78–82.

^c Function in AT–AC spliceosomes. See Tarn, W.-Y., and Steitz, J. A. (1997) *Trends Biochem. Sci.* **22**, 132–137 and Fournier, M. J., and Maxwell, E. S. (1993) *Trends Biochem. Sci.* **18**, 131–135.

^d γ -Monomethyl cap.

^e Required for 3'-end formation in histone mRNAs.

^f SnoRNAs with long complementarities to rRNA. C and D sequences are present. See Bachelierie, J.-P., Michot, B., Nicoloso, M., Balakin, A., Ni, J., and Fournier, M. J. (1995) *Trends Biochem. Sci.* **20**, 261–264 and Nicoloso, M., Qu, L.-H., Michot, B., and Bachelierie, J.-P. (1996) *J. Mol. Biol.* **260**, 178–195.

^g Polyadenylated, noncoding.

^h Participate in 2'-O-ribose methylation.

in eukaryotes. This provides an RNase P-dependent cleavage mechanism, which is alternative to action of other nucleases.⁵²⁵

The most abundant protein in the fibrillar regions of the nucleus, where the early stages of pre-rRNA processing occur, is **fibrillarin**.^{541,548,549} Many of the snoRNAs are closely associated with this protein. Fibrillarin is also well known as an autoantigen, which can induce formation of destructive antibodies that cause **scleroderma** (Chapter 31, Section F).

Modification guide RNAs. A second group of snoRNAs function in methylation, pseudouridine formation, and other RNA modifications (Section 6).^{174,525,541,548} These snoRNAs have long sequences complementary to highly conserved regions of pre-rRNA, enabling them to form helical regions that may guide the docking with modification enzymes. Many of them also contain characteristic conserved sequences: **C**, 5'-UGAUGA; **D**, 5'-CUGA; **H**, 5'-AnAnnA; and 5'-ACA. Sequences C and D are present in snoRNAs that act as methylation guides, while the H and ACA sequences characterize guide RNAs for pseudouridine formation.¹⁷⁴

Transcription and processing of snRNAs and snoRNAs. In higher organisms each of the snRNAs has several genes,⁵³⁸ e.g., there are 50–60 U1 genes in the human haploid genome. However, in yeast there are often single copies.⁵⁴¹ All of the snRNA genes, except for that of U6,⁵⁵⁰ are transcribed by RNA polymerase II. The transcripts, which are capped at the 5'-end but are not polyadenylated, pass into the cytoplasm, where they undergo further processing and become associated with proteins. The 3' ends are trimmed, the 7-methylguanosine of the cap is methylated further, and methylation may occur on other bases as well.⁵⁵¹ SnoRNAs are not capped. It was a great surprise to discover that many of the snoRNA genes lie within introns that occur in abundantly expressed genes for functionally unrelated proteins.^{173,541,548}

Patients with the autoimmune disease systemic **lupus erythematosus** make autoantibodies directed against the Sm proteins of snRNP particles.^{552,553} Antibodies from different patients vary in their specificities, making these antibodies a useful tool in the isolation and study of snRNAs and their protein complexes.^{552,554}

3. Processing of 5S RNA and tRNAs

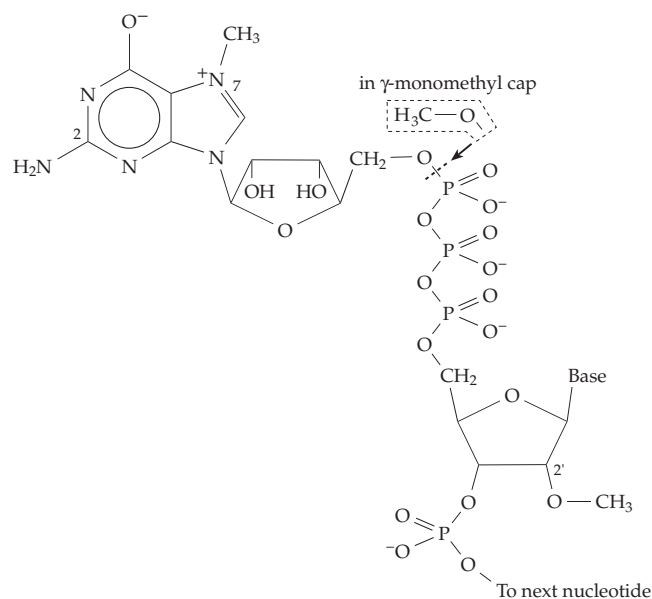
The genes for 5S ribosomal RNA and all of the tRNAs are transcribed by RNA polymerase III. In the yeast genome the 5S RNA genes are located in the spacers between the transcriptional units containing the other rRNAs. However, in animals the 5S RNA

genes are separate from the other rRNA genes and are not located in the nucleolus. In *Drosophila* ~500 copies of the 5S RNA gene are located in the right arm of chromosome 2. In *Xenopus* ~400 genes are active in somatic cells and another 24,000 only in oocytes. These are arranged as large transcriptional units, each containing several thousand copies of the 120-bp 5S RNA gene separated by 720-bp spacers. Cleavage and trimming are required to form the final product, but there are usually no introns to be removed.

Most eukaryotic tRNAs are formed from monomeric precursors, each gene acting as a transcriptional unit. Processing is similar to that in bacteria (Section A, 7). Eukaryotic RNase P usually cleaves the 5' end,^{555–558c} and another enzyme cuts at the 3' end.^{542,556} The 3' CCA sequence of the mature tRNA is usually not present in the primary transcript but is added.^{559,559a,b} As in bacteria (p. 1620) extensive modification of bases also occurs in the tRNA precursors of eukaryotes.^{235,560–562} Many tRNA genes contain introns, which must be removed by splicing (Section 5).

4. Messenger RNA, Caps, and Polyadenylate Tails

The first processing event (Eq. 28-6) for most of the pre-mRNA and snRNA transcripts made by RNA polymerase II is addition to the 5' end of a “cap,” a terminal structure containing 7-methylguanosine from which a proton has dissociated to form a dipolar ion.^{563–565} The cap structure may be abbreviated 5'-m⁷G(5')pppNm—. The 5' terminal ribose is often methylated on O2', as shown below. More complex caps are methylated at additional sites, e.g., the guanine may be dimethylated on the 2-NH₂ group.⁵⁵¹ Most snRNAs, including the U1–U5 and U7–U13 snRNAs, have such 2,2,7-trimethylguanosine



caps.^{551,566} Many viral transcripts, including those of the much-studied vaccinia virus, have similar caps.⁵⁶⁴ However, U6 and some other snRNAs, which are transcribed by RNA polymerase III, have γ -monomethyl (me-ppp) caps⁵⁶⁷ or undergo a series of additions and deletions of uridylylate residues at the 3' ends.^{566,568}

Cap synthesis occurs as follows. The 5' end of an RNA transcript initially contains a triphosphate group arising from the fact that a nucleotide triphosphate serves as the primer in initiating transcription. The terminal phospho group is removed by a triphosphatase leaving a diphosphate, which is then guanylated by GTP (Eq. 28-6).⁵⁶⁹ The capped transcripts are exported from the nucleus, after which additional methylation may follow.⁵⁷⁰

The cap structure affects several processes.^{565,571} A family of cap-binding proteins recognize the structure and may facilitate splicing as well as export from the nucleus.^{571a} The cap is very important for ribosome binding and initiation of translation (Chapter 29). The trimethylated caps of snRNAs, on the other hand, may be signals for retention in the nucleus where they function.⁵⁶³ Following capping is the often elaborate process of splicing to remove introns (Section 5).

A poly(A) “tail” consisting of ~250 residues of adenylic acid is added next by poly(A) polymerase, a component of an enzyme complex that also cleaves the RNA chains.^{545,571b} Most eukaryotic mRNA is polyadenylated with the exception of that encoding histones. The function of the poly(A) is unclear. It is needed for transport of mRNA out of the nucleus, but it does confer a greatly increased stability to the mRNA in the cytoplasm where the adenylate units are gradually removed.^{307,308} In contrast, in chloroplasts and plant mitochondria polyadenylation is required for rapid degradation of mRNA.^{571c,d} Polyadenylation may also increase the efficiency of translation.⁵⁷² Polyadenylation occurs rapidly within ~1 min after transcription is completed.

Mature mRNA molecules vary in lifetime.^{573,574} Some last for hours or days. Among the latter are mRNAs of maternal origin that accumulate in oocytes and are utilized during the early stages of embryonic development.^{575,575a,b} Other mRNAs, e.g., transcripts of the *c-fos* and *c-myc* proto-oncogene products, have half-lives of 30 min.^{573,576} Some mRNA molecules are degraded while attached to ribosomes in response to recognition of the synthesized peptide (Chapter 29). Longer lived mRNA molecules may be protected by RNase inhibitors.⁵⁷⁷ Hydrolytic removal of caps often initiates degradation, and Sm-like protein complexes participate.^{578–579a}

5. Splicing

An essential modification of the precursor forms

of large rRNA, tRNA, and mRNA molecules is the splicing out of intervening sequences. This occurs prior to polyadenylation of mRNA and is usually slow, the half-life of introns varying from a few seconds to 10–20 min.⁵⁵¹ Splicing occurs by at least four distinctly different pathways.^{47,580}

Self-splicing RNA. The precursor to the 26S rRNA of *Tetrahymena* contains a 413-nucleotide intron, which was shown by Cech and coworkers to be self-splicing, i.e., not to require a protein catalyst for maturation.^{581,582} This pre-rRNA is a ribozyme with true catalytic properties (Chapter 12). It folds into a complex three-dimensional structure which provides a binding site for free guanosine whose 3'-OH attacks the phosphorus at the 5' end of the intron as shown in Fig. 28-18A, step *a*. The reaction is a simple displacement on phosphorus, a transesterification similar to that in the first step of pancreatic ribonuclease action (Eq. 12-25). The resulting free 3'-OH then attacks the phosphorus atom at the other end of the intron (step *b*) to accomplish the splicing and to release the intron as a linear polynucleotide. The excised intron undergoes

a third transesterification reaction, of uncertain significance (step *c*), to form a circular polynucleotide and a short displaced 15-residue oligonucleotide. The *Tetrahymena* pre-rRNA intron is a member of a group of similar **Group I introns**, many of which are found in fungal mitochondrial pre-mRNA and pre-rRNA.⁵⁸³ All are excised by a similar mechanism. Many are self-splicing, but others require a protein catalyst.^{584–585a} A similar splicing sequence is involved in removal of a 1017-nucleotide intron from the thymidylate kinase gene of phage T4 and other introns in T-even phage. The later are among the relatively rare introns in prokaryotic systems.⁵⁸³

All Group I introns have several small conserved sequences, which suggest a common folded tertiary structure as is indicated in Fig. 28-19A. The conserved sequences are labeled **A**, **B**, **9L**, **2**, **9R**, and **9R'**. **A** is paired with **B**, **9R** with **9R'**, and **9L** with **2**. The sites of chain cleavage at the 3' and 5' ends of the intron are indicated by the heavy arrows. They are evidently selected by formation of the double-stranded regions.^{586–588}

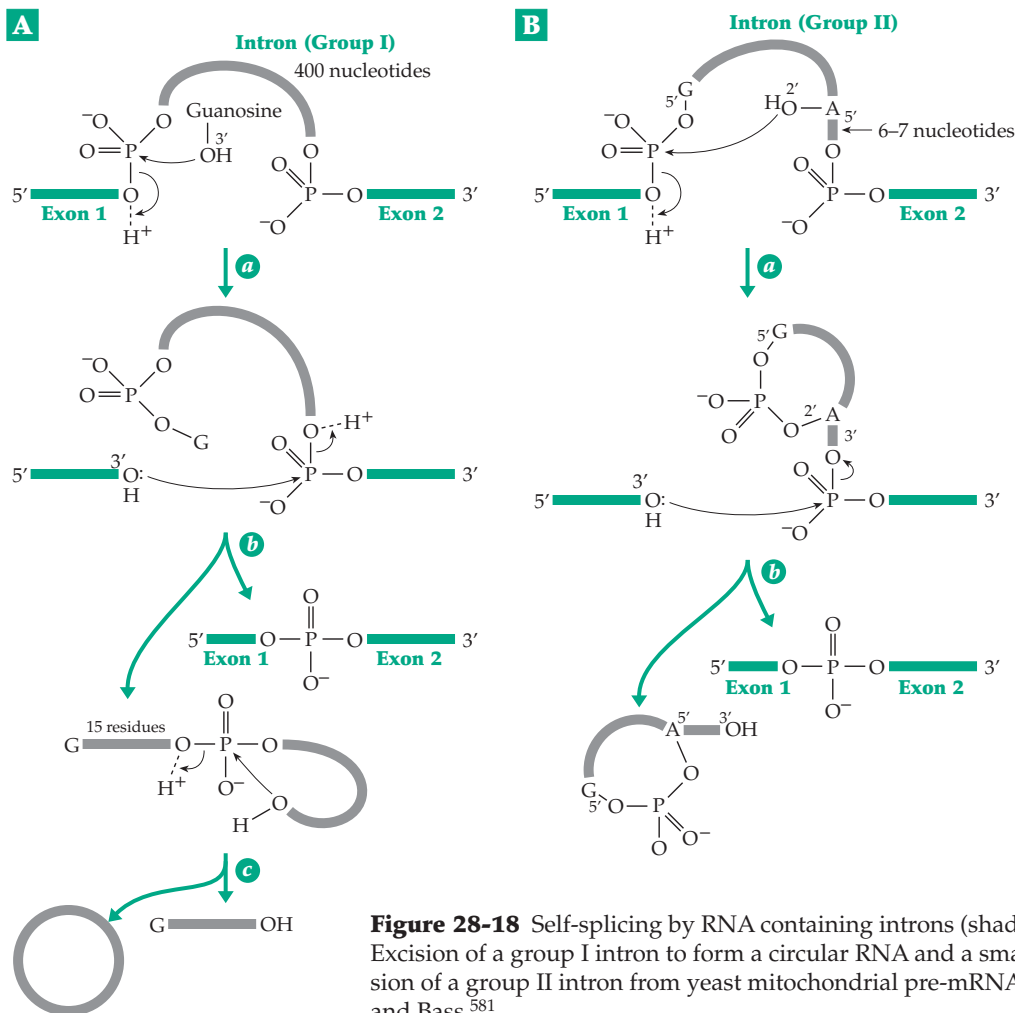
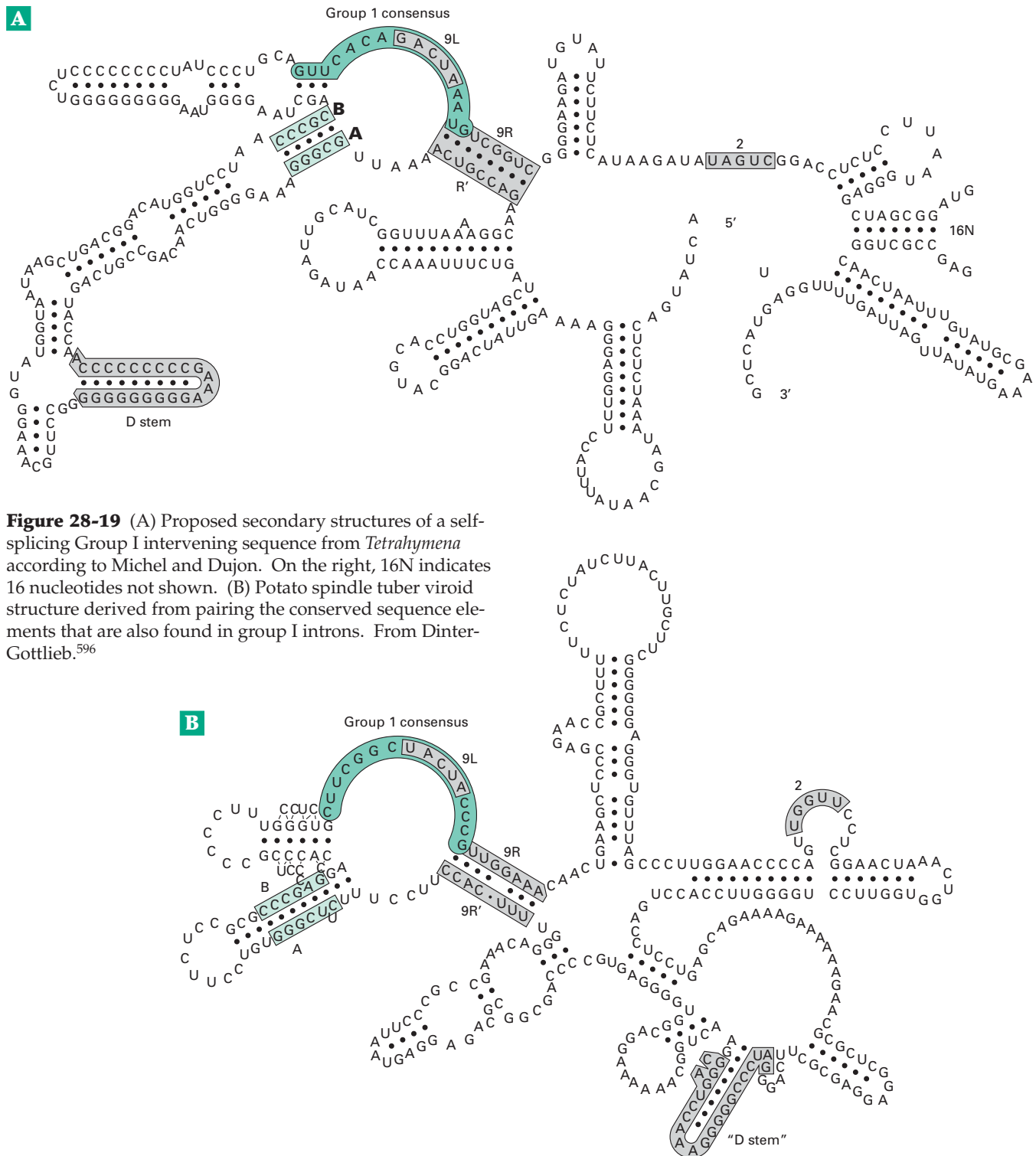


Figure 28-18 Self-splicing by RNA containing introns (shaded) of groups I or II. (A) Excision of a group I intron to form a circular RNA and a small oligonucleotide. (B) Excision of a group II intron from yeast mitochondrial pre-mRNA as a circular RNA. See Cech and Bass.⁵⁸¹

Another type of intron (Group II) also undergoes self-splicing.^{589–590d} The best known example is the last intron in the yeast mitochondrial pre-mRNA. The splicing pathway shown in Fig. 28-18B is similar chemically to that of the group I introns. However, the initial attack is not by free guanosine, but by the 2' OH of

an internal adenosine, the intermediate product having a **lariat structure** with a loop at the end. Otherwise, processing is similar to that of group I introns. The same pathway with lariat formation is followed by the more widely used removal of introns from pre-mRNA in spliceosomes (see Fig. 28-22).



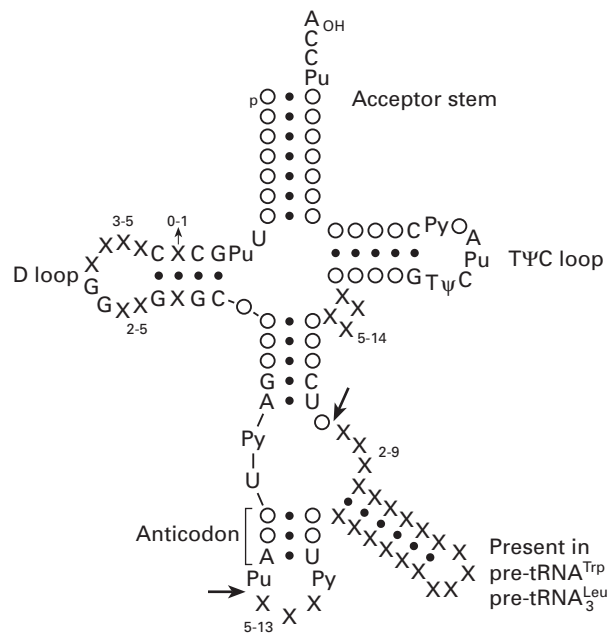


Figure 28-20 Composite structure representing several tRNA precursors arranged in a similar secondary structure (see also Fig. 5-30). The arrows indicate splice points. Variable positions are designated (O) for the mature tRNA and (X) for the intervening sequence and also in loops where insertions or deletions occur. From Ogden *et al.*⁶⁰³

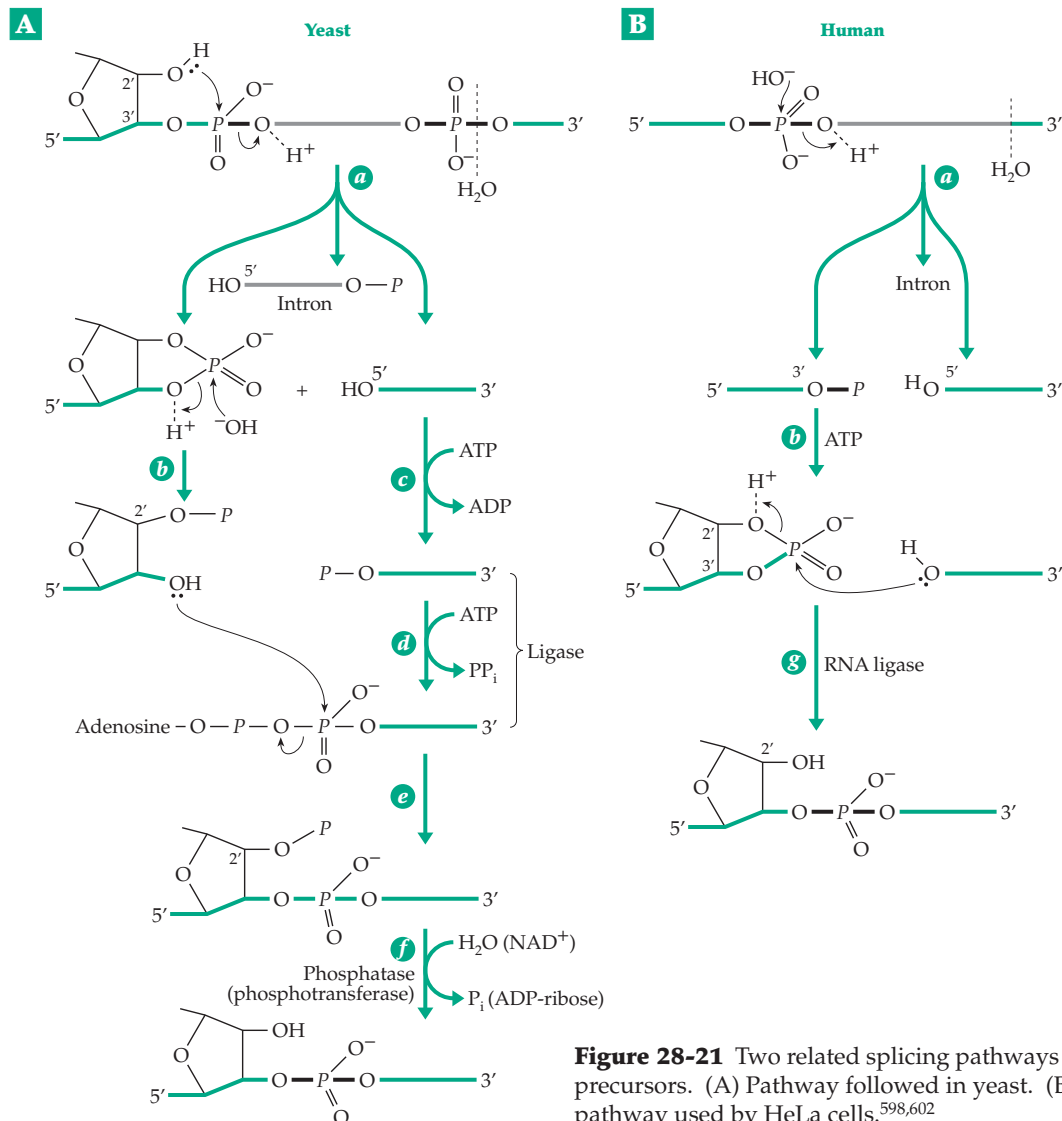


Figure 28-21 Two related splicing pathways for tRNA precursors. (A) Pathway followed in yeast. (B) Related pathway used by HeLa cells.^{598,602}

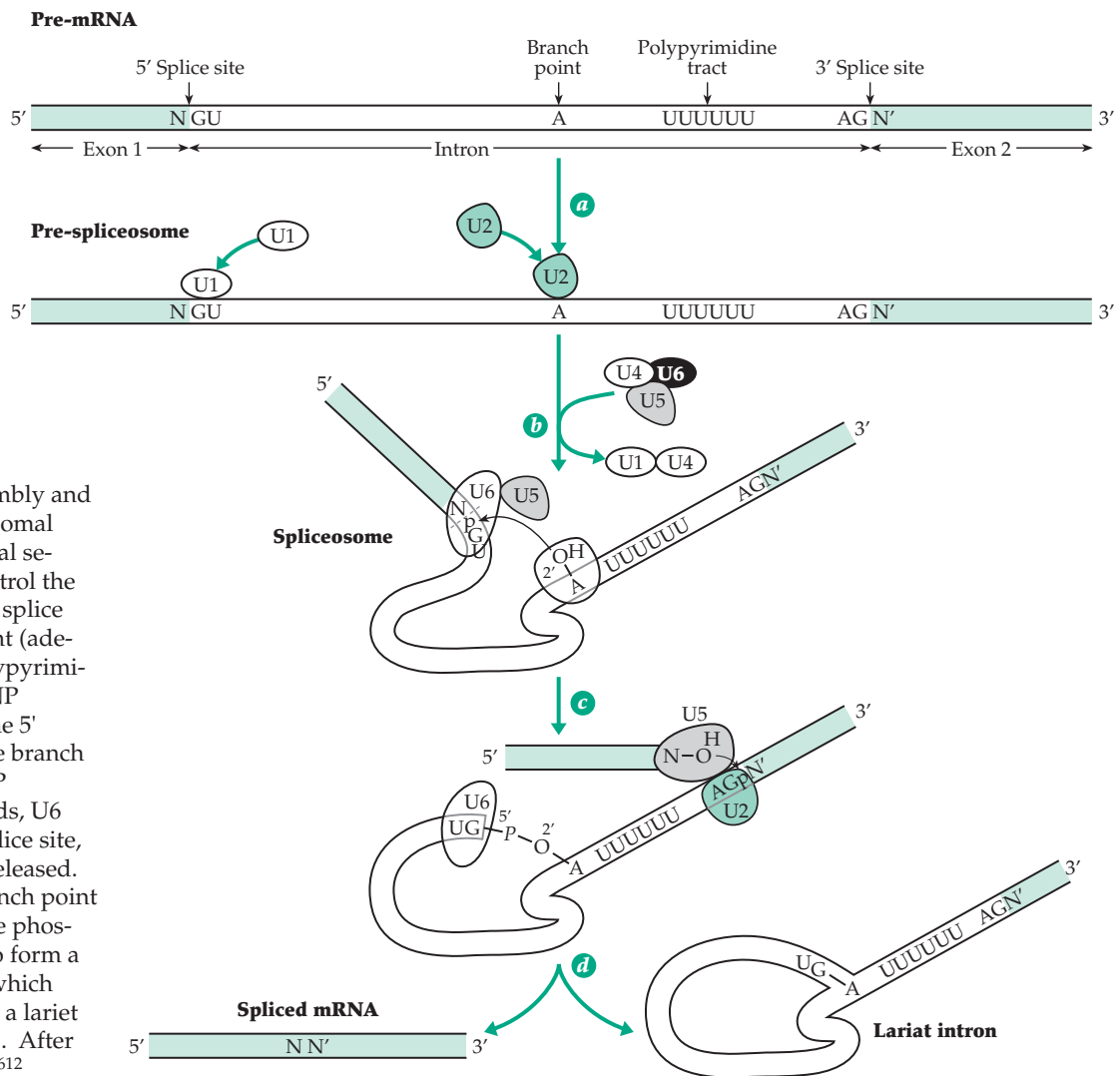


Figure 28-22 Assembly and action of the spliceosomal complex. Four special sequence elements control the process: the 5' and 3' splice sites, the branch point (adenosine A), and a polypyrimidine tract. The snRNP particle U1 locates the 5' splice site and U2 the branch point. The tri-snRNP U4•U6•U5 then binds, U6 recognizing the 5' splice site, and U1 and U4 are released. The 2'-OH of the branch point adenosine attacks the phosphodiester linkage to form a lariat intermediate, which releases the intron in a lariat form in the final step. After Valcárcel and Green.⁶¹²

Relationship to viroids and virusoids. The smallest of viruses are the naked ~250–460 nucleotide single-stranded circular RNA molecules called viroids (Chapter 5).^{591–593} Viroids are closely related to group I introns. The conserved paired sequence characteristic of the group I introns are also present in viroids (Fig. 28-19B), and it looks as if viroids may be “escaped introns.” Another group of “plant satellite RNAs” or virusoids replicate only with the help of larger RNA viruses. These satellite RNAs are replicated by a rolling-circle mechanism.^{593a} The resulting long RNA molecules are self-cleaving, cutting themselves to form the unit length satellite RNAs.^{594,595} These are the simplest known self-cleaving RNA molecules. They have been discussed in Chapter 12.

Pre-tRNAs. In the removal of type I introns the formation of specific stem and loop structures directs the splicing reactions (Fig. 28-18).^{47,597} Stems and loop structures already exist in tRNA precursors. Cleavage sites are usually located just to the 3' side of the anti-

codon as is seen in the pre-tRNA molecule and in the composite structure shown in Fig. 28-20. The chemistry of the splicing process^{205,598–601} is shown in Fig. 28-21. Cleavage at the two splice sites excises the intron. Then the two ends are rejoined. In yeast cleavage at the 5' end of the intron is by a multimeric enzyme with a pancreatic ribonuclease-type of action that leaves a 2',3'-cyclic phosphodiester (Fig. 28-21A, step *a*) that is opened hydrolytically (step *b*). Another hydrolytic cleavage, catalyzed by a different subunit of the enzyme complex, occurs at the 3' end of the intron. The resulting 5' OH is phosphorylated by GTP (step *c*),⁶⁰¹ and the resulting 5'-phospho group reacts with ATP to displace PP_i and to form a transient adenosine-5'-diphosphate terminus (step *d*). This reacts with the 3' OH formed at the 5' splice site (step *e*) to create a phosphodiester linkage between the two pieces of RNA. A phosphotransferase then removes the superfluous 2' phosphate (step *f*).^{601a} Steps *d* and *e* are catalyzed by an **RNA ligase**.^{598,599}

In HeLa cells and presumably in normal mammalian nuclei, the initial cleavage at the 5' end of the intron leaves a 3' phospho group (Fig. 28-18B, step *a*), which is cyclized, probably in an ATP-dependent process, to the 2',3'-cyclic phosphate (step *b*). This is ligated to the other piece of RNA by a direct displacement on the cyclic phospho group (step *c*).⁶⁰²

The spliceosome. The hnRNA of nuclei, which includes all of the pre-mRNA, is associated with proteins, which sometimes form very large 200S particles.⁶⁰⁴ After limited cleavage with nucleases they tend to sediment in the 30S–40S range and to contain a variety of proteins.^{605,606} Some of the proteins may have been involved in control of transcription.⁶⁰⁶ Others participate in splicing. The smaller snRNP particles then appear to come into the nucleus and displace much, but not all, of the protein present in the pre-mRNA ribonucleoprotein particles.⁶⁰⁵

The 50–60S spliceosome complexes, with their protein and RNA components, are reminiscent of ribosomes. Although smaller in size, they can be visualized by electron microscopy.⁶⁰⁷ Each spliceosome is constructed from the four snRNP particles and additional proteins (Fig. 28-22).^{539,608–612} Each spliceosome can accommodate ~500 nucleotides of pre-mRNA. As the pre-mRNA moves through the spliceosome the splice sites, which have only weakly conserved consensus sequences, must be located. The two ends of the introns, which may be much longer than the average 137-nucleotide exon, must be brought close together in the spliceosome.⁶¹³ The exact splice sites are usually located by an invariant **GU** at the 5' end and **AG** at the 3' end. The sequence of the first 18 nucleotides of U1 snRNA is largely complementary to that of the 5' splice site, which has the consensus sequence AG:**GU**RAGU, where the colon marks the junction. The **GU** is invariant.

After the U1 snRNP binds to the pre-mRNA (step *a*, Fig. 28-22)⁶¹⁴ the U2 snRNP binds to another almost invariant sequence CUR**ACU** found 20 to 55 nucleotides upstream of the 3' junction.^{608,615–617} The A in this sequence becomes a branch point. It is brought close to the 5' splice site with the aid of a preassembled complex of snRNPs U4, U6, and U5. In this complex U4 and U6 are tightly paired, additional proteins are also present,^{618–621} and enhancers may be located in adjacent exons.⁶¹⁷ Upon binding of U6 to the 5' splice site, the U1 and U4 snRNPs are released (step *b*, Fig. 28-22) and the 2'-OH of the branch point adenosine attacks the backbone phosphorus atom (step *c*) at the 5' splice junction forming a lariat intermediate. The 3' end created at the 5' junction must now be held and brought close to the 3' splice junction, which is located with the aid of U5 snRNP.⁶²² The 3' splice junction, utilized in the second splicing step (step *d*, Fig. 28-22) has the consensus sequence (T/C)N(C/T)AG:G.

The first splicing step is dependent upon a divalent metal ion, but the second is not.⁶²³ Both steps appear to be in-line nucleophilic displacement reactions.⁶²⁴ Additional **splicing factors** are needed for formation of the U4, U6, U5 complex and its function in the second splicing step, which also appears to require ATP.^{621,622}

A small fraction of eukaryotic mRNA introns are characterized by **AU** and **AC** (rather than GU and AG) ends. The spliceosomes that act on these introns contain modified snRNAs U4 and U6, which are designated U4_{atac} and U6_{atac}. They also require U11 and U12 snRNPs.^{534,625,626}

Since pre-mRNAs usually contain many introns, a series of splicing events must occur. These apparently take place consecutively beginning at the 5' end. Similar splicing pathways are followed in yeast, higher plants, insects,⁶²⁷ and mammals.

Alternative splicing pathways. RNA that contains many introns can undergo splicing in more than one way. Many examples of alternative splicing have been discovered.⁶¹² The mammalian isoenzyme forms of pyruvate kinase called M₁, M₂, L, and R are all tetramers of 60-kDa subunits. The M₁ and M₂ forms are encoded by a single gene. The two mRNAs contain 1593-nucleotide coding regions, which are identical except for a 160-nucleotide sequence that determines the amino acid sequence in a region responsible for intersubunit contact. The difference between the M₁ and M₂ forms involves a choice of two alternative exon regions, one or the other of which is omitted during splicing.⁶²⁸ Other examples have been found in human collagen,⁶²⁹ in fibronectin,⁶³⁰ in neuropeptide formation (Chapter 30),⁶³¹ and among human proline-rich salivary proteins,⁶³² cytoskeletal tropomyosin,⁶³³ platelet-derived growth factor,⁶³⁴ coagulation factor X,⁶³⁵ and porphobilinogen deaminase.⁶³⁶ Alternative splicing is very common in transcripts of viral DNA (Section E).

Alternative splicing could have arisen accidentally, but it is controlled by proteins. Best known is the **alternative splicing factor (ASF or SF2)**.^{612,637} It was first recognized by its function in *Drosophila melanogaster*, where the sex of individuals is determined by alternative splicing of an mRNA.⁶³⁸ In addition to ASF other serine- and arginine-rich **SR proteins** participate in alternative splice site selection.⁶¹²

Trans splicing. Every mRNA in trypanosomes has, at its 5' end, a short 35-nucleotide sequence that is not encoded in the transcribed gene. It was found that for each mRNA molecule two transcripts are formed and are spliced together, always with the 5' piece from a short SL (spliced leader) gene being joined to each of the others.⁶³⁹ This trans splicing has since been observed in many plants, animals, and protists.⁶⁴⁰

Among these are the nematode *Caenorhabditis*,^{639,641} flatworms,⁶⁴² yeast,⁶⁴³ plant chloroplasts and mitochondria,^{640,644} and mammalian cells.⁶⁴⁵

6. Modification and Editing of RNAs

Both mRNA and rRNA undergo rapid methylation of selected residues. About 1–3 internal adenylate residues per kb are methylated at their N⁶ positions.⁶⁴⁶ These are usually the central adenylates in the sequences GAC or AAC. The methylated sites are not uniformly distributed but are clustered, sometimes in the 3' untranslated ends of the RNA. Many more residues (55 in yeast and ~100 in vertebrates) in rRNA are methylated on selected ribose 2'-OH groups.^{173, 535c-e, 647–648a} About an equal number of uridine residues are converted to pseudouridines (Eq. 28-3). Methylation sites are apparently selected by the fibrillarin-associated snoRNAs U32–U40 (Table 28-3). These **methylation-guide** snoRNAs contain 10- to 14-nucleotide sequences that are complementary to segments of the rRNA that contain the methylation sites, and evidently provide rigid helical regions that are targets for the methylase action. The snoRNAs contain the previously mentioned C and D sequences. Methylation occurs on the ribose of the nucleotide that is base-paired with the fifth nucleotide that is upstream of the D sequence in the snoRNA. Each snoRNA directs methylation of a different ribose.¹⁷³ In a similar manner the snoRNAs containing the ACA motif located three nucleotides upstream of their 3' ends appear to direct the conversion of uridines to **pseudouridines**, as in Eq. 28-3.

Editing of RNA transcripts. Interpretation of the genetic code utilized by the mitochondria of trypanosomes and other kinetoplastid protozoa was confounded by the discovery that the DNA sequences of many genes, including the COIII cytochrome oxidase gene present in the kinetoplast maxi circle DNA, do not appear to encode the correct amino acid sequence. In fact, the RNA transcripts are extensively edited, mostly by insertion of multiple uridine residues at many positions^{649,650} and by occasional deletions at others. Editing of some mRNAs causes 45% of the message to be rewritten.⁶⁵¹ Several additional kinds of editing were soon discovered, not only in protozoa but also in mammals, plants, and archaea.^{652,652a}

A frequent editing change is the hydrolytic deamination of a C to a U residue.^{652b} For example, human apolipoprotein B is synthesized in two forms: apoB100, a full-length 512-kDa protein made in the liver and used for transport of cholesterol and triglycerides, and a shorter 241-kDa form, apoB48, used in absorption of dietary lipids (Chapter 21, Section A1). ApoB100 is synthesized from a full-length mRNA, but

apoB48 is made according to a shortened mRNA in which a glutamine codon (CAA) has been converted by editing to the translation stop codon UAA.^{652,653} A special enzyme deaminates only cytidine 6666 of the mRNA.⁶⁵⁴ C to U editing occurs in chloroplasts and mitochondria of plants.^{655–657} In *Arabidopsis* mitochondria 456 different C to U conversions have been identified in mRNAs.⁶⁵⁶

Deamination of adenosine residues produces inosine, which occurs in brain mRNA once in ~17,000 ribonucleotides.⁶⁵⁸ Some ionotropic glutamate receptors in the brain have subunits translated from inosine-containing mRNAs. A glutamine codon CAG is edited to CIG, an arginine codon. The arginine codon AGA is converted to the glycine codon IGA at another site, ATT is converted to ITT and TAC to TIC. All of these changes affect the properties of the glutamate-activated ion channel.⁶⁵⁹ The adenosine deaminases involved in these editing events are usually specific for double-stranded helical segments of RNA, e.g., for the stems in stem-loop structures.^{660–662} At least one human adenosine deaminase not only binds to RNA but also contains a DNA-binding domain specific for Z-DNA.⁶⁶³

Returning to the trypanosomes and their relatives, mitochondrial RNAs undergo extensive insertion and deletion of U's. The editing site is located by a **guide RNA** (gRNA), which directs the hydrolytic cleavage of the chain and either the addition of U's to the 3' cut end by transfer from UTP or hydrolytic deletion of U's from the 3' cut end. The chain ends are then rejoined by an RNA ligase.^{172,664–667c} The functional significance of the editing of kinetoplast mRNA is uncertain. However, at least some of the edited mRNA is translated to give proteins that are presumably used.⁶⁶⁸ While trypanosomes usually insert only U's, the slime mold *Physarum polycephalum* may insert dinucleotides such as AA, AU, CU, or GU⁶⁶⁹ and may also add nucleotides at the 3'-ends of RNAs.^{669a}

A to G editing occurs in RNA of *Drosophila*.⁶⁷⁰ Yet another type of editing has been observed in viral RNA from paramyxovirus. The virally encoded RNA polymerase sometimes “stutters” reading the same template base two or more times, with a resulting insertion of a base.⁶⁷¹ Editing of transcripts usually serves an essential biological function, creating the correct sequence for translation of the mRNA and often generating multiple isoforms of proteins.

Finishing the transcripts. Additional modifications must be made to some mRNAs, and there will doubtless be many surprises as the details are worked out. One detail, which was discovered in the 1980s, is the specific function of snRNA U7 in recognition of the 3' end of pre-mRNAs for histones. The U7 RNA apparently base-pairs with a sequence near the 3' end cleavage site, acting as a cutting guide.^{47,672,673}

Newly synthesized mRNA emerges from the nuclear pores as nucleoprotein complexes containing as many as ten different proteins. A major component is a 78-kDa polypeptide thought to be associated with the poly(A) tail. These ribonucleoproteins are sometimes stored for long periods of time, for example, in mature seed embryos and in amphibian oocytes.^{575a,674} They may also travel rapidly for long distances, e.g., down nerve axons⁶⁷⁵ or from cell to cell in plants via transport in the phloem.⁶⁷⁶

E. Transcription of Mitochondrial, Chloroplast, and Viral Genes

In the compact 16-kb chromosomes of mammalian mitochondria, the genes are tightly packed against one another (Fig. 18-3).^{677,678} Most genes are transcribed using heavy H strands as templates and specialized bacteriophage-type RNA polymerases encoded in nuclear DNA.⁶⁷⁹ A single promoter in the D loop region (Fig. 18-3) is used to make a long transcript from the entire H strand.⁶⁸⁰ The transcript is then cut precisely by mitochondrial RNase P at the 5' ends of the tRNAs.⁶⁸¹ Similarly precise cleavage must occur at the 3' ends because there are often no nucleotides or only one to a few nucleotides separating adjacent genes. The nucleases involved differ from those used in nuclear tRNA 3' processing.⁶⁸² In animal mitochondria tRNA genes sometimes overlap by one nucleotide. Polyadenylation of the transcripts provides a form of editing that is required to create a UAA translational termination signal to which termination proteins bind.⁶⁸³ The tRNA genes lack the 3' terminal CCA, which must be added. The lighter L strand has its own promoter, also located in the D loop. Both promoters contain the conserved nonanucleotide sequence 5'-ATATAAGTA. The pre-mRNAs created by these cleavages are not capped but are polyadenylated. Since the promoters are simple and the transcription factors few, mitochondrial transcription is controlled largely by mRNA stability, translation, and posttranslational events.⁶⁸⁴

The 70- to 100-kb circular mitochondrial DNA molecules of yeast and of higher plants contain more genes than do animal mitochondria, but most of the increased size is accounted for by intergenic spacers and by a few long introns. All yeast mitochondrial genes except for that of tRNA^{Thr} are transcribed from a single strand. About 20 different primary transcripts have been identified in *Saccharomyces cerevisiae*. These originate at several points in the genome but always at the sequence 5'-ATATAAGTA, the 3'A corresponding to the 5' nucleotide of the transcript.⁶⁷⁸ One of these sequences is located at the origin of replication, suggesting the possibility that a normal RNA transcript provides the primer for DNA replication in yeast mitochondria.⁶⁷⁷

The 100- to 160-kb chloroplast genomes (Chapter 23, Section E,2) also have many prokaryotic features. They encode ~50 proteins as well as the tRNAs and rRNAs. Promoter and terminator sequences resemble those of bacteria and protein sequences are often homologous to those in bacteria. This applies, for example, to the α , β , and β' subunits of RNA polymerase.⁶⁸⁵

1. Viral Transcription and Replication

Because viruses contain small genomes, study of transcription of viral DNA and of replication of RNA viruses has played an important role in helping us to understand transcription in eukaryotes.^{47,686-688} An example is the discovery of the virus SV40 enhancer, which has been discussed in Section C,4. Study of viral life cycles is also essential to future progress in fighting viral diseases. Each of the many different viruses has its own often very complex life cycle. Only a few details can be given here. For lucid summaries see Voyles.²⁵⁹

Eukaryotic DNA viral genomes, like that of phage λ , usually contain early and late transcriptional units. The small papovaviruses, such as SV40 and polyoma virus, have 5.2 kb genomes. Like the small RNA viruses, they make use of overlapping genes and alternative RNA processing. In SV40 DNA there are two overlapping promoters called **early-early** and **late-early**. The first of these contains a TATA sequence, and both promoters also depend upon a 21-bp repeat segment as well as the SV40 enhancer. At least two proteins, one that binds to the enhancer and one that binds to the 21-bp repeat, are needed for initiation of early transcription. The early-late promoter lacks the TATA sequence but requires the 21-bp repeat and enhancer. A 94-kDa encoded protein called the **large T-antigen** (Chapter 27, Section C,10) is one of the regulators of transcription as well as of DNA replication.^{689,690} This protein is also sufficient to transform rodent cells in culture. Although predominantly nuclear it is also inserted into the membrane where it acts as an antigen.

The large icosahedral adenoviruses cause respiratory infections in humans and attack and may cause cancer in many other vertebrate species including birds and amphibians. The 35.9-kb genome of human adenovirus-2 encodes at least 30 proteins, 10 of which appear in the virion. One of these is covalently linked to the 5' end of the DNA. As with smaller DNA viruses extensive use is made of alternative splicing of the transcribed RNA. There are at least six early transcriptional units, each with its own promoter. A variety of mRNAs are created using the various coding segments and a complex array of regulatory sequences control transcription. A 32-kDa phosphoprotein transcription factor designated E1A is encoded by a "pre-early"

gene. It is required along with host-encoded transcription factors for viral transcription.^{691,692}

The late region of the adenovirus genome encodes structural proteins for the virus coat. Most of its transcripts begin about 16.5% of the way along the 36.5-kb dsDNA. However, the initial transcripts are cleaved at several different positions to yield a series of different 3'-poly(A)-terminated transcripts. In an exceedingly complex process the transcripts undergo splicing out of genes at their 5'-ends so that the final mRNAs typically code for single proteins. At the 5' ends the cap is joined to short segments from the original mRNA 5' end.^{687,688,693}

Viruses SV40, polyoma, and some strains of adenoviruses are oncogenic in some species and cause transformations of cells in culture. Transformed cells always contain integrated viral DNA. That of SV40 can be incorporated at many different sites in the host genome. The integrated DNA does not always include the complete SV40 genome, and parts of the DNA may be inverted, deleted, or scrambled. Integration is not an essential part of the viral life cycle and has no effect on the infective properties of the viruses.⁶⁸⁸ Cells transformed by adenoviruses usually also contain only a fragment of the viral genome in their DNA. However, one small set of genes from early region 1A is present in all transformed cells. The encoded proteins appear to be modulators of transcription and may cause cancer by promoting uncontrolled transcription of certain genes.⁶⁹⁴

2. Replication of RNA Viruses

The RNA (+) strands present in many RNA viruses often serve immediately after infection as a messenger RNA. However, replication requires formation of (–) strands of viral RNA from which new (+) strands can be transcribed for assembly into new virus particles. Other RNA viruses contain (–) strands of RNA or double-stranded RNA and, therefore, have significantly different life cycles.²⁵⁹

Small RNA viruses. The human polio virus, the common cold virus (rhinoviruses, Fig. 7-15), and other picorna viruses have 7.2- to 7.5-kb genomes with considerable homology (50% between the polio and rhinoviruses) and similar overall structures.⁶⁹⁵ The polio genome encodes eight different proteins, one being a small 22-residue peptide that becomes covalently linked to the 5'-end of the RNA through a phosphodiester linkage to the side chain of a tyrosine. Cleavage of this linkage by a host enzyme allows the viral RNA (+) strand, which is polyadenylated at the 3' end, to serve as an mRNA for synthesis of a single large 220-kDa polyprotein. This is cleaved by a host protease at several Gln-Gly bonds to form several proteins. These

include the 22-residue RNA-linked peptide, two capsid proteins, and a capsid precursor protein. The latter is cleaved during capsid assembly by a viral protease at an Asn-Ser bond to give two more capsid proteins. These four proteins have masses of 7, 26, 29, and 32 kDa. A viral protease, a large 58-kDa replicase, and a 37-kDa protein of unknown function are also cut from the polyprotein. A host protein initiates cleavage of the polyprotein, but the virally encoded protease later takes over this function.

One of the best understood of the many viral pathogens of plants is the tobacco mosaic virus (Fig. 7-8). Its 6.7-kb positive strand RNA encodes a replicase, coat protein, and at least one other protein.⁶⁹⁶

Influenza viruses. These negative-stranded viruses are classified into types A, B, and C, but it is only type A that infects nonhuman species including birds, horses, pigs, seals, mink, and whales.^{697–699} Type A influenza viruses have also caused the great pandemics such as those in 1918–1919 and in 1968. Influenza viruses are surrounded by a lipid bilayer in which the virally encoded **hemagglutinin** and a **neuraminidase** (p. 186) are embedded. The inside of the bilayer is coated with a matrix protein and within this coat eight pieces of RNA of total length 13.6 kb are coiled together with a basic nucleoprotein. Also present are ten molecules each of three other proteins. The eight pieces of RNA vary in length from 900 to 2500 nucleotides. Seven of them encode one each of the seven virion proteins. One encodes an additional nonstructural protein while the smallest piece, using overlapping nucleotide sequences, encodes two nonstructural proteins.⁶⁹⁷

The existence of a fragmented genome evidently underlies the ability of influenza A viruses to undergo rapid changes in antigenic behavior. If a cell is coinfecting with two strains of virus, the eight fragments act as independent chromosomes, which can be reassorted into new combinations in the progeny viruses. As a consequence, it is difficult to develop safe, live virus vaccines. A large reservoir of infection among migratory water birds and other animals facilitates the appearance of new strains and their rapid spread throughout the world.⁶⁹⁸

The first step in the replication of influenza viruses, which takes place in the cytoplasm, is the synthesis of (+) strands that can serve both as mRNA for synthesis of proteins and as templates for synthesis of new (–) strands. Three of the capsid proteins form the required RNA polymerase. This “transcriptase” is primed preferentially by 5'-capped 10- to 13-nucleotide segments of RNA that have been cut by a viral nuclease from host mRNAs.⁷⁰⁰ The mRNAs made from viral RNA are polyadenylated and are translated by the host cell's ribosomes. However, some transcripts are used as templates to form viral (–) strands, which

are not polyadenylated and which contain uncapped pppA at the 5'-ends.

HIV-1 and other retroviruses. Because of their association with viral oncogenes (Chapter 11) and because of the **human immunodeficiency virus** (HIV-1) and the AIDS epidemic a great deal of attention is focused on retroviruses.^{701–701b} Each retrovirus particle contains *two* identical single-stranded (+) RNA molecules, which may be as long as 10 kb. Their unique characteristic is that they induce synthesis of DNA, which must be integrated into the host genome before new viral (+) strands are transcribed. Retroviruses may sometimes cause cancer and may carry oncogenes (Chapter 11). Study of the **Rous sarcoma virus** (RSV), which infects chickens, and of the related **avian myeloblastosis virus** (AMV) and of HIV has revealed a common structure and a complex life style that are largely shared by all known retroviruses.

The organization of retroviruses^{687,688,702} always includes a sequence of genes designated *gag* (glycoprotein antigen core proteins), *pol* (polymerase), and *env* (envelope) (Fig. 28-23). These are often followed by an oncogene.⁷⁰³ In RSV this is the *src* gene (Chapter 11). At each end of the retrovirus gene sequence is a short direct repeat labeled R in Fig. 28-23. In RSV the R sequence is 21 nucleotides in length⁶⁸⁸ and in HIV (see Fig. 28-23)⁷⁰⁴ it is 98 nucleotides long.^{705,706} The 5' end of the viral RNA is capped, and the 3' end is polyadenylated. The dsDNA of the integrated form of the virus (Fig. 28-23) is longer and at each end is bounded by **long terminal repeats** (LTRs). Each LTR consists of a sequence, designated U3, that is present next to R at the 3' end of the viral RNA. In the LTR this is followed by sequence R and then by U5, a unique sequence that came from the 5' end of the viral RNA. Each ds-LTR begins and ends with a short inversely repeated segment:

5'-TGT — ACA in RSV
5'-CTG — CAG in HIV

The integrated provirus is always bounded by a sequence of host DNA that is repeated without inversion at the opposite end. For RSV this is a 5-bp sequence.

The LTRs in RSV are 569 bp in length, and those in HIV are 634 bp in length, 83 bp coming from U5, 98 from R, and 453 from U3.⁷⁰⁶ The LTRs themselves often contain promoter and other control elements and even entire genes. The organization of a retrovirus (Fig. 28-23) reflects the complex mode of replication, which is presented in simplified form in Fig. 28-24. The key enzyme is the RNA-directed **reverse transcriptase** (Fig. 27-12).^{707–709} The initial synthesis of DNA by this enzyme is primed by a tRNA. RSV uses tRNA^{Trp} and HIV tRNA^{Lys} for this purpose.^{710–712} The 3' end of the tRNA, including the nucleotides

forming both the acceptor stem and the stem of the TΨC loop (Fig. 5-30), must relinquish its normal base-pairing to form ~18 Watson–Crick base pairs with a **primer binding site** (labeled PB in Fig. 28-24) near the 5' end of the retroviral DNA. Because synthesis of the (–) strand of the retroviral DNA begins so close to the 5' end of the template (Fig. 28-24), only a short piece of DNA, including sequences U5' and R' complementary to U5 and R, can be formed. For replication to continue RNA must be removed from hybrid regions. This is accomplished by the **RNaseH** activity of the reverse transcriptase.^{708,709,713,713a} After removal of the RNA the primer tRNA must undergo a **strand transfer**, in which it shifts from the 5' end of the viral RNA template to the 3' end (Fig. 28-24), utilizing pairing between the right-hand R sequence of the template and the R' sequence of the growing cDNA copy.^{703,711,714} This transfer is sometimes to the second of the pair of identical RNA molecules in the virus, providing a way of increasing diversity by recombination. A second strand transfer of the growing (+) strand is needed to complete the dsDNA, which now contains the two identical LTRs. The 5' and 3' ends of the template RNA are doubtless held close together to facilitate strand transfer.

Integration of the dsDNA into the host DNA can occur at many places. The mechanism of integration probably resembles that used by phage λ (Fig. 27-27) and accounts for the duplication of host sequences at the two ends of the integrated virus. A virally encoded integrase catalyzes the process (see also Chapter 27, Section D,3).^{715–717} It is the integrated virus that is transcribed to form new (+) viral RNA strands.

Integrated retroviruses are usually transcribed as full-length RNA copies, which may or may not have introns spliced out. The smaller spliced pieces encode the *env* and other genes such as *src* (Fig. 28-25). The *gag-pol* region is translated as a polypeptide that is cleaved into a number of pieces. These include four proteins of the virus core (encoded by *gag*), the reverse transcriptase with its associated RNaseH, and an **integrase**^{718,718a} (all encoded by *pol*).²⁵⁹ There is also an **aspartic protease** only 99 residues in length within *pol* (Fig. 28-25; Chapter 12, Box 12-C). The promoter and control region for transcription is located in the U3 region and is placed into a position where it can function only upon synthesis of the first LTR. The gene *env* encodes the major viral envelope protein and is translated from a spliced mRNA (Fig. 28-25).

Accessory regulatory genes. HIV and some related retroviruses such as HTLV-1 (which causes rare T-cell leukemias)⁷¹⁹ are distinguished from other retroviruses by a marked increase in the rate of DNA transcription within infected cells as compared with uninfected cells. This is thought to be a result of synthesis of virally encoded proteins that are trans-acting

Figure 28-23 Comparison of the forms of a retrovirus. (A) Infective RNA (+) strand. (B) The double-stranded DNA form. (C) The DNA form integrated into the host DNA. LTR, long terminal repeats.

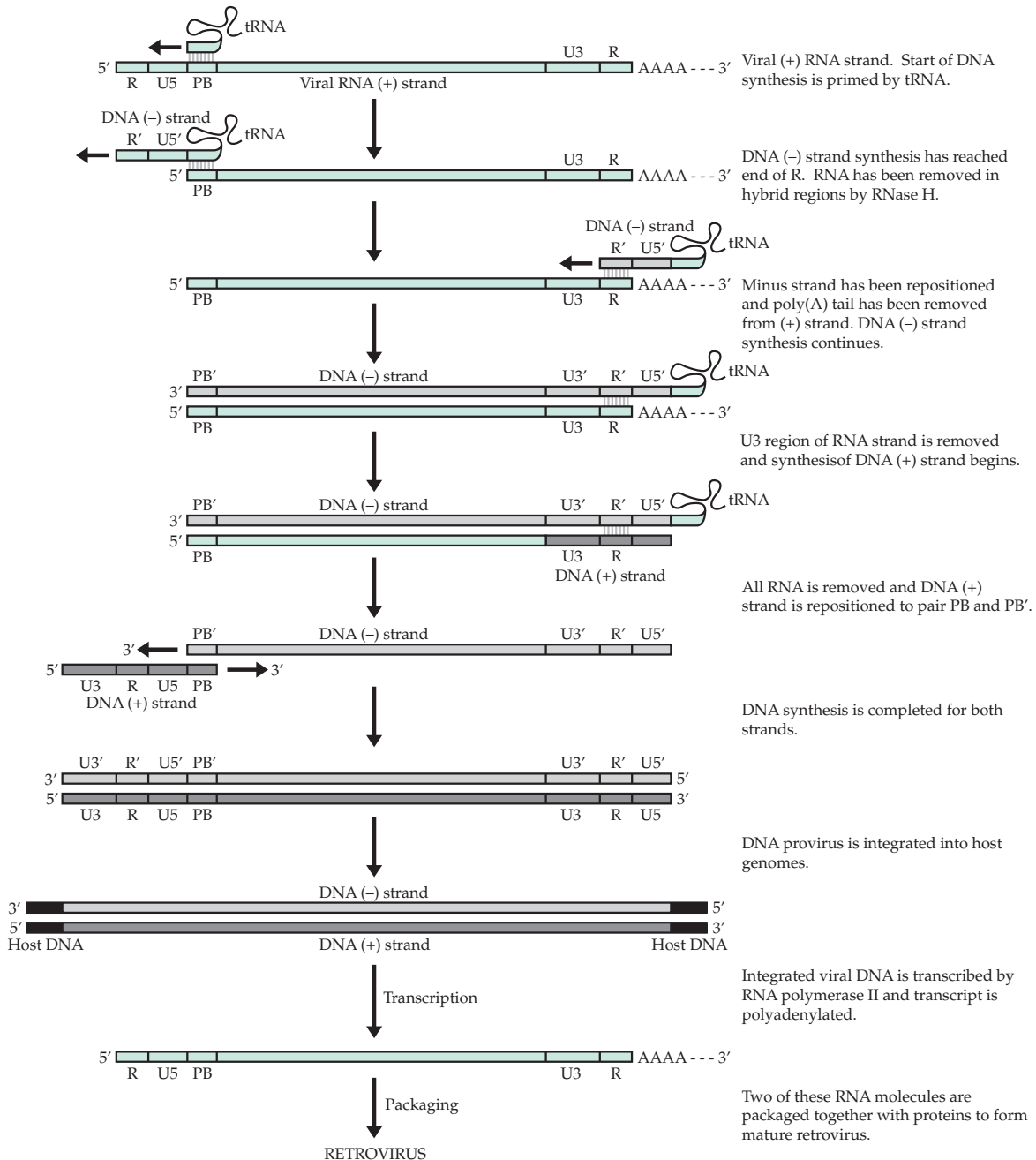
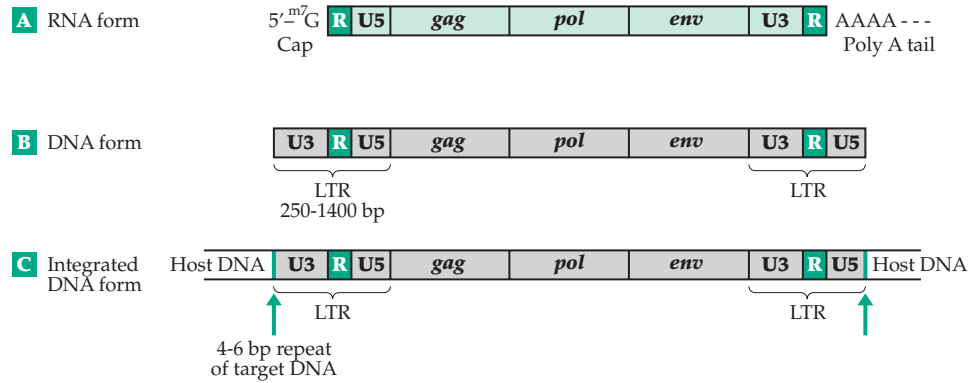


Figure 28-24 Simplified scheme for replication of the RNA genome of a retrovirus. See Sugden.⁷⁰³ PB, Primer-binding site.

regulators of transcription. The HIV genome map (Fig. 28-26) shows the positions of the nine recognized genes marked in the three reading frames. In addition to the *gag*, *pol*, and *env* genes there are genes for six accessory proteins: Tat, Rev, Vif, Vpr, Vpu, and Nef.⁷²⁰⁻⁷²²

Transcription is initiated at the promoter in the 3' LTR. This contains a TATAA sequence, an SP1 binding site, and an enhancer that binds transcription factor NF- κ B (Fig. 5-40). The full-length 9-kb transcript contains, according to Frankel and Young,⁷²² the following

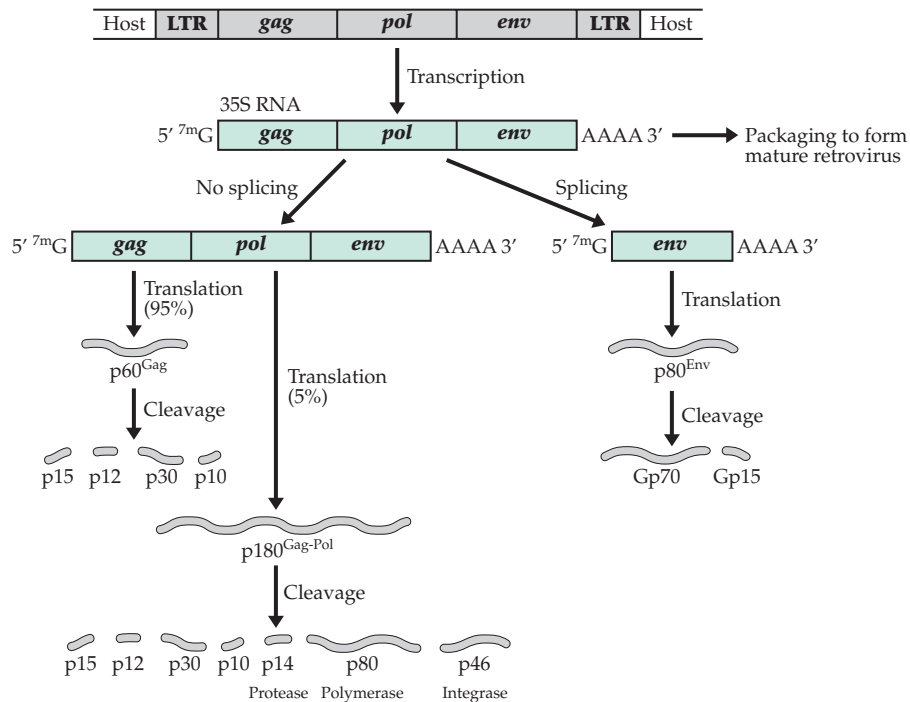


Figure 28-25 Gene expression from a typical retrovirus that has been integrated into a host's genome. This figure illustrates how a variety of proteins are encoded by a single rather short piece of DNA. After Voyles.²⁵⁹

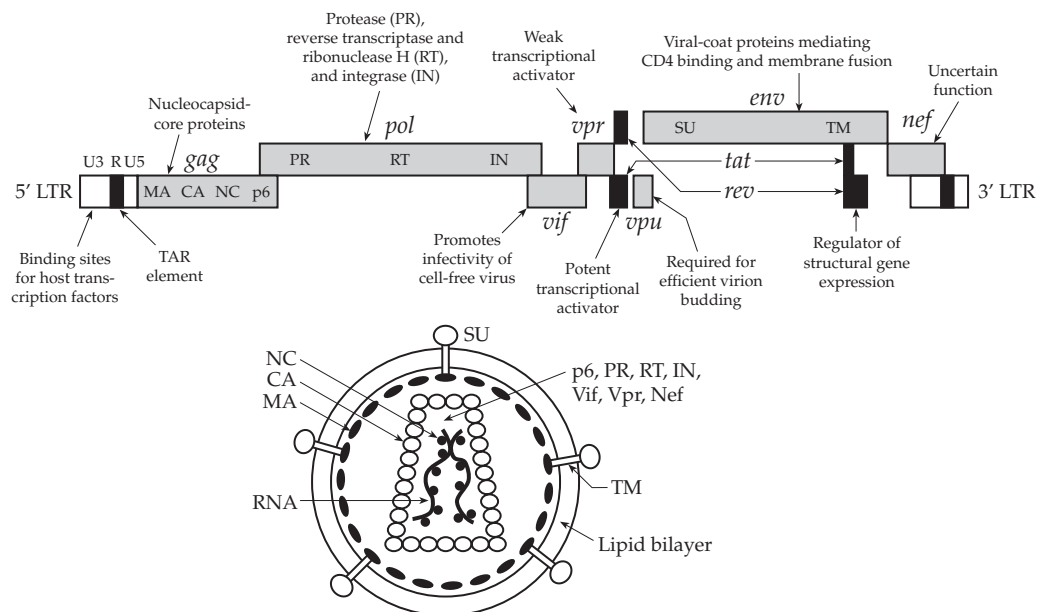


Figure 28-26 Simplified genetic map of the AIDS virus HIV-1. All three reading frames are utilized to encode nine genes, which give rise to 15 proteins. After Frankel and Young.⁷²²

BOX 28-C SYNTHETIC ANTIVIRAL COMPOUNDS

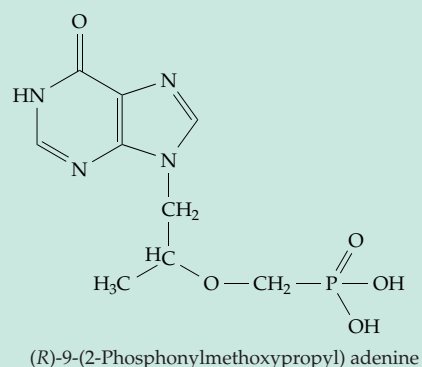
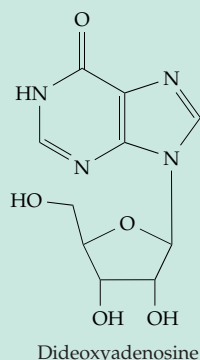
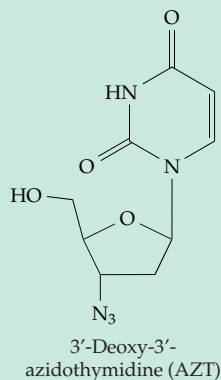
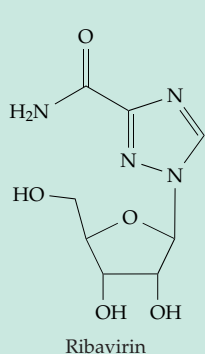
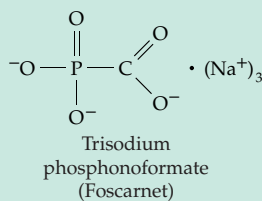
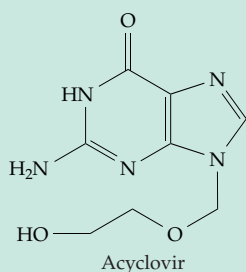
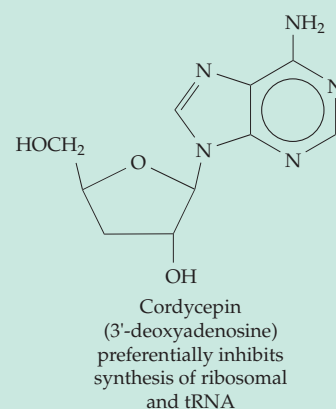
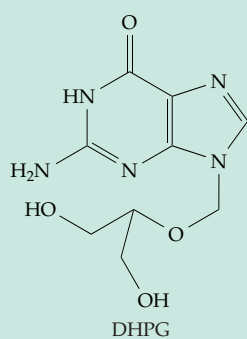
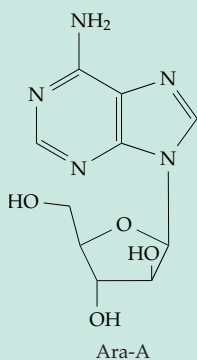
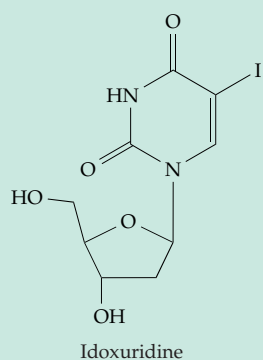
Most bacterial infections can be treated successfully with antibiotics, but the development of satisfactory antiviral agents has been slow. Yet we are susceptible to many dangerous virus diseases, and viruses also take a huge toll among domestic animals and plants.^a The first antiviral drug, 5-iodo-2'-deoxyuridine (idoxuridine), was introduced in 1962 and was used for 20 years by ophthalmologists to treat serious eye infections by the herpes simplex virus (HSV).

More recently 9- β -D-arabinofuranosyladenine (Ara-A) has become a preferred drug in treatment of ocular herpes infections. This compound, which is a naturally occurring antibiotic, can also be administered intravenously for life-threatening infections such as herpes encephalitis.^a Ara-A is quite toxic but the guanine derivative 9-(2-hydroxyethoxymethyl) guanine (acyclovir) is less so. Another acyclic 2'-deoxyguanine analog,

9-(1,3-dihydroxypropoxymethyl)-guanine (DHPG), is more soluble, more potent, and has a broader range of effectiveness.^{a-c}

One of the first effective drugs against RNA viruses was ribavirin (1- β -D-ribofuranosyl-1,2,4-triazole-3-carboxamide), an analog of guanosine. It has a broad range of action and is used to treat severe viral pneumonia and bronchitis caused by respiratory syncytial virus (RSV). This common virus usually is mild but may cause death in infants and children. Ribavirin may also be of some value in the treatment of AIDS.

How do these compounds exert their antiviral effects? The nucleoside analogs are usually phosphorylated to the corresponding mono-, di-, and triphosphate derivatives by cellular enzymes. Thus, Ara-A yields Ara-ATP, which inhibits the herpes virus-encoded DNA polymerase. Ara-A may also enter the viral DNA. In addition Ara-A may inhibit



BOX 28-C (continued)

polyadenylation of virally induced mRNA. Some analogs, such as acyclovir, undergo little conversion to the monophosphate by cellular kinases but are efficiently phosphorylated by herpes virus-encoded thymidine kinase. Thus, acyclovir does little damage to uninfected cells.^a Ribavirin 3'-monophosphate may inhibit IMP dehydrogenase (Fig. 25-16, left) thereby interfering with GTP production. At the same time ribavirin triphosphate competes with GTP to inhibit virally encoded RNA polymerase.^a

Phosphonoformate is a pyrophosphate analog and inhibits both DNA polymerases and reverse transcriptase. However, toxicity may prevent long-term treatment of AIDS patients. Amantadine has a narrow antiviral specificity. It specifically inhibits initiation of the replication of influenza virus RNA of type A (but not of type B). Active only against retroviruses, 3'-azidothymidine is a reverse transcriptase inhibitor, which acts by a chain termination mechanism. It was synthesized in the early 1960s but only recently has been used in treatment of AIDS victims. More recently a series of 2',3'-dideoxynucleosides, such as dideoxyinosine, have also been used.^d Acyclic phosphonates, such as phosphonylmethoxypropyladenine, avoid the need for metabolic phosphorylation of the drug.^e

Development of synthetic antiviral compounds is hardly beyond its infancy.^f Serious problems must be overcome with most of these compounds. Toxicity (sometimes carcinogenicity), development of resistance by viruses, and enzymatic destruction limit the utility of most drugs. For example, adenosine deaminase destroys Ara-A quite rapidly. With our rapidly advancing knowledge of viral life

cycles and protein and nucleic acid structures many new drug targets have been identified.^g Among the targets for HIV are the reverse transcriptase,^h protease,ⁱ and integrase.^{j,k} Computer-assisted design, as well as new techniques of synthesis and screening, have allowed development of many non-nucleoside inhibitors.

Oligonucleotide phosphoramidates and other triplex-forming compounds may be designed to bind to specific DNA targets.^l

^a Robins, R. K. (1986) *Chem. Eng. News* **64**, 28–40

^b Cheng, Y.-C., Grill, S. P., Dutschman, G. E., Nakayama, K., and Bastow, K. F. (1983) *J. Biol. Chem.* **258**, 12460–12464

^c Biron, K. K., Fyfe, J. A., Stanat, S. C., Leslie, L. K., Sorrell, J. B., Lambe, C. U., and Coen, D. M. (1986) *Proc. Natl. Acad. Sci. U.S.A.* **83**, 8769–8773

^d Sandberg, J. A., and Slikker, W., Jr. (1995) *FASEB J.* **9**, 1157–1163

^e Tsai, C.-C., Follis, K. E., Sabo, A., Beck, T. W., Grant, R. F., Bischofberger, N., Benveniste, R. E., and Black, R. (1995) *Science* **270**, 1197–1199

^f Mitsuya, H., Yarchoan, R., and Broder, S. (1990) *Science* **249**, 1533–1544

^g Richman, D. D. (1996) *Science* **272**, 1886–1888

^h Althaus, I. W., Chou, J. J., Gonzales, A. J., Deibel, M. R., Chou, K.-C., Kezdy, F. J., Romero, D. L., Palmer, J. R., Thomas, R. C., Aristoff, P. A., Tarpley, W. G., and Reusser, F. (1993) *Biochemistry* **32**, 6548–6554

ⁱ Rosin, C. D., Belew, R. K., Walker, W. L., Morris, G. M., Olson, A. J., and Goodsell, D. S. (1999) *J. Mol. Biol.* **287**, 77–92

^j Robinson, W. E., Jr., Reinecke, M. G., Abdel-Malek, S., Jia, Q., and Chow, S. A. (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 6326–6331

^k Goldgur, Y., Craigie, R., Cohen, G. H., Fujiwara, T., Yoshinaga, T., Fujishita, T., Sugimoto, H., Endo, T., Murai, H., and Davies, D. R. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 13040–13043

^l Giovannangeli, C., Perrouault, L., Escudé, C., Gryaznov, S., and Hélène, C. (1996) *J. Mol. Biol.* **261**, 386–398

essential regions, whose positions may vary somewhat from one isolate of the virus to another:

A complex pattern of splicing produces more than 30 mRNAs.⁷²³ When viral RNAs are first produced most are doubly spliced, allowing the split genes *tat* and *rev* to be expressed by synthesis of Tat and Rev. The *trans*-activator protein Tat is a small 86-residue cysteine-rich protein, which binds the Tar stem-loop structure and greatly stimulates transcription.^{724–725a}

The 116-residue *rev* (regulator of expression of virion genes) gene product is also a transactivator, which is needed for rapid production of singly spliced (4 kb) or unspliced (9kb) *gag-pol* mRNA required for formation of virus structural proteins.^{726,727} The effect of Rev is probably on transport from the nucleus rather than on splicing.

Transcription is repressed by the 206-residue N-terminal myristoylated protein, Nef, a phosphoprotein

that associates with cytoplasmic membranes.^{728–729a}

It has been difficult to learn its exact function, but it seems to be required for maintenance of the integrated provirus for long periods of time without extensive replication. Mutations in gene *nef* do not eliminate the ability of the virus to replicate in T lymphocytes and to kill them.

The 23-kDa protein Vif (**viral infectivity factor**) is not needed for growth but is essential for infectivity.^{729b} Other genes in HIV are *vpu*, which encodes an 81-residue integral membrane protein (**virion protein U**), and *vpr*, which encodes the 96-kDa **virion protein R**. Several possible functions have been proposed for these small proteins.^{722,729b,c}

HIV-1 is a member of the group of slow viruses or **lentiviruses**.⁷³⁰ Other lentiviruses include the human HIV-2,⁷³¹ an immunodeficiency virus that attacks cats causing leukemia,⁷³² and the human leukemia virus

Position from 5' end	Description
1–55	TAR, a 59-residue stem-loop structure; binding site for Tat
182–199	PB, primer binding site
240–350	Packaging signal. Binds envelope protein NC
248–271	Dimerization site with “kissing loop”
290	Major splice donor site, used to form all spliced mRNAs
1631–1673	Gag-Pol frameshifting region where –1 ribosomal frameshifting occurs to allow 5–10% synthesis of Gag-Pol polyprotein
7362–7596	Rev response element. Binding site for Rev
5358 and 7971	Two major splice acceptor sites; other minor sites are also used
9205–9210	Polyadenylation signal

HTLV-1.⁷³³ Another lentivirus causes two diseases of sheep, **maedi**, a pulmonary disease, and **visna**, a paralytic condition somewhat similar to multiple sclerosis.⁷³⁴ Because of its slow development there has been doubt as to the cause of AIDS, but there is now little doubt that HIV-1 is the true culprit.

Success in treating AIDS may depend upon better understanding of the complex life cycle of HIV-1,^{722,730,735} which is summarized in Fig. 28-27. The cycle begins with the binding of the virion envelope protein to the immunoglobulin-like surface protein **CD4**, which is found principally on the type T4 helper T cells (Chapter 31). Binding of CD4 to the HIV envelope proteins appears to activate the T cells to enter the cell cycle and to take up and integrate the virus. The virus infection destroys these CD4+ lymphocytes with a half-life of less than two days.⁷³⁵

A major effort is being made to devise a vaccine against HIV. However, rapid mutation of the *env* gene makes it difficult to accomplish.^{736,736a} This high rate of mutation appears to be a result of a high frequency of errors by the HIV reverse transcriptase.⁷³⁷ There is

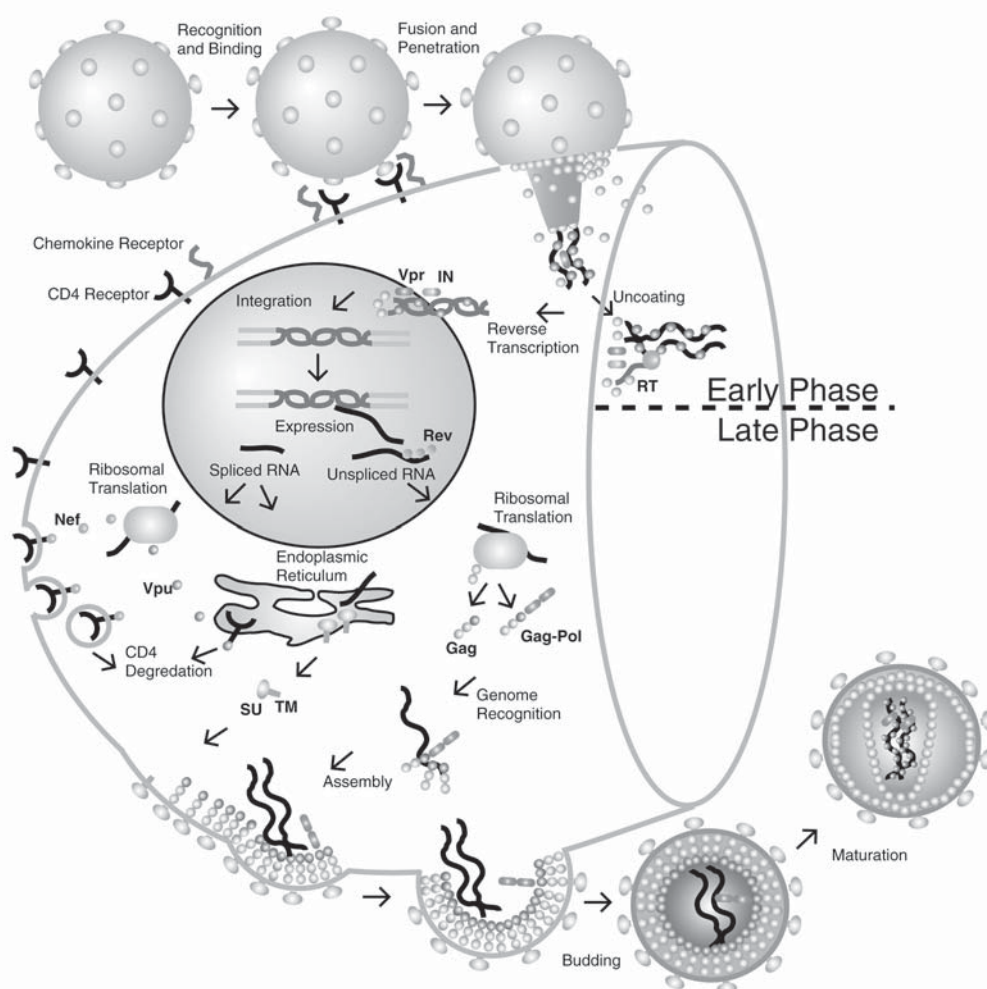


Figure 28-27 General features of the HIV-1 replication cycle. The early phase (upper portion of the diagram) begins with CD4 recognition and involves events up to and including integration of the proviral DNA, and the late phase includes all events from transcription of the integrated DNA to virus budding and maturation. From Turner and Summers.⁷³⁵

hope from the fact that some individuals are naturally resistant to HIV infection.^{738,739}

3. Retrotransposons

Transposition of DNA, which is discussed in Chapter 27, Section D.4, may seem to be a rare and relatively unimportant event in our body cells. However, transposon DNA accounts for 35% or more of the human genome⁷⁴⁰ and apparently plays a major role in evolution. Like other transposons, the DNA sequences known as retrotransposons also move about within DNA. However, they use an indirect mechanism that involves synthesis of mRNA and reverse transcription.^{740,741} The reverse transcribed complementary DNA may be inserted back into the genome at new locations. The necessary chemical reactions parallel those involved in the replication of retroviruses (Fig. 28-23, 28-24). Retrotransposons, truncated retrotransposons, and related sequences constitute as much as 16% of the human genome.⁷⁴¹

There are two classes of retrotransposons: those with long terminal repeats (LTRs) and those without (LTRs). The first group is closely related to retroviruses, but its members lack genes for envelope proteins. They do carry *gag* and *pol* genes similar to those of retroviruses (Fig. 28-3). Most retrotransposons are defective and do not move. Over evolutionary time they accumulate in the genome, sometimes to the extent that the genome size grows enormously. This

has happened often during the evolution of plants, some of which (e.g., certain lilies) have 40 times more DNA per cell than do humans (Table 1-3).⁷⁴² Although most retrotransposons are inactive, some of them occasionally jump to new locations where they may mutate a gene and may sometimes cause disease. However, their major significance is probably in facilitating evolution, perhaps including the formation of new species.⁷⁴³

The non-LTR transposons are exemplified by the 6–7 kbp LINES (p. 1539)^{741,743} and the short 90–400 bp SINES (p. 1538, Fig. 27-9).⁷⁴⁴ Mammalian genomes contain ~50,000 truncated members of the LINE-1 (L1) family and 3000–5000 full-length L1s. Only a few of these are active in our present population. The RNA intermediates that participate in retroposition of LINES are generated by RNA polymerase II, while RNA polymerase III forms the RNA intermediates for propagation of SINES. Participation of these RNAs in trans splicing processes can modify existing genes, contributing to the remodeling of the genome.⁷⁴⁴

Yet another group of mobile elements in the genome are **intein genes**, which encode protein-splicing polypeptides (see Box 29-D). Many inteins also have **homing endonuclease** activity and cleave DNA at specific insertion sequences, initiating incorporation of intein DNA into new locations in the genome.⁷⁴⁵ Group II introns, which are found in bacteria and in organelles of fungi and plants, may also act as mobile DNA elements.⁷⁴⁶

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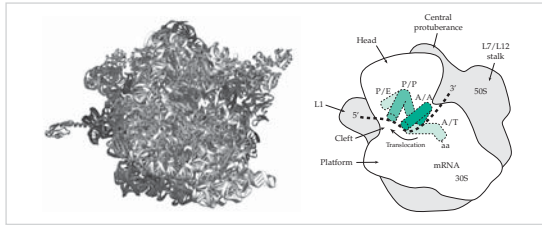
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Study Questions

(Additional questions are located on p. 1738.)

1. Describe the sequence of events involved in the initiation of transcription in *E. coli*. As part of your answer, describe those features that must be present in a gene for proper recognition and transcription by the RNA polymerase.
2. How does transcription in eukaryotes differ from that in prokaryotes?
3. In *E. coli* precise spacing between the conserved -35 and -10 (Pribnow) promoter elements has been found to be a critical determinant of promoter strength. What does this suggest about the interaction between RNA polymerase and these conserved sequences in the DNA?



The ~15,000 ribosomes in a bacterial cell synthesize over 4000 proteins following the genetic code in messenger RNA molecules. A ribosome (left) consists of two large subunits, each composed largely of ribosomal RNA, whose folded chains can be seen. About 100 proteins are bound, largely to solvent-exposed surfaces, but with extended “tails” protruding into the ribosome. The messenger RNA (mRNA; right) moves through the ribosome between the large subunits. Amino acids, activated for reaction, are carried into the ribosomes by transfer RNAs (green) which move consecutively from A/T to A/A, A/P, P/P, and P/E sites. They insert their activated amino acids into the growing polypeptide chain in the P site of the 50S subunit. Image of ribosome^{33a} courtesy of the authors.

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Boxes

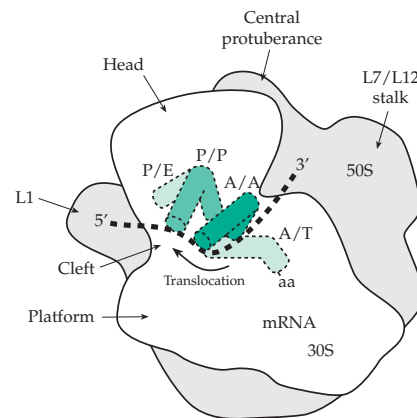
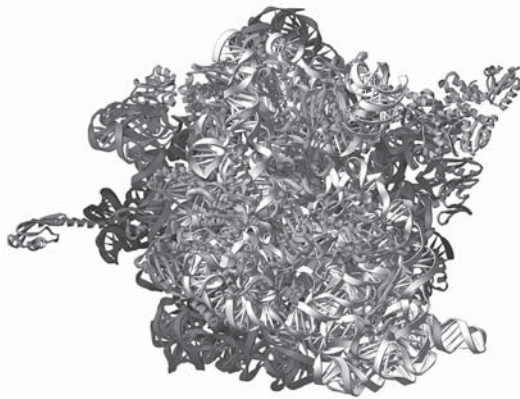
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Tables

- 1672 Table 29-1 The Composition of Ribosomes
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Ribosomes and the Synthesis of Proteins

29



The many thousands of proteins present in each cell are made within the ribosomes, which are able to read each specific mRNA that comes their way. While ribosomes appear as little more than blurred dots in most electron micrographs, the 15,000 ribosomes of one cell of *E. coli* represent one-fourth of the total mass of the cell. Eukaryotic cells contain many times more of these little molecular machines. When ribosomes were first observed in the early 1950s,^{1–3} nobody could imagine either their composition or their function. Less than 50 years later (1999) their complete three-dimensional structure was known at nearly atomic resolution, and the function of ribosomes in protein synthesis was quite well understood. However, the structure could not have been obtained without the development of a whole range of new methods.

As electron microscopy developed, the fuzzy granules in micrographs assumed well-recognized forms. By the 1970s, the approximate shapes of the two ribosomal subunits were known, locations of several ribosomal proteins had been established, and binding sites of transfer RNAs and other features of ribosomes were being mapped.^{2,4–7} The resulting picture of the ribosome structure, which is shown schematically in Fig. 29-1, is quite similar to the present-day view. Later, three-dimensional images were reconstructed from electron micrographs (electron tomography),^{8,9} and cryo-electron microscopy provided detailed images at a resolution of ~2 nm.¹⁰

In 1950, when the study of ribosomes began, no methods for determining the sequences of amino acids in proteins or of nucleotides in nucleic acids existed.¹¹ Sanger published the sequences of the two short chains of insulin in 1953, and the first transfer RNA sequence was published by Holley in 1965.²¹ Never-

theless, by 1980 the Wittmanns and coworkers in Berlin had sequenced 53 of the *E. coli* ribosomal proteins^{4,22} (Table 29-2), and the three rRNA molecules had also been sequenced.^{22a,b,c} In 1950, X-ray crystallography of proteins was still in its infancy; the structure of myoglobin was not determined until 1960. Ribosomal proteins proved hard to crystallize, the first structure being solved in 1980.²³ NMR structural analysis yielded several structures including that of L30 (Fig. 3-25A). In recent years, high-resolution structures of many additional ribosomal proteins in their free forms have been established as have the structures of most of the proteins bound into ribosomes.²⁴

The first crystals of bacterial ribosomes in a three-dimensional lattice suitable for study by X-ray diffraction at a resolution of ~1 nm²⁷ were obtained by Yonath in about 1980.^{6,25,26} Now atomic structures are being established at a resolution of 0.3–0.1 nm, or less.^{17–19,28–33g} However, such progress would have been impossible without information about ribosomes obtained from improved cryo-electron microscopy,^{10,20,33e,34–37a} phylogenetic analysis of ribosomal RNAs,^{38–39b} mutational analysis,^{40,41} neutron scattering,^{42–47} chemical and photochemical crosslinking,^{48–53} photoaffinity labeling,⁵⁴ immunological labeling,⁵⁵ chemical footprinting,^{56,57} fluorescence resonance energy transfer (FRET),⁵⁸ mass spectrometry,^{59,60} and study of the effects of toxic proteins (Box 29-A)⁶¹ and antibiotics (Box 29-B).

Why is the ribosome so large and complex? Aside from the fact that it must form the peptide linkages, it must translate the genetic code in the mRNA into the correct amino acid sequence for each of the thousands of proteins present in the cell. The process takes place

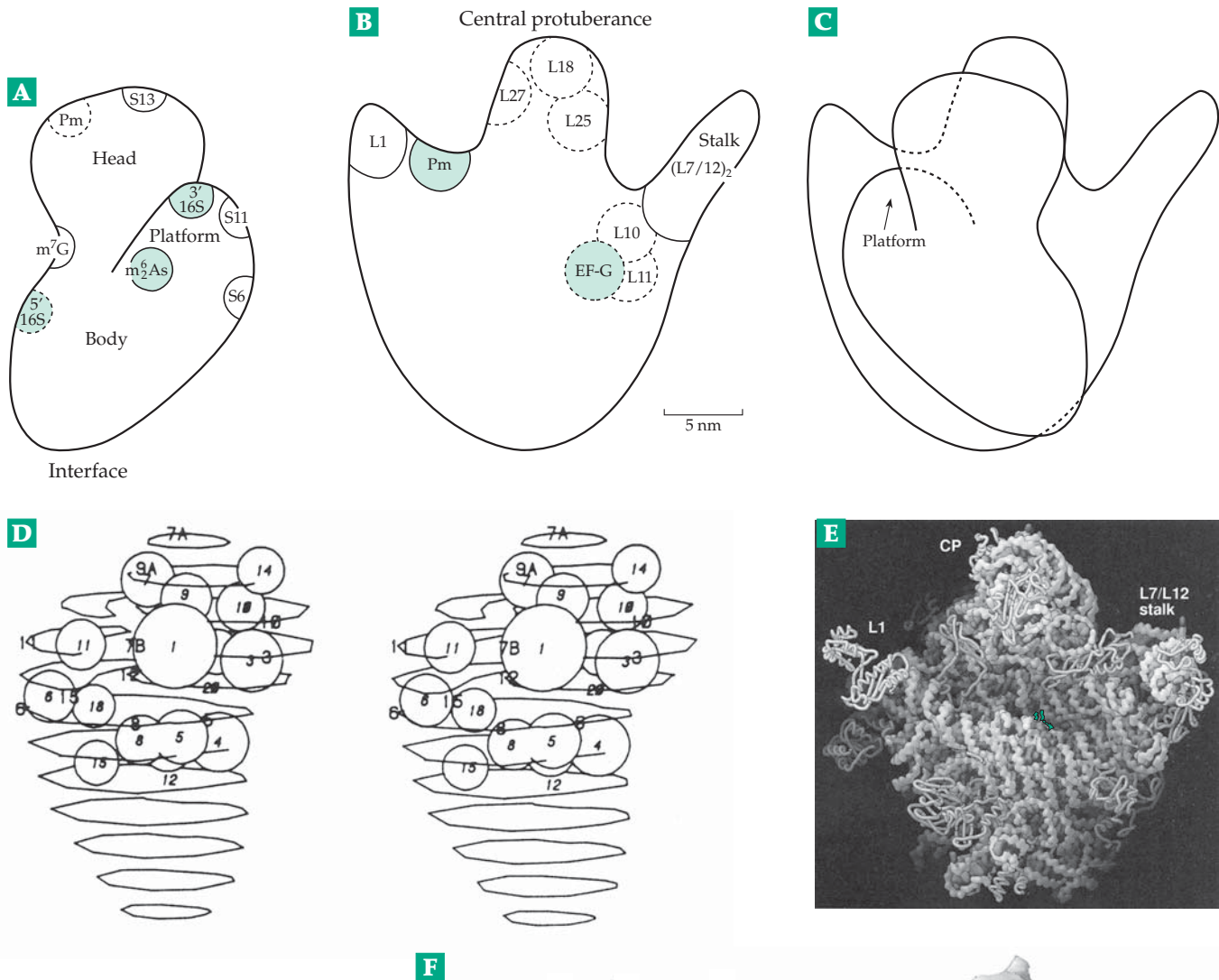
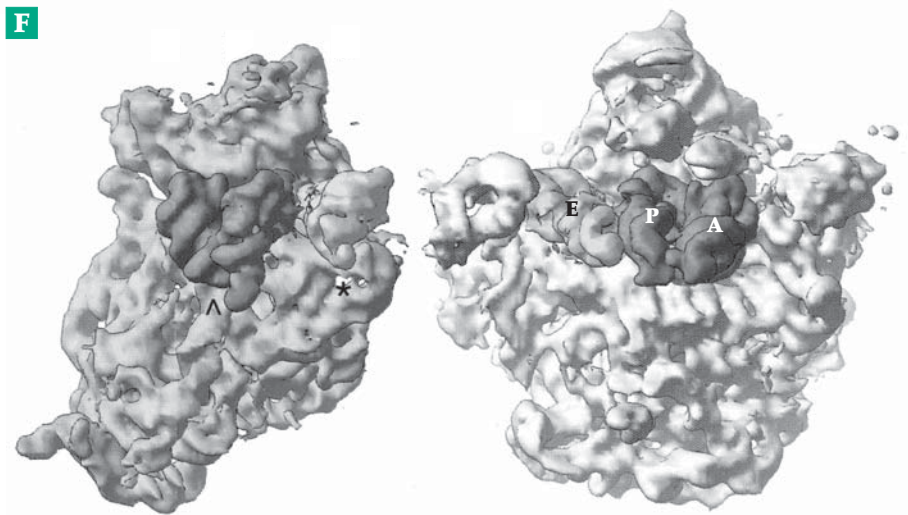
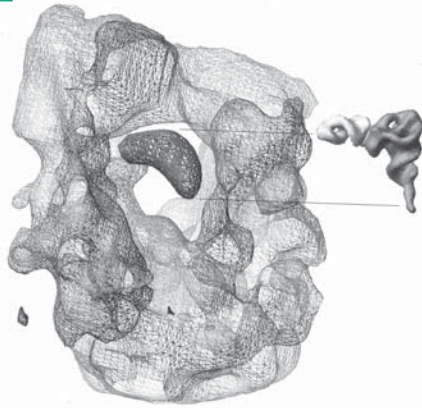
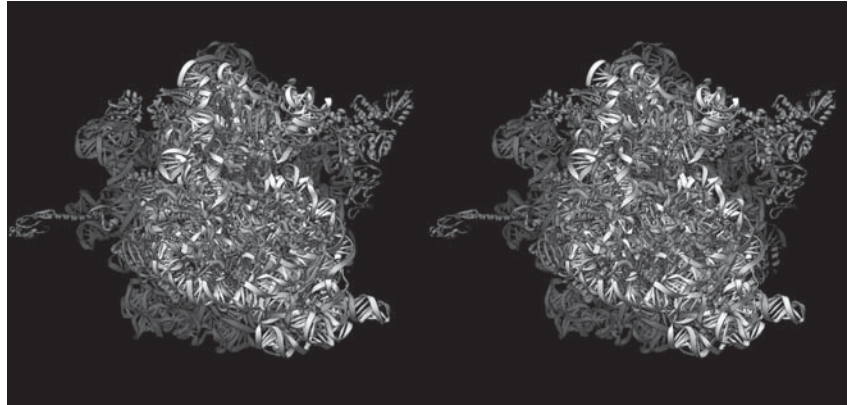
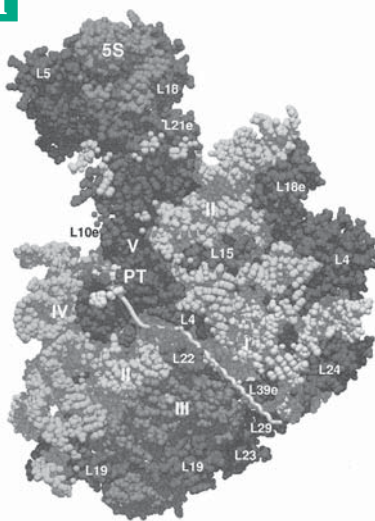
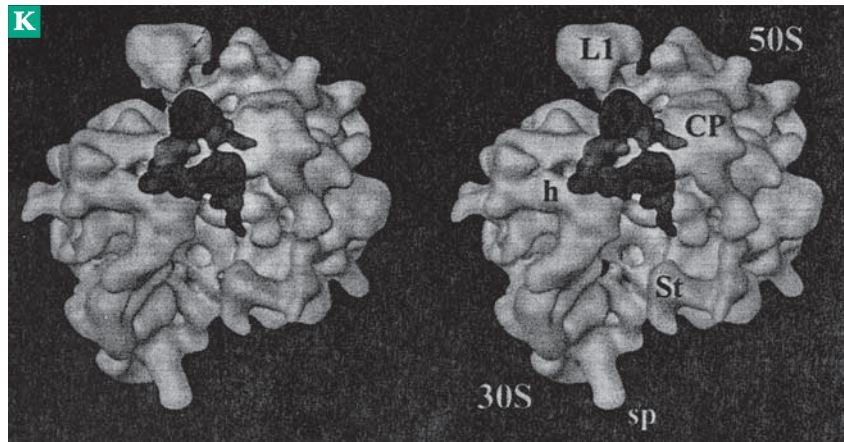


Figure 29-1 (A–C) A 1970s view of a bacterial ribosome achieved by electron microscopy and image reconstruction. These interface views show the surfaces that face each other in the 70S ribosome. Locations marked with dashed lines are on the outer (back) surfaces. From Nagano and Harel.¹² Based on shapes and data of Lake and associates.^{13,14} (A) The 30S subunit. Positions of a few proteins located by immunoelectron microscopy and three positions in the 16S RNA are marked. The puromycin binding site labeled Pm was mistakenly thought to be near the peptidyltransferase center. (B) The 50S subunit. Only a few positions of many located by a variety of techniques are marked. EF-G marks the site at which elongation factor G (see Fig. 29-12) binds. A prominent feature is the “stalk”, designated St in some of the drawings. It was early shown to be formed by two copies each of the nearly identical L7 and L12. The stalk is flexible and in many circumstances may be folded down or not visible as in the X-ray structures of F–H. (C) The 70S ribosome. The 50S subunit is oriented as in (A) while the 30S subunit has its outer face toward the viewer. (D) Stereoscopic view of a neutron scattering map of the 30S subunit of an *E. coli* ribosome. The proteins studied are represented as spheres with volumes corresponding to those of the anhydrous proteins. Also marked on the map are positions of proteins located by immunoelectron microscopy as mapped by Kahan *et al.*¹⁵ Figure from Ramakrishnan *et al.*¹⁶ Courtesy of V. Ramakrishnan. (E) Model of the 50S subunit from *Haloarcula marismortui*. From



G**H****I****J****K**

Ban *et al.*¹⁷ Courtesy of T. A. Steitz. The peptidyltransferase center is marked by the green image of the transition state inhibitor shown in Fig. 29-13. (F) Model of three tRNAs bound to a ribosome from *Thermus thermophilus* in the A (aminoacyl), P (peptidyl), and E (exit) sites. These are based on 0.75-nm X-ray data and a number of difference electron density maps. The 3'-CCA end of the A-site tRNA is not modeled but is marked “^”. Views are left, facing the inner surface of the 30S subunit; right, facing the inner surface of the 50S subunit. (G) Schematic side view of a ribosome showing a molecule of tRNA bound in the A site between the 30S and 50S subunits. The anticodon of the tRNA is base-paired with mRNA in the “decoding site” on the 30S subunit. The 3'-CCA end with attached aminoacyl group lies in the peptidyltransferase site in the 50S subunit. (F) and (G) are courtesy of Cate *et al.*¹⁸ (H) Stereoscopic view of a model of the 70S ribosome from *T. thermophilus*. The 30S subunit (lighter) is toward the viewer. Courtesy of Harry F. Noller and Albion Baucom. (I) Section through the 0.24 nm-resolution model of the 50S subunit shown in (E). The modeled path of the polypeptide chain through the exit tunnel is marked. Courtesy of Nissen *et al.*¹⁹ (J) Stereo diagram of the relative orientations of the A-, P-, and E-tRNAs and mRNA showing codon-anticodon interactions and the kink between the A and P codons. (H) and (J) courtesy of Yusupov *et al.*^{33a} (K) Stereoscopic view of tRNAs in the P site and in an overlapping P/E site as observed by cryo-electron microscopy at a resolution of 0.5 nm in an image of the 70S ribosome at 1.5 nm resolution. The anticodon arms are to the left. Two tRNA molecules are not present simultaneously but their images have been presented together. From Agrawal *et al.*²⁰ Courtesy of Rajendra Agrawal.

TABLE 29-1
The Composition of Ribosomes

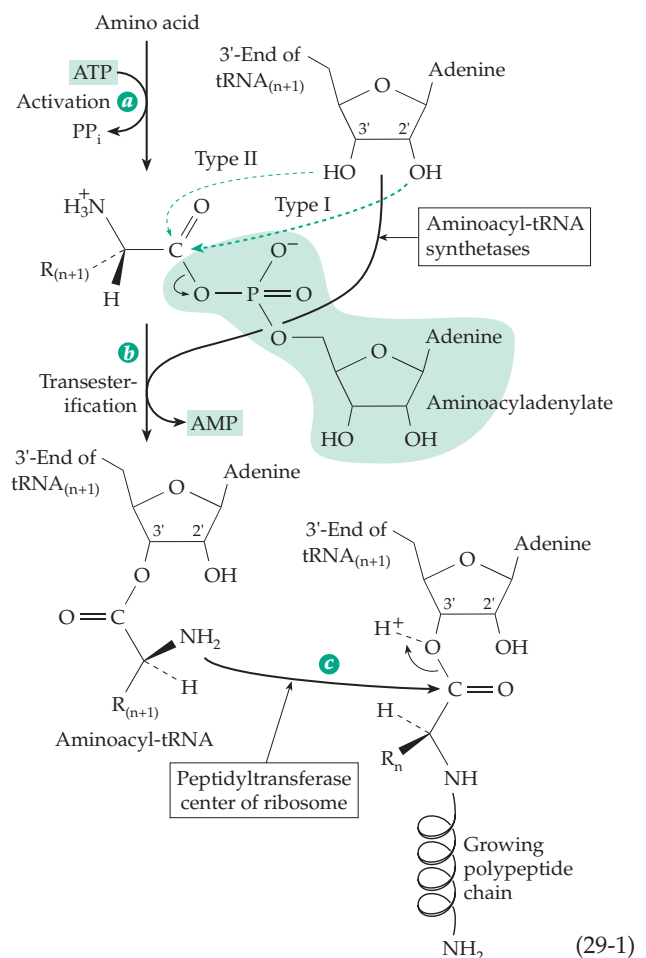
Prokaryotic ^a		Eukaryotic ^b	
Component	Mass, kDa	Component	Mass, kDa
Small (30S) subunit	850	Small (40S) subunit	1440
16S RNA	500	18S RNA	700
Proteins (21)	350 (total)	Proteins (~30)	740
Large (50S) subunit	1450	Large (60S) subunit	2800
23S RNA	950	28S RNA	1700
5S RNA	40	5.8S RNA	51
Proteins (32–34)	460	Proteins (~46)	1010
Complete (70S) ribosome	2300	Complete (80S) ribosome	4240

^a Data from Wittmann, H. G. (1982) *Ann. Rev. Biochem.* **51**, 155–183. Based on sequences of all components. Presence of spermine, K⁺, etc., may add 10%.

^b Data from Freifelder, D. (1983) *Molecular Biology*, 2nd ed., Jones and Bartlett, Boston, Massachusetts (p. 419) and Mao, H., and Williamson, J. R. (1999) *J. Mol. Biol.* **292**, 345–349.

in several steps: (1) **Initiation** of protein synthesis in bacteria requires binding of the 30S ribosomal subunit to an mRNA molecule, location of the start signal (initiation codon and nearby Shine-Dalgarno sequence, shown in Fig. 29-2 and Eq. 29-8), and binding of the initiator tRNA carrying formylmethionine. The 30S complex must then bind to the 50S ribosomal subunit. (2) **Elongation** of the polypeptide chain in the resulting complete 70S ribosome ensues with binding of the appropriate aminoacyl tRNA to the next codon in the 5' → 3' direction. Base pairs form between the anticodon of the tRNA and the mRNA codon that lies in the **aminoacyl (A) site**; the peptide bond is then formed by the **peptidyltransferase** reaction. This reaction is followed by **translocation**, movement of the initiator tRNA into an exit site at the same time that the second tRNA (together with its mRNA codon and the attached growing peptide chain) moves into the **peptidyl (P) site**. The elongation cycle is repeated until the peptide chain is complete. (3) **Termination** of translation involves release of the completed protein and preparation of the ribosomal subunits for another cycle. The entire process is powered by the hydrolysis of ATP and GTP. The ATP is utilized in a three-step process for the activation of the amino acids, which become linked to the appropriate tRNAs (Eqs. 17-36 and 29-1).^{61a} If the inorganic pyrophosphate that is formed is hydrolyzed, two molecules of ATP are required for activation of each amino acid molecule. In addition, at least two molecules of GTP are hydrolyzed to GDP and inorganic phosphate within the ribosome for each peptide linkage formed.

The pairing of codons and anti-codons required for insertion of the correct amino acid into the growing polypeptide chain is often referred to as **decoding** of the gene sequence. However, an equally important part of the decoding is the attachment of the correct amino acid to its corresponding **cognate tRNA**. This occurs in the cytoplasm and also in the nucleus.⁶² The base pairing of tRNAs and mRNA, which follows, occurs in the **decoding center** on the 30S ribosomal subunit. Both the A and P sites of tRNA-binding and the decoding center are formed by folds of the 16S RNA. The peptide bond formation takes place at the opposite ends of the tRNA molecules in the **peptidyltransferase center** of the 50S subunit (see Figs. 29-1, 29-14). As pointed out in Chapter 12 (top of p. 650), peptidyltransferase is a **ribozyme**. Its active site consists entirely of segments of the 23S RNA (see Fig. 29-14). Another important site is the **GTPase activating center**, at which the 23S



RNA interacts with specific G proteins known as **initiation, elongation, and termination factors**.

The proteins in a ribosome may help to hold the RNA into conformations that are correct for its functions. They may also catalyze conformational alterations during the various steps of the translation process. In addition, the proteins may help provide binding sites for substrate molecules and participate in regulatory activities. Both the tRNA **exit (E) site** and the **tunnel** through which the polypeptide chain leaves the ribosome are composed, in part, of ribosomal proteins.

A. The Architecture of Ribosomes

Ribosomes of *E. coli* each have a mass of $\sim 2.3 \times 10^6$ daltons and are $\sim 65\%$ RNA and 35% protein. Ribosomes of eukaryotic organisms are larger ($\sim 4.3 \times 10^6$ daltons) and consist of $\sim 50\%$ RNA and 50% protein. Under some conditions such as a low Mg^{2+} concentration complete bacterial ribosomes, called **70S ribosomes**, dissociate into two subunits of unequal size, which are known as **30S** and **50S ribosomal subunits**. The larger 50S subunit is about twice the size of the smaller one (Table 29-1). The small 30S ribosomal subunit contains the 16S rRNA, a chain of ~ 1500 – 1700 nucleotides (nt) that, if fully extended, would stretch to a length of over 500 nm. In addition to the highly folded RNA molecule, the 30S subunit contains 21 proteins, each one unique in its amino acid composition and sequence (Table 29-2). Many of these proteins, which are designated S1, S2, S3, etc., are of relatively low molecular mass. Many are strongly basic. They contain numerous lysine and arginine residues, many of which are able to interact with RNA in the ribosome. However, neutral and acidic proteins are also present. The 50S ribosomal subunit contains the ~ 2900 nt 23S rRNA, the ~ 120 nt 5S RNA, and about 31–34 proteins, two of which (L7 and L12) are present as two copies each. The composition of ribosomes is variable, but most proteins are present in a strict 1:1 ratio. Others may be lacking in some of the ribosomes. Some proteins bind to the ribosomes transiently during their function in protein synthesis as do certain proteins with functions other than protein synthesis. In both subunits the RNA molecules form the internal core. Proteins are largely found on the solvent-exposed surfaces. Some of them form the stalk and other features.¹⁷ They often have globular domains with extended tails that interact with the ribosomal RNA.

Eukaryotic ribosomes are not only larger but also (Table 29-1) contain more protein subunits than do those of bacteria: ~ 30 for the small subunit and 49 for the large subunit.⁶³ However, the number of essential proteins may be the same. Both eukaryotic ribosomal proteins and rRNA molecules are larger than those of

bacteria. Bacterial ribosomes are ~ 22 nm in diameter and ~ 30 nm in the third dimension. Eukaryotic ribosomes are of the order of 1.17 times larger in linear dimensions. Ribosomes of chloroplasts resemble those of eubacteria such as *E. coli* but contain a few more proteins.⁶⁴ Mammalian mitochondrial ribosomes also resemble those of bacteria in many respects.⁶⁵ However, their RNA chains are shorter and they contain more proteins.^{66,66a} The protein content is $\sim 66\%$ compared with $\sim 35\%$ for *E. coli* ribosomes.

1. Ribosomal RNA

The sequences of all three pieces of RNA in the *E. coli* ribosomes are known as are those from many other species. These include eukaryotic mitochondrial, plastid, and cytosolic rRNA. From the sequences alone, it was clear that these long molecules could fold into a complex series of hairpin loops resembling those in tRNA. For example, the 16S rRNA of *E. coli* can fold as in Fig. 29-2A and eukaryotic 18S RNA in a similar way (Fig. 29-4).^{38,39,67–69} The actual secondary structures of 16S and 18S RNAs, within the folded molecules revealed by X-ray crystallography, are very similar to that shown in Fig. 29-2A. Ribosomal RNAs undergo many posttranscriptional alterations. Methylation of 2'-hydroxyls and of the nucleic acid bases as well as conversion to pseudouridines (pp. 1638–1641) predominate over 200 modifications, principally in functionally important locations that have been found in human rRNA.^{69a}

Chemical modification and crosslinking.

Before high-resolution X-ray data were available, two major biochemical approaches were used to deduce the secondary structures of ribosomal RNAs.^{38,39} The first was the application of chemical reagents and enzymes that modify the RNA. Crosslinking reagents were used to establish pairs of nucleotides that lie close together in the three-dimensional structure. Cleavage by specific endonucleases was used to establish whether a region of the molecule is double-helical or single-stranded.⁶⁸ Nucleases were also used to clip out base-paired fragments, which were separated, denatured, and sequenced. This revealed both hairpin loops and pairings between regions that are far apart in the primary sequence. The ability of nucleic acid bases to undergo specific chemical reactions at positions not involved in base pairing was used to establish whether or not a given base was actually paired.^{67,69} Thus, every position in *E. coli* 16S RNA was probed by reactions of dimethylsulfate with adenine at N1 and cytosine at N3, reaction of kethoxal (Eq. 5-16) with guanine at N1 and N2', and by reaction of a carbodiimide with uracil at N3 and with guanine at N1.⁶⁷

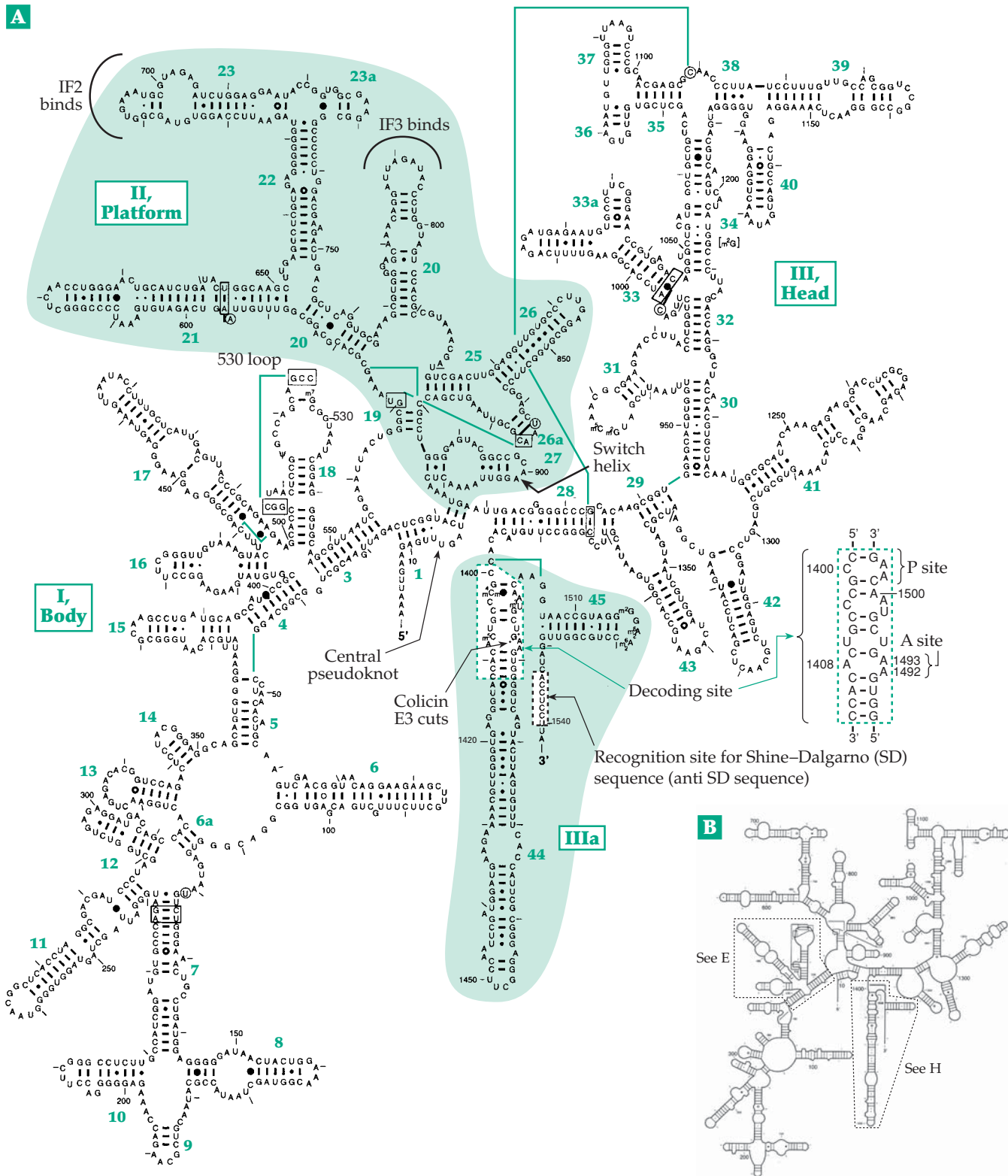
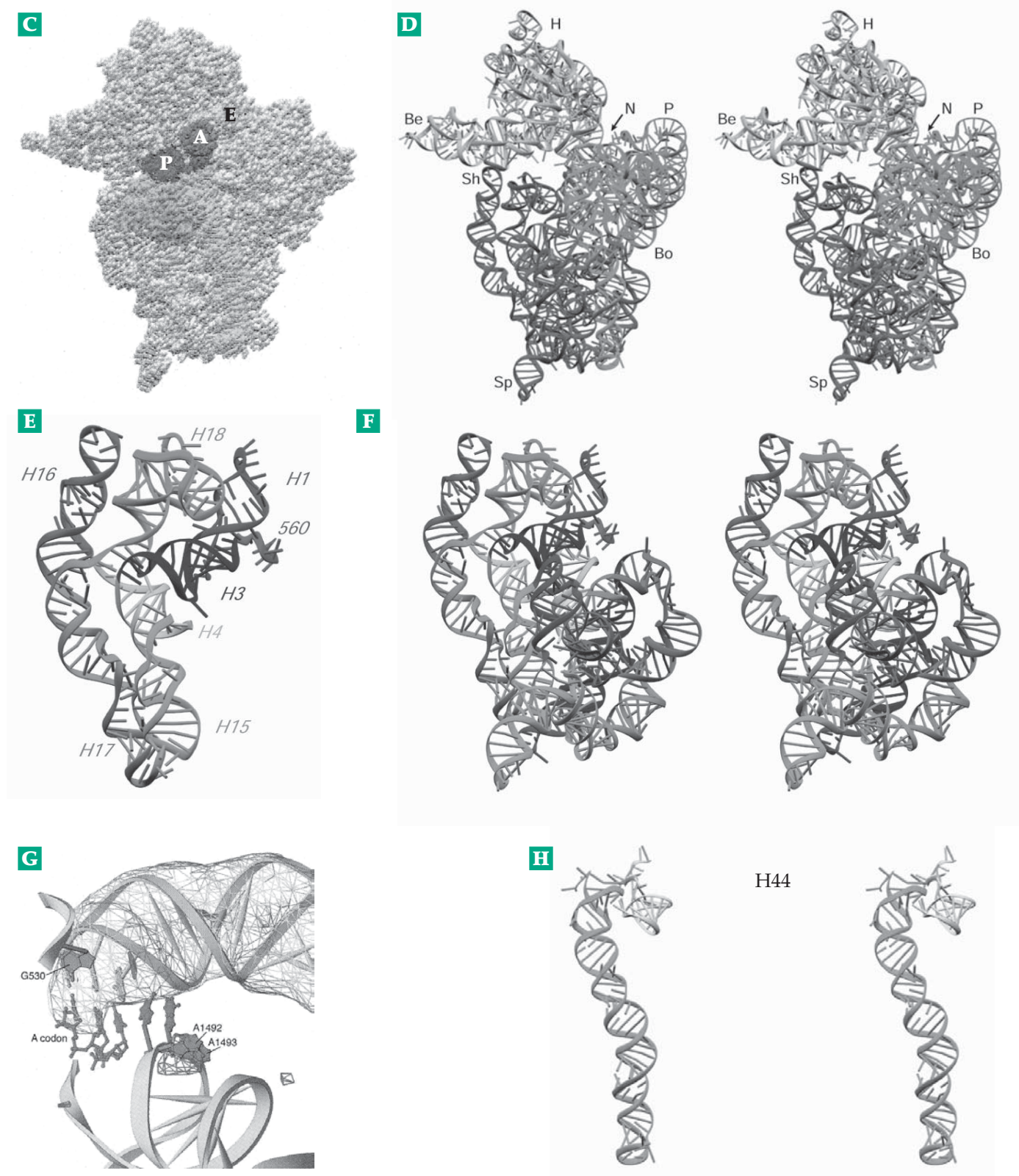


Figure 29-2 (A) Secondary structure model for the 1542-residue *E. coli* 16S rRNA based on comparative sequence analysis.^{73a} Dots indicate G•U or A•G pairs; dashes indicate G•C or A•U pairs. Strongly implied tertiary interactions are shown by solid green lines. Helix numbering according to Brimacombe. Courtesy of Robin Gutell. (B) Simplified schematic drawing of type often used. (C) Positions of the A, P, and E sites on the 30S ribosomal subunit from Carter *et al.*⁷⁰ (D) Stereoscopic view of the three-dimensional fold of the 16S RNA from *Thermus thermophilus* as revealed by X-ray structural analysis at 0.3 nm resolution. Features labeled are the head (H), beak (Be), neck (N), platform (P), shoulder (Sh), spur (Sp), and body (Bo). (E–H) Selected parts of the 16S RNA. In (E) and (F) the helices are numbered as in (A). (F) and (H) are stereoscopic views. The decoding site



is located at the upper end of helix 44. (G), (H). In (G) the electron density difference observed upon binding of tRNA into the A site is displayed as a Fourier difference map (at 0.7-nm resolution). The molecular model of the tRNA with its anticodon paired with a codon from mRNA is superimposed. Two positions of bases A1492 and A1493 are shown as they are found in the presence and absence of paromomycin. A patch of negative density can be seen near the A1492 and A1493 labels, indicating that these groups may rearrange to interact with the minor groove of the codon-anticodon helix when the A-tRNA is bound. See also p. 1690. Courtesy of Yusupov *et al.*^{33a} (D) through (F) and (H) are from Wimberly *et al.*³³ Courtesy of Venki Ramakrishnan.

Phylogenetic comparison. This technique, also called comparative sequence analysis, has proved very powerful.^{38,39,39a,b,71,72} An example is illustrated in Fig. 29-3. Here a loop from 23S RNA of *E. coli* is shown and is compared with the sequence of 26S RNA from the fungus *Physarum polycephalum*.³⁸ Wherever the latter differs from the *E. coli* sequence, the substituted base is indicated in a box. These square boxes, which are concentrated in base-paired regions, indicate compensatory changes for which there is usually a second change that preserves base pairing in a double helical region. The studies also showed clearly that bacterial 16S RNA is homologous with eukaryotic 18S RNA,^{68,69} with 17S RNA of dinoflagellates,⁷³ and also with 12S RNA of human mitochondria. Likewise, 23S RNA of bacteria corresponds to 28S RNA of eukaryotes.

Structural domains in 16S ribosomal RNA.

Three major compact structural domains, 5', central, and 3', can be distinguished in 16S RNA.³³ An extended subdomain is also present at the 3' end. These are indicated on Fig. 29-2 as I, II, III, and IIIa. The double helical segments are also numbered. Ribosomal RNA molecules must be folded into compact forms to fit into the envelope of the ribosomal subunits. The individual structural domains form independent globular cores to which several proteins apiece are bound. Domains I, II, and III form the body, platform, and head, respectively (Fig. 29-1A). Relatively minor changes in conformation accompany the incorporation of the rRNA molecules into the ribosomes.⁶⁷ On this basis, and taking account of all available data, attempts were made for many years to predict a three-dimensional structure.^{12,74-76} One of these⁷⁷ is portrayed in Fig. 5-32A as a series of cylinders representing the 45 double-helical segments suggested by the structure of Fig. 29-2A. This can be compared with the X-ray based model shown in Fig. 29-2B.

23S rRNA. The large RNA of the 50S subunit consists of six structural domains.⁵ Its secondary structure is shown in Fig. 29-4. As with 16S RNA each domain is tightly folded. However, the domains are interdigitated in such a way that they form a single monolithic structural unit.¹⁷ Nevertheless, there are distinct catalytic sites, as described in Section 4. Like proteins, which are able to undergo conformational alterations that usually involve some rearrangement in their internal hydrogen-bonding patterns, these large RNA molecules may also assume alternative conformations. Conformational changes may involve not only alternative hydrogen bonding patterns but also alternative base-pairing.^{77a} Such changes may be essential to the functioning of ribosomes⁸⁶ and may also accompany maturation of pre-rRNAs.⁸⁷ Eukaryotic 28S RNAs have basically the same structures as the

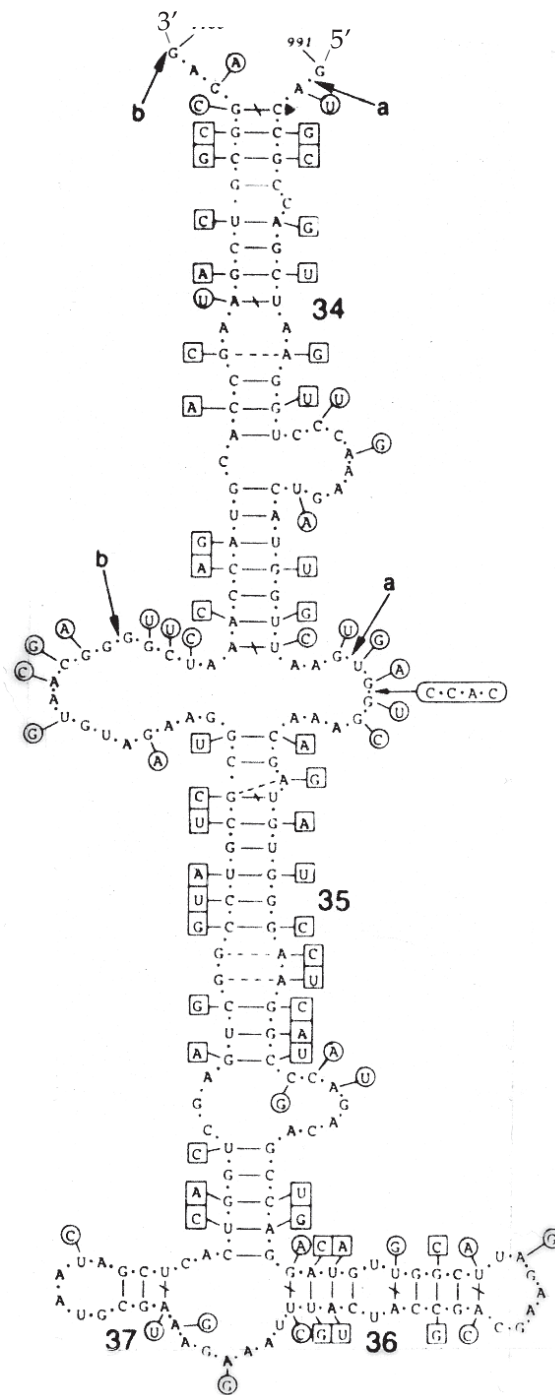


Figure 29-3 Example of phylogenetic comparisons in ribosomal RNA. The diagram shows helices 34–37 of *E. coli* 23S RNA, compared with the corresponding region of *Physarum polycephalum* 26S RNA. The diagram depicts the *E. coli* helices, with changes in the *P. polycephalum* sequences denoted by symbols in boxes on the side. Bases in square boxes are compensating; those in round boxes are mismatching or in single-stranded regions. Solid triangles denote deletions, while bases with arrows indicate insertions. Dotted lines or “crossed-out” base pairs indicate modified base-pairing in *P. polycephalum*. The letters “a” and “b” indicate the termini of RNA fragments isolated as a base-paired complex. From Brimacombe.³⁸

23S RNAs of bacteria but have been expanded by insertion of additional nucleotides at many places.^{10,36,79}

Most of the chemical activity of ribosomes occurs in the interface between the 30S and 50S subunits. Entrance and exit tunnels for both mRNA and the aminoacylated tRNAs are formed between these subunits. The mRNA apparently moves across the platform as the tRNAs move from A to P to E sites experiencing codon selection (decoding) and peptidyltransferase activity. Many loop ends from 16S RNA interact with those of 23S RNA.^{41,88}

5S rRNA. This ~120-nucleotide molecule organizes one domain of the 50S ribosomal subunit.⁸⁹ Extensive phylogenetic comparisons of 5S RNA sequences led to the secondary structure shown in Fig. 29-5.^{90,91} The three-dimensional structure, as seen in a ribosome, is also shown in this figure. Study of base-pairing possibilities suggests that 5S RNA can exist in more than one conformation.^{90,92} In a possible second conformation the sequence GUGUGGGG (residues 79–86) pairs in an antiparallel fashion with the sequence CCCCAUGC (residues 35–42), with loss of base pairing in stem 4 (Fig. 29-5). A structure somewhat similar to that of 5S RNA is probable for eukaryotic 5.8S RNA.⁹⁴ Nearly a thousand different prokaryotic and eukaryotic 5S RNA sequences have been compared.⁹⁵ From them **phylogenetic trees**, which suggest evolutionary pathways between species, have been constructed.⁹⁶ Sequences of 16S RNA have been used in a similar way (Fig. 1-5).⁹⁷

2. Ribosomal Proteins

Ribosomal proteins are soluble in concentrated salt solutions. Most of them can be dissolved without damage by buffers containing 2 M LiCl and can then be separated by electrophoresis or ion exchange chromatography and gel filtration (molecular sieving).⁹⁸ Although many of them are quite insoluble and are often unstable, all ribosomal proteins of *E. coli* (Table 29-2) have been separated and sequenced, mainly by Wittmann-Liebold and coworkers.²² The ribosomal proteins of other bacteria usually resemble those of *E. coli*.⁹⁹ The more numerous eukaryotic ribosomal proteins have also been isolated and studied individually.¹⁰⁰ Many of these 84 proteins appear to correspond directly in properties and functions to those of *E. coli*.^{100a} As with ribosomal RNAs, the sizes of the eukaryotic proteins have been expanded.¹⁰¹ Mitochondria have their own set of ribosomal proteins, which are more numerous than those of either *E. coli* or yeast.^{65–66a,102–102c} Pure individual ribosomal proteins are now produced from the cloned genes as are 16S and 23S ribosomal RNAs.

Most ribosomal proteins are folded into compact

forms, much of whose surfaces are accessible to added reagents. However, X-ray structures have revealed that parts of some proteins penetrate deeply into the RNA core.¹⁷ Much of the RNA is also accessible from the outside, and the ribosome contains ~50% of its mass as internal hydration. A ribosome usually contains only one molecule of each kind of protein with the exception of proteins L7 and L12 of the large subunit. There are two of each. Sequencing of the 120-residue proteins from *E. coli* shows that L7 is

TABLE 29-2
Ribosomal Proteins from *E. coli*^a

Proteins of 30 S Ribosomal subunits			Proteins of 30 S Ribosomal subunits		
Designation	Mass, kDa	Binding ^b	Designation	Mass, kDa	Binding ^b
S1	61.2		L1	24.6	
S2	26.6		L2	29.4	+
S3	25.8		L3	22.3	
S4	23.1	+	L4	22.1	
S5	17.5		L5	20.2	
S6	15.7		L6	18.8	+
S7					
(strain K)	19.7	+	L7	12.2	
(strain B)	17.1	+			
S8	14.0	+	L8		
S9	14.6		L9	15.5	
S10	11.7		L10	17.7	
S11	13.7		L11	14.9	
S12	13.6		L12	12.2	
S13	13.0		L13	16.0	
S14	11.1		L14	13.5	
S15	10.0	+	L15	15.0	
S16	9.2		L16	15.3	+
S17	9.6	+	L17	14.4	+
S18	8.9		L18	12.8	+
S19	10.3		L19	13.0	+
S20	9.6	+	L20	13.4	+
S21	8.4		L21	11.6	
Total mass 350 (strain K)			L22	12.2	
			L23	11.0	+
			L24	11.2	+
			L25	10.7	+
			L26 = S20	9.6	
			L27	9.0	
			L28	8.9	
			L29	7.3	
			L30	6.4	
			L31	7.0	
			L32	6.3	
			L33	6.3	
			L34	5.4	
			Total mass 460 ^c		

^a Molecular masses from Wittmann, H. G. (1982) *Ann. Rev. Biochem.* 51, 155–183

^b A plus sign indicates direct binding to ribosomal RNA.

^c Four copies of L7/L12 are assumed.

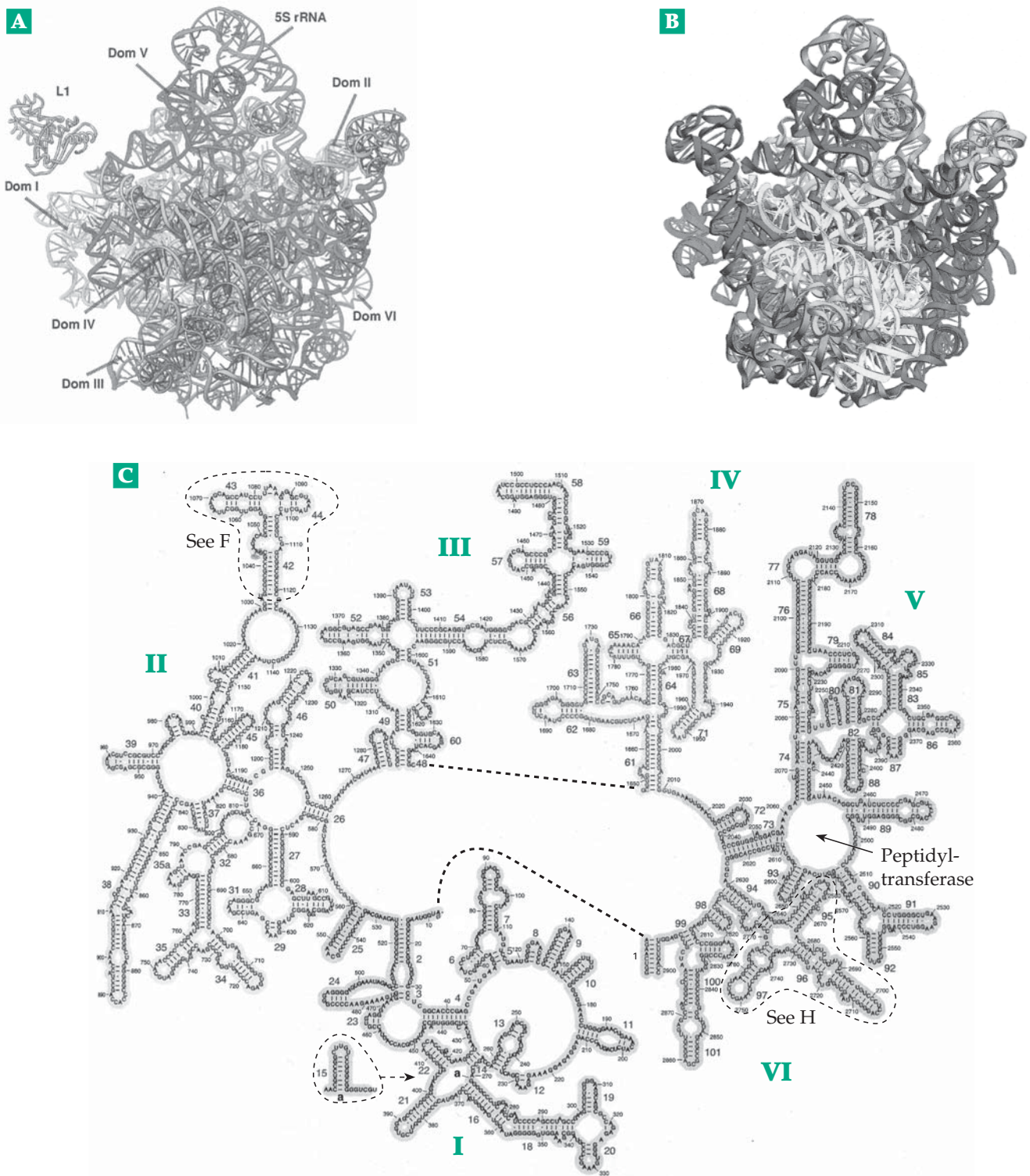
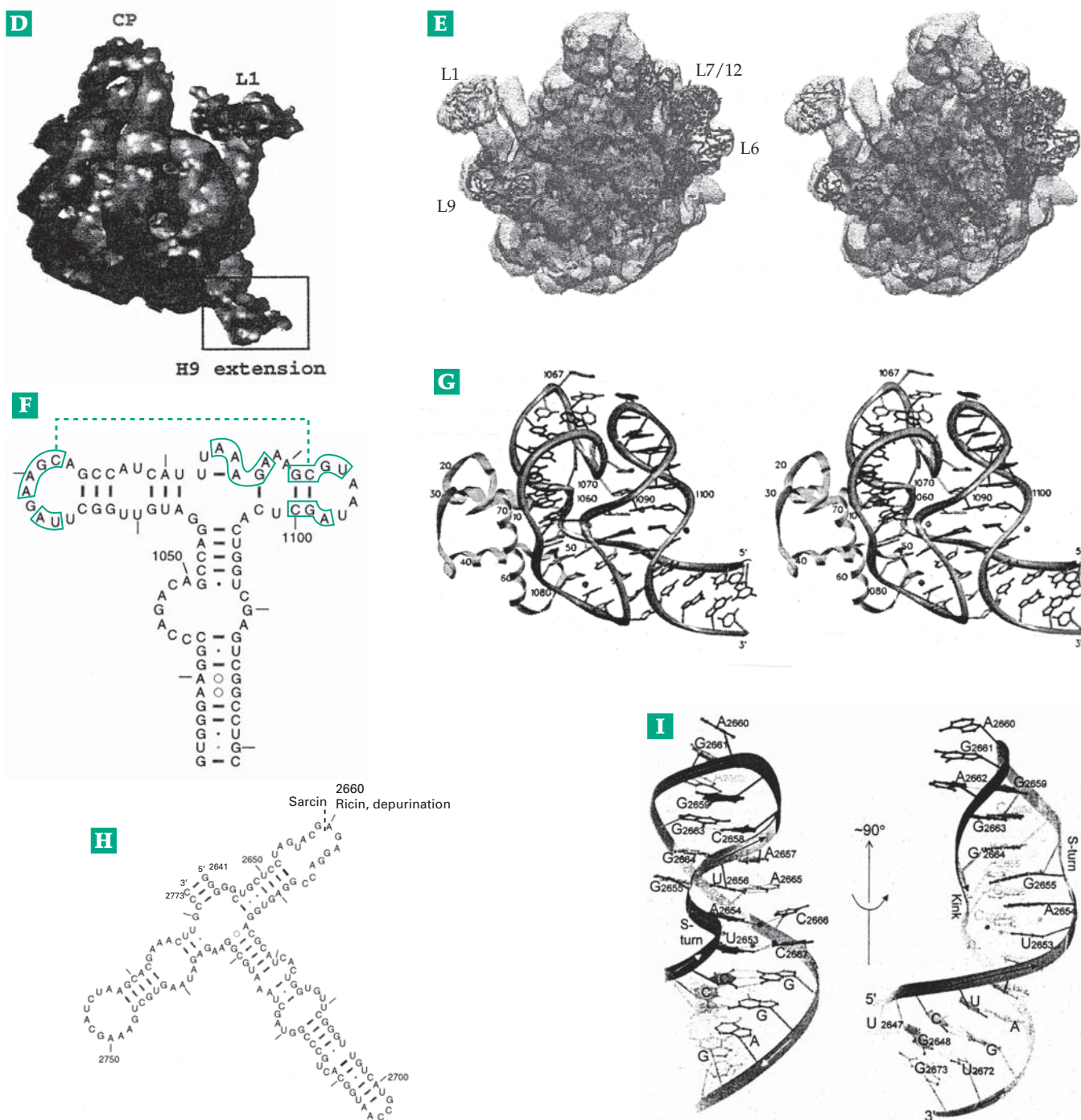


Figure 29-4 Structure of 23S–28S ribosomal RNAs. (A) The three-dimensional structure of RNA from the 50S subunit of ribosomes of *Haloarcula marismortui*. Both the 5S RNA and the six structural domains of the 23S RNA are labeled. Also shown is the backbone structure of protein L1. From Ban *et al.*¹⁷ Courtesy of Thomas A. Steitz. (B) The corresponding structure of the 23S RNA from *Thermus thermophilus*. Courtesy of Yusupov *et al.*^{33a} (C) Simplified drawing of the secondary structure of *E. coli* 23S RNA showing the six domains. The peptidyltransferase loop (see also Fig. 29-14) is labeled. This diagram is customarily presented in two halves, which are here connected by dashed lines. Stem-loop 1, which contains both residues 1 and 2000, is often shown in both halves but here only once. From Merryman *et al.*⁷⁸ Similar diagrams for *Haloarcula marismortui*¹⁷ and for the mouse⁷⁹ reveal a largely conserved structure with nearly identical active sites. (D) Cryo-electron microscopic (Cryo-EM) reconstruction of a 50S subunit of a modified *E. coli* ribosome. The RNA has been modified genetically to have an



approximately 34 nt predicted extension of helix 9 of the 16S RNA (see Fig. 29-2). The helix 9 extension, clearly visible in this image, locates that helix in *E. coli* ribosomes, which have not yet given crystals satisfactory for X-ray investigation. From Matadeen *et al.*^{79a} Courtesy of Richard Brimacombe. (E) Stereoscopic interface view of the 50S subunit of an *E. coli* ribosome with atomic structures of ribosomal proteins fitted to the cryo-EM density (semitransparent) of the 50S subunit. Protein structures are displayed as backbone tubes, and rRNA fragments in ball-and-stick format. Courtesy of Mueller *et al.*^{37a} (F) The GTPase-activating loop of 23S RNA of *E. coli*. This loop, from domain II, binds to protein L11, which shields nucleotide A1067 from methylation and prevents the binding of the antibiotics thiostrepton (Box 29-B) and micrococin. Green nucleotides are highly conserved in bacterial, chloroplast, and mitochondrial RNAs. The small loop (1054–1081) containing the thiostrepton-binding site is also part of the binding site for elongation factors EF-Tu and EF-G.^{80,81} (G) Stereoscopic view of the 58-nucleotide loop shown in (E) with the associated protein L11. Courtesy of Conn *et al.*⁸² (H) Secondary structure of the sarcin/ricin (SR domain) of the *E. coli* 23S RNA.^{83–85} The site of hydrolytic cleavage by the ribonuclease sarcin (Box 29-A) is indicated as is the site of depurination catalyzed by the plant toxin ricin (Box 29-A). (I) Three-dimensional structure of the sarcin-ricin loop. The two views are from directions 90° apart. The sites of attack by ribotoxins are at the top. Courtesy of Correll *et al.*⁸³

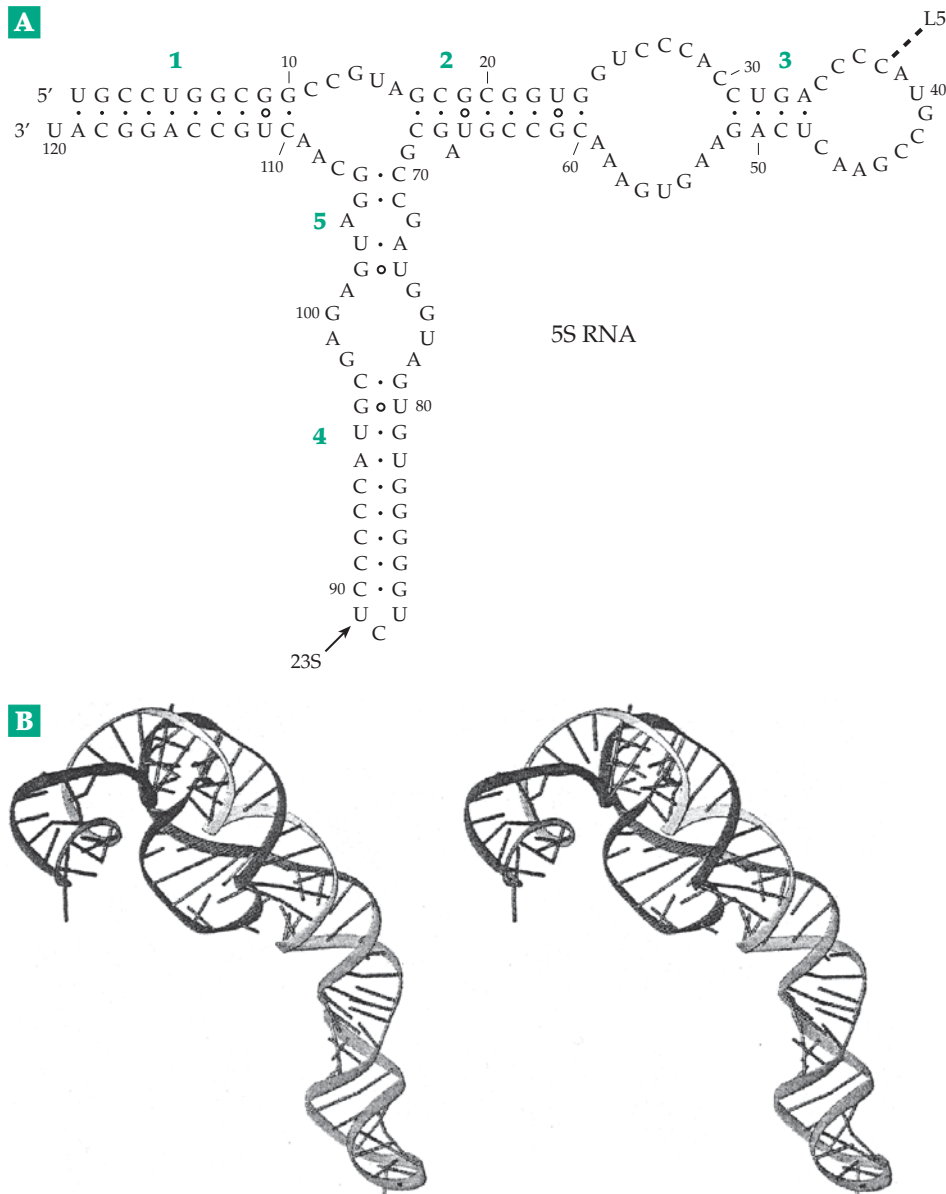


Figure 29-5 (A) Secondary structure of *E. coli* 5S RNA with five universal helical stems (labeled 1–5). This small RNA is found in the central protuberance of the 50S ribosomal subunit. See Fig 29-4A. Photocrosslinking using thiouridine-containing 5S RNA suggested a close proximity of U89 (marked by arrow) with nucleotide 2477 of the 23S RNA in the loop end of helix 89 (Fig. 29-4).⁹³ (B) Stereoscopic view of the 5S RNA as observed in ribosomes of *Haloarcula marismortui*. From Ban *et al.*¹⁷ Courtesy of Thomas A. Steitz.

N-acetylated L12. Thus, the 50S ribosomal subunit is often described as containing four copies of protein L7/12. They form the flexible stalk seen in Fig. 19-1.

Most ribosomal proteins are rich in lysine and arginine and, therefore, carry a substantial net positive charge. Proteins S20, L7/12, and L10 have over 20% alanine, while L29 is almost as rich in leucine. Proteins S10, S13, L7/L12, L27, L29, and L30 are surprisingly low (<2 mol %) in aromatic amino acids. Proteins S5, S18, and L7 have acetylated N termini while L11, L3, L7/12, L11, L16, and L33 contain methylated amino acids. L11 contains nine methyl groups.²² Protein S6 is the major phosphoprotein of eukaryotic ribosomes.^{103,104} Most ribosomal proteins have no known enzymatic activity. Although often difficult to crystallize, high-resolution three-dimensional structures are known for many free ribosomal proteins.²⁴ Most of them have shapes resembling those previously found

in globular proteins, including DNA-binding proteins. Many have extended “tails” that reach into the interior of the ribosome.^{33b} A few seem to assume a defined shape only when packed into a suitable niche in the ribosome. Proteins L7/L12 and the 60-residue L30^{105,105a} have similar folding patterns with 2–3 helices and a 3-strand β sheet. The structure of L30 of *E. coli* was deduced¹⁰⁶ by NMR methods (Fig. 3-25) and resembles that from the *Bacillus stearothermophilus* determined by X-ray diffraction.^{105,107}

Many specific parts of ribosomal RNA molecules and specific proteins within the intact ribosome were located prior to the determination of high resolution crystal structures. One major approach was the use of **immunoelectron microscopy**. Antibodies to specific ribosomal proteins or to special sites in the RNA were prepared, and electron microscopy was used to map the binding sites of the antibodies on the ribosomal

subunit surfaces.^{108,109} In this manner, the locations of numerous proteins in both the 30S and 50S subunits were identified. A few of these are indicated in Fig. 29-1A,B.⁵ In several instances more than one distinct antibody binding site was found for a given protein. Pairs of sites were sometimes 8–19 nm apart, suggesting that these proteins assumed an elongated or fibrous conformation. However, X-ray studies have established more compact structures for many of the proteins. Perhaps the ease of denaturation of the proteins led to some errors in localization with antibodies. The X-ray studies have now established exact locations for almost all of the ribosomal proteins. However, the correct identification of each protein involved extensive measurements, many of which were done prior to the availability of the X-ray structures.

A variety of crosslinking reagents have been used to locate the positions of specific proteins within ribosomes. For example, bifunctional compounds may bind covalently to two different SH groups or NH₂ groups.^{110,111} Among the many crosslinked protein pairs identified in this way are S5-S8, S7-S9, S6-S18, and S13-S19.¹¹² Crosslinking experiments on both small and large ribosomal subunits have yielded complex distance maps that helped to establish the packing relationships.¹¹³

Another important approach has been to isolate ribosomal proteins from bacteria grown in D₂O and then to reconstitute ribosomal subunits with pairs of deuterated proteins. By studying **neutron scattering** the distances between the centers of mass of these pairs could be measured. By triangulation the three-dimensional relationship of the entire group of proteins could be determined. The results of such studies^{43,47,114} for the 30S subunit are shown in Fig. 29-1D. Most of the results are in agreement with those obtained by other methods. Neutron scattering from the 50S subunit was investigated by using pairs of protonated proteins in a subunit consisting of otherwise deuterated components. This gives an increase in sensitivity.⁴⁷

3. RNA-Protein Interactions and Assembly of Ribosomes

Within bacterial cells the assembly of ribosomes is coupled to rRNA synthesis and requires only 1–2 minutes.¹¹⁵ In the laboratory both the 30S ribosomal subunits¹¹⁶ and the 50S subunits^{117–121} of *E. coli* can be completely dissociated into individual protein and RNA molecules and can be reconstituted in a functional form. This is true for both natural 16S or 23S RNA or for RNAs prepared by *in vitro* transcription. In these reassembly experiments, which were pioneered by Nomura,¹¹⁶ it was found that the order of addition of the protein is important. Some proteins bind

directly to ribosomal RNA. For example, S4, S7, S8, S15, S17, and S20 bind directly to 16S RNA.³¹ Other proteins bind only after one or more proteins have already bound and the RNA has folded properly to form a structural core (Fig. 29-6A). Domains I, II, and III each form an independent RNA-protein assembly. The lower half of domain I of the 16S RNA, from positions ~60–300, is unreactive toward single-stranded probes and may serve as one core for assembly of the ribosome.⁶⁷ Protein S20 binds to the 240–286 stem, which is in this core. Protein S4 also binds directly to 16S RNA in the 5' domain. Proteins S8 and S15 bind in the central domain and S7, which is structurally related to the DNA-binding proteins HU and IHF (Chapter 27),^{24,122} binds near the 3' end.

5'-Domain of 16S RNA. The 23-kDa protein S4, one of the largest ribosomal proteins, appears to have an important organizing role for the 5' domain.^{123,124} It binds in such a way as to protect sequences 27–47 and 394–556 of the RNA (Fig. 29-2) from chemical modification. The small loop at positions 323–330 is protected in the 30S subunit, and the residues A325, A327, A379, and G331 are universally conserved.⁶⁷ The same is true of bulge loop 505–510 and the loop sequence 518–533, which contains 7-methylguanine (m⁷G) at position 526. Reconstitution experiments also suggested that S16 binds to S4 as well as to S20. Some mutations in proteins S4 and S5 are associated with reduced fidelity of translation, while others lead to spectinomycin resistance.¹³⁴

Central domain of 16S RNA. Proteins S6, S8, S15, and S18 bind to the central domain II of 16S RNA (Fig. 29-2)^{31,31a,67,125–129} and organize the platform region (Fig. 29-6). Protein S8 binds with high affinity to regions 588–606 and 632–651 of helix 21 and plays a key role in ribosomal assembly.^{126,130,130a} S15 protects residues 655–672 and 734–751 of helix 22. The region contains functionally important conserved loops at positions 570–571, 766–768, and 811–820 as well as many individual adenines in other locations. S15 binds not only to the 16S RNA but also to the 715 loop of 23S RNA in the large subunit and to its own mRNA.¹²⁹ A Mg²⁺-dependent conformational change in the RNA seems to be important in the assembly of the central domain.¹³¹ S6 and S18 bind to 16S RNA after S15 has bound (Fig. 29-6). Proteins S11 and S21 also bind after S15.¹²⁹ S11 binds to the 690 loop of the RNA, as is illustrated in Fig. 29-6B. This loop is conserved in all three phylogenetic domains. Located in the platform of the small subunit, it protrudes into the interface to interact with domain IV of the 23S RNA and is also a site of binding of initiation factor IF3.¹³³ The mutant A649G in 16S RNA confers resistance to **pactamycin** in *E. coli*. Protein S8 is not only an important structural protein in the central domain but also

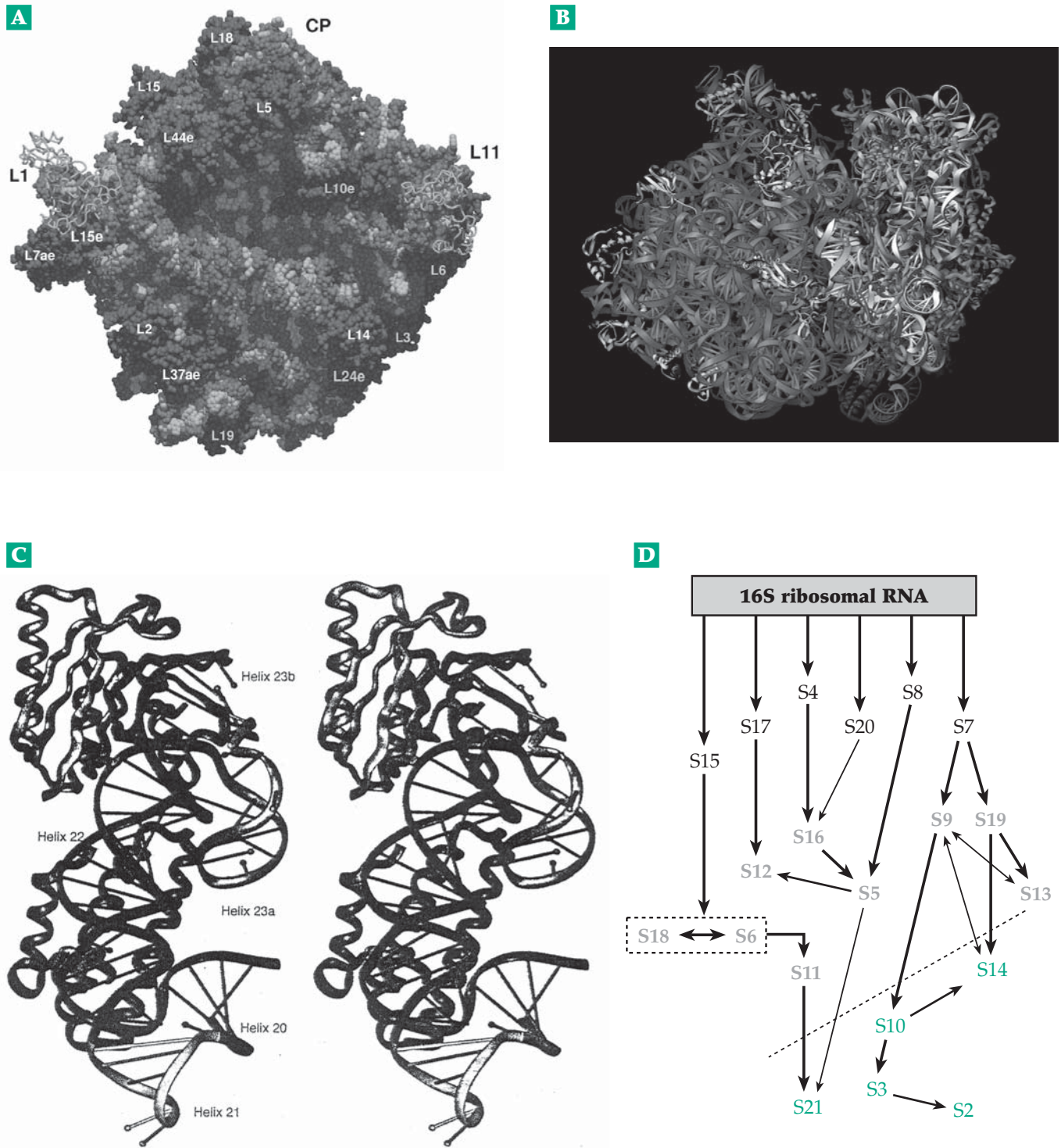
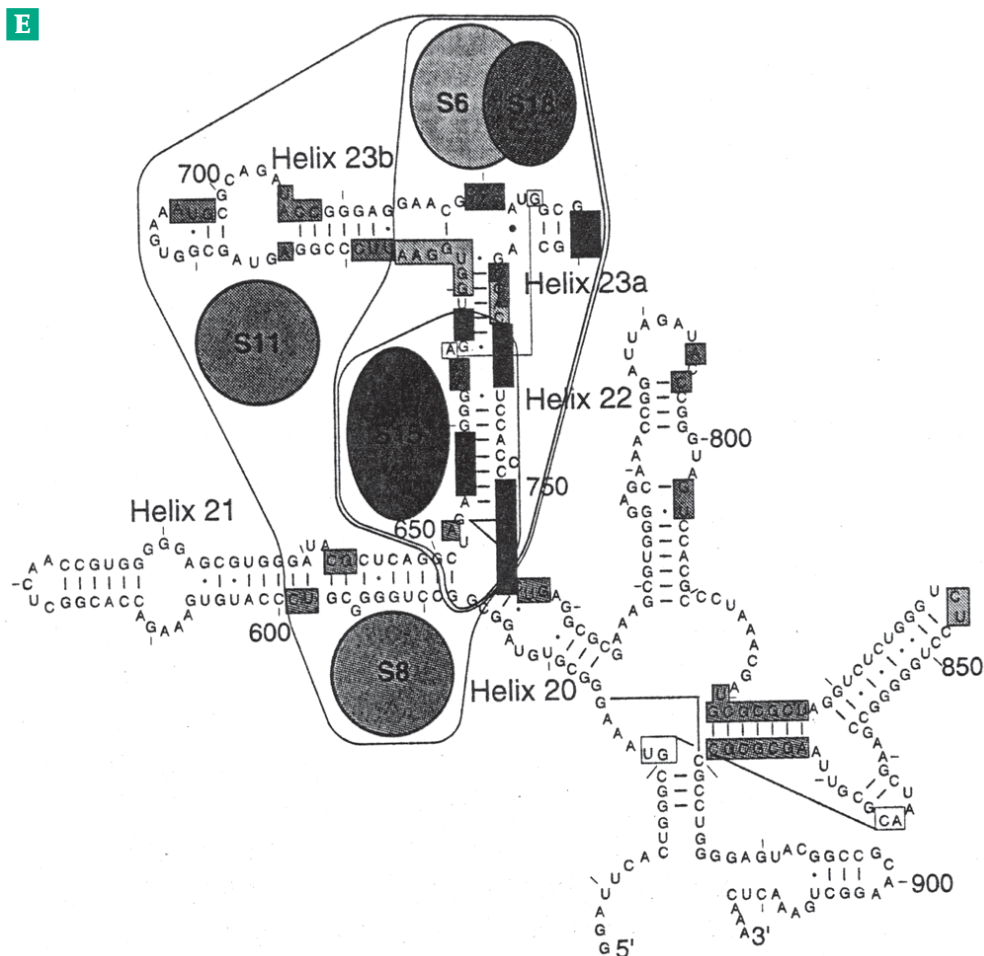
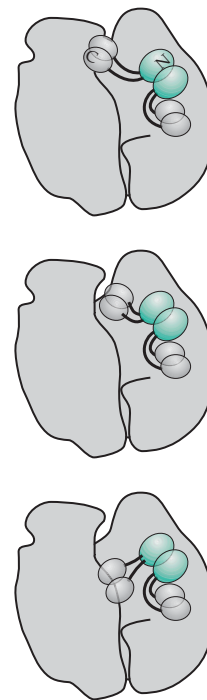


Figure 29-6 Some protein–RNA interactions within the ribosome. (A) A space-filling model of the 23S and 5S RNA with associated proteins from the ribosome of *Haloarcula marismortui*. The CCA ends of bound tRNA molecules in the A, P, and E sites are also included. The view is looking into the active site cleft. The proteins with e after the number are related to eukaryotic ribosomal proteins more closely than to those of *E. coli*.¹⁷ Courtesy of T. A. Steitz. (B) Three-dimensional structure of a 70S ribosome from *Thermus thermophilus*. The 30S subunit is to the right of the 50S subunit. Courtesy of Yusupov *et al.*^{33a} (C) Stereoscopic view of the helix 21 to helix 23b region of the 16S RNA with associated proteins S6 (upper left), S18 (upper center, front), and S15 (lower back) from *T. thermophilus*. Courtesy of Agalarov *et al.*³¹ (D) Simplified *in vitro* assembly map of the central domain of the 30S bacterial ribosome. Courtesy of Gloria Culver. (E) Contacts of proteins with the central (platform) domain of the 16S RNA component. The sequence shown is that of *Thermus thermophilus*. Courtesy of Agalarov *et al.* (F) Three drawings showing alternative location of the four copies of protein L7/L12. The N-terminal and C-terminal

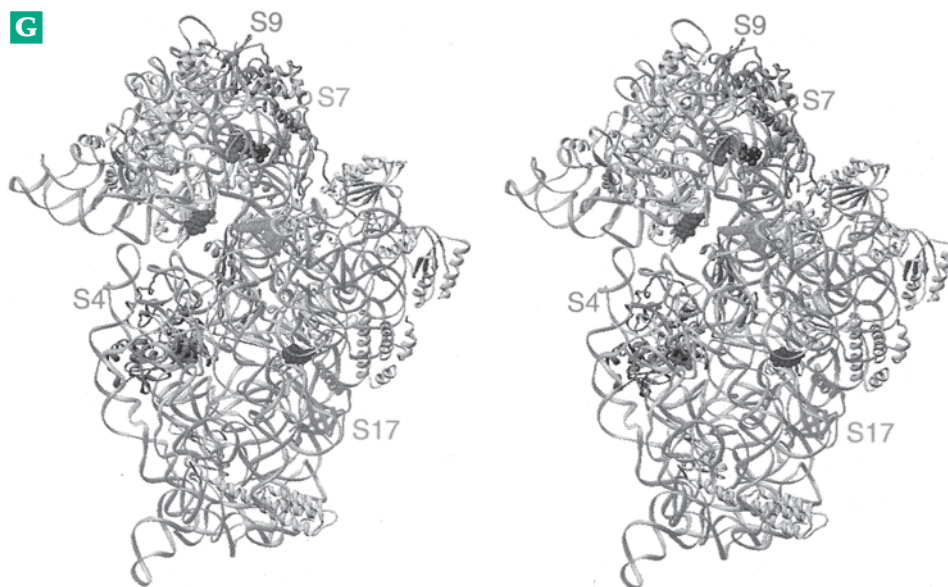
E



F



G



globular domains (labeled on one of the four molecules) are connected by a flexible region. One pair of L7/L12 molecules retains a fixed position toward the center of the 50S subunit but the C-terminal domains of

the other pair are seen to occupy three different positions. Courtesy of Montesano-Roditis *et al.*^{146a} (G) Stereoscopic view of the 30S subunit of the *T. thermophilus* ribosome with six positions at which tetracycline binds and one at which the peptide-like antibiotic **edeine** binds. From Pioletti *et al.*^{146b}

acts as a translational repressor of its own gene and of other genes of the spectinomycin-resistance operon (see Box 29-B). This operon encodes genes for ten ribosomal proteins, of both the large and small subunits.¹²⁶ S8 is a neighbor to proteins S2, S4, S5, S12, S15, and S17.

3'-Domain of 16S RNA. Domain III of 16S RNA binds proteins S2, S3, S7, S9, S10, S13, S14, and S19. Proteins S3 and S14 may be necessary for assembly of ribosomes but may no longer be needed once the 30S subunit has been correctly formed.¹³⁵ The largest of the *E. coli* ribosomal proteins is the 556-residue S1. It does not bind tightly and has sometimes been regarded as nonessential. However, mutations in the S1 gene can be lethal, and the protein seems to be essential for both initiation of translation and for elongation of polypeptide chains *in vivo*.¹³⁷ S1 behaves as an elongated molecule 22 nm in length¹³⁵ and is unusual in having an ~86-residue RNA-binding sequence repeated four times in the central and N-terminal regions. The protein possesses RNA-unwinding activity^{137,138} and may employ these four motifs in unwinding mRNA as it enters the ribosome.

23S and 5S RNAs. Reconstitution of the large ribosomal subunit reveals that proteins L3 and L24 act as assembly initiators.^{115,118} L1, L9, L20, and several other proteins (Table 29-2) also bind directly and independently to the 23S RNA. Assembly maps similar to that in Fig. 19-6A have been prepared for the 50S subunit.¹¹⁷

One of the most prominent features of the 50S subunit is the L1 protuberance, seen on the left side in Fig. 29-6A. This protuberance is formed almost entirely by protein L1, which is one of the largest ribosomal proteins. It binds to the 2105–2184 loop in domain V of the 23S RNA (see Fig. 29-14).¹³⁹ L1 has an important regulatory role in bacteria in which it represses translation of its own structural gene by binding to a region in its mRNA close to the Shine–Dalgarno sequence. The polygenic mRNA also carries the code for protein L11.¹³⁹ This is one of several examples of such autogenous regulation of translation of ribosomal proteins.^{139a} L1 also interacts in the ribosome with the 5S rRNA.¹⁴⁰ The 272-residue L2 also associates directly with 23S RNA and assists in ribosome assembly.¹⁴¹ Protein L2 is one of the structurally most highly conserved of the ribosomal proteins.^{46,142} It binds to the 1794–1865 region of domain IV of 23S RNA. Histidine 229 of this protein may play a functional role in the ribosome. The protein is elongated, and one end contacts 16S RNA.^{46,33a} Protein L9 binds to domain V of 23S RNA in the 2100–2190 region. It is an elongated molecule with two globular α/β domains separated by an α helix. This enables it to bind also to domain III, acting as a rigid strut.^{143–145}

On the right side of the 50S subunit, as viewed in Fig. 29-1, is the stalk, a pentameric protein complex consisting of two L7/L12 (*E. coli*) or (L12)₂ dimers bound to one molecule of L10.^{24,146–147} The stalk is not always seen in X-ray structures, e.g., in Fig. 29-6A, and appears to be flexible. In crosslinking experiments the N-terminal domains of L7/L12 can be linked to L10 and also to its neighbor, L11,^{82,148,149} which lies in the GTPase-activating center (Fig. 29-4F) at the base of the stalk. However, the C-terminal domains can be crosslinked to three distinctly different locations: to L11 on the platform surface, to L2 and L5 near the peptidyltransferase center, and to S2, S3, and S14 of the head and neck of the 30S subunit.¹⁴⁶ Domain I of 23S RNA, near the 5' end, binds to protein L20.¹⁵³

An independent and essential structural domain of the ribosome is formed around the 5S RNA.^{5,108,154–156} Proteins L5, L18, and L25, whose structure is similar to that of glutamyl-tRNA synthetase,¹⁵⁴ bind specifically to one loop of the 5S RNA.^{156a} Furthermore, the L5–L18–L25–5S RNA complex binds the oligonucleotide TCC. This suggests an interaction between the 5S RNA and the TΨC arm of a tRNA molecule bound to the ribosome. In addition, it has been observed that L18 + either L5 or L25 cause 5S RNA to bind to 23S RNA.

Eukaryotic ribosomal proteins. The functions of the 70–80 different eukaryotic ribosomal proteins are less well known than those of *E. coli*. In eukaryotes the assembly of ribosomes begins in the nucleus with binding of proteins to the individual ribosomal RNA precursors (Chapter 28).^{121,156b} Significant functional properties that are peculiar to eukaryotic ribosomal proteins include the following: S6 is the site of multiple phosphorylation reactions, which control initiation of protein synthesis.^{132,132a,132b} Mammalian S3 may function in the nucleus in DNA repair.¹³⁶ Eukaryotic proteins P0, P1, and P2 are homologous to *E. coli* stalk proteins L10, L7, and L12, respectively. Higher eukaryotes possess only one type of P1 and P2,¹⁵⁰ but yeast,^{150a,b} maize,¹⁵¹ and other species have multiple forms. An L7-related protein is also required for a nucleolar function in ribosomal protein synthesis, perhaps as a component of a snoRNP complex (Chapter 28).¹⁵² Rat liver L37 is involved in peptidyltransferase, but sequencing of the 111-residue protein reveals homology with *E. coli* L34 rather than with L16.¹⁵⁷ Proteins L14, L21, L24, L27, L29, and L30 bind to the 5.8S RNA of the large subunit of yeast ribosomes.¹⁵⁸

Yeast protein L30, which is not homologous to any bacterial protein, controls its own synthesis by a feedback inhibition at the mRNA splicing step. L30 binds to its own pre-mRNA near the 5' splice site, blocking completion of the spliceosome assembly (Chapter 28).¹⁵⁹

BOX 29-A THE DIPHTHERIA TOXIN AND OTHER RIBOSOME-INACTIVATING PROTEINS

Until a suitable vaccine was developed, an infection by *Corynebacterium diphtheriae* was one of the dread diseases of childhood. Despite the fact that the bacteria caused only superficial membranous lesions in the throat, the patient often died with evident damage to many organs. The cause is a potent heat-labile toxic protein,^{a-d} which the bacterium produces when infected by a temperate bacteriophage carrying the *tox* gene and when the inorganic iron of the surroundings has been largely depleted. Diphtheria toxin is a 535-residue protein with a minimum lethal dose (LD₅₀) of only 0.16 mg kg⁻¹ for the guinea pig. Tests in cell culture show that the toxin blocks incorporation of amino acids into proteins by inactivation of the eukaryotic elongation factor EF2, which is required for **translocation**, an essential step in protein synthesis in mammalian ribosomes. The toxin acts as an enzyme that transfers (with inversion at the ribose C1) an ADP-ribosyl group from NAD⁺ to a side-chain ring nitrogen of the single residue of **diphthamide** in EF2. This modified histidine is found in EF2 and, apparently, in no other protein.^d

The modified elongation factor reacts normally with GTP, but the complex so formed is unable to participate in translocation. A concentration of the toxin in the cytoplasm of 10⁻⁸ M is sufficient to promote the fatal reaction. The reaction with diphthamide parallels that of cholera toxin (Box 11-A).

The diphtheria toxin molecule^{e,f} consists of three domains, an N-terminal catalytic (C) domain (residues 1–193), a central, largely α -helical, trans-

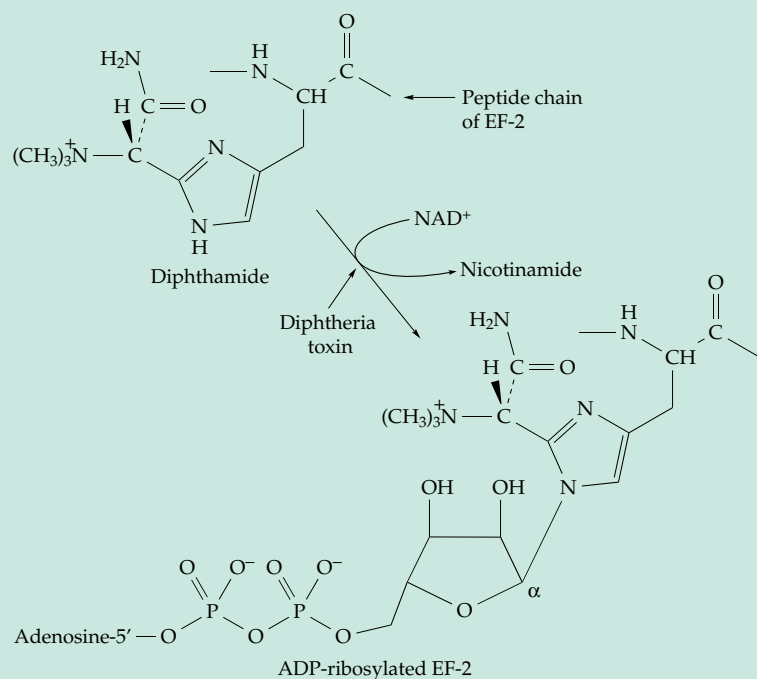
membrane (T) domain (residues 205–378), and the C-terminal receptor-binding (R) domain (residues 386–535). Before it enters a cell, the toxin molecule is “nicked” by protease activity between the catalytic and transmembrane domains, a step that is apparently necessary for binding to its receptor, which has been identified as a heparin-binding EGF-like growth factor precursor.^g The catalytic domain (also called the A fragment) enters the cell through endocytosis from coated pits. Additional proteolytic cleavage, as well as reduction of a disulfide linkage, are required for activity.^{f,h} After entering the cytosol the toxin fragment catalyzes inactivation of EF2.^f A single molecule can kill a cell. The larger 613-residue exotoxin A of *Pseudomonas aeruginosa* catalyzes the same reaction as does diphtheria toxin. It also consists of three domains: a large β -sheet-containing N-terminal region, a central α -helical domain, and a C-terminal domain. The last contains the ADP-ribosyltransferase active site.^{a,i}

What is the origin of the *tox* gene, and why is it carried by a virus? Cells do normally contain ADP-ribosyltransferases.^j The genes for such a protein may have become incorporated into a virus and, after a period of evolution, came to specify the toxic protein.

Another family of toxins attacks ribosomes in a very different way, cleaving ribosomal RNA at specific sites. One of the best known of these is the neurotoxin from *Shigella dysenteriae* (Shiga toxin). Like the cholera toxin (Box 11-A) it consists of

a single catalytic A subunit and a pentameric ring of B subunits,^k which binds to specific surface glycolipids.^l

Verotoxin, another poison from certain strains of *E. coli*, has a similar structure. Although they have very different effects and there is no detectable similarity in their amino acid sequences, the structures of the B pentamers of verotoxin and of the cholera toxin-like heat-labile enterotoxin of *E. coli* are similar.^{m,n} The A subunit of Shiga toxin hydrolyzes the N-glycosyl linkage of adenine to the ribose ring at position 4324 of 28S ribosomal RNA.^k A number of plants form very toxic lectins: **ricin** (from castor bean),^o **viscumin** (from mistletoe), **modeccin**, **abrin**,^p gelonin,^q and **volkensin**.^o The names are derived from the genus names of the plants. All appear to be glycoproteins consisting of two disulfide-linked chains, one of which is a lectin. The



BOX 29-A THE DIPHTHERIA TOXIN AND OTHER RIBOSOME-INACTIVATING PROTEINS (continued)

lectin subunits of ricin^r and of volkensin^s bind to galactose residues. The A chains are cytotoxins, which enter cells and, like Shiga toxin, inactivate 60S ribosomal subunits. The 267-residue A chain of ricin is similar to that of a pokeweed viral antigen^t and of Shiga toxin. It catalyzes the same reaction,^u the depurination of adenosine 4324. (The pokeweed toxin also catalyzes the corresponding reaction with A2660 of the *E. coli* 23S RNA.^v) Like the diphtheria toxin these toxic proteins bind to cell surface receptors, are taken up by endocytosis, and are transported through the Golgi to the endoplasmic reticulum. Their structures facilitate uptake but allow them to escape degradation in proteasomes.^w

Ricin is one of the most toxic substances known. A single molecule can inactivate over 1700 ribosomes per minute and kill the cell.^u With an LD₅₀ of only 1 µg / kg of body weight for many animals, ricin has been used as a poison by assassins. Of more importance is the attempt to couple ricin and related toxins to immunoglobulins to produce **immunotoxins** that will attack cancer cells (Box 31-A). A related goal is to design a potent inhibitor that could serve as an antidote.^x It is fortunate that most plant seeds do not contain toxins like ricin. Many plants, including such important food grains as wheat and barley, do contain ribosome-inactivating proteins similar to the A chain of ricin. However, the plants lack the B (lectin) subunits and do not enter animal cells.

A group of unusual fungal ribonucleases, which includes **α-sarcin** and **restrictocin**, are produced by *Aspergillus*. The cytotoxic nucleases enter animal cells, where they cut the 28S RNA of ribosomes, specifically on the 3' side of guanosine 4325 in the sarcin / ricin domain (see Fig. 29-4), thereby blocking protein synthesis.^{u,y,z} *Staphylococcus aureus* produces a 22-kDa toxic protein thought to be responsible for **toxic shock syndrome**.^{aa} Another toxic ribonuclease is **colicin E3** (Box 8-D), which cuts the 16S RNA of *E. coli* after nucleotide 1493 (see Fig. 29-1A).^{bb} Colicin D stops protein synthesis by cleavage of four isoaccepting tRNA^{Arg} molecules between positions 38 and 39 in the anticodon loop.^{cc}

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4. Locating Active Sites in Ribosomes

In early studies, antibodies against haptens covalently linked to ends of the 16S RNA were used to locate the 3' end of 16S RNA on the upper “platform” of the 30S subunit (Fig. 29-1A).^{5,160} The 5' terminus was found in the lower body. Two N⁶,N⁶-dimethyl-

adenosines occur at positions 1518 and 1519, about 25 residues from the 3' end of the 16S RNA (Fig. 29-2). Antibodies were used to locate this position on the lower platform. Likewise, the m⁷G at position 526 lies in the “neck” as shown in Fig. 29-1A.¹⁶¹ Taking into account known protein–RNA interactions, domain II of the 16S RNA was located in the “platform” on the

upper left side of the “body” (viewed from the “outside” as in Fig. 29-1C), while domain III is in the “head.” Recent structural studies have confirmed these biochemical localizations.^{19,29,33,33a}

The tRNA-binding sites. During protein synthesis tRNA is bound sequentially in at least three places located between the 30S and 50S subunits. These are known as the **A (aminoacyl)**, **P (peptidyl)**, and **E (exit)** sites. The latter binds deacylated tRNA before it is released from the ribosome.^{162,163} Because tRNA is such a large molecule, these sites have subsites in both 30S and 50S ribosomal subunits. When in the P site, a tRNA has its anticodon held firmly and base-paired with a codon in the mRNA in the decoding site of the 30S subunit. The CCA 3'-end with its attached peptidyl chain lies in the 50S subunit at the peptidyltransferase center. A “charged” aminoacyl-tRNA enters the A site, which is close to the 5S RNA in the central protuberance of the large ribosomal subunit, with its anticodon in the decoding site and its aminoacyl group at the peptidyltransferase site.

One end of the P site must be close to the 3' end of the 16S RNA near the two m²6A residues (Fig. 29-1A). This conclusion, which was based on photochemical linking of a hypermodified base at position 34 (see Fig. 29-7) in tRNA^{Val} with C-1400 of 16S RNA by cyclobutane dimer formation (Eq. 23-26),¹⁶⁴ has been confirmed by structural studies.^{29,33,33d} Investigation of tRNA binding, effects of mutations in ribosomal RNA, and effects of antibiotics pointed to locations of the P and A sites in both ribosomal subunits. These have been located precisely by crystallography. See Fig. 29-1FJ; 29-2C. Residue 6530 together with nucleotides 921–927, 1390–1394, and 1491–1505 of 16S RNA participate in forming the form A and P sites in the decoding center.^{33c,d,164a,b,378} The two adjacent adenine rings of A1492 and A1493 swing out from helix 44 (Fig. 29-2; see also Fig. 29-14) to form a major part of the A site. In the 50S subunit the adjacent cytosines C74 and C75 of the CCA 3'-ends of the tRNAs in the A and P sites interact respectively with G2553 of the A loop and G2252 of the P loop (Fig. 29-14B,E).^{164b-e} Tetracycline (Fig. 22-7) also binds into the A site (see Box 29-B). It can be photochemically crosslinked to proteins S18 and S4.¹⁶⁷

The peptidyltransferase site. The position was located by binding of derivatives of the antibiotic **puromycin** (Fig. 29-13). An arylazide derivative of puromycin was photochemically linked (Eq. 23-27) to proteins L23, L18/22, and L15; immunoelectron microscopy, using antibodies to the N⁶-dimethyladenosine of puromycin,^{165,166} located the binding site adjacent to the central protuberance between the 50S subunit and 30S subunit near S14.⁵ 4-Thio-dT-p-C-p-puromycin was photochemically crosslinked to G2553

of the peptidyltransferase A site (see Fig. 29-14). X-ray data provided a precise structure of the peptidyltransferase site (see pp. 1702–1704).^{166a} Studies of mutant ribosomes together with affinity labeling and cross-linking experiments pointed to the **peptidyltransferase loop** marked on Fig. 29-4 and further illustrated in Fig. 29-14.^{164a,167a,b}

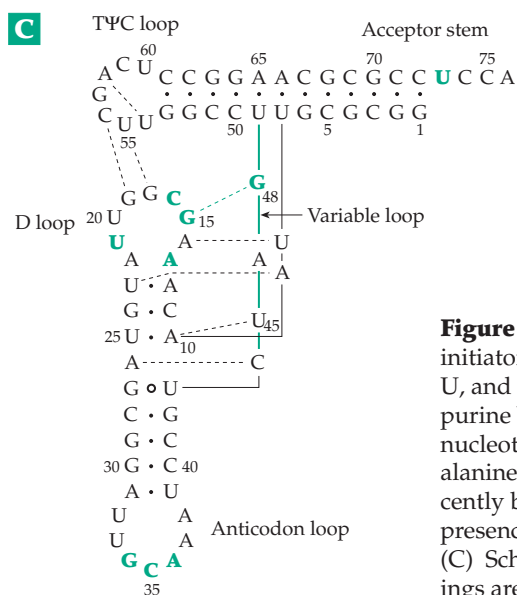
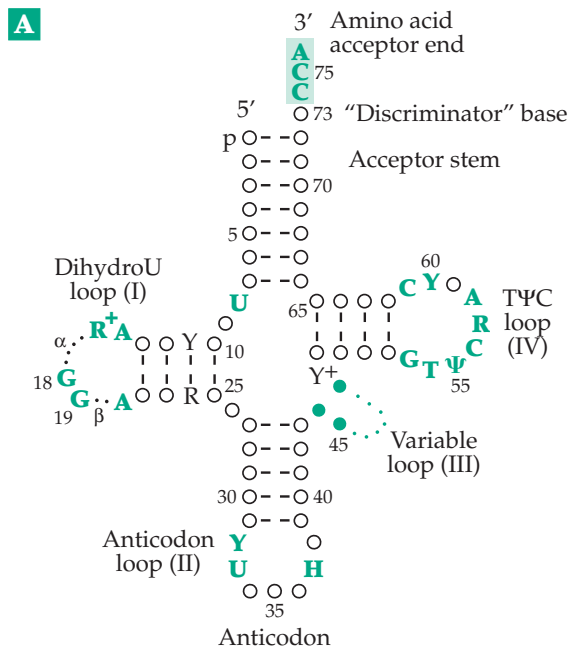
The GTPase-activating center. Also shown in Fig. 29-1C is a site for binding of the elongation factor EF-G (Section C,2). This was located, in part, because the antibiotic **thiostrepton** prevents EF-G from binding to the ribosome. Thiostrepton binds to a complex of protein L11 and a 61-fragment of the 23S RNA (positions 1052–1112; see Fig. 29-4F).⁸⁰ Another elongation factor, EF-Tu, also binds at the same site or adjacent to the EF-G site by the head of the small subunit.¹⁶⁸ An additional location of interest is the **polypeptide exit tunnel**, which brings the growing protein chain from the peptidyltransferase site out of the ribosome (Fig. 29-1I).^{5,6,33f}

B. Transfer RNAs

The small 4S tRNA molecules have masses of ~26 kDa and consist of 75 ± 5 nucleotides (Figs. 5-30, 5-31, and 29-6). The basic structures are similar in bacteria and eukaryotic cells. The need for “adapters,” to carry amino acids to the proper positions along the mRNA template, had been predicted prior to the discovery of tRNA.^{169,170} It had been expected that there would be a base sequence constituting an **anticodon**, which would fit against the proper codon at some binding site on the protein-synthesizing machinery. This is just what tRNA molecules do, but their chemistry contained many surprises.

1. Structures of Transfer RNAs

The first surprise was that these molecules are much longer than seems necessary for the formation of adapters. In addition, 10–20% of their bases are modified greatly from their original form.¹⁷¹ Another surprise was that the anticodons are not all made up of “standard” bases. Thus, hypoxanthine (whose nucleoside is inosine) occurs in some anticodons. Conventional “cloverleaf” representations of tRNA, which display their secondary structures, are shown in Figs. 5-30 and 29-7. However, the molecules usually have an L shape rather than a cloverleaf form (Figs. 5-31 and 29-6),¹⁷² and the L form is essential for functioning in protein synthesis as indicated by X-ray and other data.¹⁷³ Three-dimensional structures, now determined for several different tRNAs,^{174,175} are all very similar. Structures in solution are also thought to be



similar for the various tRNA molecules.^{176,177} One of the four hydrogen-bonded "stems" of a tRNA in the cloverleaf form terminates in the universally conserved CCA-3' **amino acid acceptor end** (Fig. 29-7), which can carry an esterified amino acid generated as in Eq. 29-1, steps *a*, *b*. The other three stems terminate in loops, which usually contain a large number of modified bases. The modifications may serve to optimize the interaction of the tRNA with other components of the protein-synthesizing machinery.¹⁷⁸ The **dihydroU loop** (loop I) contains 5,6-dihydrouridine in various amounts and in varying positions. The **anti-codon loop** (loop II) always contains the anticodon directly opposite the amino acid acceptor end in the cloverleaf drawing. On the 5' side of the anticodon at position 33 there is almost always a U (shaded in Fig. 29-7A) preceded by another pyrimidine. A hydrogen bond from the N3 proton of U-33 and a phosphate oxygen of residue 36 stabilize the U-turn that precedes the anticodon triplet (Fig. 5-31).¹⁷⁷ Next to the 3' side of the anticodon there is usually a **hypermodified** base, such as N⁶-(Δ²-isopentenyl) adenosine (Fig. 5-33) or a more complex derivative.¹⁷⁸ The **variable loop** (loop III) can range between 5 and 21 nt in length.^{177,181} The **TΨC loop** (loop IV) contains the specific nucleotide sequence for which the loop is named.

Cloverleaf and L forms. Interconversion between the cloverleaf and L forms of tRNA molecules can be pictured as in Eq. 29-2. Notice that in the L

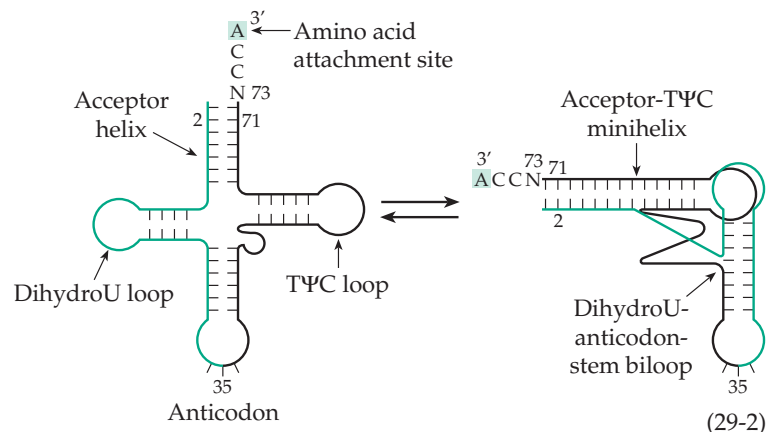
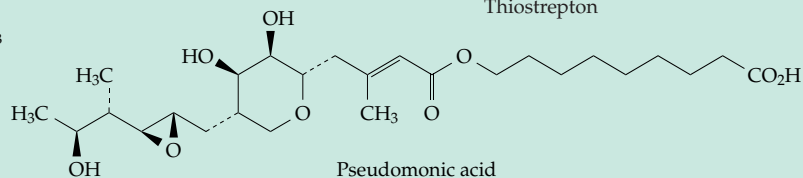
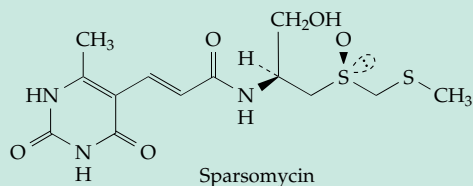
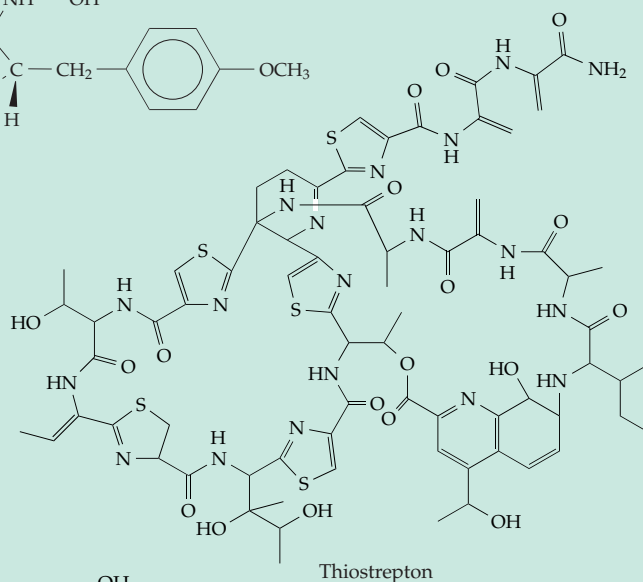
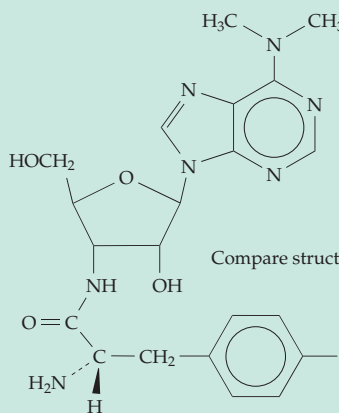
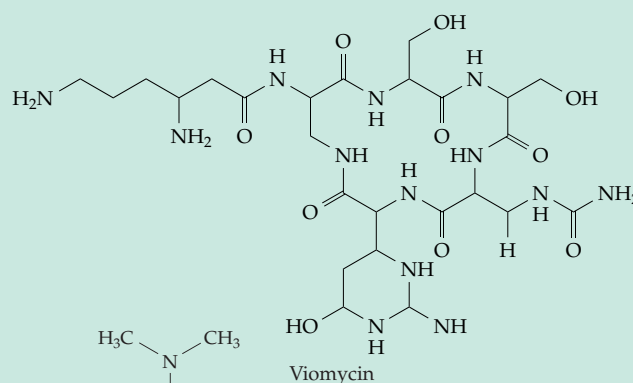
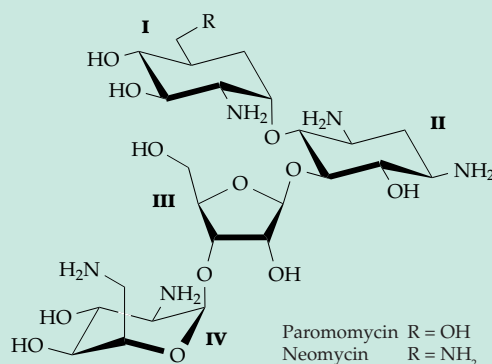


Figure 29-7 (A) Generalized cloverleaf diagram of all tRNA sequences except for initiator tRNAs numbered as in yeast tRNA^{Phe} (Fig. 5-30). Invariant bases: A, C, G, T, U, and ψ; semivariant bases: Y (pyrimidine base), R (purine base), H (hypermodified purine base). The dotted regions (α, β, variable loop) contain different numbers of nucleotides in various tRNA sequences. See Rich.¹⁷⁹ (B) L form of the yeast phenylalanine-specific tRNA^{Phe}. The structure is the same as that in Fig. 5-31 but has recently been redetermined at a resolution of 0.20 nm.¹⁷⁵ The new data revealed the presence of ten bound Mg²⁺ ions (green circles) as well as bound spermine (green). (C) Schematic representation of L form of *E. coli* tRNA^{Cys}. Some tertiary base pairings are indicated by dashed lines. No modified bases are shown. See Hou *et al.*¹⁸⁰

BOX 29-B ANTIBIOTICS THAT INHIBIT PROTEIN SYNTHESIS

Many of our most effective antibiotics act by blocking protein synthesis on ribosomes.^{a-de} The usefulness of many of these remarkable drugs in human medicine depends upon the fact that they inhibit protein synthesis on bacterial 70S ribosomes but do not seriously affect eukaryotic ribosomes. Antibiotics act at a variety of sites involving both ribosomal proteins and rRNA. Some of the resulting points of inhibition of the ribosomal elongation cycle^e are marked on Fig. 29-12. Binding sites for many antibiotics have been located by footprinting experiments, by studies of mutants, and more recently by X-ray crystallography. Exposure of bacteria to antibiotics often gives rise to antibiotic-resistant mutants. In some of these mutants a specific ribosomal protein has been altered, but in others a specific RNA base has been changed.

One of the first antibiotics to be studied was **puromycin** (Fig. 29-13), which binds to the 50S subunit and causes premature termination of peptide synthesis. A glance at its structure reveals how it can do this. It resembles in fine detail the 3' end of a tRNA molecule bearing an aminoacyl group. However, it is not an aminoacyl group, and once the growing peptide chain has been transferred onto the puromycin, further chain elongation is impossible. It was shown in 1975 that puromycin could be crosslinked to several different proteins in the ribosome by ultraviolet irradiation.^{f,g} L23 of the 50S subunit and S14 of the 30S subunit were labeled most heavily. More recently puromycin, acting on 70S ribosomes, has been shown to label protein S7 and several large subunit proteins near the peptidyltransferase center in the central domain V of the 23S RNA.^h A puromycin-derived transition state inhibitor has permitted precise identification of the peptidyltransferase site (Fig. 29-13).^{i,j} In earlier work immunoelectron microscopy on *N*-bromoacetylpuromycin-labeled 50S subunits had located the site marked Pm in Fig. 29-1A. However, this is quite far from the recently determined location of the peptidyltransferase center,^k



BOX 29-B ANTIBIOTICS THAT INHIBIT PROTEIN SYNTHESIS (continued)

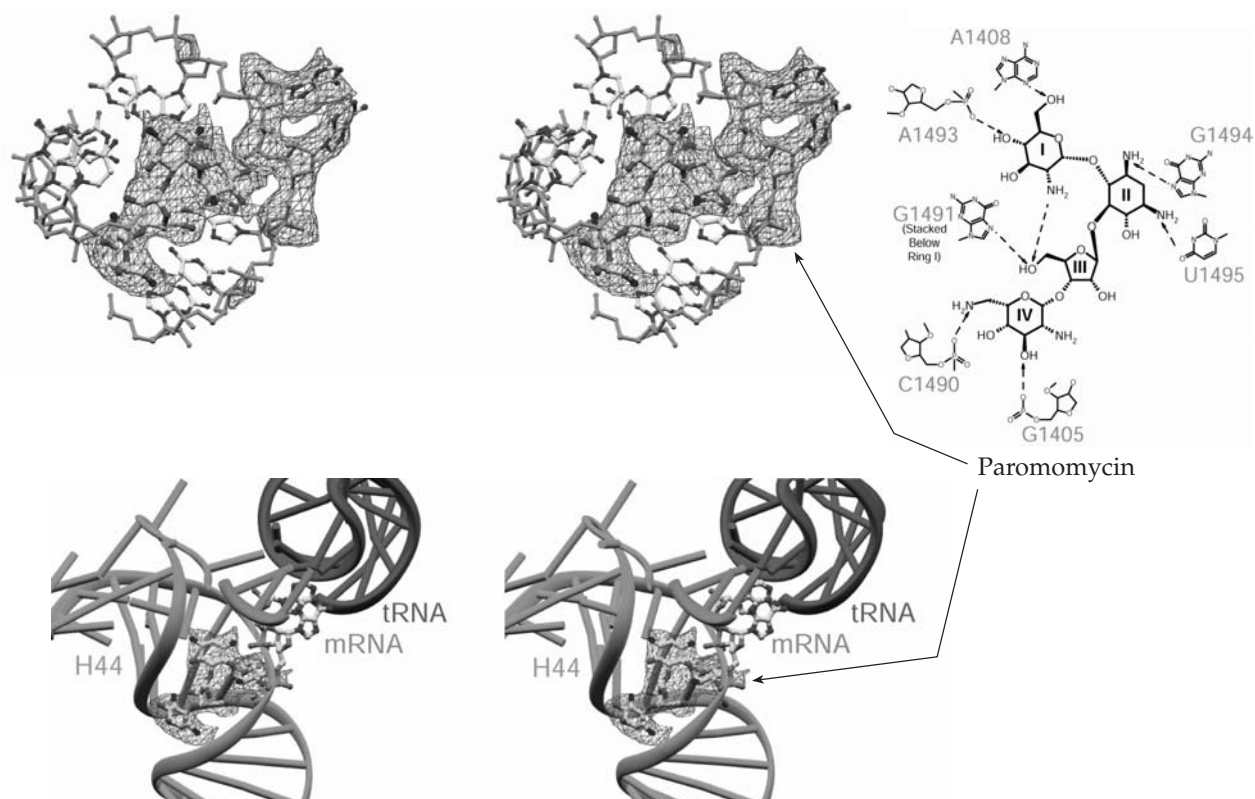
illustrating the difficulty in use of active-site labeling techniques.

A large number of other antibiotics also bind in the vicinity of the peptidyltransferase center (Fig. 29-14).^{l,m} Among them are macrolide antibiotics such as erythromycin (Fig. 21-1)ⁿ and spiramycin,^{o,p} chloramphenicol (Fig. 25-10), griseofulvin (Fig. 21-10), streptogramins,^q oxazolidinones such as linezolid,^r sparsomycin,^s and lincomycin. Erythromycin has been very useful in locating the peptidyltransferase center. For example, 23S RNA mutant G2057A, in which A has replaced the normal G, and mutants G2058G and G2058U are resistant to erythromycin.^{l,n,t} Mutant C2518U in *Halobacterium halobium* 23S RNA (C2499 in *E. coli*) is among mutants resistant to sparsomycin.^s Chloramphenicol not only blocks the peptidyltransferase but also causes an accumulation of the compound ppGpp (p. 1715).

The aminoglycoside antibiotics **streptomycin** (Box 20-B),^{c,d,u,v} the neomycins,^w **paromomycin** (see drawing below),^{c,x-z} gentamycin,^{aa} and kanamycin have one structural unit in common. They often bind to 16S ribosomal RNA in the decoding center.

However, they bind in distinctly different ways. Streptomycin causes ribosomes to misread the genetic code^{bb,cc} primarily at the first base of a codon. Thus, when poly(U) serves as a messenger RNA, the expected polyphenylalanine product contains 40% isoleucine. When a bacterial population is subjected to the action of any antibiotic, a few mutants are able to grow and survive in the presence of the antibiotic. Streptomycin-resistant mutants of *E. coli* arise at the very low frequency of $\sim 10^{-12}$. One of the genes affected (*rpsL*) was mapped at 72 min. Subsequently, it was shown that streptomycin binds to ribosomal protein S12, and that *rpsL* is the gene for this protein. Mutations in the universally conserved 2660 loop of 23S RNA in the sarcin/ricin domain lead to blockage of the elongation cycle. Bacteria containing both a G2661C mutation in their 23S RNA and also a streptomycin resistance mutation in protein S12 lose efficiency in the action of EF-Tu and die. However, they survive in the presence of streptomycin.^{dd}

Streptomycin can also be chemically crosslinked to 16S RNA,^{ee} and several aminoglycoside antibiotics including streptomycin and spectinomycin bind



Figures are from Carter *et al.*^c

BOX 29-B (continued)

to and, in footprinting experiments, protect specific nucleotides in 16S RNA.^{ff} Streptomycin binds tightly to the upper part of helix 44 of bacterial 16S RNA (Fig. 29-2). This part of the helix is in the region that binds messenger RNA. It also contains parts of the A and P sites of the decoding region of the 30S ribosomal subunit. The same antibiotic binds less tightly to the 915 region in the center of the 16S RNA.^{gg,hh} As mentioned above, some streptomycin-resistant mutants become dependent upon the antibiotic and will not grow in its absence. This streptomycin dependence sometimes results from modification in ribosomal protein S4, but the dependence can be suppressed by specific mutations in S5.ⁱⁱ It is clear that a single point mutation altering one amino acid is all that is necessary to enormously change the sensitivity of a living organism to a particular toxin, or even to make the organism dependent upon that toxin. Paromomycin also binds to the upper end of helix 44 in the major groove of the RNA (see figure in this box) and close to the streptomycin-binding site.^{c,z,jj} Binding distorts the structure of the bulge loop containing adenosines A1492 and A1493, which are markers for the A site of the decoding region. Messenger RNA is also bound at this site as shown in the accompanying figure. Gentamycin also binds in the A site.^{aa}

Hygromycin B also binds at the very top of helix 44 blocking the translocation step in the ribosomal cycle.^{kk} **Spectinomycin** binds not only to RNA but also to protein S5, as indicated by analysis of resistant mutants. The S5 structural gene *spcA* maps at 64 min, a position in a ribosomal protein operon of the *E. coli* chromosome. The 16S RNA binding site at one end of helix 34 (with protection of G1064 and C1192) is adjacent to S5 as shown by X-ray structural analysis and directed hydroxyl radical probing (see Fig. 29-2).^c The antibiotic also interferes with the translocation step of polypeptide elongation. **Kasugamycin** inhibits the binding of fMet-tRNA (initiation). In this case, resistant mutants appear in which it is not a protein subunit that has been modified but the 16S RNA. In resistant strains there is less methylation of adenosines 1518 and 1519 (Fig. 29-2) than in normal strains.^{ll}

The **tetracyclines** (Fig. 21-10) inhibit the binding of aminoacyl-tRNA at the A site in the 30S ribosomal subunit.^{kk} However, this doesn't appear to be a direct effect. Tetracyclines bind to the 16S RNA at two sites. A major site is on helix 34 near the spectinomycin site in the platform region. A second site is on helix 27, the switch helix, which plays a direct role in translocation (see Eq. 29-9).^{kk} Although the basis of the inhibition is not clear,

there are distinct differences in binding to the 16S bacterial and 18S eukaryotic RNAs that explain the high specificity of the antibiotic toward bacteria.

Another site of antibiotic action is the GTPase-activating center. This center contains a double hairpin structure in the 23S RNA, which binds to protein L11 and the L10•(L12)₄ stalk complex. Several proteins, including initiation factor IF1 and the elongation factors EF-Tu, EF-G (bacteria)/EF-2 (eukaryotes), bind to this part of the 50S ribosome (Fig. 29-4). The thiopeptide antibiotics **thiostrepton**, **micrococcin**, and siomycin^{t,mm,nn} also bind in this region. Thiostrepton acts by preventing association with the ribosome of an incoming aminoacyl-tRNA as the EF-Tu•GTP complex. Its binding site is primarily in the 23S RNA, but it probably interferes with peptide elongation by interfering with a conformational change in protein L11.^t A related cyclic peptide (GE2270A) binds to EF-Tu•GDP competing for binding of an aminoacyl-tRNA and blocking the GDP-GTP exchange.ⁿⁿ Certain mutations in the EF-Tu protein confer resistance to this antibiotic.^{oo} In a similar way kirromycin prevents release of EF-Tu from the ribosome after GTP hydrolysis.^{nn,pp,qq}

The binding site of initiation factor IF1 involves both the 30S and 50S ribosomal subunits. The large oligosaccharide antibiotic **evenimicin** protects a specific set of nucleotides in two loops near the peptidyltransferase center (Fig. 29-14).^{rr} Erythromycin,^b other macrolide antibiotics, cycloheximide (Fig. 21-10), and fusidic acid (p. 1266) all prevent translocation by stabilizing the pre-translocation complex.^{ss,tt} Fusidic acid may bind to EF-G on the ribosome, preventing an essential conformational change in this G protein.^{tt} Fusidic acid also inhibits accumulation of ppGpp. Figure 29-14 shows the locations of some mutations in *E. coli* 23S rRNA that confer resistance to erythromycins and chloramphenicol. Notice that both domains II and V are involved.ⁿ **Pactamycin** binds to helices 23b and 24a, a binding site for initiation factor IF3.^{kk}

Many antibiotics, which inhibit protein synthesis, do not bind to ribosomes but block any of a variety of vital chemical processes needed for growth. Among them are **pseudomonic acid**, which inhibits isoleucyl-tRNA synthetase from many gram-positive bacteria.^{uu,vv} **Rapamycin**, best known as an immunosuppressant (Box 9-F), inhibits phosphoinositide-3-kinase and also phosphorylation of the cap-binding protein 4G, a component of the eukaryotic initiation factor complex (Fig. 29-11).^{ww} The bacterial enzyme peptide deformylase, which is absent from the human body, has been suggested as a target for design of synthetic antibiotics.^{xx}

BOX 29-B ANTIBIOTICS THAT INHIBIT PROTEIN SYNTHESIS (continued)

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form the acceptor stem and the T ψ C arm form a single **acceptor-T ψ C-minihelix**, while the other two domains fold together to create an **anticodon-dihydrouridine** stem loop.¹⁷² New tertiary interactions, some of which are indicated in Fig. 29-7C, are formed. Mitochondrial tRNAs of metazoa often lack some elements of the cloverleaf. An extreme example is the bovine mtRNA^{Ser}, which recognizes AGY codons and completely lacks the dihydrouridine loop. This fact suggests that the L shape of tRNAs cannot be

completely invariant.^{174,176} As is shown in Fig. 29-7B, divalent metal ions such as Mg²⁺ are bound at discrete sites in tRNA molecules.¹⁷¹ The tertiary interactions in the “core” of the L form contains several stacked layers of base pairs and triplets (Fig. 29-7C). The top layer is usually the single base 59; below it in succession are the 15:48 pair (see Fig. 29-7), the 21:8:14 triplet, the 13:22 pair (see also p. 231), and then base pairs present in the dihydroU loop. Considerable variation is observed among the different tRNAs.^{182,183} The

structural features of this core may also be utilized for recognition by aminoacyl tRNA synthetases.

Initiator tRNAs. While the T ψ C sequence has been found in all bacterial and most eukaryotic tRNAs examined, it is replaced by UCG in eukaryotic initiator tRNAs. In these tRNAs the preceding two nucleotides, beginning in the stem of loop IV, are also conserved; the complete conserved sequence being GAUCG.¹⁸⁴ Other characteristics of initiator tRNAs are the absence of base-pairing between residues 1, and 72, and the presence of C rather than G at position 1, A rather than G at position 72, and CCU in place of the two dihydroU residues in loop I.¹⁸⁵ Initiator tRNAs of chloroplasts resemble those of bacteria,¹⁸⁶ whereas archaeobacteria have their own unique peculiarities.¹⁸⁷ These include the presence of a hypermodified base known as **archaeosine** (p. 1456) in position 15 of the dihydroU loop.^{188,189}

2. Pairing of Codon and Anticodon

Accurate protein synthesis depends upon both correct charging of the tRNAs and correct recognition by an anticodon in the tRNA of the complementary codon in the mRNA. A surprise was the discovery of inosine (I) in anticodons of yeast tRNA (but not in most *E. coli* tRNAs). Another unexpected finding was that fewer than 61 kinds of tRNA exist in a given cell (61 = 64 codons minus three stop codons). Consideration of these matters led Crick, in 1966, to propose the **wobble hypothesis**.¹⁹⁰ According to this proposal the first two bases at the 5' end of the codon (and at the 3' end of the anticodon) must pair in the same ways as do the bases in DNA. However, the third base pair (3' end of the codon and 5' end of the anticodon) is under a less severe steric restriction. That is, there may be some “wobble.” Crick suggested the accompanying rule for pairing of the third base. All of the observed

5'-Base in anticodon	Paired 3'-base in codon
G	C or U
C	G
A	U
U	A or G
I	C, A, or U

deviations from the AU, CG pairing of a Watson–Crick helix can be explained in this way. An anticodon with G at the 5' end can pair with codons with either C or U at their 3' end. Anticodons with C or A at the 5' end

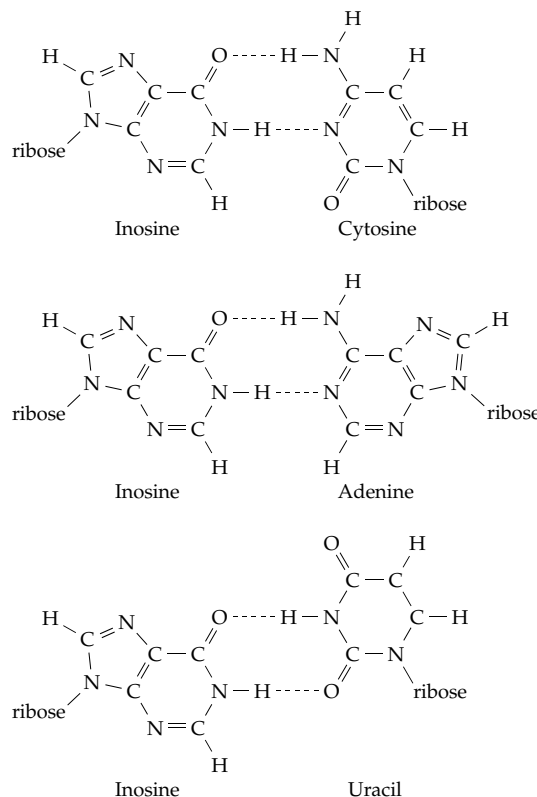


Figure 29-8 Pairing of inosine with cytosine (a Watson–Crick pair) and of inosine with adenine and uracil (wobble pairs).

pair strictly. Anticodons ending with U at the 5' end can pair with codons containing either A or G in the 3' position. Anticodons with I in the 5' position can recognize codons with any of the three bases in the third position. Comparison with Table 5-5 makes it immediately clear why fewer than 61 anticodons are needed. Many codons represent the same amino acid, and frequently the nature of the base in the 3' position of the codon is immaterial to the meaning of that codon. Thus, there is an economy in using less than the full array of anticodons. Crick showed that his proposal was chemically feasible if the spatial relationships for the wobble pair were allowed to vary from the usual ones in Watson–Crick base pairs. This is illustrated in Fig. 29-8 for binding of inosine to C (a normal Watson–Crick base pair) and to A and to U. Although the word wobble does not convey an exact meaning, the hypothesis has predicted many things correctly. For example, only three tRNAs are required to recognize the six serine codons. In fact, only three are found in *E. coli*.

The matter is made more complex by the fact that U34 in the first (5') anticodon position of tRNA is usually modified postranscriptionally.^{190a} For

example, it is usually converted to the 2-thio-5-CH₃ or 2-thio-5-CH₂-NH₂⁺-CH₃ derivative in anticodons recognizing A in the 3' position of a codon. For recognition of other bases the 5-OH, -OCH₃, or -OCH₂COO⁻ derivative is usually present. Yokoyama *et al.* attributed the selectivity to the stabilization of the C2' endo form of the ring in the former group.¹⁹¹

3. Aminoacylation of tRNAs

Discrimination between some pairs of tRNAs depends entirely on the anticodon sequence. For example, tRNA^{Met} contains the anticodon CAU. That for a minor tRNA^{Ile} is the same except that the cytosine has been posttranscriptionally modified by covalent linkage of a molecule of lysine via its ε-amino group to C2 of the cytosine. The latter base (**lysidine**) is correctly recognized by *E. coli* isoleucyl-tRNA synthetase; but, if the cytosine is unmodified, it is aminoacylated by methionyl-tRNA synthetase.¹⁹² In most instances the acceptor specificity, or **tRNA identity**, is not determined solely by the anticodon sequence. Thus, when a methionine initiator tRNA was modified to contain a tryptophan anticodon, it was only partially charged with tryptophan *in vivo*. However, when A73 of the methionine tRNA was also converted to G73, only tryptophan was inserted.¹⁹³ Nucleotide 73 (Fig. 29-7) is sometimes called the **discriminator nucleotide**.¹⁹⁴⁻¹⁹⁶ It is A in methionine and leucine tRNAs,¹⁹⁷ G in tryptophan tRNAs, and C in histidyl RNAs.¹⁹⁸ The tRNA features needed to establish its identity are sometimes referred to as its **identity-determinant set**.^{196,198a,b} This includes the anticodon and other features needed for recognition by the aminoacyl-tRNA synthetases that “charge” the tRNAs with aminoacyl groups.¹⁹⁹⁻²⁰⁴ For example, for *E. coli* tRNA^{Val} the recognition determinants are A35 and C36 of the anticodon, A73, G20, G45, and a regular A-RNA acceptor helix.²⁰⁵ All known mature tRNAs contain a 3'-CCA end on which the aminoacylation occurs. Nevertheless, alterations in this sequence still allow correct aminoacylation of some tRNAs.²⁰⁶

The aminoacyl-tRNA synthetases (amino acid: tRNA ligases) join amino acids to their appropriate transfer RNA molecules for protein synthesis. They have the very important task of selecting both a specific amino acid and a specific tRNA and joining them according to Eq. 29-1.^{175,195,207} These reactions represent the first step in the decoding of mRNAs. Organisms usually contain one aminoacyl-tRNA synthetase for each of the 20 amino acids. Each synthetase must select a specific amino acid and a correct tRNA for that amino acid. The same enzyme transfers an activated amino acid to all of the **isoacceptor tRNAs** specific for a given amino acid. Some aminoacyl-tRNA synthetases attach the aminoacyl group to the 2'-OH of

the tRNA substrate, some to the 3'-OH. The chemical mechanism is the same in both cases.

Structures. Aminoacyl-tRNA synthetases vary in size, the subunit masses ranging from 37- to 110-kDa (329–951 residues). There are monomeric species, dimers, tetramers, and (αβ)₂ mixed tetramers.^{207,208} Sequence comparisons, together with X-ray structural investigations, have shown that the enzymes can be classified into two groups, each containing ten enzymes.^{207,209,209-209b} **Class I aminoacyl-tRNA synthetases** share two consensus sequence motifs: HIGH and KMSKS. Their ATP-binding active sites are in a Rossman fold nucleotide-binding domain (Fig. 2-13).²¹⁰ The KMSKS sequence parallels the Walker sequence found in various nucleotide-binding proteins including ATP synthase.²¹¹⁻²¹³ The actual sequences vary considerably, e.g., the KMSKS sequence for a tyrosyl-tRNA synthetase is actually KFGKT.²¹¹

Class II aminoacyl-tRNA synthetases contain a different set of three “signature sequences,” two of which form an ATP-binding catalytic domain. The active site structure is built on an antiparallel β sheet and is surrounded by two helices (Fig. 29-9). Each class contains subgroups with inserted loops that form other domains. In the following tabulation the reference numbers refer to three-dimensional structural studies.

Class I

Glu,²⁰⁹ Gln,²¹⁸⁻²²⁰ Arg,²²¹
Tyr,²²² Trp,²¹²
Ile,²²³ Leu,²²⁴ Val, Cys, Met^{210,225,226}

Class II

His,²²⁷⁻²²⁹ Pro, Ser,^{230,231} Thr
Asp,^{232,233} Asn,²³⁴ Lys,^{217,235}
Phe,²³⁶ Ala, Gly,^{204,237}

The 37-kDa 334-residue subunits of the dimeric type I tryptophanyl-tRNA synthetase²³⁸ are the smallest known; the largest bacterial synthetase is an alanine-specific type II tetramer with 95-kDa 875-residue subunits.²³⁹ Gene deletions show that a much smaller core, comparable in size to that of the tryptophanyl-tRNA synthetase, is needed for amino acid activation. The synthetases share little sequence homology except for a short 11-residue part of the adenylate binding site near the N terminus.^{240,241} Some of the synthetases contain bound zinc ions.^{225,242}

Recognition of cognate tRNAs. Many attempts have been made to learn what part or parts of tRNA molecules are involved in recognition by aminoacyl-tRNA synthetases. Nucleotide sequences of isoacceptor tRNAs have been compared. Chemically modified and fragmented tRNA molecules have been studied, and many mutant tRNAs have been made. These

have often been mutants of suppressor tRNAs that place specific amino acids such as phenylalanine or alanine into a peptide at a termination codon, often the termination codon UAG (see Section C4). An alternative approach is to synthesize DNA templates, which can be transcribed *in vitro* by phage T7 RNA polymerase to give mutant tRNAs,²⁴³ or to make such tRNAs by solid-phase chemical synthesis.²⁴⁴ Although these contain no modified bases, they serve as substrates for the aminoacyl-tRNA synthetases.

The results of these efforts show that no method of tRNA recognition is universal.^{244a} In some cases, e.g., for methionine- or valine-specific tRNAs, the synthetase does not aminoacylate a modified tRNA if the anticodon structure is incorrect. Although the anticodon is 7.5 nm away from the CCA end of the tRNA, the synthetases are large enzymes. Many of them are able to accommodate this large distance between a recognition site and the active site (Fig. 29-9A). For some other tRNAs the anticodon is not involved in recognition.²⁴⁵ For yeast tRNA^{Phe} residues in the stem of the dihydrouridine loop and at the upper end of the amino acid acceptor stem seem to be critical.²⁴¹

For some other tRNAs only the acceptor helix is essential for recognition. Change of one base-pair, the pair G3 • U70 (a “wobble” pair) of an *E. coli* tRNA^{Ala} • mRNA complex to the unnatural A3 • U70, prevents aminoacylation. Conversely, a G3 • U70 pair formed with tRNAs specific for other amino acids causes them to become substrates for the alanyl-tRNA synthetase.^{241,246} Even a shortened tRNA minihelix consisting of a 7-bp acceptor stem, 6-nucleotide loop, and ACCA 3' end is a substrate for this enzyme.^{247,247a} A seryl-tRNA synthetase depends upon recognition of two base pairs in the acceptor stem.²⁴⁸ Synthetic DNA oligomers with sequences corresponding to those of *E. coli* tRNA^{Phe} or tRNA^{Lys}, and with either deoxythymidine or deoxyuridine in the positions occupied by ribouridine in the tRNAs, are also substrates for the synthetases. The affinity and reaction rates are somewhat decreased, but the ribose 2'-OH is not essential for recognition.²⁴⁹

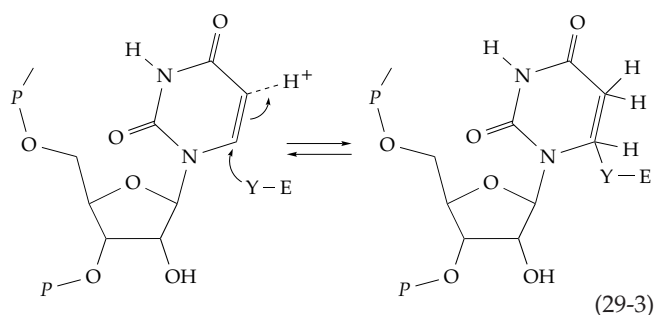
Mechanisms of reaction. Activation of an amino acid occurs by a direct in-line nucleophilic displacement by a carboxylate oxygen atom of the amino acid on the α phosphorus atom of MgATP to form the aminoacyl adenylate (Eq. 29-1, step *a*). For yeast phenylalanyl-tRNA synthetases the preferred form of MgATP appears to be the β,γ -bidentate (Δ screw sense) complex (p. 643).²⁵⁰ This is followed by a second nucleophilic displacement, this one on the C=O group of the aminoacyl adenylate by the -OH group of the tRNA (Eq. 29-1, step *b*; Fig. 29-9C). A conformational change in the protein may be required to permit dissociation of the product, the aminoacyl-tRNA. In the complex of a class I synthetase with aminoacyl

adenylate and tRNA the 3' CCA acceptor end of the tRNA is straight, but in a class II synthetase it is bent. In the two classes of synthetase the tRNAs approach the enzyme in a mirror-symmetric fashion. The 2'-OH of the terminal ribose is positioned to attack the carbonyl of the aminoacyl adenylate in class I enzymes, while the 3'-OH is positional for the attack in class II enzymes.²⁰⁷

The three-dimensional structure of *E. coli* glutaminyl-tRNA synthetase is shown in Fig. 29-9A. The active site lies against a β sheet in a structure similar to the nucleotide binding domain of a dehydrogenase (Fig. 2-13). The site of binding of a ATP is marked in Fig. 29-9A. The details of this binding to tyrosine-tRNA synthetase have been studied intensively.^{251–255} Binding of the tRNA substrates is less well understood. A large series of mutants involving 40 basic residues were prepared by Bedouelle and Winter²⁵⁶; study of these mutant enzymes, together with computer-assisted modeling, led to a proposed structure for a transition state for a complex with tRNA as is shown in Fig. 29-9B. Kinetic studies of heterodimers prepared from mutant and normal enzyme confirm that both subunits of the dimeric enzyme interact with the tRNA.

The active site of a type II synthetase is shown in Fig. 29-9C,D.^{217,217a} The expected movement of electrons in the reaction with ATP is illustrated by the green arrow in D. Both metal ions and active-site protein groups may participate as is also proposed for another type II enzyme.²²⁹

Some data suggested that a transient covalent linkage of tRNA to the synthetases may form through addition of a nucleophilic group of the enzyme to the 6' position of the uracil (or 4-thiouracil) present in position 8 of all tRNAs (Eq. 29-3).²⁵⁷ The two isoacceptors tRNA^{Tyr} species in *E. coli* contain 4-thiouracil at this position. The C=C bond in this base can be saturated by sodium borohydride reduction, which was found not only to prevent the covalent interaction with the enzyme but also to prevent aminoacylation of the tRNA. However, Eq. 29-3 probably describes a side reaction irrelevant to tRNA function.



Correcting errors. Much attention has been devoted to “proofreading” or “editing” activities of

amino acid-tRNA synthetases (see p. 482). For the majority of the enzymes proofreading is not needed.^{209a} Thus, tyrosyl-tRNA synthetase (Fig. 29-9B) mistakenly chooses phenylalanine instead of tyrosine only 5 in 10^4 times, apparently a tolerable rate of error. This enzyme, as well as a tryptophan-specific synthetase,²³⁸ depends largely upon differences in the Gibbs energy of binding to select the correct substrate. However, the discrimination between valine and isoleucine by isoleucyl-tRNA synthetase poses a more difficult problem. It is apparently solved, in part, by a “double sieve” editing mechanism,^{223,224,258,259} which is described briefly on p. 482. In the first sieve competitors that are larger than the substrate or are differently shaped are excluded by steric repulsion from binding in the active site. Isoleucyl-tRNA synthetase doesn’t convert leucine into an aminoacyladenylate, but it does act on the smaller valine. However, most of the resulting enzyme-bound valyl-adenylate is hydrolyzed to valine and AMP before it can be transferred to tRNA^{Ileu}. It shifts into an editing site, which is too small for the isoleucyl-adenylate, in effect passing through a second sieve.^{259a} Some of the activated

valine is transferred to tRNA and is removed in a second editing reaction.²²³ Misactivation of threonine or some other amino acids by valyl-tRNA synthetase is corrected in an analogous fashion.^{259b,c}

A similar editing process prevents isoleucyl-, leucyl-, and methionyl-tRNA synthetases from attaching L-homocysteine to tRNAs.^{260–263} In this case, instead of hydrolysis the editing site catalyzes conversion of the homocysteinyladenylate into homocysteine lactone. Naturally occurring mutations in tRNA molecules can sometimes have serious consequences. For example, a human mutation is responsible for a fragile mitochondrial isoleucine tRNA and serious cardiomyopathy and opthalmophegia (see also Box 18-B).^{263a}

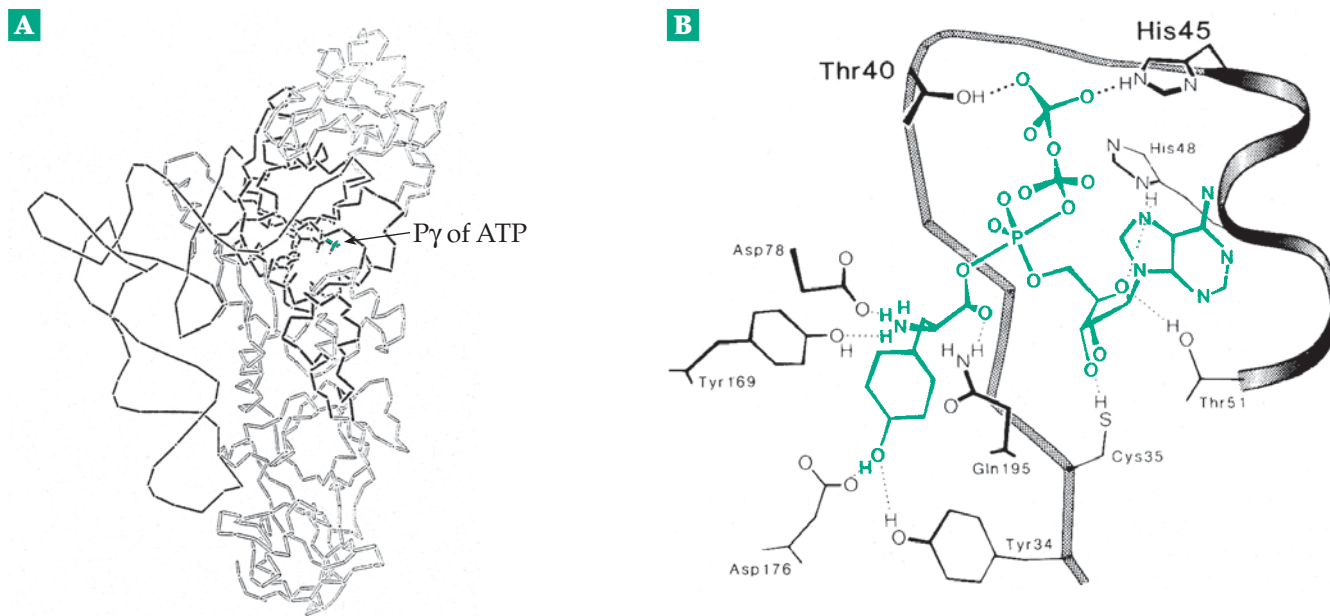
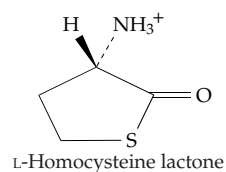
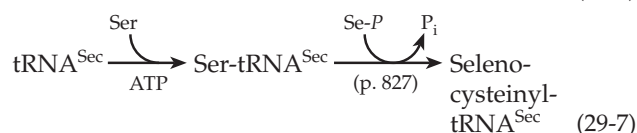
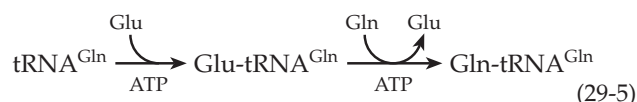
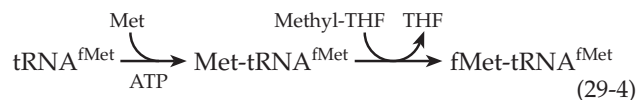


Figure 29-9 Selected views of aminoacyl-tRNA synthetase structure and action. (A) Alpha-carbon trace of the type I *E. coli* glutamyl-tRNA synthetase. The phosphate backbone of tRNA^{Gln} is shown in black; ATP is shown in the active-site cleft. The canonical dinucleotide fold domain near the N terminus is shaded. Two structural motifs (black), proposed to link the active site with regions of the protein-RNA interface involved in tRNA discrimination, are indicated. The α helix (top) connects tRNA recognition in the minor groove of the acceptor stem with binding of the ribose group of ATP. The large loop (center) connects anticodon recognition by the two β -barrel domains (bottom) with sequences flanking the MSK sequence motif, which interacts with the phosphates of ATP. From Perona *et al.*²¹⁴ Courtesy of Thomas A. Steitz. (B) The active site structure of tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* with a penta-coordinate transition state structure modeled.²¹⁵ From Leatherbarrow *et al.*²¹⁶ (C) Schematic representation of the active site of the lysyl-tRNA synthetase showing potential hydrogen bonding interaction in the ternary complex with lysine and ATP. The invariant motif 2 Arg 262 plays a key role in the recognition of the lysine carboxylate and the ATP α phosphate, while the invariant motif 2 Arg 480 binds the

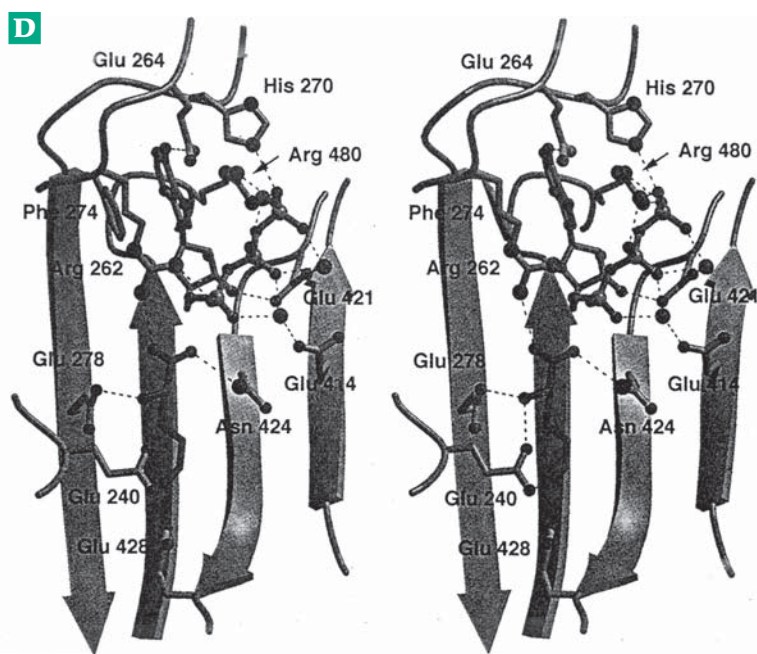
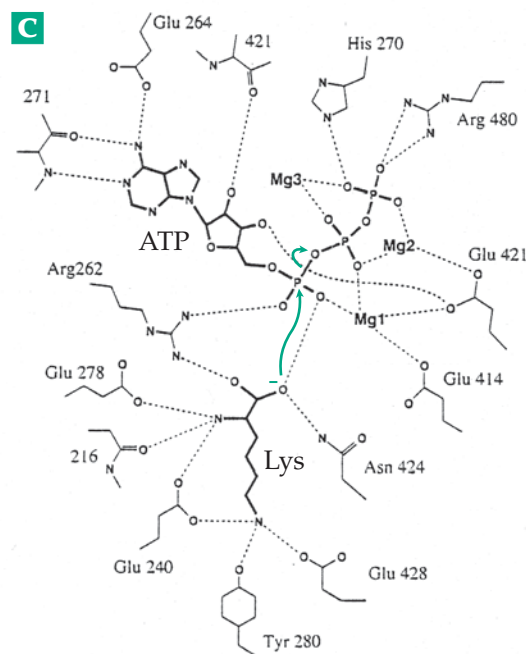
Proofreading involves kinetic as well as thermodynamic considerations.^{264–266} The relative rates of a hydrolytic reaction and the competing activating reaction must always be considered. These ratios can be strongly affected by conformational changes, which may occur in several steps (see also Section C,2).

Essential modification reactions of aminoacyl-tRNAs. In bacteria the initiator tRNA needed to start the synthesis of a polypeptide is initially aminoacylated by methionine, but the methionyl-tRNA^{fMet} must then be *N*-formylated by transfer of a formyl group from *N*¹⁰-formyltetrahydrofolate (Fig. 15-18; Eq. 29-4).^{267,268} In gram-positive bacteria and in archaea, mitochondria, and chloroplasts the glutamine-specific tRNA^{Gln} is charged with glutamate to form Glu-tRNA^{Gln}. The latter is converted by action of an ATP-dependent amidotransferase (see Eq. 24-22) to the necessary Gln-tRNA^{Gln} (Eq. 29-5).^{268a} In a similar way, tRNA^{Asn} in some organisms is charged with aspartate, then converted by transamidation to Asp-tRNA^{Asn} (Eq. 29-6).^{207,267,269} An important reaction, that occurs in all kingdoms of life, is the charging of the special tRNA^{Sec} with serine and conversion of the product into selenocysteinyl-tRNA^{Sec} (p. 827; Eq 29-7).^{267,270}



The introduction of selenocysteine into proteins at selected stop codons using this tRNA is described in Section C,5.

Examination of the complete genome sequences of methanogens revealed an apparent lack of cysteinyl-tRNA synthetase. However, prolyl-tRNA synthetase does correctly aminoacylate the tRNAs for both proline and cysteine in these archaeobacteria.^{271–272a}



γ phosphate of the ATP. A number of conserved residues in the motif 2 loop (residues 264–271) assume an ordered conformation only upon ATP binding. The positions of the Mg^{2+} sites are indicated. (D) View of the active site of the type II lysyl-tRNA synthetase showing the conformations of the substrates lysine and ATP before the first step of the reaction takes place. The ATP molecule is located on one side of the central β sheet of the C-terminal domain, with the adenine ring sandwiched between a conserved phenylalanine (Phe 274) and the motif 3 arginine residue (Arg 480). The pyrophosphate moiety is bent toward the adenine placing the α phosphate in the correct position for nucleophilic attack of the lysine carboxylate oxygen. The hydrogen bonding and electrostatic interactions between the substrates and some of the key residues, including the invariant motif 2 arginine (Arg 262), are shown. The three Mg^{2+} ions (green) involved in catalysis are included. (C) and (D) are from Desogus *et al.*²¹⁷

Additional functions of aminoacyl-tRNA synthetases. The primary function of these enzymes in protein synthesis is well known, but they have a whole range of other activities.^{273,274} In *E. coli* the large alanyl-tRNA synthetase can repress transcription of its own gene by binding to a palindromic sequence in the control region of the gene.²⁷⁵ Expression of some genes, such as that for threonyl-tRNA synthetase, is regulated at the translational level.^{273,274,276} In mammalian cells the formation of the threonine-specific synthetase appears to be regulated by a phosphorylation–dephosphorylation mechanism.²⁷⁷ Other synthetases participate in mitochondrial RNA splicing²⁷⁸ and in aminoacylation of tRNA-like 3' ends of viral genomes (see Fig. 28-24) and of N termini of certain proteins.²⁷⁹ For example, an arginyl group may be transferred onto the N terminus of a protein, marking it for rapid degradation.²⁸⁰ Under conditions of apoptosis, tyrosyl tRNA synthetase is hydrolytically cleaved to form two different cytokines.²⁷⁴ Phenylalanyl-tRNA synthetase is a DNA-binding protein.²⁸¹ Within the nucleus newly synthesized tRNAs are checked before being exported to the cytoplasm. Only tRNAs with mature 5' and 3' ends are exported. In both *Xenopus* oocytes^{281a} and in *S. cerevisiae*^{281b} the tRNAs are also tested prior to export, using aminoacyl-tRNA synthetases, to ensure that they are functional.

Many proteins have structures related to those of aminoacyl-tRNA synthetases.^{282,283} For example, asparagine synthetase A functions via an aspartyl-adenylate intermediate (Chapter 24, Section B), and its structure resembles that of aspartyl-tRNA synthetase.²⁸⁴ The *his G* gene of histidine biosynthesis (Fig. 25-13) encodes an ATP phosphoribosyltransferase with structural homology to the catalytic domain of histidyl-tRNA synthetase.²⁸⁴ The reason is not clear, but some aminoacyl-tRNA synthetases, especially the histidyl-tRNA synthetase, are common autoantigens for the inflammatory disease **polymyositis**.^{285,286}

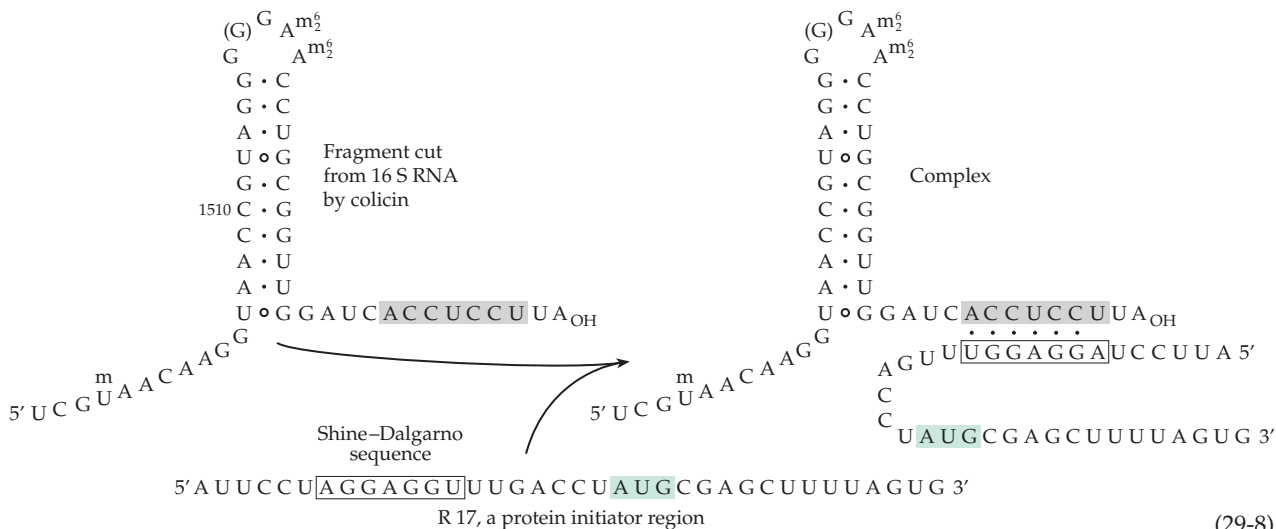
C. Protein Synthesis: The Ribosome Cycle

Initiation (Figs. 29-10 and 29-11), elongation (Fig. 29-12), and termination are three distinct steps in the synthesis of a protein. A variety of specialized proteins are required for each stage of synthesis. Their sequential interaction with ribosomes can be viewed as a means of ensuring an orderly sequence of steps in the synthesis cycle. The rate of protein formation will depend upon the concentrations of amino acids, tRNAs, protein factors, numbers of ribosomes, and kinetic constants. The formation of specific proteins can also be inhibited by **translational repressors**, proteins that compete with ribosomes for binding to target mRNAs.²⁸⁷

1. Initiation

For most polypeptide chains initiation begins with one of the three **initiation codons**, most commonly the methionine codon AUG. When properly placed in an mRNA chain, GUG may also serve as a bacterial initiation codon. In such cases, it codes for methionine rather than for valine. Occasionally UUG, AUU, ACG, and perhaps other codons can initiate translation.^{288,289} This is less frequent in eukaryotes than in bacteria. The sequence of bases preceding the initiation codon must also be important for recognition of the “start” signal.

In *E. coli* polypeptide chains are always initiated with the amino acid **N-formylmethionine**. Some bacteria can apparently live without the ability to formylate methionyl-tRNA,²⁹⁰ but most eubacteria as well as mitochondria and chloroplasts use formyl-methionine for initiation. In a few cases, both among bacteria and eukaryotes, initiation can sometimes occur with other amino acids.²⁹¹ The first step is the alignment of the proper initiation codon correctly on



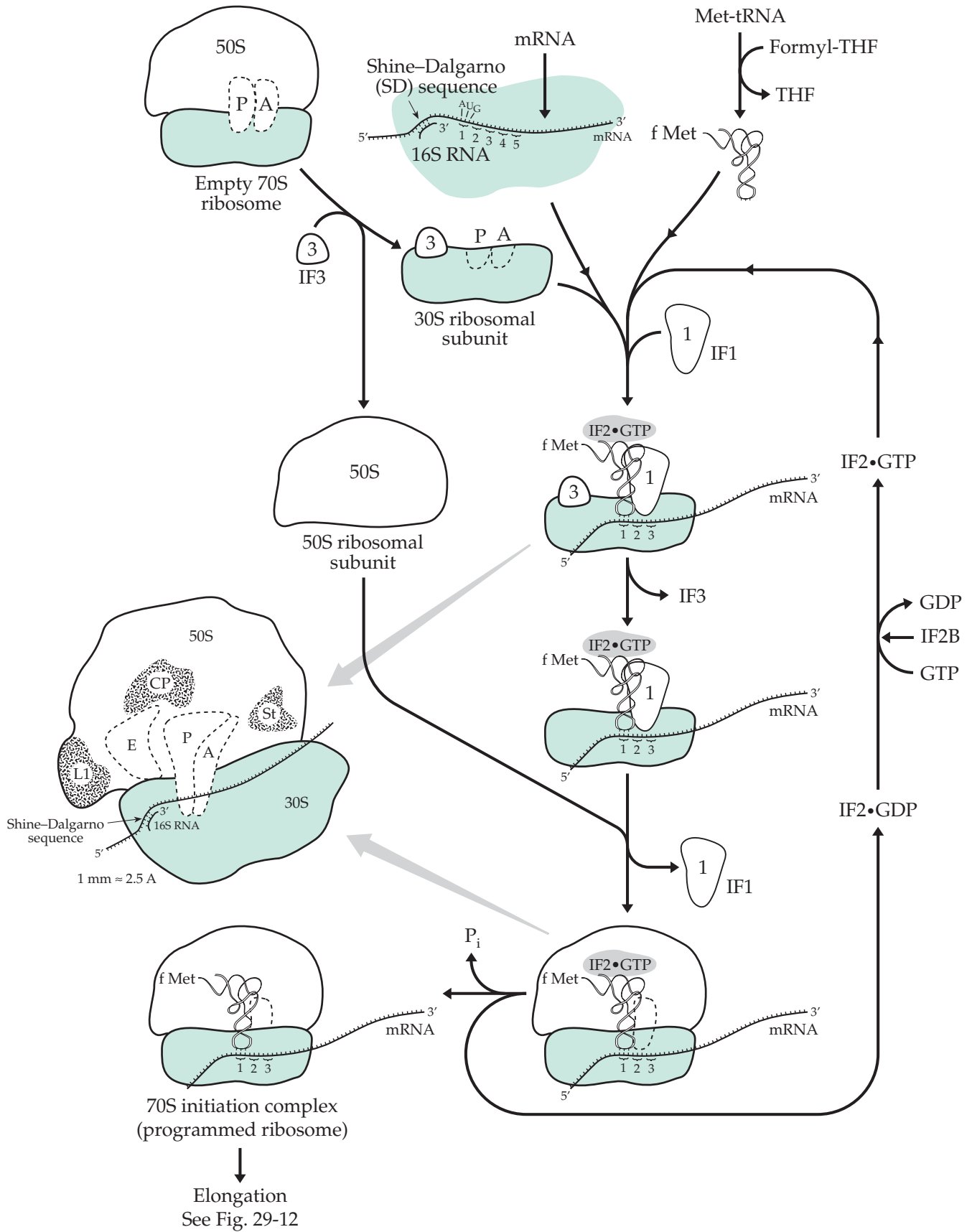


Figure 29-10 Initiation of protein synthesis on bacterial ribosomes. Images are not drawn to scale. Some details are indicated on the larger scale image at the left.

the ribosome and the binding to it of a molecule of initiator tRNA carrying *N*-formylmethionine.^{268,292,293} The process by which this occurs is relatively complex, partly because it is essential for the ribosomes to distinguish the true initiation codon from the many AUG codons in internal positions in the message. In bacteria recognition of the initiation codon is assisted by base pairing between the conserved sequence ACCUCCU at positions 1534–1540 at the 3' end of the 16S RNA (Fig. 29-2A) and the complementary **Shine-Dalgarno sequence** AGGAGGU, which is found near the 5' end of most mRNA molecules.⁵³ This is illustrated in Eq. 29-8 for a messenger RNA in the form of the A protein initiator region from the R17 phage RNA.^{294,295} Ribosomal protein S1 also seems to be required for this binding.^{135,295a}

Prokaryotic initiation factors. In addition to the ribosomal proteins, the initiation factors **IF1**, **IF2**, and **IF3**, whose molecular masses are 9.5, 9.7, and 19.7 kDa, respectively,^{70,296} are essential. They coordinate a sequence of reactions that begins with the dissociation of 70S ribosomes into their 30S and 50S subunits. Then, as is shown in Fig. 29-10, the mRNA, the initiator tRNA charged with formylmethionine, the three initiation factors, and the ribosomal subunits react to form 70S **programmed ribosomes**, which carry the bound mRNA and are ready to initiate protein synthesis. IF2 is a specialized G protein (Chapter 11), which binds and hydrolyzes GTP. It resembles the better known elongation factor EF-Tu (Section 2). The ~172-residue IF3 consists of two compact α/β domains linked by a flexible sequence, which may exist as an α helix.^{296a–298} Its C-terminal domain binds to the central domain of the 16S RNA near nucleotides 819–859 (Fig. 29-2). When bound it protects nucleotides in the 690 loop from chemical modification²⁹⁷ and induces a conformational change in the loop.^{297a} Binding of IF3 prevents association of the 30S and 50S subunits, assuring the cell of a supply of free 30S subunits for translational initiation. It also promotes the binding of the other two factors: IF1 and IF2.²⁹⁹ Binding of IF2, as its GTP complex, stimulates the binding of fMet-tRNA in the adjacent P site.³⁰⁰ Another function of IF2, in cooperation with IF1, may be to remove peptidyl-tRNAs with short polypeptide chains under conditions in which such peptidyl-tRNAs accumulate to abnormal levels.³⁰¹ However, the order of binding, which is implied in Fig. 29-10, has been hard to establish.

IF1, which is essential to the viability of bacteria, binds and partially occludes the A site of the ribosome, preventing the initiator fMet-tRNA from incorrectly occupying the 30S A site.^{70,296a,302} Binding of IF1 also causes the functionally important bases A1492 and A1493 of 16S RNA (Fig. 29-2) to be flipped out of helix 44 and to bind to pockets in IF1. This induces further

long-range conformational changes.⁷⁰ It has also been hard to establish whether the charged initiator tRNA binds into the P site before or after the mRNA binds to the 30S subunit. Some evidence supports the latter possibility,²⁹⁶ which is indicated in Fig. 29-10. In any case, an important step is the specific base-pairing of the initiator tRNA with the first AUG start codon. IF3 seems to be essential for this pairing process, which establishes the correct reading frame for translation of the genetic message.³⁰³ A proofreading function at this step is often attributed to IF3.³⁰⁴ Intact ribosomes bind charged tRNAs tightly in the P site. Perhaps the initial binding to the 30S subunit is loose enough to allow the mRNA, which ties itself to the 3' end of the 16S RNA via Eq. 29-3, to move back and forth until the correct reading frame is located. Then a conformational change occurs and locks the initiator tRNA in place. This change also weakens the binding of IF3, which dissociates from the complex, allowing the 50S subunits to rejoin the complex. The ribosome-binding domain of IF3 is homologous to spliceosome protein U1A (Chapter 28).³⁰⁴

The hydrolysis of GTP during initiation is essential as is shown by the fact that 5'-guanylmethylene diphosphonate, a GTP analog containing a methylene bridge between the terminal and central phosphorus atoms (see p. 558), can substitute for GTP in all steps up to and including the binding of the 50S ribosome. However, it cannot function in the final step because it cannot undergo hydrolysis. Why is GTP hydrolysis needed? It may provide energy for the conformational rearrangement of ribosomal components, or it may simply be required for release of the IF2•GDP complex. For example, IF2•GTP may bind to the ribosome with a high affinity, but IF2•GDP only weakly. Remember that G-proteins exist in at least two conformations, one stabilized by GDP and another by GTP (Chapter 11). When the hydrolysis of the bound GTP is incorporated into a reaction sequence, it provides a Gibbs energy change that may be needed to drive the reactions. In this case, it ensures that the charged initiator tRNA is firmly bound and ready to initiate translation.

Some information about spatial arrangements of the ribosomal proteins involved in initiation was provided by the fact that antibodies against proteins S19 and S21 block the formation of a complex with fMet-tRNA, while antibodies against S2, S18, and S20 block the binding of IF3. Crosslinking experiments showed that IF2 and S19 are close together and that IF3 is close to S12 (Fig. 29-1A).

Initiation of protein synthesis in eukaryotes.

Most eukaryotic mRNAs have a 5' cap (p. 1642) and lack a Shine–Dalgarno sequence. Otherwise, initiation follows a pattern similar to that in bacteria but more complex.^{305–308} There are at least ten eukaryotic

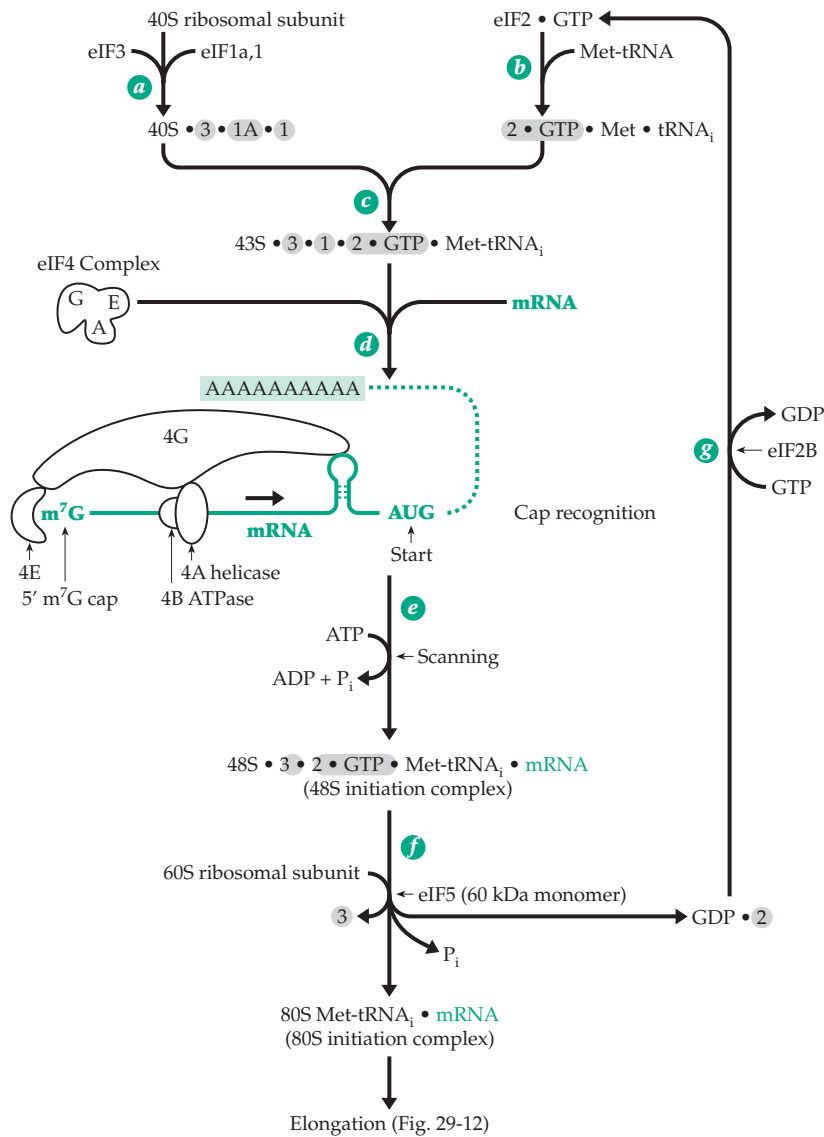


Figure 29-11 Initiation of eukaryotic protein synthesis. ①, ②, ③ = eIF1,2,3

initiation factors (eIFs), some composed of several peptides.³⁰⁹ Hydrolysis of both ATP and GTP is required to form the initiation complex. Cap-binding proteins help to locate the 5' end, but the first initiation codons occur at greatly varying distances from the cap. Ribosomes apparently conduct a systematic scanning beginning at the 5' end cap of the mRNA to locate the first initiation codon.^{305,310,311} Initiation of translation in eukaryotes is also often subject to controls that are more complex than those in bacteria.^{308,312–314a} At least 25 polypeptides are involved.³¹⁵ Specific functions of only a few of these are described here. Functions of some are unknown or uncertain, and new proteins such as the stimulatory factor 4H^{316,317} have been discovered recently.

The first initiation step is the dissociation of idle

80S ribosomes into their 40S and 60S subunits. This depends upon the ~700-kDa eIF3, a complex of 5–11 peptides of mass 30 to 170 kDa each, which binds to the 40S subunit (Fig. 29-11, step a).^{306,311,318–320} In a separate reaction (step b) the charged initiator tRNA (Met-tRNA_i) is bound by the G protein eIF2,^{321–324} an αβγ mixed trimer whose α subunit not only binds GTP but is also the site of regulation by a phosphorylation–dephosphorylation mechanism.^{325,326} As is indicated in Fig. 29-11, GDP-containing eIF2, released from ribosomes upon formation of the initiation complex, does not bind the charged initiator tRNA. The bound GDP must first be exchanged for GTP, a process that requires a five-subunit guanine nucleotide exchange factor eIF2B (Fig. 29-11, step g).^{327–328a} However, if the α subunit of eIF2 becomes phosphorylated, the nucleotide exchange and consequently the initiation of protein synthesis, is retarded.^{329–330c} In reticulocytes protein synthesis stops rapidly if there is a deficiency of heme. This appears to occur via a **heme-sensitive eIF2α kinase** whose catalytic activity is inhibited by heme.^{322,327,331} The [NAD⁺]/[NADH] ratio may also be a factor in controlling the nucleotide-exchange GEF-catalyzed reaction.³³²

The ternary complex of eIF2 with GTP and Met-tRNA_i binds to the 40S complex (Fig 29-11, step c). If the ribo-trinucleotide AUG, the initiation codon, is added to this complex, it is converted to a 43S initiation complex.³²¹ However, with natural mRNAs several additional protein factors as well as ATP are needed (Fig. 29-11, step d). Factor 4F (eIF4F) is a large complex of several components known as factors 4A, 4B, 4E, and the large 220-kDa 4G (formerly EIF4γ or p220).^{333–334a} Factor 4G is a multifunctional adapter or scaffold that apparently organizes the complex and coordinates various control mechanisms.^{335–338} Factor 4E is a cap-binding protein, which recognizes and binds tightly to the 7mG cap present on most mRNAs.^{306,339–341a} It also binds to 4G. Factor 4A, an ATPase, acts together with 4B as an RNA helicase to unwind the mRNA and remove hairpin loops^{342,343} during the scanning to locate the initiation codon (step e). Kozak suggested that the 43S ribosome scans from the cap at the 5' end and stops at the

first initiation codon, which is usually the AUG found within the sequence (A/G)NNAUGG.

AUG codons in other positions, known as **internal ribosome entry sites** (IRES),^{311,344–347} and, more rarely, non-AUG codons can also initiate translation with lower efficiency.³⁴⁸ Thus, mechanisms exist for synthesis of small amounts of proteins of varying lengths and of proteins that are encoded in any one of the three reading frames.^{305,349–350a} Even circular RNAs can serve as mRNAs by this mechanism.³⁵¹ It is significant that, as shown in Fig. 19-11, factor 4G, the large subunit of eIF4, also binds to the poly(A) tail present on the 3' terminus of most mRNAs. This binding, which seems to be essential for rapid initiation,^{352–354} is mediated by yet another protein, the **poly(A)-binding protein**. The importance of this protein in the human body is emphasized by its identification as **ataxin-2**, the protein defective in type 2 spinocerebellar ataxia (see Table 26-4).³⁵⁴ The significance to the regulation of initiation is not clear, but the poly(A) binding may favor reuse of the mRNA, which may be translated repeatedly under conditions of rapid growth.

The last initiation step (step *f*, Fig. 29-11) is the reaction of the 60S ribosomal subunit with the 48S initiation complex to form the 80S initiation complex. Initiation factors 3, 4C, the eIF2•GDP complex, and inorganic phosphate are all released in this process, which is promoted by IF5. This monomeric ~60-kDa protein^{355,356} also stimulates conversion of the GTP bound to IF2 into GDP and P_i. IF5 is unique as the only known protein containing **hypusine**, N^ε-(4-amino-2-hydroxybutyl)lysine, a posttranslationally modified lysine. It occurs only at position 50 in the 17-kDa protein.^{356–358} Hypusine is not present in eubacteria but is essential for viability of both eukaryotes and archaeobacteria³⁵⁸ and is present within an invariant 12-residue sequence.

2. Elongation of Polypeptide Chains

Once the initiating fMet-tRNA of bacteria or the eukaryotic Met-tRNA_i is in place in the P site of a ribosome and is paired with the initiation codon in the mRNA, peptide chain growth can commence. Amino acid residues are added in turn by insertion at the C-terminal end of the growing peptide chain. Elongation requires three processes repeated over and over until the entire peptide is formed.

1. Codon-specific binding of a charged tRNA bearing the next amino acid at the A site (decoding).
2. Formation of the peptide bond. This process transfers the growing peptide chain from the tRNA in the P site onto the aminoacyl-tRNA in the A site.
3. Translocation of the peptidyl tRNA from the A site

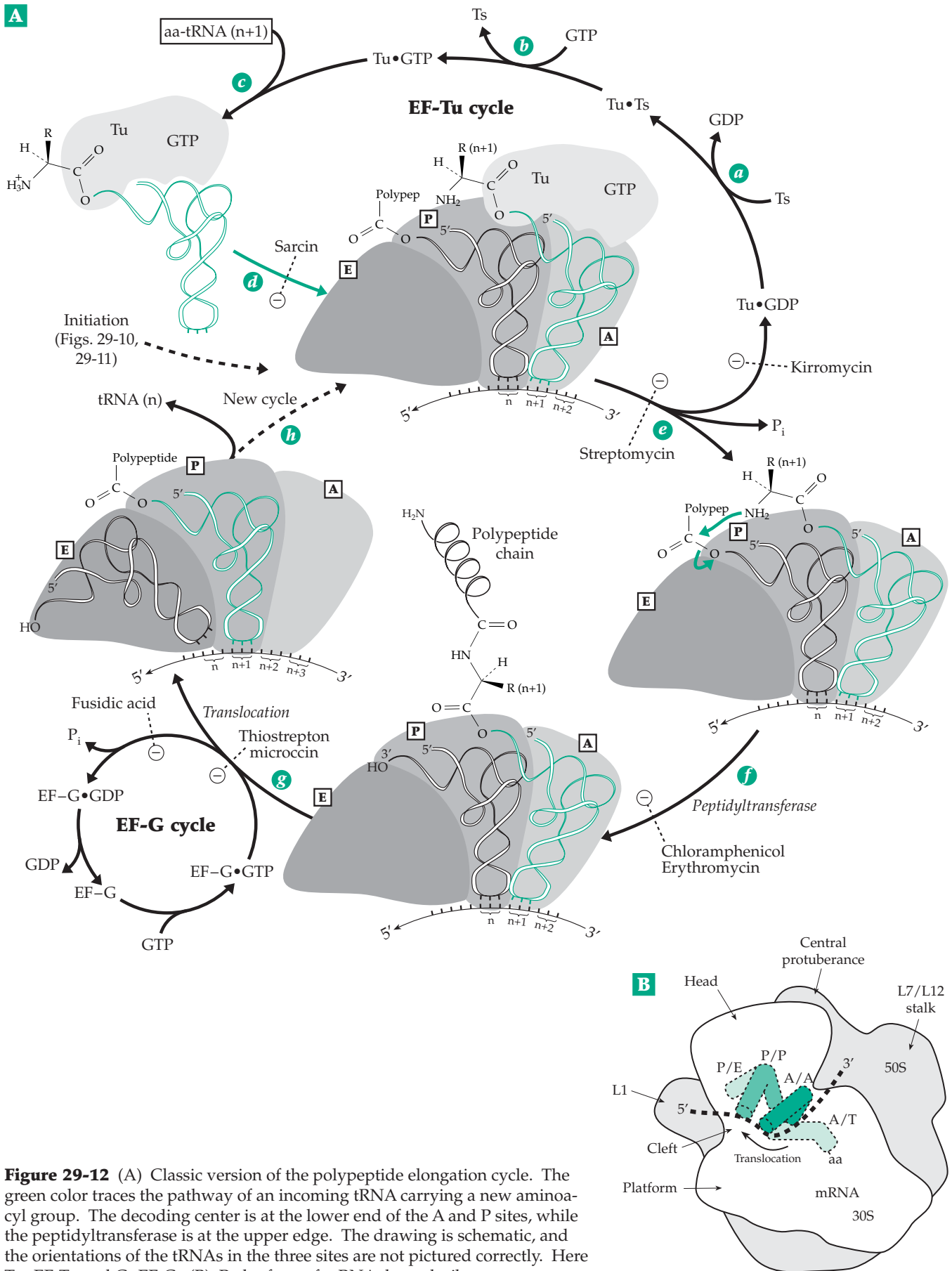
to the P site. This process also involves movement of the used tRNA from the P site into the exit site and simultaneous movement of the mRNA to bring the next codon into place in the A site. Both the release from the A site and translocation require energy. This is provided by the hydrolysis of GTP, one molecule for each of the two processes.^{359,360}

The elongation cycle for *E. coli* is shown in Fig. 29-12. That for eukaryotic ribosomes is similar except that 40S and 60S subunits are involved in formation of the complete 80S ribosome.

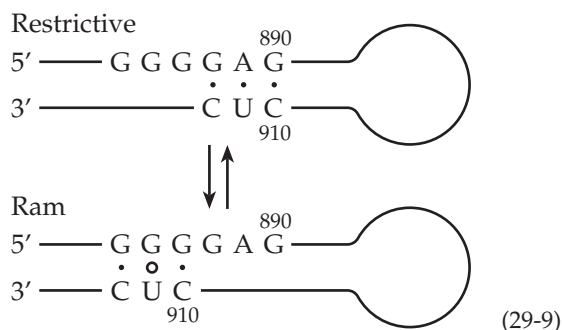
Codon-specific binding of an aminoacyl-tRNA (decoding). The binding of an aminoacyl-tRNA to the A site of the 70S or 80S initiation complex depends upon a protein called **elongation factor Tu (EF-Tu or eIF1** in eukaryotes), which is present as a mixed dimer with a second protein, **EF-Ts**. In *E. coli* EF-Ts is a stable 35-kDa protein, while Tu is a 43-kDa soluble protein present in a large excess over Ts. Tu is one of the most abundant soluble proteins in bacterial cells and accounts for about 5% of the total protein. Most of the tRNAs in a bacterial cell are present as complexes with Tu. Tu may also have functions other than in protein synthesis and is found associated with the plasma membrane as well as with ribosomes.

Factor Tu is a G protein. It not only carries the aminoacyl-tRNAs into the A site on ribosomes, as shown in Fig. 29-12, but also binds and hydrolyzes GTP during the elongation cycle.^{361–368} Factor Ts is a nucleotide exchange factor that catalyzes the exchange of GDP bound to Tu for GTP.³⁶⁹ This is shown in Fig. 29-12 (steps *a* and *b*). The GTP/GDP-binding site of EF-Tu is located in the N-terminal portion. Eukaryotic eEF-Tu is also called EF1α or EF-TA. As isolated from various sources it has a molecular mass ranging from 50 to 53 kDa. Like the bacterial counterpart, it is abundant.^{368,370–373} Like bacterial EF-Tu it exists largely as a complex with a more abundant nucleotide exchange factor EF1β. The complex tends to be bound to actin filaments.^{373,374} Fungal eEF-Tu contains mono-, di-, and trimethylated lysine at up to 16 positions.³⁷¹

EF-Tu will bind to any aminoacylated tRNA other than tRNA^{f-Met}, the initiator tRNA^{374a} (step *c*, Fig. 29-12), and carry it to the ribosome (step *d*), where it binds into the A site. There it is selected if it forms a proper base pair with the mRNA codon in the A site or is rejected if it does not. This decoding process involves both an initial step and a proofreading step. The aminoacyl-tRNA binds both to the decoding site in the 16S RNA and to the peptidyltransferase site in the 23S RNA. (See discussions on p. 1687.) The decoding site is on the platform at the upper end of helix 44 (Fig. 29-2). Nucleotide G1401 plays a crucial role.³⁷⁵ When one of the isoacceptor species of *E. coli* tRNA^{Val} is irradiated with ultraviolet light, the



5-(carboxymethoxy)uridine at position 34 in the anticodon becomes crosslinked specifically with C-1400 of the 16S rRNA in the 30S subunit.^{5,376} This nucleotide lies in the deep cleft, in the decoding region, between the neck and the platform of this ribosomal subunit (Fig. 29-1). Various crosslinking and protection experiments^{377,378} show that other helix 44 residues bind the tRNAs in both the P and A sites. A1492 and A1493 form part of the A site, while C1400 is in the P site.^{378,379} Also strongly affecting tRNA binding and decoding is the nearby **switch helix** in the 900 region of the 16S RNA. This helix readily undergoes a shift between two hydrogen-bonded configurations (Eq. 29-9).^{378,378a,380}



Judging by the effects of mutations in 16S RNA or in proteins S5 and S12 that favor one or the other conformation, the restrictive conformation gives a greater fidelity in translation than the “ram” (ribosomal ambiguity) conformation.³⁸⁰ This loop is near the central pseudoknot in the 16S RNA and is involved in binding S5 and S12 as well as streptomycin (Box 29-B), all of which affect fidelity of protein synthesis. As mentioned on p. 1687 the adenine rings of residues A1492 and A1493 move out to interact with the CCA-3' ends of the tRNA (Fig. 29-14).

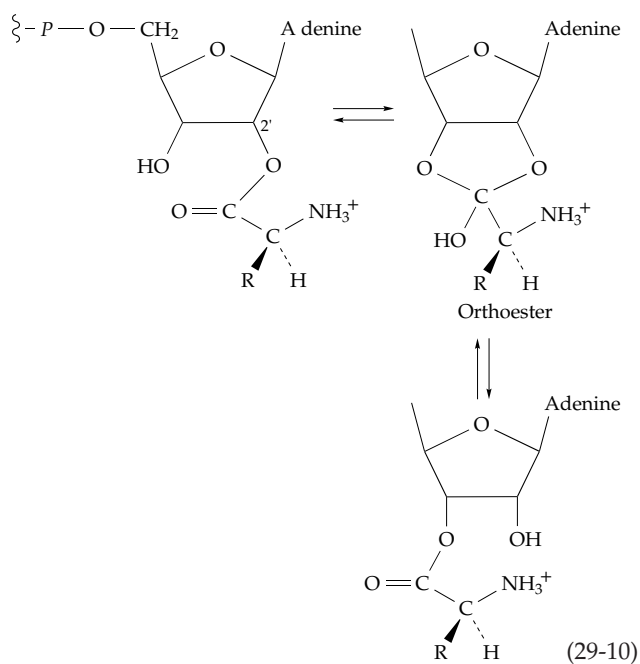
The location of binding of EF-Tu on ribosomes has been established directly by cryo-EM. It binds both to the L7/L12 stalk and to the body of the 50S ribosomal subunit.³⁸¹ The other end of the P site is at the peptidyltransferase locus and has been photochemically labeled by azide derivatives of aminoacyl groups bound to a tRNA.³⁸² The labeling is primarily in the 50S subunit of *E. coli* ribosomes and involves the central loop of domain V (residues 2043–2625) of the 23S RNA. Residues U2584 and U2585 are major sites of crosslinking (see Fig. 29-14). The presence of nearby sites of mutation leading to resistance to chloramphenicol or erythromycin^{383,384} (Box 29-B) served to confirm the central loop as part of the peptidyltransferase. Domain II of 23S RNA is also involved, and there is evidence that the unique sequence UGG at positions 807–809 may also interact with the CCA end of tRNA in the P site.³⁶¹

Bound Tu leaves a “footprint” at positions 2655

and 2661 in the sarcin/ricin loop of domain VI (Fig. 29-4H) when chemical probes are applied.³⁸⁵ From a thermodynamic viewpoint, the hydrolysis of GTP to GDP and P_i during the functioning of EF-Tu is unnecessary, but it appears to drive a conformational change needed to bring the reacting groups together or as part of a proofreading mechanism.^{385-385d} The hydrolysis of GTP appears to follow codon-anticodon recognition between the tRNA and mRNA in the A site (see Figs. 29-2G and 29-14). EF-Tu • GDP has a greatly reduced affinity for an aminoacyl-tRNA and dissociates, leaving the latter firmly bound into the A site.^{385b-d} Simonson and Lake proposed that binding of a tRNA into the A site is *preceded* by binding into a **D site**. After the initial binding the anticodon of the tRNA wing from the D site into the A site as a result of a conformational rearrangement of the base stacking within the tRNA.^{385e}

The peptidyltransferase reaction. It has been difficult to establish whether the 2' or the 3' of the terminal adenosine of tRNA carries the activated aminoacyl or peptidyl group. Rapid equilibration between the two via an orthoester may occur (Eq. 29-10), and EF-Tu of *E. coli* binds to either the 2' or 3' isomer.^{386,386a} However, reaction of the 3'-aminoacyl-tRNA at the peptidyltransferase site is probable.^{387,387a,b}

The peptidyltransferase reaction resembles that of the proteases (Chapter 12, pp. 649, 650), with a tetrahedrally bonded intermediate probable (Fig 29-13A). As is shown on pp. 649–650, the catalytic acid has been proposed to be the N3 atom of adenosine 2486 (2451 in *E. coli*) in the *H. marismortui* 23S RNA. This is in the central loop of domain V (Fig. 29-14). However, replacement of A2451 with G, U, or C did



not totally destroy peptidyltransferase activity in *Thermus aquaticus*, nor did replacement of the essential G2447 with A, U, or C. Polacek *et al.* suggest that the ribosome may need only to hold the reacting aminoacyl and peptidyl groups attached to tRNA in the correct positions for reaction and that no other catalysis is necessary.^{387c} However, A2451G, U, or C mutant ribosomes have very low activity and organisms with such mutations are often not viable.^{33f,167b} The pH rate profile for peptidyltransferase activity indicates a pK_a of ~ 7.5 in the RNA.^{33f,387d} This is consistent with the view that A2451 may serve as a catalytic base. There

has been considerable discussion about the pK_a . Can it be assigned to A2451? As mentioned on pp. 751–753, many enzymes have a broad pH region of maximum velocity over which catalytic groups of quite different microscopic pK_a s (pp. 305–307) may function. For ribosomal RNAs, as for proteins, tight bonding between ionized groups in a substrate–catalyst complex may lock in an overall protonation state of the macromolecule. However, a proton may jump from one group to another within the complex (e.g., as in Fig. 29-14D) to provide a set of tautomeric species in a pH-independent equilibrium. Among these some

will be on the catalytic pathway. One may arise by deprotonation of the reacting $-NH_2$ group of the aminoacyl-tRNA (Eq. 29-1, step c). Conformational changes,^{387e} which may be induced by proton movements, may also be encompassed within the array of pH-independent equilibria.

A careful stereochemical analysis has led to the conclusion that for all of the different aminoacyl groups to be able to react in the same way at the peptidyltransferase site and to all generate trans amide linkages, the torsion angles ϕ and ψ of the resulting peptide must be approximately those of an α helix.³⁸⁸ Thus, the peptide emerging from the ribosome exit tunnel may be largely helical.

Elongation factor EF-G and translocation. The third step in the elongation sequence on ribosomes (Fig. 29-12, step g) depends upon **EF-G**, a monomeric GTP-binding protein with a sequence homologous with that of other members of the G protein family. It apparently utilizes the Gibbs energy of hydrolysis of GTP to GDP to drive translocation of the peptidyl-tRNA from the A site to the P site (Fig. 29-12) and of the previously utilized (deacylated) tRNA to the exit site. EF-G binds to the 50S ribosomal subunit at the base of the L7/12 stalk as indicated in Fig. 29-1.^{392,393} It competes with EF-Tu, which binds in nearly the same location.⁵ EF-G is a large five-domain GTPase. Domain 1 contains the GTPase site and resembles other G proteins, and domain 2 has some similarity to the

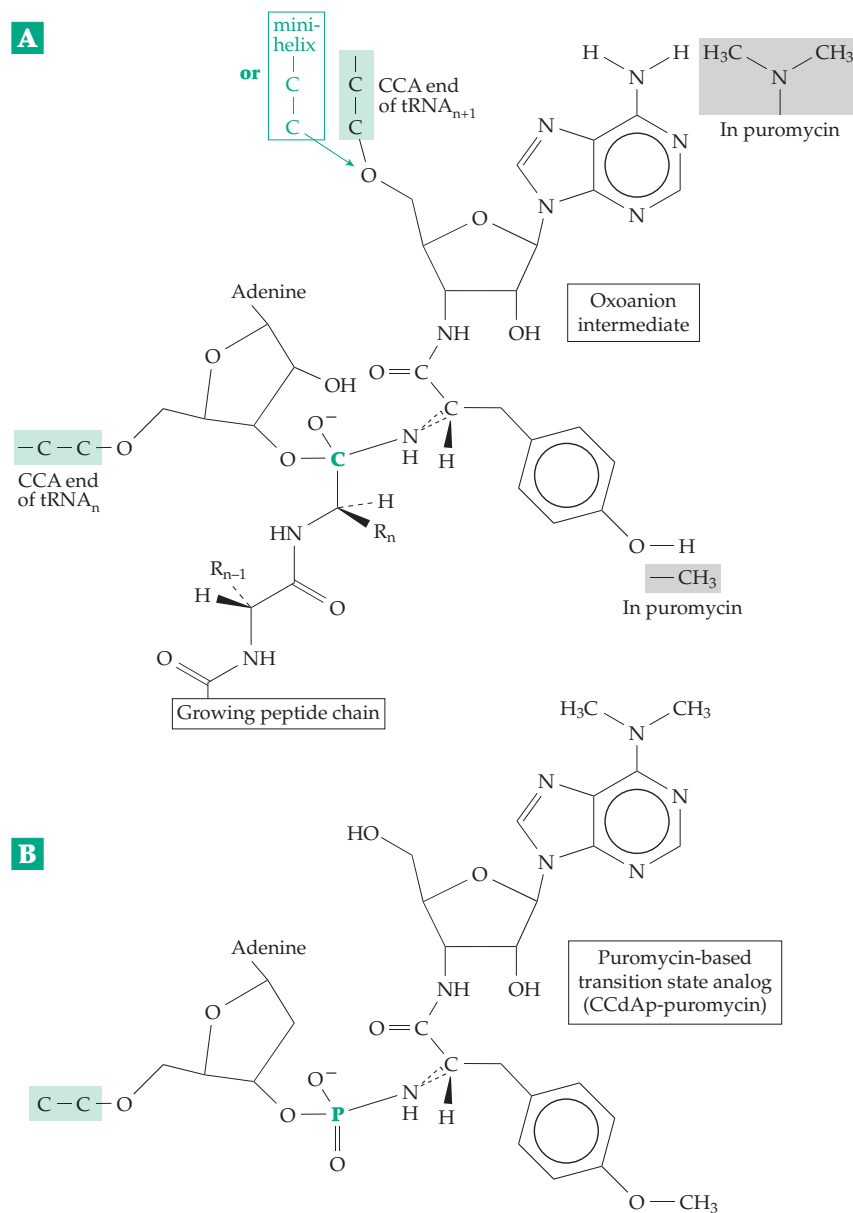


Figure 29-13 (A) Structure of expected intermediate with tetrahedral C-atom in peptidyltransferase reaction with a tRNA, with a minihelix analog, or with the antibiotic puromycin. (B) Transition-state (or bisubstrate) analog formed with puromycin and a mimic of the CCA end of a tRNA. See Box 29-B.

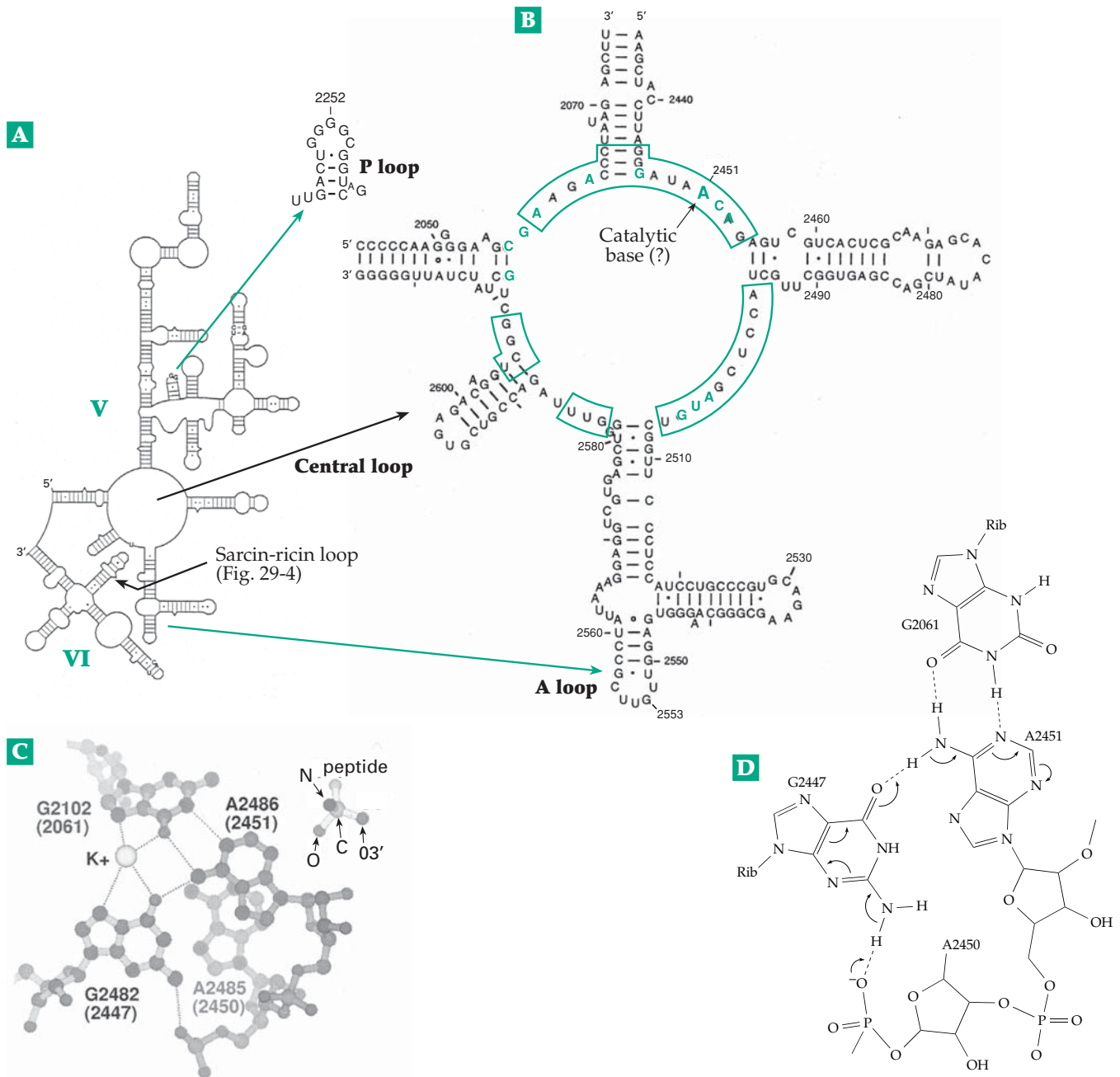
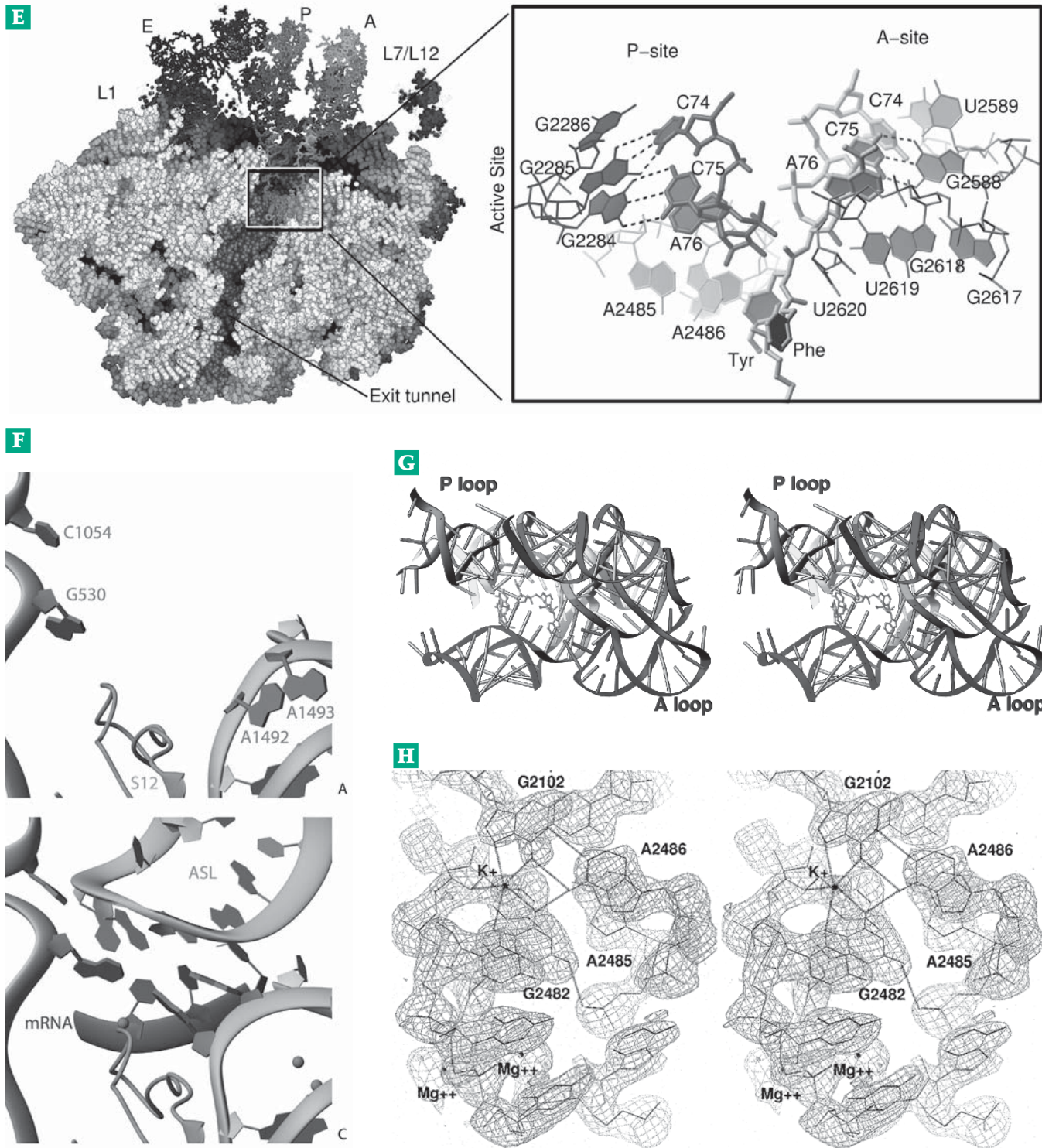


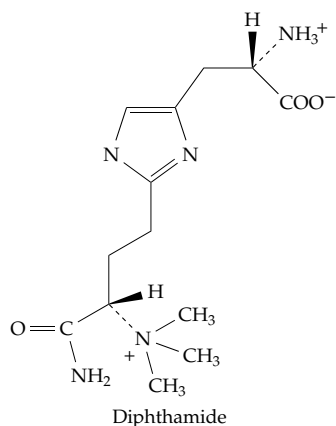
Figure 29-14 The ribosomal peptidyltransferase center in the 23S RNA. (A) Secondary structure map for domains V and VI of *E. coli* 23S RNA. From Samaha *et al.*³⁸⁹ (B) Sequences of the P loop and A loops and of the central loop of domain V and of the 23S RNA of *Halobacterium halobium* with numbering according to the *E. coli* sequence. Sequences within the green boxes are highly conserved in eubacteria, chloroplast, and mitochondrial RNAs. See Kloss *et al.*³⁹⁰ and Garret.⁸⁰ Sites of mutations that confer resistance to erythromycin (G2057, A2058, and C2611) and chloramphenicol (G2057, G2447, A2451, C2452, A2503, and U2504) are indicated. See Douthwaite and Aagaard.³⁸³ (C) A skeletal representation of the peptidyltransferase active site which is depicted more completely in the stereoscopic view in (H). Included is the peptidyl group on the 2'-end of a tRNA. (D) Schematic region of domain V showing the hydrogen-bonding interactions of the catalytic base A2486 (2451, *E. coli*) with neighboring bases and also locations of nearby K^+ and Mg^{2+} ions. (C, D, G, H) are from Nissen *et al.*¹⁹ (E) Interaction of the CCA 3'-ends of ribosome-bound tRNAs (at top) with the large ribosomal subunit. This is a cutaway view with tRNAs in the A, D, and E sites. The ribosome is sliced to show the acceptor ends of the tRNAs in the entrance to the peptide tunnel. Enlarged section shows interactions of the CCA 3'-ends in the P and A sites. The CCA end of a deacylated tRNA is shown in the P site, while a peptidyl-tRNA analog (CCA puromycin-phenylalanine-caproic acid-biotin) is shown in the A site. Bases of the 23S rRNA are numbered as in *H. marismortui*. Bases A2486 and U2620 (corresponding to A2451 and U2585) are closest to the newly formed peptide bond. From Schmeing *et al.*^{33f} (F) Discrete states of the A site of the 30S subunit, as deduced from four different crystal structures. The tRNA, anticodon stem-loop (ASL), A-site mRNA codon, P-site mRNA, protein S12, and



important bases involved in conformational changes are shown. A few elements of the 16S RNA such as helix 44 (lower right), 530 loop (mid and lower left), and helix 34 (upper left) are also shown. At the top, the native 30S subunit. A1492 and A1493 have been stacked in the interior of H44 and G530 is in the *syn* conformation. C1054 is shown in the upper left corner. Below, when the codon and cognate tRNA-ASL bind in the A site, A1492 and A1493 flip out to monitor the codon-anticodon interaction, and G530 switches to the *anti* conformation to interact with A1492, the anticodon in the second position, and the codon in the third. Two Mg^{2+} ions are visible near the region vacated by A1492 and A1493 in the interior of helix 44, and one is located close to the ribose of the codon in the wobble position. From Ogle *et al.*^{33d} Courtesy of Venki Ramakrishnan. (G) Three-dimensional structure of the active-site region of domain V including the P loop³⁹ⁱ and A loop and active-site region. (H) Three-dimensional structure of the active-site region showing the hydrogen-bonding interactions of the catalytic base A 2486 (2451, *E. coli*) with neighboring bases and also locations of nearby K^+ and Mg^{2+} ions. (C, G, H) are from Nissen *et al.*¹⁹

corresponding domain in EF-Tu. GTP hydrolysis appears to induce within EF-G a conformational change that is coupled directly to a rapid mechanical movement within the ribosome.^{394,394a} The large domain 4 protrudes from the molecule and by its shape mimics a molecule of tRNA, and the complete EF-G molecule bears a striking resemblance to an EF-Tu•aminoacyl-tRNA complex.^{395,396} It leaves chemical footprints around position 1067 (the thiostrepton-binding site) in domain II of the 23S RNA and in the universally conserved loop around position 2660.³⁸⁵

Factor EF-G from eukaryotes (eEF2) is similar to the bacterial protein, but its interaction with the larger eukaryotic ribosomes seems to be more complex. For example, interaction with the ribosomal stalk is more extensive.³⁷ EF2 contains a single modified histidine called **diphthamide**.³⁹⁷ This amino acid is not found in other proteins but is always present in eukaryotic EF2 and also in EF-G from archaeobacteria. It is the site of modification by diphtheria toxin (Box 29-A).



The ribosomal translocation process is quite complex. As the tRNAs move from A to P to E sites on the 16S RNA platform, the mRNA must also move in discrete single-codon steps. The acceptor stems of the tRNAs in the A and P sites must react at the appropriate times in the peptidyltransferase center. Study of protection from chemical probes suggests that tRNAs sometimes lie with the anticodon loop in the A site of the small ribosomal subunit, while the acceptor stem is in the P site of the large subunit (an A/P site as illustrated in Fig. 29-12B). Each aminoacyl-tRNA enters as a complex with EF-Tu and may initially bind with its anticodon in the A site and the acceptor stem with attached EF-Tu in a transient T site, the composite state being A/T. After loss of EF-Tu the acceptor stem can move into the A site to give an A/A state. The peptidyltransferase reaction itself necessarily involves movement at the acceptor stems by 0.1 nm or more. However, additional movement of ~1 nm is needed to move the two tRNAs into states A/P and P/E, respectively. Movement of the mRNA then moves the

anticodon loops of the peptidyl-tRNA into the P/P state and of the deacylated tRNA completely into the exit site (Fig. 29-12B).^{86,397a} Translocation may occur at different times in the 30S and 50S subunits. The pathway of the mRNA through the ribosome is known^{397b,c} and is indicated approximately in Fig. 29-12B.

EF-G seems to be the motor protein that drives translocation in the 30S subunit. When it is not attached to a ribosome, the EF-G•GTP complex is very stable, but in its functioning location GTP is hydrolyzed rapidly. This occurs prior to translocation^{398,398a} and presumably causes an internal alteration in the ribosome that energizes it for the translocation step. G proteins usually undergo large changes in conformation when GTP is hydrolyzed (Chapter 11). A very large change is observed for EF-Tu, but such a change has not been seen for EF-G. However, large conformational changes in the ribosome do evidently accompany translocation.³⁹⁹ The hydrolytic activity of EF-G is stimulated strongly by its binding to the L7/L12 stalk proteins.⁴⁰⁰ Eukaryotic EF2 like EF-G binds to the stalk proteins (P1, P2, P0 complex) and also to domains of 28S rRNA equivalent to the bacterial 1067 and sarcin / ricin loops.⁴⁰¹ However, EF-G and EF2 are not functionally interchangeable.

Translocation occurs slowly even in the absence of GTP. However, it is greatly enhanced by GTP hydrolysis.^{394a,402} Even empty ribosome, without tRNAs, but in the presence of GTP and EF-Tu and EF-G, hydrolyze GTP. The ribosome may sequentially bind EF-Tu, then EF-G, oscillating between two differing states.⁴⁰³ The movement of mRNA through the ribosome has been plotted using a variety of immunochemical, crosslinking, and chemical footprinting methods.^{52,404–407}

A third elongation factor, eEF3, which is an ATPase, is required by yeast and fungi.^{408–410} The 1044-residue yeast protein may be required for ATP-dependent release of deacylated tRNA from the exit site.

Polyribosomes. Under suitable conditions ribosomes isolated from cells are found to sediment together in clusters, often of five or more. These **polyribosomes** (or **polysomes**), which can be seen in electron micrographs (Fig. 28-5), are held together by chains of mRNA. Polyribosomes arise because a single mRNA molecule is being translated by several ribosomes at once. As the 5' terminus of the mRNA emerges from one ribosome, it may soon combine with another and initiate translation of a second peptide chain, etc. The length of the mRNA determines how many ribosomes are likely to be associated in a polyribosome.

Rates of synthesis of ribosomes and of proteins.

In a rapidly growing yeast cell with a generation time of ~100 min there are nearly 200,000 ribosomes. Almost 200 new ribosomes must be formed in one minute. Each

of the 150 tandemly repeated ribosomal RNA genes must be transcribed into the 4560 nucleotides of one ribosome in less than one minute. The ~150 nuclear pores must import nearly 1000 ribosomal proteins per minute and must export ~25 ribosomal subunits per minute.^{410a} The ribosomes that are formed can at 37°C add 14–17 amino acids per second to a growing polypeptide chain,^{410b,c} while eukaryotic ribosomes can add 2–4 amino acids per second.^{410d,e}

3. Termination of Polypeptide Synthesis

A ribosome faithfully translates the genetic message, adding amino acids to the polypeptide chain until a stop codon is reached. Then a **termination** or **release factor** acts, probably by binding directly to the stop codon on the mRNA in the A site.^{411–413b} In *E. coli* termination factor **RF1**, a 47-kDa protein, recognizes UAA or UAG, while **RF2**, a very similar protein,^{414,415} recognizes UAA or UGA. There are several hundred molecules per cell of these release factors. They not only recognize the stop codons but also catalyze the hydrolytic removal of the peptidyl chain from the tRNA in the ribosomal P site. They bind into the A site, where they may interact with mRNA bases in addition to those of the stop codon.^{415,416} Hydrolytic release of the polypeptide chain from the tRNA in the P site may represent a change in specificity of the peptidyltransferase center induced by binding of a release factor. Genes are often terminated by a succession of two stop codons. Thus, there is a safety factor that prevents translation from continuing in case the first stop codon is missed. An example is provided by the *I* gene of the *lac* operon of *E. coli* (Fig. 28-2), which has a second stop codon in phase with the TGA codon marked in the figure and located five codons further “downstream” (to the right). A third release factor **RF3**⁴¹⁶ is a GTP-binding protein resembling EF-G. It is not essential to life for *E. coli*, but it accelerates the release of RF1 or RF2 and is needed for rapid growth.^{417,418} Eukaryotes contain one release factor **eRF1**, which recognizes all three termination codons, and a second release factor **eRF3**, which binds and hydrolyzes GTP.^{413,413b,418a,419}

Just as elongation factor EF-G mimicks the aminoacyl-tRNA•Tu complex, release factors RF1 and RF2, in their shapes, mimick molecules of tRNA.^{419a} One domain of human eRF1 has an anticodon-recognition domain and a conserved GGQ sequence in a second domain, which mimicks the amino-acceptor arm of tRNA.^{419,419b} Mutations in either eRF1 or eRF3 affect translational accuracy and may allow “read-through” of stop codons. In yeast (*S. cerevisiae*) a 685-residue subunit of eRF3 has an N-terminal domain, that like the human **prion protein** (Box 29-E) is capable of being transformed into a self-seeding amyloid-like conformation. In the yeast the

formation of amyloid aggregate leads to depletion of the termination factor and increased readthrough.^{420–422}

Recycling factors. Even though release factors remove the completed polypeptide chain, a ribosome is not ready for reuse until the deacylated tRNA in the P site is removed and the mRNA is released. This depends upon **ribosome recycling factor (RRF)** together with EF-G. The recycling factor is also a tRNA mimic.^{419a,423–424b} It may bind into the empty A site, and in an action similar to that of the translocation step of elongation remove the P site tRNA.^{417,423–425} However, probing with hydroxyl radicals indicates a different mode of binding.^{425a}

4. Preventing and Correcting Errors in Translation on the Ribosome

The wrong amino acid is inserted into most positions in a protein about one time in $\sim 10^4$, a frequency^{361,426,427} of $\sim 10^{-4}$. However, in *E. coli* misreading of certain codons is observed more often. For example, AAU (Asn) is read as AAA (Lys) with a frequency⁴²⁸ of $\sim 5 \times 10^{-3}$. Misreading also depends upon adjacent codons, i.e., the codon context.⁴²⁹ Having all of the tRNAs charged with the correct amino acids, as discussed in Section B3, is a first essential for accurate translation. A second is finding the correct location of the initiation codon and binding of the aminoacyl-initiator tRNA into the P site. The decoding process by which the correct aminoacyl-tRNA is brought into the A site is still not fully understood. It has often been proposed that (as in DNA replication; Chapter 27, Section C.2) the fidelity of this process depends upon two consecutive recognition steps.^{265,266} The first is the binding of the complex of EF-Tu•GTP and the charged tRNA to the ribosome. The second may be associated with the conformational change that locks the aminoacyl-tRNA into the A site and perhaps sends to the peptidyltransferase center a signal that the correct codon-anticodon pairing has been achieved. Some checking is done in the first step. For example, many of the 380 possible mischarged forms of aminoacyl-tRNAs that may have escaped previous proofreading steps (pp. 1695–1696) are rejected because they bind too loosely or too tightly to EF-Tu.^{429a,b} Codon-anticodon base pairing may also be checked in the P site^{429c} after translocation. The P site is buried deep in a cleft in the RNA of the large subunit. It is designed to hold the mRNAs in a kinked conformation (Fig. 29-1E) with the codons in the A and P sites oriented differently. Some tRNA residues required for high-fidelity participate in imposing this geometry. Mutants in either of the major rRNAs or in tRNAs can lead to loss of fidelity in base pairing and sometimes to excessive frame-shifting.^{429c} During the

proofreading process a mispaired aminoacyl-tRNA may be allowed to dissociate and be replaced by a new one. Certain mutations, such as those in ribosomal protein S12 that lead to streptomycin resistance, cause greatly increased fidelity of protein synthesis. However, these mutations slow bacterial growth,^{426,430} perhaps because some misreading is necessary for synthesis of minor essential proteins.

A strictly hypothetical way in which an alteration in hydrogen bonding could be used to signal the achievement of correct base pairing is illustrated in Fig. 29-15. As indicated by the curved arrows, the approach of a negatively charged group could induce an electron pair to move from the ring nitrogen on the right side. If the pairs of hydrogen bonds were correct, a concerted flow of electrons could take place across the base pair and out into group H-Y and beyond through the postulated tautomeric chain. If the base pair were not correctly formed, the signal could not be transmitted, except during an occasional mispairing with a minor tautomer. Note that another reciprocal electron transfer in the opposite direction to that shown in the figure is also possible through the same base pair. Similar tautomeric shifts are possible for all legitimate base pairs. Initiation of a signal of the type shown could also occur by the addition of some nucleophile to a purine or pyrimidine ring, e.g., to C-6 of the cytosine ring as in Eq. 29-3. In ribosomes such electronic signals could be passed in turn through each of the base pairs involved in codon-anticodon recognition and also through other base pairs formed within loops of ribosomal RNA. If group H-Y is connected by a suitable chain of hydrogen bonds that passes through the active site of the peptidyltransferase, coupling between the recognition signal and the formation of

the transition state might be accomplished. Since changes in hydrogen bonding can trigger conformational alterations, the sensing of correct hydrogen bonding could increase the rate of the peptidyltransferase reaction as has been observed experimentally.²⁶⁶ Base pairing in both the A and P sites may be sensed in similar ways. The observation that rRNA residues hydrogen-bond with groups in the minor grooves of base pairs^{33d} seems to be consistent with the proposal of Fig. 29-15.

Codon usage. The usage of codons in specific mRNAs is not random.⁴³² For example, in a glyceraldehyde-3-phosphate dehydrogenase gene of yeast >96% of the 1004 codons make use of only 25 of the 61 possible coding triplets. Genes undergoing most rapid transcription are more highly biased toward these 25 than are other genes.⁴³³ Many other evolutionary factors have affected usage. These include the need for translational accuracy.⁴³²

Dealing with lost peptidyl-tRNAs and broken transcripts. Many problems arise during protein synthesis. For example, a peptidyl-tRNA may become detached from a ribosome. In *E. coli* this seems to happen most frequently with peptidyl-tRNA^{Lys}. A 193-residue **peptidyl-tRNA hydrolase** is essential for life!^{434–436} It releases the tRNA for reuse, recycling all peptidyl-tRNAs other than formylmethionyl-tRNA. Perhaps the enzyme is essential because detached peptidyl-tRNAs are toxic, but it is more likely to be to avoid a shortage of free tRNA^{Lys}.⁴³⁴

If a ribosome starts to synthesize a protein using a broken mRNA, it will reach the end of the mRNA but will not find a termination codon. The peptidyl-tRNA will eventually fall off, but the ribosome will be stalled temporarily. Eukaryotes try to prevent this problem by sending only intact mRNAs to ribosomes,^{436a,b} but bacteria have a ribosome rescue system that also tags the partially formed protein on the stalled ribosome for rapid proteolytic degradation.^{437–440c} Bacteria synthesize a special 362-residue RNA that resembles a tRNA but also contains a short mRNA-like module that codes for the 11-residue peptide tag AANDENYALAA. This hybrid tRNA-mRNA, which is designated tmRNA (or *ssrA* RNA), mimicks tRNA^{Ala} and is recognized and charged by alanyl-tRNA synthetase. The resulting tmRNA^{Ala} binds into the A site of the stalled ribosome, undergoes the peptidyltransferase and translocation steps (Fig. 29-16). The old mRNA is released, the mRNA-like sequence of tmRNA becomes seated, and translation of the new tail sequence follows. The tail sequence is similar to C-terminal sequences that are known to mark other proteins for rapid proteolytic degradation. An associated protein SmpB is also required for functioning of the tmRNA system.^{438a,b}

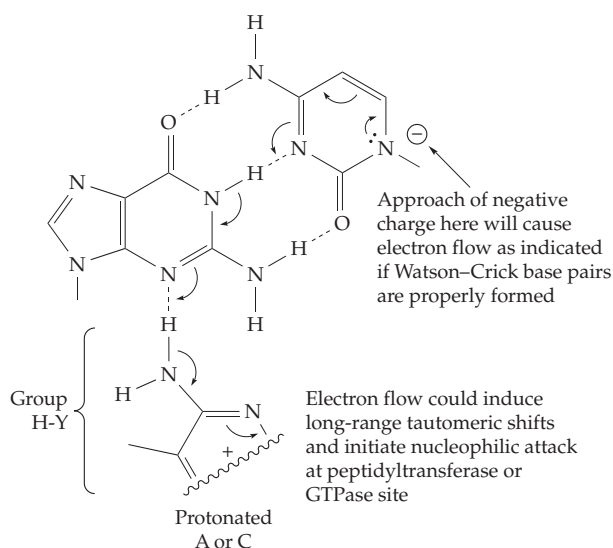


Figure 29-15 Hypothetical scheme by which an electronic signal might be sent through a base pair to initiate the peptidyltransferase reaction. See also Metzler.⁴³¹

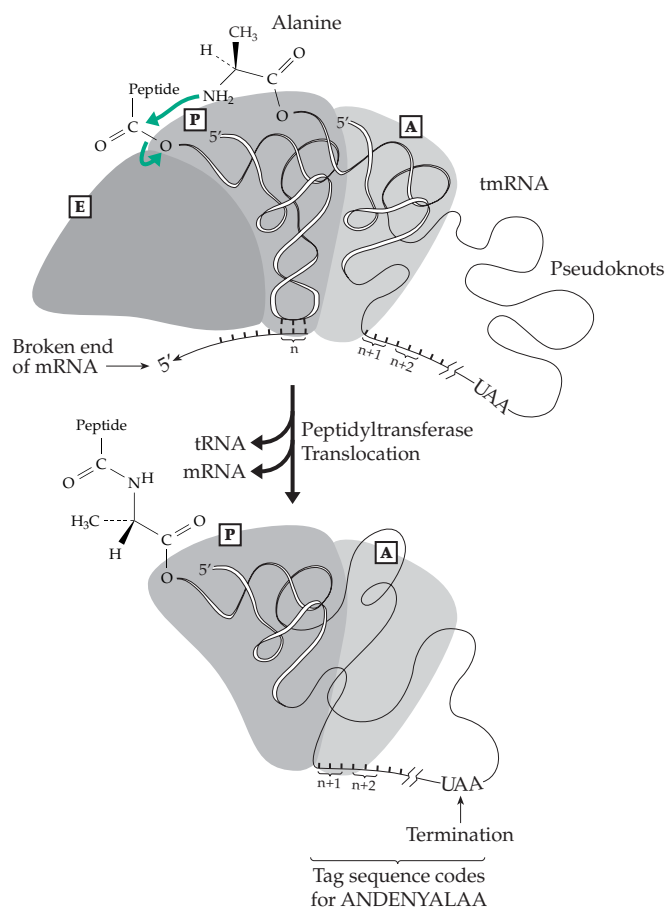


Figure 29-16 Schematic diagram of the tmRNA structure and its function in the rescue of ribosomes stalled at the end of a messenger RNA that has been broken and has lost its in-frame termination codon. After it binds into the ribosomal A site the tmRNA, which has been charged with alanine, undergoes the peptidyltransferase reaction and translocation to the P site. Then it lays down its mRNA-like coding sequence, which is used by the ribosome to add ten more amino acids to form the 11-residue C-terminal degradation signal AANDENYALAA. This induces rapid degradation of the imperfect protein that has been formed.^{436a}

5. Suppressor Genes

The suppression of nonsense mutations by suppressor genes has been discussed in Chapter 26. The chemical nature of these genes was discerned, in part, from experiments involving transfer of suppressor gene *supF(su3)* into the DNA of a bacteriophage. This DNA was found to specifically hybridize with a minor transfer RNA, tRNA₁^{Tyr}. Subsequent investigation showed that *sup F* is a structural gene for this tRNA, and that in it the normal 5'-GUA-3' (Tyr) anticodon has been replaced with CUA. The latter can pair with the chain termination codon 5'-UAG-3' (the *amber* codon) permitting the ribosome to insert tyrosine at the site of chain termination signals introduced in *amber* muta-

tions. It may seem puzzling that a tRNA, which prevents chain termination, does not prevent synthesis of other essential proteins within the bacterium. However, suppression is typically less than 30% efficient. Hence, many protein chains terminate normally. Since two chain termination signals are often present in a gene, most protein synthesis in the presence of the small amount of suppressor tRNA present is concluded normally. Premature chain termination caused by selected *amber* mutations will be partially inhibited, permitting the cell to make enough of the missing proteins to survive. The nucleotide sequence of a further mutated *supF* tRNA and of its longer precursor is shown in Fig. 28-10. Several other suppressor genes have also been identified as specific tRNA structural genes.⁴⁴¹

A suppressor of frame-shift mutations in *Salmonella* is a tRNA containing at the anticodon position the nucleotide quartet CCCC instead of the usual CCC triplet anticodon.^{442,443} It has eight unpaired bases in the anticodon loop instead of the usual seven. Other frame-shift repressor tRNAs have been identified in *E. coli*,⁴⁴⁴ *Salmonella*, and yeast.⁴⁴⁵ Not all suppressor genes encode tRNAs. For example, a UGA suppressor from *E. coli* is a mutant 16S rRNA from which C1054 has been deleted.⁴⁴⁶ A general nonsense suppressor in yeast is homologous to yeast elongation factor EF-1 α as well as to *E. coli* EF-Tu.⁴⁴⁷

Among other suppressor genes present in eukaryotic organisms⁴⁴⁸ are mammalian genes encoding serine tRNAs that are *opal* (UGA) suppressors. These and other eukaryotic suppressor tRNAs have specific and important normal functions in cells. For example, a specific kinase phosphorylates the *opal* suppressor seryl-tRNA to its phosphoserine derivative.⁴⁴⁸ This suppressor tRNAs may sometimes be responsible for introducing phosphoserine at specific positions in proteins. An *opal* suppressor is also used for the introduction of selenocysteine. An *amber* suppressor is used by some methane-forming Archaea to introduce **pyrrolysine** into specific sites in methyltransferases. In pyrrolysine the epsilon amino group of lysine is joined by an amide linkage to a derivative of pyrroline-5-carboxylate (p. 1374).^{448a,b}

Selenocysteine (Sec) Selenocysteine is incorporated into a small number of proteins in species from all three kingdoms of life by a suppressor tRNA^{Sec} that reads certain UGA codons, which are marked as representing selenocysteine.^{449,450} The selenocysteinyl-tRNA is made from a seryl-tRNA (Eq. 29-7) as described further in Chapters 16 and 24. In *E. coli* selenocysteine is present in three proteins, all formate dehydrogenases. The archaeon *Methanococcus jannaschii* contains genes for seven selenocysteine-containing proteins. Only one Sec-containing protein has been found in the nematode *Caenorhabditis elegans* and none in the yeast

Saccharomyces cerevisiae. However, there are at least 14 in the human body.^{451,452} One of these, selenoprotein P, contains ten selenocysteine residues.^{452,453} Products of four special genes are needed for incorporation of selenocysteine into *E. coli* proteins.^{454,455} *Sel C* encodes the special *tRNA*^{Sec}, which becomes charged with selenocysteine.⁴⁵⁶ *Sel D* encodes selenophosphate synthetase and *Sel A* selenocysteine synthetase (Eq. 29-7). *Sel B* encodes a special elongation factor, which resembles EF-Tu but has an extra domain that binds to an mRNA segment known as the **SECIS** (selenocysteine insertion sequence).^{457–461a} The SECIS sequence follows the 3' end of the UGA termination codon. It is a 40-nt segment that is able to form a stem-loop structure. However, in archaea and in eukaryotes the SECIS sequence lies at the end of the selenoprotein gene in the 3' nontranslated region. It may be some distance away and may function by a foldback mechanism. It recodes the entire message, acting on any in-frame UGA codon.⁴⁶¹ In mammals a special SECIS-binding protein SBP2 is also required.⁴⁶²

Expanding the genetic code. Suppressor tRNAs can also be created artificially and are being used in protein engineering. *Amber*, *ochre*, or *opal* chain termination mutations can be introduced readily at many points in a protein (Chapter 26). Suppressor tRNAs can be made that will then place any one of the possible amino acids into most of the mutated positions.⁴⁶³ Synthetic amino acids not normally found in proteins can also be incorporated using such tRNAs.^{464–467} The TAG(UAG) *amber* stop codon is often used together with a genetically engineered tRNA. In early experiments these techniques were used to create hundreds of mutant forms of the *lac* repressor protein (see Chapter 28, p. 1606). Since then a variety of additional approaches have been explored. Transfer RNAs have been engineered to recognize four-base codons such as AGGU and CGGG.^{468,469} Organisms such as *Micrococcus luteus*, in which not all of the available triplet codons have been utilized, allows development of a mutation system using an unassigned codon rather than a stop codon.⁴⁶⁸ A general method for site-specific incorporation of any amino acid or amino acid analog requires a suppressor tRNA that is not aminoacylated by any aminoacyl-tRNA synthetase present within the host cell, and also an aminoacyl-tRNA synthetase that acts only on the suppressor tRNA and no other tRNA in the cell.⁴⁷⁰ Several such systems are being developed.^{470–472} Another idea is to utilize a 65th codon-anticodon pair, one depending upon a new synthetic nucleoside that can be incorporated into mRNA.⁴⁷³

Another possible application of suppressor genes is *in vivo* suppression of undesirable termination codons. An example comes from a β^0 thalassemia caused by mutation of lysine codon CAG to UAG. By changing the anticodon of a human *tRNA*^{Lys} gene to

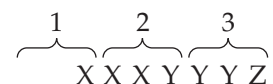
an *amber* suppressor, premature termination of globin chains was suppressed in an *in vivo* test.^{474,475}

6. Read-Through Translation and Ribosomal Frameshifting

If termination codons are not recognized efficiently by termination factors, synthesis continues past the termination codons and new longer protein chains are made. This **read-through translation**^{429,476} may sometimes be accidental, but it is also used by cells to form several important proteins. For example, the 14-kDa coat protein of bacteriophage Q β is elongated by read-through during translation of the RNA about 4% of the time. This produces a 38-kDa protein known as A₁, which has an extra 200 amino acid residues at the C terminus and is essential for formation of infectious virus particles.

A similar situation is met with retroviruses (Chapter 28) whose coat protein gene *gag* is fused to the reverse transcription polypeptide precursor gene *pol* (Fig. 28-26).⁴⁷⁷ In fact, the polypeptide gene overlaps the 3' end of the *gag* gene. Read-through translation allows synthesis of the polypeptide about 5% of the time. However, the *pol* gene is written in a different reading frame: –1 with respect to the *gag* gene. For example, in the HIV genome (Fig. 28-26) the *pol* gene begins at nucleotide 1638 in the –1 reading frame with respect to the *gag* gene. In both the Rous sarcoma virus and HIV a polypeptide that is formed contains peptide sequences from both *gag* and *pol* genes.^{478,479} This fusion of the two proteins is accomplished by a frameshift, which occurs on the ribosome as it operates in the region of overlap of the *gag* and *pol* genes. This mechanism allows synthesis of relatively small amounts of the enzymes encoded by *pol* but large amounts of structural proteins encoded by *gag*.⁴⁸⁰ Many other examples of essential ribosomal frameshifting are known.^{429,477,481–483} For example, the gene for *E. coli* ribosomal release factor RF2 has a UGA termination codon at position 26, but the coding sequence for the protein continues in the +1 frame.⁴⁸⁴

Ribosomal frameshifting can be accounted for by more than one mechanism. It can occur when a four-base anticodon is present in a suppressor tRNA. It can result from incorrect base pairing. If a tRNA slips over by one nucleotide, a single base in the mRNA can be left unpaired with the reading frame being shifted +1. However, most frameshifts are in the –1 direction and occur at specific locations in the mRNA, i.e., they are **programmed frameshifts**.^{484a} These often occur at “slippery sites”⁴⁸⁵ including the following mRNA sequences in which three codons are marked:



This sequence is followed closely by an element of secondary structure, most often a pseudoknot.^{486–492} Eukaryotic frameshifts are almost always in the –1 direction, the exception being found in the mammalian mRNA for **antizyme**, a negative regulator of ornithine decarboxylase (Chapter 24, p. 1382).⁴⁹³ The frame-shift occurs at an initially in-frame termination codon (UGA), which is followed by a pseudoknot.

Most translation is terminated at this stop codon, but frameshifting, which is induced by a high polyamine concentration, allows read-through and synthesis of the antizyme protein. In rare cases frameshifting may lead to **translational bypass** of some codons on the mRNA. Such a case is found in a bacteriophage T4 mRNA for which the *E. coli* ribosomes bypass 50 nucleotides in order to complete the synthesis of a

BOX 29-C NONRIBOSOMAL PEPTIDE SYNTHESIS

Many small biologically active peptides, including hormones and some antibiotics, are synthesized on ribosomes as precursor proteins, which are cut into small pieces and may then be modified in a variety of ways. However, many other peptides including many antibiotics are made without use of ribosomes by large polyfunctional synthetases. The first of these, gramicidin S synthetase, was described by Lipmann and coworkers in 1971.^a It is discussed on p. 994 as is the mechanism of synthesis. It is now recognized that these enzymes are modular and have much in common with fatty acid synthetases (Fig. 17-12 and p. 1186) and polyketide synthetases (Fig. 21-11). They are able to link not only the amino acids found in proteins but also modified and unusual amino acids. They may also join one or more α -hydroxy acids to a peptide to form a depsipeptide, and they may contain modules that carry out modification reactions such as methylation, acylation, or glycosylation.^{b–e} Because of their modular nature they are attractive proteins for genetic engineering.^{b,f,g}

Each synthetase module contains three active site domains: The **A domain** catalyzes activation of the amino acid (or hydroxyacid) by formation of an aminoacyl- or hydroxyacyl-adenylate, just as occurs with aminoacyl-tRNA synthetases. However, in three-dimensional structure the A domains do not resemble either of the classes of aminoacyl-tRNA synthetases but are similar to luciferyl adenylate (Eq. 23-46) and acyl-CoA synthetases.^h The **T-domain** or **peptidyl carrier protein domain** resembles the acyl carrier domains of fatty acid and polyketide synthetases in containing bound phosphopantetheine (Fig. 14-1). Its –SH group, like the CCA-terminal ribosyl –OH group of a tRNA, displaces AMP, transferring the activated amino acid or hydroxy acid to the thiol sulfur of phosphopantetheine. The **C-domain** catalyzes condensation (peptidyl transfer). The first or **initiation module** lacks a C-domain, and the final **termination module** contains an extra termination domain. The process parallels that outlined in Fig. 21-11.ⁱ

A few of the products of nonribosomal peptide

synthesis are gramicidin S (Fig. 2-4), enniatins, bacitracins, and tyrocidines (p. 994),^{b,e} vancomycin (Box 20-H),^j actinomycin (Box 28-A),^k the siderophore yersiniabactin,^l surfactin (Fig. 2-4),^{m,n} and cyclosporin (Box 9-F).^{o,p} The δ -(L- α -amino adipoyl)-L-cysteiny-D-valine synthetase,^{q,r} which forms the precursor to penicillin and cephalosporins (Box 20-G), also belongs to this group of enzymes as do synthetases that make cyclooctadepsipeptides with antihelminthic activity^s and many other compounds.^{t–v}

^a Lipmann, F. (1971) *Science* **173**, 875–884

^b Mootz, H. D., Schwarzer, D., and Marahiel, M. A. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 5848–5853

^c Linne, U., and Marahiel, M. A. (2000) *Biochemistry* **39**, 10439–10447

^d Guenzi, E., Galli, G., Grgurina, I., Pace, E., Ferranti, P., and Grandi, G. (1998) *J. Biol. Chem.* **273**, 14403–14410

^e Belshaw, P. J., Walsh, C. T., and Stachelhaus, T. (1999) *Science* **284**, 486–489

^f Symmank, H., Saenger, W., and Bernhard, F. (1999) *J. Biol. Chem.* **274**, 21581–21588

^g de Ferra, F., Rodriguez, F., Tortora, O., Tosi, C., and Grandi, G. (1997) *J. Biol. Chem.* **272**, 25304–25309

^h Pfeifer, E., Pavela-Vrancic, M., von Döhren, H., and Kleinkauf, H. (1995) *Biochemistry* **34**, 7450–7459

ⁱ Cane, D. E., Walsh, C. T., and Khosla, C. (1998) *Science* **282**, 63–68

^j Trauger, J. W., and Walsh, C. T. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 3112–3117

^k Pfennig, F., Schauwecker, F., and Keller, U. (1999) *J. Biol. Chem.* **274**, 12508–12516

^l Miller, D. A., and Walsh, C. T. (2001) *Biochemistry* **40**, 5313–5321

^m Weinreb, P. H., Quadri, L. E. N., Walsh, C. T., and Zuber, P. (1998) *Biochemistry* **37**, 1575–1584

ⁿ Reuter, K., Mofid, M. R., Marahiel, M. A., and Ficner, R. (1999) *EMBO J.* **18**, 6823–6831

^o Lawen, A., and Traber, R. (1993) *J. Biol. Chem.* **268**, 20452–20465

^p Hoffmann, K., Schneider-Scherer, E., Kleinkauf, H., and Zocher, R. (1994) *J. Biol. Chem.* **269**, 12710–12714

^q Shiau, C.-Y., Byford, M. F., Aplin, R. T., Baldwin, J. E., and Schofield, C. J. (1997) *Biochemistry* **36**, 8798–8806

^r Kallow, W., Kennedy, J., Arezi, B., Turner, G., and von Döhren, H. (2000) *J. Mol. Biol.* **297**, 395–408

^s Weckwerth, W., Miyamoto, K., Iinuma, K., Krause, M., Glinski, M., Storm, T., Bonse, G., Kleinkauf, H., and Zocher, R. (2000) *J. Biol. Chem.* **275**, 17909–17915

^t Milne, J. C., Roy, R. S., Eliot, A. C., Kelleher, N. L., Wokhlu, A., Nickels, B., and Walsh, C. T. (1999) *Biochemistry* **38**, 4768–4781

^u Gaitatzis, N., Kunze, B., and Müller, R. (2001) *Proc. Natl. Acad. Sci. U.S.A.* **98**, 11136–11141

^v Gewolb, J. (2002) *Science* **295**, 2205–2207

topoisomerase subunit.^{494–495a} Ribosomal protein L9 may play a role in the bypass process.

7. RNA Viruses

The RNA-containing bacteriophages are convenient sources of relatively simple mRNA molecules, whose sequences can be studied.⁴⁹⁶ The genetic information for these viruses is carried by RNA molecules

consisting of only 3500–4500 nucleotides and which may contain only four genes (p. 247). The RNA from phages f2, R17, MS2, and the more distant Q β have been studied intensively.^{483,497}

Parts of the 2569-nucleotide sequence for the RNA of phage MS2⁴⁹⁸ are shown in Fig. 29-17. The 5' end (upper left center) still bears the triphosphate group of the initiating GTP. Following a number of hairpin loops there is a ribosome-protected region, which begins with the initiation codon GUG for the A protein

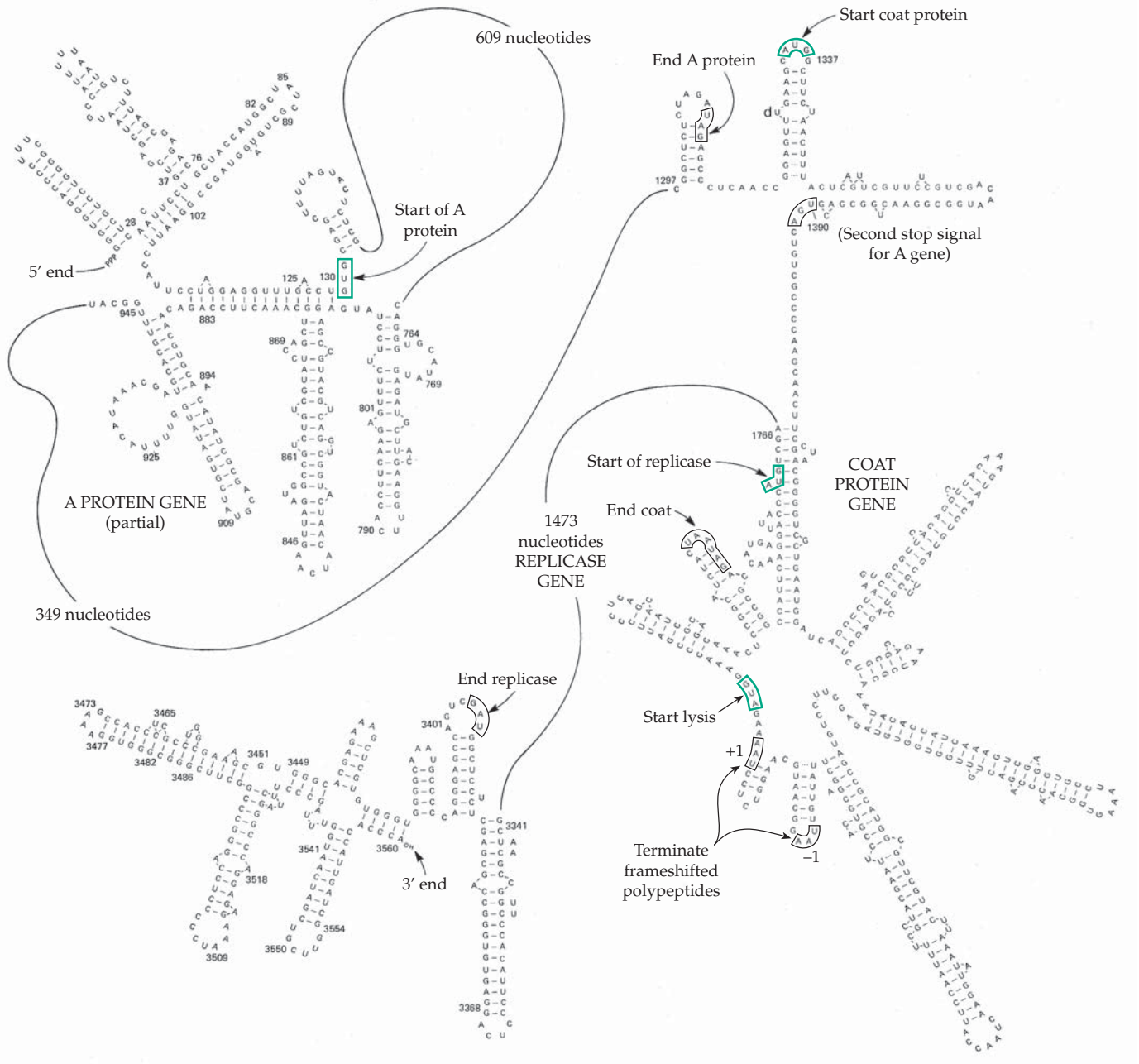


Figure 29-17 Partial sequence and secondary structure model of RNA of bacteriophage MS2. The initiation and termination codons for each of the three genes (A protein, coat protein, and replicase) are enclosed in boxes as is the second stop signal that is in-frame for the A protein gene but out-of-frame for the coat protein gene. The entire coat protein gene is shown but less than one-third of the entire sequence is given. From W. Fiers and associates.^{499–501}

(enclosed in a box). Here is some of the first direct evidence ever obtained that GUG as well as AUG is a biologically important initiation codon. Following the initiation codon the nucleotide sequence codes exactly for the established amino acid sequence of the protein. The termination codon UAG is also enclosed in a box in the figure. Following this is a short intergenic region, which includes one side of a hairpin loop with the initiator codon AUG for the next gene at the end. The nucleotide sequence following this codes exactly for the experimentally established sequence end of the coat protein.⁴⁹⁹ One other feature of the sequence shown is the UGA termination codon in a box shortly after the beginning of the coat protein gene (at position 1390). This termination signal is out of phase with the initiator codon AUG; hence, it does not represent a termination point for the coat gene. However, it is in phase with the UAG termination codon for the A protein. In the presence of various host amber (UAG) suppressor genes, the A protein is elongated and terminated at this UGA codon.

The coat gene, containing only 390 nucleotides, is shown in its entirety. The secondary structure proposed resembles a flower.⁴⁹⁹ The gene ends with a double stop signal UAAUAG. Following an intergenic sequence of 36 nucleotides the long replicase gene starts with an AUG codon. It ends at position 3395 leaving an untranslated segment of 174 nucleotides at the 3' end.

Initially it was thought that MS2 RNA contained only the three genes mentioned in the preceding paragraphs, but later it was found to have an additional gene required for lysis of the host cell.⁴⁸³ The initiation codon for this gene begins at position 1678 (Fig. 29-17) in the +1 reading frame. There is a UAA stop just two codons before this in the same frame and another UAA stop codon in the -1 frame beginning at position 1652. As a result of these stop codons any reading frameshift during synthesis of the coat protein yields mistranslated proteins that are terminated at these codons. There is no Shine-Dalgarno sequence to bind ribosomes for initiation in this region, but because the initiation codon for the lysis (L) protein is nearby, reinitiation occurs and the L protein is made in the relatively small amounts needed. This arrangement permits efficient use of the RNA by making use of overlapping genes. It also ensures that enough coat protein has been synthesized to make new virus particles before the L protein accumulation causes lysis.⁴⁸³

Many viral RNAs that are formed within eukaryotic cells lack a 5' cap. They depend upon internal ribosomal entry sites (IRESs). This has been studied most with picorna viruses.^{338,502,503} These viruses not only initiate translation at discrete sites in uncapped RNA but carry out a proteolytic cleavage of initiation factor 4G (Fig. 29-11), which seems to be necessary for initiation of viral-RNA translation.^{338,504} The IRES

region of hepatitis C viral RNA contains a complex pseudoknotted secondary structure that is necessary for initiation.^{346,505,506} Cryo-EM reveals a pronounced change in the 40S ribosomal subunit structure when the viral IRES binds.³⁴⁶ Some RNA viruses of plants have complex secondary structures in the untranslated 3' region that promote efficient initiation of translation.^{506a}

8. Other Functions of Ribosomes

In addition to making proteins, ribosomes also participate in regulatory mechanisms that influence the entire cell. One such mechanism is seen in the **stringent response**.⁵⁰⁷⁻⁵¹⁰ Many amino acid-requiring auxotrophs of *E. coli* and other bacteria, when deprived of an essential amino acid, respond by decreasing their production of ribosomal RNA, ribosomal proteins, purine nucleoside triphosphates, lipids, and other essential materials. However, mutations in the gene *rel* (relaxed) lead to continued production of rRNA even in the absence of an essential amino acid. (The stringent response is "relaxed.") It was observed that the **guanosine polyphosphates ppGpp** and **pppGpp**, originally termed MS or "magic spot" compounds, accumulate in stringent (*rel*⁺) strains to a concentration of ~1 mM but not in relaxed (*rel*⁻) strains. Guanosine polyphosphates are synthesized on the ribosomes by transfer of a pyrophospho group from ATP (Eq. 29-11):



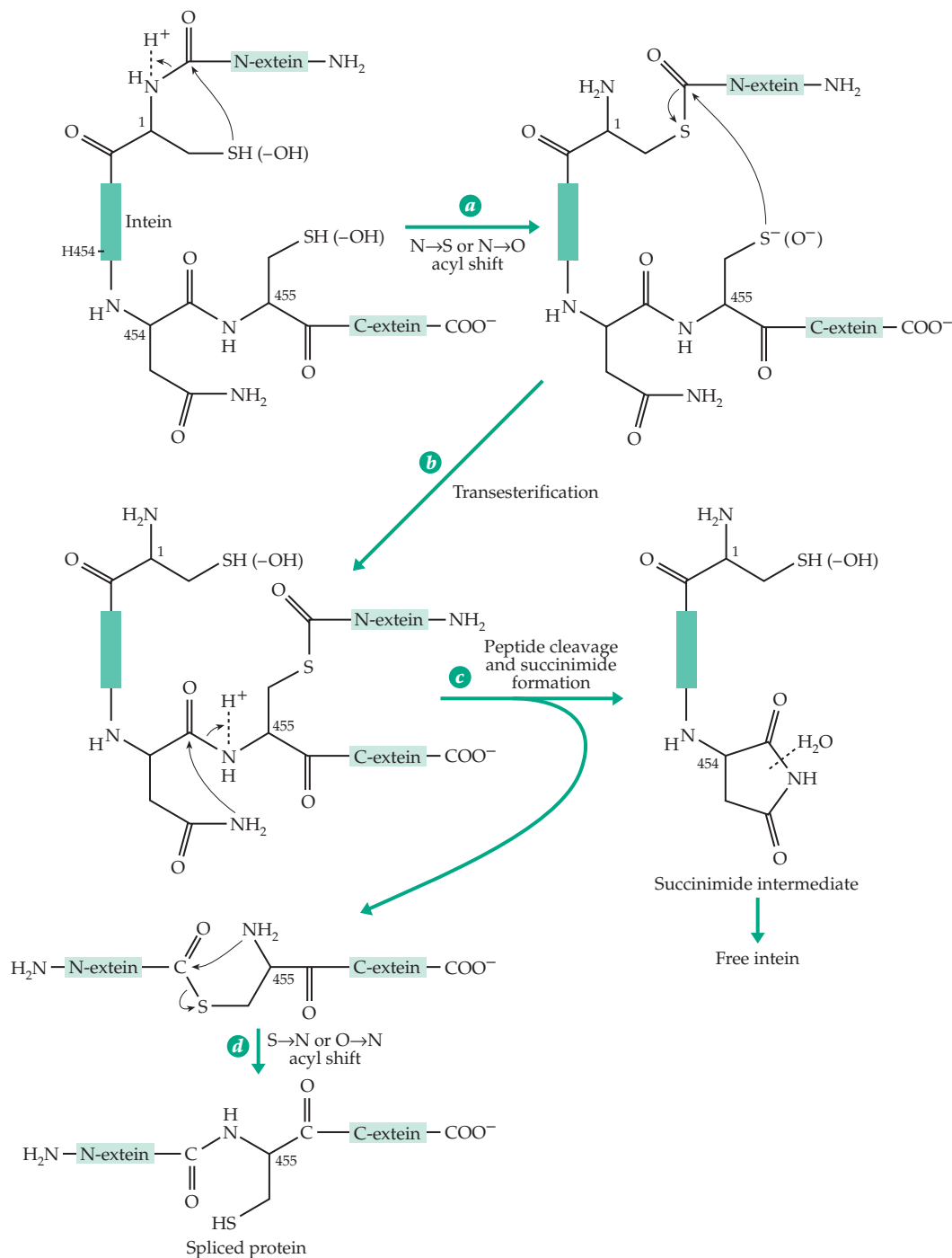
The reaction is catalyzed by the 84-kDa ppGpp synthetase (**stringent factor**), which is encoded by the *rel* gene and is present only in stringent strains.^{511-512a} It binds to ribosomes and becomes active only if mRNA is bound to the ribosomes and if codon-selected uncharged tRNA is present in the A sites. A second ppGpp synthetase (PSII) is encoded by gene **spoT**, which also codes for a ppGpp hydrolase.⁵¹⁰ The presence of an uncharged tRNA in the ribosomal A site is expected during amino acid starvation. The stringent factor competes with elongation factor EF-G for its ribosomal site.⁵¹³

The most important effect of accumulating ppGpp may be to bind to an allosteric site on RNA polymerase.⁵⁰⁹ The ppGpp-polymerase complex appears to be inefficient in initiating transcription of genes for rRNA, other stable RNAs, and ribosomal proteins. However, it stimulates expression of various amino acid biosynthetic genes and catabolic genes, perhaps via the "discriminator sequence" (Chapter 28; p. 1608). This is not the only effect of ppGpp. The fidelity of translation is decreased when amino acid concentrations fall and

BOX 29-D PROTEIN SPLICING, INTEINS, AND HOMING ENDONUCLEASES

Like self-splicing RNAs, which excise introns from their chains, a few proteins are able to splice out segments of their own chains as **inteins**. The surrounding protein sequences can be referred to as **exteins**. Over 100 self-splicing proteins are known. They are found in all kingdoms of life.^{a-d} The inteins, which are excised, are typically 50 kDa in size but range from ~360 to over 500 residues.

The mechanism of splicing is related to the chemistry of pyruvoyl enzyme activation (Eq. 14-41), succinimide formation from asparagine residues (Eq. 2-24), and protein carboxymethylation (Box 12-A). The intein always contains serine or cysteine in its N-terminal (1)-position and asparagine in its C-terminal position. The latter is always followed by cysteine, serine, or threonine in the N-terminal



BOX 29-D (continued)

position of the C-extein. The penultimate residue in the intein is usually (~90 %) histidine, which is thought to play a catalytic role. Other residues in the catalytic domains, which form the ends of the inteins, may also participate in catalysis.

One of the first inteins discovered was found in the 119-kDa precursor to a subunit of a vacuolar ATPase of yeast.^{a,c} In this 50-kDa intein Thr 72, His 75, and His 197 may have catalytic functions.^d The intein is spliced out to form the 69-kDa subunit. The splicing mechanism, which is illustrated for this intein, is shown in the accompanying equations.^{b,d-g} Step *a* is an N → S or N → O acyl shift. This is followed by transesterification (step *b*) which involves either thioesters (as illustrated) or oxygen esters. Formation of a succinimide intermediate (step *c*) releases the intein and the spliced protein. The latter must undergo an S → N or O → N acyl shift (step *d*), and the succinimide in the extein must be hydrolyzed to complete the process.

Why do cells ever splice proteins? It isn't clear. However, a curious fact is that many inteins are **homing endonucleases**.^{h-k} The genes for these nucleases are often present in introns in mRNA, and the homing endonuclease often cuts DNA in such a way as to initiate movement of its own gene (Chapter 27). The endonuclease itself is found in the center of the intein between the two end domains, which contain the catalytic centers for the splicing reaction.

A few cases are known in which proteins undergo *trans* splicing. For example, the *dnaE* gene of *Synechocystis*, which codes for DNA polymerase III,

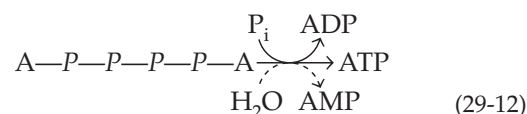
is actually two partial genes that are 745 kb apart and on opposite strands of the DNA. One of the partial genes codes for a protein containing the N-terminal splice site for an intein, and the other gene codes for a polypeptide containing the C-terminal splice site. Evidently the two splicing domains associate and then catalyze the splicing sequence in the usual way. Split inteins have become very useful in protein engineering because they can be used to join various polypeptide sequences.^{k-m} They have also provided an efficient system for purification of specific proteins.^{b,n}

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ribosomal action slows. However, ppGpp apparently binds to the ribosome and slows the binding of the aminoacyl-tRNA•EF-Tu•GTP complex. This allows more time for rejection of mispaired tRNAs and increases the fidelity of translation.⁵¹⁴ Under conditions of nutrient starvation the accumulating ppGpp may promote enzymatic degradation of unneeded proteins^{512a,b} and may also induce programmed cell death.⁵¹⁵

Another “alarmone” that regulates both transcription and DNA replication and other cell functions is diadenosine tetraphosphate (Ap₄A). Effects of Ap₄A and related compounds have been discussed in Chapter 28 (p. 1635). These compounds affect many biological events including replication, growth, and differentiation.⁵¹⁶ However, the synthesis of Ap₄A is a reaction not of ribosomes but of an aminoacyl-tRNA synthetase. An enzyme-bound aminoacyl adenylate carries out adenylation of ATP rather than amino-

acylation of tRNA, especially when Zn²⁺ is present. Ap₄A is abundant in blood platelets, where it is stored in dense granules.⁵¹⁷ Both Ap₂A and Ap₃A accumulate as granules in myocardial tissues,⁵¹⁸ and Ap₅A and Ap₆A are also present in adrenal chromaffin cells, in blood platelets, and in synaptic vesicles.⁵¹⁹ These compounds are catabolized by hydrolases or in lower eukaryotes by phosphorylases. For example, Ap₄A may be converted into ATP + AMP or converted into ATP and ADP (Eq. 29-12).^{516,520}



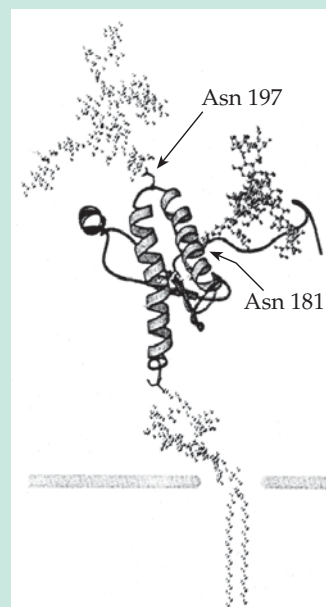
A quite different role of ribosomes is to regulate the life span of certain mRNA molecules. The best studied example is the mRNA for the microtubule

BOX 29-E PRIONS AND AMYLOID DISEASES

The mysterious prions (proteinaceous infective agents), which are described briefly on p. 248, are under intensive investigation. Prion diseases affect fewer than one in 100,000 persons, but there is fear of a possible epidemic. Furthermore, there is a close relationship of prions to a large family of **amyloid diseases**. The most frequent of these is Alzheimer disease, which is estimated to affect one-third of people over 85 years of age in the United States.^{a,b}

Prion diseases include **scrapie** of sheep and goats, **bovine spongiform encephalopathy (BSE or mad cow disease)**, **chronic wasting disease (CWD)** of deer and elk, and the human diseases **kuru**, **Creutzfeldt-Jakob disease (CJD)**, **Gerstmann-Sträussler-Scheinker syndrome (GSS)**,^c and **fatal familial insomnia (FFI)**.^{a,d-f} The diseases have a variety of symptoms that include dementia, ataxia (loss of muscular coordination), insomnia, and behavioral problems. All involve some loss of neurons, which may or may not be indicated by a sponge-like appearance of the brain. A characteristic feature of prion disease is the appearance of **amyloid (starch-like) plaques**, which consist of fibrils of insoluble protein.^a Exhaustive attempts failed to identify a virus particle or an associated DNA or RNA. On this basis, Stanley Prusiner suggested that the diseases are transmitted by pure proteins.^a All of the diseases seem to involve the same protein, which is known as the **prion protein (PrP)**. It is encoded by a single-copy gene on human chromosome 20.^g The amino acid sequence of the C-terminal region of PrP is highly conserved among all animals. However, there are more than 20 known human genetic variants, and a second prion protein has been found in mice.^h The function of the normal cellular prion protein (**PrP^C**) is unknown, but it appears to be a copper ion carrier, which may be essential to proper synaptic function.^{i,j} “Knockout mice” lacking PrP are resistant to prion disease^{j,k} but may not be completely healthy.

If it were not for the diseases, PrP^C might be viewed as just another cell surface glycoprotein. Determination of its three-dimensional structure has been difficult, but use of NMR spectroscopy and modeling has given a nearly complete picture, which is shown in the accompanying drawing.^{d,l-n} The 250-residue (~220 residues after removal of N- and C-terminal signal sequences) has a long N-terminal tail, a glycosylated globular domain, and a C terminus that is anchored in the outer membrane of neurons by a glycosylphosphatidylinositol (GPI) anchor similar to that shown in Fig. 8-13. The globular domain contains three α helices, a small β sheet, and two glycosylation sites. These last carry



typical N-linked, branched sialic acid-containing oligosaccharides with a total of 52 or more sugar residues. The N-terminal 120 amino acid residue “tail” appears to be largely unstructured. However, it contains five octapeptide repeats with the consensus sequence PHGGGWGQ, each able to bind one Cu^{2+} or Mn^{2+} ion.^{d,i,o,p}

How can this ordinarily harmless protein become a killer? The prion is a 20- to 30-kDa hydrophobic particle, which is thought to arise from PrP^C by a conformational alteration in which the α helices are largely changed into a β structure. The new conformer is often designated PrP^{Sc} or PrP-res. The latter abbreviation arises from the fact that native PrP^C can be completely hydrolyzed to small fragments by proteinase K, but PrP^{Sc} contains a 142-residue extremely resistant core (residues 90–231), which is not hydrolyzed and is over 80% β sheet.^q Evidently the PrP^{Sc} form is able to associate to form a “seed” that, when conditions are favorable, can induce the conformational change in other molecules spreading the PrP^{Sc} form throughout the brain and even into tissues of the immune system.^r With prion diseases and other amyloid diseases the body may be able to fight off the process by normal proteolytic turnover of the prion protein.

About 85% of all cases of prion disease are **sporadic CJD**. These are thought to arise by spontaneous conversion of PrP^C to PrP^{Sc}. Inherited (familial) forms of CJD, GSS, and FFI are also known. A series of point mutations as well as expansion of the octapeptide repeats^s account for the various diseases, which have an autosomal dominant inheritance. At least 23 pathogenic mutations have been

BOX 29-E (continued)

reported.^{t,u} The point mutations occur at several locations, some of them adjacent to the glycosylation sites.^m These mutant proteins may be more readily converted to the less soluble PrP^{Sc} type structure, initiating the disease process. However, a mutant with a stop codon (TAG) in the place of the tyrosine 145 codon loses its C-terminal anchor and is degraded rapidly in the proteasomal pathway.^t

The infectious forms of prion diseases are more puzzling. They account for less than one percent of all cases. Attention was first focused on kuru, a disease of the Fore people of New Guinea. In earlier times they practiced a ritualistic cannibalism of brain tissue that apparently propagated the disease, which is now nearly extinct. Of present concern are over 100 cases of a “new variant” form of CJD, some involving teenage persons and young adults, which have been reported in Europe.^{u,v} This disease may have originated in sheep, then jumped to cattle, where it was spread by the ingestion of prion-contaminated meat and bone meal.^a In addition, more than 120 cases of CJD have arisen from injection of prion-contaminated human growth hormone. Other cases have been traced to contaminated surgical instruments, to tissue grafts, and to use of contaminated human pituitary gonadotrophin.^a

A hard-to-understand aspect of the “protein-only” theory of prion diseases is the existence of various “strains” of prion proteins. These do not involve differences in amino acid sequence but differences in the conformations of the PrP^{Sc} forms and in the glycosylation patterns.^{d,m,w} How can there be several different conformations of the same protein, all of which seed the conversion of normal PrP into differing insoluble forms? In spite of this puzzle, support for the explanation of strain differences comes from a yeast prion system, which involves transcription termination factor eRF3.^{x-z} In this system, which involves a prion whose insoluble form can be redissolved by guanidine hydrochloride,^{aa} differing strains have also been described.^{y,bb,cc} Nevertheless, the presence of the various strains of animal prions, as well as observed vaccination of inbred mice against specific strains,^{dd} may be more readily understood if the disease is transmitted by an unidentified virus rather than by a pure protein.^{r,u,ee,ff} In fact, the diseases have not been successfully transmitted by truly virus-free proteins synthesized from recombinant DNA.^{ee}

What are the prospects for a cure for prion diseases? Several compounds show some effect in slowing accumulation of amyloid plaques,^{d,v,gg,hh} but suitable drugs have not been developed. Prevention is the best cure, but more needs to be

known about the basic biology of the disease transmission before effective strategies for prevention can be developed.^u

What is the nature of the insoluble forms of the prion protein? They are hard to study because of the extreme insolubility, but the conversion of α helix to β sheet seems to be fundamental to the process and has been confirmed for the yeast prion by X-ray diffraction.ⁱⁱ It has been known since the 1950s that many soluble α -helix-rich proteins can be transformed easily into a fibrillar form in which the polypeptide chains are thought to form a β sheet. The chains are probably folded into hairpin loops that form an antiparallel β sheet (see Fig. 2-11).^{jj-ll} For example, by heating at pH 2 insulin can be converted to fibrils, whose polarized infrared spectrum (Fig. 23-3A) indicates a **cross- β structure** with strands lying perpendicular to the fibril axis.^{jj,mm} Many other proteins are also able to undergo similar transformation. Most biophysical evidence is consistent with the cross- β structure for the fibrils, which typically have diameters of 7–12 nm.^{ii,ll,nn} These may be formed by association of thinner 2 to 5 nm fibrils.^{oo} However, β -helical structures have been proposed for some amyloid fibrils^{pp} and polyproline II helices for others.^{qq}

A wide range of human diseases involving amyloid deposits are known. These include not only the prion diseases and the neurodegenerative diseases, Alzheimer, Parkinson, and the polyglutamine repeat diseases (Table 26-4),^{rr,ss} but also **systemic amyloidoses**.^{tt} Among the latter are deposits of transthyretin,^{uu} the 37-residue **amylin** that develops in the β cells of the pancreas in type II diabetes,^{vv} mutant forms of lysozyme,^{ww} and of β 2 microglobulin,^{xx} and gelsolin.^{yy} A serum protein amyloid P, a calcium-binding protein, is usually also a component of amyloid deposits.^{zz}

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BOX 29-E PRIONS AND AMYLOID DISEASES (continued)

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proteins tubulin.^{521–523} Accumulating α and β tubulin subunits act in a feedback loop to induce the degradation of their mRNA. They do this by binding to the N-terminal sequence MREI of β tubulin as it is emerging from ribosomes. This binding allosterically activates an endonuclease that cuts the polysome-bound mRNA. A similar mechanism for tubulin mRNA may involve the MREC N-terminal sequence of that protein.

An unexpected finding was that **phosphatidylserine synthase** of *E. coli* is tightly bound to ribosomes.⁵²⁴ This enzyme, which incorporates serine into phospholipids according to step *h* of Fig. 21-4, is responsible for synthesis of the principal membrane

lipid of *E. coli*. Its localization on ribosomes may be linked to the joint regulation of the synthesis of proteins and lipids.

D. Processing, Secretion, and Turnover of Proteins

The concepts of processing and turnover of proteins have been introduced in Chapter 10, and many details have been presented in other chapters. However, as we complete our discussion of protein synthesis, it is appropriate to discuss processing further. As

polypeptide chains leave the ribosomes via the exit channels, they may follow several different paths. They may enter the cytosol and fold quickly into a compact form. This may require only a few seconds, whereas the translation process in the ribosome may take many seconds. The folding will therefore be **cotranslational**.⁵²⁵ Depending upon the N-terminal signal peptide the protein may later unfold and pass through a membrane pore or **translocon** into the endoplasmic reticulum (ER), a mitochondrion, chloroplast, or peroxisome. Wherever it is, it will be crowded together with thousands of other proteins. It will interact with many of these, and evolution will have enabled some of these to become chaperones (discussed in Chapter 10).⁵²⁶

A single rapidly growing cell of *E. coli* may contain $\sim 2.3 \times 10^6$ soluble polypeptide chains of ~ 2600 different types with an average length of ~ 317 residues and average mass of ~ 35 kDa.⁵²⁷ These are chaperoned in a variety of ways,^{528,529} not only in the cytoplasm but in the periplasm (see p. 364).^{530,531} The three chaperones **trigger factor** (TF), DnaK, and GroEL participate in folding newly formed proteins.^{525a,b} TF is a prolyl-*cis-trans* isomerase (Box 9-F), which associates with the large ribosomal subunit with a 1:1 stoichiometry.^{525b-d} DnaK and related chaperones hold and protect newly formed polypeptides in extended conformations, while the GroES-GroEL chaperonins assist folding within their internal cavities (Box 7-A).^{525b,e-g} **Prefoldin**, a 90-kDa complex,^{525b} has a special function in chaperoning microtubule subunits. A Type II chaperonin also assists the folding of actin and tubulin.^{525h} Chaperones assist not only in the folding of proteins but also in translocation into the ER (e.g., by the Hsp 70 homolog BiP), and into mitochondria and other organelles.^{532,533} **Co-chaperones** are additional proteins that act as selective agents to direct proteins to a particular chaperone. For example, the DnaJ protein is a scanning factor for the Hsp 70 chaperone DnaK. As is described on p. 518, it catalyzes ATP-dependent association of a substrate with the binding cavity of DnaK.⁵³³ A chaperone whose function has long remained elusive is an abundant secreted glycoprotein known as **clusterin**.⁵³⁴ It seems to have a protective function in protection against stress.

How are the possible choices for newly formed proteins made? Much seems to depend upon the amino acid sequences at the ends of the polypeptide chains. As they emerge from a ribosome, some N-terminal signal sequences bind to recognition proteins. One such protein labels the ends of proteins destined for secretion into the vesicles of the ER. This protein ensures that the protein end binds to the **signal recognition particle** (SRP), enters a translocon pore, and undergoes cotranslational passage into the periplasmic space in bacteria or the ER in eukaryotes. Cotranslational modification reactions also occur both in the

cytosol and in the ER vesicles. These too influence the choice of destinations as do additional signal sequences in the polypeptide chains. Proteins may be directed to the various organelles, to residence in membranes,⁵³⁵ or to secretion into the external medium. It was somewhat surprising to discover that under some circumstances most newly synthesized proteins are degraded in proteasomes.⁵³⁶ Cotranslational degradation of proteins with imperfect ends may account for some of this.⁵³⁷⁻⁵³⁹ In addition, imperfect proteins that are retained in the ER may be sent back into the cytoplasm for degradation in proteasomes.⁵⁴⁰

1. Cotranslational and Posttranslational Processing

The modifications that lead to the presence of fully functional proteins in their proper locations begin while peptide chains are still emerging from the ribosomes.⁵⁴¹ In bacteria and in eukaryotic organelles the first of these modifications is hydrolytic removal of the N-formyl group by an Fe²⁺-dependent **deformylase** leaving the N-terminal methionine.⁵⁴²⁻⁵⁴³ Deformylases are present in eukaryotes as well as in bacteria, making the deformylase a less attractive target for antibiotic design than has sometimes been proposed.^{544,544a} When the chain is only 20–30 residues long, the terminal methionine that remains after deformylation may be removed by a ribosome-associated methionine aminopeptidase.⁵⁴⁵ The methionine is usually removed if followed by P, G, A, S, or T and is retained if followed by K, R, L, I, F, or N. With other amino acids removal is variable.⁵⁴⁶ A ribosome-bound N-acetyltransferase may acetylate the N terminus either before or after removal of Met.⁵⁴⁷ Approximate rules for eukaryotic cells are⁵⁴¹:

N-terminal D, E, N	Acetylation without removal of Met
N-terminal P, V, C	Removal of Met; no acetylation
N-terminal G, A, S, T	Removal of Met followed by acetylation
Other N termini	No modification

An example is provided by actin, which contains acetyl-Met-Asp, acetyl-Met-Gln or acetyl-Met-Cys-Asp at the N terminus immediately after synthesis. Then, within ~ 15 min the acetyl-Met is cleaved off, and the next terminal residue is acetylated.⁵⁴⁸ N-Acylation of nascent peptides by fatty acyl groups can also occur cotranslationally. For example, 14-carbon myristoyl groups are added in amide linkage to the N-terminal glycines of many cellular and virally encoded proteins.^{535,549,550} This may take place on the ribosomes,⁵⁵¹

but it is often not clear whether the modification is cotranslational or posttranslational. The same may be said of many other “posttranslational” alterations, many of which may begin on a nascent polypeptide chain. Fatty acyl groups (mainly palmitoyl) may form thioester linkages with cysteine side chains.⁵⁵² This often occurs near the C terminus (see p. 559). Other C-terminal modifications include prenylation (p. 559)⁵⁵³ and attachment to diacylglycerols via thioester linkages to cysteine (pp. 402, 428)^{553a} or to glycosylphosphatidylinositol glycan anchors (Fig. 8-13; Eq. 20-23).

The addition of an N-terminal myristoyl group to a protein causes a relatively permanent alteration as does methylation of histidine, lysine, or arginine side chains.^{554,555} So do hydroxylation, vitamin K-dependent carboxylation (Eq. 15-55), and many other alterations. In contrast, glycosylation, phosphorylation, and sulfation produce reversible alterations. Sometimes, as in the conversion of proenzymes to active enzymes, a modification step is used to generate a catalytic activity. In other instances, as in the processing of glycoproteins in the Golgi, the major function of the modification reaction seems to be one of directing a protein to the correct intracellular location.

2. Forming Functional Proteins

Proenzymes and other precursor proteins are often almost totally inactive until they are activated by some alteration that occurs when they reach their destination in a cell or in the body. Cleavage of the polypeptide chains of proenzymes, covalent attachment of coenzymes,^{556,557} oxidation (Eq. 16-57) or halogenation (Eq. 25-6) of tyrosine or tryptophan⁵⁵⁸ side chains, and oxidation of cysteine in the sequence CTPSR to formylglycine (in sulfatase formation; Eq. 12-44)⁵⁵⁹ are only a few of many modifications needed to form functional proteins. Sometimes, as in activation of chymotrypsinogen, a single simple modification creates the active protein. In other cases modification may be quite complex. For example, although many polypeptide antibiotics are formed by nonribosomal synthesis (Box 29-C), some are created on ribosomes and may require extensive subsequent alteration. An example is **microcin**, a 69-residue peptide antibiotic formed by some strains of *E. coli*. Eight Gly-Cys and Gly-Ser pairs in a pre-microcin chain are cyclized to thiazole and oxazole rings. Then the 69-residue antibiotic is cut out from the precursor and secreted into the medium.^{560,561} The 22-residue antibiotic **epidermin** is one of a family of **lantibiotics** that contain lanthionine as a characteristic component. Biosynthesis involves dehydration of serine and threonine residues, sulfide (thioether) bridge formation, oxidative decarboxylation, and removal of a leader peptide.⁵⁶²

3. Translocation through Membranes

The processes by which proteins are selected for secretion into the periplasmic space of bacteria or into the vesicles of the endoplasmic reticulum of eukaryotic cells are similar and have been discussed in Chapter 10 (pp. 519–521). However, some details are still being worked out. The first step in translocation is binding of the N terminus of a protein that is emerging from a ribosome to the **signal recognition particle**.^{563,564} The core of this particle has a universally conserved structure consisting of two proteins and an RNA molecule.^{565–571b}

Bacteria	Eukaryotes
4.5S RNA (~114 nt)	7S RNA (~295 nt)
Protein Ffh	Protein SRP54
Protein Ftsy (SRP receptor) ⁵⁷²	Protein SR α

All of these proteins are GTPases, and in eukaryotes SR α is associated with a third protein SR β , which is also a GTPase.⁵⁷³ Either protein Ffh or SRP54 recognizes the N terminus of the protein that is to be translocated, chaperoning it to the receptor Ftsy⁵⁷⁴ or SR α , where it may be anchored to the translocating pore (**translocon**). Hydrolysis of GTP by both proteins accompanies the recognition process. The eukaryotic SRP is more complex, containing six proteins and a larger RNA than in bacteria.^{568,575–576} One domain of the 7S RNA is homologous to the bacterial 4.5S RNA, while an additional domain is closely related in its sequence to that of the highly repetitive *Alu* sequences in DNA (Fig. 27-9). However, the significance of this similarity is unclear.

In eukaryotic cells binding of SRP54 induces a transient retardation of translocation, an **elongation arrest**, while the SRP complex binds to its receptor SR α . This 70-kDa peripheral membrane protein is tightly associated with the 30-kDa integral membrane protein SR β . Binding to this receptor leads the nascent polypeptide chain from the ribosome directly into the Sec61 translocon,^{33f,576a} which consists of α , β , and γ subunits and a central aqueous pore. The ribosome apparently also becomes bonded firmly to the translocon until synthesis of the polypeptide chain has been completed.^{573,577,578} The translocon complex also contains additional components^{563,577,579} including the **leader peptidase** (signal peptidase, p. 620)^{563,580–582a} and the **oligosaccharyltransferase** of ER membranes (Eq. 20-21).^{563,583} The latter transfers an oligosaccharide from a lipid carrier onto certain asparagine side chains of polypeptides entering the ER. This and other glycosylation reactions help to keep the polypeptide moving to its correct destination, whether it be in some membrane surface, a lysosome or other organelle, or a secretion vesicle (Chapter 20). Furthermore,

some proteins are translocated by a mechanism that doesn't depend upon SRP but utilizes a different complex, which consists in yeast of proteins Sec62, Sec63, Sec71, Sec72, and Kar2 (the chaperone BiP present in the ER lumen).⁵⁷⁷

Translocation of most bacterial proteins occurs posttranslationally rather than cotranslationally.⁵⁸⁴ After recognition by SRP the polypeptide chains are transferred to chaperone complexes.⁵⁸⁵ Some proteins are escorted to the folding compartment of the GroEL-GroES chaperonin (Box 7-A). Those that are to be secreted are often chaperoned by the protein **SecB**.⁵⁸⁶ Genetic analysis shows that for *E. coli* the secretion of many proteins requires the products of genes *secA*, *secB*, *secD*, *secE*, *secF*, *secG*, and *secY*. Gene *secA* encodes a 92-kDa cytoplasmic ATPase protein (**SecA**), SecB is a 64-kDa homotetrameric chaperone that prevents folding of preproteins prior to export, and SecY is a 42-kDa integral membrane protein.^{587,588}

A complex of the three transmembrane proteins **SecYEG** forms the translocation channel in *E. coli* membranes. **SecY** and **SecE** are essential components in most bacteria and are homologous to components of the eukaryotic **Sec61p** translocon complex.⁵⁸⁹ From their sequences SecY, SecE, and SecG are predicted to have ten, three, and two transmembrane sequences, respectively. Additional accessory proteins in the complex are designated SecD, SecF, and yajC. Their functions are uncertain, and they are not essential for transport. The driving force for translocation is provided by the peripheral ATPase SecA.^{588,590} For many proteins the signal sequence, which is usually positively charged, stays on the negatively charged cytoplasmic surface of the membrane, while SecA in an ATP-dependent process pushes a loop of protein through the membrane. Hydrolysis of a second ATP molecule seems to be required to release SecA, allowing it to reload with ATP and to assist the next 20–30 residue polypeptide sequence to be translocated.^{591,592} Surprisingly, SecA is also an ATP-dependent RNA helicase.⁵⁹⁰ The significance of its apparent ability to translocate along either an RNA or a polypeptide chain is unclear. The protonmotive force provided by the membrane potential is another important factor in the translocation of many proteins.^{591,593–596} Yet another factor is the lipid composition of the membrane. Non-bilayer lipids seem to be required for efficient transport.⁵⁹⁷

As is mentioned in Chapter 10, bacteria have additional mechanisms of polypeptide transport. A recently recognized Sec-independent pathway is used by *E. coli* and many other bacteria to secrete proteins that contain the twin-arginine motif RRX $\phi\phi$, where ϕ is a hydrophobic amino acid, in their N-terminal signal sequences. Proteins encoded by genes *tatABC* are required by this **Tat pathway**.^{594,598,599} Related pathways have been identified both in mitochondria and

in chloroplasts. Small peptides may pass out of the periplasmic space through the porins in the outer membrane of gram-negative bacteria. However, larger proteins require conduit molecules such as the **TolC** channel-tunnel, which directly connects an inner membrane translocon with a channel in the outer membrane of *E. coli* cells.^{600,601}

Eukaryotic cells also have additional transport mechanisms. One of these is an ABC transporter (p. 417) known as the **transporter associated with antigen processing (TAP)**. It carries small polypeptides generated by proteasomes from the cytosol into the ER for export and binding to **MHC Class I** molecules and subsequent presentation to the immune system (Fig. 31-15).^{540,602}

4. Translocation into Organelles

Most of the proteins of mitochondria are encoded in nuclear DNA and are synthesized on cytoplasmic ribosomes. Mitochondria do not utilize proteins homologous to those of the bacterial Sec system but have their own set of transport proteins.^{603–605} These proteins, which include an outer membrane complex (Tom) and an inner membrane complex (Tim), are discussed in Chapter 18 (see Fig. 18-4). Perhaps these specialized mitochondrial proteins are needed because transport into the mitochondrial matrix is in an opposite direction to the transport out through bacterial membranes.

The transport of proteins into chloroplasts also occurs by more than one mechanism. An SRP-dependent pathway may be needed only for insertion of proteins into membranes.⁵⁹⁴ Other proteins, among which are the 23-kDa and 16-kDa photosystem II proteins (Chapter 23), enter by a pathway related to the Tat pathway of bacteria. In thylakoids this pathway is directly dependent upon the large pH difference (Δ pH) across the thylakoid membrane. In contrast to the bacterial Sec pathway, the Δ pH pathway seems to be able to transport completely folded proteins.

Proteins destined for peroxisomes have their own targeting signals. One of these (**PTS1**) is the sequence SKL at the C terminus. A second signal (**PTS2**) is an N-terminal nonapeptide (R/K)(L/V/I)X₅(H/Q)(L/A).^{606,607}

5. Membrane Proteins

Some proteins enter membranes immediately after synthesis. The translocon channel is not required. However, in *E. coli* an additional protein **YidC** is needed.⁶⁰³ Homologs of this protein are found in mitochondria (**Oxa1** protein) and in thylakoid membranes of chloroplasts (**Alb3** protein).⁶⁰⁸ These proteins may function in cotranslational insertion. If a protein carries a

positively charged N-terminal region, it will tend to stick to the negatively charged cytoplasmic surface of a cell membrane. This “positive inside” rule (p. 401)⁶⁰⁹ is strong for bacterial proteins but somewhat weaker for eukaryotic cells. A second topological rule is that hydrophobic segments of proteins will be attracted to membrane surfaces and can enter the membrane (perhaps via translocon pores) as loops (Fig. 29-18).⁵⁹⁶ Passage of the loop out through the membrane will be facilitated if negatively charged groups are present in the loop and are acted upon by the membrane potential (pp. 401,402).^{595,610,611} If the entire polypeptide chain follows the loop out through the membrane, the protein will be anchored to the inside of the cell membrane with its C terminus outside. On the other hand, if the C terminus also has a positively charged cluster nearby, a membrane associated **leader peptidase** (or signal peptidase) may cut the loop past the signal sequence on the outside of the membrane leaving the bulk of the protein with its N terminus outside (Fig. 29-18B). How are **polytopic** integral membrane proteins with multiple cytoplasmic and external loops formed? Hydrophobic signal sequences are not always at the N terminus of a polypeptide chain. Suitable **internal signal sequences** may be found in the sequences that form the transmembrane helices, e.g., those present in the many 7-helix receptors found in a membrane. This suggests the possibility that successive loops may be translocated. If the N terminus is allowed to pass through a translocon in one of the steps, the topology of Fig. 11-6 or of Fig. 23-41 will result.⁶¹¹

Genetic methods have also been applied to study the insertion of coat subunits of phage M13 into the plasma membrane of *E. coli*.^{612–614} The subunits are stored in the plasma membrane waiting to form a cylindrical shell about a viral DNA molecule as it is

extruded from the bacterium.⁶¹² The rod-like subunits (Fig. 7-7) have their N termini in the periplasmic space and their C termini in the cytoplasm. Each end carries a cluster of electrically charged residues, mostly negative at the N terminus and positive at the C terminus. Insertion into the membrane occurs only if the membrane has its normal membrane potential with a positive external surface charge and a negative internal charge, complementary to the charges on the coat subunit. Insertion does not occur unless the leader peptide with its positively charged N-terminal cluster and the C-terminal positive cluster are both present.^{610,615} This suggests insertion by the loop mechanism of Fig. 29-18A. Genetic studies of the *E. coli* leader peptidase revealed that this protein also has an internal signal sequence, which becomes inserted into the membrane and which is not cleaved.^{612,616,617} The final orientation of the mature enzyme is indicated in Fig. 29-18C.

Targeting of proteins to specialized domains of a membrane are less well understood. These include caveolae and lipid rafts, domains that are high in cholesterol and sphingolipids and which function in endocytosis and in cell signaling. A recent proposal is that proteins with hydrophobic surfaces needed in these domains become coated with a lipid “shell” before entering the membrane.^{617a}

6. Secretion of Proteins

Cells continuously secrete materials via small cytoplasmic vesicles, which in eukaryotes arise largely from the Golgi apparatus (pp. 425–427; Fig. 20-8). The vesicles of this **constitutive pathway** may have diameters of ~50 nm. They carry phospholipids, proteins, and other constituents for incorporation into the plasma membrane of the cell.^{618,619} In addition, there are

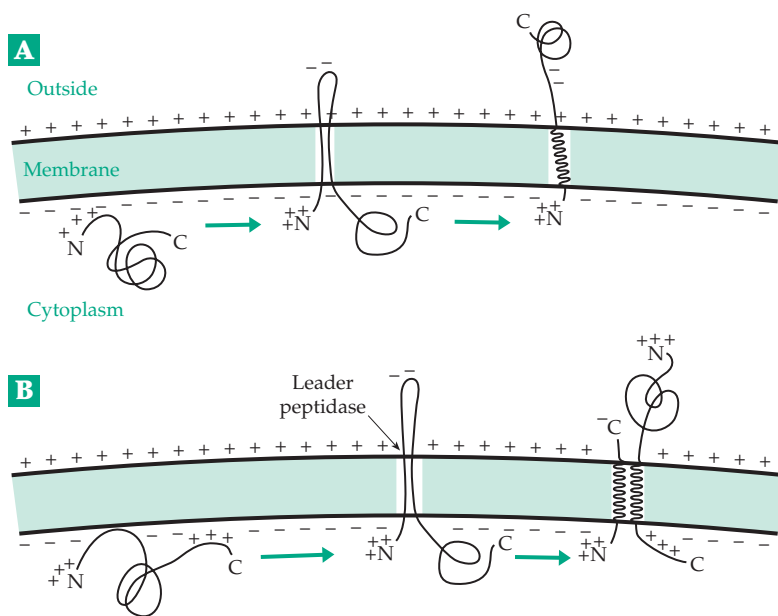
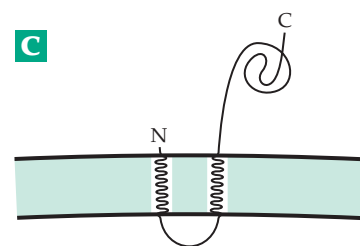


Figure 29-18 (A) Proposed mechanism for insertion of a loop of polypeptide chain through a translocon pore in a membrane with the positively charged N terminus anchored to the negatively charged inner membrane surface. (B) Cleavage of a polypeptide loop formed as in (A) by a leader peptidase to give a polypeptide chain anchored by a positively charged cluster near its C terminus. (C) Membrane topology of the *E. coli* leader peptidase. The active site is in the periplasmic domain. See Tschantz *et al.*⁵⁸⁰



BOX 29-F THE YEAST TWO-HYBRID SYSTEM FOR IDENTIFYING PROTEIN-PROTEIN INTERACTIONS

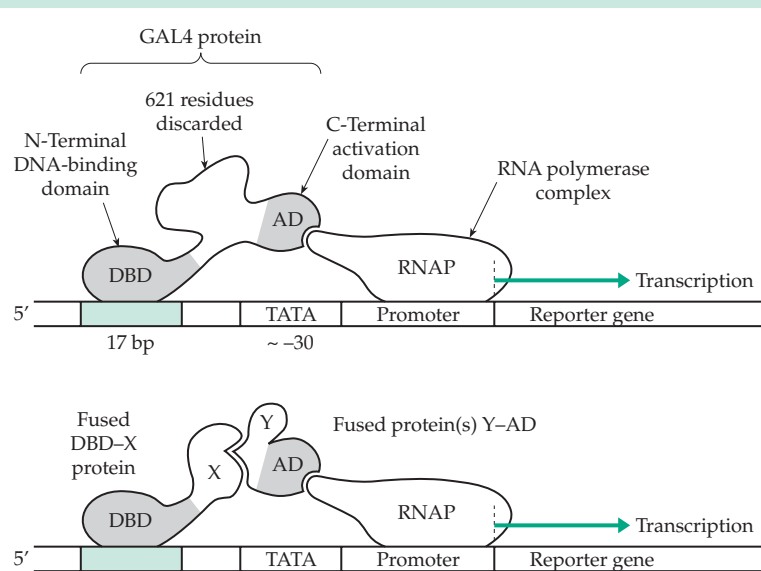
Many techniques including ultracentrifugation, chemical crosslinking, and X-ray crystallography are used to identify interactions between proteins. However, for study of the entire proteome new approaches are needed. One new technique that has already been widely applied is the yeast two-hybrid system.^{a-f} In its original form^{a,b} a transcriptional activator is utilized together with a reporter gene, e.g., a fused *GAL1-lac2* gene that when transcribed yields β -galactosidase. This enzyme then cleaves a chromogenic substrate (see p. 1494) to give a blue color. The 881-residue transcription factor GAL4 (p. 1630) is often used in two-hybrid systems. It binds upstream of the TATA sequence in the promoter regions of genes that code for enzymes of galactose catabolism. The N-terminal **DNA-binding domain** of GAL4 binds to a specific 17-bp palindromic sequence in the DNA, while the acidic C-terminal **activation domain** activates transcription by interacting with the RNA polymerase complex bound to the promoter (see figure). The GAL4 activator seems to be quite flexible and is able to activate transcription, even if the distance from its binding site on DNA to the transcription initiation site is varied considerably. The two-hybrid system was constructed by cloning separately the pieces of DNA that code for residues 1–147

of the DNA-binding domain of the GAL4 protein (DBD in the figure) and for residues 768–881 of the activation domain (AD). The intervening nucleotides coding for the remaining 620 residues of GAL4 are discarded. The two-hybrid system tests whether a protein X, sometimes called the **bait**, binds or otherwise interacts strongly with another protein (Y, often called the **prey**) or with a series of other proteins ($Y_1, Y_2 \cdots Y_n$). To carry out the test the gene for protein X is fused with that for the DNA-binding domain of GAL4. When expressed in a living yeast cell the hybrid protein DBD–X will be formed. The gene fusion must be in-frame to ensure a correct structure for the X portion. Likewise, genes for protein Y, or for a series $Y_1, Y_2 \cdots Y_n$, are fused in-frame to the gene fragment carrying the GAL4 activation domain. Y–AD hybrids will be formed. The test is made using a strain of yeast in which the GAL4 gene has been replaced

with a hybrid *GAL1-lacZ* reporter gene. If both DBD–X and Y–AD are present, and if they interact strongly (bind tightly), transcription of the reporter gene will be activated, and a blue colony will grow from the yeast cell. It is useful to create the hybrids with protein X, or proteins $X_1 X_2 \cdots$ (the baits) in a haploid strain of one of the two yeast mating types *MATa* or *MAT α* (pp. 20, 1574) and the proteins Y (the prey) in a strain of the other mating type. Mating of the two strains will produce diploid cells that express both the DBD–X and Y–AD hybrids.

Many variants of the two-hybrid system have been devised.^d For example, a green fluorescent protein reporter can be used.^g Because significant biological protein–protein interactions often require that three or more proteins interact,^d hybrid systems involving more than two proteins have been developed. Two-hybrid systems for bacteria have also been devised.^h A virtue of the two-hybrid methods is that they work with undenatured, if not totally natural, proteins. This is in contrast to widely used methods that involve separation of denatured proteins on gels or columns.

The most popular two-hybrid systems utilize microarrays.^{d,f} In the simplest approach hybrid DBD–X is tested against a library of Y–ADs



Top: The yeast GAL4 protein interacts with the RNA polymerase complex to activate transcription of a suitable reporter gene. Bottom: Two hybrid proteins, one containing the DNA-binding domain of GAL4, fused to protein X and the other containing protein Y fused to the activation domain of GAL4, are present in a cell. If X and Y bind strongly to each other, activation domain AD will be held close to the RNA polymerase and will activate transcription.

BOX 29-F THE YEAST TWO-HYBRID SYSTEM FOR IDENTIFYING PROTEIN-PROTEIN INTERACTIONS (continued)

prepared by random cleavage of DNA of known sequence. For example, the entire genome of the gastric pathogenic bacterium *Helicobacter pylori* was cut into ~1000 nucleotide pieces. These were cloned into plasmids in *E. coli*, then into yeast. Over 10 million *E. coli* clones provided a final two million independent yeast colonies, which carried the Y-ADs (prey). The genomic DNA fragments were also used to prepare 285 DBD-X (baits) from 261 genes. In a series of two-hybrid screening tests more than 1200 different protein-protein interactions connecting 47% of the proteome were detected.ⁱ

The complete yeast (*S. cerevisiae*) has been probed using at least two large-scale two-hybrid investigations. Uetz *et al.*^j generated a large set of ~6000 genetically engineered yeast colonies, each one expressing just one of the possible Y-AD hybrid proteins (prey) derived from the ~6000 gene products identified in the yeast genome. These strains were distributed into microtiter plates and were individually crossed with 192 strains of yeast, each of which expressed a single DBD-X hybrid. This simple automated array screening identified 281 interacting pairs. In a second approach, the cells producing the Y-AD prey hybrids were mixed to give a single library. This was then screened against nearly all of the possible DBD-X hybrids in a large-scale automated procedure. The two approaches together detected 957 probable interactions involving 1004 different proteins.^j In an independent study, using similar approaches but different cloning vehicles, Ito *et al.*^{k,l} identified 4549 two-hybrid interactions among 3278 proteins. Of these 841 interactions were judged to be most relevant (core). Surprisingly, only 135 were identical to those found by Uetz *et al.* The significance and possible reasons for this disparity have been discussed.^{f,j,l}

Interpretation of results of these studies is still difficult. Results of two-hybrid methods become more useful if they can be coordinated with other approaches. For example, computational methods can predict interactions from genome sequences alone.^{m,n,o} More than 45,000 interactions have been predicted among yeast proteins.^m Reliable identification of such motifs as DNA-binding domains and Ca²⁺-binding domains can complement two-hybrid analysis.ⁿ The yeast genome is predicted to contain 162 coiled-coil sequences and at least 213 unique interactions between them.^o Examination of sequences of protein families in the Protein Data Bank (PDB) led to prediction of 8151 interactions of 664 types between protein families in yeast.^p

Improved experimental procedures of other types can also complement two-hybrid methods. Among these are formaldehyde crosslinking with immunoprecipitation,^q methods that couple mass spectrometry and crosslinking,^r and detection of intermolecular nuclear Overhauser enhancements in protein-protein complexes.^s Phage display methods (see Fig. 3-16) have been developed as another method of detecting protein-protein interactions^e as has fluorescence resonance energy transfer (FRET; p. 1291).^d Evanescent wave methods, e.g., surface plasmon resonance (Box 3-F) are increasingly being used to quantify protein-protein interactions. These may be combined with single-hybrid methods in inexpensive and rapid micro-devices.^d

^a Fields, S., and Song, O.-K. (1989) *Nature (London)* **340**, 245–246

^b Chien, C.-t., Bartel, P. L., Sternglanz, R., and Fields, S. (1991) *Proc. Natl. Acad. Sci. U.S.A.* **88**, 9578–9582

^c Finley, R. L., Jr., and Brent, R. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 12980–12984

^d Mendelsohn, A. R., and Brent, R. (1999) *Science* **284**, 1948–1950

^e Allen, J. B., Walberg, M. W., Edwards, M. C., and Elledge, S. J. (1995) *Trends Biochem. Sci.* **20**, 511–516

^f Oliver, S. (2000) *Nature (London)* **403**, 601–603

^g Shioda, T., Andriole, S., Yahata, T., and Isselbacher, K. J. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 5220–5224

^h Joung, J. K., Ramm, E. I., and Pabo, C. O. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 7382–7387

ⁱ Rain, J.-C., and 12 other authors. (2001) *Nature (London)* **409**, 211–215

^j Uetz, P., and 19 other authors. (2000) *Nature (London)* **403**, 623–627

^k Ito, T., Tashiro, K., Muta, S., Ozawa, R., Chiba, T., Nishizawa, M., Yamamoto, K., Kuhara, S., and Sakaki, Y. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 1143–1147

^l Ito, T., Chiba, T., Ozawa, R., Yoshida, M., Hattori, M., and Sakaki, Y. (2001) *Proc. Natl. Acad. Sci. U.S.A.* **98**, 4569–4574

^m Marcotte, E. M., Pellegrini, M., Ng, H.-L., Rice, D. W., Yeates, T. O., and Eisenberg, D. (1999) *Science* **285**, 751–753

ⁿ Gallet, X., Charlotiaux, B., Thomas, A., and Brasseur, R. (2000) *J. Mol. Biol.* **302**, 917–926

^o Newman, J. R. S., Wolf, E., and Kim, P. S. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 13203–13208

^p Park, J., Lappe, M., and Teichmann, S. A. (2001) *J. Mol. Biol.* **307**, 929–938

^q Orlando, V. (2000) *Trends Biochem. Sci.* **25**, 99–104

^r Bennett, K. L., Kussmann, M., Björk, P., Godzwon, M., Mikkelsen, M., Sorensen, P., and Roepstorff, P. (2000) *Protein Sci.* **9**, 1503–1518

^s Clore, G. M. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 9021–9025

regulated pathways for storage and release by exocytosis of hormones, neuropeptides, and neurotransmitters. The last are secreted from both small (~50 nm diam.) synaptic vesicles and larger **dense core vesicles** (>100 nm diam.),^{620,621} which are discussed in Chapter 30, Section B.8. In every case specialized proteins (discussed on pp. 427 and 521 and in Chapters 20 and 30) are involved.^{622–624}

7. Protein Folding

Before a protein can function its polypeptide chain must fold into its own native tertiary structure.^{624a} This folding is influenced by many surrounding proteins, by the state of glycosylation of side chains,⁶²⁵ and by other posttranslational modifications, by the presence of cis amide linkages in unfolded or folded forms (pp. 82, 83; Box 9-F),⁶²⁶ and by possibility for formation of disulfide bridges (pp. 521, 522).^{626a–628} The prediction of the folding pattern of proteins from the amino acid sequence remains a major goal of protein chemistry. In principle, a protein fold can be predicted from the DNA sequence of the genes, with proper allowance for effects of posttranscriptional modification. This goal once seemed intractable, but two things have provided new hope. (1) The speed and power of computers is still increasing. This not only has allowed more rapid calculations but also has led to improvement in experimental methods. (2) Methods for studying folding, which include mass spectroscopy,⁶²⁹ NMR spectroscopy,^{630–633} and optical methods,^{634,635} have become more rapid and more sensitive.⁶³⁶ As a consequence, we have an abundance of new data.

Anfinsen, in 1963, proposed that the three-dimensional structure of a protein is in its lowest Gibbs energy state when present in its natural environment.^{637,638} However, there is a problem with this suggestion (the “Levinthal paradox”; p. 82). Even if a polypeptide chain occupies only two of the lowest energy regions of the Ramachandran diagram (Fig. 2-9), a 100-residue protein would have ~10³⁰ possible conformations. If a folding protein checked all of these conformations at a realistic rate of ~10¹¹ s⁻¹, it would take ~10¹¹ years to fold. Furthermore, in a test tube of protein, which would contain at most 10¹⁸ molecules, each of the molecules would probably have a different conformation.⁶³⁹ In fact, most proteins fold reliably to the same final structure in less than a second, and some in a millisecond.^{639–643} It is also true that proteins, under altered solvent conditions, can misfold into totally “incorrect” structures.⁶⁴⁴ Most can assume an amyloid structure under some conditions (Box 29-E). One clear conclusion is that folding is not totally random but follows a **folding pathway**, which is dictated by the sequence. Nevertheless, experimental

data indicate that there is an ensemble of related structures at each stage of folding.⁶³¹

We cannot answer the question posed by Anfinsen’s hypothesis. Does the native state have a minimum value of the Gibbs energy? Nevertheless, it is observed that proteins usually behave as if folded, unfolded forms are in a true thermodynamic equilibrium, and that this equilibrium is attained rapidly. The difference ΔG between a folded and a denatured protein is only 21–63 kJ mol⁻¹, which shows that folded proteins are only marginally more stable than are unfolded polypeptide chains.⁶⁴⁵ The value of ΔG of unfolding as a function of temperature T is given by Eq. 29-13, where $\Delta H(T)$ and ΔC_p are the changes in enthalpy and heat capacity upon unfolding.^{645,646}

$$\Delta G(T) = \Delta H_m (1 - T/T_m) - \Delta C_p [(T_m - T) - T \ln(T/T_m)] \quad (29-13)$$

ΔH_m is the enthalpy change at T_m , the midpoint of the thermal unfolding curve (the “melting” temperature). The temperature of maximum stability T_s occurs when $\Delta S = 0$ (Eq. 29-14). T_s is usually between -10°C and

$$T_s = T_m \exp(-\Delta H_m / [T_m \Delta C_p]) \quad (29-14)$$

35°C. For staphylococcal nuclease (Fig. 12-29) $T_s = 18^\circ\text{C}$ and $T_m = -19^\circ\text{C}$ and $+57^\circ\text{C}$, i.e., *the protein is denatured by either cooling below 18°C or heating above 57°C*, a behavior that is common for many proteins. Cold denaturation is observed whenever the unfolded state has a higher heat capacity than does the folded state.⁶⁴⁷

Can we predict the Gibbs energy of unfolding from the protein sequence? To do this it is necessary to utilize experimental data on known proteins to obtain a series of terms (Eq. 29-15) that can be summed to give the ΔG°_u , the standard Gibbs energy of unfolding.

$$\Delta G^\circ_u = \Delta G^\circ_{\text{charge}} + \Delta G^\circ_{\text{hyd}} + \Delta G^\circ_{\text{conf}} + \Delta G^\circ_{\text{vW}} + \Delta G^\circ_{\text{Hbond}} \quad (29-15)$$

The terms refer to summations of all of the charge-charge, hydrophobic, configurational, van der Waals, and hydrogen-bonding interactions between both main-chain and side-chain atoms.^{638,646,648} Such computations are formidable and are uncertain, especially for electrostatic (charge-charge and hydrogen-bonding) interactions.^{649–652} Both the folded and denatured state must be considered,⁶⁵³ as must the heat capacities⁶⁵⁴ and configurational entropies.⁶⁵⁵

While we tend to think of proteins as having fixed structures, conformational changes are basic to life. Many proteins are very flexible and in part disordered.^{655a} At the same time proteins can be misfolded leading to amyloid formation (Box 29-E) and to other diseases.^{655b–e}

8. Completing the Cycle: Proteolytic Degradation of Proteins

Like all other body constituents proteins must undergo breakdown as well as synthesis. Regular turnover of all proteins is essential, and defects in the process may lead to amyloid deposits (Box 29-E). Turnover and degradation of proteins depends upon a variety of proteases, many of which are discussed in Chapter 12, Section C. Because of the changing needs of body cells, specific proteins turn over at widely varying rates. Some enzymes, hormones, and regulatory proteins have half-lives of only a few minutes while others may function for months or years. How can the regulation of the breakdown of thousands of different proteins be controlled? The answer seems to lie in the amino acid sequence. Just as the sequence determines the location that a protein occupies in the cell and its folding pattern, it also determines the turnover rate.

Much of the breakdown takes place in the cytosol in proteasomes (Box 7-A) and is controlled by the ubiquitin system (Box 10-C), which selects the proteins for degradation; control of the system is quite complex.^{656–661} One aspect depends upon the N-terminal amino acid of the substrate protein. Defective proteins often have N-terminal destabilizing amino acids such as phenylalanine, leucine, aspartic acid, lysine, and arginine (p. 527). If an internal lysine is also present, the protein may be conjugated with ubiquitin and degraded rapidly. Many metabolic processes, such as the cell cycle (Fig. 11-15), are controlled by protein degradation.^{538,662} In some cases an arginine residue is transferred onto the N terminus of a protein by an **arginyl-tRNA protein transferase**. This creates a better substrate for ubiquitination and rapid degradation.^{280,663} In other cases proteolytic cleavage uncovers an arginine or other destabilizing residue and speeds hydrolysis.⁵³⁸ While ubiquitination often initiates the degradation of proteins, it also helps to direct proteins to specific locations within a cell.^{663a} Rapid degradation of a ubiquitinated protein may require hydrolytic **deubiquitination** by a metalloprotease, which is a subunit of the 26S proteasome lid (Box 7-A). This allows the ubiquitin to be recycled and also directs the deubiquitinated protein into the proteasome.^{663b}

As mentioned on p. 1854, an important function of proteasomes is formation of short antigenic peptides for use by the immune system.⁶⁶⁴ Inhibition of proteasome activity reduces or prevents antigen presentation (Chapter 31).^{665,666} In this immune surveillance system mature proteins of host cells are cut up and checked for self-identity. The checking also includes the rapidly degraded imperfect proteins and foreign proteins from invading organisms or viruses.^{667,667a}

Lysosomes, which contain more than 50 proteases, lipases, glycosidases, and other hydrolases, also play a

major role in protein degradation.⁶⁶⁸ Their importance is emphasized by the range of lysosomal deficiency diseases (Table 20-1).⁶⁶⁸ Lysosomes also function in the process of **autophagy**, by which cells can sacrifice a whole section, organelles and all, by walling off a large vesicle or **autophagosome** and fusing it with a lysosome.⁶⁶⁹ In such a way a tadpole can resorb its tail while becoming a frog. We have now come full circle: our proteins have been converted back to the amino acids and other small molecules derived from them. The amino acids can be reutilized or can be catabolized, depending upon the needs of the organism.

E. Proteomics

The vast amount of data on protein structures and improved methods of predicting structures^{670–672} have led to development of new areas of science variously designated as genomics, proteomics, transcriptomics,^{673,674} and bioinformatics.^{675–677} These fields encompass all of the methods for sequence determination, observation of gene expression, protein synthesis by cells, and mathematical analysis of resulting data. Proteomics includes new approaches to polypeptide separation⁶⁷⁸ and identification,^{678,679} sequencing at the attomole level,^{680–681} and comparison of sequences between species.^{682–684} Protein separation by liquid chromatography,⁶⁸⁵ capillary electrophoresis,⁶⁸⁶ or two-dimensional gel electrophoresis^{687,688} can be followed by mass spectrometry of intact proteins or of proteolytic fragments.^{689,690} Microarrays on proteome chips can be used to observe production of thousands of proteins simultaneously.^{691–694} Structural genomics centers have been established for rapid determination of protein structures using NMR^{695,696} and X-ray methods.^{697–699} If each center determines 200 or more structures per year there will soon be 16,000 new structures, enough to allow us to predict much about all the rest.⁶⁹⁹ Then we can study all the important remaining details for millions and millions of proteins.

Current efforts to understand the structures, conformational movements, and functions of these molecules range from the classification of nearly 10,000 different protein folds⁷⁰⁰ to investigation of the dynamics of single protein molecules.⁷⁰¹ Well known motifs such as β sheets and α helices are studied with the goal of more accurate predictions of structure and better understanding of interactions between proteins in solution and in membranes. For example, one natural topology is the β barrel, which may contain 8 to 22 strands (e.g., see Fig. 8-20). These cylindrical proteins are abundant in outer membranes of gram-negative bacteria.^{702–704} The partial electrical charges at the edges of the β sheets (see Fig. 2-11,B) may interact to help stabilize the barrels. In contrast, soluble β barrel-containing proteins are designed to avoid edge-to-

edge interactions, which could cause aggregation of the proteins.⁷⁰⁴ A recently discovered membrane-protein motif is an α **barrel**, which is composed of 12 α helices stacked side-by-side with side-chain groups fitting together in a knobs-in-holes fashion.⁷⁰⁵ An example is the TolC protein of *E. coli*. A trimer of 428-residue subunits forms a long cylinder, which is a 12-stranded β barrel at one end and an α barrel at the other.⁷⁰⁵ More common are transmembrane α helices, many of which are present in 7-helix receptors (e.g., Fig. 11-6). Relatively accurate prediction methods are now available for these structures,^{706,707} but there are still uncertainties about mechanisms of transmission of signals across the membranes.

Predictions of structures of more complex proteins from their amino acid sequences presents a major challenge.⁷⁰⁸ Assignment of domains within the protein is a first step.^{709–712} Regions of probable helix or β -strand structure can be recognized but it is difficult to predict the exact lengths of the helices and the structures of

connecting loops and strands. These depend upon many factors including the possible formation of ion pairs⁷¹³ and of locks at the ends of strands created by van der Waals interactions.⁷¹⁴ There are also circular proteins.⁷¹⁵ Composite structures such as those of silks (Box 2-B) have surprising properties. Both silkworm,^{716,717} and spider silks^{718,719} undergo marked changes in properties upon spinning of the random coil forms of the proteins found in silk glands into the drawn fibers.

Whether we discuss silk, proteins embedded in membranes, or soluble complexes of cytosolic proteins, we must ask questions about interactions. A first step is to identify interactions^{720–730} among proteins either *in vitro* or in living cells.⁷³¹ Proteomic methods, which include the yeast two-hybrid method (Box 29-F), are widely used for this purpose. It is possible to identify large sets of interacting proteins, to identify disease states, to observe effects of drugs, and to compare metabolism among species.

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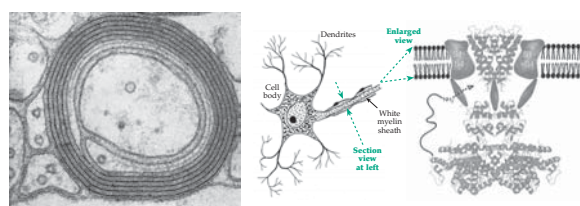
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Study Questions
for chapters 28 and 29

1. Describe the role of sigma factors (σ) in transcription by prokaryotic RNA polymerases. What is the effect of the release of σ from the holoenzyme once transcription has been initiated. How would a mutation that prevents a σ factor from dissociating from core RNA polymerase affect the rate of transcription?
2. Explain how histidine biosynthesis is controlled in *E. coli*, a bacterium that has no *his* repressor.
3. One mechanism of transcriptional control in prokaryotes, especially of several operons controlling the biosynthesis of amino acids, is **attenuation**. Briefly describe the mechanism of attenuation. How does the supply of amino acid in the cell affect the process?
4. Is transcription attenuation likely to be an important mechanism of transcriptional regulation in eukaryotic cells?
5. Discuss two main DNA-recognition motifs found in eukaryotic transcription factors. Describe their structures, indicate how they bind to DNA, and discuss how each specifically recognizes its DNA binding site.

Study Questions

6. How can a DNA enhancer sequence located as many as several thousand base pairs from a gene transcription start site influence transcription even if its orientation is reversed?
7. Some eukaryotic DNA viruses code for two or more mRNA transcripts of differing lengths from the same region on the DNA. Suggest an explanation. How do you expect the two translation products of these mRNAs to differ?
8. High salt concentrations weaken the interaction of histones with DNA but have little effect on the binding of many regulatory proteins. Explain this observation in terms of the modes of interaction of the two types of protein.
9. Discuss the changes that must be made in a typical eukaryotic structural gene to allow its protein product to be synthesized in bacteria.
10. List the different types of covalent modification that may be made to tRNA. To ribosomal RNA. To messenger RNA.
11. List various small RNAs and their functions within cells.
12. Some amino acids utilize only one codon of the 64 in the genetic code. Other amino acids use as many as six codons (Tables 5-5, 5-6). What advantages to a cell is provided by utilization of several codons for a single amino acid?
13. In what ways is the genetic code not quite "universal?" What is meant by "editing" of mRNA?
14. Why is it necessary to have "adapters" in the form of tRNAs to read the genetic code during translation?
15. Most nonsense suppressor genes are mutants of tRNA genes. In view of this fact, how can cells survive the presence of such mutations?
16. Explain how the protein synthesizing machinery is able to differentiate the initiation AUG codon from an internal AUG (methionine) codon in prokaryotes. How is this accomplished in eukaryotes?
17. The amino acid sequence of a mature protein sometimes differs from that deduced from the DNA nucleotide sequence of the structural gene for that protein. Discuss three ways by which this may occur.
18. Write out in detail, using structural formulas, the chemical mechanism of synthesis of an aminoacyl-tRNA and of incorporation of the aminoacyl group into a peptide chain being formed by a ribosome.
19. a) Calculate the minimum number of ATP equivalents consumed in the biosynthesis of a 300-amino acid *E. coli* protein, having the N-terminal sequence Ala-Ser-Val-Tyr, from the free amino acids.
b) Much of this energy involves hydrolysis of GTP. What is the role of this GTP hydrolysis in protein synthesis?
20. How do the polypeptide products produced in the presence of (a) puromycin and (b) streptomycin differ from polypeptides synthesized in the absence of these inhibitors? Explain your answer.
21. What is the significance to protein synthesis of each of the following?
Shine-Dalgarno sequence
Signal recognition particle
proteasome
22. How can useful antibiotics that act on ribosomes kill bacteria but not people?
23. Compare termination of translation in bacteria and in eukaryotes.
24. List some types of error that are likely to be made during protein synthesis. What mechanisms have cells developed to deal with these?



Center: Diagram of the cell body of a neuron with dendrites and a short section of its long myelinated axon (see Fig. 30-8). Left: Electron micrograph of a thin section through an axon showing the myelin sheath formed by the wrapping of the plasma membrane of a neuroglial cell around the axon (see p. 390 and Fig. 30-9). Right: Model of a voltage-regulated K⁺ channel in the cell membrane of an axon. The pore, which is formed from four α -subunits, is represented by that of the bacterial pore shown in Fig. 8-21. Also shown is an inner cytoplasmic activation gate consisting of four β -subunits, which are proposed to form ball-and-chain devices that can close the pores in response to voltage changes. From Zhou *et al.* See Fig. 30-18.

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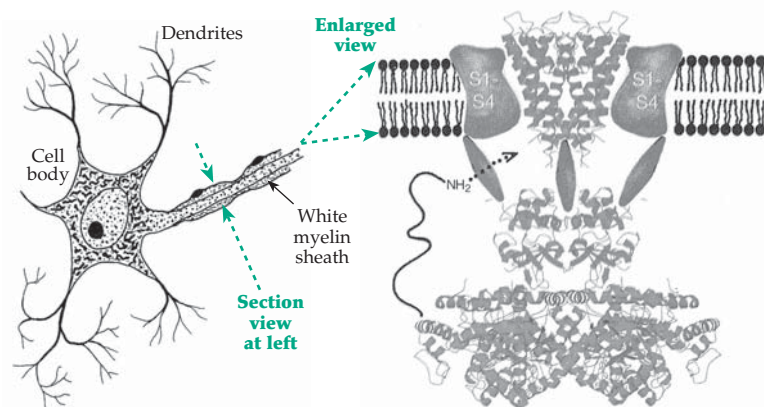
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Chemical Communication Between Cells

30



The regulation of growth and metabolism of complex multicellular organisms depends heavily upon chemical messages sent between cells. This includes secretion of hormones into the circulatory system,¹⁻³ chemical transfer of information through communicating cell junctions, and passage of signals between neurons in the brain. This chapter deals with these matters and also with communication between organisms, i.e., with the biochemistry of ecological relationships. Embryonic development and differentiation of tissues also require communication between cells as does the functioning of the immune system. These topics are considered in Chapters 31 and 32.

A. The Hormones

The term hormone has traditionally been applied to substances synthesized in and secreted by one tissue and which act to influence distant target organs or tissues. However, many peptide hormones also act as neurotransmitters, passing across very short gaps between cells. In addition, many chemical messengers, including the peptide growth factors, act more locally. Looking at lower invertebrates as simple as *Hydra*, we find peptides resembling our own hormones and neurotransmitters. These are secreted by neuroendocrine cells of *Hydra* and diffuse throughout the body. In higher animals hormones regulate the concentrations of nutrients such as glucose and of ions such as Ca^{2+} and phosphate in the blood. They control the volume and osmotic pressure of body fluids, as well as digestion, growth, reproduction, and responses to stress.

1. Receptors, Feedback Loops, and Cascades

Every hormone must have one or more receptors, most of which are proteins. These may be found embedded in the outer surface of the plasma membrane, in the cytoplasm, or in the cell nucleus. Binding of a hormone to its receptor often elicits both a rapid response and a slower one. For example, we have seen that glucagon, adrenaline, and vasopressin bind to cell surface receptors and promote the synthesis of cyclic AMP (Fig. 11-4). The cAMP induces rapid chemical modifications of many proteins. Some of these may diffuse into the nucleus and affect transcription of genes, a slower response. Insulin (Chapter 11, Section G) also exerts both rapid and slower responses.

Receptor types. Many different kinds of protein can serve as hormone receptors. Some of these are discussed in Chapter 11. The most abundant are the G protein-coupled 7-helix receptors^{4-5c} such as that of a β adrenergic receptor pictured in Fig. 11-6. Glucagon, adrenaline, ACTH, and gastrin are a few of the hormones that bind to receptors of this type. Similar receptors respond to light (rhodopsin; Chapter 23) and over 1000 different 7-helix receptors respond to smell and taste. The G proteins and their controlling cycles, Eq. 11-10,⁵⁻⁷ have also been considered in Chapter 11. The reality of the dissociation and reassociation of the α and $\beta\gamma$ subunits in response to binding of a hormone has been demonstrated in living cells by the use of fluorescence resonance energy transfer (FRET).⁸ Not all receptors activate G proteins. One large group of membrane-associated receptors have single transmembrane helices but require dimerization to be effective.

The bacterial chemoreceptor (Figs. 11-8 and 19-5) has a very small ligand-binding domain and a larger internal domain that activates a histidine kinase. Many growth-factor receptors, including the insulin receptor (Figs. 11-11, 11-12), have internal domains with protein tyrosine kinase activity.

In contrast, steroid hormones, thyroxine, and retinoids bind to internal receptors. In 1968, Gorski *et al.*⁹ and Jensen *et al.*^{10,11} proposed independently that steroid hormone receptors in the cytoplasm bind incoming steroid molecules and after an “activation” step carry the hormone into the nucleus, where the hormone–receptor complex would bind at many sites in the chromatin inducing transcription of selected genes.¹² Doubt has been cast on the assumption that the steroid hormone receptors must bind hormone initially in the cytoplasm. However, the role of steroid receptors in regulating transcription is well established (see discussion in Chapter 22, Section E,5; Chapter 28, Section C,6).

Feedback loops. Maintenance of a steady state within an organism depends upon numerous negative-feedback loops. Hormones assist in adjusting reaction rates to maintain a steady state when conditions are changed. For example, blood glucose rises after a meal. This increase is sensed in the pancreatic beta cells (pp. 998, 999), which release insulin. The released insulin promotes uptake of glucose by cells and its conversion into glycogen and lipid stores. When the glucose level falls, inhibitory mechanisms that decrease insulin release are allowed to operate.

Similar regulatory loops can be traced for nearly all hormones. Sometimes they involve several stages and involve sensing devices in the central nervous system. In such cases neural impulses stimulate the **hypothalamus** of the brain (Fig. 30-1) to release **neurohormones**, which travel to the anterior lobe of the pituitary gland. The pituitary, in turn, releases hormones such as **corticotropin** (adrenocorticotrophic hormone, **ACTH**), which stimulate the adrenal cortex to release its hormones. The latter exerts feedback inhibition upon the hypothalamus to decrease the secretion of ACTH by the pituitary. Steroids also participate in feedback loops to the hypothalamus.¹³ Using ³H-labeled hormones or fluorescent analogs, it has been possible to locate specific brain cells sensitive to a given hormone by autoradiography.¹⁴

A characteristic of hormonal effects is that they are seldom unique, and are often balanced by counter-acting effects of other hormones. For example, both glucagon and adrenaline promote the release of glucose from liver glycogen into the bloodstream. The glucocorticoids stimulate the rate of production of glucose from other body constituents (Chapter 11). Growth hormone tends to increase glucose levels by inhibiting utilization of sugar by tissues. On the other hand, insulin acts to promote uptake of glucose by tissues and a more efficient utilization. The thyroid

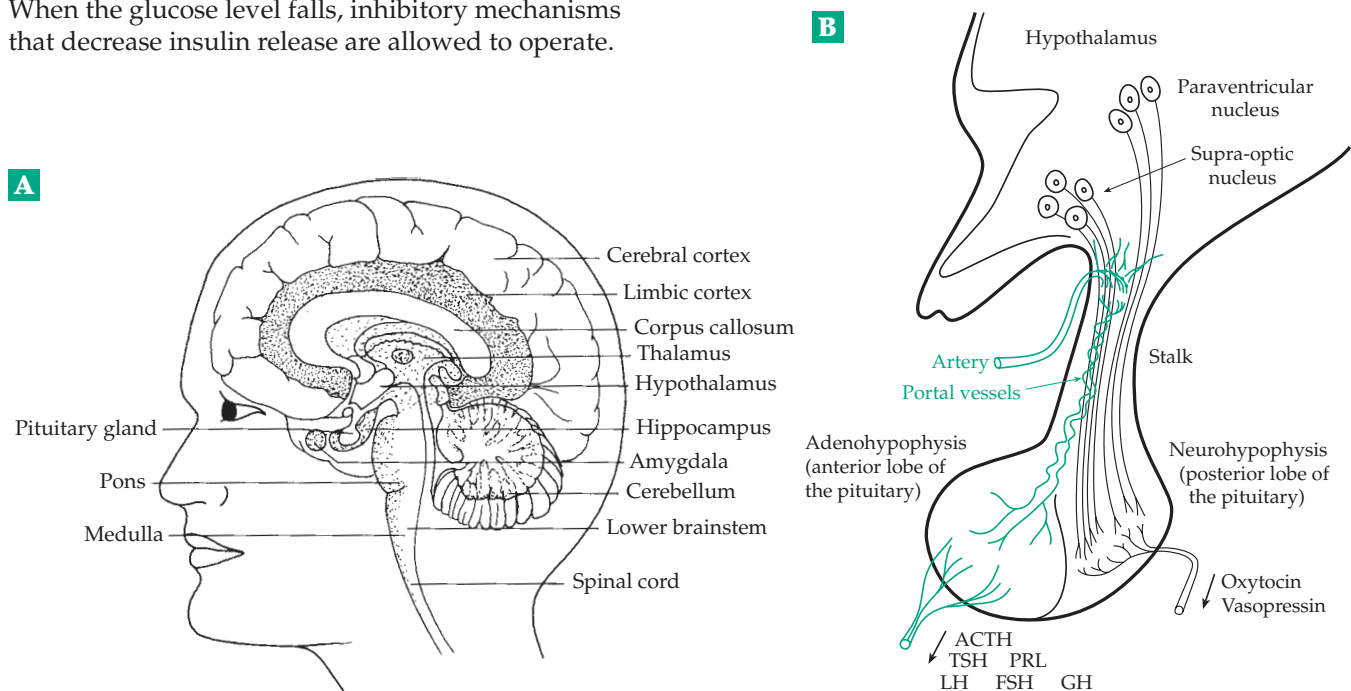


Figure 30-1 (A) Median sagittal section of the human brain. From Maya Pines.¹⁵ (B) Drawing illustrating the synthesis of peptide hormones in the hypothalamus and transport via portal blood vessels into the anterior lobe of the pituitary gland or via nerve tracts into the posterior lobe.¹⁶

hormone increases the overall rate of metabolism of cells and also tends to promote a decrease in blood glucose.

Signaling cascades. As we have seen in Chapter 11, hormones frequently elicit the synthesis of second messengers such as cAMP, inositol phosphates, or diacylglycerol. This not only provides amplification of the initial hormonal signal but also allows a single hormone to control a “domain” of many metabolic processes. Each of these processes, in turn, can influence others. Many processes are affected by several different hormones and by more than one second messenger. Since we are far from knowing how many hormones exist and how many second messengers are released, the network of regulatory interactions within cells may be one of overwhelming complexity. An abbreviated version of a mitogen-activated kinase (MAP kinase) cascade^{4,17} is shown in Fig. 11-13. These cascades are not only initiated but also are propagated by a series of phosphorylation reactions catalyzed by more than 1000 protein kinases encoded by the human genome.¹⁸ Together with more than 500 protein phosphatases, which are often joined together as a bifunctional protein (p. 545),^{19,20} they form a complex branching network of interactions.^{21,22} These help to control responses not only to hormones but also to varying metabolite concentrations and physical stimuli.

Second messengers as hormones. The same compounds that serve as intracellular second messengers sometimes act as hormones. Tomkins suggested²³ that cAMP and some other small molecules serve as “symbols” indicating a metabolic need. For example, in bacteria ppGpp (p. 1715) serves as a symbol of nitrogen or amino acid deficiency. In cells ranging from those of bacteria to animals, cAMP is a symbol for carbon-source starvation. In *E. coli* cAMP levels increase during carbon-source starvation and stimulate the initiation of transcription of many bacterial operons (Chapter 28). In *Dictyostelium discoideum* (Box 11-C) cAMP is released by cells, when substrate depletion occurs. In this instance, the cyclic nucleotide acts as a hormone transmitting a signal to other cells.

Whereas cAMP is sometimes used by lower organisms as a hormone, its metabolic lability makes it unsuitable for higher animals. Thus, in our bodies the hormones glucagon and adrenaline carry a message to cell surfaces, where binding to receptors stimulates cAMP production. This, in turn, leads to mobilization of metabolic stores such as those of glycogen and triglycerides, just as if these cells had also been subjected to acute starvation. Tomkins suggested that hormones are produced by “sensor” cells in direct contact with environmental signals and travel to and activate more sequestered “responder” cells. The picture can be generalized further by realizing that

neurotransmitters are largely derivatives of amino acids. These amino acids may have originally served as intracellular symbols reflecting changes in environmental amino acid concentration but were later utilized in short-range intercellular communication within the nervous system.

The vertebrate hormones. The principal established vertebrate hormones are listed in Tables 30-1 and 30-2. Also given are references to other parts of the text, where specific hormones are discussed. The hormones can be divided into four groups on the basis of chemical structure: (1) peptides and proteins, (2) derivatives of the aromatic amino acids, (3) steroids and prostaglandins, and (4) volatile compounds such as NO and CO. The most numerous are the peptide hormones, many of which also act as neurotransmitters. Peptide hormones, e.g., those with insulin-like effects, function in all phyla of the metazoa, and hormone-like molecules are found in bacteria.²⁴

2. Hormones of the Pituitary Gland (Hypophysis) and Hypothalamus

Connected to the brain by a stalk (Fig. 30-1), the pituitary gland releases at least ten peptide or protein hormones that regulate the activity of other **endocrine** (hormone-producing) glands in distant parts of the body. The pituitary is composed of several distinct parts: the anterior lobe (**adenohypophysis**), a thin intermediate portion (**pars intermedia**), and a posterior lobe (**neurohypophysis**). Each has its own characteristic endocrine functions.

The anterior lobe of the pituitary secretes a series of ten or more peptide hormones ranging in size from the ~20-residue β -melanotropin to the ~200-residue growth hormone (somatotropin). Several of these contain a common heptapeptide unit, which is marked in green in the following structure:



Structure of α -melanotropin from pig, beef, and horse

Not only this heptapeptide but also the entire amino acid sequence of **α -melanotropin** is found within the sequence of **corticotropin** (Fig. 30-2), which has an additional 29 amino acids at the C-terminal end.²⁵ The same heptapeptide was also found in the **lipotropins**. The explanation is, in part, that several of these hormones arise from a single 31-kDa precursor protein called **prepro-opiomelanocortin**.^{25,26} It contains an N-terminal signal sequence that is removed shortly after synthesis, as well as pairs of adjacent basic residues (Arg-Arg, Arg-Lys, Lys-Arg and Lys-Lys) at a number of places (Fig. 30-2). After removal of the

signal sequence, further cleavage is thought to occur within the secretory vesicles by proteases, which cut either on the carboxyl side of these basic pairs or between them.^{25,27,28} The same precursor is made in both anterior and intermediate lobes and is rapidly cut to ACTH, β -lipotropin, and an N-terminal part. In the

intermediate lobe the ACTH is then cleaved at the Lys-Lys and Arg-Arg pairs to form α MSH and another peptide called corticotropinlike intermediate lobe peptide (CLIP). Beta lipotropin is degraded rapidly in the intermediate lobe and more slowly in the anterior lobe to γ -lipotropin and the opioid peptide β -endorphin

TABLE 30-1
Peptide and Protein Hormones of Vertebrates

Source and name of hormone	No. residues	Principal site of action	References Chapter, Section
A. Pituitary gland (hypophysis)			
1. Adenohypophysis (anterior portion)			
Corticotropin (ACTH) ^a	39	Adrenal cortex, adipose tissue	Fig. 2-4
β -Melanotropin (β melanocyte-stimulating hormone, β -MSH) ^a	18–22	Skin	Fig. 30-2
β -Lipotropin (β -LPH) ^a	91	Precursor of β -MSH and β -endorphin	Fig. 30-2
γ -Lipotropin (γ -LPH) ^a	58	Precursor of β -MSH	Fig. 30-2
β Endorphin ^a	31	Brain	
Somatotropin (growth hormone, GH)	~200	All tissues	
Prolactin (mammotropin)	~200		
Thyrotropin (thyroid-stimulating hormone, TSH) ^b		Thyroid	Ch 25, B2
Follitropin (follicle-stimulating hormone, FH) ^b		Ovaries, testes	
Lutropin (luteinizing hormone, ICSH or LH) ^b		Ovaries, testes	
2. Pars intermedia (intermediate portion)			
α -Melanotropin (α -melanocyte-stimulating hormone, α MSH) ^a	13	Skin	Fig. 30-2, pp. 1742, 1748
3. Neurohypophysis (posterior portion)			
Oxytocin (ocytocin)	9	Uterus, mammary glands	Fig. 2-4
Vasopressin (antidiuretic hormone)	9		Fig. 2-4
B. Pancreas			
Insulin	51	All cells	Fig. 7-17, Ch 11, G
Glucagon	29	Liver, adipose tissue	Ch 11, D
C. Ovary (corpus luteum)			
Relaxin	—	Pelvic ligaments	p. 1746
D. Thyroid			
Calcitonin (thyrocalcitonin)	32	Bones, kidney	Ch 25, B2 Box 6-D
E. Parathyroid			
Parathyrin (parathyroid hormone)	84	Bones, kidney	Box 6-D
F. Kidney			
Erythropoietin		Bone marrow	
Renin (an enzyme)		Adrenal cortex	p. 621; Box 22-D

^a Arise by cleavage of pro-opiomelanocortin.

^b Related two-subunit ($\alpha\beta$) proteins with a common β subunit for these three hormones, for human chorionic gonadotropin (hCGH), and for mitogen-regulated protein (proliferin).

(Section B,10). Precursor proteins have been identified for many other peptide hormones, even those with very short chains.^{29–30a} Proteolytic cleavages and other processing reactions occur within the secretory pathways of organisms from yeast to humans.^{30b,c}

Many pituitary hormones have a pyroglutamate (5-oxoproline) residue at the N terminus (e.g., see Fig. 2-4). This presumably arises by attack of the terminal $-NH_2$ group on the amide carbon of an N-terminal glutamine side chain with displacement of NH_3 (Eq. 10-10).²⁹ The C terminus is often an amide of the carboxyl group with ammonia, which usually arises from a peptide chain containing one additional glycine residue at the C terminus (Eq. 10-11). The processing of peptide hormones doesn't end with their synthesis. They are usually degraded quickly or are converted into derivatives with weaker hormonal activity.

Pituitary growth hormone and related hormones. The pituitary growth hormone (**somatotropin**)³¹ and **prolactin** are 22- to 23-kDa proteins, which share homology also with human **placental lactogen**, a lactogenic hormone secreted by the placenta,³² and with a growth factor called **mitogen-regulated protein** (or proliferin).^{33–35} The polypeptide chain of the 191-residue porcine³⁶ and human³⁷ growth hormone folds into an antiparallel four helix bundle (similar to that in Fig. 2-22) but with two long irregular connecting strands. The high degree of homology among somatotropins of many other species indicates a near identity of three-dimensional structures. However, biological function is species-specific. Humans and monkeys respond only to growth hormone from primates. The interaction of an aspartate side chain at position 171 with arginine 43 of the receptor protein may account for some of this specificity.³⁸ The receptor is a member of a large superfamily of receptor proteins with single transmembrane helices and extracellular domains similar to that of tissue factor (Fig. 12-18) and also, in some respects, to immunoglobulin domains.^{39,40} Receptors for growth hormones bind in specific ways to two molecules of receptor protein.^{41,42}

Human growth hormone produced in bacteria is used to help very short children to grow. Bovine growth hormone produced in bacteria is used to increase milk production from cows.⁴³ However, this

use may be damaging to the cows.⁴⁴ Some humans produce too much growth hormone, often as a result of tumors in the pituitary. The resulting condition of **acromegaly**⁴⁵ causes excessive bone growth and many other problems. Growth hormone has a broad range of other effects, e.g., mimicking the action of insulin.⁴⁶

Lactogenic hormones also have 4-helix bundle structures, and the prolactin receptor structure resembles that of growth hormone as well as those of a large cytokine family.^{47,48} Prolactin affects placental development during pregnancy. However, during the latter half of pregnancy the placenta has a dominant endocrine effect, synthesizing both progesterone and the placental lactogen.^{32,49}

The pituitary glycoprotein hormones. The thyroid-stimulating hormone **thyrotropin** (TSH), together with **folitropin** (FH) and **lutropin** (LH; Table 30-1), form a family of related ~28-kDa dimeric glycoproteins in which each subunit has a three-loop structure stabilized by a characteristic “cystine knot.”⁵⁰ Also included in the family is the placental **chorionic gonadotropin**,^{51,52} which is found only in human beings and a few other species. LH has a central role in promoting both spermatogenesis and ovulation by stimulating synthesis of steroid hormones in the testes and ovary, respectively.⁵³ Human chorionic gonadotropin (hCG) is also essential for maintenance of pregnancy and acts by stimulating the ovaries to secrete required steroid hormones.

All of these glycoprotein hormones are $\alpha\beta$ dimers, and within a single species the subunits of TSH, FH, LH, and CG are identical. However, the β subunits are all different.^{54,55} In the human there are at least six genes or pseudogenes for the hCG β chain in a cluster that also contains a single LH β chain gene.⁵⁴ The hormones undergo glycosylation and sulfation in the Golgi before secretion.⁵⁶ The hormones bind to 7-helix receptors, which are coupled to formation of cAMP or inositol trisphosphate.⁵⁷ Mutations in LH may cause male infertility,⁵⁸ while mutations in the corresponding receptor may cause male precocious puberty.⁵⁹

Hypothalamic releasing hormones. As was mentioned previously, the anterior lobe of the pituitary

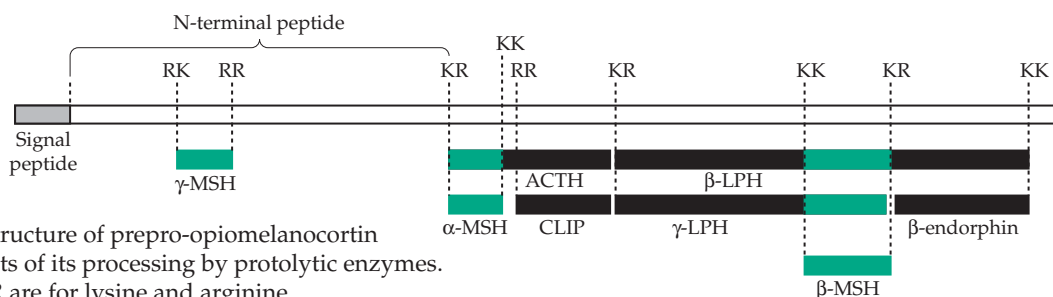


Figure 30-2 Schematic structure of prepro-opiomelanocortin and of some of the products of its processing by proteolytic enzymes. The abbreviations K and R are for lysine and arginine.

releases its hormones in response to at least nine neurohormones known as **releasing hormones** or releasing factors.^{16,60,61} They are secreted in minute quantities by the hypothalamus into a special portal vein that carries them directly to the pituitary where they exert their effects (Fig. 30-1B). As is indicated in Fig. 2-4 and in Table 30-3, several releasing factors are small peptides, but others are quite large. **Thyrotropin-releasing hormone** (THR, thyroliberin)⁶² is a tripeptide; but human **growth hormone-releasing hormone** (somatoliberin) is a 44-residue peptide. Both are synthesized as larger proteins, which are cleaved and processed to form the mature C-amidated hormones.^{63,64} **Corticotropin-releasing hormone** (CRH; CRF; corticoliberin), the 41-residue ACTH-releasing factor, is also cut from the much larger prepro-CRF.^{65,66} Release of both LH and FSH is stimulated by a single **gonadotropin-releasing**

hormone (GnRH).^{5a,10,67,68} The releasing factors bind to 7-helix G-protein coupled receptor.⁶⁸ Both the releasing factor and gonadotropin are released into the appropriate parts of the bloodstream in a pulsatile fashion emphasizing the neural origin of their release.^{69,70}

The hypothalamus also synthesizes **release-inhibiting factors**.⁶⁷ One of these, **somatostatin** (Table 30-3), inhibits release of somatotropin, thus counteracting the effect of the growth hormone releasing hormone. Somatostatin acts both in the pituitary and also in the pancreas, where it inhibits the release of both insulin and glucagon.^{71,72} The result is a lowering of blood glucose. This suggested a new approach to the treatment of diabetes. However, because of the many other effects of somatostatin⁷³ and its rapid degradation it has not been useful clinically. Nevertheless, hundreds of analogs of somatostatin have been synthesized, some of which may be of practical

TABLE 30-2
Nonpeptide Vertebrate Hormones

Type, source, and name of hormone	Principal site of action	References Chapter, Section
A. Amino acid derivatives		
1. Thyroid		Ch 25,B,2
Thyroxine and triiodothyronine	Most cells	
2. Adrenal medulla		Ch 11
Adrenaline, noradrenaline (epinephrine, norepinephrine)	Most cells	
3. Pineal gland		
Melatonin	Melanophores	Fig. 25-12
4. Nerves and other cells		
Serotonin (5-hydroxytryptamine)	Arterioles, central nervous system	Fig. 25-12
B. Steroids and prostaglandins		
1. Testes		Ch 22, F
Testosterone	Most cells	Fig. 22-11
2. Ovaries		
Estrogen (estradiol-17 β)	Most cells	Fig. 22-11
3. Corpus luteum		
Progesterone	Uterus, mammary glands	Ch 12; Ch 28
4. Adrenal cortex		Fig. 22-11
Corticosterone, cortisol	Most cells	Fig. 22-11
Aldosterone	Kidney	Fig. 22-11
5. Various tissues		
Prostaglandins	Smooth muscle	Ch 21, D
C. Volatile hormones		
1. Nitric oxide, NO	Endothelium, brain	Fig. 21-8
2. Carbon monoxide, CO	Brain	

value.⁶⁷ The biological activity of somatostatin resides largely in the sequence FYKT at positions 6–10, a sequence that is thought to form a beta turn (Fig. 30-3). Much of the rest of the molecule can be left off and the disulfide bridge moved up as far as positions 6–11 with retention of high potency. Human somatostatin is synthesized initially as a 116-residue precursor.⁷⁴

A 56-residue peptide, which is formed from the 10-kDa precursor to GnRH, inhibits secretion of prolactin.⁷⁵ Inhibition of FSH release is accomplished by feedback inhibition. Hormones known as **inhibins** are produced in the gonads and act to inhibit release of FSH from the pituitary.⁷⁶

Vasopressin and oxytocin. In contrast to the large peptide hormones made in the anterior lobe of

the pituitary are **vasopressin** and **oxytocin**, which are secreted from the neurohypophysis, the posterior lobe.⁶⁰ The neurohypophysis consists of neural tissue, whose secretions are directly controlled by the central nervous system. In fact, the cell bodies of the secretory neurons are located in specific nuclei of the hypothalamus (Fig. 30-1B). About 4000 vasopressin-secretory neurons and a similar number of oxytocin neurons are present in the neurohypophysis of the rat.¹⁶ Vasopressin is a major regulator of blood volume and pressure,⁷⁷ and its secretion is influenced by stress. It increases the water permeability of the kidney collecting duct cells by inducing translocation of aquaporin proteins from intracellular storage vesicles into the apical plasma membrane.⁷⁸ Vasopressin binds to

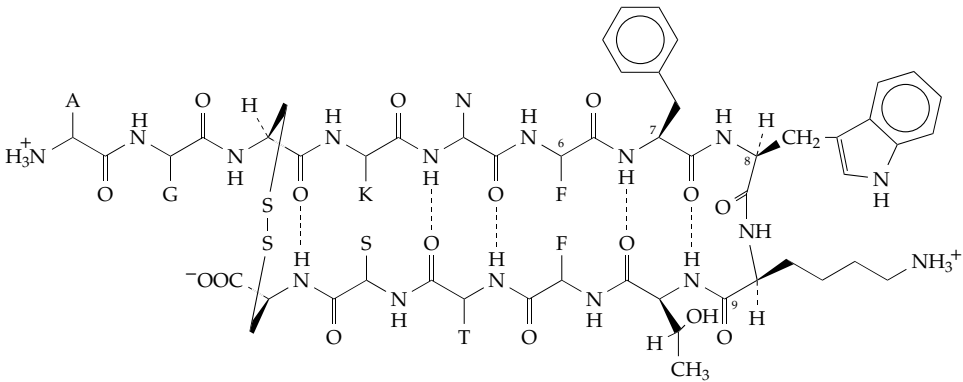


Figure 30-3 Possible secondary structure of somatostatin with a beta turn at residues 7–10 and a disulfide bond between positions 3 and 14. The true conformation would have the plane of the beta sheet puckered and twisted.

TABLE 30-3
Releasing and Inhibiting Hormones from the Hypothalamus

Name	Number of amino acid residues	Sequence ^a
Thyrotropin-releasing hormone (thyroliberin, TRH)	3	pEHP-NH ₂
Gonadotropin-releasing hormone (GnRH, LH- and FSH-releasing hormone)	10	pEHWSYGLRPG-NH ₂
GH-releasing hormone (somatoliberin)	44	
Corticotropin-releasing hormone (corticoliberin, CRH)	41	
MSH-releasing factor (melanoliberin) ^b	5	CYIQNC └─S─S─┘
Somatostatin (GH-release-inhibiting hormone)	14	AGCKNFFWKTFTSC └──S─S──┘
MSH-release-inhibiting factor ^b	3	PLG
Prolactin-releasing factor		
Dopamine (prolactin-release-inhibiting factor)		

^a Standard one-letter abbreviations are used. pE is pyroglutamyl (5-oxoprolyl) and –NH₂ at the right indicates a C-terminal *carboxamide*.
^b Ring and tail fragments of oxytocin.

G-protein-coupled receptors.⁷⁹ A defect in the type 2 vasopressin receptor leads to the condition of nephrogenic **diabetes insipidus** in which the body fails to concentrate the urine.^{77,80} Oxytocin acts on smooth muscles of the uterus during childbirth and triggers the release of milk from the mammary glands.⁸¹ The latter response is partially controlled by the suckling of the infant, which induces the nervous system to release oxytocin into the bloodstream.

Hormones related to oxytocin and vasopressin occur in most vertebrates, the compound **vasotocin** shown in Fig. 30-4 being the most common. Substitution of phenylalanine for isoleucine at position 3 gives arginine vasopressin, the vasopressin found in our bodies. Structure of oxytocin and related hormones⁸² are also shown in Fig. 30-4. Like somatostatin, vasopressin and oxytocin may also form antiparallel pleated sheet structures with β turns. The structural requirements for hormone activity have been studied intensively. Both the macrocyclic hexapeptide ring and the tripeptide side chains are necessary for maximal activity.⁸³

The gene for arginine vasopressin is that of a 166-residue precursor protein carrying a 19-residue signal sequence at the N terminus.⁸⁴ This sequence is followed by that of vasopressin, then after a GKR linker by the 95-residue **neurophysin II**. Finally, after one additional arginine there is a 39-residue glycopeptide. Oxytocin originates in a parallel way from its own precursor.⁸⁵

The 93- to 95-residue neurophysins act as carriers for vasopressin and oxytocin, forming specific complexes with them. Neurophysins contain 14 cysteine residues, which form seven disulfide bonds. There is a striking similarity in sequence between the neurophysins, snake venom toxins, a wheat germ lectin (agglutinin), a ragweed pollen allergen, and a small plant protein called hevein. On the basis of the alignment of cysteine residues, Drenth proposed⁸⁶ that all of these proteins have a disulfide-linked core whose structure is shown in Fig. 30-16.

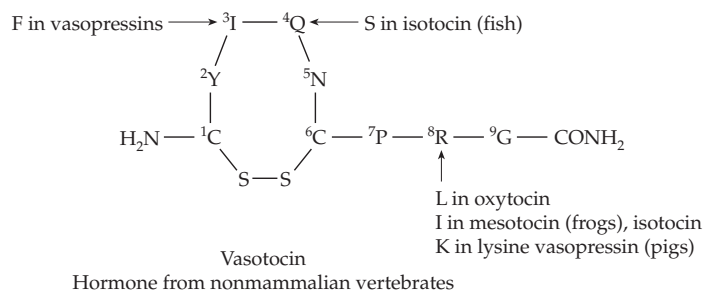


Figure 30-4 Structure of the nonmammalian hormone vasotocin and of related hormones including oxytocin and vasopressin.

Melanocortins. The melanocortin peptides, which are derived from pro-opiomelanocortin as indicated in Fig. 30-2, are formed in varying amounts in the pituitary, in two brain nuclei, and in some peripheral tissues.²⁵ In some animals α -MSH arises primarily in an intermediate part of the pituitary. The hormone has a direct effect on the melanocytes (Box 8-F) causing darkening of the skin. In addition, the various melanocortin peptides (ACTH, α -, β -, and γ -MSH) bind to five different types of receptors. These have been linked to the control of energy homeostasis, appetite, and obesity in both mice and humans.⁸⁷⁻⁸⁹ The sequence His-Phe-Arg-Trp (of the green-shaded sequence in Fig. 30-2) is essential for binding. In keratinocytes α -MSH may form a 1:1 complex with tetrahydrobiopterin,⁹⁰ the coenzyme for tyrosine hydroxylase, and a regulator of tyrosinase, an essential enzyme for melanin formation (Fig. 25-6).

3. Pancreatic and Related Hormones

The functions of the 51-residue insulin (Figs. 7-17 and 7-18) are discussed in Chapter 11. Its actions begin early in life. Mammalian preimplantation blastocysts already show a response to insulin. The glucose transporter GLUT1 is present at the earliest stages; synthesis of GLUT2 and GLUT3 (p. 416) begins at the eight-cell stage. However, the insulin-regulated transporter GLUT4 is not present in the blastocyst. A newly discovered insulin-regulated GLUT8 may function during preimplantation development.⁹¹ Although the secretion of insulin is of primary importance to the regulation of the glucose concentration in mammals, it is still not clearly understood.⁹² The beta cells have insulin receptors and other components of the insulin signaling system such as the insulin receptor substrates IRS-1 and IRS-2 and phosphatidylinositol 3-kinase (PI3-K). The sensing of glucose by the beta cells is also not yet well understood.⁹³ This lack of knowledge has made it difficult to improve the treatment of diabetes. A new approach is to engineer non-beta cells to secrete a steady supply of insulin. Such a possibility has been demonstrated in mice using gut K-cells.⁹⁴

The **insulinlike growth factors** (IGF-I and IGF-II) are produced in many different tissues and promote growth of other cells (see Section 6). **Relaxin**,^{94a} which is produced in the corpus luteum of ovaries during pregnancy, is responsible for inducing widening of the birth canal during the late stages of pregnancy and inhibits contraction of uterine muscle, perhaps by decreasing the activity of the kinase that phosphorylates the 20-kDa light chains of myosin.⁹⁵ Relaxin is found throughout the animal kingdom, even in the protozoan *Tetrahymena*.⁹⁶ Its

structure is apparently identical in pigs, whales, and in a primitive tunicate.⁹⁷ In human males relaxin is apparently produced in the prostate, where it may function as a sperm motility factor.⁹⁷ Relaxin, IGF-I, and IGF-II are all structurally homologous to proinsulin and contain the characteristic 3-disulfide structure of insulin. The IGF-I receptor structure resembles that of the insulin receptor (Fig. 11-11) and also that of the epidermal growth factor (EGF) receptor.⁹⁸

Glucagon belongs to a family that also includes the gastrointestinal hormones **secretin**, **gastrointestinal inhibitory peptide (GIP)**, **vasoactive intestinal peptide (VIP)**, and **glicentin** (Table 30-4). The function of glucagon in regulation of the blood glucose level is considered in Chapter 17, but the hormone may have other effects. A complex processing pathway converts 14- to 16-kDa preproglucagons into the active hormone.^{99,100} Proglucagon is processed to glucagon in the pancreas, but in the endocrine L cells of intestinal mucosa it yields glicentin, a polypeptide containing the entire glucagon sequence, and other products.^{101-102a} Glucagon receptors generate both cAMP and Ca^{2+} as second messengers.¹⁰³

The 27-residue secretin stimulates secretion of bicarbonate into the pancreatic juice and inhibits gastric secretion of acid. The 28-residue VIP is found throughout the gastrointestinal tract of mammals and birds as well as in the brain and the lungs. It is a potent vasodilator and may be the major relaxant of pulmonary smooth muscle.¹⁰⁴ It has been reported totally absent from lungs of asthma patients.¹⁰⁵ The gastrointestinal inhibitory peptide (GIP) is larger than VIP but also has a close homology with glucagon (Table 30-4).¹⁰⁶

The 36-residue **pancreatic polypeptide** is a hormone of uncertain functions. The crystalline polypeptide has at the N terminus an 8-residue collagenlike helix that lies parallel to a C-terminal α helix (Fig. 30-5). The overall shape resembles that of both insulin and glucagon.^{107,108} This PP-fold includes also neuropeptide Y, which is considered in the next section,¹⁰⁹ and neuropeptide YY.¹¹⁰

4. Gastrointestinal and Brain Peptides

The largest endocrine gland in the body is the gastrointestinal tract, which produces a profusion of peptide hormones, many of which are also found in the brain.^{111,112} Indeed, a majority of the known verte-

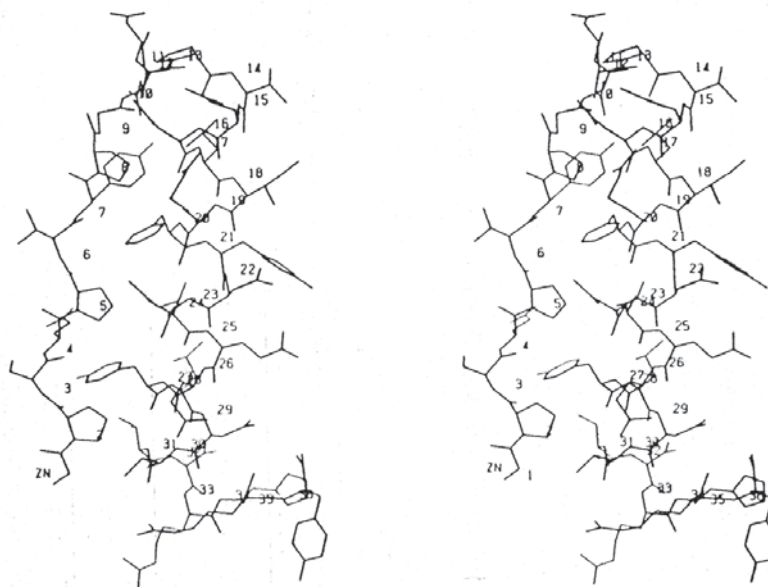


Figure 30-5 Structure of the avian pancreatic polypeptide, a small globular protein. From Blundell *et al.*¹⁰⁷

brate peptide hormones occur in the brain.^{82,112,113} For example, glucagon has been found in the brainstem and hypothalamus.¹¹⁴ Many of these peptides, or closely related ones, are also found in lower invertebrates.¹¹⁵ For example, the 10-residue **Hydra head activator** is also present in mammalian brain.^{115,116} The concentrations of these peptides in the brain is very low (10^{-12} – 10^{-15} M).

Gastrin is produced in the lower portion of the stomach and regulates the secretion of acid as well as growth of the gastrointestinal mucosa.^{112,117} It may also function as an Fe^{3+} carrier.^{117a} The shorter gastrin 17 as well as the longer 34-residue gastrin 34 are both active as is a synthetic pentapeptide with the hormone's C-terminal sequence.¹³ The family of **pancreozymin-cholecystokinins (CCK)** are 8- to 58-residue peptides produced in the upper intestinal tract. They have a 4-residue amidated C-terminal sequence in common with gastrin. This tetrapeptide has some biological activity, but eight residues are required for full activity as is conversion of the tyrosine at position seven from the C terminus to an O-sulfate ester. Both gastrin and CCK molecules are partially converted to sulfate esters.^{118,119} Most regions of the brain contain CCK peptides in amounts exceeding those of other neurotransmitters. These arise (in pigs) from a 114-residue preproCCK.¹²⁰⁻¹²² The sulfated insect neuropeptide, **leukosulfakinin** (Table 30-4), is homologous to gastrin and CCK.¹²³ Another hormone **gastrotropin**¹²⁴ is produced by cells of the intestinal mucosa in the distal ileum and stimulates gastric secretion.

Motilin, a 22-residue intestinal neuropeptide, stimulates motor activity of the gastrointestinal tract.^{125,126} **Bombesin** was first isolated from frog skin but probably also functions in both the intestinal tract and the brain. It has a powerful hypothermic effect.^{127,128} A mammalian homolog of bombesin, the 27-residue **gastrin-releasing peptide** (GRP), is found throughout the gastrointestinal and pulmonary tracts as well as the central nervous system.^{120,129} Bombesin-like material, possibly GRP, is produced by some cancers and may serve as an autocrine growth factor.¹³⁰ The 29-residue **galanin** was originally isolated from porcine intestine but is found throughout the central nervous system. It may function as a neurotransmitter or modulator.¹³¹ The 15-residue **guanylin** is an important regulator of epithelial transport in the intestine and probably in other tissues.^{132,133} The active hormone, which is cut from the 99-residue proguanylin, contains two disulfide bonds. Guanylin receptors activate guanylate cyclase with production of cyclic GMP, which functions in the regulation of intestinal fluid and electrolyte absorption. The 18- or 19-residue heat stable enterotoxins of some strains of *E. coli* bind to and activate the guanylin receptors. The resulting overproduction of cGMP causes severe diarrhea (see also Box 11-A).¹³³

Neuropeptides Y (NPY) and YY are 36-residue amidated peptides that are members of the pancreatic polypeptide (PP) family (Fig. 30-5). NPY is produced both in the peripheral nervous system and in the brain,^{110,134} where it is one of the most abundant neuropeptides. Another member of the PP family is **seminalplasmin**, a regulator of calcium ion transport in bovine sperm.¹³⁵ NPY is best known for its stimulation of appetite. It also inhibits anxiety and increases memory retention. It has a vasoconstrictive effect on blood vessels, participating in cardiovascular regulation.^{136,137} Peptide YY is formed in endocrine cells of the intestine, while NPY is formed in neurons of the parasympathetic system.¹³⁸ Both participate in regulation of fluid and electrolyte secretion. Both are found in other vertebrate species.¹³⁹

NPY is one of the most important of several regulators of feeding behavior of animals. PYY₃₋₃₆, another member of the neuropeptide Y family, suppresses appetite by antagonizing the action of NPY.^{139a,b} A large variety of hormonal effects seem to be involved in control of appetite.^{139b,140,141} There are both short-term and long-term mechanisms. For example, when introduced into the gut of rats prior to feeding, CCK and various other gastrointestinal peptides decrease the amount of food eaten.¹⁴⁰

Much attention has been focused on the 146-residue cytokine **leptin**, a hormone produced by adipose tissue.¹⁴¹⁻¹⁴⁴ Leptin, which is sometimes described as the antiobesity hormone, was recognized by mutations of the *obese* gene (OB) or of the OB receptor in

genetically obese mice. When food is scarce, the fat cells shrink and decrease their secretion of leptin. The decrease is sensed by receptors in the hypothalamus, which signal for increased NPY secretion and decreased secretion of α MSH. NPY increases appetite, while α MSH has an opposing role of blocking feelings of hunger.^{144-145a} Nevertheless, there are doubts that leptin's primary role is control of obesity.^{146,147}

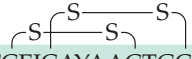
The 13-residue **neurotensin** was first isolated from the hypothalamus but is more abundant in cells of the ileum.¹³ It induces gut contraction, lowers blood pressure, and has a variety of other effects.^{127,148}

Substance P (SP; Table 30-4) has been regarded as a possible neurotransmitter for some time¹²⁷ but is also found in the digestive tract. It is the most abundant of a family of five neurokinins (or tachykinins). Others include neurokinin A (substance K), neurokinin B, neuropeptide K, and neuropeptide γ . They have a common C-terminal sequence FXGLM-NH₂.^{149,150} Substance P is thought to be involved in the perception of pain, and mice lacking a substance P receptor appear to have reduced sensitivity to pain.^{151,152} Substance P as well as the related substance K are derived from two large precursor proteins, which appear to arise as a result of alternative modes of splicing of mRNA.²⁹

5. Other Mammalian Peptide Hormones

The action of the 32-residue thyroid hormone **calcitonin**¹⁵³ has been described in Box 22-C. This calcium-regulatory hormone is produced in the thyroid C cells from a precursor having an extra 82 residues at the N terminus and 16 residues at the C terminus. The same gene gives rise in neural tissues to a neuropeptide, possibly a neurotransmitter, called **calcitonin gene-related polypeptide** (CGRPP).^{29,154} The 84-residue **parathyrin** (parathyroid hormone) is present in secretion granules as a 90-residue prohormone containing six extra residues at the N terminus. The primary biosynthetic product **preproparathyrin** contains an additional 25 residues.¹⁵⁵ An N-terminal 34-residue fragment of the hormone, when injected subcutaneously daily, causes an increase in bone density in persons with osteoporosis.¹⁵⁶⁻¹⁵⁷ The hormone acts via a G-protein-coupled receptor in bone and kidney (see Box 22-C).^{158,159} A calcium ion receptor, which binds Ca²⁺ cooperatively, acts as a sensor that regulates release of the parathyroid hormone to regulate the serum Ca²⁺ concentration.¹⁶⁰ A 141-residue **parathyroid hormone-related protein** has an N-terminal sequence homologous with that of parathyroid hormone, eight of the first 13 residues being identical. It is secreted by a variety of cells and serves as a growth factor.¹⁶¹

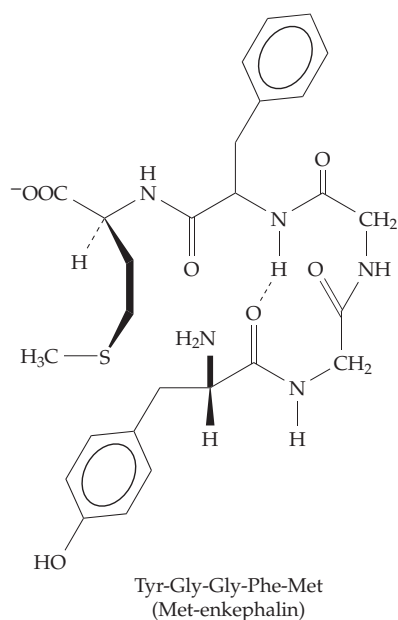
TABLE 30-4
Some Pancreatic and Gastrointestinal Hormones and Neurohormones and Their Sequences

Name and source	No. of residues	Sequence ^a
Glucagon	29	HSQGTFTSDYSKYLDSRRRAQDFVQWLMNT
Secretin (pancreas)	27	HSDGTFTSELSRLRDSARLQRLQLV-NH ₂
Vasoactive intestinal peptide	28	HSDAVFTDNYTRLRKQMARKKYLSILN-NH ₂
Gastrointestinal inhibitory peptide (GIP)	43	YAEGTFISDYIAMDKIRQQDFVNWLLAQ-Q ⁴³
Glicentin	100	A proglucagon containing the entire glucagon sequence in residues 64–92
Pancreatic polypeptide	36	
Neuropeptides Y (NPY) and YY	36	
Gastrin (stomach)		
Gastrin-17	17	pEGPWLEEEEEAYGWMDF-NH ₂ ^b
Cholecystokinin, CCK or pancreozymin (gallbladder, pancreas), many forms exist		
CCK 58	58	
CCK 8	8	DYMGWMDF-NH ₂
Motilin (porcine)	23	FVIFTYGE LQRMQE KERNKGQ
Bombesin	14	pEQRLGNQWAVGHLM-NH ₂
Gastrin-releasing peptide	27	
Galanin	29	
Guanylin	15	
Neurotensin	13	pELYENKPRRPYIL
Substance P	11	RPKPQQFFGLM-NH ₂
Physaelemin (frog skin)	11	pEADPNKFYGLM
Neurophysins	93 – 95	
β-Endorphin	31	YGGFMTSEKSQTPLVTLFKNAIIKNAHKKGQ
Dynorphin	17	YGGFLRRIRPKLKWDNQ
Met-enkephalin	5	YGGFM
Leu-eukephalin	5	YGGFL
Angiotensin II	8	DRVYIHPF
Bradykinin (BK)	9	RPPGFSPFR
Lys-bradykinin (kallidin)	10	KRPPGFSPFR
Sleep peptide	9	WAGGDASGE
Atrial natriuretic hormone	28	
Chemotactic factors		
for neutrophils	3	f-MLF
for phagocytes	4	TKPR
Speract	10	GFDLNGGGVG

^a Standard one-letter abbreviations; pE, 5-oxoprolyl; f-, formyl; -NH₂, C-terminal carboxamide.

^b Y-12 may be sulfated.

Endogenous opioid peptides. Extensive processing is also involved in formation of analgesic opioid peptides, which are present naturally in the brain (see also Section B). The formation of β -endorphin in the hypothalamus from prepro-opiomelanocortin (Fig. 30-2) has already been mentioned. Prior to the discovery of β -endorphin, the pentapeptides **Met-enkephalin** and **Leu-enkephalin** (Table 30-4) were discovered and were found to compete with opiate drugs for receptors in the brain. The larger β -endorphin, which contains the Met-enkephalin sequence at its N terminus, is a far more potent opiate antagonist than are the enkephalins. Since the Met-enkephalin sequence within β -endorphin is not flanked by basic residues, it apparently is normally not released. Two other recently discovered brain peptides are **endomorphin-1** (YPWF-NH₂) and **endomorphin-2** (YPFF-NH₂). They are also potent agonists for the opioid receptors, especially the μ receptor (see Section B,10).^{161a,161b}



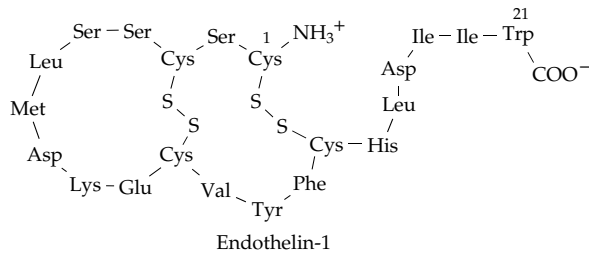
Both Met-enkephalin and Leu-enkephalin have their own pro- and prepro forms.²⁹ Bovine **prepro-enkephalin A** is a 268-residue protein containing a 20-residue signal sequence and four sequences of Met-enkephalin and one of Leu-enkephalin, each flanked by pairs of basic residues. There are also Met-enkephalin-Arg-Gly-Leu (YGGFMRGL) and Met-enkephalin-Arg-Phe sequences. Not all of these are cut out cleanly, and other peptides such as Met-enkephalin-Arg-Arg-Val-NH₂ are also found in brain. **Proenkephalin B** contains three copies of Leu-enkephalin contained within longer peptides. One of these, **β -dynorphin** (Table 30-4), is also a potent opioid compound. The enkephalins are thought to act as neurotransmitters, which are rapidly degraded after their release by two or three membrane-bound

peptidases.^{29,162} Attempts are being made to design inhibitors that might inactivate these enzymes allowing buildup of enkephalin concentrations with a resultant analgesic effect.

ATP, ADP, and adenosine. Usually regarded as a strictly intracellular compound, ATP is also released into extracellular space. There the ATP, as well as ADP and adenosine, have a variety of local hormonal functions. ATP receptors are found in many tissues and are present in some nerve synapses.^{163–166} ATP is one of the substances that induces sensations of pain.^{164,167,168} It may affect secretion of saliva,¹⁶³ signal a full urinary bladder, induce a feeling of warmth,¹⁶⁷ and have functions in the immune system, in platelet clotting,¹⁶⁶ and as a neurotransmitter. Adenosine has been recognized for many years as an extracellular signaling molecule, a local hormone that can arise by breakdown of ATP or by secretion from cells.^{169,170} At least four types of receptor are present in the human body.^{170–174} Adenosine is thought to modulate neural responses in many tissues. It may be involved in sleep,¹⁷² in regulation of serotonin transport,¹⁷¹ and in control of appetite.^{171,174} Extracellular ADP appears to have a role in controlling bone osteoclasts (p. 441).¹⁷⁵

Kinins. These hormones are small peptides that induce contraction of smooth muscles, lower blood pressure (Box 22-D), and increase vascular permeability.¹⁷⁶ They also have a function in contact-activated blood coagulation. The most important human kinins are the nonapeptide **bradykinin**^{177,178} and the related decapeptide **lysine-bradykinin** (Table 30-4). Other forms such as Met-Lys-bradykinin and Ile-Ser-bradykinin (T-kinin) are also known. The precursors to the kinins, the **kininogens**,¹⁷⁶ are cleaved by the protease **kallikrein** (Fig. 12-17) or by kallikreinlike enzymes to form the kinins. Kinins are suspected of being important producers of pain in inflammatory conditions such as arthritis.^{176a}

Endothelins. Endothelial cells of blood vessels produce **endothelins** that cause vascular smooth muscle contraction and a rise in blood pressure.^{179–183} Three human genes code for the closely related endothelins-1, -2, and -3. A 203-residue preproendothelin-1 is processed to form the 39-residue prohormone called **big endothelin-1**. Some of this peptide is secreted and circulates in plasma, where it may have various hormonal functions. Cleavage of the prohormone by a cellular metalloprotease yields endothelin-1, a 21-residue peptide held in a looped configuration by two disulfide bridges. It is homologous to a group of neurotoxins that includes the α -scorpion toxins and ω -conotoxin.¹⁸¹ These toxins act on voltage-dependent ion channels. Endothelin-2 is produced largely in the kidneys and intestine, while endothelin-3 is found in



high concentrations in the brain. Type A endothelin receptors are 7-helix G-protein-coupled proteins, which activate phospholipase C with generation of inositol 1,4,5-trisphosphate and diacylglycerol (Ins- P_3 ; Figs. 11-4, 11-9). The Ins- P_3 causes release of Ca^{2+} , while diacylglycerol mediates mitogenic responses.

Opposing the effects of the endothelins, which act slowly, is a fast-acting endothelium-derived **relaxing factor**, which has been identified as nitric oxide, NO. It is discussed in Chapter 18, Section F, and in Section 7 of this chapter. Also affecting blood pressure is the potent vasorelaxant **atrial natriuretic factor**. This 28-residue peptide, which is discussed in Box 22-D, is produced by the cardiac atria and stimulates the excretion of Na^+ and of water by the kidneys.¹⁸⁴ It also promotes hydrolysis of lipids within human adipocytes.¹⁸⁵

Peptides as attractants. Small peptides as well as larger polypeptides serve to attract cells within the human body and other multicelled organisms. Both unicellular and multicellular organisms also use peptides as pheromones. The human immune system depends upon hormonelike **chemotactic factors**. Neutrophils are attracted by such peptides as formyl-Met-Leu-Phe,^{186–188} which have a bacterial origin, while the basic tetrapeptide Thr-Lys-Pro-Arg activates the phagocytic polymorphonuclear leukocytes and macrophages.¹⁸⁹ Larger 8- to 10-kDa proteins known as **chemokines** (chemotactic cytokines) attract leukocytes to sites of inflammation (Fig. 30-6).^{190–192} Some proteins serve as pheromones. Examples range from the 40-residue mating pheromones of protozoa of the genus *Euplotes*^{193,194} to the 17-kDa sex pheromone of the female hamster.¹⁹⁵ The decapeptide **speract** (Table 30-4) is produced by sea urchin eggs and stimulates the respiration of spermatozoa.¹⁹⁶ Similar factors probably function in fertilization of human ova.

6. Protein Growth Factors and Cytokines

The pituitary growth hormone is only one of a large family of protein growth factors that are secreted by cells and which promote the growth of other cells.²⁰⁰ Many of the growth factors are also described as **cytokines**, local protein hormones that conduct cell-to-cell communication to regulate growth, development,

and differentiation.^{197,201–203} Among the first growth factors to be recognized were the **insulinlike growth factors** (IGF or somatomedins), mitogenic peptides isolated from plasma. They share some of the metabolic effects of insulin but are less active. On the other hand, they are much more active than insulin²⁰⁴ in promoting cell growth and proliferation of cells.²⁰⁵ The abundant IGF-I (somatomedin C), a 70-residue single-chain basic peptide with a sequence and three-dimensional structure homologous to that of proinsulin,^{206,207} is considered a major mediator of the action of the pituitary growth hormone (GH, somatotropin). Studies in cell culture suggest that GH may induce differentiation of cells, and that IGF-I may then cause a rapid proliferation of the newly differentiated cells.²⁰⁸ The homologous 67-residue IGF-II may have a similar function in fetal development.²⁰⁹ The cell surface receptor for IGF-I is similar to the insulin receptor, but IGF-II receptor is structurally different. It is a monomeric 250-kDa protein; and although it is a substrate for a tyrosine kinase, it has no kinase activity of its own.²¹⁰

The 53-residue **epidermal growth factor** (EGF or **urogastrone**) is found in human urine and in very high concentration in the submaxillary salivary glands of male mice. Like the pancreas these glands contain both endocrine and exocrine tissues. EGF is synthesized in mice as a 1217-residue precursor, which contains not only the EGF sequence but also seven other related sequences.²¹¹ Related growth factors include transforming growth factor- α (TGF- α), neuregulins,^{212–214} betacellulin, and epiregulin, all of which promote growth of epithelial cells and are involved in wound healing.²¹⁵ The EGF and related peptide chains are each crosslinked by three disulfide bridges. The three-dimensional structure of EGF, deduced from NMR measurements, contains largely β structure and loops and is organized into two domains in a “mitten shape.”²¹⁶ The receptor for EGF is a 1186-residue transmembrane glycoprotein. Its relationship to cellular oncogene *c-erbB* has been discussed in Chapter 11, Section H. The extracellular glycosylated N-terminal region of the receptor contains the EGF-binding site.^{217–220} It also contains two cysteine-rich repeat sequences homologous to one of those in the insulin receptor A chain (Fig. 11-11). The cytoplasmic C-terminal part of the EGF receptor contains a 250-residue tyrosine-specific protein kinase sequence. Following dimerization the EGF receptor phosphorylates tyrosine residues in various proteins including itself (autophosphorylation).^{212,219,221} The receptor is also phosphorylated on Thr 654 and other residues through the action of the Ca^{2+} - and phospholipid-dependent diacylglycerol-activated **protein kinase C** (Fig. 11-9).²²² Serines 1002, 1046, and 1047 may also become phosphorylated, perhaps resulting in desensitization of the receptor.²²³

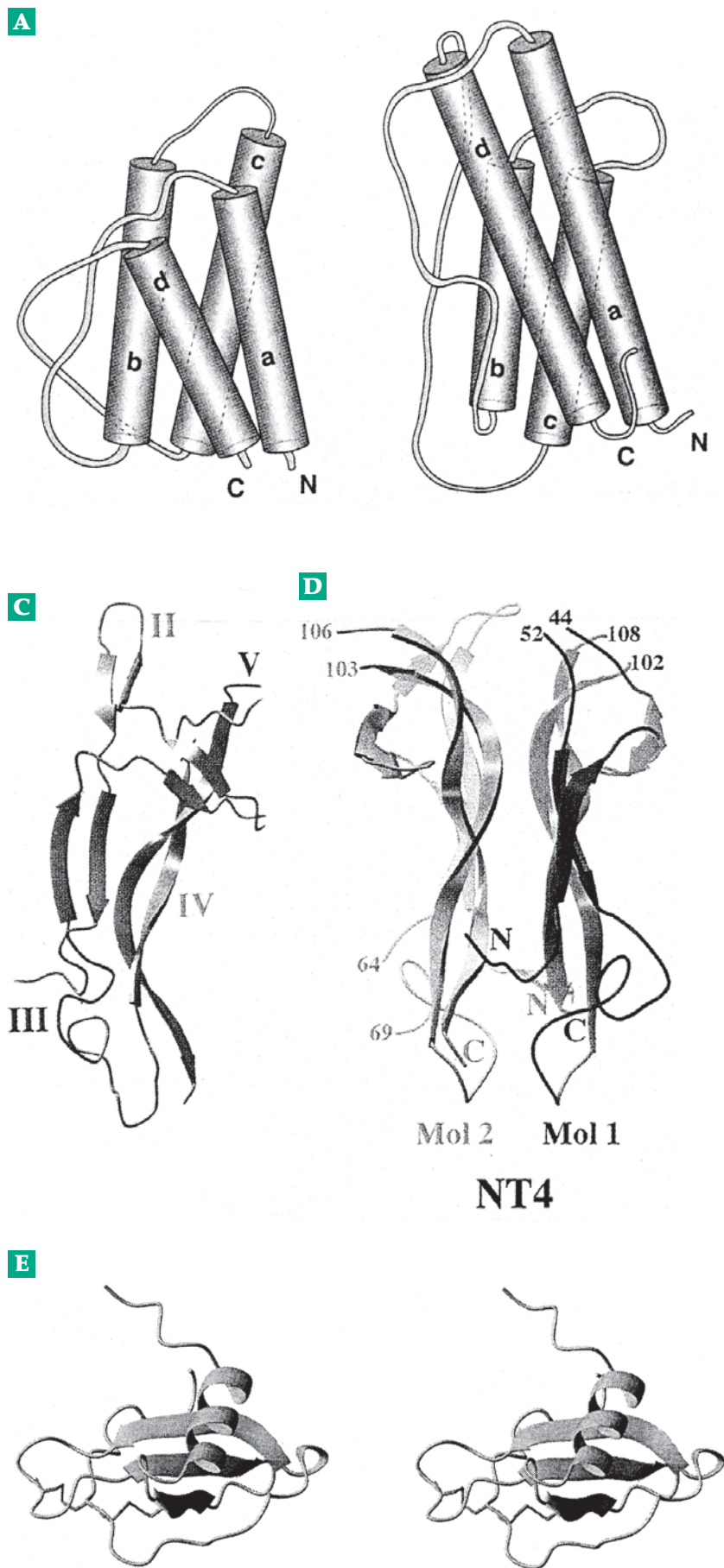


Figure 30-6 Structures of some cytokines and other growth factor proteins. (A) Schematic drawing of representative structures of “short” and “long” helical cytokines. Note the difference in the topology of the connection between helices A and B in the two models. The short cytokines (left) include interleukins-2 and -4. Human growth hormone has a long cytokine structure. Both groups include various colony-stimulating factors. (B) Schematic representation of the disulfide knot topology of cystine knot cytokines. The cysteine residues are numbered I to VI in order of occurrence. See Davies and Wlodawer.¹⁹⁷ (C) Ribbon drawing of a monomer of nerve growth factor (NGF). (D) Ribbon drawing of a dimer of the closely related neurotrophin 4 (NT4). (C) and (D) are from Robinson *et al.*¹⁹⁸ (E) Stereoscopic ribbon drawing of the chemokine **eotaxin-3**. From Ye *et al.*¹⁹⁹

Binding of EGF to its receptor produces within minutes an increased transcription rate for the prolactin gene and other nearby genes.²²¹ The urinary form of EGF, urogastrone, is an inhibitor of ulcer formation. It is found in relatively large amounts in the urine of pregnant women (who tend not to develop ulcers).

Platelet-derived growth factor (PDGF)²²⁴ is released from the α -granules of blood platelets during clot formation and is thought to stimulate the growth and mitosis in fibroblasts that is necessary for wound healing.²²⁵ It consists of two chains, A and B. The 31-kDa precursor of the A chain is encoded by the cellular oncogene *c-sis* (p. 571).²²⁴ The **PDGF receptor** is another transmembrane glycoprotein with a C-terminal tyrosine kinase domain. However, its construction differs from that of the insulin or EGF receptors. The external part of the single-chain receptor appears to contain five immunoglobulinlike domains (see Figs. 2-16 and 12-18).²²⁶ Binding of PDGF to the receptor causes responses within minutes.^{217,226} These include activation of the tyrosine kinase, hydrolysis of phosphatidylinositides, increases in the levels of cAMP and of Ca^{2+} , and increased transcription of a group of genes. The last include the proto-oncogenes *c-myc* and *c-fos*, which encode proteins that regulate transcription. The PDGF receptor is part of a recognized autocrine stimulatory loop in cells infected with a virus carrying the *v-sis* oncogene.²¹⁷ The oncogene product resembles PDGF and binds to the PDGF receptors of the cell producing the *v-sis* product. In this way the cancer cell stimulates its own growth.

Transformation of kidney fibroblasts into cancer-like cells can be induced by the concerted action of PDGF, an analog of EGF, and the **transforming growth factor** (TGF- β).^{225,227,228} The latter is one of 30 or more related growth factors that have numerous functions in normal tissues.²²⁹ Platelets produce a relatively large amount of the 25-kDa TGF- β , and it too may be involved in wound healing. TGF- α is a smaller protein with a structure resembling that of EGF.^{227,228} While TGF- β inhibits epidermal cell growth, TGF- α stimulates growth. It is found in elevated levels in the skin lesions of **psoriasis** (Box 8-F) and may be the cause of the excessive epithelial growth in that disease.²³⁰

There are at least nine **fibroblast growth factors** (FGFs). Originally found in brain, they act on many cells including the endothelial cells that line blood vessels.²³¹ Basic FGF²³² and acidic FGFs²³³ have homologous sequences²³⁴ and are also related to the lymphokine interleukin-1. **Vascular endothelial cell growth factor** (VEGF), which is similar to PDGF, is essential for maintenance of the endothelium. The FGFs and VEGF as well as TFG are potent **angiogenic factors** needed for growth of blood vessels.^{235–238} These proteins are important not only to normal blood vessels but also to invasive tumors that must develop

blood vessels in order to grow. Excessive production of angiogenic factors may also be a factor in eye diseases including the retinal deterioration caused by diabetes.²³⁹ Another protein, **angiogenin**, is a ribonuclease,²⁴⁰ which is discussed on p. 648.

There are four closely related transmembrane FGF receptors and subforms that arise by alternative mRNA splicing.^{241–243} The receptor structures include three external immunoglobulinlike domains and an internal tyrosine kinase domain at the C terminus. Mutations in FGF receptors are associated with a variety of skeletal defects and other hereditary problems.^{241,244} For example, the Gly380Arg substitution in the transmembrane segment of FGF receptor 3 is the major cause of **human dwarfism** (achondroplasia).²⁴⁵ The fibroblast growth factors, as well as other proteins such as the IGFs, HGF, and TGF- β , bind not only to their receptors but also to heparan and heparin. This binding appears to be a major factor in controlling the availability of the growth factors.^{242,246,247}

The **nerve growth factor** (NGF) was identified over 40 years ago by Rita Levi-Montalcini²⁴⁸ on the basis of its activity in promoting the profuse outgrowth of neurites from embryonic neurons (Fig. 30-7). The 118-residue monomer consists largely of three β -hairpin loops, which are held together by three disulfide bridges that form a “cystine knot.” The C15–C80 disulfide passes through a ring formed by the C58–C108 and C68–C110 disulfide bridges (Fig. 30-6B,C).²⁴⁹ A similar folding pattern and disulfide core are found in TGF- β 2 and also in several other **neurotrophins**, growth factors involved in the development and survival of neurons (Fig. 30-6D).^{198,250–251a} NGF may also have a more general function in promoting tissue repair.²⁵² Like EGF nerve growth factor is most abundant in the submaxillary glands of male mice. Larger oligomers containing bound Zn^{2+} are present in mouse submaxillary glands. Two different receptor proteins, one of which is a tyrosine kinase, are present on many cell surfaces.^{198,251,253} The glial cells, which lie between the neurons, have their own growth factors.²⁵⁴

Bone formation and resorption are influenced by several protein factors. For example, IGF-I stimulates formation of bone, but EGF promotes breakdown.²⁵⁶ Additional **bone-derived growth factors** and **morphogenetic factors** also have been described.^{256,257} A **cartilage-inducing** factor has been identified as TGF- β .²⁵⁸

A group of glycoproteins function as hematopoietic growth regulators in the development of blood cells.^{259–264} The 166-residue cytokine **erythropoietin** is the primary regulator of red blood cell formation in mammals.^{260,264,265} At least four glycoprotein **colony-stimulating factors** (CSF) promote proliferation of granulocytes and macrophages.^{259,266,267} The lymphocyte-produced **lymphokines** include the **interleukins** and other proteins. Two species of

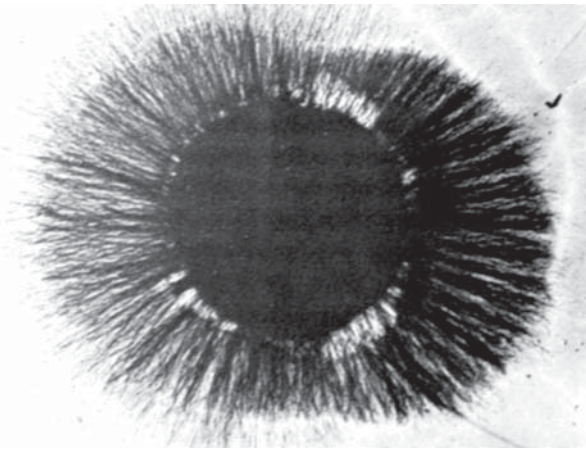


Figure 30-7 Effect of one ng of nerve growth factor in promoting the production of neurites in a chick embryonic sensory ganglion. From Frazier *et al.*²⁵⁵

interleukin-1 (IL-1) serve as mediators of inflammation.^{267,268} They induce proliferation of T lymphocytes and fibroblasts, bone resorption, release of acute phase proteins (Section C), breakdown of cartilage, and fever.

Interleukin-2 (T-cell growth factor; Fig. 30-6A) is secreted by some activated T-lymphocytes. This 133-residue largely helical protein is involved in generation of cytotoxic T-cells, stimulation of interferon release, and of release of a B-cell growth factor.²⁶⁹ Considerable excitement has accompanied the possibility of activating lymphocytes with IL-2 produced from cloned genes in bacteria to increase their ability to kill cancer cells. However, IL-2 is toxic, and this is limiting its use. See also Chapter 31, Section C.

IL-3 is one of the colony-stimulating factors, which stimulates the growth of many types of blood cells.²⁷⁰ Other lymphokines include one derived from T helper cells, which activates resting T lymphocytes thus amplifying an immune response. Others (Chapter 31) are **α -interferon** and the neurotrophic factor (autocrine motility factor) **neuroleukin**.²⁷¹ It acts in monomeric form, but as a dimer it seems to be identical to the enzyme phosphoglucose isomerase.²⁷² While most hormones regulating growth and differentiation seem to be large peptides or proteins, **bursin**, which induces differentiation of lymphocytes, is the amidated tripeptide Lys-His-Gly-NH₂.²⁷³ The corresponding differentiation hormone for T lymphocytes is the 49-residue **thymopoietin**, a hormone of the thymus gland.²⁷⁴

Tumor necrosis factor (TNF, also called cachetin) is a 157-residue hormone secreted by macrophages. It is a mediator of inflammatory responses including fever, shock, and **cachexia**, the wasting of the body during chronic diseases including cancer. TNF was isolated as the causative agent of cachexia and also as

a factor produced in acute bacterial infections, which sometimes caused death of tumor cells and spontaneous recovery from cancer. In the latter case, it is the lipopolysaccharide (Fig. 8-30) and other bacterial endotoxins that induce the release of TNF by macrophages. Its extreme toxicity has prevented immediate harnessing of the tumor-killing potential of TNF. One function of TNF is regulation of transcription factor NF- κ B (Fig. 5-40) in neutrophils and macrophages,²⁷⁵ a key part of the inflammatory response. TNF also mediates programmed cell death (apoptosis)²⁷⁶ and has been linked to obesity-induced insulin resistance.²⁷⁷ The cell surface TNF receptors have a variety of modular structures consisting of various disulfide-linked subdomains.²⁷⁶

This long list of vertebrate peptide growth and regulatory hormones is not complete. The biological actions of these hormones are also complex. Growth factors usually have pleiotropic effects, which may involve many tissues as well as many regulatory systems. Are there any simplifying generalizations? Loret *et al.*²⁷⁸ point out that some growth factors such as IFG-1 and EGF are ubiquitous, affecting virtually all tissues. Others, such as PDGF and thrombin (Fig. 12-17), are more localized in their effects. Some, such as the lymphokines, are more specialized. For one group of hormone receptors the effects are mediated by tyrosine kinases and internalization of the receptors. Another group of receptors activate G proteins and, in turn, adenylate cyclase or phospholipase C. The regulatory domains of the various receptors overlap, a property that allows different tissues to respond differently to hormonal stimuli. The result is the network of interactions that makes the body so sensitive and responsive.

7. Nonpeptide Mammalian Hormones

Most nonpeptide hormones have been considered in other places in the book as indicated in Table 30-2. Because of their importance in the brain adrenaline, noradrenaline, serotonin, and melatonin are also dealt with in Section B,9 of this chapter.

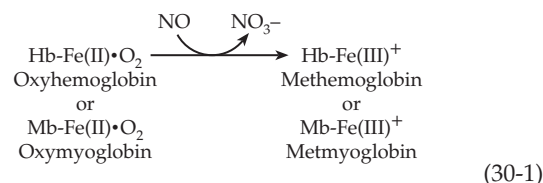
The volatile hormones nitric oxide (NO) and carbon monoxide (CO). The free radical molecule nitric oxide, commonly abbreviated as either $\bullet$ NO or simply NO, is formed by hydroxylation of guanidine groups of arginine (Eq. 18-65). First recognized as the **endothelium-derived relaxing factor**,²⁷⁹ NO has received increasing attention because of its involvement in a broad range of physiological processes. These include regulation of blood pressure through effects on smooth muscles of the vascular endothelium, regulation of several aspects of the innate immune system (Chapter 31), and neurotransmitter functions

both in the brain and in the peripheral nervous system.^{279–281} Roles for NO in bacteria, other microorganisms, and plants have also been discovered.^{281,282} These often involve regulation of transcription.²⁸³ As mentioned in Chapter 18, Section F,2, many of the effects of NO are a result of activation of soluble guanylate cyclase (p. 561).^{283–285a} In the endothelium other hormones, such as the endothelins (p. 1750), atrial natriuretic factor, and bradykinin (Box 22-D), cooperate in the regulation of NO synthase.²⁷⁹ Neuronal NO synthase functions in the brain in olfaction and in formation of memory. In the peripheral system it mediates penile erection^{284,285–286a} and plays a variety of roles in the enteric nervous system.²⁸⁷ Neuronal NO synthase is often localized to synaptic regions by binding to tissue-specific proteins.²⁸⁸ NO may also regulate cellular respiration by inhibition of cytochrome *c* oxidase.²⁸⁵

In high enough concentrations NO is toxic. It is formed of phagocytic cells and utilized in the killing of ingested pathogens.²⁷⁹ It also contributes to the inflammatory response of tissues.^{289,290} Even the firefly's flash is triggered by a pulse of NO.²⁹¹ The dangerous **stonefish**, whose sting causes death within six hours, apparently utilizes NO to kill its victim. A 148-kDa lethal protein (stonustoxin) in its venom induces rapid formation of NO, which causes a fatal drop in blood pressure.^{291a}

Like carbon monoxide, NO binds tightly to many metal centers within a cell.^{292,293} This has added greatly to the problem of understanding the mechanisms of its action. NO also reacts rapidly with thiol groups of proteins and of small molecules such as glutathione.^{294,295} Because of its importance in the regulation of blood pressure, reactions of NO with hemoglobin and the related myoglobin have been studied intensively.^{296–300a} NO binds to hemoglobin 1000 times more tightly than either O₂ or CO, preferentially occupying the hemes of the α subunits.²⁹⁸ Because there is so much hemoglobin in red blood cells, at most one NO per hemoglobin molecule can react. This allows as much as one NO to be carried to tissues along with three O₂ molecules. If the NO could be released in the capillaries, it would activate guanylate cyclase. The resulting cGMP would induce relaxation of smooth muscles and reduce blood pressure.²⁷⁹ However, tight bonding of NO to deoxyhemoglobin would prevent this release. A plausible possibility (with experimental support) is that NO is not bound to Fe but to the SH group of the conserved cysteine 93 of a β subunit of hemoglobin as SNO (*S*-nitrosothiol) hemoglobin. The NO may bind initially to the iron atom of an α subunit, but then be transferred to the nearby β Cys 93 (p. 359) to form the SNO-Hb.²⁹⁶ NO may then move from SNO-Hb to thiol groups in the tissues. Recent evidence suggests that the transfer occurs first to an SH group in the anion exchange **AE1**

(p. 420).²⁹⁹ An alternative explanation, which does not involve SNO-Hb, is that hemoglobin Fe-NO is converted to **nitrite** via oxidation of the iron to form a methemoglobin subunit (Eq. 30-1), and that it is nitrite which serves as the endothelial relaxing agent.³⁰⁰



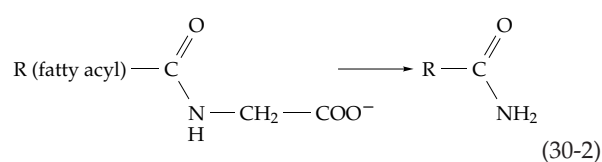
The function of the monomeric myoglobin has often been assumed to be participation in facilitated diffusion of O₂. Although this is an important function under some circumstances, an additional role for myoglobin may be to scavenge NO, via Eq. 30-1, preventing its buildup to dangerous levels. The metmyoglobin produced can be reduced by methemoglobin reductase.^{301,302} A different situation is met by the parasitic nematode *Ascaris*, whose hemoglobin binds O₂ so tightly that it can't serve as an O₂ carrier. It may serve as an **NO-activated deoxygenase**, again using Eq. 30-1 to remove O₂, which can be toxic to the nematode.³⁰³ Free myoglobin can also react with NO to form heme-NO and heme-nitroxyl complexes.^{304,305}

Like NO, CO also binds tightly to heme iron and is able to activate guanylate cyclase.³⁰⁶ CO is formed in the human body by the action of heme oxygenases (Fig. 24-24). Synthesis of heme oxygenase-1 (HO-1) in smooth muscle is induced by a low oxygen tension (hypoxia). The resulting elevated level of CO not only may produce increased vasodilation, but also may inhibit synthesis of vascular smooth muscle cells.³⁰⁷ Heme oxygenase-2 (HO-2) is found in the brain, where it is colocalized with soluble guanylate cyclase.³⁰⁸ Some other organisms have a more active CO metabolism. The CO oxidation system of *Rhodospirillum rubrum* is activated by a CO-sensing heme protein, which acts as a transcriptional regulator. The CO binds to the heme iron, apparently inducing a conformational change that allows the protein to bind to its target DNA sequence.^{309,310}

Hormonal lipids. We have already considered a number of hormones that are not water-soluble but may have to be transported by carrier proteins to their sites of action. These include retinoic acid (Box 22-A), metabolites of vitamin D (Box 22-C), and the platelet-activating factor (Box 8-A). The last functions in the brain³¹¹ as well as in blood. Hormonal lipids also include the prostaglandins (Fig. 21-7), leukotrienes, and lipoxins (Fig. 21-8). These are products of the eicosenoid cascade or network, which is activated by receptors linked to phospholipase C (Fig. 11-9). Ceramide formed by hydrolysis of sphingomyelin initiates

additional responses.^{312,313} Sphingolipids may also be important mediators of apoptosis.³¹⁴

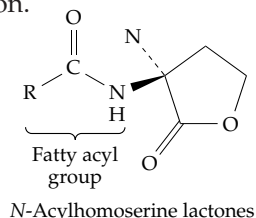
The sleep-inducing oleamide (p. 382) modulates signaling by serotonin-dependent and Gaba-dependent neurons and blocks gap junction signaling in brain glial cells.^{315–318} Oleamide is one of a family of fatty acid amides found in human plasma. One of these, found also in the brain, is **anandamide** (arachidonoyl-ethanolamide). It is an endogenous activator of the brain cannabinoid receptors.^{318–320} The 22-carbon **erucamide** (*cis*-13-docosenamide) stimulates growth of blood vessels.³¹⁶ The fatty acid amides are apparently synthesized from corresponding acylglycines (Eq. 30-2) by the action of the peptidylglycine α -amidating enzyme using the mechanism of Eq. 10-11. See also pp. 1792, 1793.



The fatty acid amides are destroyed by an integral membrane protein, a **fatty acid amide hydrolase**.^{321,322}

8. Nonvertebrate Hormones and Pheromones

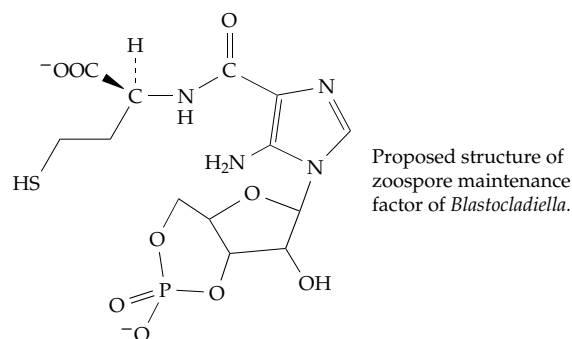
The chemical signals that are passed between bacteria and other microbial cells often resemble hormones of vertebrates. Thus, some bacteria secrete peptide mating pheromones. The sequence of an octapeptide²⁷⁰ of this type from *Streptococcus faecalis* is given in Table 30-5. Many bacteria utilize “quorum signaling.” They do not secrete signaling molecules until they sense that there are enough of them to be effective if they act in unison. Then they all secrete an inducer. The best-known example is the induction of bioluminescence of *Vibrio fischeri* (Eq. 23-49). Long-chain fatty acyl derivatives of L-homoserine lactone act as secreted inducers.^{323,324} A lactonase hydrolytically inactivates the inducer to avoid excessive accumulation.



Depending upon the types of bacteria and the specific response a variety of different fatty acyl groups may be present in the inducer. Other responses include the formation of bacterial film (biofilms) on a surface and release of virulence factors that induce attack on a host.

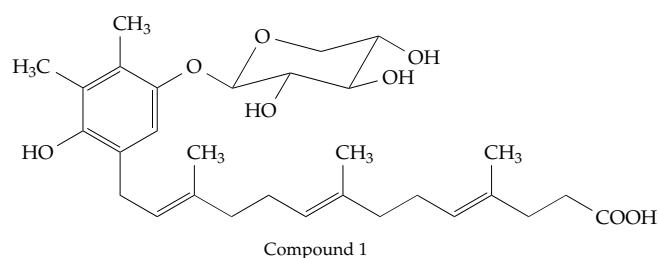
Sexual conjugation in yeast is also induced by pheromones (mating factors).^{325–327} Yeast cells of mating type **a** synthesize the 12-residue mating factor **a** which contains a C-terminal cysteine methyl ester S-alkylated with a *trans,trans*-farnesyl group (Table 30-5). Cells of type **α** synthesize a 13-residue factor **α** .^{327a} Cells are attracted to the pheromone produced by cells of the opposite type. The **tremmerogens**, sex hormones of certain basidiomycetes, have related structures (Table 30-5).³²⁸

Peptides are not the only fungal hormones. The water mold *Blastocladiella* releases a **zoospore maintenance factor**, a cyclic phosphate derivative of 5'-phosphoribosyl-5-aminoimidazolecarboxamide.³²⁹ It is similar to the succinocarboxamide, which is an



intermediate in *de novo* synthesis of purines (Fig. 25-15). Substitution of homocysteine for L-aspartate in step g of that sequence could generate a precursor to the zoospore maintenance factor.

Male sperm cells of the alga *Chlamydomonas allensworthii* (Fig. 1-11) are attracted to female gametes by a pentosylated isoprenoid quinone.³³⁰

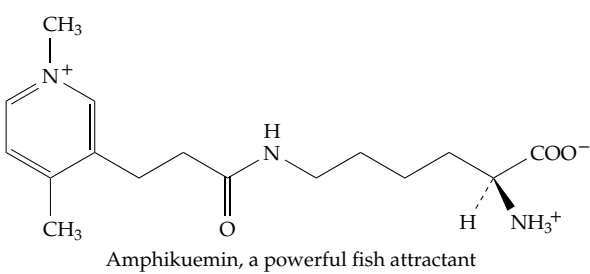


Neurohormones of invertebrate animals. The Cnidaria (coelenterates) have the simplest known nervous system. Simple amidated tetrapeptides, some of which are also found in molluscs, are among their neurotransmitters.³³¹ EGRFamide and L-3-phenyllactyl-LRNamide are found in some sea anemones.³³² When a hydra (Fig. 1-13) is cut into two pieces, one containing a head and one a foot, each piece reforms the missing end. The decapeptide **head activator** (Table 30-5) diffuses upward from the foot end and induces formation of a head. Similarly, a hormone produced by head cells

may induce growth of a new foot end. Glutathione, flowing out from the hydra's prey after wounding by a nematocyst, is the feeding attractant for *Hydra vulgaris*.^{282,333} Termination of the response is dependent upon nitric oxide in this primitive invertebrate.

Certain large anemones enter into a symbiotic relationship with fishes, which recognize chemical signals from the anemones and are also chemically protected from the anemones' stings. One of the fish attractants, **amphikuemin**,³³⁴ is effective at a concentration of 10⁻¹⁰ M.

A variety of peptide neurohormones are produced by molluscs. Among these are the sea snail *Aplysia*, which is studied because of its simple nervous system and giant neurons.^{335,336} Proteolytic processing of precursors within single cells often yields neurohormones specific to those cells. The structures of two **small cardioactive peptides** secreted from single neurons³³⁷ are shown in Table 30-5. A 4.4-kDa egg-laying hormone is formed in at least eight processing steps.³³⁸⁻³⁴¹ Among the peptides identified in the freshwater snail *Lymnaea* are many that are characteris-



tic of mammalian pituitary gland, pancreas, brain, and intestinal tract. These include TRH, ACTH, α MSH, arginine vasopressin, oxytocin, calcitonin, gastrin, gastrointestinal peptide, glucagon, insulin, Met-enkephalin, pancreatic poly-peptide, secretin, somatostatin, substance P, and vasoactive intestinal peptide. Also present are FMRF amide and arginine vasotocin.³⁴²

The cardioacceleratory peptide FMRFamide (Table 30-5), which was discovered in 1977, was the first in a large series of related neuropeptides that are found in organisms ranging from the nematode *Caenorhabditis*

TABLE 30-5
Some Microbial and Invertebrate Peptide Hormones

Source name	Number of residues	Sequence ^a
<i>Streptococcus faecalis</i> sex hormone	8	FLVMFLSG
Yeast mating factor a	12	YIIKGV(L)FWD farnesyl ^b PAC-OCH ₃
Tremerogen A-10	10	EHDPSAPGNGYC farnesyl ^b -OCH ₃
<i>Hydra</i> head activator	10	pEPPGGSKVIF
Antho-RF amide (sea anemone)	4	pEGRF-NH ₂
Small cardiovascular peptides (<i>Aplysia</i>)		
SCP-A	11	ARPGYLAFFRM-NH ₂
SCP-B	9	MNYLAFFRM-NH ₂
FMRFamide (coelenterates, molluscs)	4	FMRF-NH ₂
(octopus)	7	YGGFMRF-NH ₂
Shrimp blanching hormone	8	pELNFSPGW-NH ₂
Fidler crab pigment-dispersing hormone	18	NSSELINSILGLPKVMNDA-NH ₂
Proctolin (cockroaches)	5	RYLPT
Myotropic neuropeptide (cockroaches)	11	EQFEDYGHMRF-NH ₂ ↑ sulfate ester in leukosulfakinin
Adipokinetic hormone (locust)	10	PELNFTPNWGT-NH ₂
Crustacean cardioactive peptide	9	EPFCNAFTGC-NH ₂ └─S-S─┐

^a One-letter abbreviations, pE, 5-oxoproline; -NH₂, C-terminal carboxamide.
^b Alkylated on S.

elegans to vertebrate animals. At least 18 genes in *C. elegans* encode 53 distinct FMRFamide-related peptides. Disruption of one of these genes causes hyperactivity, uncoordination, and other behavioral difficulties in the nematodes.³⁴³ Several groups of FMRFamide-related peptides have been found in *Drosophila*³⁴⁴ and other insects.³⁴⁵ One group has the C-terminal FLRFamide and another HMRFamide. Among the latter are sulfate esters such as the cockroach neuropeptide shown in Table 30-5. The sequence of the sea urchin sperm chemoattractant **speract** (sperm attractant peptide-1; SAP-1) is shown in Table 30-4. This is one of a family of egg-associated peptides that stimulate sperm metabolism and mobility. The DNA sequence that codes for speract predicts formation of a 296-residue protein that contains four speract sequences plus six related decapeptide sequences, each separated by a single lysine residue.³⁴⁶ Many other SAP peptides, some containing the unusual amino acid *o*-bromo-L-phenylalanine,³⁴⁷ are formed.

Hormones of insects and crustaceans. Peptide neurohormones of insect brains³⁴⁸ include the pentapeptide **proctolin** (Table 30-5), which was first isolated from the cockroach and has since been found in crustaceans and in mammalian brain. It has been traced to specific insect neurons.³⁴⁹ A nonapeptide neurohormone from the shore crab does not resemble any other known vertebrate or invertebrate hormone.³⁵⁰

The prawn *Pandalus borealis* changes its body color by means of movable pigment granules. The neurosecretory octapeptide **blanching hormone**³⁵¹ (Table 30-5) controls the process. This is a member of a larger family of peptide hormones with related functions such as the 18-residue **pigment-dispersing hormone** of the fiddler crab³⁵² and similar hormones in insects.³⁵³

One of the insect neurohormones, the **activation hormone**, controls the secretion of the corpora allata, paired glands that synthesize the **juvenile hormone** (Fig. 22-4) in insect larvae. While the structure of the juvenile hormone varies somewhat with species, it is usually a polyprenyl ester. A specific binding protein provides the hormone with protection from degradative enzymes. However, in the tobacco hornworm an esterase, able to hydrolyze the protein-bound juvenile hormone, is produced at the start of pupal differentiation.³⁵⁴ The exact mechanism of action of juvenile hormones has been difficult to determine. However, it affects polyamine synthesis.^{355,356}

The **mandibular organ-inhibiting hormone** of the crab *Cancer pagurus* produces two neurohormones that inhibit the secretion of methyl farnesoate, which is thought to function as the juvenile hormone in crustaceans.³⁵⁷

A related role of the insect corpora allata is to store and release the **prothoracicotropic hormone**, a

peptide neurohormone formed in a single neurosecretory cell of the brain.^{358,359} The steroid hormone **ecdysone** (Fig. 22-12) is secreted by the insect's prothoracic gland. Also known as the **molting hormone**, ecdysone is required for the periodic replacement of the exoskeleton of the larvae.^{359a} It induces molting in crayfish and other arthropods and appears to be needed by such members of lower phyla as schistosomes and nematodes. It also controls the biting behavior of mosquitos.³⁶⁰ In addition to α -ecdysone, the 20- and 26-hydroxyecdysones and 20,26-dihydroxyecdysone have been identified in insects.³⁴⁸ It has been suggested that different ecdysones may function at different stages of insect development.

Ecdysone stimulates the synthesis of RNA in tissues. Visual demonstration of the effect is provided by its action on polytene chromosomes of fly larvae (Fig. 26-14).³⁶¹ Fifteen minutes after the application of ecdysone, a puff is induced on one band of the chromosome; a second puff forms at a later time while a preexisting puff diminishes. Thus, like steroid hormones in mammals, ecdysone appears to have a direct controlling effect on transcription. The cuticle-shedding process (**ecdysis**) is initiated by the brain peptide **eclosian**. However, the brain may be responding to the **ecdysis-triggering hormone**, a peptide that is secreted by a series of epitracheal glands located in various segments of the body.³⁶²

Adipokinetic hormones control metabolism of insects during long-distance flight.^{359,363} In the migratory locust these hormones consist of a pair of related octapeptides and a decapeptide (Table 30-5). The hormones stimulate triacylglycerol lipase in the insects' fat bodies, induce release of carbohydrates from body stores, and affect many other aspects of metabolism.³⁶³ Insects also have hormones of the insulin family, proteins consisting of disulfide-linked A and B chains as in insulin. The silkworm *Bombyx mori* has 38 genes for the insulinlike **bombyxins**, which are synthesized in the brain.³⁶⁴

Insects produce many different types of sex attractant pheromones (e.g., see p. 382). By 1995 more than 300 structures had been determined for pheromones from >1600 insect species.³⁶⁵

9. Plant Hormones

Plants possess a kind of circulatory system by which fluids are transported from the roots upward in the xylem and downward from the leaves through the phloem. Many compounds are carried between cells in this manner, while others are transported across cell membranes and against concentration gradients by active transport. A number of compounds that move between cells in either of these two manners have been classified as hormones.³⁶⁶⁻³⁶⁹ The major plant

hormones consist of five compounds or groups of compounds: **auxins** (p. 1446), **gibberellins** (Eq. 22-5), **cytokinins** (Fig. 5-33), **abscisic acid** (Fig. 22-4), and **ethylene** (Fig. 24-16). A number of other plant regulators, some involved in defensive reactions, are sometimes also described as hormone. These include the **brassinosterols** (Fig. 22-9) and related compounds,^{366,367,369,370} **jasmonic acid** (Eq. 21-18), **salicylic acid** (Chapter 25, Section B,7), bacterially produced **lipooligosaccharides** such as the NOD factors (Box 20-E), and polypeptides such as **systemin**. The sugars glucose and sucrose have hormonelike functions as does light, which controls many plant functions.³⁶⁸

Plant hormones have multiple and overlapping functions, which are exerted predominantly by repression of gene expression. This makes it difficult to discuss their functions briefly. Most studied are the auxins, of which the principal member is **indole-3-acetic acid** (Fig. 25-12). This compound, whose biosynthesis is discussed on p. 1446, has been implicated as a controlling agent for cell division and cell elongation. In this capacity auxin influences a great variety of plant processes. Produced principally by growing shoots, auxin diffuses down the stem aided by special **efflux carriers**^{371,371a} and inhibits the growth of lateral buds. However, the hormone stimulates the growth of stems, thus establishing the apical dominance of the tip of a plant. Other hormones also have an influence. Auxin is well established as the controlling agent in phototropism, the tendency of a plant to bend toward the sun.

A sensitive test for auxin, which is dependent on the bending of the coleoptile of *Avena sativa* (the common oat) in response to the hormone, allows detection of as little as three pmol of auxin. Using this assay, it was shown that auxin is transported laterally away from the illuminated side of plants, causing the darker side to elongate more rapidly. Both membrane-associated and soluble binding sites, which may represent natural auxin receptors, have been identified, and auxin response elements have been located in DNA. The membrane-bound receptors may regulate an ATP-dependent proton pump, while the soluble receptors may act to regulate gene transcription.

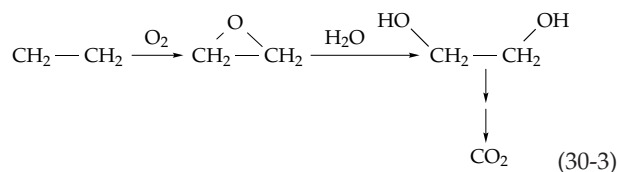
The gibberellins together with brassinosterols are active in helping to determine the *form* of plants. There are 66 different known gibberellins, which are synthesized (Eq. 22-5) in mature leaves and are transported downward. Their very active effect in stimulating RNA synthesis in dwarf varieties of vegetables suggests that gibberellins also serve as gene activators to promote RNA synthesis. A possible function in the **geotropic** response of plant roots is suggested by the presence of higher concentrations of gibberellins in the upper half of horizontal roots than in the lower half.³⁷² On the other hand, auxin has long been known to have a higher concentration on the lower side of the

root,^{371a} and it has been assumed to inhibit elongation (in contrast to stimulation in stems). Because they are structurally and in physical properties somewhat similar to sterols, gibberellins have been assumed to act by a steroid hormonelike mechanism. The brassinosterols, which are true sterols, may also be expected to bind to soluble receptors and to regulate transcription.

The cytokinins are isopentenyladenosine derivatives (Fig. 5-33), which may be hydroxylated or substituted in the 2 position by a methylthio group.^{373,374} Cytokinins are synthesized in roots and translocated upward to other parts of the plant. They may originate in part from degradation of cytokinin-containing tRNA, but there is evidence that they may also be synthesized independently.³⁷⁴⁻³⁷⁶ The N⁶-isopentenyladenine in tRNAs is generated by transfer of the isopentenyl group from Δ^2 -isopentenyl pyrophosphate. The role of cytokinins may be at the level of gene transcription, but it has been difficult to identify signaling sequences.³⁶⁸ Their hormonal influence on plants appears to be independent of their function in tRNA. The most striking effect of cytokinins in solution is on differentiation of plant cells (Chapter 32).

The terpenoid abscisic acid (Fig. 22-4) is synthesized by degradation of a carotenoid precursor. It is formed by plants in response to *stress* from low temperature, high salinity, or drought. Abscisic acid appears to block the growth-promoting effects of hormones such as the gibberellins and cytokinins. It is sometimes regarded as a general gene repressor, which prepares plants for dormancy. Synthesis of abscisic acid occurs in response to the short day and long night pattern of the fall. It often opposes the action of gibberellins. The signaling pathway for abscisic acid apparently involves release of Ca²⁺ ions and formation of cyclic ADP-ribose (cADPR; p. 564). Among other effects this induces the closing of stomata in leaf surfaces.³⁶⁸

Ethylene not only hastens the ripening of fruit but also tends to promote senescence in all parts of plants. Its signaling mechanisms are the best-known for any plant hormone.^{368,369} The synthesis and action of ethylene are discussed in Chapter 24, Section D,4. Ethylene is metabolized slowly in plants by oxidation to ethylene oxide. The latter is hydrolyzed to form ethylene glycol, which is metabolized further to CO₂ (Eq. 30-3).



A postulated flowering hormone, **florigen**, has not been isolated.³⁶⁶ Flowering seems to be controlled by a variety of different hormonal effects.³⁶⁹ Jasmonic

acid (jasmonate) and salicylic acid act as plant defense signals.^{377,378} Salicylic acid activates a large number of transcription factors, which induce resistance to a variety of pathogenic organisms, a response referred to as **systemic acquired resistance**. See also Chapter 31, Section G. Among other compounds synthesized as part of the systemic acquired resistance of plants are proteins known as **systemins**.³⁸² Initially discovered in tomatoes, systemins have been discovered in more than 100 other species of plants. Jasmonic acid emitted from tomato plants acts as a pheromone that attracts wasps to attack caterpillars that feed on the tomato plants.³⁷⁹ In fact, wounding by herbivores may stimulate emission of a variety of volatile compounds that may attract predators to the attacking herbivores.^{380,381}

Many other compounds influence plant growth. Among them are the vitamins, thiamine, pyridoxine, and nicotinic acid, which are synthesized in the leaves and transported downward to the roots. Since they promote growth of roots, they are sometimes referred to as **root growth hormones**. However, they are nutrients universally needed by cells. Various compounds secreted by other organisms can either stimulate or inhibit growth of a given plant. Some are powerful toxins. Others, such as the previously mentioned NOD factors, and evidently also the riboflavin degradation product **lumichrome** (Box 15-B), are beneficial.³⁸³ These **plant growth regulators** may be produced by other plants, by microorganisms, or by fungi.^{384,385} Much use is made in agriculture of synthetic growth regulators.

10. Secretion of Hormones

In Chapter 11 the effects of binding of hormones to cell surface receptors have been emphasized. Equally important are the mechanisms that control the secretion of hormones. The topic of exocytosis has been considered briefly in Chapter 8, Section C.6 and aspects of the Golgi in Fig. 20-8 and associated text. Both hormones and neurotransmitters are secreted by exocytosis of vesicles. Cells have two pathways for secretion.^{386,387} The **constitutive pathway** is utilized for continuous secretion of membrane constituents, enzymes, growth factors, viral proteins, and components of the extracellular matrix. This pathway carries small vesicles that originate in the trans-Golgi network (TGN; Fig. 20-8). The **regulated pathway** is utilized for secretion of hormones and neurotransmitters in response to chemical, electrical, or other stimuli.

Many neurotransmitters are packaged into **small synaptic vesicles** ~50 nm in diameter. These may originate from large endosomes rather than from the Golgi. They are usually recycled and refilled repeatedly.³⁸⁶ Secretion of hormones, and of some neurotrans-

mitters, occurs via **large dense-core vesicles** of ~100 nm diameter. These originate from the TGN and are not recycled. They are prominent in chromaffin cells and other cells that secrete large amounts of a signaling molecule. Secretion of hormones and that of neurotransmitters have several common features. Indeed, hormones of the hypothalamus, neurohypophysis, and the adrenal medulla are secreted by specialized neurons. However, while hormones are often carried in the bloodstream, neurotransmitters are most often secreted into the very small volume of a single synapse. The exocytosis must occur very rapidly from a small number of SSVs.

A common feature, and also a puzzle, of vesicular signaling is the nearly universal response to calcium ions. Exocytosis is usually triggered by a rise in the concentration of Ca^{2+} , and most receptor signaling also leads to an increase in cytosolic Ca^{2+} .³⁸⁸⁻³⁹¹ The puzzle lies in the ability of cells to use a common mechanism for so many specific purposes. This topic is considered further in Section B.8. There are also many other factors that can control exocytosis. Recent evidence suggests that NO may play a role.³⁹²

B. Neurochemistry

The nervous system, a network of neurons in active communication, reaches its ultimate development in the 1.5 kg human brain.^{149,393-396} Many invertebrates, such as leeches,^{396a} crayfish, insects, and snails, have brains containing no more than 10^4 to 10^5 neurons,^{396b,397,398} but the human brain contains $\sim 10^{11}$. Each of these neurons interconnects through **synapses** with hundreds or thousands of other neurons. The number of connections is estimated to be as many as 60,000 with each Purkinje cell of the human cerebellum. There may be many more than 10^{14} synapses in the human brain.^{399,400}

In addition to neurons, the brain contains 5–10 times as many **glial** cells of several types. The neuroglia occupy 40% of the volume of brain and spinal cord in the human. Some glial cells seem to bridge the space between neurons and bloodcarrying capillaries. Others synthesize myelin. Some are very irregular in shape.

1. The Anatomy and Functions of Neurons

Although neurons have many shapes and forms, a common pattern is evident.^{400a} At one end of the elongated cell (Fig. 30-8) is a series of **dendrites**, thin fibers often less than 1 μm in diameter. The ends of the dendrites form synapses with other neurons and act as receivers of incoming messages. Additional messages come into synapses on the **cell body**, while

the **axon** serves as the output end of the cell. The axon, a long fiber of diameter 1–20 μm , is also branched. As a consequence, the nervous system contains both highly branching and highly converging pathways. Many of the axons are wrapped in a myelin sheath (Fig. 30-9; pp. 390 and 1767).

The ends of the fine nerve fibers are thickened to form the **synaptic knobs**, which make synaptic contacts with dendrites on cell bodies of other neurons. In most instances the arrival of a nerve signal at the **presynaptic** end of a neuron causes the release of a transmitter substance (neurohormone). The transmitter passes across the 10–50 nm (typically 20 nm) **synaptic cleft** between the two cells and induces a change in the electrical potential of the **postsynaptic** membrane of the next neuron (Fig. 30-10).^{149,401} Excitatory transmitters usually cause **depolarization** of the membrane. By this we mean that the membrane potential, which in a resting neuron is -50 to -70 mv (Chapter 8), falls to nearly zero often as a consequence of an increased permeability to Na^+ and a resultant inflow of sodium ions. The resulting **postsynaptic**

potential (really a drop in the potential difference) is propagated to the cell body and axon and under appropriate circumstances may initiate an **action potential**. This is a narrow spikelike region of depolarization that travels down the axon at a constant velocity and with undiminished intensity (Fig. 30-11).

A characteristic of many neurons is an *all-or-none response* or firing. An action potential passes down the axon only if there is sufficient depolarization. In general, a stimulus must reach a neuron through *more than one synapse* before the neuron will fire. Furthermore, neurons are often *inhibitory*, releasing transmitters that counter the excitatory synapses and tend to prevent firing. Inhibition is important in damping out small excitations; thus sharpening the response of the nervous system toward strong stimuli. Another characteristic of basic importance to the operation of the brain is that neurons fire at longer or shorter intervals depending upon the strength and duration of the stimulus. The stronger the stimulus to a given neuron, the more rapid the train of spikes that passes down the axon. Thus, the brain functions to a large extent in

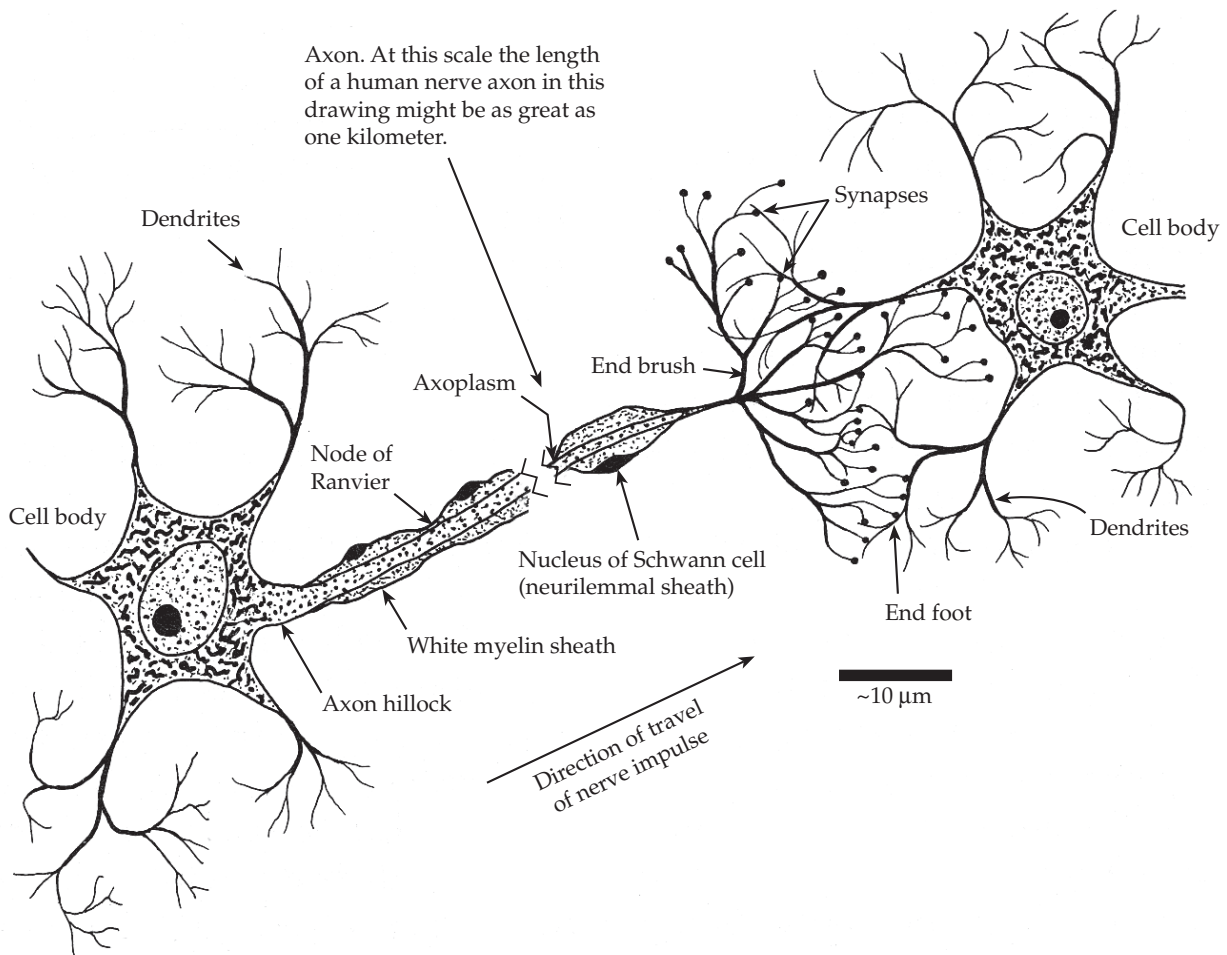


Figure 30-8 Schematic drawing of a neuron (after Brand and Westfall,⁴⁰¹ p. 1192).

Figure 30-9 Micrograph of a section through an axon of a neuron from rat brain. The structure of the myelin sheath can be seen clearly. The growing lip of cytoplasm (X) from a neuroglial cell is advancing around the axon process (NF) and insinuating itself into the space between the plasma membrane of the axon and the membrane that limits the thin layer of cytoplasm (Y) left behind by the growing lip during its previous turn. This cytoplasmic layer disappears as the inner leaflets of its plasma membrane fuse to form the major dense line of the myelin sheath. This process is occurring at the point indicated by the single arrow. The outer leaflet of the plasma membrane surrounding the lip fuses with its own outer leaflet laid down on the previous turn. The two outer leaflets thus give rise to the less dense intermediate line of the sheath (double arrow). The cell body from which the investing cytoplasmic sheet originated cannot be seen in this micrograph, but cytoplasm within the lateral margins of the sheet does appear (X'). The micrograph, by A. Hirano and H. M. Dembitzer, originally appeared in *J. Cell Biol.* **34**, 555 (1967), where a more complete explanation of myelin sheath formation is provided. Figure copied from Porter and Bonneville.⁴⁰² Courtesy of Mary Bonneville.

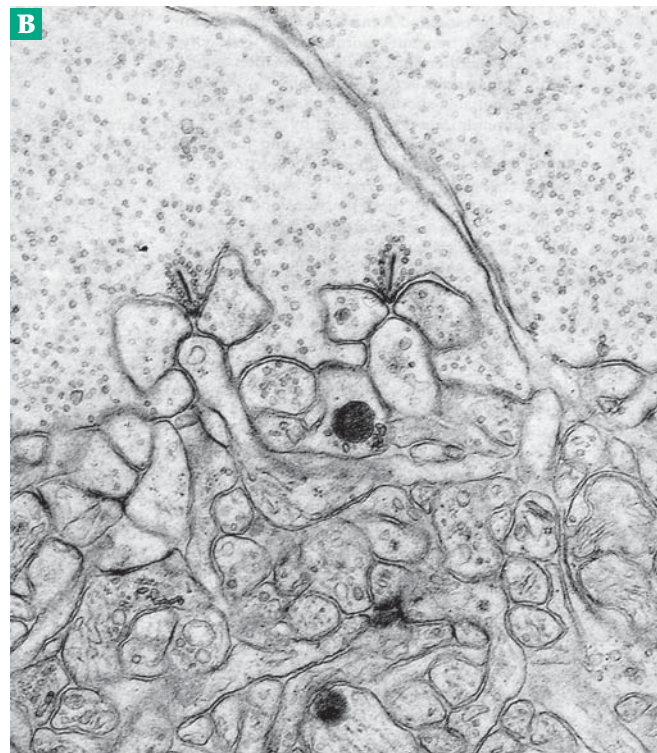
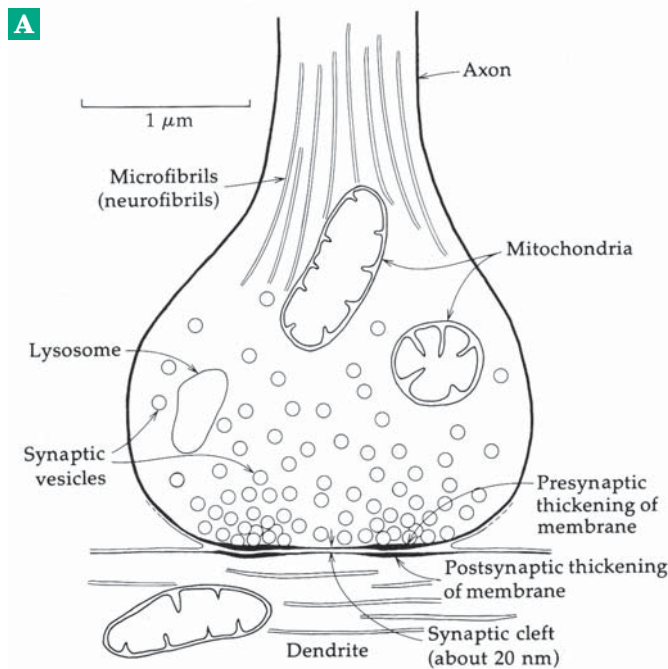
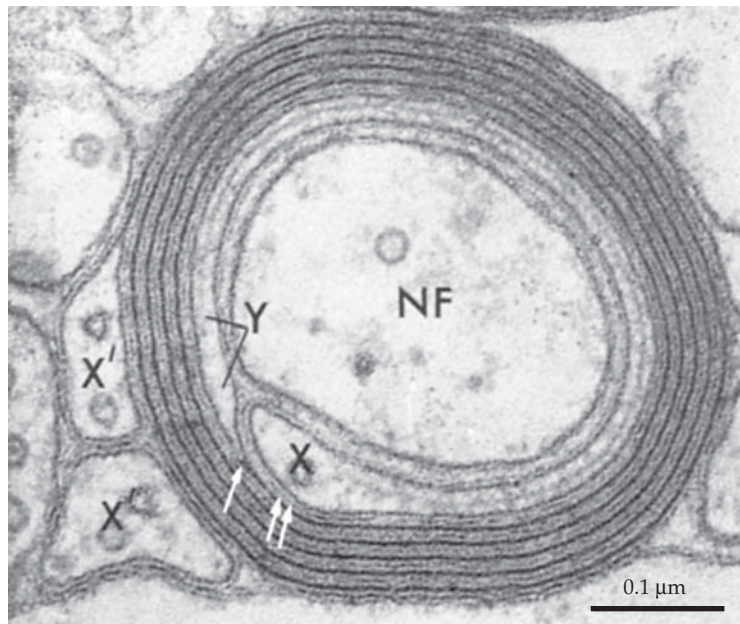


Figure 30-10 (A) Schematic drawing of a synapse. (B) Electron micrograph showing the synaptic junctions in the basal part (pedicle) of a retinal cone cell of a monkey.⁴⁰³ Each pedicle contains synaptic contacts with 12 triads, each made up of processes from a bipolar cell center that carries the principal output signal and processes from two horizontal cells that also synapse with other cones. A ribbon structure within the pedicle is characteristic of these synapses. Note the numerous synaptic vesicles in the pedicle, some arranged around the ribbon, the synaptic clefts, and the characteristic thickening of the membranes surrounding the cleft (below the ribbons). Micrograph courtesy of John Dowling.

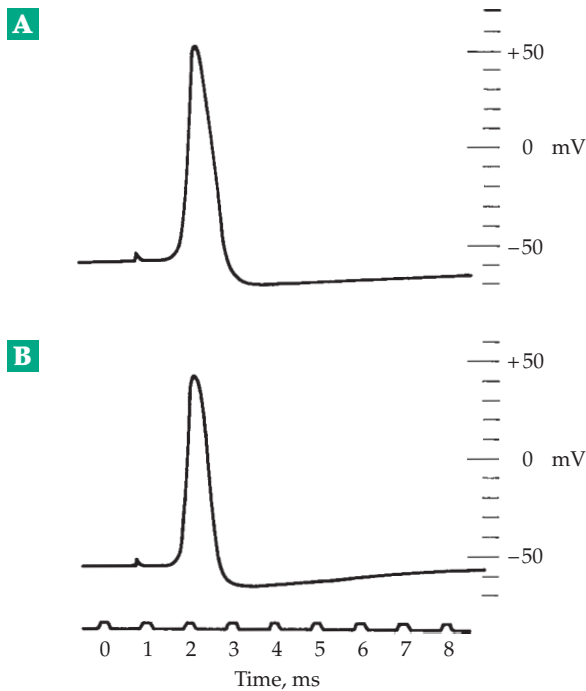


Figure 30-11 (A) Action potential recorded with internal electrode from extruded axon filled with potassium sulfate (16°C). (B) Action potential of an intact axon, with same amplification and time scale (18°C). The voltage scale gives the potential of the internal electrode relative to its potential in the external solution with no correction for junction potential. From A. Hodgkin, *Conduction of Nervous Impulses*, 1964. Courtesy of Charles C. Thomas, Publisher, Springfield, Illinois.

decoding trains of impulses. The frequency of the impulses from neurons varies from a few per second to a maximum of about 200 s^{-1} in most nerves (up to 1600 s^{-1} in the Renshaw cells of the spinal cord). The maximum frequency is dictated by the refractory period of $\sim 1 \text{ ms}$ (Section B,3).

Although the concepts of neuronal function outlined in the preceding paragraphs have been accepted for many years, more recent discoveries require that they be modified somewhat. Dendrites seem to be able to transmit information as well as to receive it. Furthermore, while information is certainly transmitted long distances by spike action potentials, shorter neurons and dendrites may communicate extensively by exchange of chemicals through low resistance gap junctions, also called **electric synapses** (Chapter 1). Small changes in membrane potential transmitted through these junctions may alter the behavior of adjacent neurons. Chemical transmitters do not always have an electrical effect on postsynaptic neurons but may influence metabolism or gene transcription.

2. Organization of the Brain

The anatomy of the brain is quite complex, and only a few terms will be defined here. The **cerebrum**, which is made up of two hemispheres, accounts for the largest part of the brain. The deeply folded outermost layer, the **cerebral cortex**, consists of **gray matter**, a mass of cell bodies, and fine unmyelinated nerve fibers. Beneath this lies a layer of **white matter** made up of myelin-covered axons connecting the cerebral cortex with other parts of the brain. The two cerebral hemispheres are connected by the **corpus callosum**, a band of $\sim 2 \times 10^8$ nerve fibers. Remarkably, these fibers can be completely severed with a relatively minimal disruption of the nervous system. In the past the corpus callosum was sometimes cut to control almost incessant epileptic seizures that could not be prevented by drugs. The “split-brain” patients suffered relatively little disability as long as both eyes functioned normally. Studies of these patients provided some insights into the differing functions of the two hemispheres of the cerebrum.³⁹⁵

Deeper in the cerebrum lie the **basal ganglia**, which include the caudate, lenticular, and amygdaloid nuclei. The lenticular nuclei are further divided into putamen (an outer portion) and the globus pallidus. The putamen and caudate nuclei together are known as the **striatum** (Fig. 30-12). The lower lying subthalamic nuclei and substantia nigra are sometimes also included in the basal ganglia.

The outer parts of the cerebrum, including the basal ganglia, make up the telencephalon. Deep in the center of the brain is the diencephalon consisting of the **thalamus** (actually two thalami), **hypothalamus**, **hypophysis** (Figs. 30-1, 30-13), and other attached regions. A major structure at the back of the brain is the **cerebellum**. Like the cerebrum, its cortex is highly folded. The 30 billion neurons of the cerebellum are organized in a highly regular fashion.^{393,404} The interconnections of the seven types of neurons present in this part of the brain have been worked out in fine detail.

The basal part of the brain or **brain stem** consists of the medulla oblongata and the pons. While the bulk of the tissue consists of myelinated nerve tracts passing into the spinal cord, synaptic regions such as the olivary nucleus are also present.

The brain, which must function in a chemically stable environment, is protected by a tough outer covering, the **arachnoid membrane**, and by the **blood-brain barrier**^{406,407} and the **blood-cerebrospinal barrier**. Both of these barriers consist of tight junctions similar to those seen in Fig. 1-15A. They are formed between the endothelial cells of the cerebral capillaries and between the epithelial cells that surround the capillaries of the **choroid plexus**. The choroid plexus consists of capillary beds around portions

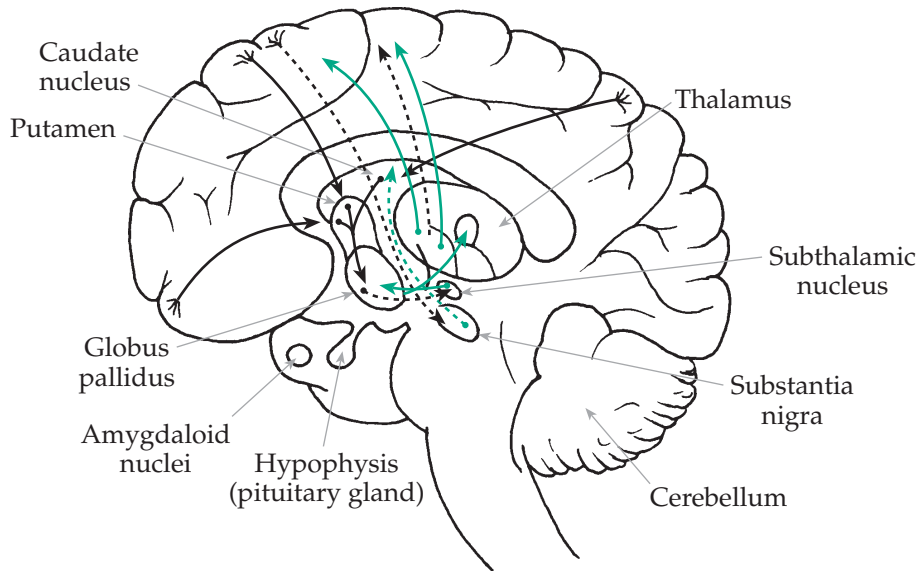


Figure 30-12 Diagram illustrating some of the major interconnections of the “extrapyramidal system” of the brain. Arrows indicate major direction of projections. The nigrostriatal (substantia nigra to striatum) and related neuronal pathways are indicated with dashed lines. After Noback and Demarest,⁴⁰⁵ pp. 182 and 183.

of the fluid-filled **ventricles** deep in the interior of the brain. They serve as a kind of “kidney” for the brain assisting in bringing nutrients in from the blood and helping to keep dangerous compounds out.⁴⁰⁶

3. Neuronal Pathways and Systems

Consider a message originating with a nerve receptor in the skin or in another sense organ. A nerve signal passes via a **sensory neuron (afferent fiber)** upward toward the brain. It may pass through two or more synapses (often through one in the spinal cord and one in the thalamus) finally reaching a spot in the sensory region of the cerebral cortex. From there the signal in modified form spreads through the **inter-neurons** of virtually the entire cortex. In each synapse, as well as in the cortex, the impulse excites inhibitory fibers that dampen impulses flowing through adjacent fibers. Likewise, if a given impulse is not strong enough, it will itself be inhibited before reaching the

cortex. Among the important sensory neurons are those from the seven million cone cells and 100 million rod cells of the eye. The nerve signals pass out of the retina by way of a million axons from retinal ganglion cells reaching, among other parts of the brain, the **visual cortex** (Fig. 30-14).⁴⁰⁸

The neuronal events that occur within the cerebral cortex are extraordinarily complex and little understood.⁴⁰⁹ In what way the brain is able to initiate voluntary movement of muscles is obscure. However, it is established that the signals that travel out of the brain down the **efferent fibers** to the muscles arise from large **motor neurons** of the **motor cortex**,⁴¹⁰ a region that extends in a band across the brain and adjacent to the sensory cortex (Fig. 30-14). The axons of these cells form the **pyramidal tract** that carries impulses downward to synapses in the spinal cord and from there to the **neuromuscular junctions**. These are specialized synapses at which acetylcholine is released, carrying the signal to the muscle fibers themselves. Passing over the cell surface and into the

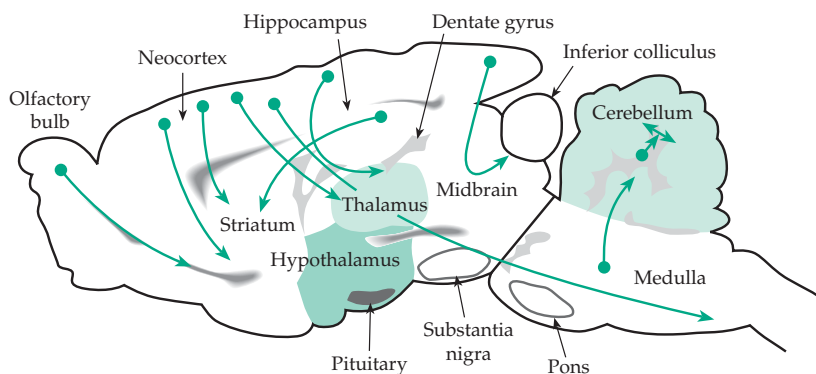


Figure 30-13 Section through a rat brain. This brain, which has been very widely used in neurochemical studies, appears superficially to be quite different from the human brain (Fig. 30-1), which is characterized by its large cerebral cortex. However, basic pathways are the same. Some major pathways for glutamate-secreting (glutamatergic) neurons are marked by arrows. Most of these originate in the neocortex (outer layers of the cerebral cortex) and the hippocampus. From Nicholls.¹⁴⁹ Courtesy of David G. Nicholls.

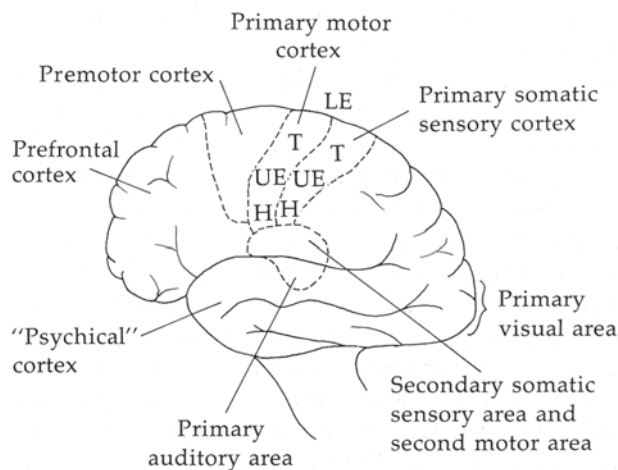


Figure 30-14 The location of several functional areas of the cerebral cortex. The representation of body parts on the primary motor and somatic sensory cortices include the head (H), upper extremity (UE), trunk (T), and lower extremity (LE). After Noback and Demarest,⁴⁰⁵ p. 193.

T tubules (Chapter 19, Section B,4; Fig. 19-21), a wave of depolarization initiates the release of calcium and muscular contraction.

At the same time that the motor neurons send signals to the muscles, branches travel into other parts of the brain including the olivary nuclei, which send neurons into the cerebellum. The cerebellum acts as a kind of computer needed for fine tuning of the impulses to the muscles. Injury to the cerebellum leads to difficulty in finely coordinated motions. Input to the Purkinje cells arises from the climbing fibers, which originate in the inferior olive of the brain stem. Each climbing fiber activates a single Purkinje cell, but the dendrites of each Purkinje cell also form as many as 200,000 different synapses with parallel fibers that run across the cortex of the cerebellum (Fig. 30-15). The parallel fibers receive input from many sources via a complex series of mossy fibers and granule cells and influence the firing of the Purkinje cells. The output from the Purkinje cells is entirely inhibitory. It is transmitted via synapses in the cerebellar nuclei to neurons that lead back to the cerebral cortex, into the thalamus, and down the spinal cord.⁴¹¹ The pathway to the cortex completes an inhibitory feedback loop, of which there are many in the nervous system. For details see Llinás⁴⁰⁴ and Nicholls.¹⁴⁹

In addition to the **somatic motor system** that operates the voluntary (striated) muscles via the pyramidal tract, there is the **autonomic system**, which controls the involuntary (smooth) muscles, glands, heartbeat, blood pressure, and body temperature. This system has its origins in both the cerebral cortex and

hypothalamus. It is subdivided into two systems, the **sympathetic** and **parasympathetic** systems, which are anatomically distinct. The sympathetic system is geared to the fight and fright reactions. Its **postganglionic fibers** (those below the ganglia in the spinal cord) liberate norepinephrine (noradrenaline) and include the adrenal medulla, which consists of specialized neurons, the **chromaffin** cells. The parasympathetic system has to do more with homeostasis and maintenance of body systems. Biochemically it is characterized by the release of acetylcholine as a transmitter substance.

The hypothalamus, a four gram portion of the brain, receives a great deal of biochemical attention because of its function in the autonomic nervous system, in homeostasis, and in endocrine secretion. Its liberation of neurohormones that stimulate the hypophysis has already been considered in Section A,3. The hypothalamus is also involved in the regulation of the body temperature, of water balance, and possibly of glucose concentration.

Two other systems of importance in the brain are the **reticular system** and the **limbic system**. The former is the mediator of the sleep–wake cycle and is responsible for characteristic waves in the electroencephalogram. The limbic system is the mediator of **affect** or mood and of **instincts**. It is anatomically complex with centers in the amygdala, other subcortical nuclei, and the limbic lobe of the cortex. The limbic cortex forms a ring lying largely within the longitudinal fissure between the two hemispheres. It includes the olfactory cortex, the **hippocampus**, a region associated with formation of conscious memories, and other evolutionarily older regions of the cerebral cortex. Within the limbic lobe are the **pleasure centers**. When electrodes are implanted in these regions, animals will repeatedly push levers that are designed to electrically stimulate these centers. There are also **punishing centers**, whose stimulation causes animals to avoid further stimulation.

4. The Propagation of Nerve Impulses

Although the chemical basis of the conduction of nerve impulses via an action potential is not entirely clear, the electrical events have been described with precision. If the permeability of a membrane toward sodium ions is increased in a local region, sodium ions flow through the membrane into the cell neutralizing the negative charge inside and depolarizing the membrane. Such depolarization leads to propagation of an electrical signal of diminishing intensity over the surface of the membrane in a manner analogous to the flow of electrical current along a coaxial cable. It is thought that local increases in Na^+ permeability of the plasma membrane often trigger nerve impulses. Other

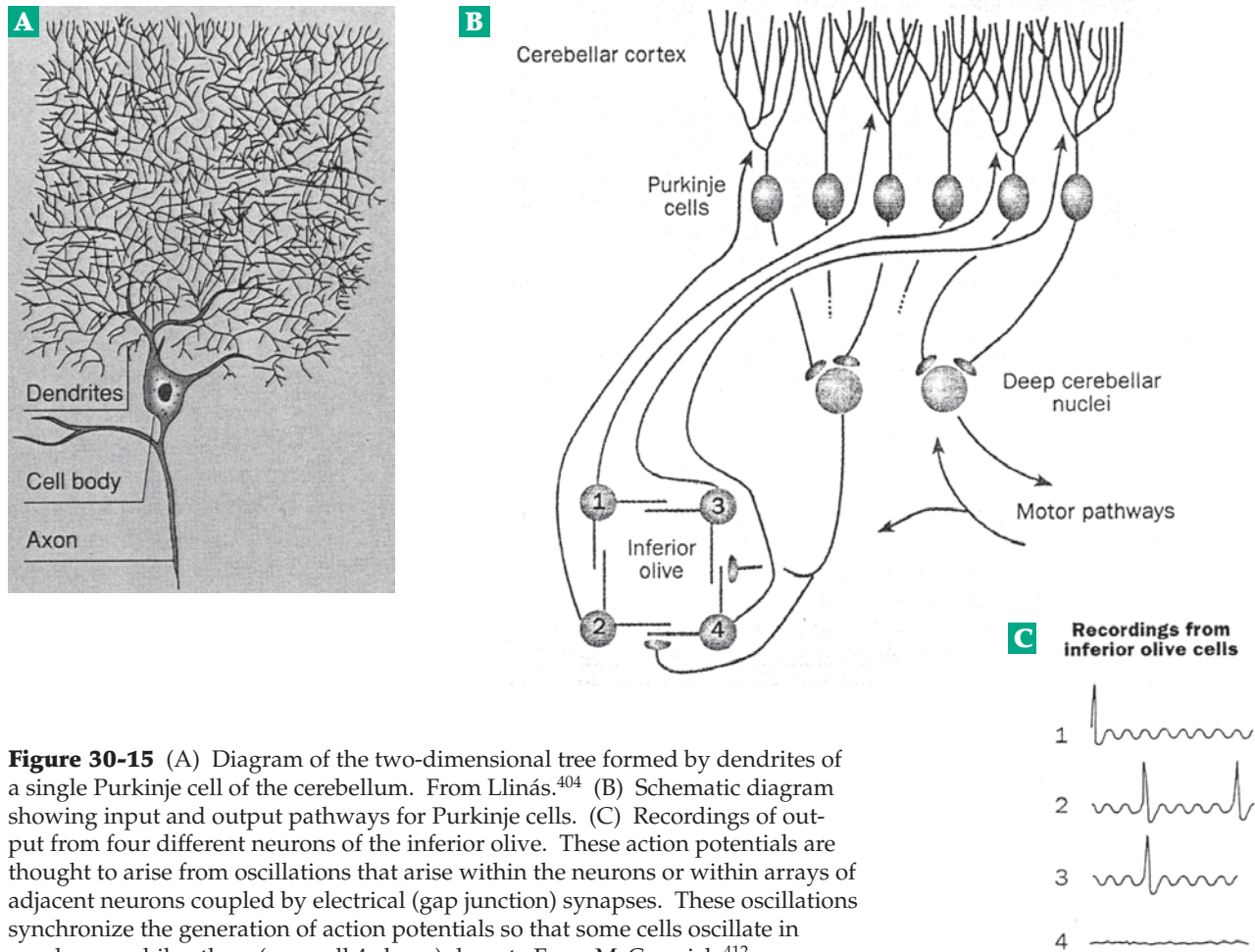


Figure 30-15 (A) Diagram of the two-dimensional tree formed by dendrites of a single Purkinje cell of the cerebellum. From Llinás.⁴⁰⁴ (B) Schematic diagram showing input and output pathways for Purkinje cells. (C) Recordings of output from four different neurons of the inferior olive. These action potentials are thought to arise from oscillations that arise within the neurons or within arrays of adjacent neurons coupled by electrical (gap junction) synapses. These oscillations synchronize the generation of action potentials so that some cells oscillate in synchrony while others (e.g., cell 4 above) do not. From McCormick.⁴¹²

ions such as Ca^{2+} may also play a role. While the kind of passive transmission of electrical signals that results from a local depolarization of the membrane is suitable for very short nerve cells, it cannot be used to send signals for long distances. Most nerve axons employ the more efficient action potential. This is an impulse that passes along the axon and for a short fraction of a second (~ 0.5 ms in mammalian nerves) changes the membrane potential in the characteristic way shown in Fig. 30-11. Initially, the negative potential of 50–70 mV drops rapidly to zero and then becomes positive by as much as 40–50 mV, after which it returns to the resting potential. The remarkable thing about the action potential is that it is propagated down the axons at velocities of 1–100 m/s without loss of intensity.

To establish the chemical basis of the action potential, A. L. Hodgkin and A. F. Huxley in the 1950s devised the **voltage clamp**, a sophisticated device by which the transmembrane current can be measured while using a feedback mechanism to fix the membrane potential at a preselected value.^{413–417} Using the voltage clamp the membrane conductance could be measured as a function of the membrane potential

and of time. It was found that immediately after a decrease in membrane potential was imposed with the voltage clamp, the permeability of the membrane toward sodium ions rose rapidly. Since an increased sodium ion permeability automatically leads to depolarization in an adjacent region of the membrane, a self-propagating wave is established and moves down the axon. The voltage clamp studies also revealed that after a fraction of a millisecond the permeability to potassium ions also increases. At the same time the sodium ion permeability decreases again, and the normal membrane potential is soon reestablished. However, during an **absolute refractory period** of ~ 0.5 ms no other nerve impulse can be passed. The sequence of events during passage of the nerve impulse can be described as the opening of sodium channels followed by the opening of potassium channels, and then by a closing of the channels in the same sequence. The results of these investigations led Hodgkin and Huxley to propose equations that quantitatively describe the action potential and that predict the observed conduction velocities and other features of nerve impulses.

A special feature of nerves that are designed to transmit impulses very rapidly is the presence of the wrapping of **myelin** (Fig. 30-9). As can be seen in this figure, the extracellular surfaces of the consecutive wraps bind tightly together, and the cytoplasm of the cell interior is squeezed out to form the compact myelin sheath.⁴¹⁸ Mutations in the integral membrane proteolipid protein (p. 401) are associated with a variety of defects in myelin formation. Some of these are severe, for example, leading to loosely wrapped myelin.^{419,420} The proteolipid protein is encoded by an X-linked gene. The most abundant protein in peripheral nerve myelin is the integral membrane **peripheral myelin glycoprotein P₀**. It is encoded by an autosomal gene for which 29 known defects account for a variety of human diseases,^{421–422a} including an autoimmune inner ear disease.⁴²³ The extracellular domain of P₀, like many other cell adhesion molecules (p. 407), has a structure related to that of immunoglobulins. Four molecules of P₀, each of which carries a single immunoglobulin domain, associate via these domains in a kind of square donut that protrudes from the outer cell surface. There it can interact with four similar donuts from the apposed cell surface, zipping up the cell–cell interface by a kind of Velcro action.^{422,424,425} Protein P₀ accounts for 50% of the total protein of peripheral myelin, but the **myelin basic protein**, which constitutes 20% of the total protein, is also essential.⁴²⁶ This protein exists as a variety of forms that arise from differential splicing of its mRNA and extensive posttranslational modification. Deimination of arginine side chains to form citrulline residues has been associated with development of the autoimmune disease **multiple sclerosis**.^{427,428} **Peripheral myosin protein 22** is a 160-residue polypeptide with four membrane-spanning helices. It accounts for 2–5% of the myelin protein and is the site of defects that cause the demyelinating **Charcot–Marie–Tooth disease** and other serious human diseases.^{428a,b}

The axon is effectively insulated from the surrounding medium by the myelin sheets except for special regions, the **nodes of Ranvier**, which lie at 1- to 2-mm intervals along the nerve. The nerve impulse in effect jumps from one nerve to the next. This **saltatory conduction** occurs much more rapidly (up to 100 m/s) than conduction in unmyelinated axons. It depends upon Na⁺ and K⁺ channels that are concentrated in the nodes of Ranvier.

5. Ion Conducting Channels

What is known about the channels through which Na⁺ and K⁺ flow during nerve excitation? That the channels for the two ions are separate was shown by the fact that **tetrodotoxin** (found in the puffer fish)^{429,429a} and **saxitoxin** of dinoflagellates, as well as

scorpion toxins (see Fig. 30-16), exert their toxic action by blocking the Na⁺ channels while having no effect upon conductance for K⁺. At the same time the K⁺ channels can be blocked by certain quaternary ammonium salts. Since the binding constants for the toxins are high ($K_f \sim 3 \times 10^8 \text{ M}^{-1}$ for tetrodotoxin), it is possible to titrate the sodium channels. The number is usually quite small, about 10–400 Na⁺ channels / μm^2 of surface⁴³⁰ (the same surface area contains 2×10^6 phospholipid molecules). However, membranes in the nodes of Ranvier of mammalian nerve fibers⁴³¹ contain $\sim 12,000$ channels / μm^2 . Note that the ion channels described here are not the same as those in the ion pump, i.e., the Na⁺,K⁺-ATPase (Fig. 8-25). In some neurons the number of conduction channels for Na⁺ appears to be ten times less than the number of pumping channels, i.e., of Na⁺,K⁺-ATPase.^{432,432a}

Since the number of ion-conducting channels is small, the rate of sodium passage through the open channels must be extremely rapid and has been estimated as $\sim 10^8$ ions / s.⁴³³ This is within an order of magnitude of the diffusion-limited rate (Eq. 9-30). On this basis it is clear that the channels cannot act by means of ionophoric carriers but form pores that can be opened and closed (**gated**) in response to changes in the membrane potential. They are **voltage-sensitive ion channels**.^{433,434} The channels are selective for specific ions and the selectivity parallels that of sites in some cation-exchange resins such as those containing carboxylate groups. This suggested that the inside surface of the channel might contain one or more carboxylate groups from protein side chains as well as other polar groups. A Na⁺ ion approaching the channel entrance might exchange some of its hydration sphere for ligands from the channel surface. The differing affinity of the “ion exchange” sites for various cations could ensure that it is predominately Na⁺ that passes through the channel. Anions could be excluded by electrostatic repulsion. Recent structural studies have allowed these speculations to be replaced with experimental findings as described in the following paragraphs. They have revealed that the selectivity mechanism are similar for Na⁺ and Ca²⁺ channels.

The sodium ion channel of the electric eel.

Making use of the binding of radioactively labeled specific toxins to identify them, the subunits of the sodium channel proteins were purified from several sources including the electrical tissue of the electric eel *Electrophorus electricus*,^{437–439} heart and skeletal muscle, and brain.^{440–441b} In all cases a large ~ 260 -kDa glycoprotein, which may be 30% carbohydrate, is present. The saxitoxin-binding protein from rat brain has two additional 33–36 kDa subunits with a stoichiometry of $\alpha\beta_1\beta_2$. The *Electrophorus* α subunit consists of 1820 residues,⁴³⁷ while rat brain contains α proteins of 2009

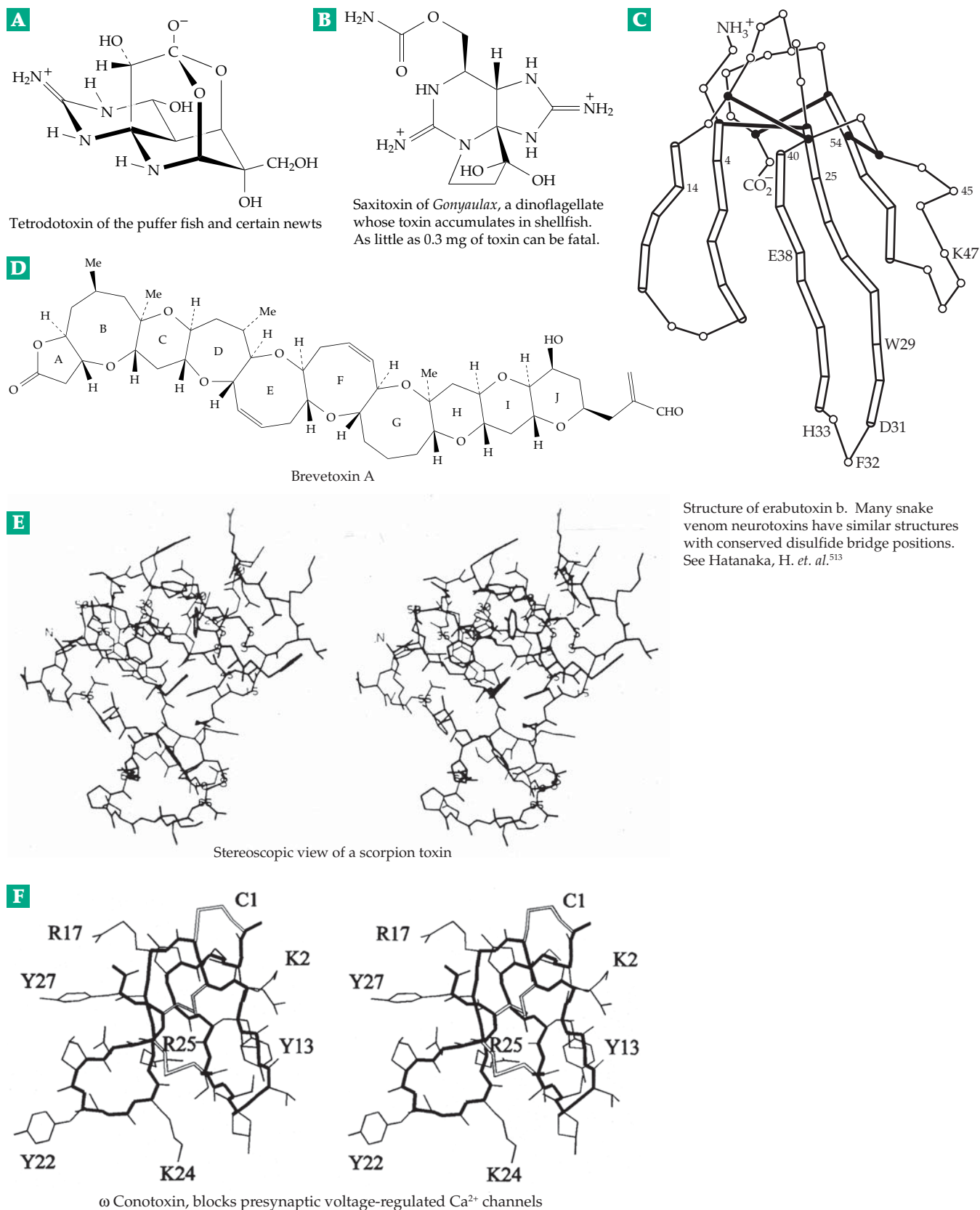


Figure 30-16 Structures of some neurotoxins that affect ion channels. Other neurotoxins include the Na⁺, K⁺-ATPase inhibitor ouabain (Fig. 22-12), batrachotoxin (Fig. 22-12), and picrotoxin (Fig. 22-4). The structure of a scorpion toxin is from Almassy *et al.*,^{494a} that of ω conotoxin is from Pallaghy *et al.*,⁴³⁵ and that of brevetoxin is redrawn after Shimizu *et al.*⁴³⁶

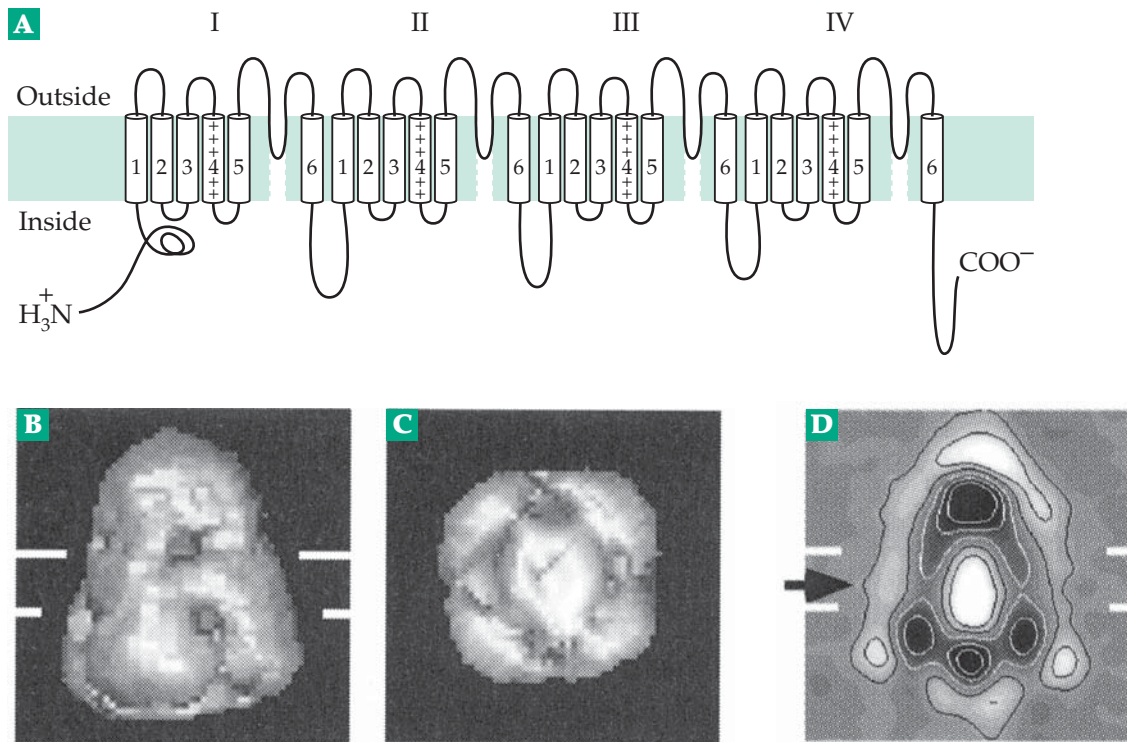


Figure 30-17 (A) Two-dimensional map of the ~ 260 -kDa α subunit of the voltage-gated Na^+ channel from the electric eel *Electrophorus electricus*.^{438,441} (B) Image of the sodium channel protein obtained by cryo-electron microscopy and image analysis at 1.9 nm resolution. In this side view the protein appears to be bell-shaped with a height of ~ 13.5 nm, a square bottom (cytoplasmic surface) ~ 10 nm on a side, and a hemispherical top with a diameter of ~ 6.5 nm. (C) Bottom view of the protein. (D) Axial section which cuts the bottom, as viewed in (C), approximately along a diagonal. From Sato *et al.*⁴³⁸ Notice the cavities (dark) and domain structures (light). The black arrow marks a constriction between upper (extracellular) and lower (cytoplasmic) cavities. White lines indicate approximate position of the lipid bilayer. From Sato *et al.*⁴³⁸ Courtesy of Chikara Sato.

and 2005 residues, respectively, for Na^+ channels designated I and II.⁴⁴¹ In fact, mammals contain ten distinct Na^+ channel genes.⁴⁴² In every case the channel proteins contain four consecutive homologous sequences of about 300 residues apiece. Within these the hydro-pathy plots (see Fig. 2-30) suggest that each homology region forms six membrane-spanning helices as shown in Fig. 30-17A.⁴⁴¹ The four sets may then fold together into a square arrangement that provides a pore somewhat familiar to that of the voltage-gated K^+ channel (see Fig. 30-18). The three-dimensional structure of the sodium channel protein, based on cryo-electron microscopy, appears to be complex. The central channel may resemble that of Fig. 30-18, but there also seem to be smaller peripheral channels (Fig. 30-17).⁴³⁸ Bacteria also contain Na^+ channels but they are tetramers of smaller subunits, resembling in this respect bacterial K^+ channels (Figs. 30-18).^{442a,b}

How do the “gates” to ion channels open? Presumably some part of the channel protein senses the change in potential and undergoes an appropriate alteration in conformation that opens the gate.⁴³⁴ The current carried by the ions flowing out through a small

number or even a single channel can be measured with tiny **patch electrodes** having openings $\sim 1 \mu\text{m}^2$ in area. These are pressed against the nerve membrane, where they form a tight seal. With such a small patch of membrane surface the electrical noise level is low, and it is possible to measure the conductance of the pore.^{149,443} From such measurements it was found that a single pore can allow $>10^8$ ions to pass through in one second. Another thing that is apparently measured with patch electrodes is a small **gating current**, which precedes the opening of the channels by ~ 0.1 ms. This has been interpreted as a flow of ~ 6 charges across the membrane or the movement of a larger number of dipoles needed to open the gate. One possibility is that a loss of the electrical field from the surface charges on the bilayer induces a rearrangement of charges on protein side chains within the bilayer or induces changes in interactions between two or more dipoles. Such changes could trigger conformation alterations within the proteins, allowing the channel to switch from open to closed.

Recordings with single channels indicate that after a sodium channel is open for a random length of time

it spontaneously closes and passes into a third state, an “inactive” state from which it cannot reopen during the refractory period. After the membrane is repolarized it can function again.^{444a}

Calcium ion channels. Immediately after the Na^+ pores open as a result of membrane depolarization, voltage-sensitive Ca^{2+} channels also open. These allow a rapid influx of Ca^{2+} , which can trigger many processes including the secretion of neurotransmitters within the synapses.^{434,444} There are several types of voltage-sensitive Ca^{2+} channels.^{444a,b} The most abundant type are specifically inhibited by dihydropyridines and are called **dihydropyridine-sensitive** or L-type channels.^{434,445–445b} They are most numerous in the transverse tubular membranes of skeletal muscle where they appear to form a complex with the very large calcium release channels, the **ryanodine receptors** (Fig. 19-21 and associated discussion).^{446,446a} These channels appear to have a structure similar to that of the Na^+ channels.⁴³⁴ Calcium channels are also discussed on p. 422 and on pp. 1114–1115. Calcium ions play a central role in cell signaling and there are a large number of different calcium channels in bacteria, plants, and animals. Many of these are coupled to specific receptors.^{445b,447} Some are involved in controlling intracellular stores.^{447–449} Some release Ca^{2+} in response to mechanical movement and function in feeling, hearing, maintaining balance, and cardiovascular regulation. Plants sense wind and gravity, and microorganisms sense changes in osmotic pressure with the aid of these channel proteins.^{450–452}

Potassium ion channels. Several types of K^+ -selective cation channels have been recognized on the basis of electrophysiological and pharmacological studies.¹⁴⁹ More recently, the cloning of channel genes has permitted the study of the proteins by X-ray crystallography. The first structure determined^{452a} was that of the *Streptococcus lividans* K^+ channel (designated KcsA; Fig. 8-21). There are three large structural families of K^+ channel proteins.^{453–455a} One group consists of voltage-regulated (K_v) channels, such as those involved in the action potential of neurons. Like the *S. lividans* channel, they are tetramers whose predicted structure contains six transmembrane helices per subunit with a pore-forming loop (P region) between helices 5 and 6. This is just what is seen in the *S. lividans* channel and in one-fourth of the much larger Na^+ channel protein (Fig. 30-17A). Furthermore, all known potassium channels, from bacteria to human beings, have the conserved sequence GYGD in the C-terminal half of the P region.⁴⁵³ A great variety of K_v channels are known. There are ~70 genes for these channels in the *Caenorhabditis elegans* genome.⁴⁵⁶ One of the first K_v channel genes to be cloned was from a *Drosophila* mutant known for its

neurological defect as *shaker*. Its structure (Fig. 30-18), which is based in part on modeling from the KcsA channel, has the ion selective filter with the conserved sequence **TVGYG** in the expected location. At the cytoplasmic end of the pore is an additional structure not found in the KcsA channel. This is the **inactivation gate**, so called because it accomplished the rapid self-inactivation of the K^+ channels during passage of the action potential (Fig. 30-18A). This is one of the factors necessary for recovery and repolarization of the axon membrane. The inactivation gate is composed of N-terminal ~130 residue “T1” domains of the α subunits together with parts of the β subunits, which are associated as a tetramer beneath the channel in the cytoplasm (Fig. 30-18B). Various experimental data including mutational analysis suggest that small ball-like domains at the N termini of the β subunits block the channel.^{456–458} Zhou *et al.* propose that the N termini unfold into an extended conformation, passing through “windows” between the T1 domains and the channel and allowing the $-\text{NH}_3^+$ ends to bind into the central cavity in the channel.⁴⁵⁹ The same site can be blocked by well-known quaternary amine inhibitors such as tetraethylammonium, tetrabutylammonium ions, or tetrabutylantimony, an analog used for X-ray crystallography.

The T1 domain of the channel not only participates in control of the ion flux but also stabilizes the pore complex.^{459a} Among the various K^+ , Ca^{2+} , and Na^+ channels the regulatory β subunits are quite variable in their structures and mechanisms of gating.^{459b} Some β subunits have bound NADH. A speculative possibility (p. 737) is that the rapid interconversion of the positively charged thiazolium ion and negatively charged thiolate ion forms of thiamin (Eq. 7-19) plays some role in nerve conduction, e.g., voltage sensing.

Some questions about ion channels have been hard to answer. For example, how are small cations allowed to flow rapidly through a very small opening in a 2–3 nm thick nonpolar core of a membrane?^{460,461} From basic electrostatic principles ΔG for transfer of an ion to the center of a membrane has been estimated as ~160 kJ/mol, a high thermodynamic barrier to transport. A solution to this problem apparently lies partly in the fact that at the center of the lipid bilayer the ion channel contains a cavity large enough (~0.5 nm diameter) to hold about 50 water molecules. Cations tend to enter this cavity, and X-ray studies have shown that the electron-dense Rb^+ does occupy the cavity. A second stabilizing factor is that four helices have their negative (C-terminal) ends pointing toward the cavity. Although the electrostatic effect of these helix dipoles (Fig. 2-20A) might be regarded as negligible, computations indicate that within the low dielectric bilayer the stabilizing effect of the helices becomes significant.^{460,461}

How are the pores in these channels opened and closed? Different channels are gated in different ways.

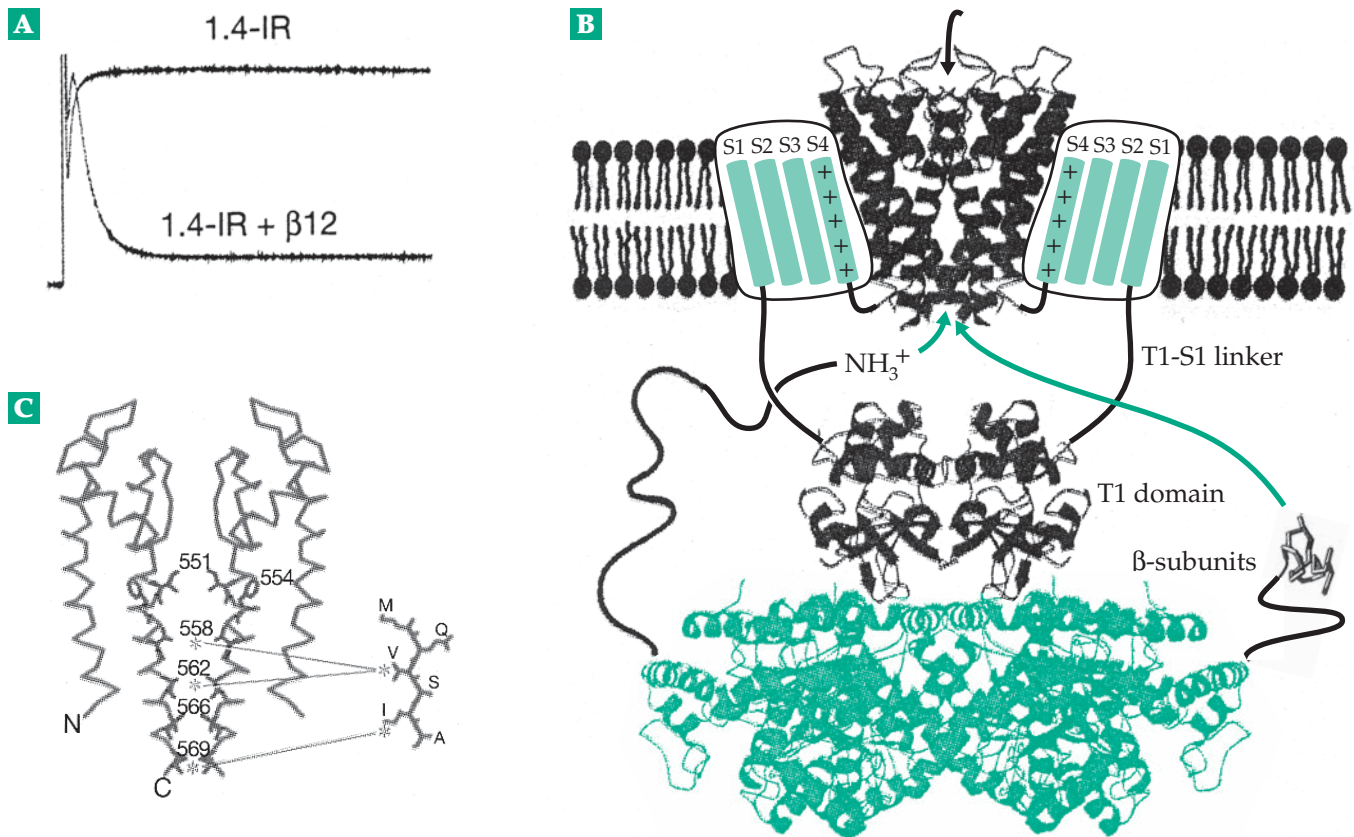


Figure 30-18 (A) K^+ currents recorded from *Xenopus laevis* oocytes carrying cloned genes of *Drosophila* shaker K^+ channels under two-electrode voltage-clamp conditions. Trace 1.4-IR was obtained from a cell expressing channels that lack the inactivation gate. Trace 1.4-IR + β_{12} , obtained from a cell expressing β subunits as well, shows rapid self-inactivation. (From Zhou *et al.*⁴⁵⁹) (B) Composite model of a voltage-dependent K^+ channel. The pore structure in the α subunit is represented by the KcsA channel (Fig. 8-21). The structure of the T1- β complex is from Gulbis *et al.*^{458a} The drawing is modified from that of Zhou.⁴⁵⁹ (C) Ball-and-stick view of the selectivity filter showing positions of four bound K^+ ions. Two of the four TVGYG peptide strands of the conduction pore are shown. Courtesy of Roderick MacKinnon.

The KcsA channel, which is mostly closed at neutral pH, responds by opening at a low external pH.⁴⁶² Using methods of spin labeling and EPR spectroscopy, Perozo *et al.* found small translational and rotational movements of the helices that form the pore (Fig. 30-18). These may alter the diameter of the pore, opening or closing it.⁴⁶³ How do the electrostatic sensors control the process? The details are uncertain, but the sensor is thought to lie in a conserved sequence of arginine and lysine residues interspersed with hydrophobic amino acids in transmembrane helix 4 of the channel protein (Fig. 30-18; see also Fig. 30-17).⁴⁵⁶ How do potassium pores select K^+ over Na^+ or Ca^{2+} ? One factor is that Na^+ is more heavily hydrated than K^+ (p. 311). This allows K^+ to pass through the channel more readily than Na^+ .⁴⁶⁴ Potassium ions travel through the 1.2-nm-long selectivity filter at a rate of $\sim 10^8 \text{ s}^{-1}$ in consecutive steps of dehydration, movement, and rehydration occurring in $\sim 10 \text{ ns}$.^{464a-d} The process is catalyzed by polypeptides and may depend

upon competition between a state in which a ring of four hydrogen-bonded peptide groups is formed and a state in which the four carbonyl groups coordinate a K^+ ion.^{464d}

Belonging to the same structural group as the K_v channels are Ca^{2+} -regulated K^+ channels.^{465,466} Some bacterial channels are controlled by binding of Ca^{2+} ions to a "gating ring" on the intracellular membrane surface.^{466a} A mammalian channel is controlled by a complex of calmodulin with the intracellular end of the α subunits of the channel^{466b} and others.^{453,467} A second large group of K^+ channels, containing seven subfamilies, are the **inward rectifying** (Kir) channels.^{455,468} They are tetramers of 360- to 500-residue polypeptide chains, each chain forming two transmembrane helices with a P region between them.^{453,469} These channels support a large conductance when K^+ ions flow out from a cell but only a small conductance when they flow in.⁴⁷⁰ Kir channels are subject to a variety of controls, which include effects of pH.^{471,472}

Some are inhibited by ATP,^{473–474b} and others by eicosanoids⁴⁷⁵ or inositol hexaphosphate.⁴⁷⁶ Some of the ATP-sensitive channels contain an ABC transporter subunit and are binding sites for sulfonylureas and other drugs. See discussion on p. 421. A number of human disorders in Kir channels have been identified.⁴⁶⁸ The human Kir channels participate in regulation of resting membrane potentials in K⁺ homeostasis, control of heart rate, and hormone secretion.⁴⁶⁸ A third group of K⁺ channels are dimeric, but each subunit contains two tandem P regions and 4–8 transmembrane helices.⁴⁵⁵

Chloride channels and the ionic environment of neurons.

All cells contain voltage-gated chloride channels, which are encoded by the *Clc* genes mentioned on pp. 420, 421.^{477,477a} Recently crystal structures^{477a–c} have revealed chloride channels formed in single polypeptide chains arranged as dimers. The selectivity filter involves stabilization by the positive ends of α -helix dipoles. The importance of the corresponding proteins to the human body is shown by the existence of several specific diseases arising from mutations in their genes (p. 420).^{478,479} A calcium-regulated Cl[−] channel is also present⁴⁸⁰ as is the ATP-gated CFTR channel (Box 26-A).^{480a} In addition, other ligand-gated Cl[−] channels, such as γ -aminobutyrate receptor channels (Section B,9), are found in the central nervous system.⁴⁸¹ A glutamate-gated chloride channel in invertebrate organisms is the site of action of the antihelminthic and insecticidal compound **ivermectin**.^{481a}

The significance of ion channels can be better appreciated by considering the ionic environment of nerve axons.¹⁴⁹ Mammalian neurons have roughly the following millimolar concentrations of ions in the cytosol and in the external medium. (The concentration gradients for the much-studied squid axon are substantially higher.^{149,482}) The membrane potentials that could arise from each one of these concentrations, according to Eq. 8-2, are also given.¹⁴⁹ In a resting

	Cytosol	Extracellular	E_m (mV)
K ⁺	150	5.5	−90
Na ⁺	15	150	+60
Ca ²⁺	10 ^{−4}	1.5	+270
Cl [−]	9	12.5	−70

neuron the K⁺ potential dominates with an observed membrane potential of $\sim$ −80 mV. Some K⁺ channels are open and the K⁺ and Cl[−] concentrations are nearly in Donnan equilibrium across the membrane. The Na⁺ and Ca²⁺ channels are closed, and the sodium and calcium pumps keep the internal concentrations of these ions low.

When an action potential is propagated, a wave of depolarization moves along the axon, changing the membrane potential suddenly to a less negative value. When it reaches $\sim$ 50 mV the Na⁺ channels open, allowing sodium ions to flow into the cell causing further propagation of the wave of depolarization. After $\sim$ 1–2 ms the Na⁺ channels begin to deactivate. At the same time the slower K⁺ channels open allowing potassium ions to flow out and to repolarize the membrane, the membrane potential sometimes transiently reaching more negative values (hyperpolarization) than the $\sim$ 80 mV resting potential. Action of the Na⁺,K⁺-ATPase then restores the original state. The finely tuned properties and sequential opening and closing of the channel proteins are essential to the conduction of nerve impulses.

The existence of voltage-gated ion channels in bilayers are not limited to nerve membranes. They are present to some extent in all cell membranes. Even the paramecium has at least seven kinds of Na⁺, K⁺, and Ca²⁺ channels.⁴⁸³ Channels may also be formed by many peptide antibiotics. Among them are the human defensins (Chapter 31) and the $\sim$ 20-residue **alamethicin**. Six to eleven of the mostly helical monomers of that antibiotic assemble to form a single voltage-dependent channel.^{484,484a} The bacterial toxin colicin E1 (Box 8-D) forms voltage-dependent channels within bacterial membranes.⁴⁸⁵

Receptor-associated ion channels. Many neurotransmitters, including acetylcholine and glutamate, act to open ion channels that are part of the receptor protein or of a tight complex of proteins.^{149,486} Such **ionotropic receptors** are responsible for most rapid neuronal action. For example, binding of acetylcholine to its receptor in the neuromuscular junction causes the release of Ca²⁺ ions from the exterior into the muscle fibers. Binding of glutamate to its ionotropic receptor in a synaptic ending of a dendrite causes an influx of ions into the cytoplasm, initiating an action potential in the dendrite. In most instances the properties of the receptor channel favor the rapid flow of Ca²⁺ ions into the cytoplasm.

Many other receptors are 7-helix transmembrane proteins, which activate guanine nucleotide G proteins (Chapter 11, Section D, 3). The G proteins couple some receptors directly to Ca²⁺ channels; they couple other receptors to adenylate cyclase and cyclic AMP-activated channels and yet others via phospholipase C to K⁺ channels and indirectly to Ca²⁺ channels (Fig. 30-19). All of these G protein coupled receptors are referred to as **metabotropic receptors**. A single synapse often contains both ionotropic receptors and metabotropic receptors. The ionotropic receptors induce a rapid (<1 ms) response, while the metabotropic receptors act more slowly. However, in most cases the final effect is the release of calcium ions into the cytoplasm

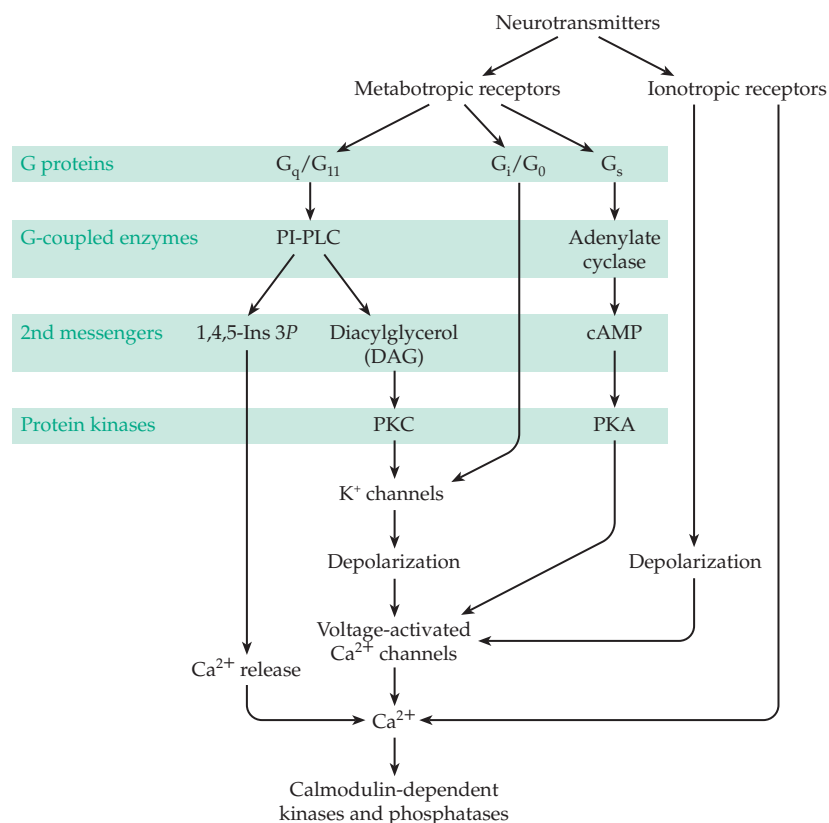


Figure 30-19 Major signaling pathways from metabotropic and ionotropic receptors in neurons. Various G proteins control the signaling from metabotropic receptors using phosphatidylinositol-specific phospholipase C (PI-PLC) and adenylate cyclase or acting directly on K⁺ ion channels. Adapted from Fig. 5.1 of Nicholls' *Proteins, Transmitters, and Synapses*.¹⁴⁹

(Fig. 30-19). The rapid response may be initiation of an action potential, while the slow response may be activation of calmodulin-dependent kinases and phosphatases.¹⁴⁹

6. A Plethora of Neurotoxins

Bacteria, protozoa, and venomous animals synthesize numerous toxins that are used to kill their prey or to defend themselves. Sea anemones, jellyfish, cone snails, insects, spiders, scorpions, and snakes all make potent and highly specific neurotoxins. Plants form a host of alkaloids and other specialized products, some of which are specifically neurotoxic and able to deter predators. More than 500 species of marine cone snails of the genus *Conus* synthesize a vast array of polypeptide toxins (**conotoxins**),^{487–489} some with unusual posttranslational modifications.^{490,491} The slow-moving snails are voracious predators that use their toxins, which they inject with a disposable harpoon-like tooth,⁴⁹² to paralyze fish, molluscs, or worms.⁴⁹³

The targets for natural biological toxins include ion channels and receptors for transmitters. At least four parts of the voltage-gated sodium channels are binding sites for extremely toxic natural products.^{494–499} **Tetrodotoxin** (Fig. 30-16),^{496,497} which is found in the puffer fish, certain newts,^{429a} and venom of the blue-ringed octopus, and also the shellfish poison **saxitoxin** (Fig. 30-16) block the entry of sodium ions into the channels.⁴⁹⁸ **Batrachotoxin** (Fig. 22-12) and related lipophilic compounds such as **veratridine** increase sodium permeability by blocking the channels permanently open. **Pyrethroid insecticides** (p. 1237) prolong the time that the sodium channels stay open after excitation. Some **scorpion toxins** (Fig. 30-16),^{494,499} which all have a hydrophobic core made from a short α helix and a three-strand antiparallel β sheet,^{500–502} and **sea anemone toxins**^{495,503–505} also stabilize the open conformation of the Na⁺ channels. Other smaller ~4-kDa scorpion toxins block K⁺ or Cl[−] channels or other receptors.^{500,506,507} Some are most toxic to insects and others to mammals.⁵⁰⁰ Although their three-dimensional structures resemble those of scorpion toxins, the amino acid sequences of anemone toxins show no homology.⁵⁰⁵ The most potent poison produced by the red tide organism, the dinoflagellate *Gymnodinium breve* (Fig. 1-9), is **brevetoxin A** (Fig. 30-16).^{436,508} It selectively opens one class of sodium channels.⁴⁹⁵

Venoms of **cobras**, **sea snakes**, and pit vipers contain several 6- to 7-kDa proteins that bind to acetylcholine receptors (Fig. 30-23) of the postsynaptic neurons, preventing binding of the neurotransmitter and opening of the ion channels.^{509,510} All of these toxins contain four disulfide bridges and share with certain plant proteins a folding pattern that has been called the **toxin-agglutinin fold**^{511,512} (Fig. 30-16). These toxins include **erabutoxin a** (Fig. 30-16) from a sea snake^{513,514} as well as the 74-residue toxin **bungarotoxin a** (from the banded Krait). This toxin, which has been used to titrate acetylcholine receptors in neuromuscular junctions, is a member of the *long neurotoxin* group, which contains 71–74 residues and five disulfide bonds.⁵¹⁵ Other *short neurotoxins* are 60–62 residues in length with four disulfide bridges.⁵¹⁶ Cobra toxins contain both neurotoxins and **cardiotoxins**, which have somewhat similar structures but quite different modes of action.^{517,518} In contrast, **crotoxin**

from the venom of a South American rattlesnake^{510,510a} and **β -bungarotoxin**⁵¹⁹ consist of 13-kDa phospholipases A₂ complexed with smaller 7.5-kDa proteins. They act at the presynaptic membranes of selected neurons by blocking neurotransmitter release.⁵²⁰

The seven types of **botulinum toxin**^{521–523a} and the **tetanus toxin**⁵²⁴ are the most neurotoxic substances known. Only 10⁸ molecules are sufficient to kill a mouse. Both toxins are zinc proteases, which block presynaptic transmitter release by cleaving specific synaptic vesicle proteins (see p. 1780 and Fig. 30-20).^{522,523,525–528} They bind initially to ganglioside in the neuromuscular junction, one subunit then being internalized as with the diphtheria toxin (Box 29-A). Botulinum toxins specifically enter motor neurons,^{521,528a} while tetanus toxin is taken up via synaptic vesicle endocytosis⁵²⁹ by both peripheral and central neurons. Retrograde axonal transport carries the toxin into the central nervous system and across synaptic clefts into cholinergic interneurons, which are poisoned.

The black widow spider produces the 130-kDa **α -latrotoxin**, which causes massive release of acetylcholine, norepinephrine, dopamine, and GABA from synaptosomal endings.^{530,531} The small **anatoxin-a** or “very fast death factor” (Fig. 30-22), which is synthesized by various cyanobacteria, antagonizes both muscarinic and nicotinic acetylcholine receptors.⁵³² Cone snails synthesize mixtures of the 13- to 17-residue conotoxins (Fig. 30-16).⁴⁹³ They cause rapid paralysis of fish permitting the snails to prey on the much faster fish. They bind to a variety of targets, which include Na⁺, K⁺, and Ca²⁺ channels,^{435,492} and acetylcholine,^{533,534} and glutamate⁴⁹⁰ receptors. One of the toxins is a 17-residue peptide containing five residues of γ -carboxyglutamate and is also notable for the fact that intercerebral injection of less than one microgram of the toxin induces a prolonged sleeplike state in mice.^{490,493} The venom of *Conus geographicus* is so toxic that two-thirds of human stinging cases are fatal.

The most deadly nonproteinaceous toxin known, **palytoxin**, is also the most complex structure ever established without the aid of X-ray crystallography.^{535,536} It is produced by marine zoanthids of the genus *Palythoa* and has the molecular formula C₁₂₉H₂₂₃N₃O₅₄.

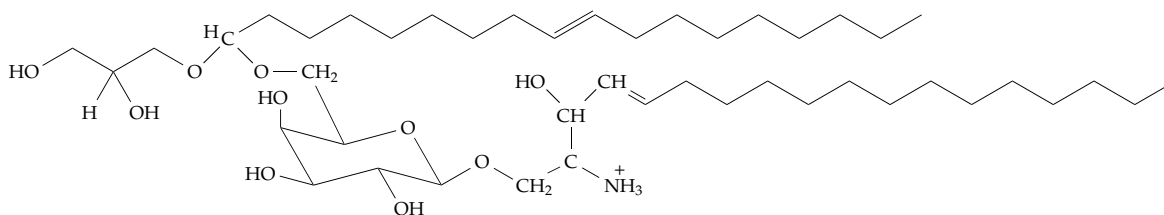
7. Neuronal Metabolism

The brain has a very high rate of metabolism. Although accounting for only 1/50 of the body mass its utilization of energy amounts to 1/5 of the basal metabolism. This is ~20 watts and is nearly constant day and night. It reflects the unusually active metabolism of neurons, a major part of which can be attributed to the sodium–potassium ion pumps in the membranes and to the maintenance of the excitable state.^{536a} The source of energy for these processes is the ATP that is

utilized to drive the ion pumps and thereby to maintain the membrane potential needed to drive the action potentials. The ATP is formed largely by oxidative metabolism of glucose and, to a lesser extent, of acetoacetate. The large surface area of the axons as well as the frequency with which they transmit nerve impulses accounts for the high rate of metabolism.

Another factor peculiar to neurons doubtless contributes also to their rapid metabolism. The nucleus and most of the ribosomes are found in the cell body. Although few ribosomes are seen in axons and dendrites^{536b}, many proteins are needed in high concentrations within the axons and synaptic endings. Among these are enzymes catalyzing synthesis and catabolism of neurotransmitters and membrane proteins. If an axon is cut, the separated synaptic endings soon atrophy, an observation that long ago suggested that essential materials, which may include mRNAs,⁵³⁷ might flow from the cell body. It has now been established experimentally that many materials do move at the rate of 0.3–3 mm / day from the cell body down the axon.⁵³⁸ More remarkable is **fast axonal transport** by which proteins and other materials move at rates of up to 5 μ m / s (0.4 m / day). This transport is specifically blocked by vinblastine (Box 7-D) and batrachotoxin (Fig. 22-12). As has been pointed out in Chapter 20, an ATP-hydrolyzing protein chemically related to the myosin heads functions together with microtubules to provide a kind of miniature railway that moves materials along the microtubules. Transport is sometimes in the opposite direction, i.e., from the synaptic endings to the cell body. This **retrograde axonal transport** may be of importance in altering neuronal properties in response to electrical activity at synaptic endings. It also provides a means of recycling materials originally sent in the other direction.

Brain cells appear to transcribe an unusually large fraction of the genome.^{539–541} About 20% of the DNA of human brain was found to hybridize with mRNA formed by brain cells. In other tissues about half this amount of DNA appears to be transcribed. A related observation that seems surprising is the absence of common electrophoretic variants of enzymes in the brain.⁵³⁹ However, brain cells synthesize specialized isoforms of many proteins, e.g., of the G proteins (p. 558), the cytoskeletal protein 4.1 (Fig. 8-14),⁵⁴² and transglutaminase.⁵⁴³ Unusual lipids, such as the cationic acetal of a galactosylcerebroside shown above,⁵⁴⁴ are also formed. Adult rat brain contains about 30,000 different kinds of polyadenylated messenger RNA,⁵⁴⁰ much of which lacks the poly(A) tail.⁵¹³ Many of these mRNAs contain a specific 82-nucleotide sequence within at least one of their introns. Sutcliffe *et al.* suggest that this is an **identifier sequence** instructing brain cells to express these genes.^{540,545} However, the sequence is also found in genes transcribed in other tissues, and its significance is not clear.^{546,547}



8. Synapses and Gap Junctions

Like the micro-transistors in a computer chip, synapses are the devices by which the brain operates. Synapses process and integrate information from many input channels, send signals on to other neurons, and store information. The information is not stored in digital form, but as chemical alterations in the synapses themselves.^{482,548,549} Synapses are formed when axons, growing in response to a chemical trail, reach their destinations and send out branches, each with a bulbous terminal knob (**bouton**). When these boutons meet receptive regions on dendrites of another axon, synapses are formed.⁵⁵⁰ The synapse is a very firm connection with a thin, tight synaptic cleft through which signaling takes place. It is surrounded in part by astrocytes or other glial cells (Fig. 30-20A,C).

With the advent of electron microscopy, the fine structure of synaptic contacts became evident. The synaptic knobs were often found to contain vesicles of ~30–80 nm diameter, which were later shown by chemical analysis and staining procedures to contain the neurotransmitters (Fig. 30-10). In the case of the acetylcholine-releasing synapses (**cholinergic synapses**) each 80-nm vesicle contains ~40,000 molecules of acetylcholine,⁵⁵¹ the concentration in the vesicle being of the order of 0.5 M. To show that the acetylcholine released at a synapse stimulated the postsynaptic membrane to initiate an impulse, the technique of **electrophoretic injection** or **microiontophoresis** was developed.⁵⁵² By using ultramicrocapillaries a small pulse of current, e.g., 3×10^8 amp for 1 ms, can be used to inject electrically a compound directly into a synaptic cleft. The results may be observed with separate recording electrodes, one of which is inserted into an axon or a muscle fiber. By this means it was shown that amounts of acetylcholine comparable to those released at the large synapses of the neuromuscular junction do cause muscles to contract.

How does the release of neurotransmitter occur? That the release is “quantal,” i.e., involving the entire content of a vesicle, was established from the observation of **miniature end-plate potentials**. These are fluctuations in the postsynaptic potential observed under conditions of weak stimulation of the presynaptic neuron. They reflect the random release of neurotransmitter from individual vesicles.⁵⁵³ Normally, a strong impulse will release on the order of 100–200

quanta of transmitter, enough to initiate an action potential in the postsynaptic neuron.

A synaptic vesicle cycle. The number of synaptic vesicles in a single synapse in the brain varies from fewer than 100 to several hundred. In specialized synapses there may be thousands. However, at any moment only a fraction of the total are in the “active zone,” often aligned along the presynaptic membrane (Fig. 30-20A) or in specialized ribbons such as those in Fig. 30-10B. The vesicles are normally reused repeatedly, undergoing a cycle of filling with neurotransmitter, translocation to the active zone, ATP-dependent priming, exocytosis with release of the neurotransmitter into the synaptic cleft, coating with clathrin, endocytosis, and acidification as outlined in Fig. 30-20B.^{554–557} The entire cycle may be completed within 40–60 s to avoid depletion of active vesicles.^{558,559} A key event in the cycle is the arrival of an action potential at the presynaptic neuron end.

The accompanying depolarization of the membrane at the synaptic ending permits a rapid inflow of calcium ions through a voltage-gated calcium channel.^{444,560} Within less than 0.1 ms the transient increase in intracellular $[Ca^{2+}]$ triggers the release of the contents of the vesicles. About four calcium ions are needed to release one clathrin-coated vesicle (Fig. 30-20A,B). The membrane fusion required for transmitter release involves cytoskeletal proteins of the synaptic endings as well as specialized proteins that are present in the membranes of the synaptic vesicles (Table 30-6). In fact, every step in the cycle depends upon specialized proteins.³⁸⁷

Synaptic vesicles can be isolated in large quantities. Their composition is well known, and the proteins have been studied intensively. Indeed, much of what we know about exocytosis and vesicular transport has been learned from investigation of synaptic vesicles.^{554,561,562} A small synaptic vesicle of 35 nm diameter will contain ~10,000 phospholipid molecules in its membrane and only about 200 protein molecules, at least one of which must be a 13-subunit vacuolar type proton pump (Fig. 18-14). This pump acidifies the vacuole, allowing uptake of a neurotransmitter. Although many different proteins may be found in synaptic membranes, only about 15, which are listed in Table 30-6, are found in all synaptic vesicles and appear essential to function.

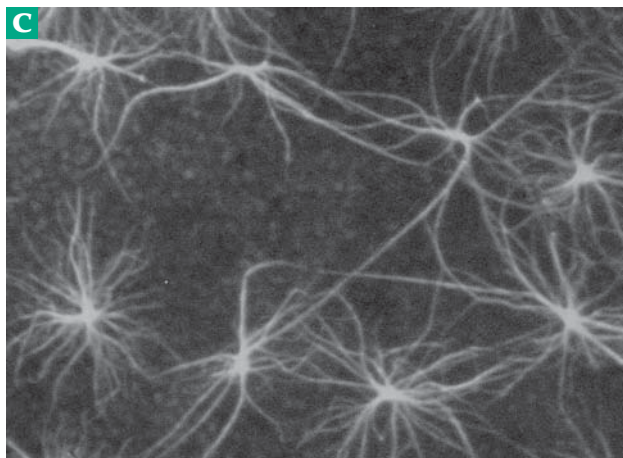
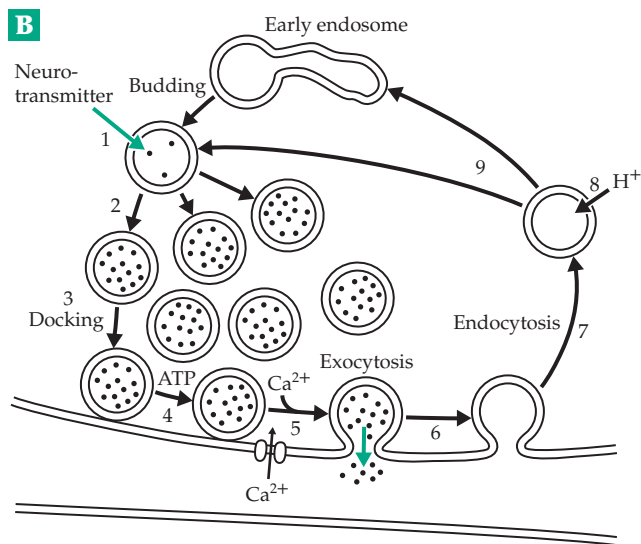
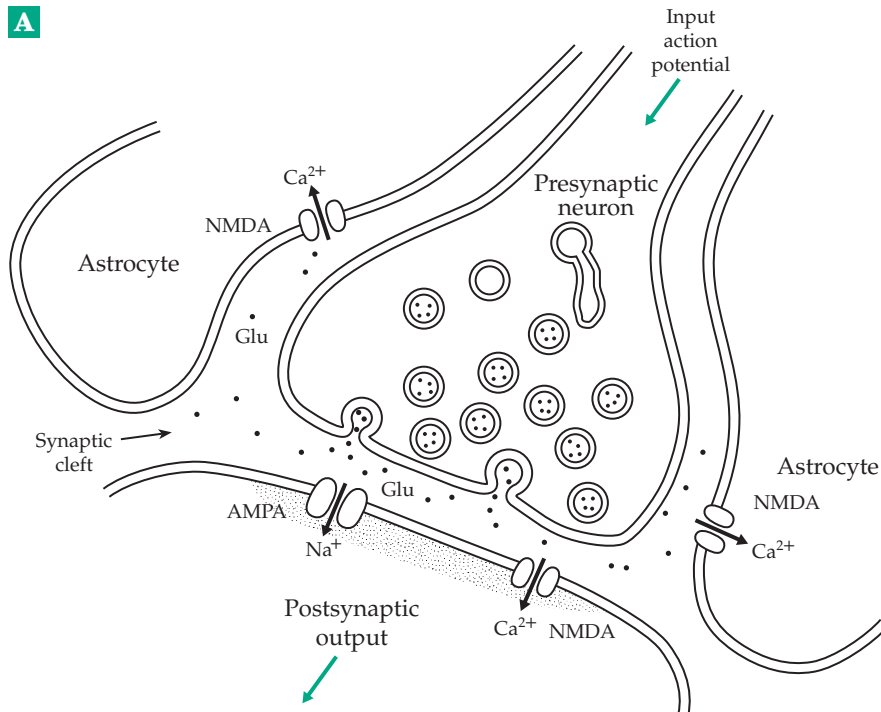
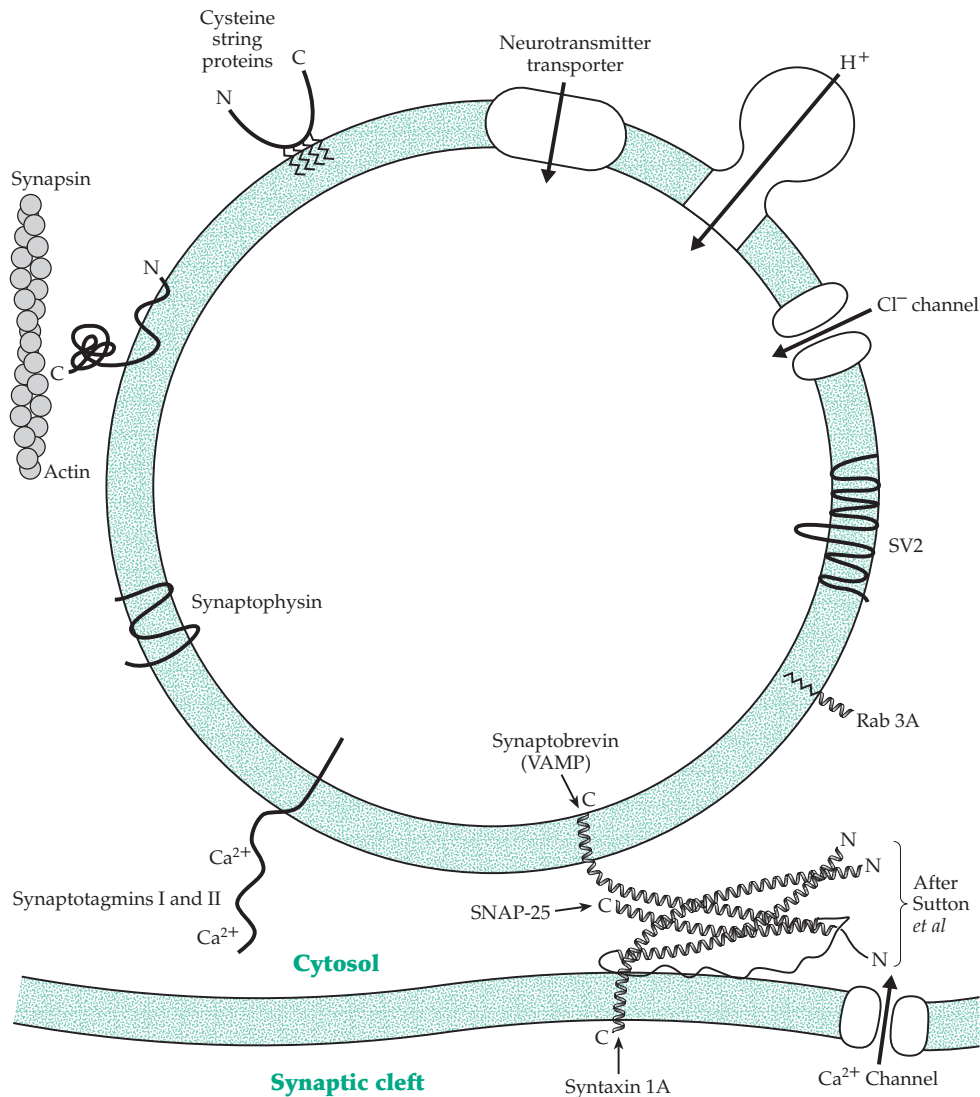


Figure 30-20 (A) Schematic drawing of a fast glutamatergic synapse. An action potential arrives at the synapse, depolarizing the presynaptic membrane and allowing calcium ions to enter the cytoplasm via voltage-gated Ca^{2+} channels. The Ca^{2+} ions induce exocytosis of small synaptic vesicles from the “active zone” near the membrane, releasing glutamate into the synaptic cleft. After diffusing rapidly across the narrow ~50-nm synaptic gap the glutamate binds to its receptors on the ending on a dendrite from a second (postsynaptic) neuron. Glutamatergic synapses usually have two types of receptor, NMDA and AMPA (see Fig. 30-24 and text). Both are ligand-gated ion channels, which release Ca^{2+} and Na^{+} into the cytosol of the postsynaptic ending depolarizing its membrane and possibly initiating an action potential. (B) The synaptic vesicle cycle. The synaptic vesicles, which are formed by budding from an early endosome, are filled with neurotransmitter (1). The filled vesicles are then transported to the active zone near the presynaptic membrane (2), are “docked” on the membrane surface (3), and undergo ATP-dependent priming (4). Binding of four Ca^{2+} ions induces exocytosis and rapid release of the neurotransmitter (5). The empty vesicles receive a clathrin coat (6) and undergo endocytosis (7) and uptake of protons (8) to acidify the content in preparation for a second round of neurotransmitter uptake. Alternatively the vesicle can fuse with an endosome as part of the cycle. After Südhof and Scheller.⁵⁵⁴ (C) Small section of brain stained to reveal the astrocytes whose extensions form synapses not only with neurons, as in (A), but also with capillary blood vessels.¹⁴⁹ From Kimelberg and Norenberg.⁵⁶⁴ Micrograph from Andreas Karschin, Heinz Wässle, and Jutta Schnitzer. (D) Illustration of some proteins essential to the synaptic vesicle cycle. Several are integral membrane proteins. Synaptotagmins contain Ca^{2+} -binding domains and may serve as calcium sensors. The vesicle is portrayed as if docked to the presynaptic membrane by interaction of the SNARE proteins synaptobrevin, syntaxin, and synaptotagmin. The 4-helix bundle is as portrayed by Sutton *et al.*⁵⁶³

D



The synaptic vesicles, which are formed by budding from early endosomes, take up neurotransmitters using one of the transporters (step 1 in Fig. 30-20B). Transmitter uptake is G-protein dependent⁵⁶⁵ and is driven by the proton electrochemical gradient generated by a vacuolar type (V-type) ATPase (Chapter 18).^{149,566} The filled vesicles move into the active zone where they undergo an ATP-dependent priming of uncertain nature.^{555,567} Exocytosis (step 5 in Fig. 30-20B) requires membrane fusion, and it is possible that partial fusion occurs during the priming steps. Priming is also thought to involve interaction between vesicle-associated v-SNAREs and synaptic membrane-associated t-SNAREs (p. 521).^{556,563} A major v-SNARE has been identified as **synaptobrevin**, which is also known as **VAMP** (vesicle-associated membrane protein).^{563,568,568a} The C-terminal-anchored synaptobrevin is inserted into the plasma membrane of neuronal and neuroendocrine cells prior to endocytosis and budding of the synaptic vesicles.⁵⁶⁸ The target

t-SNAREs have been identified as the synaptic plasma membrane proteins **syntaxin**^{568b} and **SNAP-25**.^{569–573} Syntaxin is an integral membrane protein, whereas SNAP-25 is anchored by palmitoylation.⁵⁷¹ These proteins bind together to form a synaptobrevin•syntaxin•SNAP-25 complex, which forms a four-helix bundle as shown in Fig. 30-20. Synaptobrevin and syntaxin each contribute one helix, while SNAP-25 provides two; all four have a mutually parallel orientation.^{563,574,574a} The helix bundle is so tight that it has a high melting temperature and is resistant to proteolytic cleavage. Nevertheless, the helical domains of both synaptobrevin and syntaxin are sites of very specific cleavage by the zinc proteases of tetanus and botulinum toxins.^{527,563,570} Cutting of the protein chains by these toxins prevents proper formation of the four-helix bundle and prevents release of neurotransmitter. It is thought that the complex, which probably forms at several points on the periphery of the docked synaptic vesicle, is essential for membrane fusion.

Other proteins are also needed. All cell fusion processes seem to require regulatory proteins that are essential to neurotransmission in the nematode *C. elegans*. Two of these are encoded by the nematode genes *unc-13* and *unc-18*. The corresponding mammalian proteins **munc-13** and **munc-18** interact with syntaxin and are essential for exocytosis of synaptic vesicles.^{572,575} An ATPase is also needed for correct functioning of the SNARE complex⁵⁷⁴ as are other additional proteins.⁵⁷⁰

Details of the control of exocytosis are also uncertain. **Synaptotagmin I**, which contains two Ca^{2+} -binding domains, is probably the sensor that detects the rapid influx of Ca^{2+} that initiates exocytosis.^{576–578b}

It binds several Ca^{2+} ions via a β -sandwich motif that contains five aspartate side chains at its tip. This motif is conserved in a large family of synaptotagmins. A possibility is that Ca^{2+} -synaptotagmin complexes may self-associate to form a protein ring around the site where the fusion pore forms.⁵⁷⁶ Synaptotagmin I also interacts with both syntaxin and with **neurexins**, proteins related to laminin (Fig. 8-33) and present in numerous variant forms in nerve endings. Neurexins are also targets for the α -lathrotoxin of the black widow spider.^{531,579} Other proteins that may participate in membrane fusion include the unique **cysteine string proteins**, which in *Drosophila* contain 13 cysteine residues, 11 of which are palmitoylated.^{580,581} Nitric

TABLE 30-6
Some Proteins Important to the Formation and Functioning of Synaptic Vesicles^a

1. Synaptic vesicle proteins

Synapsins Ia, Ib, IIa, IIb	Peripheral, abundant
Rab3, rabphilin	Rab 3 has lipid anchor
Cysteine string proteins (CSP)	Ca^{2+} -binding
Synaptotagmins	Single transmembrane helix; Ca^{2+} receptor N terminus in vesicle
Synaptobrevins (VAMPs) ^b	SNARE proteins, C termini in vesicle
Synaptophysins, synaptogyrin	Integral membrane protein
SV2 A, B, C	Integral membrane protein, Cl^- transporter
SCAMPS 1 and 4	Integral membrane protein
SVOP	Integral membrane protein
Vacuolar H^+ pump	13 subunits
Cytochrome 561	H^+ generator
Neurotransmitter transporters	For acetylcholine, glutamate, GABA/glycine, catecholamines, ATP
Ancillary transporters	Zn^{2+} , Cl^-

2. Presynaptic membrane proteins

Syntaxin ^b	t-SNARE
SNAP-25 ^b	t-SNARE
Munc-13	
Ca^{2+} channel	
Agrin	
Neurexin	
Actin and microtubules	In dendrites

3. Postsynaptic specializations

Receptors	e.g., NMDA, AMPA
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^a Based on data of Südhof and Scheller: Südhof, T. C., and Scheller, R. H. (2000) in *Synapses* (Cowan, W. M., Südhof, T. C., and Stevens, C. F., eds), pp. 177–215, Johns Hopkins Univ. Press, Baltimore, Maryland and Südhof, T. C. (1995) *Nature (London)* **375**, 645–653.

^b Targets for clostridial toxins, tetanus, botulinin.

oxide NO may be involved in a late stage of exocytosis,³⁹² and phospholipase D1 may also be required.⁵⁸²

Presynaptic nerve terminals may contain as few as a hundred vesicles, which must be recycled rapidly after exocytosis in order to allow for repetitive firing.^{558,559,583} Several proteins are needed for endocytosis (step 7 in Fig. 30-20). These include **endophilin I**,⁵⁸⁴ the vesicle transport ATPase **NSF**,⁵⁷⁴ GTPases,⁵⁶⁵ and the soluble NSF attachment protein α -SNAP (which is not related to SNAP-25).⁵⁸⁵

Functions of some other abundant proteins of synaptic vesicles have not yet been accurately defined. The **synapsins** are abundant peripheral membrane ATP-binding proteins with multiple phosphorylation sites and variable C-terminal domains that interact with cytoskeletal proteins such as actin microtubules, microfilaments, and spectrin.^{554,561,586,587} Another abundant protein is **synaptophysin**, an integral membrane protein found in all synaptic vesicles.^{554,561,588} Other proteins are discussed by Südhof and Scheller.⁵⁵⁴ The small G protein **rab 3** together with the Ca^{2+} -binding protein **rabphilin** participate in a G-protein cycle that helps to drive exocytosis.⁵⁵⁴ Synaptotagmin, as well as clathrin assembly proteins bind inositol hexaphosphate (InsP_6 ; Fig. 11-9), which undergoes active turnover in synapses. This suggests a role for InsP_6 in the endocytosis steps of the synaptic vesicle cycle.⁵⁸⁹ The brain is rich in zinc ions. Much of the Zn^{2+} is bound into zinc finger domains of transcriptional regulators, but much is also present in a relatively free form within synapses of the hippocampus, cerebral cortex, and other regions.^{590,591} Zinc ions may function as a neuromodulator in glutamatergic synapses.⁵⁹¹

What does a neurotransmitter do at the postsynaptic membrane? In the case of acetylcholine in neuromuscular junctions the principal action appears to be one of opening sodium channels and thereby depolarizing the postsynaptic membrane. If enough nerve impulses arrive, an action potential will be initiated in the postsynaptic neuron. In other cases, the first response may be activation of a protein kinase either directly or by opening a channel for Ca^{2+} , which indirectly regulates protein kinases and phosphatases.⁵⁹² Thus, a complex cascade may be activated. See also Fig. 30-19.

The postsynaptic nerve ending, which is usually the tip of an axonal dendrite, has its own set of proteins, which varies to some extent with the nature of the neurotransmitter. In excitatory cells the plasma membrane of the postsynaptic neuron is thickened to ~30–40 nm to form the “**postsynaptic density**,” a disc-like structure of clustered receptors of two types, which extends ~30 nm into the cytosol.^{593,594} Only single receptor channels are indicated in Fig. 30-20, but many receptors are present in the clusters^{594,595} as are other specialized proteins. One of these, designated

PSD-95, was found to associate with the NMDA receptor using the yeast two-hybrid system (Box 29-F).⁵⁹⁴ Neuronal nitric oxide synthase may also be present.

The large neuromuscular junctions, which contain clusters of acetylcholine receptors, have wider synaptic clefts (> 40 nm), which contain basal lamina, a dense network of collagen fibrils together with the heparan sulfate proteoglycan **agrin** (p. 437). Agrin activates a muscle-specific kinase MusK, which phosphorylates the acetylcholine receptors inducing clustering of the receptors together with other proteins embedded in the plasma membrane and binding to the cytosolic protein **rapsyn** (see Fig. 30-23B).^{596,597} Agrin is also a component of **immunological synapses**, which are important in lymphocyte development (Chapter 31).^{596,598,599} The neuromuscular junction is formed between two cell types, a neuron and a muscle myotube. Both contribute proteins, which include a muscle-specific laminin.⁶⁰⁰

Astrocytes and other glia. Although the glial cells greatly outnumber neurons, they were long regarded simply as glue, as implied by the name glia. We know now that the several types of glial cells have functions in many different aspects of brain chemistry.^{149,564,601–605} The oligodendrocytes generate myelin sheaths around many brain neurons. Macrophages that invade the brain differentiate into **microglia** that serve as part of the innate immune system (Chapter 31). **Bergmann glia** of the cerebellum help guide axons during brain development. The astrocytes have many processes, which not only contact synapses directly (Fig. 30-20A,D) but also form contacts with capillary blood vessels. They often contain receptor ion channels of the same types as are found in postsynaptic membranes (see Fig. 30-20A) and respond to Ca^{2+} influx as do neurons.^{602–603a} Glia often take up neurotransmitters and ions from synapses in order to prepare for consecutive nerve impulses. Glia may also control the number of synapses formed,^{604–604b} and they may have other roles in brain development. For example, an iodothyronine deiodinase (Eq. 15-60) is expressed primarily in neonatal brain, where it supplies thyroid hormone essential to brain development.⁶⁰⁵

Gap junctions in synapses. Not all neurons communicate via chemical synapses. Gap junctions, which are found in both neurons, astrocytes, and other cells, serve as **electrical synapses**. Thus, heart cells are all electrically coupled together by gap junctions.^{606–608} Gap junctions are formed with the aid of hexameric **connexons**, which are present in each of the opposed membranes and are aligned one with the other (Fig. 1-15F,G).^{607,609,610} There may be thousands of connexons in a single gap junction, which resemble ion channels in appearance but contain pores ~1.5 nm in diameter. They are formed from 26- to 43- kDa

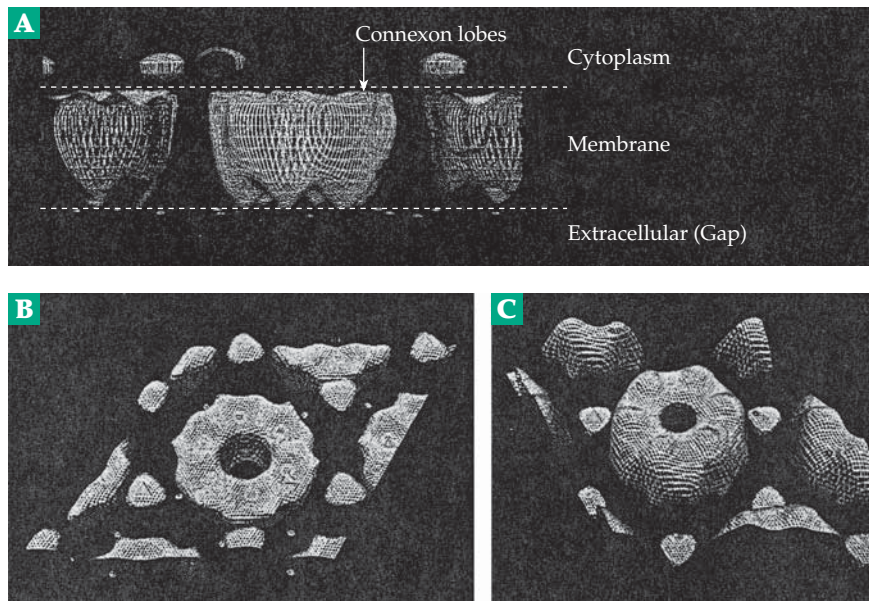


Figure 30-21 Images of gap junction connexins obtained by electron crystallographic methods at a resolution of 1.6 nm. (A) Cross-section. The thickness of the (43 x 6) kDa hexameric connexin is 5.0 nm. (B) View of the connexin from the cytoplasmic side. (C) View from the extracellular side. From Perkins, Goodenough, and Sosinsky.⁶⁰⁹ Courtesy of Guy Perkins.

protein subunits of the multigene family of **connexins**.^{610–611a} Each gap junction consists of a pair of hexameric rings of connexins (Fig. 30-21), one ring from each of the two juxtaposed membrane surfaces.⁶⁰⁹ Defects in connexins cause inherited deafness, neuropathy, malignancy, and cataract formation.^{612–613a} The connexin subunits each contain four transmembrane helices and are related structurally to the peripheral myosin protein 22, the myelin proteolipid (p. 1767), and the protein stargazin (p. 1901), which is involved in synapse formation in the brain.^{428a}

Another type of channel has been recognized quite recently. An ion channel, which regulates Mg^{2+} ion transport in kidney tubules, forms within the tight junctions that seal the extracellular space between cells (Fig. 1-15B). A protein **paracellin** forms channels through the tight junction protein complexes that surround the cells.^{614,615}

9. Neurotransmitters

Studies of neuromuscular junctions of the autonomic nervous system as early as 1904 led to the suggestion that adrenaline might be released at the nerve endings. Later it was shown that, while adrenaline does serve as a transmitter at neuromuscular junctions in amphibians, it is primarily a hormone in mammals. Nevertheless, it was through this proposal that the concept of chemical communication in synapses was formulated. By 1921, it was shown that acetylcholine is released at nerve endings of the parasympathetic system, and it later became clear the motor nerve endings of the somatic system also release acetylcholine.

Acetylcholine is an established neurotransmitter

because it meets five important criteria: (1) a synthetic mechanism exists within the presynaptic neuron; (2) a mechanism of storage (in vesicles) is evident; (3) the transmitter is released in proportion to the strength of the stimulus (frequency of firing); (4) postsynaptic action of the transmitter has been demonstrated directly by microiontophoresis; and (5) an efficient means for inactivation of the transmitter is present. The same five criteria must be met by other compounds if they are to be considered as transmitters.

At present, in addition to acetylcholine, glutamate, and γ -aminobutyrate (GABA), glycine, norepinephrine (norepinephrine), and dopamine and 5-hydroxytryptamine (serotonin) are regarded as established transmitters. Other probable (**putative**) or possible **candidate transmitters** are also known. Aspartate, taurine, and a large number of peptides (Tables 30-1, 30-4) are under consideration.

Some transmitters, including noradrenaline, dopamine, serotonin, and various neuropeptides, are sometimes called **neuromodulators** rather than neurotransmitters. These compounds may not initiate a nerve impulse but may act on adenylate cyclase to increase or decrease cAMP levels and protein kinase activity. They may also diffuse through the extracellular space to influence a region of the brain greater than a single synaptic cleft. However, the distinction between transmitters and modulators is not exact.

For many years it was assumed that a single neuron released only a single transmitter. We know now that this is incorrect.⁶¹⁶ For example, enzymes in neuromuscular junctions synthesize not only acetylcholine but also catecholamines, taurine, and GABA.⁶¹⁷ Some synapses in the central nervous system release both glycine and GABA.⁶¹⁸

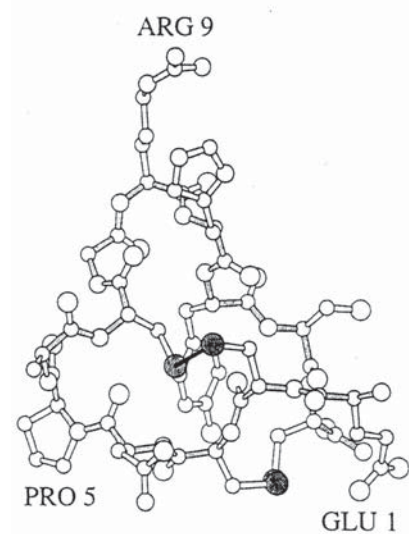
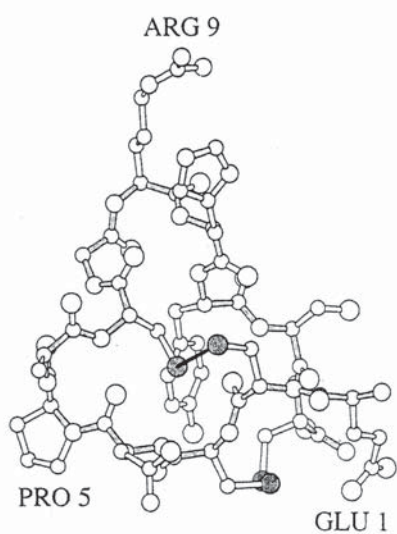
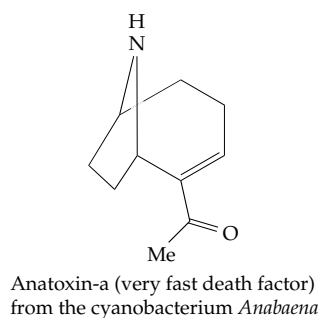
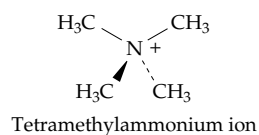
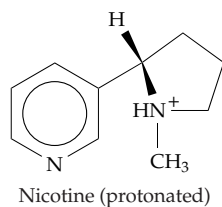
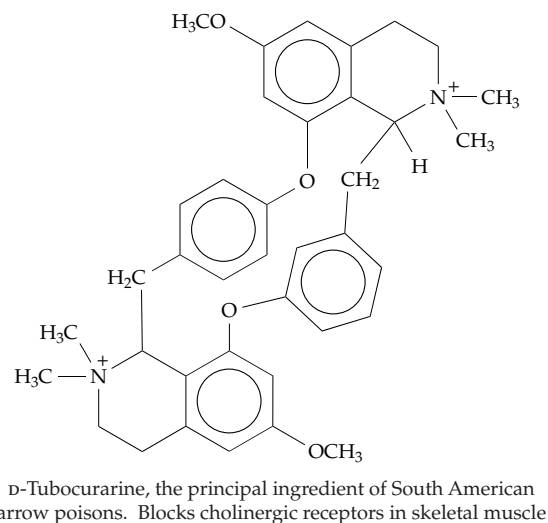
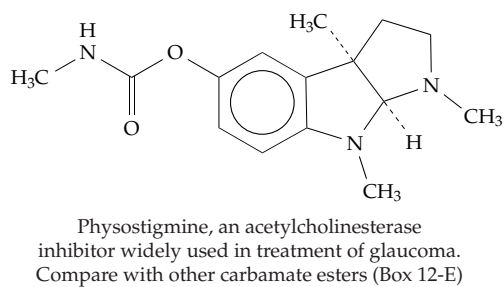
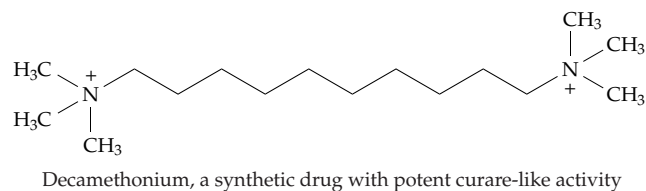
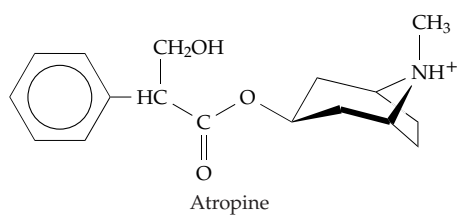
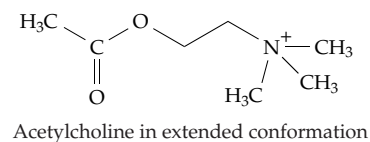
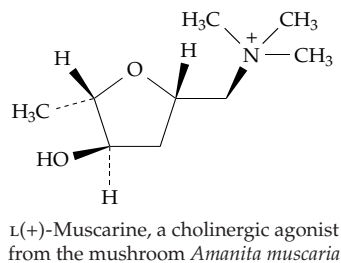
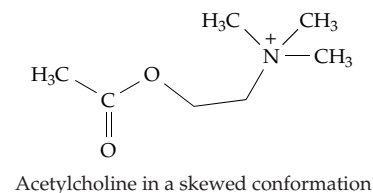


Figure 30-22 Some inhibitors of cholinergic synapses. The structure of conotoxin GI is from Guddat *et al.*⁵³³ Courtesy of A. B. Admundson.

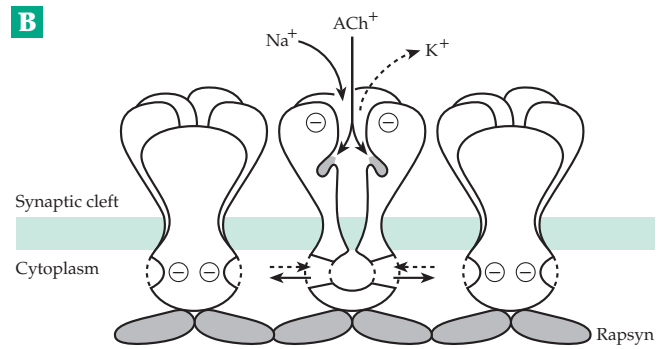
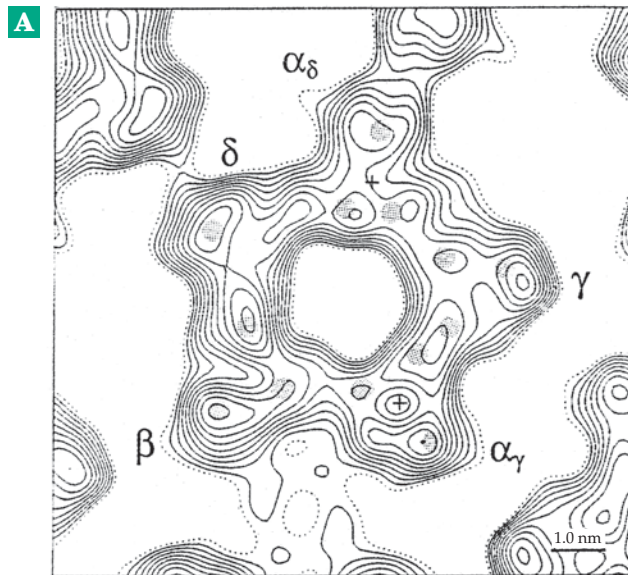


Figure 30-23 The nicotinic acetylcholine receptor from the *Torpedo* ray. (A) The mouth of the receptor channel viewed from the synaptic cleft based on reconstruction from cryo-EM images. Addition of acetylcholine, which binds to the two α subunits, induces small rotations in the five subunits of the $\alpha_2\beta\gamma\delta$ complex causing the channel to open. From Unwin.⁶⁴⁰ (B) Architecture of the subsynaptic membrane and the acetylcholine receptor. The binding of acetylcholine and the

movement of cations through the open channel is illustrated. Cations that leave the cytoplasm may be filtered through narrow openings that lead into the central channel, which is formed by transmembrane helices. Negatively charged amino acid residues may help exclude anions from the region of the pore. From Miyazawa *et al.*⁶²⁴ (A) and (B) Courtesy of Nigel Unwin. (C) Stereoscopic ribbon drawing of one subunit of a pentameric acetylcholine-binding protein, which mimics the receptor structure. Disulfide bonds are shown in a ball-and-stick form. The N terminus in a receptor would point toward the synaptic cleft and the C terminus would continue at the bottom into the transmembrane helix. Courtesy of Brejc *et al.*⁶²⁷

Cholinergic receptors and their agonists and antagonists. Among the acetylcholine-releasing (cholinergic) neurons are the motor neurons that form synapses at neuromuscular junctions, the preganglionic neurons of the entire autonomic system, and the postganglionic neurons of the parasympathetic system. There are also many cholinergic synapses within the brain. In contrast, in insects neuromuscular transmission is mediated by glutamate while acetylcholine is the principal neurotransmitter in the central nervous system.⁶¹⁹

Important in the study of neurotransmitters is the identification of specific agonists, which mimic the action of a transmitter, and of antagonists, which block the action of the transmitter. Two groups of compounds influence acetylcholine-secreting neurons, leading

to the classification of these neurons either as **muscarinic** (activated by muscarine; Fig. 30-22) or **nicotinic** (stimulated by nicotine). The muscarinic receptors, which are found in many autonomic neurons, are specifically inhibited by **atropine** and **decamethonium** (Fig. 30-22). The nicotinic synapses occur in ganglia and skeletal muscle. They are inhibited by curare and its active ingredient **D-tubocurarine** (Fig. 30-22) and by the protein snake venom **α -bungarotoxin**. This toxin has been used to titrate the number of acetylcholine receptors in the motor end plate of the rat diaphragm. About 4×10^7 receptors per end plate (or $13,000/\mu\text{m}^2$) were found.⁶²⁰

Nicotinic receptors (nAChRs; Fig. 30-23) of the type found in neuromuscular junctions are most frequently isolated from the electric organs of the electric

eel *Electrophorus* or from electric fish of the genus *Torpedo*. They have been studied more intensively than any other receptor.^{621–626a} They contain four kinds of subunit with a stoichiometry $\alpha_2\beta\gamma\delta$ and molecular masses of 39, 48, 58, and 64 kDa respectively. The amino acid sequences of the four proteins contain homologous regions, some of which are thought to represent membrane-spanning segments of the peptides. These receptors are ligand-gated ion channels and are closely similar to GABA_A and GABA_C receptors, to glycine receptors, and to 5-hydroxytryptamine (serotonin) receptors of the 5-HT₃ type. Parts of their amino acid sequences are also homologous to those of both the voltage-gated Na⁺ channels and gap junctions,^{433,627} suggesting that the transmembrane domain may resemble that of Fig. 30-18.⁶²⁸ However, notice the difference in symmetry. Acetylcholine binds to the two α subunits (Fig. 30-23). Neurotoxins may bind at several sites.⁶²⁹ Some indication of the possible function of the various subunits comes from studies of the neuromuscular junction in which the different subunits are degraded at different rates with half-lives of from one to ten days. During development fetal ϵ subunits are replaced by adult γ subunits. Perhaps more rapid changes in receptor composition are sometimes needed.⁶³⁰

Similar nAChRs are also found in the brain.^{621,631,632} However, they are not identical but have at least 17 differing amino acid sequences ($\alpha 1$ – $\alpha 10$, $\beta 1$ – $\beta 4$, γ , δ , and ϵ). The neuromuscular junction receptor (muscle type) from fish is described as $(\alpha 1)_2 \bullet \beta 1 \bullet \gamma / \epsilon \bullet \delta$.⁶²⁶ The brain contains homopentamers of subunits $\alpha 7$, $\alpha 8$, and $\alpha 9$ as well as various heteropentamers. The various forms possess different affinities for acetylcholine and for antagonists such as nicotine.^{633,634} In the brain the highest affinity for nicotine is shown by an $\alpha 4\beta 2$ form, which represents over 80% of the nAChR in mammalian brain.^{634,635} Knockout mice in which the $\beta 2$ subunit gene has been deleted lose their sensitivity to nicotine.

Conductance measurements showed that the nicotinic receptors contain channels permeable to Na⁺ and other cations and that they are acetylcholine-gated ion channels. Construction of a three-dimensional image from electron micrographs at various angles of tilt shows a tube with approximate pentagonal symmetry and a narrow channel through the center (Fig. 30-23).^{622,624,636} Acetylcholine binds to sites on the two α subunits ~3 nm away from the ion channel. An allosteric change opens the channel, allowing cations (largely Na⁺) to flow out, depolarizing the membrane. There are at least four structural states in the channel opening-and-closing cycle.^{637,638} The three-dimensional structure has been modeled using an acetylcholine-binding protein of known structure from a snail^{626,627,639} as a mimic of the cytoplasmic nicotine-binding domain of the receptor. The structure

of one subunit of the binding protein is shown in Fig. 30-23C. This protein, which is secreted into synapses by glial cells, may provide a buffering action by binding the acetylcholine. Although the most rapid effect of acetylcholine binding to the nicotinic receptor is depolarization of the postsynaptic membrane, other slower effects follow. Thus, protein kinases are activated and phosphorylate the receptor as well as other proteins.⁶⁴¹

After a pulse of transmitter is released, it must be removed or inactivated quickly to prepare the synapse for arrival of a new nerve impulse. This is accomplished in two ways in cholinergic synapses. The first is via hydrolytic destruction by acetylcholinesterase^{642–645} (pp. 634–637; Eq. 12-25). This esterase and the related butyrylcholinesterase⁶⁴⁶ are present in the synaptic membrane itself. The second mechanism is energy-dependent transport of acetylcholine into the neuron for reuse. Since much of the transmitter is hydrolyzed, new acetylcholine is synthesized by transfer of an acetyl group of acetyl-CoA to choline.⁶⁴⁷

In the central nervous system muscarinic acetylcholine receptors are more abundant than nicotinic receptors. They consist of single-chain proteins of mass ~70 kDa. They are not ion channels but are 7-helix receptors homologous in sequence with β -adrenergic receptors (Fig. 11-6) and with rhodopsin.⁶⁴⁸ Five different subtypes (M1–M5) have been characterized. The M1, M3, and M5 receptors are coupled to the G_q/G₁₁ family of G proteins (pp. 557–558), and M2 and M4 are coupled to G_i/G_o proteins.^{649–651} Their effects are slower and of longer duration than those of the nicotinic receptors. It has been difficult to assign functions to the individual types. Most regions of the brain contain more than one type, but they are thought to be involved in locomotion, learning, memory, thermoregulation, and cardiac and pulmonary functions. Many drugs, some of which are used in treatment of Parkinson and Alzheimer diseases, epilepsy, and asthma, affect muscarinic receptors. The M2 receptors predominate in the heart where they help to regulate the beating frequency and atrial contractility. Sudden infant death may sometimes result from a defect in muscarinic receptors.⁶⁵² Knockout mice lacking M2 receptors also have problems with movement control, body temperature, and pain responses.⁶⁵¹ Mice lacking M3 receptors are lean with very low levels of serum leptin and insulin.⁶⁵³ Many of the muscarinic receptors activate adenylate cyclase, while others are coupled to the phosphoinositide cascade. Some indirectly activate K⁺ channels.⁶⁵⁴ Muscarinic receptors are also studied in insects, but it is difficult to correlate the insect and mammalian receptors.⁶⁵⁵

Amino acids as neurotransmitters. The concentrations of **glutamate** and of its decarboxylation product **γ -aminobutyrate** (GABA) are high in all regions

of the brain. The two compounds are generated sequentially in the γ -aminobutyrate shunt, a pathway that accounts for a quantitatively significant part of the total metabolism of the brain (Fig. 17-5). Because they are present in all parts of the brain in high concentrations, there was initially reluctance to accept glutamate and GABA as neurotransmitters. However, it is now accepted that L-glutamate is the major excitatory transmitter in the central nervous system.^{656–658} It seems to be responsible for nearly all of the very fast acting nerve impulses in the brain. At the same time GABA is recognized as the most important inhibitory transmitter. The role of glutamate as an excitatory transmitter was first established for the neuromuscular junction of arthropods.⁶⁵⁹ Although it is a constituent of all animal tissues, the concentration of glutamate is much higher in brain than in other tissues, and it is higher in neurons than in glia. Microiontophoretic application of either glutamate or aspartate to the brain cortex leads to very strong excitatory responses.

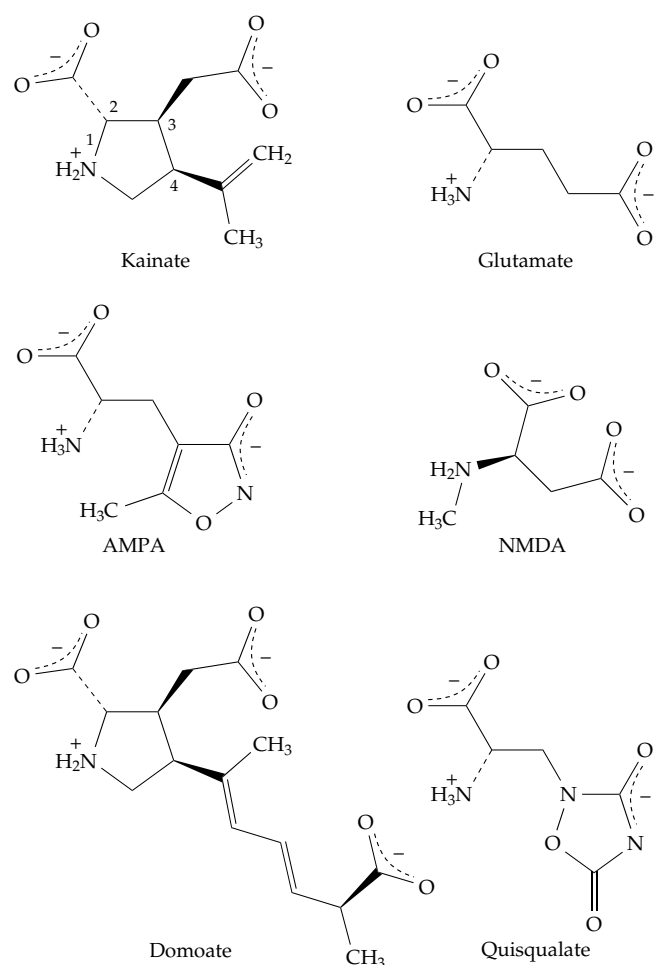


Figure 30-24 Chemical structures of some agonists of ionotropic glutamate receptors (iGluR).

Three subtypes of ionotropic glutamate receptors (iGluR) are named for the specific agonists **α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid** (AMPA), **N-methyl-D-aspartate** (NMDA), and **kainate**. The receptors resemble the acetylcholine receptor in containing a cation channel.^{149,660–662} In addition, there are 7-helix **metabotropic glutamate receptors**, which are coupled to G proteins.^{663,664} The AMPA receptors were in the recent past called **quisqualate** receptors, because they are also activated by the agonist with that name. The toxic domoate (Fig. 30-24) also binds to kainate receptors. Both domoic acid and kainic acid are terrible convulsant toxins. They are formed by two different red algae. Domoic acid accumulates in contaminated mussels and causes shellfish poisoning. The ionotropic glutamate receptors, which may be stimulated by either glutamate or aspartate, are directly linked to the opening of cation channels. Their activation may also induce the inositol phosphate cascade and slower Ca^{2+} -dependent changes. A peculiarity of the high-conductance NMDA channels is that they are blocked by Mg^{2+} in a voltage-dependent manner. They do not open unless the frequency of nerve impulses is high or some other factor causes membrane depolarization.⁶⁵⁶

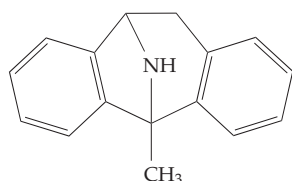
The AMPA receptors, which are thought to be the predominant mediators of fast excitatory transmission in the brain,⁶⁶⁵ are oligomers (probably tetramers^{666,666a}) of 950- to 1500-residue protein subunits. These subunits have large N-terminal domains in the synaptic cleft. There are probably three transmembrane helices and a membrane-associated loop similar to those depicted in Fig. 30-17A. A long C-terminal tail protrudes into the cytosol, while a large loop between transmembrane regions extends from the outer membrane surface, joining with the N-terminal domain to form the ligand-binding site, the structure of which resembles those of bacterial periplasmic binding proteins.^{661,665,667} Four related AMP receptors, designated GluR1, 2, 3, and 4, have been identified. Related kainate receptors, whose properties overlap those of AMPA receptors, are designated GluR5, 6, and 7.⁶⁶² Although AMPA receptors are essential for fast signal transmission they lose sensitivity rapidly (on a millisecond time scale) as a result of conformational alterations.^{667a} Many factors, including inhibition by polyamines,^{667b} affect these receptors. However, brief high-frequency activation of some AMP receptors leads to a long-lasting increase in efficiency, termed LTP, which is important to learning (see p. 1801).^{666a}

The NMDA receptors are heterooligomers with two type of subunits. The NR1 (or ζ) subunits exist as a series of at least eight splice variants. The NR2A, B, C, and D (ϵ series) are encoded by four different genes.^{668,669} NR1 is regarded as the principal subunit and NR2 as a regulatory subunit. As with the AMPA receptors⁶⁷⁰ the oligomeric NMDA receptors are

anchored at appropriate locations in the postsynaptic membrane by scaffolding proteins containing PDZ domains (Table 7-3).⁶⁷¹ The C-terminal domains of the ϵ subunits are unusually long and participate in anchoring. NMDA receptors are found not only in neurons but also in astrocytes (Fig. 30-20), where they are thought to have important signaling functions.^{672,673} These include regulation of Ca^{2+} flow, in part via gap junctions.^{603a}

The N-terminal domain of the NR1 subunit of the NMDA receptor contains a glycine-binding site.⁶⁷⁴ Full activity of the receptor requires a **coagonist** bound in this site. Surprisingly, **D-serine** seems to be the normal coagonist, at least in some sites.^{675,676} This newly recognized neurotransmitter is synthesized from L-serine by a pyridoxal phosphate-dependent decarboxylase and is destroyed by the flavoprotein D-amino acid oxidase. Associated with NMDA receptors are clusters of **ephrin receptors**, proteins that bind the glycosylphosphatidylinositol (GPI)-anchored proteins known as ephrins in presynaptic membranes. Binding of ephrins to their postsynaptic receptors activates tyrosine kinases and enhances the influx of Ca^{2+} ions.^{676a,b}

Specific inhibitors of NMDA channels include a 27-residue "spasmodic" conotoxin,⁴⁹⁰ 2-amino-4-phosphonobutyrate, related longer chain aminophosphonates, and the following potent anticonvulsant drug, which is able to penetrate the blood-brain barrier.⁶⁷⁷



(+)-5-Methyl-10,11-dihydro-5H-dibenzo[a,d]cyclohepten-5,10-imine

Metabotropic glutamate receptors have been classified into eight types (mGluRs1–8).^{678–680a} Group I (mGluRs1–5) are selectively activated by 3,5-dihydroxyphenylglycine; Group II (mGluR2 and mGluR3) are activated by L-2-(carboxycyclopropyl)glycine; and Group III (mGluR4 and mGluR 6–8) are activated by L-2-aminophosphonobutyrate. They are all 7-helix G-protein-coupled receptors with external ligand-binding domains that resemble those of bacterial periplasmic binding proteins.⁶⁸⁰ Splice variants for at least mGluR1 are known.⁶⁷⁸ Metabotropic glutamate receptors are neuromodulatory but nevertheless play essential roles in the cerebellum and other parts of the brain. For example, mice deficient in the mGluR1 protein have severe problems with motor coordination and learning.^{681,682} Metabotropic glutamate receptors may participate in calcium sensing and signaling.^{683,684}

Synaptosomal particles have a high-affinity proton-

dependent uptake system for glutamate.⁶⁸⁵ Glutamate and aspartate may also be taken up from the synaptic cleft by neurons or by glial cells, which then transfer the glutamate into neurons for reuse.^{686,687} Five distinct mammalian transporter genes have been cloned.⁶⁸⁸ They are driven by concentration gradients of Na^+ and K^+ across the membrane.^{689,690} However, some serve as glutamate-gated chloride ion channels.^{691,691a}

Excitotoxicity. As essential as glutamate is for brain function it is toxic in excess. Excessive stimulation of the NMDA receptors, which occurs during convulsions, strokes, or traumatic injury and which can accompany anoxia or hypoglycemia, causes neuronal death.^{660,692–694} Blocking these receptors with the above-mentioned anticonvulsant drug or aminophosphonates has a remarkable protective effect against the neurotoxicity of the accumulating glutamate.^{658,677} Vitamin E and **tocotrienols** (Fig. 15-24) may also be protective.⁶⁹⁵

The inhibitory neurotransmitter gamma-aminobutyrate (GABA). Glutamate, aspartate, and cysteic acid are all potent exciters, but their decarboxylation products γ -aminobutyrate (**GABA**), β -alanine, and taurine are inhibitors as is also glycine. Of these GABA is the most important.⁶⁹⁶ Its concentration in the brain is high and varies at least threefold in different parts of the brain. It is hardly present elsewhere in the body. GABA and GABA-binding sites are found in 30–50% of the nerve endings. The function as an inhibitory transmitter has also been demonstrated in inhibitory neurons present in the peripheral nervous system of arthropods. Virtually every neuron in the brain is to some extent subject to inhibition by GABA.^{697,698} Glial cells also have GABA receptors.

The receptors for GABA are divided into type A, which are blocked by **bicuculline**,⁶⁹⁹ and type B, which are stimulated by **baclofen** (Fig. 30-25).⁶⁹⁸ The GABA_A receptors are the major sites of fast synaptic inhibition in the central nervous system.⁷⁰⁰ They are structurally related to the nicotinic acetylcholine, glycine, and serotonin type 3 (5-HT₃) receptors. Cloning has revealed 16 different mammalian subunits: $\alpha 1$ – $\alpha 4$, $\beta 1$ – $\beta 3$, $\gamma 1$ – $\gamma 3$, δ , ϵ , π , and Φ .^{701–704a} The oligomeric receptors are ligand-gated chloride ion channels^{481,705} as are also glycine receptors. These receptors are clustered in synaptic membranes, apparently anchored in part by their β subunits⁷⁰⁶ and scaffold proteins such as the microtubule-binding **gephyrin** (from the Greek word for bridge)^{701,707} and a small ~14-kDa GABA receptor-associated protein.⁷⁰⁸ A novel serine protein kinase is also associated with GABA receptors.⁷⁰³

Whereas excitatory transmitters lead to depolarization of the postsynaptic membrane, inhibitory transmitters cause **hyperpolarization**, apparently by increasing the conductance of K^+ and Cl^- . The result is

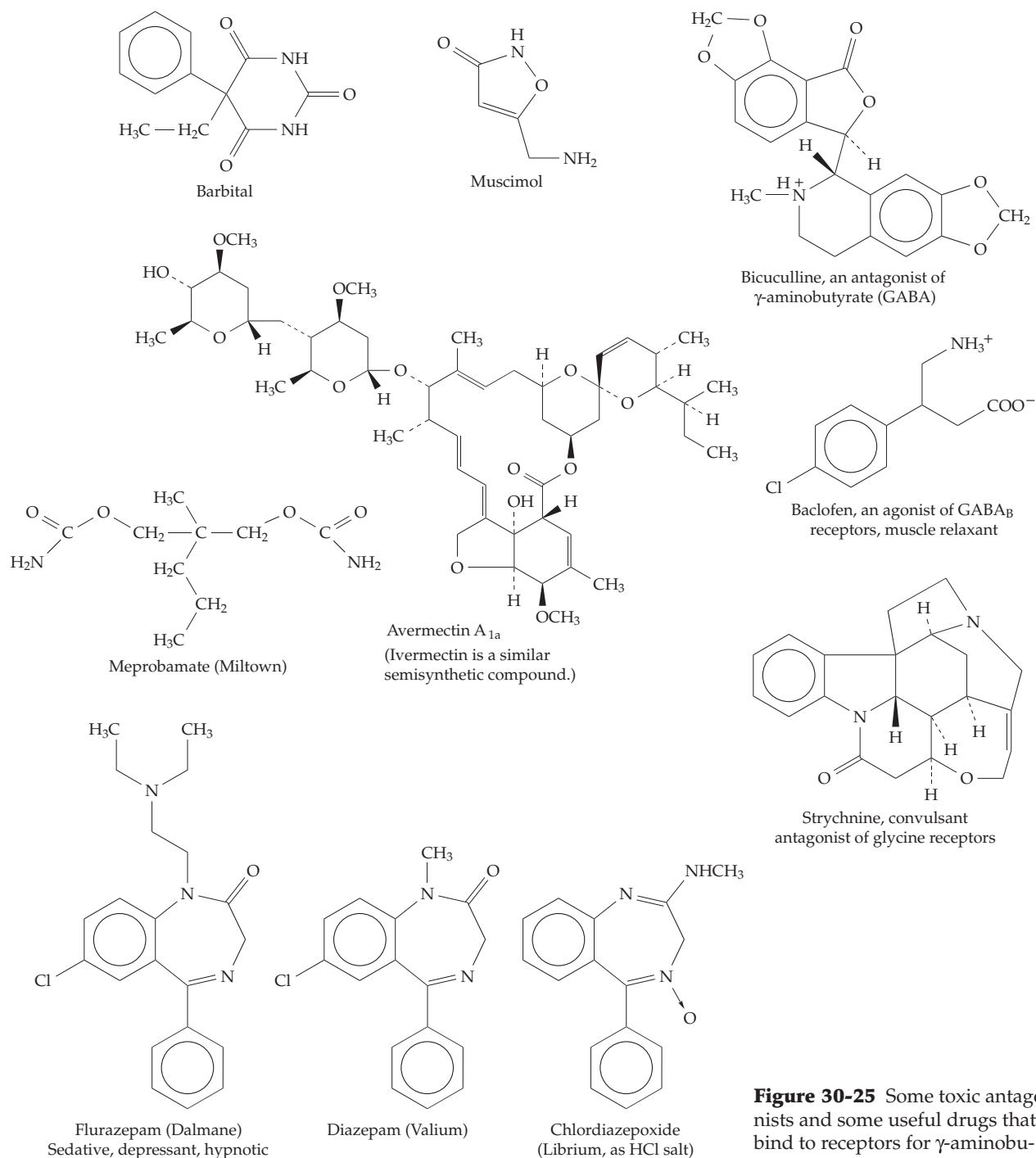


Figure 30-25 Some toxic antagonists and some useful drugs that bind to receptors for γ -aminobutyrate (GABA) or for glycine.

that it is more difficult to excite the postsynaptic membrane in the presence of, than in the absence of, these transmitters. GABA-dependent interneurons also contain the calcium-binding **parvalbumin** (Fig. 6-7), which suggests that a Ca^{2+} -dependent process is involved.⁷⁰⁹

The GABA_B receptors resemble metabotropic glutamate receptors.^{710,711} They are 7-helix G-protein coupled proteins, which activate adenylate cyclase.

They tend to dimerize, and maximum activity is observed for heterodimers of GABA_B1 and GABA_B2 receptors.^{712,713} They are often coupled to inward rectifying K^+ channels.⁷¹⁴

The GABA receptors provide binding sites for a great variety of toxins and drugs.⁴⁸¹ These include barbiturates, anesthetics, antianxiety drugs, and the insecticides such as toxaphene, cyclodienes, and pyrethroids.⁴⁸¹ **Diazepam, chlordiazepoxide, and**

flurazepam^{700,702,715–717} (Fig. 30-25) are antianxiety drugs and muscle relaxants, which, during the 1970s, were the most frequently prescribed drugs in the United States.⁷¹⁶ Binding of benzodiazepines to GABA receptor-chloride channels enhances the effect of GABA. The drugs induce relaxation but can interfere with memory, reduce concentration, and cause physical clumsiness. They may also intensify the effects of alcohol and can be addictive.⁷¹⁸

Specific antagonists for GABA_A receptors include the alkaloid convulsants bicuculline (Fig. 30-25)⁶⁹⁹ and **picROTOXIN** (Fig. 22-4) and the convulsant terpenoid compound **thujone** (Fig. 22-3), which is present in the wormwood plant *Artemisia absinthium*. Thujone is present in the liqueur absinthe, which was the national drink of France in the late 19th century but, because of its toxicity, has been illegal in most countries since ~1915.⁷¹⁹

GABA enters synaptic vesicles via a vesicular GABA transporter, an integral membrane protein whose gene has been found in *Caenorhabditis elegans*.⁷²⁰ Termination of GABA neurotransmission is accomplished by rapid Na⁺-dependent uptake into neurons for reuse and uptake into glial cells.^{721,722} Excess GABA is continuously oxidized to succinic semialdehyde by GABA aminotransferase⁷²³ in the GABA cycle of Fig. 17-4. Notice the manner in which this cycle incorporates synthesis of both of the neurotransmitters glutamate and GABA. Glutamine also functions in neurons, perhaps serving as a buffer for glutamate.

The hereditary triple-repeat disease Huntington's chorea (**Huntington disease**), with an incidence of 5–10 per 100,000 persons, affects principally persons of age over 40 and is associated with a deficiency of GABA in basal ganglia.⁷²⁴ The cortex is also affected. Severe neurologic symptoms arise as a result of premature death of neurons in the basal ganglia. Convulsions may also arise because of a deficiency of GABA in the brain.

Glycine. Glycine appears to be the most important neuroinhibitor in the spinal cord and brainstem. It is present at concentration of 3–5 mM in the spinal cord and in the medulla but is low in the cerebral cortex. **Strychnine** (Fig. 30-25) is a specific antagonist of glycine receptors in spinal synapses.⁷²⁵ Ivermectin (Fig. 30-25) also blocks glycine Cl[−] channels.⁷²⁶ A mutant mouse called *spastic* is deficient in glycine receptor function. A small dose of strychnine produces an effect on a normal mouse that resembles the effect of this mutation.^{727,728} A similar disorder affects some Hereford calves.⁷²⁹ Strychnine-binding studies have suggested a deficit of glycine receptors in human spasticity and in the loss of motor control associated with **Parkinson disease** and **amyotrophic lateral sclerosis**.⁷²⁵ A human **startle disease**, which causes an exaggerated muscular response to unexpected

stimuli, also results from reduced glycinergic neurotransmission.⁷³⁰

Most glycine receptors are Cl[−] ion channels that open in response to transmitter binding.⁷²⁵ The strychnine-binding subunit shows significant homology with the nAChR proteins,⁷²⁵ and the overall structures resemble those of GABA receptors and of nAChRs.^{731,732,732a} Human $\alpha 1$ – $\alpha 4$ and β subunits have been identified.^{733,734} Two integral membrane glycine transporters are known.^{735–737}

Anesthetics. Several types of neurotransmitter receptors provide binding sites for anesthetics. Some anesthetics are molecules of moderate size, e.g., **barbiturate** derivatives, while others, such as **diethyl ether** or **halothane** (CF₃CHClBr), are very small. The latter is one of the most widely used inhalation anesthetics. Both Mg²⁺ and Mn²⁺ are also powerful CNS depressants and can cause general anesthesia. It has often been proposed that the effectiveness of anesthetics is related to solubility in lipids, but it has been difficult to pinpoint a site of action. Now it is clear that specific synaptic proteins often provide the binding sites for anesthetics. Important among these are the glycine receptors.^{715,738,739} GABA receptors^{740,740a} and kainate glutamate receptors may also bind anesthetics.⁷⁴¹

Adrenergic synapses: the catecholamines. The three closely related tyrosine metabolites, **dopamine**, **noradrenaline**, and **adrenaline**, known collectively as catecholamines, are important products of neuronal metabolism.^{149,393} Dopamine and noradrenaline serve as neurotransmitters. Catecholamine-containing neurons are found throughout the brain, including the cortex and cerebellum regions. Very large dopamine-containing neurons are present in the brains of gastropod molluscs.⁷⁴² In the human brain a prominent series of dopamine neurons run from the substantia nigra to the caudate nuclei and putamen of the striatum, the **nigrostriatal** pathway (Fig. 30-12).^{149,743,743a} In many invertebrates **octopamine**,^{744–746} which is synthesized via tyramine (Fig. 30-26), apparently functions in place of noradrenaline. Note the precursor–product relationship between dopamine, noradrenaline, and adrenaline. The synthetic pathways to these neurotransmitters involve decarboxylation and hydroxylation, types of reaction important in formation of other transmitters as well. The most important process for terminating the action of released catecholamine transmitters is reuptake by the neurons. High-affinity uptake systems transport the catecholamine molecules back into the neurons and then into the synaptic vesicles. The uptake is specifically blocked by the drug **reserpine** (Fig. 25-12).^{746a} The dopamine transporter is a major binding site for cocaine (see Fig. 30-28).^{747–751} Catecholamine transmitters are catabolized by two enzymes. One is the

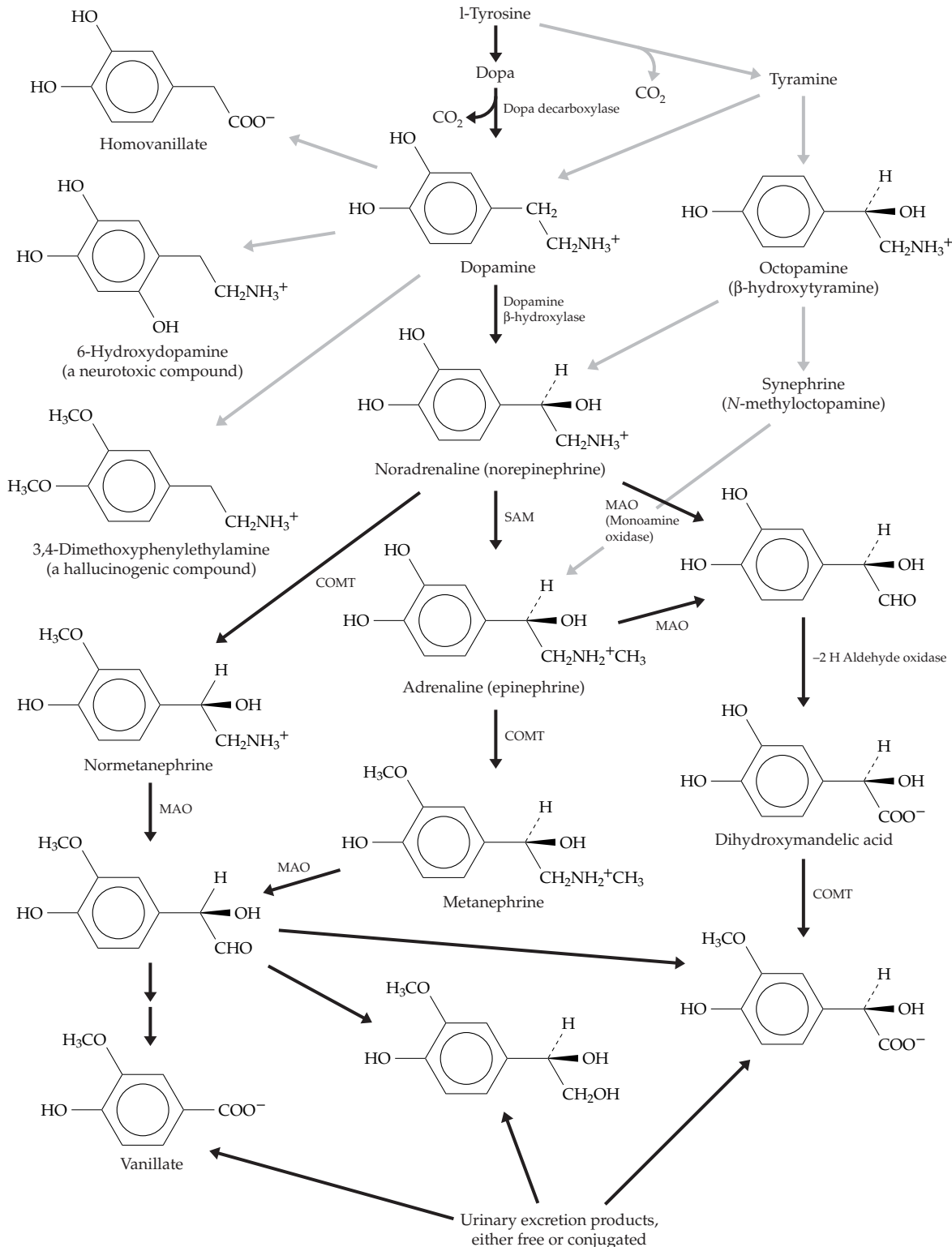


Figure 30-26 Some pathways of metabolism of the catecholamines. See also Fig. 25-5.

flavoprotein **monoamine oxidase (MAO;** Chapter 16), an enzyme present within the mitochondria of neurons as well as in other cells in all parts of the body.⁷⁵² The second enzyme is **catechol-O-methyltransferase**

(**COMT**; Eq. 12-3), which is found in postsynaptic membranes as well as in liver, kidney, and other tissues. It apparently provides the principal means of inactivating circulating catecholamines. In a process

that occurs in all organisms sulfo groups are transferred from PAPS (Eq. 17-38) onto hydroxyl groups of catecholamines, steroid compounds, and proteins (p. 659). Sulfation of catecholamines is relatively specific to humans.^{752a,b}

Both adrenaline and noradrenaline stimulate smooth muscles throughout the body and have a hypertensive effect. Their postsynaptic receptors are 7-helix transmembrane proteins (Fig. 11-6). A comparison of the effects of various analogs led to the classification of these receptors into classes α , α_2 , β , and β_2 , which are discussed briefly on pp. 553–555. The α receptors, which are structurally closely related to rhodopsin,^{753,754} are coupled via Gq / 11 proteins to a phosphoinositide-activated phospholipase C (Figs. 11-9, 30-19).⁷⁵⁵ They usually provoke an excitatory response. However, in intestinal smooth muscles they are inhibitory. Adrenaline is usually more active at α receptors than is noradrenaline. A specific antagonist

is **dibenzamine** (Fig. 30-27). The β receptors usually induce muscular relaxation but cause myocardial stimulation. Noradrenaline is usually more active than adrenaline. In most cases the β receptors of the postsynaptic membrane respond to the neurotransmitter by causing a hyperpolarization of the cell membrane and inhibition of nerve impulses. A specific antagonist is **propranolol** (Fig. 30-27). The β receptors are coupled via proteins of the G_s family (pp. 557, 558). The β_2 receptors have received special attention because of their importance to heart and pulmonary functions. Both heart failure and asthma are associated with poor β_2 receptor function.^{756,757} The β_2 receptors affect many other processes including insulin action.⁷⁵⁸ Intense efforts are being made to understand them at the structural level.^{757,759,760} Of special interest are the mechanisms by which receptors are desensitized after passage of impulses, a process that often involves multiple phosphorylation reactions⁷⁶¹

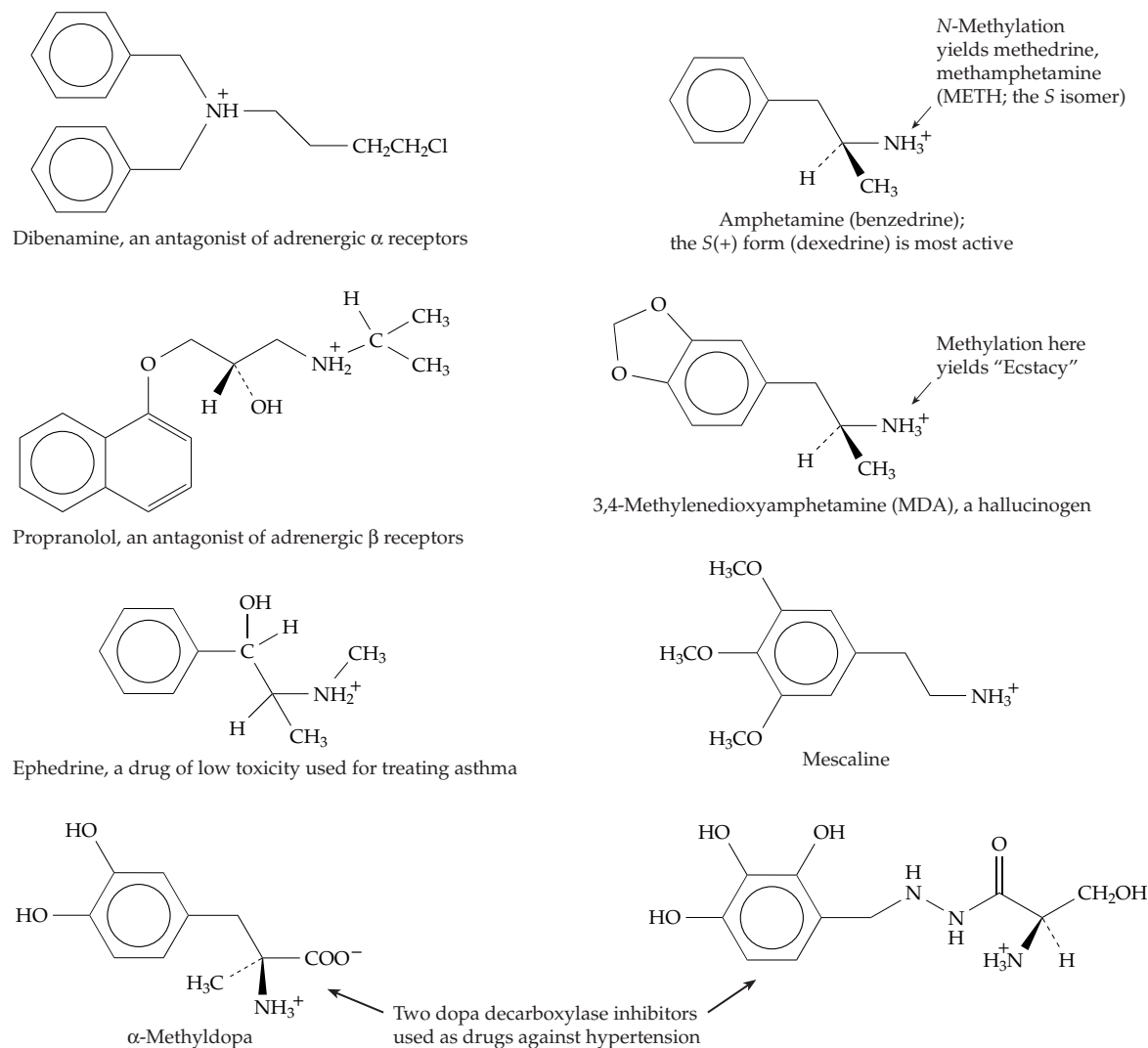


Figure 30-27 Some agonists and antagonists of adrenergic synapses (shown as cations in most cases).

as well as interaction with **arrestin** (Fig. 23-43) and receptor internalization.⁷⁶²

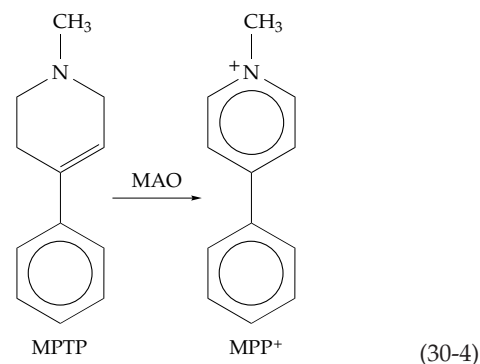
Attention has been focused on dopamine because of its relationship to neurological diseases and to addiction (discussed in Section 10). Dopamine receptors constitute a large family, which are classified into two main subfamilies. The D₁ subfamily consists of D_{1A} and D_{1B} (D₅) receptors and the D₂ subfamily of D₂, D₃, and D₄ receptors.^{763,764} The D₁ receptors, which are prominent in the prefrontal cortex and also in the striatum, are more abundant than the D₂ receptors, which are also present in the striatum and the pituitary and are targets for antipsychotic drugs such as **haloperidol** (Fig. 30-33).⁷⁶⁵ The recently discovered and less numerous D₃ receptors are present in only a few regions of the brain. However, a deficiency of D₃ receptors may also be involved in addiction, schizophrenia, and Parkinson disease.^{766,767}

The role of the catecholamines as transmitters in the sympathetic nervous system and in the peripheral ganglia has been well established, but the function in the central nervous system is less clear. Catecholamines are present in varying quantities throughout the brain, and histochemical techniques^{149,768} have made it possible to visualize both dopamine and noradrenaline-containing neurons by the green fluorescence produced from reaction with formaldehyde or glyoxylate.⁷⁶⁹ The reactions are presumably analogous to those in Fig. 25-10. Another method for tracing dopamine receptors in the central nervous system is through labeling with specific antibodies to dopamine-β-hydroxylase (Eq. 18-53), the enzyme that converts dopamine to noradrenaline, to tyrosine hydroxylase, or to other specific neuronal enzymes.⁷⁷⁰

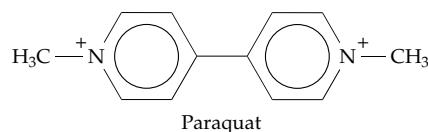
Parkinson disease. Neurons of the nigrostriatal pathway degenerate in Parkinson disease, a condition accompanied by severe tremors and rigidity. The significance of dopamine was illustrated by the finding that the precursor amino acid **L-dopa** caused dramatic improvement in many persons with Parkinson disease.⁷⁷¹ Dopamine and other catecholamines do not cross the blood–brain barrier but L-dopa does. This leads to an increase in the dopamine level in the basal ganglia of the brain, which apparently compensates for the deficiency resulting from the neuronal degeneration.

In 1982 a number of young people in California injected themselves with an illegally manufactured opiate drug that was subsequently found contaminated with *N*-methyl-4-phenyltetrahydropyridine (MPTP). Within a few days they developed irreversible symptoms of Parkinson disease. Subsequent investigation revealed that MPTP itself is not toxic but that it is oxidized by monoamine oxidase B (MAO-B) to the corresponding pyridinium derivative MPP⁺ (Eq. 30-4). It is this pyridinium derivative, or perhaps

related free radicals, that is toxic.⁷⁷² MPP⁺ is readily taken up by mitochondria and is apparently concentrated in the mitochondria of the nigrostriatal cells to a toxic level.⁷⁷³ The MAO inhibitor pargyline (Fig. 30-33) interferes with the oxidation of Eq. 30-4 and prevents development of Parkinson disease in squirrel monkeys exposed to MPTP.⁷⁷⁴ These results suggested



possible environmental causes for Parkinson disease and also a new approach to treatment.^{775,776} MPP⁺ has been marketed as a herbicide, and it has a close structural relationship to another herbicide, **paraquat**.



Many food constituents including peppermint, spearmint, and tea contain 4-phenylpyridine, another close relative.⁷⁷⁵ While administration of L-dopa to replace the deficit in the basal ganglia seemed the ideal treatment for Parkinson disease, mental deterioration is not stopped, and for some patients the drug loses its effectiveness in about three years. Based on the new information about MPTP, treatment with extra vitamin E as an antioxidant along with an MAO inhibitor is being tested as a way to prevent further damage from environmental toxins.⁷⁷⁶

Serotonin and melatonin. The indolealkyl amine serotonin (5-hydroxytryptamine, 5-HT; Fig. 30-28), is found in all mammalian brains and in invertebrates as well. Its distribution in the brain is limited, serotonin-containing neurons being found in the raphe nuclei of the brainstem from which they ascend into the brain and down the spinal cord. Serotonin-containing neurons have been traced within brains of snails using ³H-labeled serotonin.⁷⁷⁷ Studies with these simpler brains have revealed both inhibitory and excitatory responses to these neurons. Serotonin-accumulating neurons are also found in the retina⁷⁷⁸ and are widely distributed in the peripheral nervous

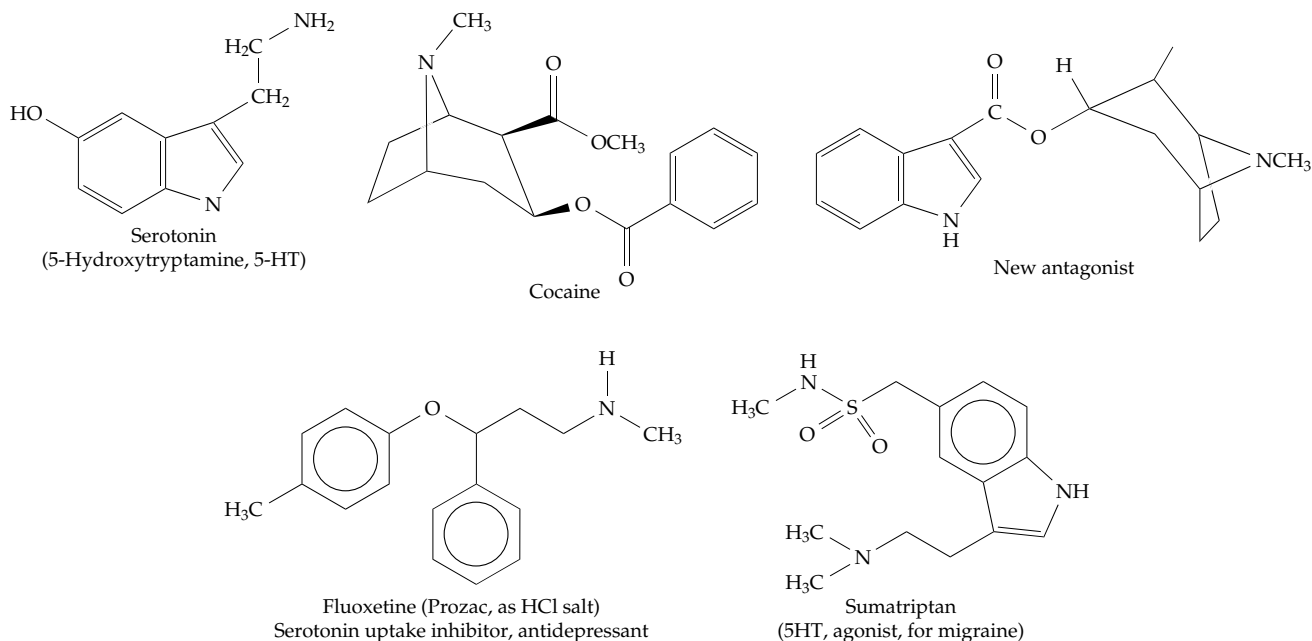


Figure 30-28 Serotonin (5-hydroxytryptamine) and some drugs that affect receptors and transporters.

system.⁷⁷⁹ Serotonin-containing granules are present in blood platelets.⁷⁸⁰

Serotonin appears to be involved in activation of pain fibers, when tissues are injured. **Cocaine** (Fig. 30-28) is a powerful pain killer and a weak antagonist of responses to serotonin, a fact that has led to the synthesis of new antagonists, such as the one in Fig. 30-28 whose structure encompasses that of both cocaine and serotonin.^{779,781} It is active at a concentration as low as 10^{-14} M and is among the most potent known drugs of any type.

Serotonin is synthesized via tryptophan and 5-hydroxytryptophan with decarboxylation of the latter (Fig. 25-12). Within the **pineal body** of the brain and in the retina, serotonin is acetylated to *N*-acetylserotonin,^{782,783} which is then *O*-methylated to **melatonin**, the pineal hormone (Fig. 25-12). A specific inhibitor of serotonin synthesis is *p*-chlorophenylalanine, and studies with this and other inhibitors suggest that serotonin is required for sleep.⁷⁸⁴

At least 14 distinct types of serotonin receptors (5-HT_{1A}, 5-HT_{1B}, 5-HT_{2A}, etc.) have been identified.^{785,786} They are present in the heart, in gastrointestinal tissues, adrenal and other glands,⁷⁸⁷ and bone⁷⁸⁸ as well as in the brain. Drugs, such as **sumatriptan** (Fig. 30-28), which activate serotonin receptors are important in the treatment of **migraine**. This common disorder of serotonin metabolism is characterized by severe or moderately severe headache and a variety of other symptoms, which are frequently preceded by a visual aura.⁷⁸⁹ Serotonin is removed from synapses via a

transporter, which also contains the binding site of the widely used antidepressant **Prozac** (Fig. 30-28) and related drugs.^{790-791b}

Serotonin and melatonin are evidently involved in maintenance of the 24-h circadian rhythm of the body (see Section 13).^{792,792a} Melatonin regulates the sexual cycle in photoperiodic animals and influences the onset of puberty.⁷⁹²⁻⁷⁹⁴ The serotonin content of the brain is influenced by the diet, being higher after a meal rich in carbohydrates. Serotonin may serve as a chemical message sent from one set of neurons to the rest of the brain, reporting on the nature of dietary intake.⁷⁸⁴ Melatonin, which can readily form free radicals, may function as part of the body's antioxidant system.^{795,796}

Other neurotransmitters. The abundant glutamate, GABA, and glycine are major neurotransmitters. Do other amino acids also function in the brain? Roles for *L*-aspartate and *D*-serine (p. 1785) have been identified, but it is very difficult either to discover or to disprove a neurotransmitter function for other amino acids. It is even more difficult for small amounts of various amines and small peptides that are present in the brain. **Taurine** (Fig. 24-25) is one of the most abundant free amino acids in animals and meets several criteria for consideration as both an inhibitory and an excitatory transmitter.^{797,798} However, its function is still uncertain (see Chapter 24). **Homocysteic acid**, formed by oxidation of homocysteine, is a powerful neuroexcitatory substance, but its concentration in the brain is very low.¹⁴⁹ *D*-Aspartate is also present

at high concentrations in the cerebellum, pituitary, pineal gland, and adrenal chromaffin cells. It appears to be a modulator of melatonin synthesis.^{799,800}

Receptors for **histamine**, which probably acts as a neuromodulator,⁸⁰¹ occur in the brain.⁸⁰² Histamine is formed by decarboxylation of histidine (p. 745)⁸⁰³ and is inactivated by histidine *N*-methyltransferase. Histamine is best known for its presence in mast cells,⁸⁰⁴ components of the immune system that release histamine during inflammatory and allergic reactions (Chapter 31). However, histaminergic neurons of the hypothalamus extend throughout the whole forebrain,⁸⁰⁵ and specific receptors have been found both in the brain and in peripheral tissues.⁸⁰⁶ Several other amines that are formed by decarboxylation of amino acids are present in trace amounts but may have im-

portant functions, some of which may be related to psychiatric disorders. These include tyramine (from tyrosine), β -phenylethylamine (from phenylalanine), and tryptamine (from tryptophan). As previously mentioned, octopamine is also present in trace amounts in mammalian brains.⁸⁰⁷

ATP, ADP, and adenosine are among the purines that are present in some synapses and activate a variety of receptors. Adenosine receptors are blocked specifically by methylated xanthines such as caffeine (Fig. 25-18) and theophylline.^{808–808b} A drug almost 10^5 times as potent as theophylline is 1,3-dipropyl-8-(2-amino-4-chlorophenyl)xanthine.⁸⁰⁹ Adenosine receptors, which are present in large numbers in the hippocampus,¹⁴⁹ form functional complexes with metabotropic glutamate receptors.⁶⁷⁸ Adenosine

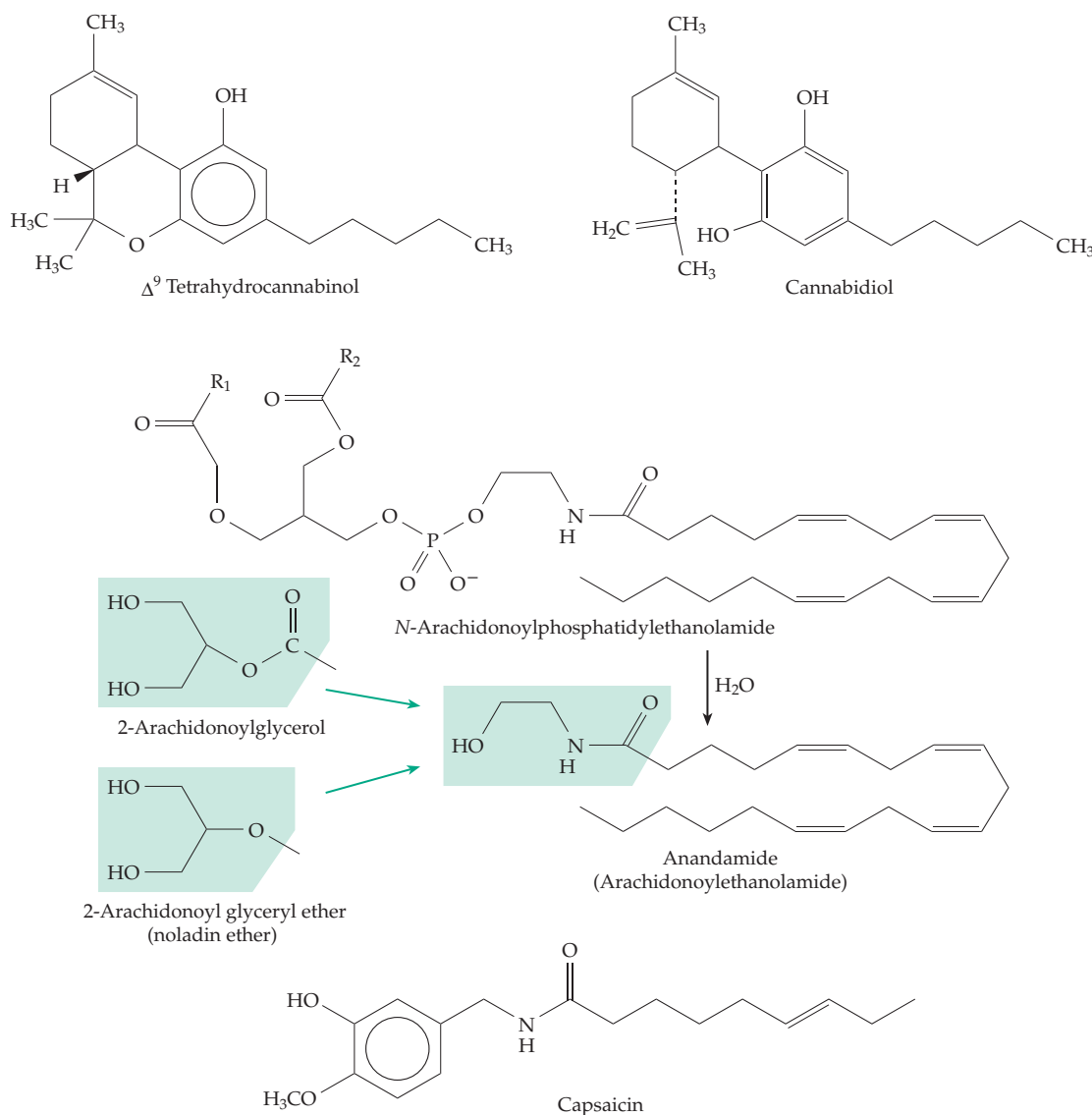


Figure 30-29 Structures of the active components of cannabis, tetrahydrocannabinol, and cannabidiol, and structures of endogenous cannabinoids and of the vanilloid lipid capsaicin.

usually has a depressive effect. Craving for chocolate is often attributed to the methylxanthines present, but it may be a result of anandamide and related compounds.⁸¹⁰

The occurrence of a variety of **neuropeptides** in the brain has been discussed in Section A. The first of these to be discovered⁸¹¹ was the 11-residue **substance P** (Table 30-4), which was isolated in 1931. Like other neuropeptides it may function either as a transmitter or neuromodulator or perhaps both. Substance P, as well as many other neuropeptides, has been localized to specific neurons. Along with somatostatin, CCK, and enkephalins, it is found in high concentrations in the basal ganglia. Enkephalin and substance P are also found in specific neural elements in the visual system of lobsters.⁸¹² In some cases a neuron contains both synaptic vesicles containing a major neurotransmitter and also vesicles containing a peptide or other cotransmitter. The peptide pituitary hormones ACTH, MSH, and vasopressin as well as the hypothalamic neurohormones may have effects on learning and behavior.⁸¹³

Lipid mediators in the brain. The brain is rich in phospholipids, glycolipids, and long-chain unsaturated fatty acids. Many signaling functions seem likely, and some are discussed in Section A,7. Prostaglandin D₂ is a major prostanoid in the brain, which induces both hypothermia and sleep.⁸¹⁴ As mentioned in Section A,7, **oleamide** also induces sleep, perhaps by modulating the effects of 5-HT receptors.^{815,816} **Anandamide** is a lipid derived by hydrolysis of the unusual phospholipid *N*-arachidonoylphosphatidylethanolamide. This is one of a recently discovered series of amides, esters, and ethers derived from arachidonic acid (Fig. 30-29).^{816a,b} They have been identified as endogenous ligands of the abundant **cannabinoid receptors**.⁸¹⁶ The latter were identified as binding sites of Δ^9 -tetrahydrocannabinol and cannabidiol (Fig. 30-29), both of which are constituents of **marijuana**. Anandamide was the first of the endogenous cannabinoids to be isolated.⁸¹⁷ However, the monoglyceride **2-arachidonoylglycerol** (Fig. 30-29) is much more abundant in brain and also activates cannabinoid receptors.⁸¹⁸ It arises by hydrolysis of a diglyceride.^{819,820} Recently 2-arachidonoyl glyceryl ether (**noladin ether**; Fig. 30-29) has been identified as another endogenous agonist of the CB₁ cannabinoid receptors.⁸²¹ A possible alternative pathway for anandamide synthesis is via an energy-dependent coupling of arachidonic acid with ethanolamine.^{822,823} The two known types of cannabinoid receptors are both 7-helix proteins coupled by G_i or G_o proteins to adenylate cyclase and to Ca²⁺ and K⁺ channels.^{824,825} The CB₁ receptors are found largely in the brain and are responsible for the psychoactive effects of cannabis, while the CB₂ receptors are more widely distributed. They seem to have a special role in cells of the immune

system, e.g., in macrophages and B cells.^{818,820,825–828} Palmitoylethanolamide has been proposed as an additional endogenous ligand for CB₂ receptors.^{820,829} Cannabinoid receptors of invertebrate immune system cells and of human monocytes have been found coupled to NO release.⁸³⁰

Cannabinoid receptors are present at extremely high levels in the basal ganglia of the brain,^{831,832} but they do not appear to be essential. Knockout mice lacking the CB₁ receptors appear normal in most respects. However, they do not respond to cannabinoid drugs and, curiously, do not become addicted to morphine as normal mice and have less severe withdrawal symptoms than normal after morphine addiction.⁸²⁶ The CB₁ receptors in the basal ganglia modulate GABA neurons that have outputs to the substantia nigra and the globus pallidus (Fig. 30-30B). The nigrostriatal neurons also secrete substance P and dynorphin, while those extending to the globus pallidus generally contain enkephalin as a cotransmitter.⁸³² These interconnections affect the dopaminergic neurons. Cannabinoids also have pain suppressing and neuro-protective effects. They may have many possible medicinal uses, which are being explored.^{833–837}

The endogenous cannabinoid compounds are lipids and are not stored in synaptic vesicles but are presumably released by enzymatic action following passage of a nerve impulse. Recent evidence suggests that the endocannabinoids are released at a postsynaptic membrane and then diffuse back to a presynaptic surface and outward to other cell surfaces where they affect signaling.^{838–840} This **retrograde signaling** in synapses of the hippocampus is thought to be involved in **long-term potentiation (LTP)**, the changes in synaptic properties that occur during learning and in the formation of memories (Section 12). A monoglyceride lipase participates in inactivation of endocannabinoids.^{840a} Anandamide is also a substrate for cyclooxygenase-2 (Eq. 21-16), whose action may lead to formation of additional immunomodulatory compounds.^{841,842} Long-chain relatives of arachidonic acid such as docosahexaenoic acid (DHA; Box 21-B) are especially high in brain lipids.^{843,843a}

Nitric oxide and carbon monoxide. The gaseous molecules NO and CO have both been found in the brain, and neuronal NO synthase (nNOS or NOS I) has been studied intensively.^{844–847} NO synthases and the functions of NO and CO are discussed in Section A7 and in Chapter 18. Complexity in understanding the role of NO in the brain arises from the fact that different isoenzyme forms of NO synthase occur in three different types of cell: nNOS in neurons, iNOS from microglial immune system cells, and eNOS from endothelial cells of capillary blood vessels.⁸⁴⁶ All three types of cells are so tightly intermingled in the brain that it is hard to interpret observed experimental

effects. Elevated Ca^{2+} concentrations that can arise from stimulation of NMDA receptors in the hippocampus seem particularly effective in activating the calmodulin-dependent nNOS. This suggests that, like the endogenous cannabinoids, NO may be a retrograde messenger in LTP.¹⁴⁹ The possibility that CO may function in a similar way also remains uncertain, as does any pathway for metabolism of CO. Certainly NO and CO generated in the brain will have some effects that arise from their very tight binding to heme groups. An example is the observed inhibition of dopamine β -hydroxylase by N_2O_3 with a resulting decrease in noradrenaline synthesis.⁸⁴⁸

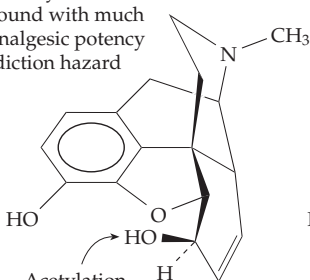
10. Some Addictive, Psychotropic, and Toxic Drugs

Humans have a long history of use of stimulant and mind-altering substances. Tea, coffee, alcohol, tobacco, opium, cocaine, marijuana, and a host of

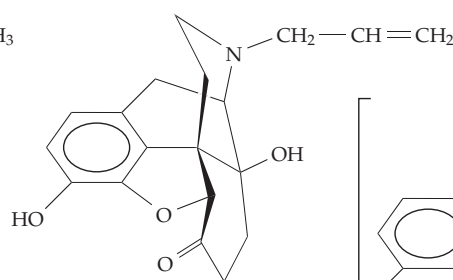
modern synthetic compounds have been used as stimulants, as medications, and for pleasurable experiences.^{849–851} Many are also addictive and sometimes lethal. Stimulant drugs such as nicotine, cocaine, methamphetamine (METH), and other amphetamines (Fig. 30-27)^{849,852–854b} can give users feelings of increased energy, well-being, and self-confidence. Nicotine enhances fast excitatory transmission⁸⁵⁵ and may sharpen memory.⁸⁵⁶ However, all are acutely toxic and are highly addictive. Amphetamines and cocaine act directly to increase the brain dopamine level causing euphoria. However, in response the dopamine receptors rapidly decrease their sensitivity. This leads to mental depression and the desire for more drug. Nicotine appears to indirectly affect the same dopamine neurons.⁸⁵⁷ The wisdom and ethics of giving hypoactive children the addictive stimulant **methylphenidate** (Ritalin; see Fig. 30-33) have been questioned.^{858,859} The depressive drugs, including **morphine** and other narcotics (Fig. 30-30), barbiturates (Fig. 30-22), and ethanol, are all strongly addic-

A

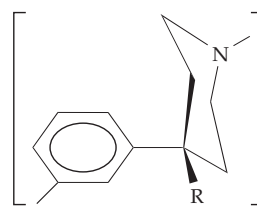
Methylation yields codeine a compound with much lower analgesic potency and addiction hazard



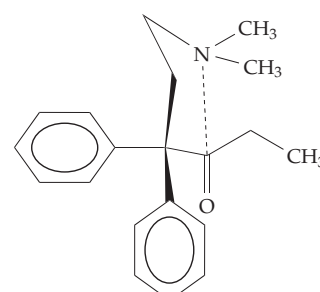
Acetylation yields heroin
Morphine (see also Fig. 25-10)



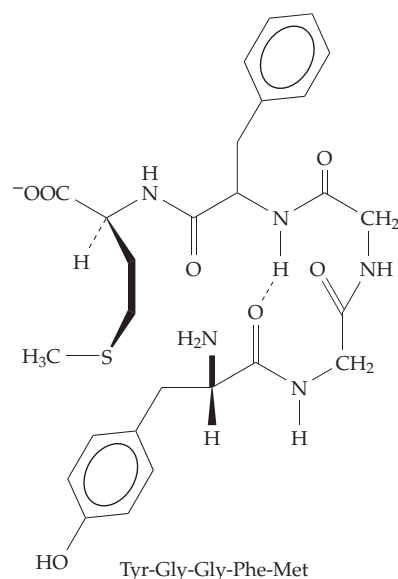
Naloxone



Structure common to many narcotic drugs



Methadone



B

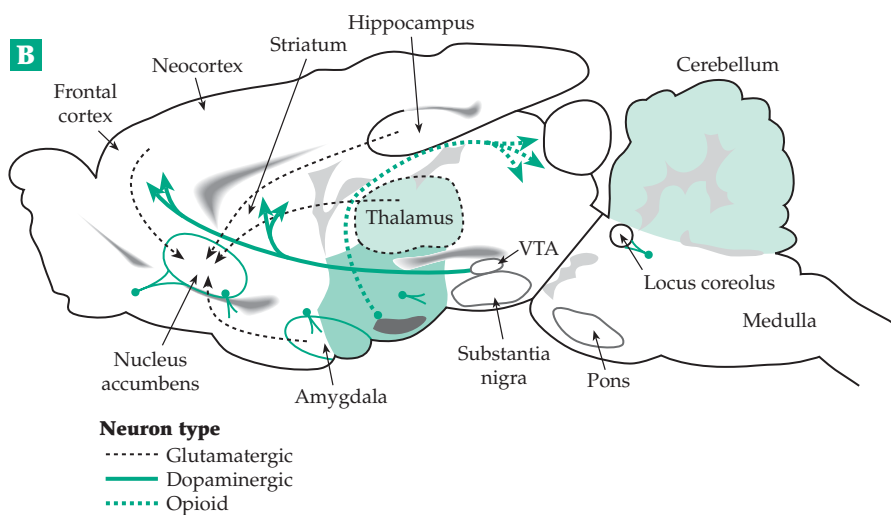


Figure 30-30 (A) The structures of morphine and of some analogs including the brain peptide Met-enkephalin. Also shown is a structure common to many narcotic drugs. (B) Diagram of a rat brain as shown in Fig. 30-13 with some aspects of the mesolimbic dopamine system emphasized. See Shulzeis and Koob.⁸⁶⁹

tive for susceptible individuals. The phenomenon is most striking in the case of the opiates. Addiction leads to physical dependence, a situation in which painful withdrawal symptoms occur in the absence of the drug. At the same time a striking tolerance to the drug is developed. The addicted individual can survive what would otherwise be a fatal dose without ill effect. Aside from the pathological hunger for the drug, an addict can function normally in almost every respect.⁸⁶⁰ Dependence develops only from frequent doses of drug over a long period of time and is not observed with cocaine or amphetamines.⁸⁶¹ Marijuana is only mildly addictive, according to some data about the same as caffeine.⁸⁶² However, this conclusion is controversial.

Opioid receptors. Direct binding of highly radioactive opiates has permitted localization of specific opiate receptors of several types.^{863–866} The three major types (μ , δ , κ) are all 7-helix receptors coupled to adenylate cyclase, K^+ and Ca^{2+} channels, and the MAP kinase cascade.⁸⁶⁶ The μ receptors bind morphine most tightly.^{867,867a} These receptors are found in various cortical and subcortical regions of the brain. Most narcotics are polycyclic in nature and share the grouping indicated in Fig. 30-30. However, the flexible molecule **methadone** binds to the same receptors.⁸⁶⁸ Among antagonists that block the euphoric effects of opiates the most effective is **naloxone** (Fig. 30-30).

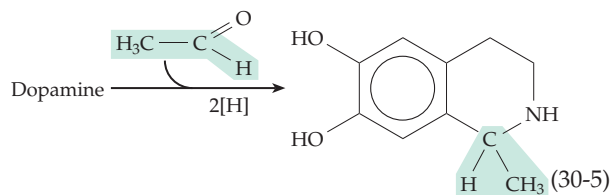
What is the natural function of opiate receptors? Opiates are the most powerful **analgesic agents** known. The existence of the **enkephalins**, **endorphins**, and **endomorphins** (Section A,5; Table 30-4) in the brain suggests that opiate drugs mimic the normal action of these peptides, which may function in controlling pain. Although opiates are powerful drugs, their efficiency in diminishing pain is directly related to their addiction potential. To date, it has not been possible to design a nonaddictive analgesic drug of the potency of morphine.

Addiction seems to follow compensatory changes in the receptor–agonist system that result from the occupation of the receptor sites by the drug. For example, studies of opiate receptors indicate that morphine acts in an inhibitory fashion, lowering the internal level of cAMP.^{861,870} The neuron then compensates by increasing the number or activity of adenylate cyclase molecules restoring the internal cAMP level. This leads to dependence upon morphine because in its absence the cAMP level rises too high. The increased number of adenylate cyclase molecules and associated receptors also accounts for the observed tolerance. It is now clear that this adaptation is complex. The properties of many synapses in various parts of the brain are altered by phosphorylation or dephosphorylation or other reactions of receptors and other synaptic proteins. Some changes are rapid, but

others are slower and involve alterations in transcriptional patterns within neurons. These changes occur in three different neuronal systems: (1) physical control systems, in which changes lead to physical dependence; (2) motivational control systems; and (3) associative memory systems.⁸⁶¹

The **mesolimbic dopamine system** is thought to be involved either directly or indirectly in addiction to many drugs. The dopaminergic neurons of this system have cell bodies in the **ventral tegmental area** (VTA) of the brain (Fig. 30-30B) and extend into the **nucleus accumbens**, a region at the base of the striatum that is thought to provide the “rewarding effects,” i.e., pleasure from drugs such as cocaine or amphetamines. There is direct experimental support for this conclusion.⁸⁷¹ Less certain is the proposal that opiates and other depressive drugs indirectly cause a similar effect in the nucleus accumbens.^{869,870,872} A more recent view is to regard addiction as an aberrant form of learning.^{861,871,873} This concept is applicable also to “behavioral addictions.”^{873a}

Ethanol. As with morphine addiction, tolerance to alcohol is developed, and a lack of ethanol produces withdrawal symptoms. The principal route of metabolism of ethanol (both ingested and the small amount of endogenous alcohol) is believed to be oxidation in the liver to the chemically reactive acetaldehyde (p. 774),^{874,875} which is further oxidized to acetate. Some theories of alcoholism assume that addiction, and possibly also the euphoric feeling experienced by some drinkers, results from a metabolite of ethanol in the brain. For example, acetaldehyde could form alkaloids (Eq. 30-5).⁸⁷⁶

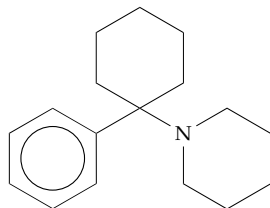


In fact, small amounts of morphine, 6-acetylmorphine, codeine, and thebaine, all opiate compounds, have been found in mammalian brain^{877,878} and have presumably arisen by the same pathway observed in plants (Fig. 25-10). However, there is no cross reactivity between morphine and alcohol in addicted mice,⁸⁷⁹ and acetaldehyde is probably not the addictive agent. Acetaldehyde is very reactive and may be responsible for much of the damage caused by ethanol.⁸⁸⁰ At a blood ethanol concentration of 20 mm a person is legally intoxicated, and large amounts of acetaldehyde may be formed and react with many amines, nucleotides, proteins, etc. Ethanol blocks glutamatergic NMDA receptors and

activates GABA receptors.^{740a} These effects may be involved in the neurodegeneration of fetal alcohol syndrome.^{881,882} Ecitotoxicity may also be a factor in alcohol damage.⁶⁹² Alcoholic liver disease may involve malnutrition as well as direct damage.⁸⁸³

Experiments with mice and rats have established a genetic propensity toward addiction to alcohol. Animals from some strains shun alcohol and become addicted only if force-fed for prolonged periods. Others, which may have low levels of neuropeptide Y in the brain,⁸⁸⁴ accept the alcohol readily and become addicted quickly. That a similar situation holds for humans is quite possible. However, a specific “alcoholism gene” has not been found.

Psychotropic or mind-changing drugs. Hallucinogenic compounds have long been a source of special fascination to many people. The presence of the indole ring in the powerful hallucinogen **lysergic acid diethylamide** (LSD; Fig. 25-12) suggests that this compound may mimic the action of serotonin. However, other experiments suggest antagonism of dopamine receptors in the striatum.⁸⁸⁵ Other hallucinogens include 3,4-methylenedioxyamphetamine (MDA; Fig. 30-27),⁸⁸⁶ a compound that damages serotonergic neurons, its derivative “ecstasy” (Fig. 30-27),^{886a} mescaline (Fig. 30-27), and phencyclidine (angel dust), a compound introduced in the late 1950s as a general anesthetic. Unfortunately, it produces a long-lasting condition resembling schizophrenia.⁸⁸⁷ A common site of action for a large variety of hallucinogens has been suggested.⁸⁸⁸



Phencyclidine (Angel dust)
a discredited anesthetic

Neurotoxins produced by the body. Some normal body constituents are neurotoxic in excess. These include **quinolinic acid** (Fig. 25-11),⁸⁸⁹ **3-hydroxykynurenine** (Fig. 25-11; p. 1444),⁸⁹⁰ and homocysteine.⁸⁹¹ Elevated levels of homocysteine are also associated with vascular disease and stroke (Chapter 24). 3-Hydroxykynurenine is a precursor to ommochrome pigments of insects and an intermediate in conversion of tryptophan into the nicotinamide ring of NAD in humans (Fig. 25-11). 6-Hydroxydopamine (Fig. 30-26), which may be formed in the body, is severely toxic to catecholaminergic neurons.⁸⁹²

Other neurotoxins can be formed from environmental pollutants. The solvent 1,4-butanediol is converted to **γ -hydroxybutyrate**, which is also a drug of

abuse.⁸⁹³ Many compounds in commercial use have not been adequately tested as neurotoxins.⁸⁹⁴

11. The Senses: Sight, Smell, Taste, Hearing, Touch, and Others

Our brains receive a continuous stream of impulses from receptors that sense light, taste and odor molecules, sound waves, touch, pain, gravitational pull, etc. Of these receptors those of vision, which are discussed in Chapter 23, may be the best known. The photoreceptors consist of rhodopsin and related 7-helix proteins embedded in membranes of the rod and cone cells (Fig. 23-40). A complex series of control mechanisms, some of which are outlined in Fig. 23-43, permit enormous amplification of the initial signal generated by a G protein and a cGMP-gated ion channel. The array of rods and cones in the retina send messages via the optic nerve to the **visual cortex**, an area of $\sim 15 \text{ cm}^2$ on the cerebral cortex surface at the back of the brain.^{149,895} The visual cortex is divided into two halves, but curiously, the right eye sends its signals to the left brain and vice versa. The image viewed by the retina can be mapped to the visual cortex. There it may reside in the form of chemical alterations in the $\sim 40,000$ neurons thought to be present in the visual cortex^{409,896–898} for a short time until it is stored in short-term working memory locations.

Receptors for the other senses, like those for sight, also consist of clusters, often in regular arrays, of 7-helix receptors. Most of these are also G protein-coupled ion channels that are controlled by cAMP or cGMP.⁸⁹⁹

Odor. Even bacteria possess something akin to our ability to taste and smell. As is discussed in Chapter 19, Section A, many bacteria are attracted to L-serine or D-ribose and are repelled by phenol. Receptor proteins in the plasma membrane are involved in sensing these compounds and in allowing bacteria to move toward food and away from danger. Many other examples of chemotaxis are known among the lower invertebrates such as *Euglena*. Chemoreceptors in *Hydra* sense glutathione that flows from the broken tissue of their prey and control the animal's feeding behavior. Related organisms respond to proline. Asparagine induces the bending of the tentacles of the sea anemone *Anthopleura*, while glutathione induces swallowing.⁹⁰⁰ Salmon return to their home streams using a memory of specific odors.⁹⁰¹

Throughout the animal kingdom the sense of smell is essential for survival. Perhaps it is not surprising that from the nematode *C. elegans* to human beings there is a largely conserved mechanism for sensing odors.⁹⁰² A large array of 7-helix G protein-coupled olfactory receptors embedded in an epithelial membrane carry signals directly into the nervous system.

In *C. elegans*, which has only 302 neurons, there are 32 chemosensory neurons and more than 100 genes for 7-helix receptors that are expressed in these neurons.^{903–905} The fruit fly *Drosophila melanogaster* has at least 59 genes for olfactory receptors.^{906,906a} Zebrafish and catfish have ~100.^{905,907} Mice and rats have ~1000 olfactory receptor genes and human beings at least 500, which account for about 1–4 % of the genome.^{905,908,909} In higher animals most receptors are coupled via G proteins, adenylate cyclase, and cAMP to ion channels in the membrane.^{905,908} Insects utilize both cAMP and Ins3-*P* in their chemosensory receptors.⁹⁰⁶ The signaling pathways parallel those of the visual receptors (Figs. 23-40, 23-43), which, however, utilize cGMP. Each gene is thought to give rise to a receptor of a specific **type** able to respond to specific structural features in an odor molecule.

Human olfactory cells are located in the **olfactory epithelium** on the upper surface of the back portion of the nasal cavity. They are neurons with chemosensory cilia similar to the rods and cones of the retina (Fig. 23-40). The cilia, which can be detached and isolated from the olfactory epithelium, contain the odorant-stimulated G-protein-dependent adenylate cyclase.^{910,911} There are ~10 million receptor cells of at least 500–1000 different types. The 10 million axons form bundles of ~5000 axons each and pass through small perforations in the skull directly into the **olfactory bulb** (at the front of the brain before the pituitary, Fig. 30-13), a distance of 3–4 cm. The cortex of the olfactory bulb is lined with ~1800 **glomeruli**. Each glomerulus is a bundle, ~0.1–0.2 mm in diameter, of synaptic endings of the neurosensory nerves coming from the olfactory epithelium with dendrites of neurons that run to the **olfactory cortex** and other regions of the brain.⁹⁰⁹ Each sensory receptor sends signals to a single glomerulus, but the glomerulus receives signals from 500 or more sensory neurons, which are not all of the same types. The glomerular cortex of the mouse is divided into four zones, each of which contains only some of the types of receptor. It seems that the cortex contains a crude “map” that relates position to the type of smell.⁹¹² The neural processing involved in the discrimination of odors is not yet clear.^{912,913} Interneurons of the olfactory bulb are unusual, being continuously discarded and replaced by new neurons that arise from neural stem cells.^{908,914} This process seems to be essential for odor discrimination but not for the sensitivity of odor detection.

Most mammals have a second olfactory apparatus, the **vomerinal organ** (VNO) or “sexual nose,” which is located on the lower surface of the nasal cavity. It is a fluid-filled cavity containing chemosensory receptors through which nasal fluid is literally pumped, when the animal seeks to maximize the sensitivity of detection.^{908,915} The VNO is especially

important to reproduction, defense, and food-seeking. A specialized set of olfactory sensory neurons that project to atypical glomeruli in the olfactory bulb utilize cGMP signaling and may also function in reproductive behavior.⁹¹⁶

The olfactory epithelia are bathed in an aqueous mucus through which odorant molecules must pass. A number of specialized proteins, including **odorant-binding proteins**, are secreted in this fluid.^{917–919} Many odorant-binding proteins are **lipocalins** (Box 21-A) and presumably assist in transporting lipophilic odorant molecules to the olfactory receptors. They tend to have a low specificity for the odorant and a weak binding affinity, properties that are consistent with this function. Pheromone-binding lipocalins encoded by ~30 genes are also found in rodent urine,⁹²⁰ where they play a similar role. In contrast, the pheromone-binding proteins of some male moths are largely α helical.^{920a} Although pheromones are not as important to human physiology, axillary odors from both males and females do apparently carry chemical signals. One well-established effect is the synchronization of menstrual cycles of women living in the same house or dormitory. Alipoprotein D apparently serves as a binding protein that carries odorant precursors that are acted on by bacteria to produce the pheromones.⁹²¹

Virtually all people lack the ability to detect some specific odors. A striking example of such an **anosmia** is the inability to smell the volatile steroid **androstenone** (5 α -androst-16-en-3-one), a constituent of perspiration, of some pork products, truffles, and celery.⁹²²

Taste. Less is known about the biochemistry of taste. The taste that we perceive is affected by odor, temperature, and physical contact. However, five primary tastes are recognized.^{923,923a}

Salty: apparently perceived by an ion-channel-linked receptor

Sour: apparently linked to an H⁺ channel

Bitter: perceived by bitter-sweet G protein-coupled receptors

Sweet: also perceived by bitter-sweet receptors

Umami: a recently recognized taste, that of glutamate

An experimental difficulty lies in the fact that there are only a few thousand taste buds in the tongue, with only 50–100 cells in a bud. They age rapidly, having a lifespan of only about ten days.⁹²⁴ There may be only 30,000–50,000 hard-to-isolate taste receptor cells on the tongue's surface.⁹²³ However, very recently published reports describe a large family of bitter and sweet receptors in mice and humans^{924–928} and in *Drosophila*.^{929,930} The sweet-sour receptors are thought to activate a G protein called **gustducin**,^{931,932} which plays a role similar to that of transducin in vision and

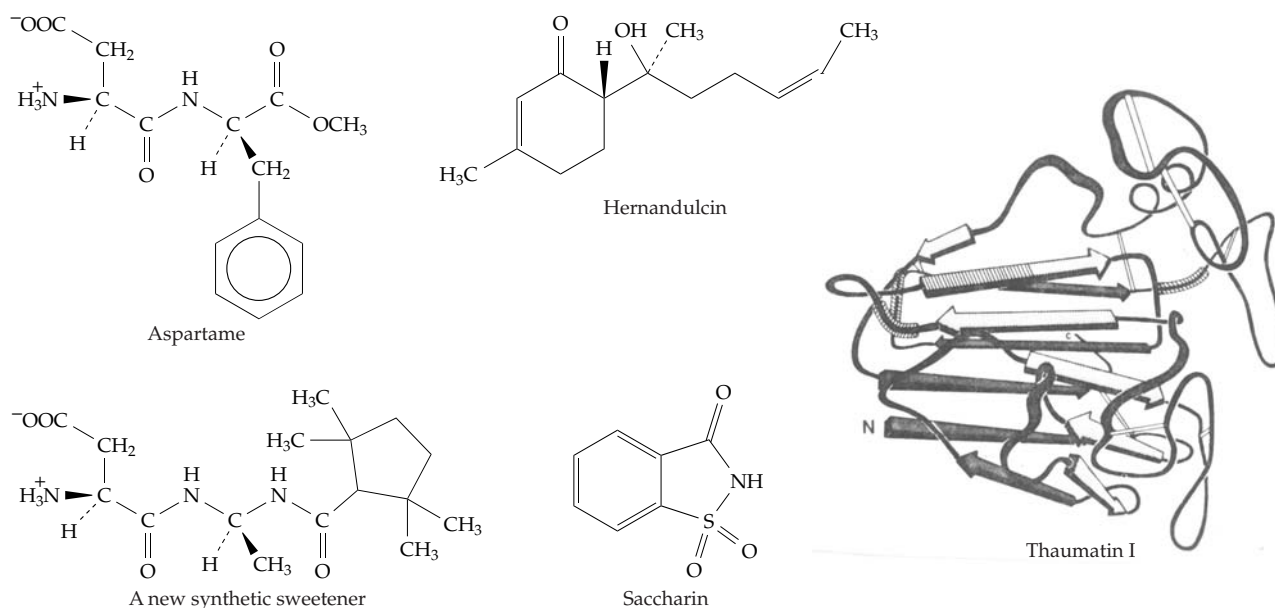


Figure 30-31 Structures of some very sweet compounds. The backbone structure of the protein thaumatin I is included. The main body of this structure consists of two β sheets forming a flattened β barrel. β Strands in the top sheet are shaded light, and those in the bottom sheet are darker. Open bars represent disulfide bonds, and the regions with sequences homologous to monellin are indicated by the hatched marks. From de Vos *et al.*⁹⁴⁰

also activates ion channels.⁹³³ Like the odor receptors, taste buds are also bathed in a special fluid. The **von Ebner's glands** in the tongue contain binding proteins,^{934,935} at least some of which are lipocalins.

The relationship between structures and sweet taste in humans has been investigated intensively, but no simple rules have been discovered (see Robyt⁹³⁶ for a discussion). Sucrose is usually perceived as very sweet. D-Fructose and D-xylose are nearly twice as sweet, but D-glucose is less sweet than sucrose. D-Galactose is usually perceived as not sweet and D-mannose as bitter. Many sucrose derivatives, in which hydroxyl groups have been replaced with Cl or other halogen, are very sweet. One tetrachloro derivative of this type is 7000 times sweeter than sucrose.⁹³⁶ Some especially sweet materials are depicted in Fig. 30-31. These include peptide derivatives^{937,938} such as Asp-Phe-OCH₃ (**aspartame**), the sesquiterpene **hernandulcin**,⁹³⁹ and **chemostimulatory proteins**. Among these are some of the sweetest substances known, the 207-residue **thaumatin**s^{940,941} and **monellin**^{942,943} present in certain tropical berries. Thaumatin is ~3000 times sweeter than sucrose on a weight basis and 10⁵ times sweeter on a molar basis. Thus, sucrose tastes sweet at a concentration of 10⁻³ M or higher but thaumatin⁹⁴⁴ at 3 × 10⁻⁸ M.

The proteins **miraculin** and **circulin** from tropical fruits modify taste. Acids taste sweet rather than sour after the tongue has been treated with either protein.^{945,946} Exposure of the tongue to artichokes often

makes water taste sweet.⁹⁴⁷ Thus, the response of taste receptors can be temporarily altered by binding of other substances, perhaps at adjacent sites on a receptor.

Pain. Receptors for pain (**nociceptors**) are spread over the body in nerve endings found in the skin, muscle, joints, and internal organs. There are several types of receptors, most of which are present in excitatory glutamatergic neurons.⁹⁴⁸ Some release substance P. Some activate tyrosine kinases and others ATP-gated ion channels. Some pain receptors are also activated by intense heat or pressure or by irritant compounds. Among the latter is capsaicin (Fig. 30-29), the active ingredient in chili peppers, and an ultra-potent compound, **resiniferatoxin**. Both capsaicin, which is 10,000 times more potent than jalapeño peppers,⁹⁴⁹ and resiniferatoxin, which is 20-fold more potent than capsaicin, bind to **vanilloid receptors**. These are ligand-gated ion channels related to the Shaker K⁺ channel (Fig. 30-18). They are nonselective but with a high permeability to Ca²⁺ and are members of the **transient receptor potential (TRP)** family.^{948,950–954} Pain seems to stimulate an increase in anandamide (Fig. 30-29), which has an analgesic effect. Nevertheless, anandamide and N-vanillyloleamide activate capsaicin receptors.⁹⁵⁵ Because the activated receptors become desensitized rapidly, capsaicin has been used in a paradoxical manner as an analgesic agent.⁹⁵¹ Sensing of temperature changes also depends upon TRP channels.^{955a,b}

Mechanoreceptors. The transduction of mechanical force into a chemical signal provides the basis for the senses of touch and hearing. Plants detect wind and gravitational force,⁹⁵⁶ and many organisms, even bacteria, respond to changes in osmotic pressure using mechanoreceptors.⁹⁵⁷ One of the best known mechanoreceptors is from *Mycobacterium tuberculosis*. It is a homopentamer whose three-dimensional structure^{450–452,958} resembles that of the nicotinic acetylcholine receptor (Fig. 30-23). A second type of mechanoreceptor is found in the inner membranes of *E. coli* and in plasma membranes of many other bacteria, archaea, and some eukaryotes.^{957a} These receptors, which are also sensitive to voltage changes, are heptamers of a 282-residue protein that forms a symmetric ion channel in the center. There is also a large cytoplasmic domain consisting largely of β structure.^{957b} How do such receptors sense mechanical stress? In bacteria they respond to stretch in the membrane induced by an increase in osmotic pressure. One suggestion is that the membrane expansion pulls apart the radially symmetric ion channel in the receptor.^{958,958a} In higher organisms transmembrane adhesion receptors and their linkage to the internal cytoskeleton provide a framework for detection of mechanical forces and linkage to mechanoreceptors.^{956,957,959}

Hearing. Movement of the **stereocilia** of the hair cells of the inner ear activates mechanoreceptors. Each stereocilium contains a core of crosslinked actin filaments, and tens to hundreds of these cilia are connected in hair bundles, which move in response to arrival of sound waves of appropriate frequencies. The movement of the stereocilia induces the opening of receptor ion channels in the hair cell membrane allowing K^+ and other ions to flow inward.⁹⁶⁰ The matter is much more complex than this because of the tuning and amplification mechanisms in the cochlea of the inner ear.^{960–962} These mechanisms allow receptors in hair cells to respond to very weak vibrations of specific frequencies. Both mechanical and biochemical mechanisms are involved. A number of specific proteins participate. Among these is a motor protein called **prestin**, which seems to be involved in the rapid changes in length and stiffness of some hair cells in the cochlea.^{962,963}

Other sets of hair cells are formed in specialized parts of the inner ear.⁹⁶⁴ The three semicircular canals detect angular acceleration in three directions, while the sac-like utricle and saccule detect linear acceleration including gravitational attraction. These two organs each contain a patch of hair cells whose tips project into a gelatinous layer, which is overlain by a field of small crystals of calcium carbonate. These little stones (**otoliths**) provide an inertial mass, which resists movement causing the hair cell tips to bend and activate mechanoreceptors to send information about balance and orientation to the brain.

While discussing vibrations we may ask whether 60 cycle electromagnetic field fluctuations caused by electrical power transmission can affect the human body? Considerable effort has been expended in addressing this question. The tentative conclusion is that such effects, if they exist, are extremely difficult to detect. However, the possibility has not been disproven.⁹⁶⁵

12. The Chemistry of Learning, Memory, and Thinking

What is known about the chemistry underlying memory, thinking, and the generation of the stream of consciousness within the brain? Nerve impulses originating in sensory receptors are sent to several regions of the brain, among which are the sensory regions of the cerebral cortex (Fig. 30-14). Memory also depends upon other regions of the brain including the hippocampus, amygdala, and cerebellum (Fig. 30-1). Learning, remembering, and thinking all require transfers of information between various neurons and between different parts of the brain. These transfers may perhaps be coordinated via endogenous electrical rhythms (brain waves).

Memory systems. Memories exist in several forms and are found in various regions that are reached by several pathways.^{966–967a} Two major forms of memory are:

Explicit (declarative, episodic):

Conscious recall of facts and events involving people, places, and things

Implicit (associative):

Nonconscious recall of motor skills, conditioned responses, etc.

Explicit memory depends upon the **temporal lobe** of the midbrain, an area that includes the hippocampus and the nearby subiculum and entorhinal cortex.^{966,968–971} Implicit associative learning and memory involve the cerebellum, amygdala, and other regions.^{972,972a}

Both types of memory possess both **short-term** and **long-term** components. Short-term memory lasts only minutes to hours, but long-term memory lasts days, weeks, and sometimes a lifetime. The difference between the two is clearly seen in individuals who have damage to the hippocampus and impairment of short-term memory. A blow to the head may cause total loss of short-term memory of associations (**amnesia**).^{969,973,974} Some persons with damage to the hippocampus may never regain their temporary memory, but long-term memories are intact, and new long-term memories may still be formed. An increasingly important tool for study of memory is brain imaging using

PET or **fMRI** (Box 30-A). These tools have become rapid and sensitive with the ability to observe regions of the brain that become activated by visual, auditory, or other stimuli.^{897,898}

The brain often needs to store information for a short period of time. For example, one can recall many details of a visual image after closing one's eyes or shifting one's gaze. The sensory images may be stored in **working memory**.^{896,975} Similarly, if one mentally multiplies two 2-digit numbers the partial product obtained by multiplying the two right most digits is temporarily stored in working memory until the next arithmetic operation is completed, etc. PET and fMRI tomography indicates that regions in the prefrontal cortex may be involved.^{976,977}

Some short-term memory appears to be stored by neurons that continue to fire after a stimulus has stopped. It has been proposed that such memory consists of reverberations of electrical activity in loops of coupled axons.^{978,979}

Implicit memory can be studied in animals. Much has been learned from the large marine snails *Aplysia* and *Hermisenda* whose simple nervous systems and large neurons have been investigated for over 40 years.^{967a,980-984} The basic chemical mechanism associated with learning in these creatures seems to be similar to those in our own brain. Olfactory memory can be studied in *Drosophila*, even though the organization of the fly's brain differs from ours.^{985-987b}

To be useful for more than a few minutes stored information must be transferred from the temporary to more permanent forms. We know that even temporary memory depends upon chemical changes in synapses. Long-term memory involves both stable chemical changes and also changes in the physical connections between neurons. Before discussing these

changes let us consider briefly the waves of nerve impulses that drive the necessary alterations.

Brain rhythms. A live brain displays characteristic oscillatory activity. Using electrodes placed on the scalp of an awake but relaxed individual, a rhythmic change in the recorded voltage with a frequency of ~ 10 Hz (**alpha waves**) can be detected. Such **electroencephalograms** (EEGs) contain other rhythms at ~ 5 – 6 (theta), ~ 40 (gamma), and ~ 200 (high frequency) Hz.^{978,988,989} More recent studies employ microelectrodes placed on individual neurons. Some cells generate spikes at frequencies as high as 800 Hz. However, the significance is uncertain.⁹⁹⁰ The 40-Hz frequency, which is prominent in the hippocampus, has aroused the most interest⁹⁹¹⁻⁹⁹⁵ because of its probable relationship to learning and memory. Psychophysical experiments have suggested that humans can store only 7 ± 2 items, such as digits in a telephone number. To remember more digits usually requires a conscious effort to place them in longer-term memory. One proposal is that the seven items in temporary memory are stored as 40-Hz oscillations and that ~ 7 such items can be stored within a single 5-Hz theta oscillation.^{978,979} Thinking rates have also been estimated as 7 ± 2 thoughts per second. This is also the same as the syllable rate in speech. This allows us to speak at the same rate that we think.⁹⁹⁶ Stuttering may be a result of lack of synchronization of thinking and speaking.

Individual cells or groups of cells are able to initiate rhythms.⁹⁹⁷ Examples are provided by the slower Ca^{2+} oscillations shown in Box 6-D,⁹⁹⁸⁻¹⁰⁰¹ by the periodic release (at ~ 155 intervals) of cAMP by cells of *Dictyostelium* (p. 20), and by the 24-h circadian cycle observed for virtually all living cells (Section 13). In simple invertebrates the source of neural rhythms appears to reside in **pacemaker neurons** that fire spontaneously at regular intervals. Their cell membranes apparently undergo a cyclic series of changes in ionic permeabilities sufficient to initiate action potentials. Three types of pacemaker output from molluscan neurons¹⁰⁰² are illustrated in Fig. 30-32. In lobsters three-neuron **pacemaker groups** provide a pyloric rhythm. In these groups the oscillation period of a pacemaker neuron is adjusted from its intrinsic value by feedback through inhibitory and electrical connections to the other cells.¹⁰⁰³ Electrical coupling seems to be basic to oscillatory cell networks.¹⁰⁰⁴ Individual neurons or small groups of neurons in our own bodies act as pacemakers for the heartbeat rhythm.⁷⁰⁵ The slow 3- to 8-Hz rhythm observed in the EEG apparently originates in pacemaker bursts from the basal ganglia.⁹⁹³ Rhythms from these endogenous pacemakers may combine with pulses from sensory neurons to evoke **conscious thought**. However, the basis of consciousness is still poorly understood.^{1005,1006}

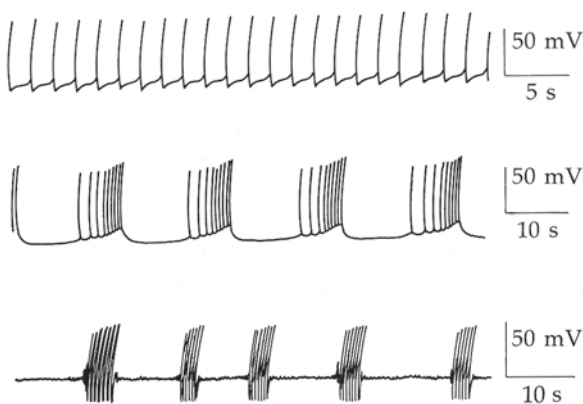


Figure 30-32 Intracellular recordings from isolated neurons of the mollusc *Aplysia*: (A) beating pacemaker, (B) bursting pacemaker, and (C) oscillating pacemaker. From Chen *et al.*¹⁰⁰²

As can be seen from Fig. 30-32, neurons send "trains" of spikes down their axons. These form synapses with dendrites, usually on dendritic spikes, of a postsynaptic cell.^{593,1007-1009} However, each such cell typically receives input from thousands of other neurons. At any moment most of these are probably "silent," but others are sending trains of impulses. Among the important questions are "How does the postsynaptic neuron know whether to fire or not?" and "What kinds of information, if any, are encoded in the trains of impulses both in the presynaptic inputs and in the output of the postsynaptic neuron?"^{1010,1011} Part of the answer to the first question is probably that firing occurs if two or more input impulses arrive synchronously,^{1010,1012-1014} and if there are not too many inhibitory impulses that damp the response. In the hippocampus a network of neurons electrically coupled via gap junctions may be synchronized to the theta and gamma brain rhythms by high-frequency (150–200 Hz) oscillations.⁹⁸⁸ See also Fig. 30-15.

Chemical changes in synapses. It has long been recognized that the synapses are the probable sites of alterations that lead to memory, whether long-term or short-term. Study of individual synapses has demonstrated the phenomena of **potentiation** (facilitation) and **depression** (habituation). Potentiation refers to the fact that a second impulse will often be transmitted through a synapse more effectively than the first, while depression refers to a decreased response to repeated stimuli. Memory may consist of potentiation and depression at specific synapses. The underlying chemical changes in the synapses are referred to collectively as **synaptic plasticity**. Chemical changes associated with short-term memory are often transient. Those associated with long-term memory are described as **long-term potentiation (LTP)** and **long-term depression (LTD)**.

Many experimental results have confirmed the chemical basis of memory. For example, learning is facilitated by administration to animals of small doses of strychnine.¹⁰¹⁵ Puromycin and other inhibitors of protein synthesis disrupt the transfer of information into long-term memory. They are especially effective during the first hour after the initial learning event.¹⁰¹⁶ Increased synthesis both of mRNA and of proteins within the cell bodies of neurons is observed.

Short-term memory is not affected by inhibitors of protein synthesis, but alteration of synaptic proteins and membranes may be induced by covalent modification of existing macromolecules.¹⁰¹⁶ One way in which this happens has been described for *Aplysia*. As the snail learns a simple gill-withdrawal reflex, the duration of the action potentials in sensory neurons is increased, and there is a greater release of transmitters. This change comes about because stimulation of the sensory neuron causes simultaneous activation of

interneurons that synapse with the sensory neurons. The interneurons release the neuromodulator serotonin, which binds to receptors in the membrane of the sensory neurons. This activates adenylate cyclase, which in turn activates a protein kinase that phosphorylates a class of open K⁺ channels. Phosphorylation causes the channels to close with a consequent strengthening of the action potential. Thus serotonin brings about presynaptic facilitation.¹⁰¹⁷ The peptide FMRF amide (Table 30-5) has the opposite effect. It causes hyperpolarization and a decrease in the duration of the action potential. It also binds to a receptor on the neuronal membrane, and presumably via a different second messenger than cAMP causes the K⁺ channels to stay open a longer fraction of the time.

Evidence that LTP is essential to learning in rats was provided by the observation that the antagonist 2-aminophosphonovalerate, which blocks the NMDA class of glutamate receptors (Fig. 30-20A), impairs both LTP and learning.¹⁰¹⁸ The potentiation is thought to result, in part, from Ca²⁺ influx through the nonselective NMDA cation channels. The increased intracellular calcium may then induce phosphorylation of various proteins with associated long-lasting changes in the postsynaptic endings.¹⁰¹⁹ A large amount of evidence favors this interpretation of LTP.¹⁰²⁰ However, it is a great oversimplification. Most studies of LTP in mammals (usually rodents) have focused on the CA1 region of the hippocampus and nearby brain regions. The excitatory axons in this organ are largely glutamatergic, and, as shown in Fig. 30-20A, the postsynaptic (dendritic) membranes contain both fast AMPA receptors and the slower NMDA receptors. Both are ionotropic. The AMPA receptor channels allow mainly K⁺ and Na⁺ to pass and are responsible for most of the nerve transmission. However, the NMDA receptors have an important controlling influence.

A generally accepted theory is that no LTP arises unless both the presynaptic and postsynaptic neurons are activated. This can happen if a presynaptic action potential activates many AMPA receptors in a synapse allowing enough flow of Na⁺ + K⁺ to depolarize the postsynaptic membrane and possibly to initiate an action potential in the postsynaptic neuron. (However, many factors, probably including influences from neighboring neurons,¹⁰²¹ will affect this outcome.) The NMDA receptors are usually blocked by extracellular Mg²⁺ ions, and their ion channels remain closed. However, when the postsynaptic membrane becomes depolarized, the Mg²⁺ dissociates, and if the NMDA receptors are also occupied by glutamate, their channels will open, permitting Ca²⁺ to enter the neuron (Fig. 30-20A). This not only enhances the probability of developing a postsynaptic action potential but is also the trigger for LTP.¹⁰²² Ca²⁺ ions have a variety of effects, one of which is to bind to calmodulin. This

activates a calcium–calmodulin-dependent protein kinase, which phosphorylates postsynaptic structural and signaling proteins to increase the synapse strength.^{641,1023–1025}

Modifications in existing proteins, such as are induced by Ca^{2+} and calmodulin, can provide LTP for a few hours, but other mechanisms must provide for longer-term effects. These require transcription of genes and protein synthesis, processes that occur in the cell bodies of neurons and may depend upon axonal transport mechanisms.^{1026,1027} Among the experimentally observed results of LTP are ultrastructural changes in synapses and in dendrites.^{1009,1028} Long-term memory is also thought to involve changes in the neocortex. Again, NMDA receptor activation seems to be involved.^{1029,1030}

LTP has been demonstrated experimentally, but does it really influence memory? Evidence that it does has been provided by clever experiments with transgenic mice. Using the Cre recombinase (Chapter 27) the NR2B subunit of the NMDA receptor was overexpressed in the hippocampus¹⁰³¹ and in the forebrain of mice.¹⁰³² This was expected to provide better synaptic strengthening than for receptors with the similar NR2A subunit. It was found experimentally that these transgenic mice were more intelligent than normal mice.

LTP is also thought to affect the presynaptic as well as the postsynaptic neuron. One way in which this may happen is for a **retrograde messenger** to pass across the synapse and induce alterations in the presynaptic cell. One proposed retrograde messenger is nitric oxide, **NO**.^{1033,1034} Neuronal NO synthase (nNOS; NOS1) contains a calmodulin binding site and is activated by Ca^{2+} . However, other substances as simple as K^+ might also be the messenger.

Long-term depression (LTD) is the *loss* of synaptic strength after passage of an impulse. There is evidence that during brain activity, including that in the hippocampus, both LTP and LTD are essential.^{1035,1036} LTD may depend upon cyclic ADP-ribose (p. 564).¹⁰³⁷ LTD, like LTP, may also spread via retrograde signaling.¹⁰³⁸

Does learning affect a few specific neurons or a large number of neurons? The rate of glucose utilization in different parts of the brain can be estimated from the rate at which labeled 2-deoxyglucose is taken up (see Box 30-A). From changes in this rate (obtained using ^{14}C -labeling) in brains of split-brain cats performing visual tasks it was estimated that from 10^{10} to 10^{11} neurons are activated.¹⁰³⁹ This supports the idea that memory is distributed over a large area of the brain, just as information about an image is stored in all parts of a hologram.

An alternative to the idea that synaptic potentiation and depression provide the chemical basis for learning is **molecular coding**. Thus, it was reported

that a 15-amino acid peptide isolated from rats trained to avoid the dark carries behavioral information. When this peptide was injected into brains of untrained rats, they also avoided the dark.¹⁰⁴⁰ This was one of several reports of transfer of learned behavior through chemical substances extracted from the brain. These ideas are hard to accept in the light of our present knowledge of the brain. However, in view of the large number of different neuropeptides known (Table 30-4) the possibility that some aspects of long-term memory may be associated with transcription of specific amino acid sequences within specific neurons should perhaps still be considered.

The complexity of the brain. A major obstacle to our understanding of the human brain is its enormous complexity.⁴⁰⁰ This problem can be appreciated if we consider the small nematode *Caenorhabditis elegans*. All of the synaptic connections among its 302 neurons had been mapped by 1986.^{400,1041,1042} There are 5000 chemical synapses and 600 electrical (gap junction) connections. There are 80 different types of K^+ -selective channels, 90 types of ligand-gated receptors, and ~1000 G-protein-linked receptors. Twenty-six of the neurons are GABAergic and are involved in three distinct behavioral motions that involve muscular contractions. Despite intensive efforts the system has been hard to understand. The brain of the macaque monkey has been described in great detail.⁴⁰⁹ Over half of its cerebral cortex is devoted to vision, and this can be subdivided into 20 functional areas. However, the human brain, with its extremely large cerebral cortex, cannot be compared accurately with the monkey brain. Whereas anatomical studies are done on postmortem human tissues, *in vivo* studies rely largely on fMRI and PET imaging (Box 30-A). The resolution of these images is now less than 1 mm, but 1 mm³ of human visual cortex contains more than 40,000 neurons!⁴⁰⁹ The microcircuits in the neocortex are still largely unknown and the tissue is of “apparently impenetrable complexity.” There may be several hundred different classes of neocortical neurons.¹⁰⁴³ The tissue is rich in GABAergic interneurons.^{1044,1045} Some fast-spiking GABAergic neurons are also connected by electrical synapses and may be involved in detecting and promoting synchronous activity.¹⁰⁴⁶

Intelligence. We must all agree that there is such a thing as intelligence, but can it be measured? In 1904, Spearman proposed the existence of a general intelligence factor *g* that could be measured as the IQ (intelligence quotient). Since then various tests have been devised that attempt to measure IQ.^{1047,1048} Most recently use of PET scan data has indicated that various types of analytical analysis lead to brain activity in the lateral frontal cortex in one or both cerebral hemispheres¹⁰⁴⁹ suggesting that this is a region important to

IQ. A question that has been raised is whether analytical intelligence, creative intelligence, and practical intelligence are correlated?¹⁰⁴⁸

Is intelligence hereditary? Both logic and observation say that heredity must be a major factor. However, it is hard to know how to measure the hereditary component.^{1050–1052} Also hard to understand is why IQ scores have been increasing about one standard deviation unit per generation.¹⁰⁴⁷ Is this really true?⁸⁴³ Does environment also influence IQ? The fact that new hippocampal nerve cells are formed continuously provides one mechanism by which learning, nutrition, and other influences may alter intelligence.

A difficult-to-explain aspect of the brain is the existence of rare **savants**, persons with amazing mental abilities in music, art, or computation but who are unable to communicate (autistic) and mentally retarded.^{1053–1055} One boy at age four could play Mozart piano sonatas flawlessly after a single hearing. A three-year old girl drew horses with lifelike perspective from memory but was unable to communicate. Some mathematical savants can instantly state the day of the week for any arbitrary date such as June 12, 1929; others rapidly identify prime numbers. They evidently use the same strategies as mathematically trained persons. Do we all have these abilities but can't have access to them? How can we explain the fact that rarely a blow to the head will convert a person into a savant?

Behavior. It may seem impossible to interpret complex behavioral patterns at the molecular level. However, the genetics of behavior is a well recognized field of investigation, and some behavioral traits have been linked to single genes. If a gene can be located cloning, sequencing, and biochemical studies may follow quickly. The behavioral genetics of lower organisms, e.g., of *Drosophila*, have provided many insights.¹⁰⁵⁶ Recently, however, the mouse has become a major object of behavioral studies.¹⁰⁵⁷ Its genome is well known, and a very large number of mutations have been mapped. The ability to prepare **knockout mice** (p. 1501) and to carry out gene transfer experiments on such animals makes them very attractive for study.

Some behavioral traits are based on simple alterations, often defects, in motor skills. For example, the following traits in mutant mice have been traced to specific brain structures and often to specific biochemical alterations.

<i>Staggerer</i>	Purkinje cell defect
<i>Vibrator</i>	Phosphatidylinositol transfer protein gene
<i>Tottering</i>	Mutation in voltage-gated Ca ²⁺ channel
<i>Lurcher</i>	Abnormality in cerebellum
<i>Weaver</i>	Gly → Ser mutation in K ⁺ channel

Knockout mice lacking oxytocin or vasopressin have altered social behavior toward other mice. Those lacking galanin seem less intelligent than normal mice, as if they had Alzheimer disease. Mice lacking neuronal NO synthase became aggressive.¹⁰⁵⁷ Human personality,¹⁰⁵⁸ language abilities,¹⁰⁵⁹ and sexual behavior all have a genetic component. However, claims that a "gay gene" has been found are not generally accepted.¹⁰⁶⁰

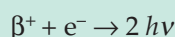
13. Circadian Cycles and Sleep

In mammals an approximately 24-h (**circadian**) rhythm controls behavior and affects many physiological functions. As previously mentioned (p. 1800), the brain has its own rhythms, which originate with pacemaker neurons. The heart beats with another neurally established rhythm. The circadian rhythm has a much longer period and, therefore, seems more mysterious. It is observable, even with single cells and for virtually all organisms.^{1061–1062a} In most instances the cycle becomes synchronized with the daily light–dark cycle with the aid of suitable light-absorbing pigments often cryptochromes (see Chapter 23, Section I,1). However, the cycle can be observed in various ways under conditions of constant light intensity and temperature. For example, the unicellular marine alga *Gonyaulax* undergoes dramatic circadian changes in the intensity of its bioluminescence. Over one 10-day period the luminescence peaked every 22.99 ± 0.01 hours.^{1063,1064} It is more difficult to measure the period for human beings (see Czeisler *et al.*¹⁰⁶⁵ for a discussion), but under suitable conditions during which time cues were missing a precise period of 24.18 hours was observed for the level of melatonin in the blood, the body temperature, and other quantities.¹⁰⁶⁵ From cyanobacteria,^{1066,1067} fungi (*Neurospora*),¹⁰⁶⁸ insects (*Drosophila*),^{1069–1071b} and frogs¹⁰⁷² to mice and people,^{1073,1074} the circadian cycle affects the organism's chemistry and behavior. Green plants likewise observe a circadian cycle.¹⁰⁷⁵

The cycle is thought to originate in feedback loops that control transcription of a small set of genes. In *Drosophila* the set includes seven genes: *period* (*per*), *timeless* (*tim*), *clock* (*clk*), *cycle* (*cyc*), *double-time* (*dbt*), *vri*, and *cryptochrome* (*cry*).¹⁰⁷⁰ Many corresponding genes have been found in mammals. For example, the mouse NPAS2 is a close relative of the *Drosophila* CLOCK protein, and the period proteins PER1 and PER2 and the cryptochromes CRY1 and CRY2 are also related to the *Drosophila* proteins.^{1076,1077} In *Drosophila* the heterodimers PER•TIM and CYC•CLK are thought to serve as DNA-binding transcription factors that repress transcription of their own genes when they reach a high enough concentration in the nucleus.^{1073,1076,1077} Because some time is required for transcription and

BOX 30-A POSITRON EMISSION TOMOGRAPHY (PET), FUNCTIONAL MAGNETIC RESONANCE (fMRI), AND OTHER IMAGING TECHNIQUES

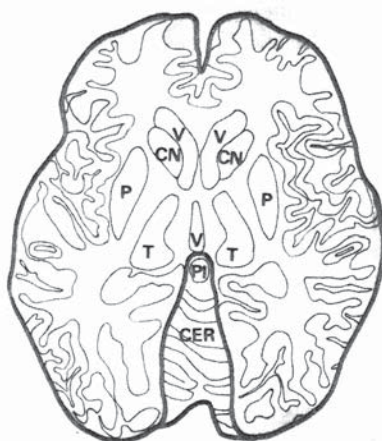
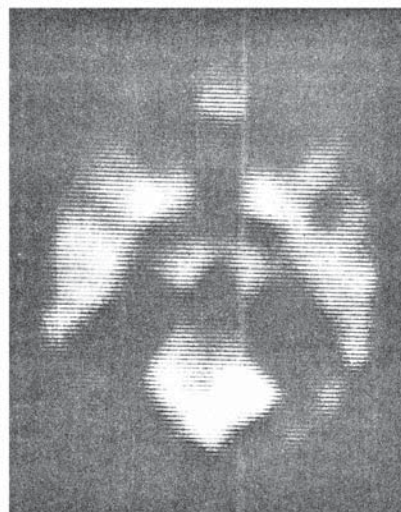
In the widely used technique of transmission computerized tomography (CT) an image of a slice through the body of a patient is obtained using X-rays. An X-ray source moves in a ring around the patient while detectors measure the intensity of the transmitted radiation and send it in digital form to a computer, which generates the desired image. A chemically more sophisticated view of the body can be obtained by positron emission tomography (PET). This technique makes use of a metabolite or drug labeled with a short-lived radioisotope that decays by emission of positrons (antielectrons). Among these are ^{11}C , ^{13}N , ^{15}O , and ^{18}F with half-lives of 20 min, 10 min, 2 min, and 110 min, respectively. The isotopes are produced in a cyclotron, and are rapidly introduced into suitable compounds, which can be injected into a bloodstream.^{a-d} An emitted positron travels only a few millimeters before undergoing annihilation with an electron to produce two high-energy (50 keV) photons (γ -rays) that travel in opposite directions and are detected by an array of scintillation detectors.



Present-day PET technology allows images to

be formed in a few seconds, and in some cases in a fraction of a second. Among the useful compounds for PET imaging is [^{18}F]2-fluoro-2-deoxy-D-glucose (see figure). This compound, which contains the longer-lived ^{18}F , is phosphorylated by hexokinase, and the resulting phosphate ester is effectively trapped in the brain. 3-Deoxy-3-fluoro-D-glucose is another useful tracer.^e One of the most useful PET measurements has been blood-flow monitored by ^{15}O -containing H_2O , which is administered into a vein in the arm.^b The ^{15}O has a half-life of only two minutes and is almost completely gone in ten minutes. However, very low doses of radioactivity are used, and several images can be obtained before the radioactivity has decayed. A common practice is to subtract images obtained after the isotope has decayed from those obtained at various times while it was still present. The technique is also useful for study of the binding and transport of hormones,^f other metabolites, drugs, and other inhibitors.^c

The NMR technique **magnetic resonance imaging (MRI)**,^{g-i} so called to avoid the word nuclear, is rapidly displacing many applications of PET scanning. MRI uses proton NMR spectroscopy to generate very sharp images based largely on the water present in tissues. These images can be made



Left: PET image of a human brain obtained using 2-[^{18}F]fluoro-2-deoxyglucose. This tomographic brain slice at the level of the basal ganglia shows the cortical gray matter and subcortical white matter. As marked on the drawing on the right: V, ventricles; CN, caudate nucleus; P, putamen; T, thalamus; PI, pineal gland; CER, cerebellum. From Rottenberg and Cooper.^p Right: fMRI image illustrating modulation of neural activity in the ventral striatum, an area of the brain associated with reward, when eye contact was made with an attractive face. The activation map shown is derived in a complex manner and is based on recorded brain activity of persons viewing images of a series of faces. It portrays the differences in neural activity when viewing images of attractive faces of either sex with the eye gaze directed at the subject and with the eyes averted. See the report of Kampe *et al.* for details.^q

BOX 30-A (continued)

to depend upon variations in T_1 and T_2 (Chapter 3) as well as upon differences in the water content. The first MRI scans required 20 minutes, but the use of more powerful magnets and more sensitive instruments has reduced the acquisition times in ultrafast MRI to ~ 0.1 s. The decay of the NMR signal from a single RF pulse is observed at several different times.^h The dynamics of blood flow and neural activity can be followed. Every technique has disadvantages as well as advantages. MRI does not use radioisotopes, but overheating of the brain must be carefully avoided. In addition, patients may suffer from uncomfortably loud noises generated by rapidly changing magnetic gradients.^h As with PET scans isotopic tracers may be used. However, most MRI scanning is done with ^1H from the solvent water. As with PET, MRI is often used to measure blood flow but with an indirect method. The Fe of deoxyhemoglobin (Hb) is paramagnetic, but upon oxygenation to HbO_2 it becomes diamagnetic (pp. 850–851), and the ^1H signal of the solvent H_2O becomes sharper. In metabolically active regions of the brain the demand for oxygenated blood is greatly increased. Perhaps surprisingly, the ratio $[\text{HbO}_2]/[\text{Hb}]$ is greater in these areas than in less active areas where a greater fraction of the hemoglobin remains unoxygenated.^g However, the exact interpretation of the ultrafast MRI images is uncertain. In **functional MRI (fMRI)**, differences in images acquired after some physiological change are recorded. For example, after a visual or other sensory stimulus a change in the MRI image of some region of the cortex will be observed (see figure).^{i–l} The technique is allowing many deductions about learning, memory, and communication pathways in the brain^{j,m} and is being used to investigate many aspects of brain disease.

Another brain imaging technique is **magnetoencephalography (MEG)**.^{c,n} It has been uniquely valuable in mapping the sensory regions of the human cerebral cortex. Looking ahead, optical methods, which include use of infrared radiation, are also under development.^o They may not be adequate for study of the human brain but can be used for smaller animals, for studies of embryonic development, etc.

- ^a Ter-Pogossian, M. M., Raichle, M. E., and Sobel, B. E. (1980) *Sci. Am.* **243**(Oct), 171–181
- ^b Raichle, M. E. (1994) *Sci. Am.* **270** (Apr), 58–64
- ^c Volkow, N. D., Rosen, B., and Farde, L. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 2787–2788
- ^d Phelps, M. E. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 9226–9233
- ^e Berkowitz, B. A., Moriyama, T., Fales, H. M., Byrd, R. A., and Balaban, R. S. (1990) *J. Biol. Chem.* **265**, 12417–12423
- ^f Berman, K. F., Schmidt, P. J., Rubinow, D. R., Danaceau, M. A., Van Horn, J. D., Esposito, G., Ostrem, J. L., and Weinberger, D. R. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 8836–8841
- ^g Shulman, R. G., Rothman, D. L., and Blamire, A. M. (1994) *Trends Biochem. Sci.* **19**, 522–526
- ^h McKinstry, R. C., and Feinberg, D. A. (1998) *Science* **279**, 1965–1966
- ⁱ Disbrow, E. A., Slutsky, D. A., Roberts, T. P. L., and Krubitzer, L. A. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 9718–9723
- ^j Ungerleider, L. G. (1995) *Science* **270**, 769–775
- ^k McCarthy, G., Blamire, A. M., Puce, A., Nobre, A. C., Bloch, G., Hyder, F., Goldman-Rakic, P., and Shulman, R. G. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 8690–8694
- ^l Chen, W., Zhu, X.-H., Thulborn, K. R., and Ugurbil, K. (1999) *Proc. Natl. Acad. Sci. U.S.A.* **96**, 2430–2434
- ^m Wagner, A. D., Schacter, D. L., Rotte, M., Koutstaal, W., Maril, A., Dale, A. M., Rosen, B. R., and Buckner, R. L. (1998) *Science* **281**, 1188–1191
- ⁿ Yang, T. T., Gallen, C. C., Schwartz, B. J., and Bloom, F. E. (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 3098–3102
- ^o Lok, C. (2001) *Nature (London)* **412**, 372–374
- ^p Rottenberg, D. A., and Cooper, A. J. L. (1981) *Trends Biochem. Sci.* **6**, 120–122
- ^q Kampe, K. K. W., Frith, C. D., Dolan, R. J., and Frith, U. (2001) *Nature (London)* **413**, 589

protein synthesis, this inhibitory feedback can lead to oscillations in the concentrations of the circadian clock proteins. Proteosomal degradation of the TIM protein may also be a factor.¹⁰⁷¹ The need for proteins encoded by other genes indicates that the matter is more complex. Individual cells or individual tissues, e.g., mammalian retinas,^{1074,1078} may independently set up circadian cycles. However, these normally become **entrained** by the daylight cycle and are reset daily as discussed in Chapter 23, Section I,1. Other factors such as temperature, activity, and food may also affect the resetting. One factor, which may be influenced by food, is the NAD^+/NADH and $\text{NADP}^+/\text{NADPH}$ ratios within cells.¹⁰⁷⁷ The circadian cycles for mam-

malian tissues are synchronized by a **master clock** that originates in neural tissues^{1062a} and specifically in a region of the hypothalamus containing the **suprachiasmatic nuclei**.^{1079–1080b}

The pineal gland appears also to play a role in maintaining the mammalian circadian cycle.^{1081–1083} The concentration of the pineal hormone melatonin (Fig. 27-11) as well as its precursor *N*-acetylserotonin and the enzyme serotonin *N*-acetyltransferase (Eq. 30-4) all fluctuate far more than do the concentrations of other metabolites during the 24-h cycle. These metabolites increase over 10-fold concentration at night and decrease by day. During the daytime the serotonin *N*-acetyltransferase, which forms the precursor, is rapidly

and apparently irreversibly inactivated, perhaps through a disulfide exchange reaction.¹⁰⁸¹ Bright light will reset the circadian cycle¹⁰⁸⁴ keeping it approximately (circa) daily. The effect of light is apparently a result of signals sent to the hypothalamus from the optic nerves. In chickens and in lower vertebrates the pineal gland may directly sense light passing through the skull.

The circadian cycle is not the only timing device used by animals. A short-term interval timer helps male doves to know how long to sit on a nest¹⁰⁸⁵ and helps all of us in timing everyday tasks.

We spend a third of our lives asleep, but our understanding of sleep from a molecular viewpoint is minimal. Sleep is essential for the life of mammals, which die if completely deprived of sleep. It has been shown that during prolonged sleep deprivation sleep-inducing material accumulates in the brain. One such substance, isolated from human urine, appears to be a peptide containing glutamate, alanine, diaminopimelic acid, and muramic acid.¹⁰⁸⁶ Thus, it resembles a fragment of bacterial peptidoglycan. Prostaglandin D₂ also induces sleep.¹⁰⁸⁷ Hayashi proposed that a balance between this substance and prostaglandin E₂, which induces wakefulness, is in part responsible for the sleep–wake cycle.¹⁰⁸⁷ More recently oleic acid amide (**oleamide**; p. 382) was identified as a sleep-inducing compound. A fact observed by everyone is that the longer one is awake the higher the probability of going to sleep. The accumulation of sleep inducers is part of a homeostatic mechanism. On the other hand, the circadian cycle probably provides the signal to awake and tends to consolidate our sleep into the characteristic 8-hour period.¹⁰⁸⁸ The melatonin level, which drops in daylight, plays a role.¹⁰⁸⁹ Release of adrenocorticotropin (ACTH) one hour before waking may also be important.¹⁰⁹⁰

During much of the night's sleep the EEG is characterized by the slow 5- to 6-Hz waves.^{989,993} However, after ~90 min there is an ~10-min period of **rapid eye movement (REM) sleep** during which the EEG resembles that of an awake person and dreaming occurs. The closed eyes move rapidly in unison, breathing is irregular, and the heart rate increases. Motor neurons are inhibited allowing only minimal body movement. Three more periods of slow-wave sleep, each shorter than the preceding one, are followed by REM sleep. The REM sleep periods become successively longer. The fourth period lasts 20–30 min and is followed by awakening. All placental and marsupial mammals follow a similar sleep pattern and all dream.⁹⁸⁹ The importance of dreaming is not obvious^{1088a} but is often thought of as a reprocessing of memory, a means of ridding the mind of unneeded memories, a process of **unlearning**. However, this is uncertain as is the relationship of sleep to learning and memory.^{1088a,b}

A number of disorders of sleep are known. Among these is **narcolepsy**, uncontrollable, sudden daytime sleepiness. It affects 1 in 2000 individuals.¹⁰⁸⁸ The same occurs in dogs.¹⁰⁹¹ After a 10-year effort at great expense the narcolepsy gene of dogs (*canarc-1*) was located by positional cloning.^{1088,1092} The corresponding human (and rat) gene was independently discovered by other investigators. It encodes a receptor for neuropeptides produced by the hypothalamus and named **hypocretins** or **orexins** for their stimulation of appetite. It seems probable that the hypocretin / orexin neuropeptides are involved in promoting wakefulness. Another sleep disorder is **familial advanced sleep phase syndrome**. Persons with this trait are “morning larks” who tend to fall asleep at ~7:30 p.m. and awake suddenly at ~4:30 a.m., about four hours in advance of a typical sleep period. A missense Ser → Gly mutation in the human period gene (*hper2*) has been found.¹⁰⁹³

Some mammals hibernate. Special blood proteins that induce hibernation apparently control the process.¹⁰⁹⁴

14. Mental Illness

Whereas many metabolic defects affect only a small number of individuals, emotional illnesses including depression, **schizophrenia**, and other **affective disorders** at one time or another afflict a large fraction of the population. Autism affects thousands of children.¹⁰⁵⁵ Parkinson disease and **Alzheimer disease** are just two of a number of degenerative neural diseases attacking older people. Less commonly, young persons contract **multiple sclerosis** and **muscular dystrophy**, which is often a disease of neuromuscular junctions.

Depression. Depression is our most common mental problem. One in four women and one in ten men will have a major depression during their lifetime.¹⁰⁹⁵ More than 15 million people in the United States are affected by severe depression in any given year and more than 30,000 may commit suicide.^{1096,1097} Worldwide psychiatric problems, mostly depression, account for 28% of all disabilities.¹⁰⁹⁸ The **biogenic amine hypothesis** states that depression results from the depletion of neurotransmitters in the areas of the brain involved in sleep, arousal, appetite, sex drive, and psychomotor activity. An excess of transmitters is proposed to give rise to the manic phase of the bipolar (manic–depressive) cycle that is sometimes observed. In support of this hypothesis is the observation that administration of reserpine precipitates depression, which may be serious in 15–20% of hypertensive patients receiving the drug. Similar effects are observed with the dopa decarboxylase inhibitor **α-methyl dopa**

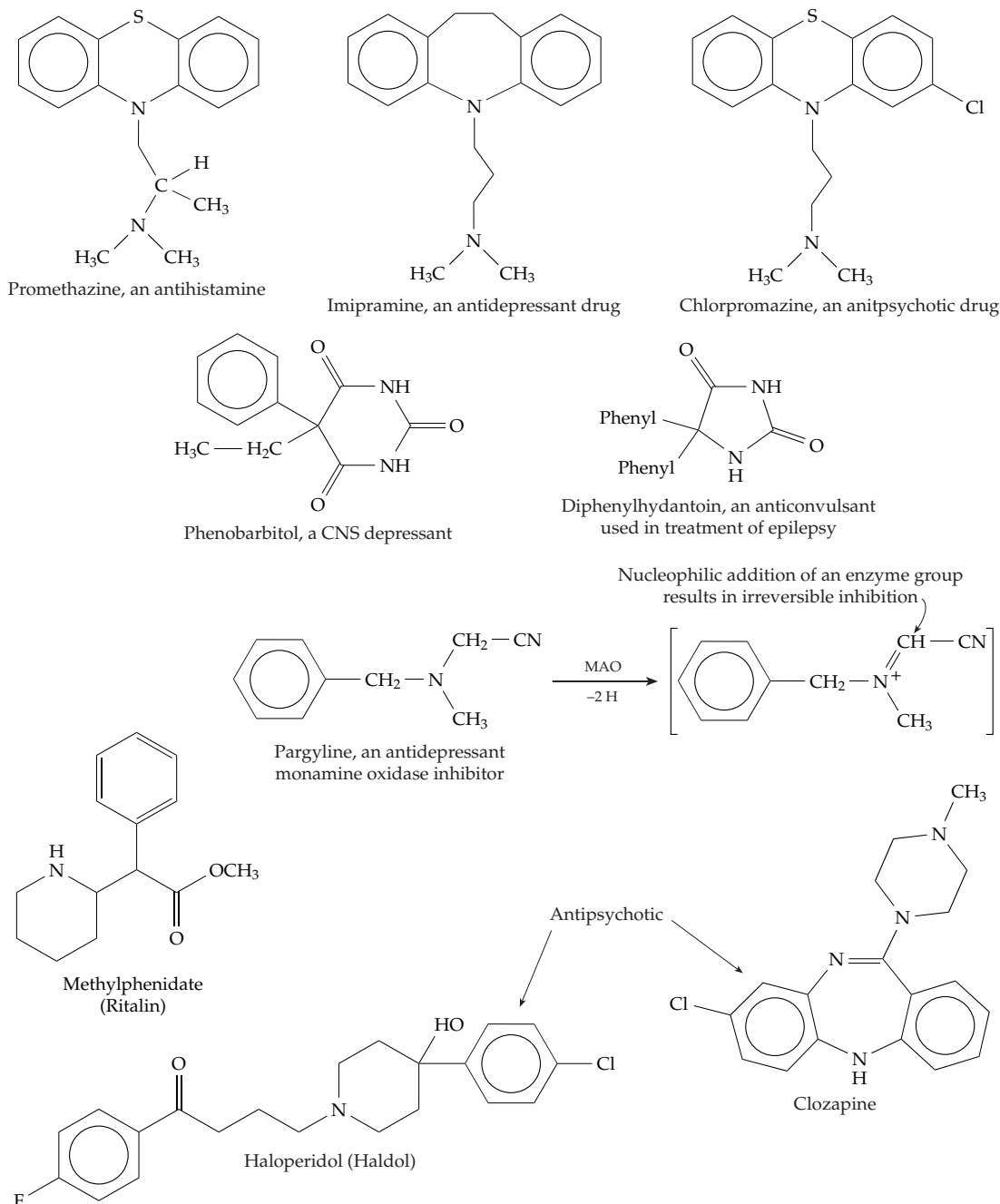


Figure 30-33 Some drugs used to treat psychiatric disorders. See also Figs. 30-25 and 30-28.

(Fig. 30-27). The fact that L-tryptophan has some antidepressant activity, but L-dopa does not, was one clue that a low concentration of serotonin (5-hydroxytryptamine) might be responsible for depression. Excessive formation of histamine¹⁰⁹⁹ and decreased formation of tyramine and octopamine¹¹⁰⁰ have also been suggested as causes of depression.

Strong support for the biogenic amine theory of depression is provided by the powerful antidepressant effect of inhibitors of monoamine oxidase. An example is **pargyline** (Fig. 30-33), which forms a covalent

adduct with the flavin of MAO.¹¹⁰¹ Although effective, this drug is somewhat dangerous. Because their monoamine oxidase activity is so low, patients taking pargyline have been killed by ingesting compounds such as tyramine, which occurs in cheese. Less easy to understand but clinically more important are tricyclic antidepressants such as **imipramine** (Fig. 30-33),^{746a} whose antidepressant action was discovered accidentally. Notice the close similarity to chlorpromazine but the greater flexibility of the central ring.¹¹⁰² Imipramine was found to block transporters of both

noradrenaline and serotonin. In 1986, the less toxic serotonin reuptake inhibitor fluoxetine (Prozac; Fig. 30-28) was introduced and is now used by many millions of people.^{1103,1104} Nevertheless, its mode of action is not entirely clear. For example, it blocks nicotinic acetylcholine receptors and may have many other effects. Interestingly, depression sometimes responds to a placebo just as well as to an antidepressant drug.¹¹⁰⁵ In addition to newer drugs related to Prozac, antagonists of substance P are also effective antidepressants.¹¹⁰⁶ MRI images of brains of depressed patients show that hippocampal volume has decreased and suggest that formation of new neurons is inhibited.^{1106a} Antidepressants seem to stimulate growth of new cells as does exercise, which also has an antidepressant effect.¹¹⁰⁷ Dietary treatment can also help.⁸⁴³ Among older people depression may be caused by deficiency of vitamin B₁₂ and can be treated by injection of the vitamin.¹¹⁰⁸ An **anxiety peptide** that may be the natural ligand for benzodiazepine receptors has been reported.¹¹⁰⁹

Another recognized type of depression is **seasonal affective disorder (SAD)**. People in far northern or southern latitudes develop this condition in the winter, apparently from lack of sunshine needed to lower the melatonin level in the morning (see Section 13). Light therapy is beneficial.¹¹¹⁰ Persons with the SAD syndrome also tend to crave carbohydrates and to stay in bed for 9–10 hours.

An effective treatment for **bipolar disorder** (manic–depressive illness) is the administration of lithium salts.^{445,1111–1113a} Inhibition of the hydrolysis of inositol phosphate by Li⁺ (Fig. 11-9) may be related to its therapeutic effect. Reduced phosphatidylinositol turnover may dampen responses to neurotransmitters.¹¹¹⁴ Li⁺ may affect gene expression in neuropeptide-secreting neurons.¹¹¹⁵ Bipolar disorder apparently has more than one cause. There are strong indications of genetic susceptibility,¹¹¹⁶ and genes that increase susceptibility have been located on chromosomes 4, 12, 13, 18, 21, and X.¹¹¹⁷

Schizophrenia. Among the most baffling of mental illnesses are the group of diseases known as schizophrenia. They involve thought disorder, disturbance of the affect, and withdrawal from interactions with other people. Hallucinations and paranoid feelings are common.⁸¹³ In some cases a striking loss of gray matter in some areas of the brain is revealed by MRI scans.^{1117a,b} The schizophrenias are of varying degrees of severity and shade continuously into the affective or mood disorders, which include manic-depression and depression. As many as one person in a hundred is affected by schizophrenia.^{1118–1119a} There is a complex genetic susceptibility.^{1119,1120} One theory about the persistence of the genes favoring schizophrenia is that they are also associated with creativity.¹¹²¹

A revolution in the treatment of the schizophrenias, as well as in thinking about mental illnesses, took place following the synthesis, in 1950, of the antipsychotic drug **chlorpromazine** (Fig. 30-33). At about the same time the effect of the *Rauwolfia* alkaloid reserpine (Fig. 25-12) in calming mentally disturbed persons was rediscovered. The Indian plant *Rauwolfia* had been used for centuries in Hindu medicine for the same purpose. The tricyclic phenothiazines such as promethazine (Fig. 30-33) earlier had been found to have powerful **antihistamine** activity. It was the search for better antihistamine drugs that led to the synthesis of chlorpromazine.¹¹²² As many as 250 million people throughout the world were treated with chlorpromazine and related drugs in the 20 years following its discovery before newer and safer drugs (e.g., **clozapine**; Fig. 30-33) were developed.¹¹²³ What does chlorpromazine do? A possible clue comes from the fact that it sometimes induces serious “extrapyramidal” side effects including tremors and other symptoms of Parkinson disease. This suggested that chlorpromazine may block dopamine receptors in the corpus striatum, thereby precipitating a functional deficiency of dopamine.¹¹²⁴ If so, it is possible that schizophrenia may result from an overactivity of dopamine neurons, perhaps including some of the same neurons that are hypoactive in Parkinson disease. Supporting this view is the observation that amphetamines (Fig. 30-27), which may substitute for dopamine, worsen the symptoms of schizophrenia and in very high doses induce striking schizophrenialike symptoms in normal individuals.

A stereotyped compulsive behavior is induced both in humans and in laboratory animals by amphetamines. This provided the basis for a method that has been used to measure the action of drugs on amphetamine-sensitive centers of the brain. A lesion in the nigrostriatal bundle on one side of a rat brain was made by injection of a neurotoxic compound such as 6-hydroxydopamine. This caused degeneration of dopamine-containing neurons on one side of the brain. When rats that had been injured in this way were given amphetamines, they developed a compulsive rotational behavior. Administration of chlorpromazine and several other antipsychotic drugs neutralized this behavior and in direct proportion to the efficacy in clinical use, an observation that also supports the theory that schizophrenia involves overactivity of dopamine neurons.

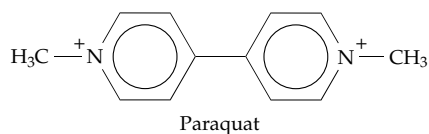
If schizophrenia results from an elevated dopamine content of the brain, the fault may lie with either an oversupply or a reduced rate of metabolism of dopamine. The possibilities of reduced activity of monoamine oxidase or of dopamine β-hydroxylase have both been suggested. The plasma level of the dopamine metabolite **homovanillic acid** (Fig. 30-26) is elevated in schizophrenia and is correlated with the

severity of the illness,¹¹²⁵ suggesting the hypothesis of a decreased rate of metabolism. Possible defects in dopamine receptors may be at fault.¹¹²⁶

Chlorpromazine may also act on brain cholinergic neurons.¹¹²⁷ Blockage of muscarinic acetylcholine receptors in the brain by belladonna alkaloids such as atropine (Fig. 30-22) has often been used in treatment of Parkinson disease. Apparently antagonizing acetylcholine action is to some extent functionally equivalent to increasing dopamine concentrations. There is evidence that suggests a role for cholecystikinin (CCK) in development of schizophrenia. CCK-containing neurons interact with dopaminergic neurons in the mid-brain.¹¹²⁸ GABA neurons in the prefrontal cortex may be faulty.¹¹²⁹ Excessive glutamate may also induce schizophrenia. The schizophrenia-like symptoms induced by phenylcyclidine (p. 1796) are eased by antagonists of metabotropic glutamate receptors. This suggests another possible therapy.¹¹³⁰ Among other suggested causes of schizophrenia are dysregulation by **retinoids**¹¹³¹ and action of retroviruses.¹¹³² Demyelination in portions of the prefrontal cortex may disrupt neural connectivity.^{1132a} Recent genetic evidence points to a possible defect in proline dehydrogenase which reduces Δ^1 -pyrroline-5-carboxylate to L-proline (Fig. 24-9).^{1132b}

Numerous theories of mental illness have embodied proposals that a toxic metabolite is produced in abnormal quantities. An example is **6-hydroxy-dopamine** (Fig. 30-26), which is known to damage dopamine-containing neurons.¹¹³³ Overactive methylation of catecholamines has also been suggested as a cause of mental disorders.¹¹³⁴ The hallucinogen **3,4-dimethoxyphenylethylamine** (Fig. 30-26) has been identified in urine during acute schizophrenic attacks, but the variability is so high that no definite conclusion has been reached. N-Methylation of serotonin yields **bufotenin** (N-methylserotonin) and **N-dimethylserotonin**, known hallucinogenic agents. Enzymatic synthesis of the latter by human brain and other tissues has been demonstrated,¹¹³⁵ and administration of tryptophan and methionine to schizophrenic patients exacerbates their illness.

Another theory of mental illness postulates endogenous alkaloid formation. Aldehydes formed by oxidation of catecholamines as well as formaldehyde and acetaldehyde are present in tissues in small amounts. Condensation with amines could generate Schiff bases and alkaloids as in Fig. 25-10. This "plant chemistry" is spontaneous and can apparently take place in the brain, where it may have a potent effect.



Incubation of tryptamine derivatives with 5-methyl-tetrahydrofolic acid and an enzyme preparation from brain gives **tryptolines**. Dopamine and its derivatives form related tetrahydroisoquinolines such as the product that arises from reaction with acetaldehyde (see Eq. 30-5). This product has been found in elevated amounts in alcoholics (who synthesize excess acetaldehyde), in phenylketonurics, and in L-dopa-treated patients with Parkinson disease.¹¹³⁶

Epilepsy. The brain disorders known as epilepsies affect 1–2 % of the population worldwide. Characteristic of epilepsies are recurrent **seizures**, sudden brief changes in behavior caused by the simultaneous, disordered firing of large numbers of neurons in the brain. Many seizures are thought to be initiated in specific areas of the cerebral cortex. For example, seizure-induced firing of neurons in the thumb area of the right motor cortex will be accompanied by rhythmic jerking in the left thumb. More than 40 different types of epilepsy are known.^{1137–1138a}

GABA is the principal inhibitor neurotransmitter, and one cause of epilepsy may be a deficiency in GABA formation from glutamate. The brain contains two isoforms of glutamate decarboxylase, designated GAD65 and GAD67, in accordance with their molecular masses in kDa. They are encoded by separate genes.^{1139,1140} GAD67 is formed mainly in cell bodies of neurons, binds its cofactor PLP tightly, and is essential to survival of young mice. GAD65 is associated mainly with nerve termini, where it is anchored, apparently by association with other proteins to the membranes of synaptic vesicles.¹¹⁴¹ It binds PLP weakly. Some convulsive agents such as 1,1-dimethylhydrazine are thought to act by interfering with PLP-dependent enzymes (Box 14-C) among which is GAD.^{1140,1142,1143} Convulsions are one of the most striking symptoms of a severe vitamin B₆ deficiency. A zinc deficiency can also cause convulsions, apparently because pyridoxine kinase is a Zn•ATP-requiring protein and the rate of synthesis of PLP is too slow to supply apo-GAD with the PLP needed for GABA synthesis. The PLP in GAD65 undergoes rapid substrate-dependent transamination to pyridoxamine phosphate (see Chapter 14), which must be replaced by new PLP.^{1143,1144}

Epilepsy may arise also from defects in a GABA transporter¹¹⁴⁵ or receptor.¹¹⁴⁶ One form of epilepsy is a triple-repeat disease of cystatin B (Table 26-4). Mutation in potassium channels,¹¹⁴⁷ glutamate receptors,¹¹⁴⁸ absence of neuropeptide Y,¹¹⁴⁹ and absence of L-isoaspartyl / D-aspartyl O-methyltransferase (Box 12-A)¹¹⁵⁰ have all been associated with epilepsy.

Neurodegenerative diseases. As many as 5% of persons of age 65 and 20% of those of age 80 are afflicted with the progressive senile dementia known

as **Alzheimer disease**. The condition is characterized by a gradual loss of memory and of the abilities to speak, think, or take care of one's self. Histologically Alzheimer disease is marked by the accumulation within neurons of **paired helical filaments**. These filaments, of 10 nm diameter, twist about each other to form a helix with an 80-nm pitch. The helices aggregate to create **neurofibrillary tangles**. The tangles are composed largely of a highly phosphorylated form of the microtubule-associated protein **tau** (p. 372)^{1151–1155a,b} together with phosphorylated neurofilaments, apolipoprotein E (p. 1183), and other materials. The tangles are found in the cell bodies, axons, and dendrites of neurons in the hippocampus, amygdala, cerebral cortex, and other areas of the brain. Tangles may also be present in Parkinson disease, in the nearly extinct **Guam disease**,^{1156,1157} and in some types of prion disease.¹¹⁵⁸ Outside of the diseased neurons are numerous, spherical **amyloid plaques**. Their principal component is a 40- to 43-residue fragment called **amyloid β -protein (A β)**, which appears to be toxic to neurons. A β is cut from a larger **amyloid precursor protein (APP)**.^{1154,1159–1160a} The APP gene is a member of a family of 16 related genes found in many organisms including nematodes, flies, and mammals. In humans the APP gene is found on chromosome 21, the chromosome that is present in three copies in **Down syndrome**. People with Down syndrome who live into their late thirties or beyond develop Alzheimer disease,¹¹⁵³ presumably from excessive synthesis of APP. Both APP and its cleavage product A β are formed by nonneuronal cells throughout the body. However, the A β plaques form only in the brain, and the APP gene is essential for life. The rare **familial British dementia** resembles Alzheimer disease in producing amyloid plaques and neurofibrillary tangles. They appear principally in the cerebellum and arise from a different precursor protein.^{1161,1162}

There are many other neurodegenerative diseases, some with a high incidence, and others rare. They include **Parkinson disease** (p. 1790), **Huntington disease** (Table 26-4), **spinal muscular atrophy (SMA)**; a leading hereditary cause of infant mortality),^{1162a,b} amyotrophic lateral sclerosis (**ALS**), prion diseases (Box 29-E), **ataxias**, and other diseases caused by triple-repeat DNA sequences (Table 26-4) and X-linked adrenoleukodystrophy (ALD; p. 945).¹¹⁶³ In the last, membrane function is disrupted. Although these diseases arise from a variety of causes many of them have in common amyloidosis, the deposition of insoluble proteins in or around neurons.^{1163a}

Parkinson disease, some cases of Alzheimer disease, and some types of prion disease are accompanied by the presence of **Lewy bodies** within the cytoplasm of neurons and also in nearby glia. These deposits consist largely of a dense core of fibrils of **α -synuclein**, a small 140-residue protein abundant in various parts

of the brain.^{1164,1164a,b} Mutations in the α -synuclein gene are associated with autosomal-dominant inheritance of early-onset Parkinson disease.^{1165,1166} Just as tau tends to be associated with microtubules, α -synuclein may function in cooperation with microfilaments.¹¹⁵⁸ Studies of an autosomal-recessive form of inherited juvenile Parkinson disease led to mutations in a large (>1 Mbp) gene on chromosome 6. It encodes the 465-residue **Parkin**.^{1167,1168} Parkin is an E3 ubiquitin ligase (p. 524), which ubiquitinates α -synuclein.^{1168,1169} This finding suggests that abnormally slow degradation of synuclein may be an important cause of Parkinson disease.

One of the triple-repeat polyglutamine diseases discussed in Chapter 26 (Table 26-4) is Huntington disease. The defective **Huntingtin** is a cytosolic protein that normally protects neurons but fails when the polyglutamine sequence becomes too long.^{1170,1171} Neurons of the cerebral cortex and striatum die, apparently by apoptosis.¹¹⁷² Huntingtin interacts with p53, with a CREB-binding protein, and with an EGF receptor suggesting that it functions in regulation of transcription.^{1172–1173a} One of the genes whose transcription is regulated is that of the neurotrophin known as **brain-derived neurotrophic factor (BDNF)**.¹¹⁷⁰

Many approaches have been taken in therapy of Parkinson disease. As mentioned on p. 1790 enhancing dopamine production by administration of L-dopa or by use of MAO inhibitors is a standard treatment. Experimental gene therapy with a glial cell line-derived neurotrophic factor also appears promising.^{1174,1175}

Another aspect of neurodegeneration involves oxidative damage. A clue comes from amyotrophic lateral sclerosis (**ALS**), which struck down the New York Yankees baseball player Lou Gehrig, after he had started 2130 consecutive games over a 15-year period. ALS (Lou Gehrig disease) is the most prevalent of more than 70 diseases that cause loss of motor neurons.¹¹⁷⁶ As pointed out on p. 1075, the cause of a rare hereditary form of ALS is a defect in superoxide dismutase, which appears to promote excessive formation of free radicals.¹¹⁷⁷ However, this interpretation is uncertain.¹¹⁷⁸ Parkinson disease induced by the compound MPTP (Eq. 30-4) may also arise as a result of free radical damage.¹¹⁷⁹ Among possible effects, MPTP may induce apoptosis.¹¹⁸⁰ Both oxidative damage and apoptosis may be factors in Alzheimer and other neurodegenerative diseases as well.^{1181,1182}

In every disease in which an abnormal protein is found there must be pathways of processing the protein to generate its functional form and pathways for degradation. These pathways are being investigated for all of the neurodegenerative diseases and none more intensively than for Alzheimer disease. The amyloid precursor protein APP is an integral

membrane glycoprotein with a large ~687-residue extracellular N-terminal portion, which resembles a cell surface receptor. It contains both a protease inhibitor-like domain and a zinc-binding region, which can be phosphorylated. It binds heparin and collagen as well as other proteins.^{1183,1184} Rare mutations that cause early-onset familial Alzheimer disease are found in the APP gene. Some of these mutations alter the regulation of pre-mRNA splicing. Splicing generates eight different APP isoforms vaying in length from 677 to 770 residues. The properties of the isoforms vary. For example, if the 18 residues of exon 15 are spliced out a new motif for posttranslational modification is created by fusion of exons 14 and 16. The newly created sequence ENEGSG is recognized by a xylosyltransferase, which initiates formation of the terminal unit for glycosaminoglycan formation. The resulting proteoglycan is known as **appican** (p. 1154).¹¹⁸⁵ The precursor protein APP is transported down axons to the nerve endings and is proteolytically cleaved to form the insoluble amyloid deposits. Alzheimer disease may occur when there is excessively rapid proteolysis of the precursor or if there is a failure to metabolize the amyloid protein.¹¹⁸⁶ The folding and glycosylation reactions of APP occur in the ER and the Golgi, but the major problem in Alzheimer disease appears to be in subsequent proteolytic processing, some of which may occur as the APP is being transported through the Golgi to the cytoplasmic membrane. The protein may be cleaved at three sites by enzymes known as α -, β -, and γ -secretases as is indicated in Fig. 30-34, in which the protein is represented as an unfolded "stick."^{1154,1187-1189a} Most of the mutations in APP that cause Alzheimer disease are near these three cleavage sites. As indicated in Fig. 30-34, cutting at the α site liberates into the extracellular space the large N-terminal portion as a soluble protein called APPs α .^{1190,1191} It is thought to have a protective effect on neurons. If this cleavage occurs the protein is not cut at the β site and fragment A β is not formed. However, if cleavage by β -secretase (beta-site APP-cleaving

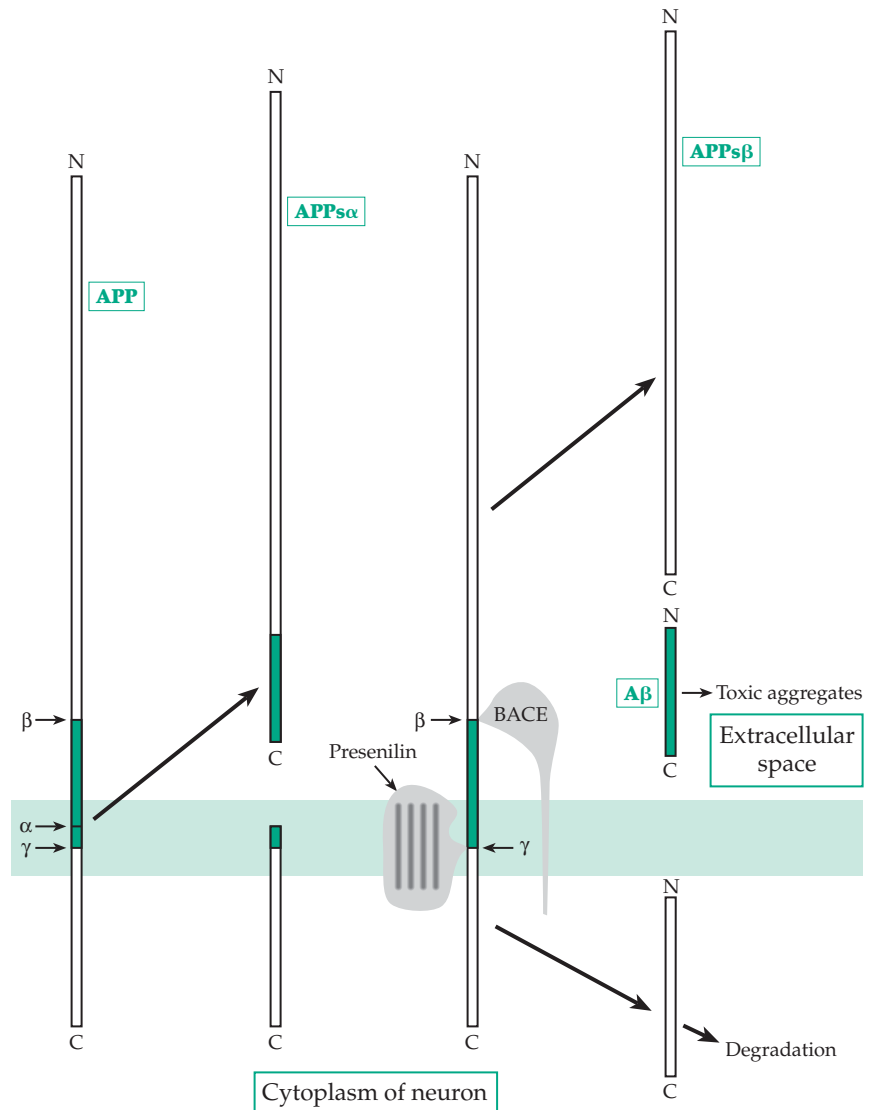


Figure 30-34 Cleavage of the amyloid precursor protein APP with liberation of amyloid A β protein. The proteins are represented as sticks (not to scale) but in reality contain both intracellular and extracellular globular domains.

enzyme or BACE) occurs first and is accompanied by or followed by cleavage at the γ -site, A β is liberated (Fig. 30-34). The β -secretase is an integral membrane protein, which carries a pepsinlike domain in its luminal (or extracellular) part.¹¹⁹²⁻¹¹⁹⁴ Since the A β peptide in an aggregated form appears to be toxic to neurons, a logical therapy for Alzheimer disease may be to block either the β - or γ -secretase.^{1195,1196}

The γ -secretase has been difficult to locate but has been identified as a result of other rare familial forms of Alzheimer disease. These are caused by mutations in genes for proteins known as **presenilin-1** (on chromosome 14) and **presenilin-2** (on chromosome 1).^{1188,1197,1198} The presenilins are integral membrane proteins with multiple transmembrane helices.

They have been regarded as regulators of γ -secretase, but there is much evidence that the presenilin molecules may be cleaved proteolytically and that the C- and N-terminal domains formed in this way may associate to form an unusual aspartyl protease. It, too, is a target for inhibitors.¹¹⁹⁸ The picture is made more complex by the fact that presenilins form complexes with other proteins. These include a newly discovered protein **nicastatin**,^{1199–1200a} a large 709-residue transmembrane glycoprotein. Nicastrin not only seems to modulate presenilin action but also participates in an important developmental process via the highly conserved **Notch pathway** (Chapter 32).¹²⁰⁰ Many other proteins are found in the amyloid plaques of Alzheimer disease. Among them are acetylcholinesterase,¹²⁰¹ proteoglycans,¹²⁰² hydroxyacyl-CoA dehydrogenase,¹²⁰³ GM1 ganglioside,¹²⁰⁴ apolipoprotein A-1,¹¹⁸⁷ and lithostatine.¹²⁰⁵

What are the possible adverse consequences of accumulation of the A β protein? It may cause inflammation by activation of **microglia**,¹¹⁵⁷ which may cause damage by release of NO.¹²⁰⁶ A β may induce death of neurons by apoptosis.^{1201,1207–1209} A defect in proteosomal degradation may be a factor.¹²⁰⁸ Both A β and the prion protein may promote oxidative damage. The brain derives most of its energy from oxidative metabolism, a major source of damaging radicals. Mitochondria are found in dendrites as well as cell bodies.¹²¹⁰ Methionine residues in glycine-rich parts of the A β and prion proteins are suspected as centers of free radical formation.^{1202,1211}

Both amyloid plaques and the tangles of protein tau-containing paired helical filaments are typically present in Alzheimer disease. Which comes first? Some hereditary neurodegenerative diseases are known in which tau filaments are present in neurons and sometimes also in glia.^{1203,1212,1213} Since mutations in tau don't lead to Alzheimer disease whereas mutations affecting APP do, it is often assumed that the primary defect in the disease is with APP and that accumulating A β induces the observed changes in tau. However, this is by no means certain. Six different isoforms of tau (the longest with 441 residues) are created by alternative splicing of the mRNA. During its normal functioning tau is phosphorylated and carries an average of 2–3 phospho groups. In Alzheimer disease the level of tau is greatly increased (4- to 8-fold) and the molecules carry 5–9 phospho groups.^{1204,1213a} It is this hyperphosphorylated tau that forms the paired helical filaments and tau tangles, which appear to clog the slender neurons.

What does tau do normally? Although it has been studied for many years, its exact functions are elusive. However, the role of the microtubules in axonal transport is well established. The tau isoforms may play a functional role in this process. The hyperphosphorylated tau of Alzheimer disease doesn't promote proper assembly of microtubules and may interfere with axonal transport of materials along the microtubules (see p. 1119).^{1214,1215} Alzheimer disease may reflect an imbalance between the phosphorylation and dephosphorylation processes. Another possible problem with tau may be slow isomerization of prolyl linkages because of a deficiency of a prolyl *cis-trans* isomerase (Box 9-F).¹²¹⁶

Until 1993 **apolipoprotein E** was best known for its central role in plasma lipoproteins and cholesterol transport (Fig. 21-1). However, one of the three common alleles of the apoE gene confers a significant risk of development of Alzheimer disease.^{1217,1218} A high blood cholesterol level is also correlated with increased risk.^{1219,1220} Membrane abnormalities in mitochondria have been associated with Alzheimer disease.¹²²¹ Also related to membranes and lipid metabolism, **vitamin E** appears to combat Alzheimer disease.^{843,1218}

Environmental and nutritional factors may also affect the development of Alzheimer disease and other mental illness. Aluminum frequently accumulates in the neurons containing neurofibrillary tangles.^{1206,1222} Copper and zinc ions can cause the amyloid A β to aggregate. However, Zn²⁺ may actually protect against neurotoxicity.¹²²³ The amino acid β -N-methylamino-L-alanine, a constituent of the toxic seeds of a type of palm (*Cycas circinalis* L.), may have induced both ALS and Guam disease, a condition resembling Parkinson disease, in a population in Guam that traditionally used these seeds as food.¹¹⁵⁷

Can neurodegenerative diseases be prevented or delayed? Much evidence suggests that the answer is yes. Rare early onset forms pose a special problem, but for most of us maintaining an active life style, using our minds, and choosing a good diet with adequate amounts of vitamins and essential ω 3 fatty acids (Box 21-B) may be very helpful.⁸⁴³ New methods of treatment are being tested. Antiinflammatory drugs are helpful,^{1218,1218a} and even vaccination against A β and other amyloid proteins appears possible.¹²²⁴ Is it possible that antibodies and phagocytic cells can clear the cobwebs from our brains?

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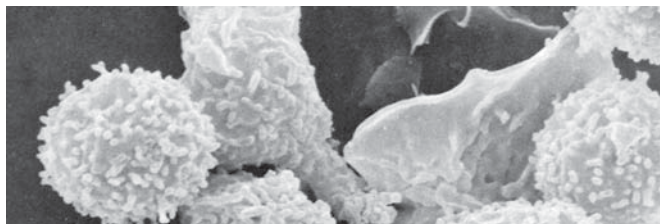
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Study Questions

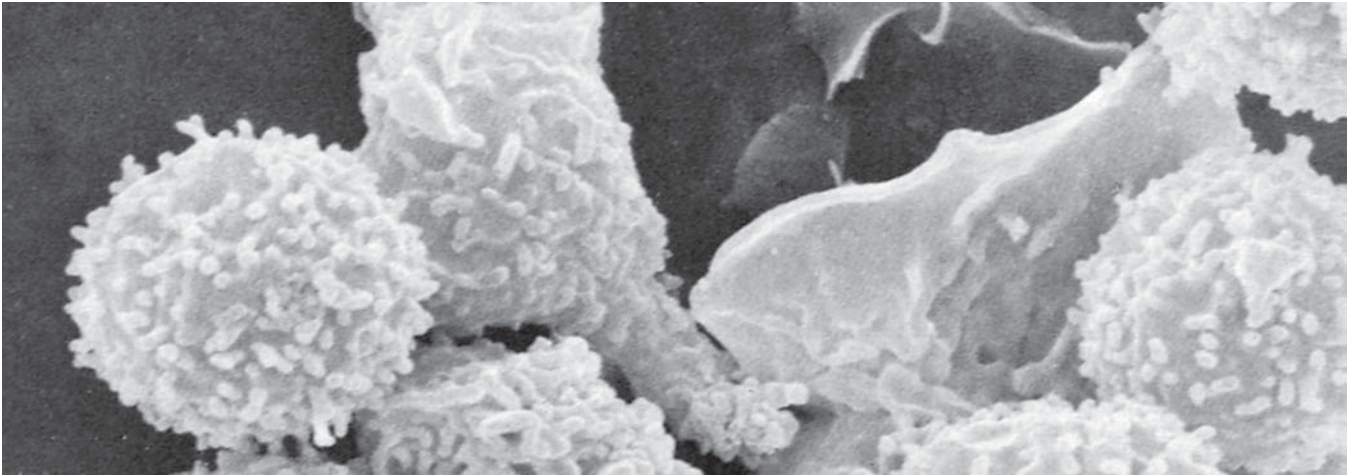
1. Compare the sensing of signals and responses to signals in liver, muscle, or other tissue with signaling in the brain.
2. Compare signaling by ionotropic receptors, metabotropic receptors and gap junctions.
3. List major neurotransmitters in the brain. In what other locations do these compounds function?
4. Compare addiction to gambling with addiction to cocaine. Are they similar on a biochemical basis? What about addiction to Internet games, chatting, pornography, compulsive overeating, etc.^{873a}
5. Can extracts of leaves of *Ginkgo biloba* counteract age-related neurological disorders?¹²²⁵



The large flat cell, a portion of which is seen here, is a macrophage which has ingested bacterial proteins and is displaying peptide fragments on its surface. Some of the small, spherical T lymphocytes (T cells) seen here, interact with the macrophage, recognize an antigen, and respond by becoming helper T cells. They can then stimulate B lymphocytes (B cells) to multiply and produce antibodies. See Fig. 31-11 for an enlarged view. Scanning electron micrograph courtesy of Morton H. Nielsen and Ole Werdelin, University of Copenhagen.

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Our bodies are under constant attack by viruses, bacteria, protozoa, and metazoan parasites. Persons born without an immune system adequate to fight off these invaders die very quickly unless heroic measures are taken. We have learned to cooperate with our immune systems by immunizing ourselves against some bacteria and viruses. At other times we may fight a stubborn battle with our own defense systems against allergic reactions and a variety of autoimmune responses.¹

A. Locations and Organization of the Immune System

The immune system has many components, many of which are dispersed throughout the human body (Fig. 31-1). Lymphocytes, which are a foundation of the immune system, constitute little more than 1% of the body's mass. However, this represents $\sim 10^{12}$ cells of several types, about 10 times more than there are neurons in the brain. These 10^{12} cells make antibodies and T-cell receptors, both of which are thought to have $\sim 10^{15}$ different peptide sequences.² And this is only the beginning of the complexities.³⁻⁸

Immune responses have often been described in terms of **humoral** and **cellular** components. The humoral response involves the small circulating **B lymphocytes** (B cells), the **antibodies** (immunoglobulins), and proteins of the **complement** system. The cellular response is mediated by another group of small lymphocytes, the **T lymphocytes** (T cells). They resemble B cells in appearance but have quite different functions. However, newer knowledge has provided a somewhat different description of the body's defense

systems, which can be classified into three levels. (1) The skin and internal mucous membranes, which are resistant to infection and have antibacterial properties, provide the first level of defense.^{2,9} (2) A fast-acting **innate immune system** can respond within a few minutes to breaches in the barriers provided by the tough outer skin and the glycoproteins of mucous surfaces and provides a second level.^{2,10-14} (3) A slower **adaptive** (acquired) part of the immune system leads to synthesis of antibodies and to long-term immunity, providing the third level (Table 31-1). Both B and T lymphocytes together with **antigen-presenting cells (APCs)** are necessary for the selection and development of immunoglobulin structures appropriate for attack on an invading organism.

The innate immune system utilizes **phagocytic cells** including neutrophils, monocytes, and macrophages¹⁵ to ingest and kill invading organisms. Basophils, mast cells,¹⁶ eosinophils, and other cells release inflammatory mediators, which attract additional lymphocytes and affect their development.¹⁷ Specialized T lymphocytes called **natural killer (NK) cells**¹⁸ may also attack foreign cells (Table 31-2). The innate system is ancient and has apparently evolved to recognize molecular structures that are foreign to the host but are characteristic of pathogens. These structures, which are described as **pathogen-associated molecular patterns (PAMPs)**, include those of lipopolysaccharides of bacterial cell walls (Fig. 8-30), mannans (p. 175),¹⁹ other carbohydrates of surface layers,²⁰ oxidized phosphatidylcholines,^{20a} bacterial flagellins,²¹ various posttranscriptionally modified proteins, teichoic acids (p. 431), etc. However, they do not include patterns characteristic of host cells. Those have been avoided during evolution of the system. Although

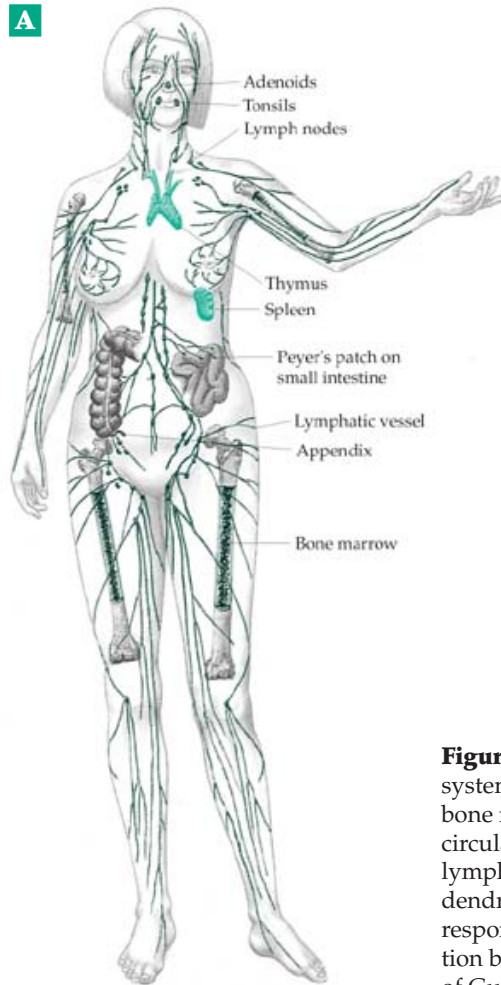
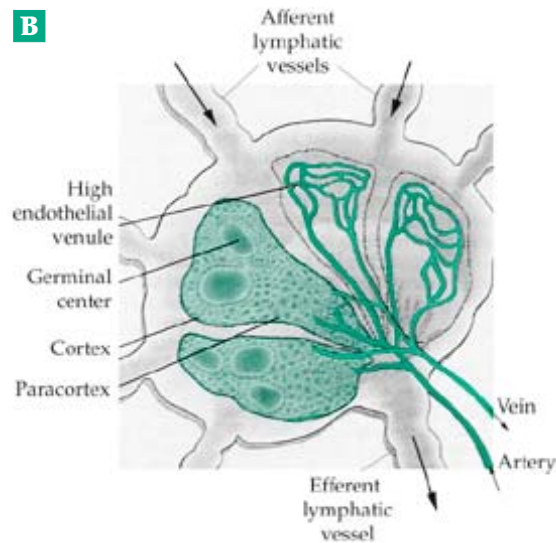
A**B**

Figure 31-1 (A) Locations of the primary and secondary tissues of the immune system. The primary lymphoid organs are the thymus, which makes T cells, and the bone marrow, which forms B cells. After moving from these organs into the blood circulation the cells reach one of the secondary lymphoid organs, which include lymph nodes, spleen, tonsils, and Peyer's patches on the small intestine. Immature dendritic cells are found in body tissues including skin and mucous membranes and respond to foreign proteins by inducing attack by T lymphocytes and antibody formation by B cells. (B) Schematic drawing of a lymph node. From Nossal.¹ Courtesy of Gustav J. V. Nossal.

T lymphocytes are major mediators of the innate immune response they are under control of the **dendritic cells (DCs)**, which are found in "immature" forms in tissues throughout the body.^{9,22–26a}

The immature DCs are phagocytic cells that act as "immunological sensors." They recognize various PAMPs, which act as **danger signals**,^{11b} using what are known as **toll-like receptors (TLRs)**.^{23,24,27,27a,b} They are also the most active APCs. Their proteasomes cleave proteins, both of the host and of invading organisms, into short peptides. These peptide fragments are displayed on the APC surfaces for recognition by T lymphocytes and for activation of adaptive immune responses. Some "autoreactive" B cells are also part of the innate immune system^{28,29} as are the IgA antibodies present in mucous membranes.³⁰ The innate immune system of insects resembles that of vertebrates. The toll-like receptors of the latter are named for their resemblance to the Toll receptors of *Drosophila*, which are utilized in resistance to fungi. In both mammals and insects the innate immune system activates responses via the transcription factor NF- κ B. However, many details of the signaling pathways differ.^{30a–c}

The innate system also provides for synthesis of small antibiotic peptides called **defensins**^{12,31–33} as well as larger proteins. Some of these proteins constitute the **complement system**, while others are described as **acute-phase reactants**. Some defensins are also **cytokines**, which attract lymphocytes.

The innate system is of special importance during early infancy. Prior to birth and for at least 4–12 months after birth a child's immune system is poorly developed. It may not become fully competent until age ~5.^{34,35} During the prenatal period maternal antibodies are transferred to the child. IgG crosses the placenta and enters the fetal circulation. Breast milk provides IgA, which remains largely in the child's gut, as well as other protective proteins. UNICEF and the World Health Organization recommend breast-feeding to two years or beyond.³⁴

While the innate immune system provides for immediate and direct attack on invaders, it also provides information to the slower adaptive system. Genes both for immunoglobulins and for the T-cell receptors of the adaptive system undergo extensive rearrangement during development of an individual.

TABLE 31-1
The Two Major Branches of the Immune System

	Innate (natural)	Adaptive (acquired)
Cells	Dendritic cells Phagocytic cells (neutrophils, monocytes, macrophages) Cells that release inflammatory mediators (basophils, mast cells, eosinophils) Natural killer (NK) cells	Dendritic cells B lymphocytes (B cells) T lymphocytes (T cells) Other antigen-presenting cells, e.g., macrophages
Molecular components	Antibacterial peptides (defensins, complement, acute-phase proteins)	Immunoglobulins
Receptor genes	Fixed in genome	Complement proteins Encoded in gene segments; rearrangement necessary
Recognition	Conserved molecular patterns	Small molecular groups (epitopes)
Immunogenic memory	Absent	Present
Self–nonself discrimination	Perfect	Imperfect
Action time	Immediate	Delayed

This provides potential defensive proteins directed at almost every imaginable invader. It also ensures that every individual has a set of proteins that labels its own cells as “self,” and that virtually every individual on earth has cell surface proteins different from those of every other person. In both the innate and adaptive responses the immune system must carefully distinguish “self” from “nonself.”^{36,37} In the innate system this discrimination developed during evolution of the host and its pathogens. In the adaptive system it depends upon interaction of the T cells with surface molecules, primarily those of the **major histocompatibility complex** (MHC).

Another basic characteristic of immune responses is the development of **immunologic memory**.^{38–40} This is exemplified by the fact that vaccination can sometimes impart immunity for a person’s lifetime. If a foreign protein is injected into an animal, after a lag period of 2–5 days the animal will synthesize antibodies against this foreign antigen. This is called a **primary adaptive immune response**. If after a few days or weeks a second injection of the same protein is made, a much more rapid synthesis of additional antibodies occurs. This **secondary immune response** may take place within hours and will last longer than

the primary response. It is a manifestation of immunologic memory.

1. Development of Lymphocytes and Other Specialized Cells

Both the B cells and T cells arise in the fetal liver or bone marrow (Fig. 31-1) from pluripotent stem cells. In birds the B cells develop in a special organ, the bursa of Fabricius. Mammalian B cells complete their differentiation into mature **lymphocytes** within the bone marrow. However, the T cells must travel to the **thymus**, where they complete their maturation. The T lymphocytes include the previously mentioned NK cells as well as the somewhat similar **cytolytic T cells** and **immunoregulatory T cells**. The latter are further characterized as **helper T cells**⁴¹ or **suppressor T cells**. The adaptive response requires cooperation of helper T cells in many instances. The mature B and T cells leave the bone marrow and thymus, which are known as the **primary lymphoid tissues**, and enter the blood circulation. Following “homing” signals⁴² they take up residence in a variety of locations

in the lymph nodes, spleen, adenoids, tonsils, and Peyer's patches. The last are small clusters of lymphoid cells in the wall of the intestine. All of these tissues, which are referred to as **secondary lymphoid tissues**, are the sites in which the adaptive immune system is developed.

TABLE 31-2
Cells of the Immune System^a

Type	Functions
B lymphocytes	
Plasma cells	Antibody synthesis
Memory B cells	Immunologic memory
T lymphocytes	
Cytolytic	Destroy infected and malignant cells
Helper cells	
Type 1 (T _H 1)	Participate in activation of B cells
Type 2 (T _H 2)	
Memory T cells	Immunologic memory
Natural killer (NK) cells	Destroy infected and malignant cells with pore-forming protein perforin and cytotoxic granules
Dendritic cells	
Interdigitating	Antigen recognition and processing
Follicular	Antigen presentation
Microglia	Defensive network in brain ^b
Cells that release inflammatory mediators	
Mast cells and basophils	Possess high-affinity receptors for IgE May secrete histamine, prostaglandins, leukotrienes Important in allergies
Eosinophils	Weakly phagocytic, secrete cationic proteins, reactive reduced oxygen species, leukotrienes, prostaglandins, cytokinins
Phagocytic cells	
Neutrophils	Acute inflammatory response
Monocytes	
Macrophages	Carry receptors for carbohydrates not normally exposed on surfaces of cells in vertebrates e.g., mannose; kill engulfed organisms with •O ₂ ⁻ , HOCl, NO, cationic proteins and peptides, lysozyme Antigen processing and presentation

^a General reference: Delves, P. J., and Roitt, I. M. (2000) *N. Engl. J. Med.* **343**, 37–49.

^b Streit, W. J., and Kincaid-Colton, C. A. (1995) *Sci. Am.* **273** (Nov), 54–61.

An important component of the immune system that was neglected until recently is located in the mucous membranes and the skin.^{9,30,43,43a} The mucosal surfaces of airways and the gastrointestinal tract provide the point of entry for many diseases. Both internal and external body surfaces are protected by

dendritic cells, whose immature forms in skin are called Langerhans cells. See figure in Box 8-F.^{22,26a}

Eight different types of cells of the immune system (Table 31-2) develop by differentiation of pluripotent stem cells^{44–48} as indicated in Fig. 31-2. Dendritic cells^{26a} (which are not shown in this figure) may be formed from monocytes but may also arise by other routes.²² The development of the various cells takes place under the influence of a number of **hemopoietic regulators**. Among these are the protein hormone **erythropoietin** and various interleukins and colony-stimulating factors.^{44,45,49}

2. Triggering an Immune Response

When a foreign antigen enters the body the B cells, with receptors of appropriate specificity and present in the lymph nodes, are stimulated to divide repeatedly and to produce a large clone of **plasma cells**. These contain a highly developed ER and actively synthesize and secrete immunoglobulins. One activated B cell may produce 10 million antibody molecules per hour.¹ This B-cell response occurs within a network of **follicular dendritic cells** in the **germinal centers** of the lymph nodes (Fig. 31-1B).^{2,50,51} The antibodies are “adaptor” molecules, which bind to antigenic proteins often on surfaces of invading microorganisms. Another part of the antibody binds to one of several **effectors** systems. These immobilize microorganisms, induce phagocytosis, activate the complement system, carry antibodies across placental membranes, etc.^{52,53}

Induction of a T-cell response is more complex and is very demanding.^{22,47,54} Antigenic peptide frag-

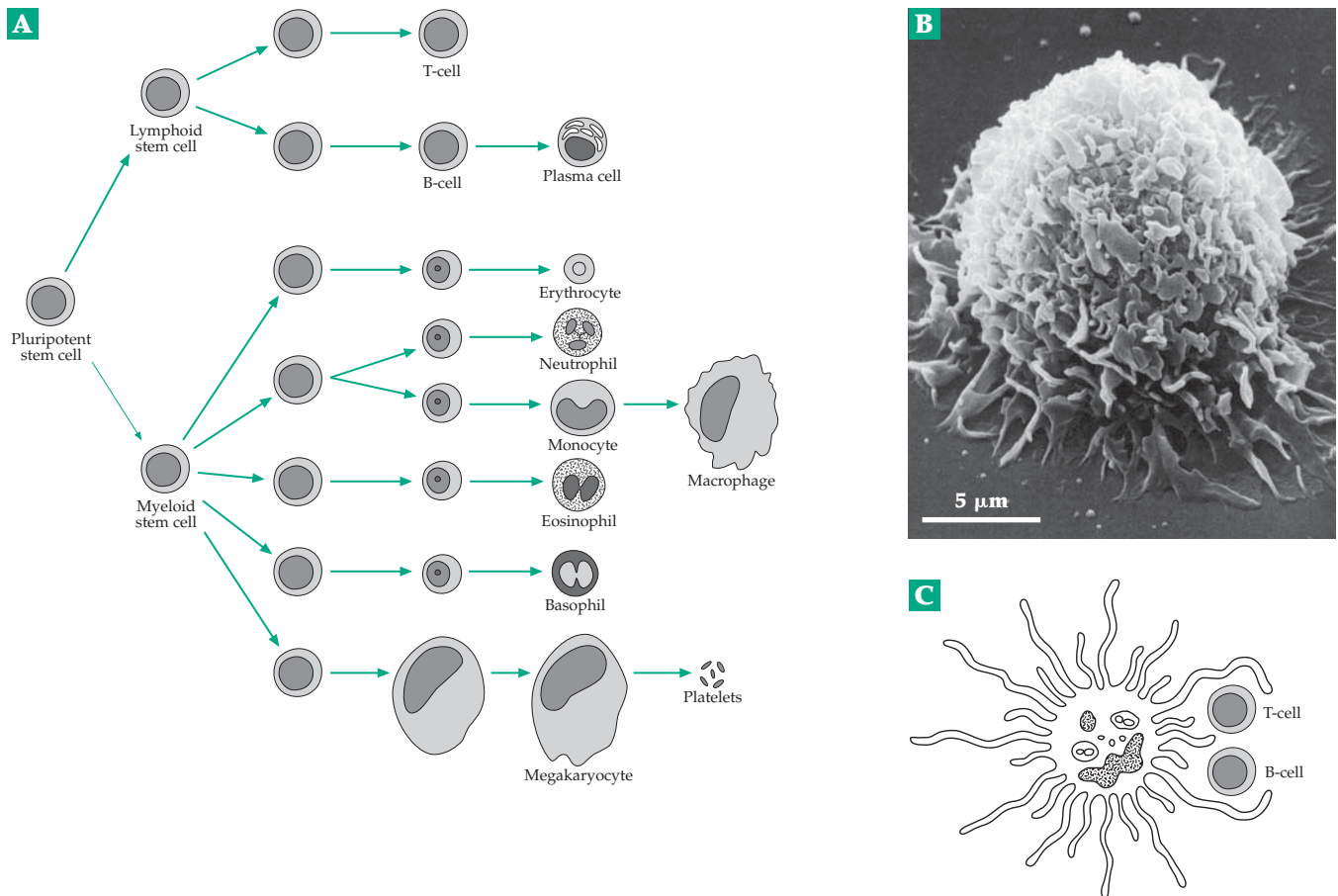


Figure 31-2 (A) Development of eight types of blood cells including those of the immune system from pluripotent (multipotential) stem cells. The cells develop under the influence of a variety of protein growth factors. Some steps, e.g., maturation of B cells, involve complex rearrangements in the DNA of the cell. (B) Scanning electron micrograph of a macrophage, a large motile cell that plays a key role in the immune system. It moves by means of its surface “ruffles.” It actively phagocytoses both pathogens and waste materials and is also one of the cells that releases the hormones known as colony-stimulating factors. Micrograph courtesy of Shirley G. Quam. Drawings courtesy of David W. Golde and Judith C. Gasson.⁴⁴ (C) Schematic drawing of a dendritic cell. Redrawn from Banchereau and Steinman.²² Both macrophages and dendritic cells present antigens for recognition by T cells and synthesize cytokines, which affect lymphocyte development.

ments from infected or malignant cells anywhere in the body must be recognized by an appropriate receptor on a T cell that is circulating in the blood. Only a few T cells with receptors for any given antigen specificity exist.^{54,55} The foreign antigen fragment must bind to a protein of the **major histocompatibility complex (MHC)** while present within a dendritic cell or other APC. The resulting MHC•antigen complexes pass through the ER and the Golgi to the outer cell surface in a rather complex process. Recognition of the antigen that is “presented” in this manner on the APC surface is accomplished with the aid of $\sim 10^{15}$ different receptor proteins (**T-cell receptors**) on the T-cell surfaces. When a T-cell receptor is occupied by an HMC•antigen complex of appropriate specificity the T cell is activated to participate in adaptive immunity. However, some T cells, notably $\gamma\delta$ T cells, like antibodies bind to antigen directly.^{56,56a,b} The recogni-

tion process occurs in an **immunological synapse**, which has elements of similarity to neurological synapses.^{54,57–58a}

B. The Immunoglobulins (Antibodies)

1. Molecular Structures

There are five classes of antibodies or immunoglobulins.^{59,60} The first three, IgG, IgM, and IgA, are quantitatively the most significant, but IgD and IgE are also important. For example, the content of IgE is elevated in allergic persons. The basic structure of all of the immunoglobulins is that of a quasi-symmetric dimer composed of a pair of light chains and a pair of heavy chains whose lengths vary among the different

Symbol	Mass (kDa)	Formula
IgG	150	$\kappa_2\gamma_2$ or $\lambda_2\gamma_2$
IgM	950	$(\kappa_2\mu_2)_{2,5}$ or $(\lambda_2\mu_2)_{2,5}$
IgA	320	$(\kappa_2\alpha_2)_n$ or $(\lambda_2\alpha_2)_n$
IgD	180	$\kappa_2\delta_2$ or $\lambda_2\delta_2$
IgE	190	$\kappa_2\varepsilon_2$ or $\lambda_2\varepsilon_2$

classes of immunoglobulins. Two classes of **light chains**, κ and λ , are found in human antibodies. The **heavy chains** are designated γ , μ , α , δ , and ε (see accompanying tabulation). Both IgM and IgA contain an additional J chain.

Treatment with mercaptoethanol splits the disulfide linkages holding the chains together, permitting preparation of monomeric light and heavy chains. When peptide chains of the immunoglobulins were hydrolyzed enzymatically, the resulting peptide fragments were found to be extremely heterogeneous. They were mixtures of many different kinds of peptides. The result was not unexpected, for it had long been recognized that the body contains millions of different antibodies, with binding sites specific for different antigenic determinants. It had been unclear how different binding sites could be formed, but the heterogeneity in amino acid sequence suggested the correct answer: Each antibody has its own sequence.

Progress toward understanding of the detailed structure of antibodies came when it was recognized that patients with tumors of the lymphatic system, e.g., the bone marrow tumors **multiple myeloma**, produced tremendous quantities of homogeneous immunoglobulins or parts thereof. Similar tumors were soon discovered in mice and provide a ready source of experimental material. The **Bence-Jones proteins** that are secreted in the urine of myeloma patients were found to be light chains of immunoglobulins. Sequence determinations showed that each Bence-Jones protein was homogeneous, even though no two patients secreted the same protein.^{61,62} Later, intact myeloma globulins and macroglobulins (IgM) of a homogeneous kind were also obtained.

The first complete amino acid sequence of an IgG molecule was announced in 1969.⁶³ The protein contained 446 amino acids in each heavy chain and 214 in each light chain. The longer heavy chains of IgM molecules contain 576 amino acids.⁶⁴ In all of the immunoglobulins the heavy and light chains are held together by disulfide linkages, and the chains are folded into loops to form compact domains. The IgM molecule is polymerized through additional disulfide linkages to form a pentamer readily visible with the electron microscope (Fig. 31-3). The heavy chains also carry oligosaccharide units. In IgM there are five of these, as indicated in Fig. 31-4A. They contain mannose and *N*-acetylglucosamine units linked to asparagine. Other immunoglobulins (IgA, IgE, and IgG)

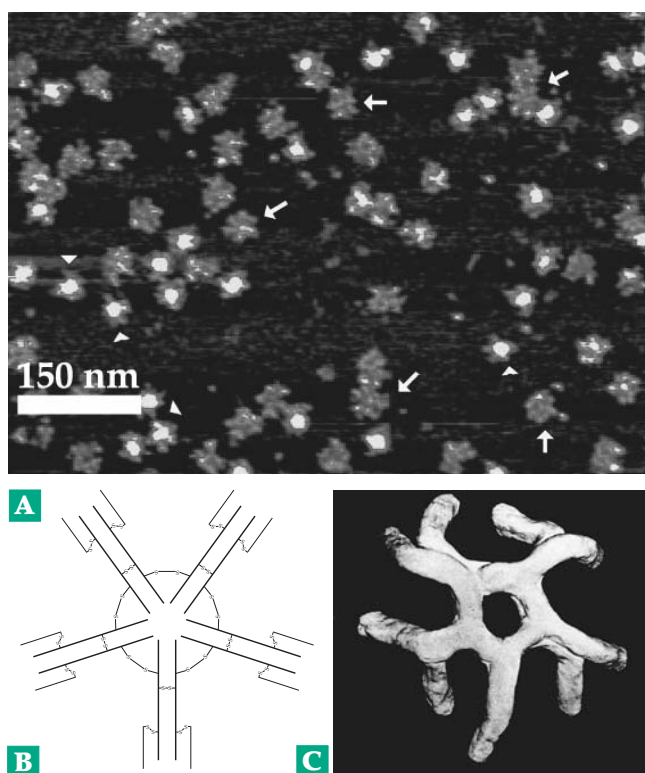


Figure 31-3 (A) Cryo atomic force (AFM) micrograph of molecules of the human immunoglobulin IgM. Courtesy of Zhifeng Shao, University of Virginia. (B) Schematic diagram. One-fifth of this structure is shown in greater detail in Fig. 31-4A. (C) Model based on earlier electron microscopic images. From Feinstein and Munn.^{64c}

contain fucose, galactose, and *N*-acetylneuraminic acid as well. In fact, almost all of the most important macromolecules that participate in innate and adaptive immune responses are glycoproteins.^{64b}

Digestion of an intact molecule of IgG with papain cleaves both heavy chains in the hinge region near the interchain disulfide bridge. This splits the molecule into three parts; two **Fab** (antibody-binding) **fragments**, each containing the N-terminal end of a heavy chain together with a linked light chain, and an **Fc fragment**. Even before it was known that IgG could be split into two Fab fragments, the antibody was known to be divalent, i.e., capable of binding with two different antigens (Fig. 31-4). The shape and overall structure of IgG molecules have been verified by electron microscopy and numerous X-ray diffraction studies.

Sequence determinations showed that in some regions of immunoglobulin molecules there is extreme variation in the amino acid sequence between one homogeneous antibody and the next; other regions have a constant sequence. The molecule can also be divided into domains. The **variable regions**, which occupy the N-terminal ends of the chains, are designated

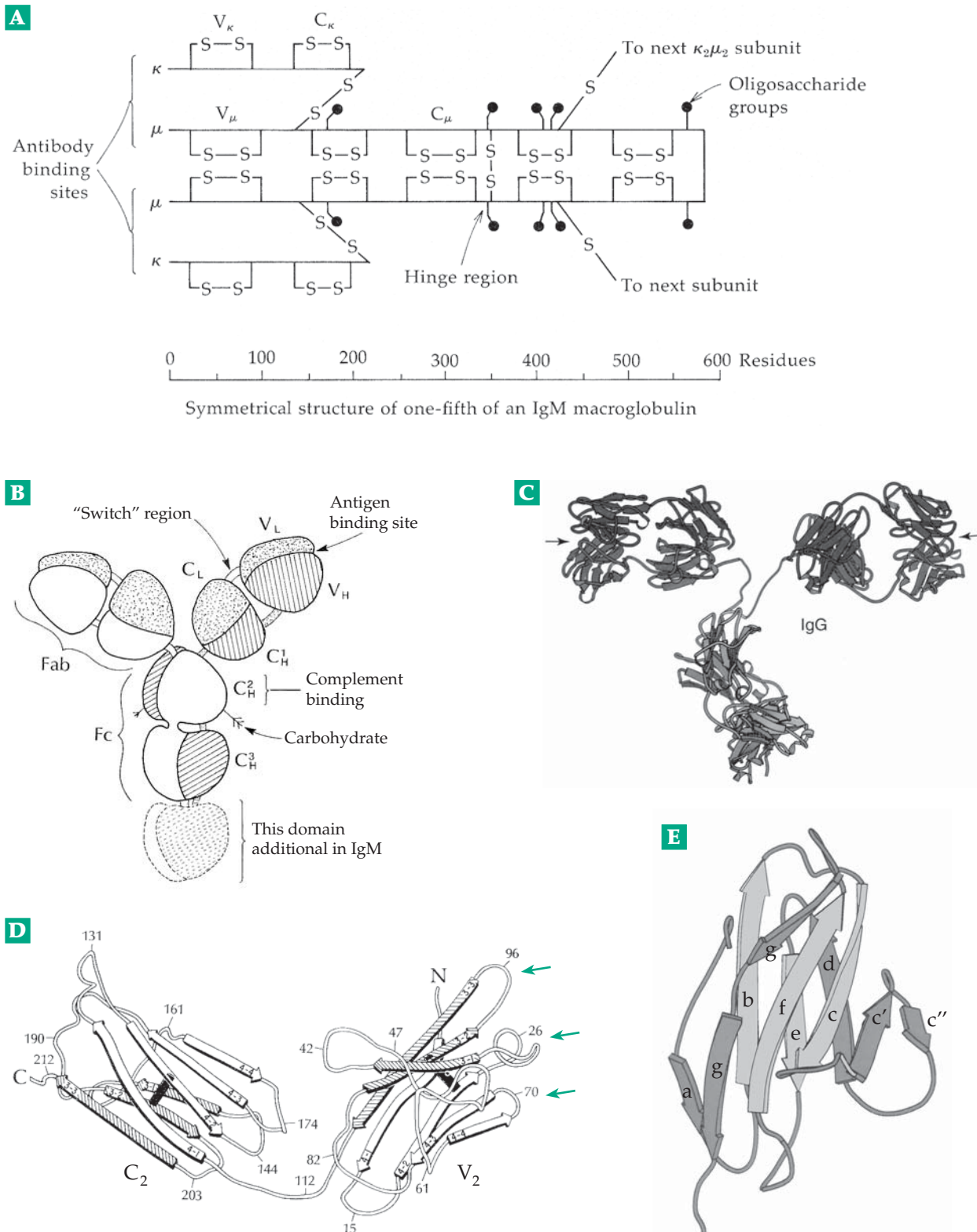


Figure 31-4 Schematic structure of one-fifth of an IgM molecule. From Putnam *et al.*⁶⁴ (A) Covalent structure. (B) Schematic three-dimensional representation. (C) Ribbon diagram of an IgG molecule. From Cochran *et al.*^{64a} (D) Folding patterns of one chain in a constant and a variable domain of a Bence-Jones protein. From Schiffer *et al.*⁶⁶ Green arrows indicate hypervariable regions. (E) MolScript drawing of the common core structure of Ig-like domains. The lighter shaded strands (b, c, e, f) form the core common to all Ig-like domains, which is surrounded by structurally more varied additional strands (darker). The front sheet has up to five strands (a, f, c, e, c'') and the back sheet up to four (a, b, e, d). Strand c'' is very flexible and is not always a part of the β sheet. From Bork, Holm, and Sander.⁶⁵ See also Fig. 2-16.

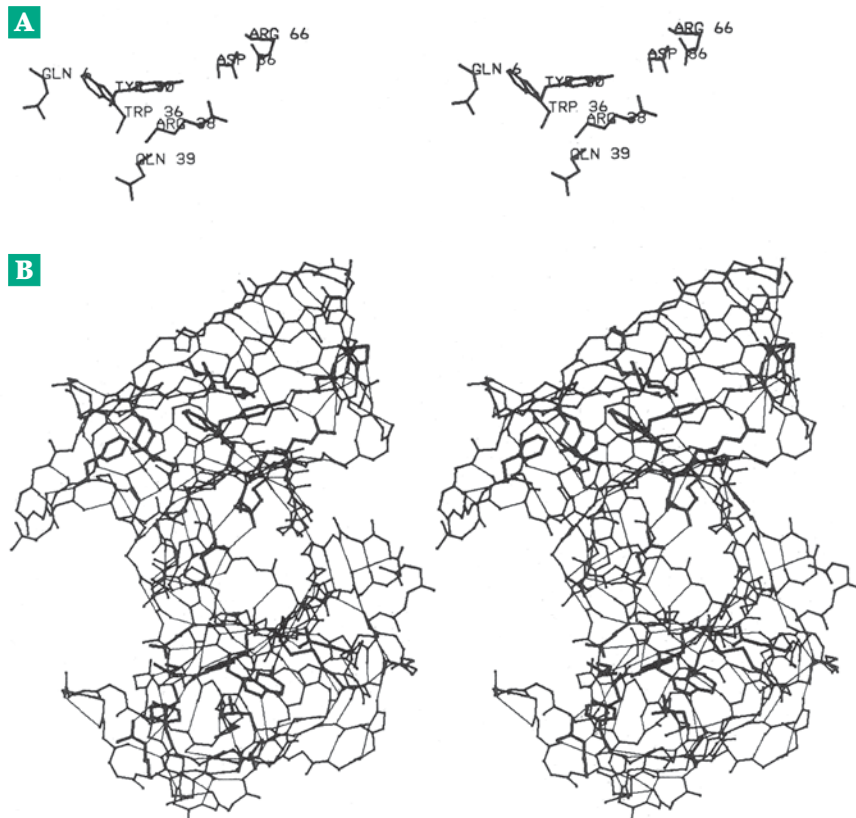


Figure 31-5 The extensive conserved hydrogen-bonding pattern in an immunoglobulin variable domain provided by polar residues buried inside the V_L and V_H domains. (A) To facilitate orientation, prominent side chains are displayed here and identified by names and numbers in the same orientation as in (B). (B) Polypeptide chain backbones of both domains are denoted by heavy lines and hydrogen bonds by light lines. In addition to the regular interbackbone hydrogen-bonding network characteristic of antiparallel β -sheets, there are hydrogen bonds provided by side-chain atoms. Note the two hydrogen bonds of Gln-38 (V_L) and Gln-39 (V_H) that span the domain-domain interface. From Novotny and Haber.⁷⁵

V_L and V_H for the light and heavy chains, respectively. The **constant regions** are C_L and C_H. Examination of the C_H region showed that much of the sequence is repeated after ~ 110 residues. In the IgG molecule the constant region of the heavy chains is made up of three such homologous domains (C_H1, C_H2, and C_H3). A fourth C_H domain is present in IgM. These facts suggest that duplication of a smaller gene coding for about 110 amino acids took place in the evolution of the immunoglobulins. Within the variable regions of immunoglobulin chains are **hypervariable regions** that form the antigen binding sites. These regions are located at the ends of the Fab fragments and involve both the light and heavy chains.^{66a}

Within all of the domains each of the two peptide chains is folded in a similar way. Seven extended lengths of chain form two mostly antiparallel sheets between which hydrophobic side chains are packed. The overall size of the unit is $\sim 4.0 \times 2.5 \times 2.5$ nm. An S-S bridge links the two sheets in the center of each domain. The folding patterns in the variable domains are somewhat more complex. Different domains are linked by segments of extended peptide chain known as hinge or switch regions.⁶⁷⁻⁶⁹ These impart a segmental mobility, which seems to be important for functioning of the molecules.⁷⁰

The exact mode of binding to Fab fragments has been established for several specific **haptens**. Haptens are small molecules having the binding properties

of antigenic determinants but unable by themselves to induce formation of antibodies when injected into animals. Binding of the hapten phosphocholine to one Fab fragment and of vitamin K to another⁶⁷ involves the hypervariable regions of both the heavy and light chains. The same is true for the binding of lysozyme⁷¹⁻⁷³ and a bacterial oligosaccharide.⁷⁴ The binding sites for the haptens or for the antigenic determinants on larger antigens are largely within the nine-stranded elliptical barrel or β sandwich formed from the two β sheets (Fig. 31-5).^{69,75} Four strands come from the V_L domain and five from the V_H domain. The barrel forms the bottom and sides of the antibody-binding site, which can also be viewed as consisting of six separate loops of peptide chain.⁷⁶ IgA molecules are similar to those of IgG but have structurally different hinge regions as well as an extra 18-residue tail-piece at the C terminus of each heavy chain.⁷⁷

As shown in Fig. 31-5, which provides a three-dimensional view of the variable domain of an Fab fragment of an immunoglobulin, there is a conserved hydrogen-bonded network even in this region. There are also "framework residues," which are highly conserved.⁷⁸ The antibody-binding site is provided largely by the three hypervariable regions present in each of the V_L and V_H domains. These are usually referred to in current literature as **complementarity-determining regions (CDRs)**.⁷⁸⁻⁸¹ Each pair of heavy and light chains is held together by a conserved

disulfide bridge.⁸² Three-dimensional structures of a substantial number of different Fab fragments have provided precise knowledge about the antibody-binding cavities and about the forces involved in binding.^{78,79,81,83} Among the established structures are those of Fab fragments specific for the following antigens: the haptens *p*-azophenylarsonate⁸⁴ and phencyclidine (p. 1798),⁸⁵ a sweet-tasting hapten,⁸⁶ triple-stranded DNA,⁸⁷ a DNA photoproduct,⁸⁸ creatine kinase,⁸⁹ staphylococcal nuclease,⁹⁰ an HIV capsid protein,⁹¹ and an EGF receptor.⁹² Structures are also known for single-domain antibodies from camels and llamas. These antibodies are naturally lacking in heavy chains but have single chains that fold back to mimic two-chain Fab fragments.^{93,94} Similar single-chain antibody domains have also been created artificially.⁹⁵

Not all proteins bind to antibodies in the usual binding cavity. **Protein G**, a cell surface protein from *Streptococcus* bonds to IgG molecules from many different species. Its binding site is on the outer surface of the heavy chain C_H1 domain.⁹⁶

2. Antigenicity

Antibodies often bind haptens or complete antigens very tightly. The association constants K_f observed for monoclonal antibodies (Box 31-A) range from 10^6 to 10^{12} M⁻¹.⁹⁷ However, most natural antibodies have a lower affinity for their antigens. When protein antigens are denatured, the binding constants often decrease by 10^{-4} to 10^{-5} . This suggested that only antigenic determinants of relatively rigid structures serve as good antigens. However, when the reaction of antibodies with proteins of well-established three-dimensional structure were studied, it was found that the best antigenic determinants are those with some segmental mobility.^{98,99} Furthermore, while some small peptides are good antigens, peptides are most highly antigenic when they can readily fold into a bend or other definite conformation.¹⁰⁰ Good antigenicity apparently requires some segmental flexibility as well as a definite conformation for the antigenic determinant.

3. Responses to Antibody Binding

Both B cells and T cells circulate throughout the body, spending only about 30 min during each cycle. They may meet and bind to an antigen in one of several different places.⁵⁰ Lymphocytes, which encounter blood-borne pathogens, usually initiate an immune response in the spleen. Responses to microorganisms in tissues are usually generated in lymph nodes. Ingested pathogens activate lymphocytes in specialized epithelial **microfold (M) cells** from which the antigen

is transported to the Peyer's patches. Responses to inhaled or intranasal pathogens arise in the tonsils and adenoids. In every case one major aspect of the immune response results from binding of antibodies to antigens.

Antibodies by themselves do not destroy bacteria or viruses, but they induce responses that do. One immediate effect of antibodies is to remove offending materials or cells from circulation. When multivalent antibodies each combine with two different cells **agglutination** occurs. The agglutinated cells or multicellular organisms can then be destroyed by phagocytes. The coating of a cell surface by IgG is one form of a process called **opsonization**, a process that marks the cell as foreign and a target for phagocytosis.^{3,53} Antibody-antigen interactions trigger several other responses as well. One of these results from the binding of protein **C1q**, a component of complement. Complement consists of a series of blood proteins that is poised to respond and to *complement* the action of antibodies in a variety of ways that are described in Section C.2 (see also Figs. 31-8 and 31-9). It has been established that it is the C_H2 domain of the Fc region of IgG that binds to C1q.¹⁰¹ The binding occurs only after antigen (but not a small hapten) binds to the immunoglobulin.

Complement C1q is only one of several types of **Fc receptor**.^{53,102,103} Others are involved in antigenic stimulation of B and T lymphocytes, macrophages, polymorphonuclear lymphocytes, and mast cells. Binding of the antibody-antigen complex to the receptors on phagocytic cells induces phagocytosis and release of oxygen metabolites, leukotrienes, prostaglandins, and other mediators of inflammation. The Fc domain mediates the uptake of antibodies from the mother's milk by young rats.^{104,105} It also is the binding site of antibodies to **protein A**, a constituent of the cell wall of *Staphylococcus aureus*,¹⁰⁶ which is also widely used as a tool in immunological studies (Box 31-C). The neonatal Fc receptor, which is related structurally to Class I MHC antigens (Section D.5), is one of three major types of Fc receptor. The other two are the receptors for Fc γ (of IgG) and Fc ϵ (of IgE). They (like their ligands) are members of the immunoglobulin superfamily. An exception is Fc ϵ R2 (also called CD23), which resembles a C-type lectin. Some Fc receptors, e.g., Fc γ R1 (CD64) and Fc ϵ R1, have a high affinity for their ligands with $K_d \sim 10^{-8}$ to 10^{-10} M. Others, such as Fc γ R2 (CD32) and Fc γ R3 (CD16), have lower affinities with $K_d \sim 10^{-5}$ to 10^{-7} M.¹⁰³ Three-dimensional structures of several Fc receptor fragments, some in complexes with Fc fragments (Fig. 31-6), are known.^{53,103,107-109} These include both IgG and IgE receptors.

It may be worthwhile to recall that many quite different proteins are members of the immunoglobulin structural family (Fig. 2-16). These include proteins

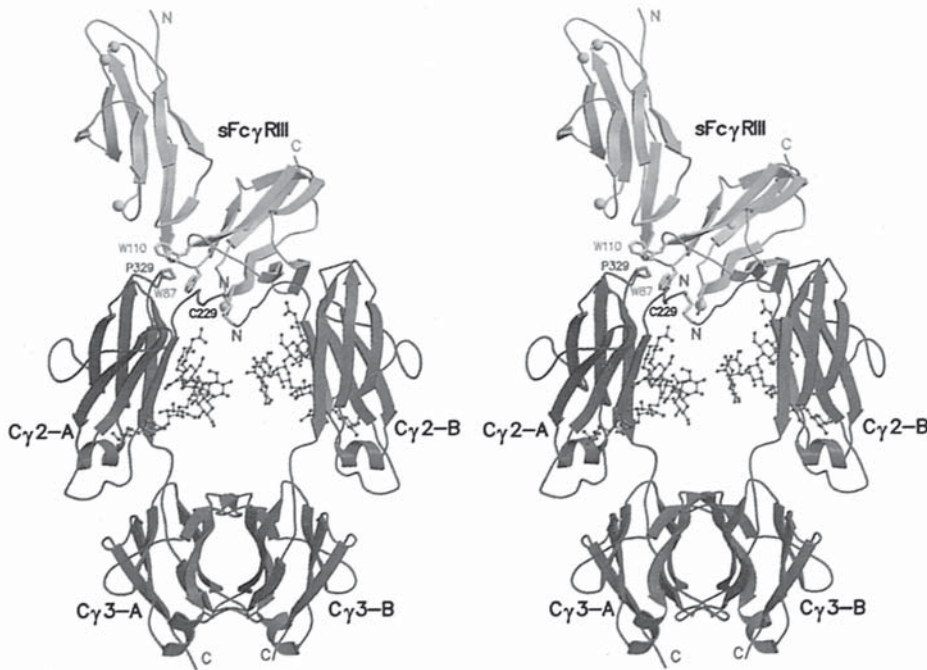


Figure 31-6 Three-dimensional ribbon representation of the structure of a complex of a soluble Fc fragment of a human IgG1 molecule. Pro 329 of the IgG and Trp 87 and Trp 110 of the Fc-receptor fragment form a “proline sandwich,” which is shown in ball-and-stick form. The oligosaccharide attached to the Fc fragment of the antibody and the disulfide bridge between the two Cys 229 residues (at the N termini of the C2 domains of the heavy γ chains) are also shown. The small spheres on the Fc receptor fragment are potential sites for N-glycosylation. From Sondermann *et al.*¹⁰⁷ Courtesy of Uwe Jacob.

encoded by 64 genes of *E. elegans*, an organism that doesn't form antibodies.¹¹⁰

4. Clonal Expansion of B Cells; Plasma Cells

The immunoglobulins are synthesized both by plasma cells and by their precursors, the B cells. Each B lymphocyte makes antibodies of specific sequence in two forms, secreted and membrane-bound or **antigen receptor**. Mature “virgin” B cells, which are responsible for the primary immune response, make largely monomeric IgM and some IgD.¹¹¹ It isn't clear why IgD should be the predominant surface immunoglobulin on most B cells.¹¹² The secreted and membrane-bound (receptor) antibodies differ in their C-terminal sequences but are otherwise the same. A B cell responds to the binding of an antigen with a shape complementary to that of its IgM antigen receptor by multiplying and differentiating. Some of the progeny B cells start to divide and begin to differentiate into clones of plasma cells that secrete IgG or into cells of the gut that secrete IgA. Some of the B cells give rise to **memory cells**, long-lived lymphocytes that can be triggered into rapid proliferation many years later if the same antigen is encountered. The B cells also undergo a shift to secretion of pentameric IgM rather than to synthesis of membrane-bound antibody.

5. Help from T Cells

The maturation of B cells is a complex process that requires the cooperation of helper T cells. The B cells

must process some antigen and present the peptide fragments for recognition by the MHC complex and T-cell receptors as described in Section D. If the antigen is recognized as foreign, the differentiation goes forward with the activated T cells secreting lymphokines that promote B-cell growth.

Before activated lymphocytes start to divide, interesting surface phenomena occur. If fluorescent antigens are allowed to bind to a lymphocyte, the cell surface is seen to be relatively evenly covered with the antibody–antigen complexes. Then after a short time the antibodies aggregate to form “patches” and begin to migrate to one side of the cell, where they eventually form a “cap.” At still longer times the cap material is engulfed by the lymphocytes. Perhaps this phenomenon simply reflects the oriented flow of liquid within membranes (Chapter 8). On the other hand, the membrane-bound immunoglobulins, like other cell surface receptors, are integral membrane proteins whose cytoplasmic C termini may be attached on the inside of the cell to the cytoskeleton, which may control the capping process. The binding of lectins sometimes triggers lymphocytes into antibody synthesis, but it is not clear how the binding of a lectin to a carbohydrate receptor can have the same effect as binding of an antigen to a surface IgM.¹¹³

C. Some Specialized Proteins of the Immune System

The immune response depends not only upon recognition of foreign antigens but also upon an extensive signaling network and upon a series of specialized

BOX 31-A MONOCLONAL ANTIBODIES

A mouse may make over 10 million different antibodies. Because of this heterogeneity it was impossible to learn antibody structures until the discovery of the myeloma proteins (Fig. 31-4D). These were produced in the bone marrow by clones of specific immunoglobulin-forming malignant cells. However, it was still not possible to obtain homogeneous antibodies to any desired antigenic determinant. The discovery of a method of forming such **monoclonal antibodies** by Milstein and Köhler^{a-c} in 1975 provided a new tool with many biochemical and medical applications.^{d-f} What Milstein and Köhler did was to immunize mice against an antigen of interest. They then fused B cells from the spleen of the immunized mouse with cultured myeloma cells. The resulting **hybridomas** grow vigorously and produce antibodies of the type dictated by the B cells. Since each hybridoma cell is derived from a single B cell, it makes a single kind of antibody. By plating out and selecting clones of hybridoma cells it is often possible to find a monoclonal antibody that binds well to a specific antigenic determinant. The hybridoma can be cultured indefinitely, producing its monoclonal antibody in any desired quantity.

A major application of monoclonal antibodies is in clinical assays for drugs, bacterial and viral products, tumor antigens, hormones, and other circulating proteins. Their use in conjunction with immunoassays (Box 31-C) has provided increased specificity and sensitivity. Another major application is to observe binding of antibodies to specific proteins by electron microscopy. The location of specific receptor proteins can be established^{g-j} as can the locations of ribosomal proteins and many other cellular components (Fig. 29-1). Monoclonal antibodies to acetylcholine receptors have been shown to induce symptoms of myasthenia gravis (Box 31-D), supporting the autoimmune origin of that disease.^h Monoclonal antibodies specific for such a small hapten as mercuric ion have been isolated.^k

Several problems have limited the wider use of monoclonal antibodies created by the hybridoma method. The antibodies are those of a mouse and are antigenic to humans.^{f,l-n} This long prevented many medicinal uses. Years of effort have gone into

attempts to “humanize” the antibodies. One approach is to introduce human immunoglobulin genes into mice. Another is to use recombinant DNA techniques to clone genes for immunoglobulin fragments and to introduce these into cells of *E. coli* in which additional genetic diversity in the antibodies arises.^{l,o} Selection of antibody fragments is often accomplished using bacteriophage display systems (Fig. 3-16).^{f,l} After selection gene fragments can be reassembled into a final form. Recently, using cloning of large pieces of the several Mbp of human immunoglobulin genes into yeast artificial chromosomes (p. 1497), it has been possible to prepare purely human monoclonal antibodies.^{f,n}

Many attempts have been made to link monoclonal antibodies specific for antigenic determinants on cancer cells to protein toxins such as ricin (Box 29-A). It is hoped that this may provide an effective way of carrying toxins into cancer cells.^{f,p-r} Therapeutic human monoclonal antibodies are already in use as antirejection drugs for kidney transplantation, for treatment of rheumatoid arthritis, Crohn disease, and for some types of cancer.^f

^a Milstein, C. (1980) *Sci. Am.* **243**(Oct), 66–74

^b Milstein, C. (1986) *Science* **231**, 1261–1268

^c Köhler, G. (1986) *Science* **233**, 1281–1286

^d Yelton, D. E., and Scharff, M. D. (1981) *Ann. Rev. Biochem.* **50**, 657–680

^e Birch, J. R., and Lennox, E. S., eds. (1994) *Monoclonal Antibodies*, Wiley-Liss, New York

^f Ezzell, C. (2001) *Sci. Am.* **285**(Oct), 36–41

^g Greaves, M. F., ed. (1984) *Monoclonal Antibodies of Receptors: Probes for Receptor Structure*, Chapman & Hall, London

^h Tzartos, S. J. (1984) *Trends Biochem. Sci.* **9**, 63–67

ⁱ Harlow, E., and Lane, D. (1999) *Using Antibodies: A Laboratory Manual*, Cold Spring Harbor Lab. Press, Cold Spring Harbor, New York

^j Goldman, R. D. (2000) *Trends Biochem. Sci.* **25**, 593–595

^k Wylie, D. E., Lu, D., Carlson, L. D., Carlson, R., Babacan, K. F., Schuster, S. M., and Wagner, F. W. (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 4104–4108

^l Marks, C., and Marks, J. D. (1996) *N. Engl. J. Med.* **335**, 730–733

^m Neuberger, M. S. (1985) *Trends Biochem. Sci.* **10**, 347–349

ⁿ Neuberger, M., and Brüggenmann, M. (1997) *Nature (London)* **386**, 25–26

^o Plückthun, A. (1990) *Nature (London)* **347**, 497–498

^p Collier, R. J., and Kaplan, D. A. (1984) *Sci. Am.* **251**(Jul), 56–64

^q Pastan, I., and FitzGerald, D. (1991) *Science* **254**, 1173–1177

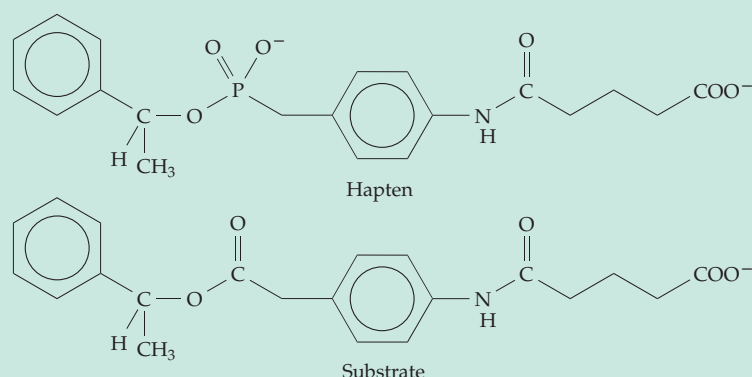
^r Oeltmann, T. N., and Frankel, A. E. (1991) *Trends Biochem. Sci.* **5**, 2334–2337

antibiotics, toxins, and hormones. Some of these, such as the defensins, are major, rapid-acting components of the innate system. Both the innate and adaptive systems utilize the complement proteins, and both

employ numerous cytokines and other signaling proteins. Plants, and perhaps also other organisms, employ gene silencing by small RNA molecules as part of their defense against viruses.^{113a,b}

BOX 31-B CATALYTIC ANTIBODIES

Both enzymes and antibodies are proteins. Antibodies consist of subunits with multiple domains, just as do some enzymes. Both enzymes and antibodies have binding sites for small molecules between domains or subunits. In view of such similarities it isn't surprising that some antibodies have catalytic properties. The possibility was suggested in 1969 by Jencks.^a He also proposed that injection of a mouse with a hapten, that resembled a transition state for an enzyme, might induce formation of antibodies complementary to the transition-state structure. These might be catalytic. By the early 1980s such antibodies were discovered.^{b–d} Some of the first catalytic antibodies (also dubbed **abzymes**) had esterase activity. The haptens used to induce antibody formation were phosphonates such as the following.^{e,f}



Using the transition-state analog shown on p. 485 a catalytic antibody with chorismate mutase activity was isolated.^g Many antibodies catalyzing additional reactions have also been found. Although they are usually less active than natural enzymes, in some cases they approach enzymatic rates. Furthermore, they may catalyze reactions for which no known enzymes exist.^h

Catalytic antibodies, like enzymes, must be isolated and purified to homogeneity before they can be studied. Initially this was done by using the hybridoma technique for isolation of monoclonal antibodies (Box 31-A). After induction of antibody formation by injecting a selected hapten into a mouse, large numbers of monoclonal antibodies had to be tested for catalytic activity. Even if several thousand different monoclonal antibodies were tested, only a few with catalytic properties could be found.ⁱ Newer methods have incorporated recombinant DNA techniques (Box 31-A) and use of combinatorial libraries and phage display.^{j–m} Incorporation of acidic or basic groups into the haptens used to induce antibody formation may yield antibodies capable of mimicking the acid–base catalysis employed by natural enzymes.^{n,o}

A sample of the types of reaction for which catalytic antibodies have been discovered or designed include the following: ester hydrolysis,^e transesterification,^p amide hydrolysis,^q serine protease-like hydrolysis,^r elimination,^{h,s} aldol cleavage,^t decarboxylation,^{u,v} deiodination by a selenium-containing antibody,^w pericyclic rearrangements,^{g,x} and the Diels–Alder reaction.^{y,z} Like natural enzymes catalytic enzymes can be mutated and engineered and can be used to study fundamental aspects of catalysis.^{aa} Fluorescent probes incorporate near active sites may provide information about mechanisms or may signal information of diagnostic significance.^{bb}

In science we must always expect the unexpected. Do antibodies all catalyze the reaction of singlet molecular oxygen $^1\text{O}_2^*$ with H_2O to form H_2O_3 and H_2O_2 ? How?^{cc}

1. Defensins and Other Antibacterial Polypeptides

Only higher vertebrates have an adaptive immune system with circulating antibodies. However, from bacteria to higher plants and human beings all of us utilize defensive polypeptides for protection. More than 500 have been identified.¹¹⁴ Many have a broad specificity, attacking both bacteria and other pathogens. Among these peptides are more than 200 bacterially produced antibiotics such as gramicidin,

tyrocidines, and colicins (Boxes 20-G, 8-D). More recently discovered are the 37- to 59-residue **bacteriocins**, formed by lactic acid bacteria.¹¹⁵ Like colicin E1 (Box 8-D) and alamethicin (p. 1774) they disrupt cytoplasmic membranes of some other groups of bacteria.

Helicobacter pylori, which is associated with stomach ulcers, forms a 38-residue antibiotic that may help protect infected persons from other bacteria.¹¹⁶ This peptide forms a simple two-helix structure and is one of a large number of simple helical antimicrobial polypeptides 40 residues or less in length. Among them

BOX 31-B (continued)

- ^a Jencks, W. P. (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York (p. 288)
- ^b Lerner, R. A., and Tramontano, A. (1987) *Trends Biochem. Sci.* **12**, 427–430
- ^c Benkovic, S. J. (1992) *Ann. Rev. Biochem.* **61**, 29–54
- ^d Lerner, R. A., Benkovic, S. J., and Schultz, P. G. (1991) *Science* **252**, 659–667
- ^e Wedemayer, G. J., Wang, L. H., Patten, P. A., Schultz, P. G., and Stevens, R. C. (1997) *J. Mol. Biol.* **268**, 390–400
- ^f Janda, K. D., Benkovic, S. J., and Lerner, R. A. (1989) *Science* **244**, 437–440
- ^g Haynes, M. R., Stura, E. A., Hilvert, D., and Wilson, I. A. (1994) *Science* **263**, 646–652
- ^h Larsen, N. A., Heine, A., Crane, L., Cravatt, B. F., Lerner, R. A., and Wilson, I. A. (2001) *J. Mol. Biol.* **314**, 93–102
- ⁱ Tawfik, D. S., Zemel, R. R., Arad-Yellin, R., Green, B. S., and Eshhar, Z. (1990) *Biochemistry* **29**, 9916–9921
- ^j Posner, B., Smiley, J., Lee, I., and Benkovic, S. (1994) *Trends Biochem. Sci.* **19**, 145–150
- ^k Janda, K. D., Lo, L.-C., Lo, C.-H. L., Sim, M.-M., Wang, R., Wong, C.-H., and Lerner, R. A. (1997) *Science* **275**, 945–948
- ^l Gao, C., Lavey, B. J., Lo, C.-H. L., Datta, A., Wentworth, P., Jr., and Janda, K. D. (1998) *J. Am. Chem. Soc.* **120**, 2211–2217
- ^m Baca, M., Scanlan, T. S., Stephenson, R. C., and Wells, J. A. (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 10063–10068
- ⁿ Kemp, D. S. (1995) *Nature (London)* **373**, 196–197
- ^o Thorn, S. N., Daniels, R. G., Auditor, M.-T. M., and Hilvert, D. (1995) *Nature (London)* **373**, 228–230
- ^p Wirsching, P., Ashley, J. A., Benkovic, S. J., Janda, K. D., and Lerner, R. A. (1991) *Science* **252**, 680–685
- ^q Thayer, M. M., Olender, E. H., Arvai, A. S., Koike, C. K., Canestrelli, I. L., Stewart, J. D., Benkovic, S. J., Getzoff, E. D., and Roberts, V. A. (1999) *J. Mol. Biol.* **291**, 329–345
- ^r Zhou, G. W., Guo, J., Huang, W., Fletterick, R. J., and Scanlan, T. S. (1994) *Science* **265**, 1059–1064
- ^s Romesberg, F. E., Flanagan, M. E., Uno, T., and Schultz, P. G. (1998) *J. Am. Chem. Soc.* **120**, 5160–5167
- ^t Karlstrom, A., Zhong, G., Rader, C., Larsen, N. A., Heine, A., Fuller, R., List, B., Tanaka, F., Wilson, I. A., Barbas, C. F., III, and Lerner, R. A. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 3878–3883
- ^u Hotta, K., Lange, H., Tantillo, D. J., Houk, K. N., Hilvert, D., and Wilson, I. A. (2000) *J. Mol. Biol.* **302**, 1213–1225
- ^v Smiley, J. A., and Benkovic, S. J. (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 8319–8323
- ^w Lian, G., Ding, L., Chen, M., Liu, Z., Zhao, D., and Ni, J. (2001) *J. Biol. Chem.* **276**, 28037–28041
- ^x Driggers, E. M., Cho, H. S., Liu, C. W., Katzka, C. P., Braisted, A. C., Ulrich, H. D., Wemmer, D. E., and Schultz, P. G. (1998) *J. Am. Chem. Soc.* **120**, 1945–1958
- ^y Romesberg, F. E., Spiller, B., Schultz, P. G., and Stevens, R. C. (1998) *Science* **279**, 1929–1933
- ^z Heine, A., Stura, E. A., Yli-Kauhaluoma, J. T., Gao, C., Deng, Q., Beno, B. R., Houk, K. N., Janda, K. D., and Wilson, I. A. (1998) *Science* **279**, 1934–1940
- ^{aa} Romesberg, F. E., Santarsiero, B. D., Spiller, B., Yin, J., Barnes, D., Schultz, P. G., and Stevens, R. C. (1998) *Biochemistry* **37**, 14404–14409
- ^{bb} Simeonov, A., and 14 other authors. (2000) *Science* **290**, 307–313
- ^{cc} Wentworth, P., Jr., Jones, L. H., Wentworth, A. D., Zhu, X., Larsen, N. A., Wilson, I. A., Xu, X., Goddard, W. A., III, Janda, K. D., Eschenmoser, A., and Lerner, R. A. (2001) *Science* **293**, 1806–1811

are the **cecropins** of insects and **magainins** and **buforins** of amphibians.^{117,117a} Many of these kill by disrupting membranes or by forming pores in membranes. However, others enter bacteria and disrupt functions of nucleic acids, enzymes, etc.^{117,118} Many antibacterial peptides have been isolated from insects,^{12,119} scorpions,¹²⁰ spiders and horseshoe crabs,^{121,122} and amphibians.¹²³ All of these organisms lack adaptive immunity but have strong innate immunity.

The human body is protected by two groups of defensins formed in the skin, in mucous membranes, in secretions of neutrophils, and other phagocytic cells. The α -defensins (Fig. 31-7) are 29–35 residues in length and are active against both gram-positive and gram-negative bacteria as well as fungi and enveloped viruses including HIV.^{12,124–125a} The β -defensins are mainly active against gram-negative bacteria and yeast. They also possess immunostimulatory activity that is important in activating the adaptive immune response.^{32,125,126} Various tissue-specific defensins have been discovered.¹²⁷ Peptides of the **trefoil family** protect the gastrointestinal tract.^{128–130} Eosinophils,¹³¹ leukocytes, and neutrophils make additional

protective proteins. One leukocyte defensin is a macrocyclic peptide, whose gene may have arisen by fusion of two segments encoding nonapeptide segments of α -defensins.¹³² Neutrophils form, in addition to defensins, α -helical peptides called **cathelicidins**, which protect skin from invasive bacterial infection.^{133,133a} Their synthesis is greatly increased after wounding. They may be among the proteins whose absence after severe burning is likely to be fatal.

Both α and β -defensins consist largely of β strands (Fig. 31-7) and are linked by three disulfide bridges. Some scorpion and insect defensins resemble scorpion toxins (Fig. 30-16) and have four S–S bridges. Fungi and green plants^{135,136} also form antimicrobial peptides. A 30-residue fungal protein is highly knotted and contains four S–S bridges.¹³⁷ Some polypeptides from the oleander and related plants are 29- to 31-residue macrocyclic structures with two S–S bridges in a **cysteine knot** structure^{138–140} (Fig. 31-7). They are exceptionally stable and protease-resistant and may have defensive activity against insects. Defensins are small polypeptides, but larger proteins are also part of the innate defense system. For example, a

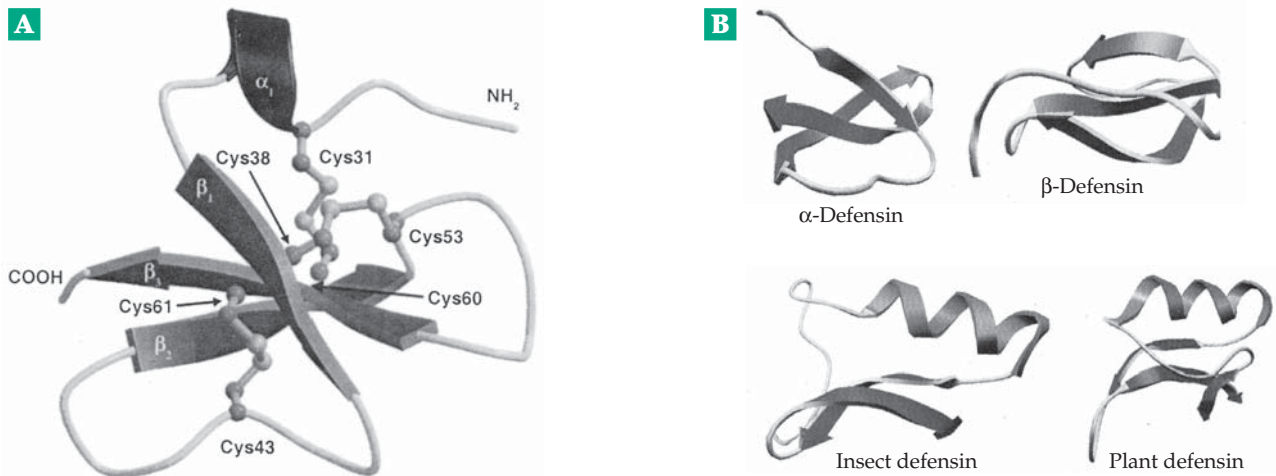
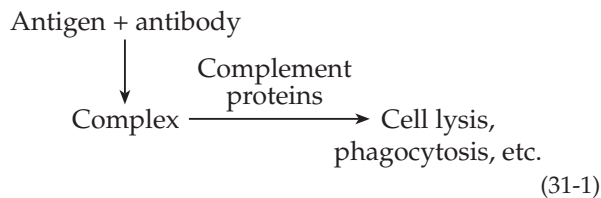


Figure 31-7 Ribbon structures of some defensins. (A) Structure of a human β -defensin showing the three disulfide bonds. From Bauer *et al.*¹³⁴ Courtesy of Heinrich Sticht. (B) Comparison of the folding patterns of four types of defensins. Mammalian α - and β -defensins are all β sheets with somewhat different arrangements of disulfide bridges. Insect and plant defensins have an α helix joined to the β sheet. Mammalian and insect defensins have three disulfide bridges, while plant defensins have four. From Hoffmann *et al.*¹² Courtesy of Jules A. Hoffmann.

93-residue protein from onion seeds resembles plant in lipid-transfer proteins.¹³⁵ Some frog skins contain a 60-residue trypsin inhibitor.¹⁴¹ Ribosome-inactivating proteins are well known (Box 29-A).

2. Complement

Complement is a group of more than 30 proteins found in blood serum, which are activated in a cascade mechanism when antibody and antigen combine^{10,111,142–148} (Eq. 31-1). This **classical pathway** for activation of complement is outlined in Fig. 31-8. The proteins involved in the cascade are designated C1 to C9. Many of them undergo proteolytic cleavage, the



cleavage products being designated by a or b, e.g., C3a and C3b. The b fragment is usually the larger of the two. There is also an **alternative pathway** that is part of the innate system. It is activated by such foreign surfaces as lipopolysaccharides of bacterial cell walls. Its special proteins are called **factors**, e.g., factor B, factor D. A third pathway, the **lectin pathway**, is activated by microbial surface mannans, which bind to a serum **mannan-binding lectin (MBL)**. This protein, a so-called defense collagen, resembles protein C1q (next paragraph).¹⁴⁹ It activates two associated

serine proteases (MASP1 and MASP2), which are able to cause cleavage of proteins C4 and C2 and possibly C3 in the classical pathway (Fig. 31-8). The ultimate effects of the action of complement include destruction of cells by lysis and activation of leukocytes, which engulf foreign cells by phagocytosis. Complement also induces the release of **chemotactic factors** that attract polymorphonuclear leukocytes and monocytes to the site involved.¹⁵⁰

The classical pathway begins with the **recognition component C1** of complement. This is a complex of three proteins, C1q, C1r, and C1s. Proteins C1r and C1s form a mixed tetramer C1r₂s₂, while C1q binds to the C_H² domain of “activated” antibodies, that is, with immunoglobulins that have combined with an antigen. It takes at least a dimer or larger aggregate of IgG to activate C1q, whereas a single molecule of the naturally pentameric IgM suffices. The mechanism by which this activation occurs is uncertain. Perhaps a change of conformation within the immunoglobulin accompanies antibody binding and is responsible for generation of a binding site for C1q. It may seem strange that haptens cannot cause complement binding, and that they do not cause detectable conformational alterations in Fab. Only multivalent antigens able to bind to more than one antibody induce complement binding. However, as we have learned in recent years, many biological responses involve transient assembly of large aggregates of different protein components. In this context, the requirement for two or more antibody molecules doesn’t seem so strange.

The 400-kDa C1q consists of a central portion of diameter 3–6 nm and length 10–12 nm to which are attached six very thin connecting strands. These are

~14.5 nm long and ~1.5 nm in diameter and terminate ~135-residue globular ends of ~6 nm diameter,^{151–153} which are thought to be the sites of combination with the immunoglobulins. The thin connecting strands have, for most of their length, a collagenlike structure with a high content of hydroxyproline and hydroxylysine. The latter is glycosylated by glucosylgalactosyl disaccharides as in collagen itself (pp. 181, 432, 433). The reason for this unusual structure is not obvious. We do know that the binding of antigens activates the complement-binding regions of antibodies, and that the activated antibodies then bind C1q. This binding in some manner activates C1q, which in turn activates C1r subunits of the C1r₂s₂ tetramer.¹⁵⁴ The latter is thought to bind at the center of C1q, while the antibodies bind at the outer ends. We don't know how the

activation message is carried from the outer arms to the center. C1q is a member of a group of collagen-like proteins that includes protein MBL (also designated MBP) and surfactant protein A (SP-A; p. 436).¹⁵⁵

Activated C1r (often designated $\overline{\text{C1r}}$ but here and in Fig. 31-8 as **C1r**) is one of five different serine proteases involved in activation of complement.¹⁵⁶ The substrate for the trypsinlike **C1r** is C1s, a proenzyme which is converted by the action of **C1r** into another trypsinlike serine protease **C1s**.^{157–160} Through a rather elaborate cascade mechanism, depicted in Fig. 31-8, the important proenzyme C2 is activated.¹⁶¹ Its active form **C2a** is a serine protease, which cleaves proteins C3 and C5 to the active forms C3b and C5b. Protein C4 is also cleaved to C4b by activated C1. C4 and C3 are also activated, and protein C5 is cleaved

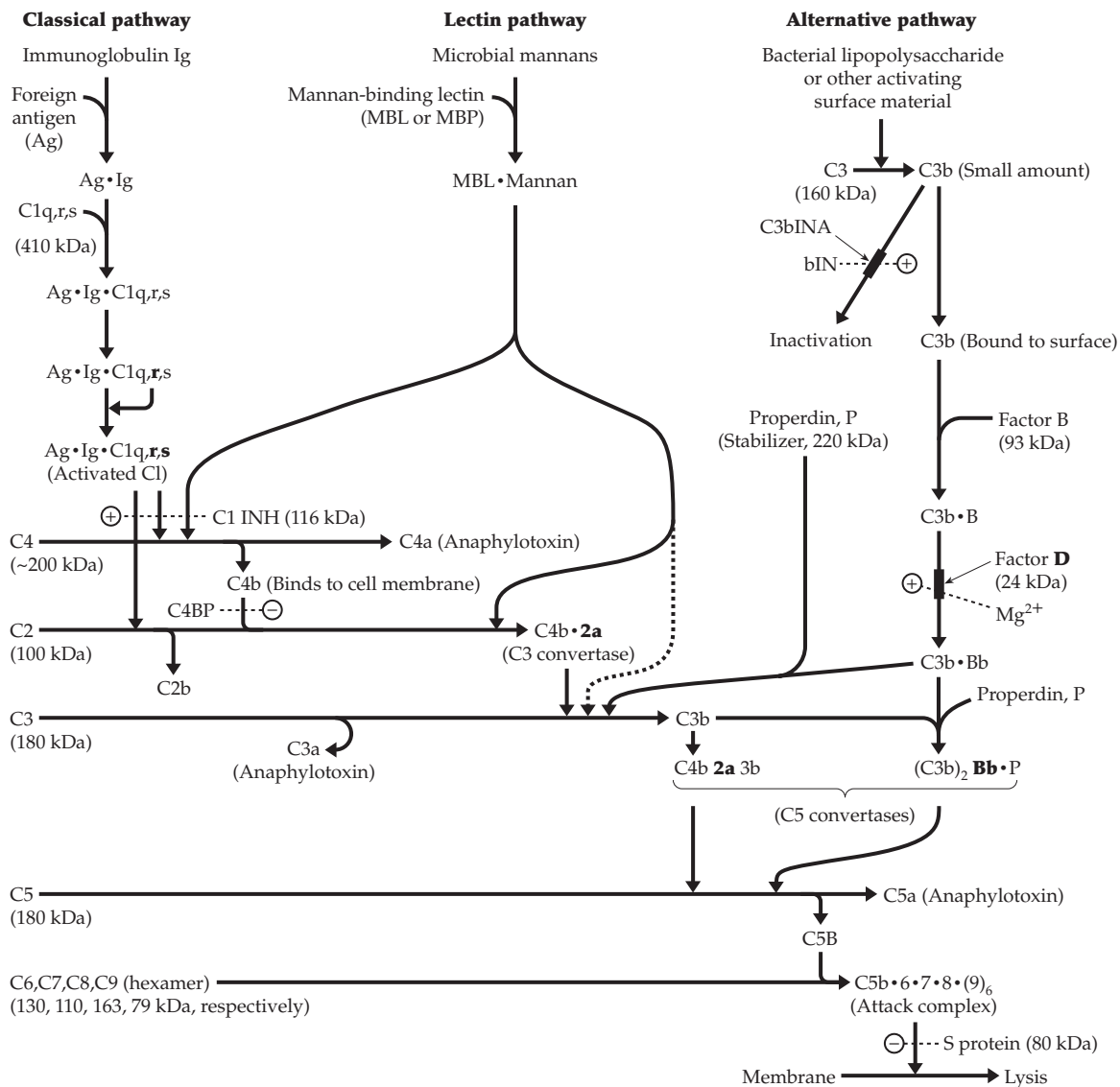


Figure 31-8 Pathways for activation of the complement system. Active proteases are designated by abbreviations in boldface.

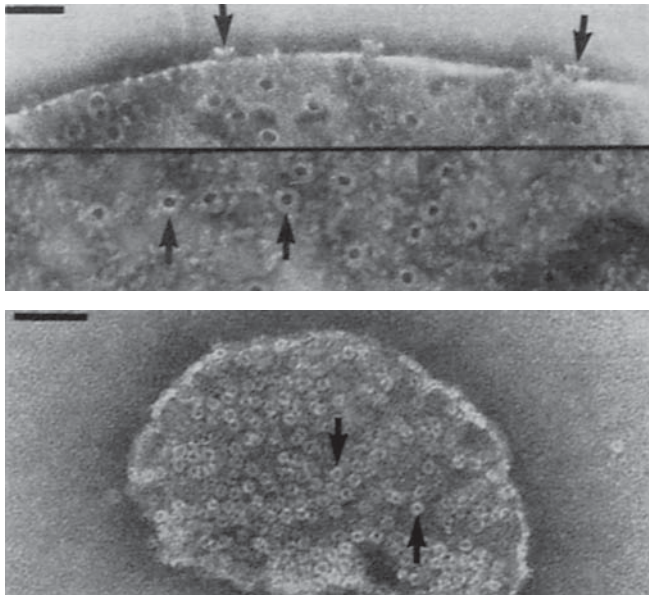


Figure 31-9 Electron micrograph of a negatively stained sheep erythrocyte lysed with human complement. The cylindrical “attack complex” embedded in the membrane is seen in the upper left frame in side projection and in the lower frames in axial projection. The top views are of a proteolytically “stripped” ghost, the lower view from a freshly lysed ghost. The inner diameter of the cylinders is 10 nm; scale bars 50 nm. From Bhakdi and Trantum-Jensen.¹⁶⁹

to C5b. The 200-kDa C4 consists of three chains, all derived by proteolytic cleavage of a precursor. It is also glycosylated and sulfated. Both C4 and C3 contain internal thiol ester linkages and act as “molecular mousetraps” (Box 12-D).¹⁶² They react to fix the proteins covalently to the assembling complement complex.¹⁶³ Protein C5b interacts with C6, C7, C8; and six molecules of C9 to generate an “attack complex,” which inserts donutlike rings into the cell membrane being attacked (Fig. 31-9).^{164,164a} Although there has been some uncertainty about the mechanism of lysis, it seems likely that it is at least partly a result of loss of ions through the holes in the donut.

In the alternative pathway of activation a small amount of C3b is formed and becomes bound to the cell surface. This binds another proenzyme factor B,¹⁶⁵ which is converted by protease factor D¹⁶⁶ to active protease **Bb**. The latter in its complex with C3b is the enzyme that cleaves C3 in large amounts and permits a rapid formation of more **Bb** and also of the complex (C3b)₂ **Bb**•P, which attacks C5 (Fig. 31-8). These complexes are stabilized by the abundant serum protein **properdin** (P).¹⁶⁷

Other components of complement are the plasma C1 protease inhibitor,¹⁶⁸ which prevents accidental activation of the system, and protease C3bINA, which

inactivates C3b. The latter depends upon accessory protein bIN. Another component, serum carboxypeptidase B (SCPB), inactivates anaphylotoxins C3a, C4a, and C5a. These small ~80-residue pieces have a variety of powerful biological activities.^{170–174} They are chemotactic factors for leukocytes and induce release of histamine from mast cells. In excess, they can cause anaphylaxis; hence their rapid degradation is essential. An excess of C5a may be present both in asthma and in rheumatoid arthritis.

The ninth component of complement, C9, is a 70-kDa 537-residue water-soluble glycoprotein, which contains a hydrophobic domain that aggregates to form the ion conducting channels.^{175,176} Proteins that closely resemble C9 and are called **perforins** or **cytolysins** are found in cytoplasmic granules of cytotoxic T-lymphocytes and natural killer cells. These ~66-kDa proteins are assembled into rings similar to those formed by C9 and may be involved in the killing action of these cells (Fig. 31-9).^{177–179} Certain pathogenic amebas, which may cause a fatal infection, also utilize a similar pore-forming protein.¹⁸⁰

Every regulatory system in the body must be prevented from overactivity or activity that is unnecessarily prolonged. This can help us understand that, just as with blood clotting (Fig. 12-17), a network of regulatory factors controls the complement system. Among these are an inhibitory C4b-binding protein (C4BP),¹⁸¹ which acts to prevent excessive formation of the C4b•C2a complex (Fig. 31-8). Complement **cofactor I** is a serine protease that cleaves both C3b and C4b into smaller pieces in the presence of **cofactor H**^{181a} or of C4BP. Its absence leads to excessive consumption of C3 and recurrent pyogenic infections.¹⁸² The **membrane cofactor protein** (MCP) stimulates this action of cofactor I in inhibiting attacks of complement on the cells that carry MCP.¹⁸³ Acting in the opposite direction is **complement receptor 2** (CR2 or CD21), which acts as a receptor for proteolytic fragment C3d. This fragment binds CR2-bearing cells to the B cell receptor, amplifying the B cell response to foreign antigens.¹⁸⁴

Complement is involved not only in attacking foreign cells but in inflammation. Unfortunately, this is sometimes accompanied by serious problems. Human diseases in which complement is thought to be involved include glomerulonephritis, rheumatoid arthritis, myasthenia gravis, and lupus erythematosus.

3. Cytokines, Interferons, and the Acute-Phase Response

The body responds in many ways to infection, injury, or cancer. These include the secretion of cytokines, interferons, and proteins of the acute phase response. These proteins, many of which are quite

small, are involved in communication with other cells, often with specialized cells of the immune system.

Cytokines, some of which are considered in Chapter 30 (Section A,6; Fig. 30-6), are small hormonelike molecules. They may stimulate, inhibit, or exhibit other effects on cells of the immune system. They often have pleiotropic effects, not acting in the same way on all types of cells.¹⁸⁵ The cytokines known as **interleukins** (IL-1, etc.) are produced by leukocytes.

Lymphokines are formed by lymphocytes and **monokines** by monocytes. Based on their functions there are four categories of cytokines.¹¹¹

- (1) Mediation of natural immunity: type I interferons, IL-1, IL-6, and more than 40 **chemokines** (small highly basic chemotactic proteins).
- (2) Regulation of lymphocytes, activation, growth, and differentiation of B and T cells: IL-2, IL-4, IL-21, TGF- β .
- (3) Regulation of immune-mediated inflammation: Interferon- γ , tumor necrosis factor (TNF), IL-5, IL-10, IL-12, and migration inhibition factor (MIF).
- (4) Stimulation of hematopoiesis (IL-3, IL-7), colony-stimulating factors (CSF; see also Chapter 32).

Cytokines all function using a group of transmembrane receptors embedded in the plasma membranes of target cells. The receptors have no tyrosine kinase activity but associate with and activate kinases known as **Janus kinases (JAKs)**. These kinases phosphorylate tyrosine side chains in their receptors, and the phosphorylated receptors activate transcription factors of the **STAT** (signal transducer–activators of transcription) group.^{186–195} The specificity of cytokine action results from a combination of receptor recognition and recognition of the various STAT molecules by different JAKs.¹¹¹ Cytokines have a variety of structures. Many are helix bundles or have β sheet structures (Fig. 30-6).

Interferons. The interferons (IFNs),^{196,197} which were discovered in 1957, are proteins secreted by leukocytes, fibroblasts, and activated lymphocytes. They inhibit replication of viruses as well as the growth of host cells and also have antitumor activity. Interferons are classified as α (from leukocytes), β (from fibroblasts), and γ (from lymphocytes). According to their affinities for the two types of known interferon receptors, interferons IFN- α , IFN- β , and the less well known IFN- ω and IFN- τ are

designated type I,^{198–201} while interferon γ (IFN- γ) is type II. At least 15 homologous 166-residue human α interferons are known.

The binding of interferons to their receptors induces a rapid increase in the transcription of particular genes and synthesis of corresponding proteins.^{196,202} One of the proteins induced is a **double-stranded RNA-activated 2'-5' A synthase**, which polymerizes ATP to a series of 2'-5' linked oligonucleotides containing triphosphates at the 5' termini.^{202–204} Double-stranded RNA is uncommon except in replicating viruses, and it is thought that the activation by dsRNA is related to establishment of an antiviral state. Another interferon-induced enzyme is the small subunit of eukaryotic protein synthesis initiation factor eIF-2. This is converted to an inactive phosphorylated form by a dsRNA-dependent protein kinase²⁰⁵ (Fig. 31-10). The protein kinase also appears to be an interferon-induced protein²⁰⁶ as is the oligo(2'-5' A)-activated RNase indicated in Fig. 31-10.²⁰⁷ Interferons have effects other than inducing the antiviral state. Thus, human IFN- β_2 is identical to a B-cell differentiation factor.²⁰⁸ Both IFN- α and IFN- β have antigrowth activity and are currently in use for treatment of some forms of cancer as well as for viral infections.²⁰⁹

Interleukin-1 (IL-1) plays a key role in the body's response to microbes and to tissue injury.^{210,211} It actually consists of three similar proteins, **IL-1 α** , **IL-1 β** , and **IL-1 receptor antagonist**. The first two are the active cytokines with a wide range of effects among

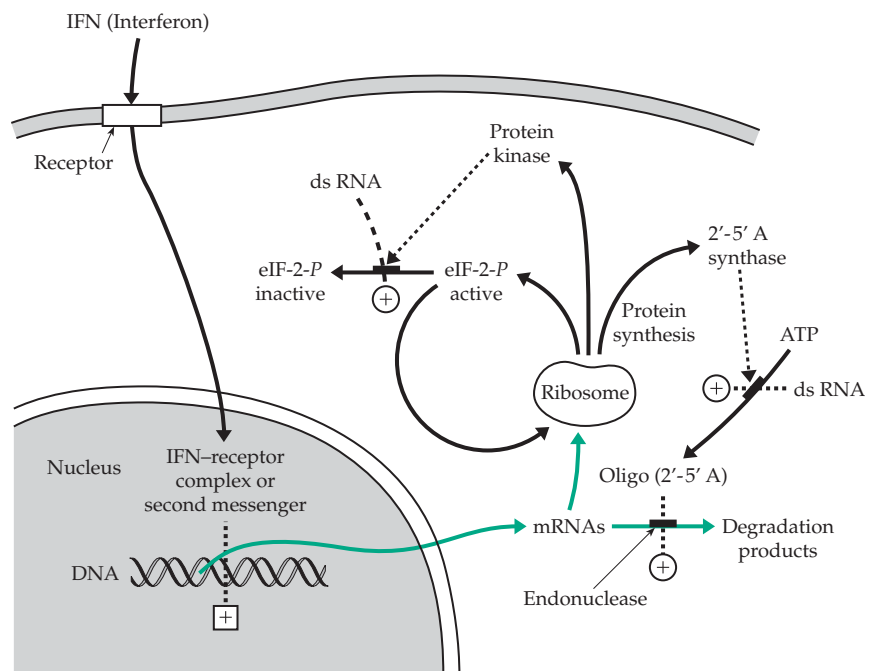


Figure 31-10 Some responses to the binding of an interferon to a cell surface receptor.

which are induction of inflammation and pain.²¹² Il-1 β is thought to be most active in promoting inflammation but only after it is cleaved by **interleukin-1 β -converting enzyme** (see p. 619).²¹³ Blocking of Il-1 receptors provides a potential new method for control of pain.²¹⁴ **Interleukin-6** (IL-6) is also needed for an optimal immune system. Its effects overlap those of

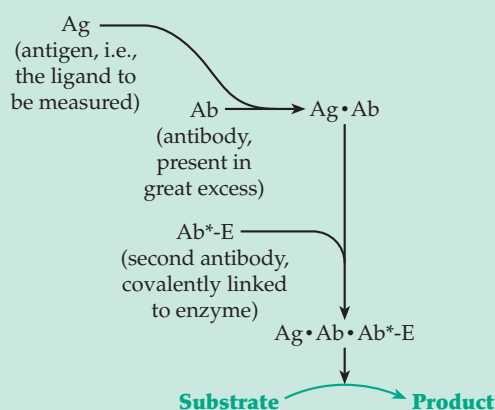
Il-1, and it has a potent activity in inducing the acute-phase response.²¹⁵ Like Il-1, it is a four-helix cytokine.

Also required by the immune response are the numerous **chemokines**. Chemoattractant molecules provide concentration gradients that direct the movement of B and T cells and other leukocytes.^{216–219} Chemokines bind to seven-helix receptors, often

BOX 31-C IMMUNOASSAYS

Among the important techniques that have permitted rapid progress in studies of hormone action is the use of specific antibodies formed against hormones, hormone-protein conjugates, or other molecules.^{a–c} The first of these techniques to come into general use was the **radioimmunoassay** (RIA),^{d–f} which was devised by Yalow and Berson.^d In one form of RIA various amounts of a sample containing an unknown quantity of hormone, e.g., insulin, are placed in a series of tubes. Additional tubes containing known amounts of the hormone are also prepared. Then a standard quantity of radiolabeled hormone (often iodinated with a γ emitter such as ¹²⁵I) is added to each tube together with a standard quantity of the specific antibody to the hormone. The solution is incubated for minutes or hours to obtain equilibrium between hormone (the antigen) and antibody-hormone complex. The antibody-hormone complex is then separated, e.g., by gel filtration or ammonium sulfate precipitation, and the radioactivity of the complex is measured. In the tubes containing higher concentrations of hormone, the labeled hormone has been diluted more, and the amount bound to antibody is less than in tubes with lower concentrations of hormone. The tubes of known concentrations are used to construct a standard curve from which the unknown concentrations can be read. As little as a femtomole of hormone (i.e., the amount present in 1 ml of a 10^{–12} M solution can be detected).^f Methods are available for virtually every pure hormone.^c

The RIA methods were made more convenient by adsorbing either the antibody or antigen to the plastic surface of a tube or depression plate. This facilitates separation of the antibody-ligand complex and washing. A variety of other immunoassays techniques have been devised. For example, in **enzyme-linked immunoabsorbent assays** (ELISA),^c the amount of adsorbed antibody-ligand complex is measured by treating the washed surface with a second antibody, which is directed against the first. The second antibody is linked covalently to an enzyme, whose activity can then be measured by a suitable colorimetric procedure. The reactions involved are as follows.^{g,h}



Variations, which avoid the use of radioisotopes, are replacing RIA. Some utilize stable isotopes. However, ¹⁴C at such low levels that there is no radioactive waste can be coupled with accelerator mass spectrometry to provide very sensitive immunoassays.ⁱ A great variety of other procedures are available. Some involve coupling to antibodies that carry fluorescent labels. Many are now automated. Often protein A from *Staphylococcus aureus* is utilized in various ways that take advantage of its ability to bind to the Fc portion of IgG from virtually all mammals. For example, it may fix antibodies to a surface or to a label.^j

^a Price, C. P., and Newman, D. J., eds. (1991) *Principles and Practice of Immunoassay*, Stockton Press, New York

^b Lindbladh, C., Mosbach, K., and Bülow, L. (1993) *Trends Biochem. Sci.* **18**, 279–283

^c Crowther, J. R. (1995) *ELISA: Theory and Practice*, Humana Press, Totowa, New Jersey

^d Yalow, R. S. (1978) *Science* **200**, 1236–1245

^e Brooker, B., Terasake, W. L., and Price, M. G. (1976) *Science* **194**, 270–276

^f Jaffe, B. M., and Behrmann, H. R., eds. (1974) *Methods of Hormone Radioimmunoassay*, Academic Press, New York

^g van Vunakis, H. and Langone, J. J. (1980) *Methods Enzymol.* **70**, entire volume

^h Langone, J. J., and van Vunakis, H., eds. (1983) *Methods of Enzymology* **92**, entire volume

ⁱ Shan, G., Huang, W., Gee, S. J., Buchholz, B. A., Vogel, J. S., and Hammock, B. D. (2000) *Proc. Natl. Acad. Sci. U.S.A.* **97**, 2445–2449

^j Surolia, A., Pain, D., and Khan, M. I. (1982) *Trends Biochem. Sci.* **7**, 74–76

without rigid specificities. However, some such as **eotaxin** (Fig. 30-6E) are more specific. Eotaxin attracts primarily eosinophils and basophils during allergic reactions.²²⁰ Interleukin-8 (IL-8) is a proinflammatory cytokine and a powerful attractant for neutrophils. Neutrophils are attracted into affected tissues, where they undergo a respiratory burst and generate toxic compounds from O₂ (pp. 1072–1074).^{221–222c}

The second group of cytokines regulate B and T lymphocytes. Among them interleukin-2 (IL-2) stands out as the major promoter of growth and differentiation of T cells. It was the first hormone of the immune system to be recognized.²²³ Both IL-2 and IL-4 have short four-helix structures (Fig. 30-6A). IL-2 is synthesized by activated T cells and binds to a multisubunit receptor complex. The latter associates with tyrosine kinases of both the Src family (p. 572) and with Janus kinases and also activates phosphatidylinositol 3-kinase.^{224–227} IL-2 promotes growth and differentiation, and clonal expansion of T cells, a key aspect of the cellular immune system. It also acts as an immunomodulator of B cells, macrophages, and NK cells. Considerable excitement has accompanied the possibility of activating lymphocytes with IL-2 produced from cloned genes in bacteria to increase their ability to kill cancer cells. However, IL-2 is toxic, and this is limiting its use.

Interleukin-4 (IL-4), cooperating with IL-21,^{227a} stimulates growth of activated B cells, T lymphocytes, and mast cells, induces formation of cytotoxic CD8⁺ T cells, and enhances formation of IgG.^{228,229} The **transforming growth factor-β** (TGF-β) is another cytokine that modulates the development of the immune system. It affects a very broad range of tissues and is discussed in Chapter 32.

Inflammatory influences. Inflammation (p. 1211), which usually accompanies infection and can also arise from allergic responses, is affected by many substances.^{229a–e} These include chemotactic factors that attract neutrophils and monocytes^{222b,229d} and the adhesion molecules that assist in the movement of lymphocytes.^{229e,f} Some epithelial tissues, such as the mucosal surfaces of the gastrointestinal tract, are maintained in a continuous very low level of inflammation. This reflects the balance between activation of the immune system and inhibition of the system by signals from microorganisms both pathogenic and commensal.^{229g,h}

The third group of cytokines^{229b,c} are among the molecules that regulate inflammation. One of these is **interferon-γ** (IFN-γ). Like the type I interferons it induces an antiviral state. However, its most important biological function lies in modulation of the immune system. It induces synthesis of both class I and class II (HLA-DR) antigens (see Section D), activates macrophages, and regulates synthesis and activity

of other lymphokines.^{230–231b} One mechanism of immunomodulation may involve induction of an aminopeptidase that participates in “trimming” of antigenic peptides that participate in antigen presentation (Section D,6). This may directly alter the T-cell response.²³² IFN-γ has a major effect on skin cell-mediated immune responses.²³³ IFN-γ exists in solution as a symmetric dimer, which binds to two molecules of its major surface receptor.^{231,234} The antiviral activity of IFN-γ is largely a result of induction of large amounts of **guanylate-binding proteins**, large 60- to 100-kDa GTPases.^{235,236}

Occasionally a well-established cancer regresses and disappears spontaneously. In the late 1800s it was observed that this sometimes happened, when a person had a concurrent severe bacterial infection. W. B. Coley pursued this lead for many years, treating cancer patients with extracts of killed bacteria, which, although highly toxic were safer than live bacteria. In the laboratory filtrates from cultures of gram-negative bacteria were shown to kill some experimental mouse tumors. The active ingredient was identified as a highly toxic and pyrogenic lipopolysaccharide (Fig. 8-28; Chapter 20, Section E).^{237,238} This lipopolysaccharide has a powerful activating effect on macrophages. More recently it was found that the activated macrophages produce a protein known as **tumor necrosis factor** (TNF-α) that can destroy tumor cells and also acts together with interferons in inducing resistance to viruses.^{239,240} A similar **lymphotoxin** (TNF-β) is secreted by lymphocytes.²⁴¹ Although highly toxic there has been hope of obtaining engineered forms of these proteins more specifically toxic to tumors.

TNF-α is identical to **cachetin**, a protein that suppresses completely the lipoprotein lipase of adipose tissue and is believed to be responsible for **cachexia**, a condition of general ill health, malnutrition, weight loss, and wasting of muscle that accompanies cancer and other chronic diseases. Nevertheless, TNF-α may be overproduced in obesity as well. It has been suggested that abnormal production of TNF-α may induce cachexia while abnormal action of the cytokine may cause obesity.²³³ Some TNF receptors have “death domains” and trigger apoptosis, while other receptors promote proliferation and differentiation via transcription factor NF-κB.²⁴²

Other cytokines with lymphocyte-regulatory functions are IL-5, IL-10, IL-12, and the **macrophage migration-inhibition factor** (MIF). IL-10 is secreted by B cells, T cells, keratinocytes, monocytes, and macrophages. It suppresses synthesis of many cytokines but stimulates growth and activity of activated B cells.²⁴³ IL-12 is formed by monocytes, macrophages, neutrophils, and dendritic cells. It activates T cells and NK cells, is a very potent stimulator of IFN-γ formation, and also inhibits angiogenesis in tumor cells. It stimulates defenses against a wide range of infectious

diseases caused by bacteria, fungi, protozoa, and worms.^{244,245} The 115-residue MIF is formed not only by the immune system but also by many other tissues. The first lymphokine to be discovered, MIF, inhibits migration of macrophages and is also a mediator of toxic shock.^{246,247} MIF is also an enzyme, a **phenylpyruvate tautomerase** (p. 692).²⁴⁷

The fourth group of cytokines are involved in hematopoiesis and control the developmental steps portrayed in Fig. 31-2. They are discussed in Chapter 32.

The **acute-phase response** consists of increased production of a group of plasma proteins in response to tissue injury or inflammation.^{229c} Important acute-phase reactants are the **C-reactive protein**,^{229a,248,249} **serum amyloid A**,^{250,251} **haptoglobin**, **hemopexin**, **α 1-acid glycoprotein**,²⁵² and **α 2-macroglobulin**.²⁵³ The C-reactive protein precipitates pneumococcal polysaccharides in the presence of Ca^{2+} . It is present in primitive invertebrates and may serve as a rudimentary immunoglobulin.²⁴⁸ Serum amyloid A is one of the apolipoproteins associated with high-density lipoproteins (Chapter 21, Section A). Its concentration may increase as much as 1000-fold during the acute-phase response, and during prolonged stress it may precipitate as extracellular amyloid fibers. This secondary amyloidosis is sometimes a severe pathological problem. The level of the general protease inhibitor α 2-macroglobulin can increase several hundredfold.

D. Organizing the Immune Response

A person's immune system must be able to respond to a large variety of foreign antigens without reacting against the individual's own tissues. The huge variety of antibodies that can be formed arise from the existence of B cells with millions of different sequences in their antibody genes. When an immune response occurs only a few B cells are stimulated to proliferate, and it is these selected clones that provide the needed specific antibodies and memory cells. However, it is not immediately obvious how we avoid a disastrous attack of the immune system triggered by the many antigenic determinants (**epitopes**) present in our own cell surfaces and macromolecules. Part of the answer is that the immune system "learns" early in life what is self and what is nonself. Thus, while foreign tissues cannot usually be grafted without rejection, cells of two immunologically incompatible embryos can be mixed at a very early stage of development, and an animal tolerant to both types of cell will develop.

A full understanding of self-discrimination is not yet available.^{253a} The adaptive system, as generally understood, is outlined in the following pages. A current view of the innate system is presented by Medzhitov and Janeway.^{11a} However, an alternative

description, the **Danger model**, is being developed by Matzinger. Her view is that the immune system is designed not so much to recognize *nonself* as to send *alarm signals* from injured tissues.^{11b} Most of the basic mechanisms of the adaptive immune system are not in dispute, but many hard-to-explain phenomena remain uncertain.

1. Coreceptors and the B-Cell Response

Early in life most B cells that would produce antibodies directed against a person's own tissues (auto-reactive B cells) are eliminated or altered to reduce their reactivity.²⁵⁴ When functional B-cell receptors do bind an antigen, the B cell will not be activated unless **coreceptors** also bind to the antigen-bearing particle. The transmembrane glycoproteins known as CD22, CD21, CD72, and $\text{Fc}\gamma\text{RIIb}$ are among the many coreceptor molecules. Coreceptors often induce tyrosine phosphorylation of internal receptor domains and attract other molecules to form a signaling complex that may release cytokines.^{255–256} The coreceptors ensure that an immune response doesn't take place without at least two signals. They also help to localize the immune response.

Among the most important factors in B-cell activation are the effects of T cells. B cells can independently mount an attack using IgMs against surface antigens. However, B-cell responses to many antigens, e.g., those present on flagella or inserted into membranes, are also dependent upon assistance from helper T cells (T_H cells).²⁵⁷ These cells also have a major role in determining the longer term fate of B cells. Upon activation B cells may survive or die via apoptosis. They may proliferate (clonal expansion) and differentiate into plasma cells or may become unreactive (**anergy**). They may become long-lived memory cells.

2. The Leukocyte Differentiation Antigens

Before discussing T-cell responses it seems appropriate to mention the nomenclature of molecules (largely glycoproteins) that have been recognized as antigens present on leukocyte surfaces. These same molecules are found on other cells, but the designation of the antigens by a **cluster of differentiation** number, such as CD1, CD4, or CD8, has provided a convenient way of distinguishing different types of leukocytes.^{258–261} For example, helper T_H cells are usually CD4^+ , carrying predominantly CD4. Cytotoxic T cells are predominantly CD8^+ .^{262,263} Both CD4 and CD8 consist largely of Ig-like domains. CD4 is a 55-kDa transmembrane protein with tyrosine kinase activity.²⁶⁴ It is a monomer containing four Ig-like domains, but CD8 is a disulfide-linked $\alpha\beta$ dimer.²⁶³

Not all CD molecules are related to IgG. Proteins are often designated by a specific name followed by a CD number, e.g., Fc γ RII / CD32, ICAM-1 (CD54).

3. Functions of T Cells

T cells carry the responsibility of identifying antigens as foreign or as belonging to self. They do this in immunological synapses (Fig. 31-11) in conjunction with the major histocompatibility complex MHC (Section 5). T cells circulate through the body searching for antigens that indicate danger to the body. To avoid being swept through the bloodstream too rapidly and to be able to enter the lymphoid organs lymphocytes form tethers with adhesion molecules such as the **selectins** (p. 188).²⁶⁶ They then roll more slowly to their destination. Within the lymphoid tissues the T cells may form synapses with activated B cells, dendritic

cells, and macrophages. Within these cells proteosomes generate a stream of peptide fragments, some of which arise from phagocytosed pathogens. These foreign peptide fragments are displayed on the cell surfaces as complexes with type II proteins of the major histocompatibility complex (MHC; see Fig. 31-13). The complexes are checked by CD4⁺ T cells, some of whose **T-cell receptors** (TCRs) will probably be complementary to the surfaces of the complex of the class II MHC protein and the foreign peptide. The T cell will recognize two things about this complex: the MHC protein is of *self* origin but the antigen is *foreign*. The CD4 on the T cell must also bind to the MHC on the surface of the antigen-presenting cell. Other costimulatory interactions may be needed as well.^{50,267} Both CD4⁺ and CD8⁺ T cells tend to bind to oligomeric **activation clusters** of receptors within the immunological synapses.^{267a} Other proteins also participate in assembly of these activation complexes.^{267b} Of

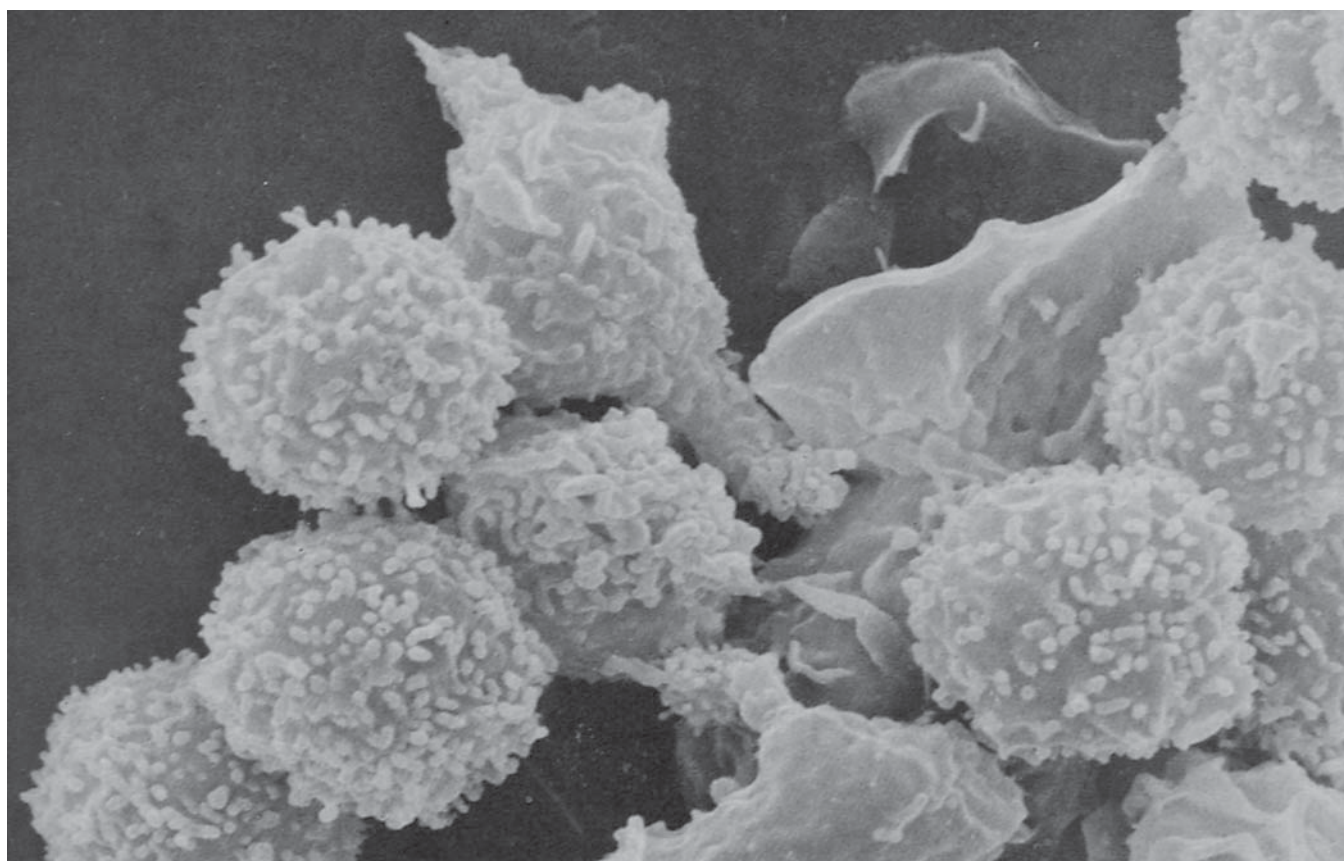


Figure 31-11 The T lymphocytes seen here are forming synapses with the large flat macrophage in the center. The macrophage is displaying antigenic peptide fragments bound to molecules of the major histocompatibility complex (MHC). Most T cells carry their own specific type of receptor. If it is complementary to a displayed antigen fragment it will bind, and the T cell will respond. Depending upon what other coreceptors are activated, it may become a T_H helper cell, or a cytotoxic T cell, or it may become inactive. Micrograph from Grey *et al.*²⁶⁵ Scanning electron micrograph courtesy of Morton H. Nielsen and Ole Werdelin, University of Copenhagen.

particular interest is the recognition of an antigenic peptide produced by a B cell. The B cell has probably already recognized and phagocytized a foreign protein and is displaying peptides from that protein on its MHC I molecules. Recognition of this peptide complex by a CD4⁺ T cell will stimulate the cell to become a T_H helper cell, which will in turn stimulate the B cell to proliferate and differentiate into a clone of as many antibody-forming plasma cells. The essential nature of the costimulation by CD4 is emphasized by the fact that infection by HIV-1, which is mediated by CD4, leads to loss of CD4 from plasma membranes and to the weakening of the immune response toward various pathogens that is characteristic of AIDS.^{268–270}

In a similar manner CD8⁺ T cells recognize peptide fragments displayed on MHC class I molecules. These fragments arise via a somewhat different pathway that forms fragments of viral proteins or proteins of other intracellular pathogens. Recognition by a CD8⁺ T cell usually converts it into a **cytotoxic (killer) T cell**, which will kill the infected cell.^{270a,b} This type of immune reaction was first recognized by the phenomena of **delayed hypersensitivity** and of **transplantation immunity**, i.e., the rejection of transplanted tissues. Both phenomena are caused by cytotoxic T cells. In delayed hypersensitivity they appear to be confused and to attack host cells.

Some very hydrophobic antigens are presented by neither a class I nor a class II MHC molecule but by members of the CD1 family, leukocyte surface proteins that are not encoded in the MHC gene region.^{266,271,272}

4. Natural Killer Cells

An additional line of defense is provided by **natural killer cells** (NK cells), a type of circulating lymphoid cell able to kill cancer cells, to participate in antiviral defenses, and to help control immune responses.^{273–276} NK cells, which utilize their own signaling pathways, are also able to use MHC class I molecules to recognize and to spare the lives of normal, healthy cells.^{277,277a,b} Partial deprivation of a night's sleep can reduce NK cell activity, damaging the cellular immune response.²⁷⁸

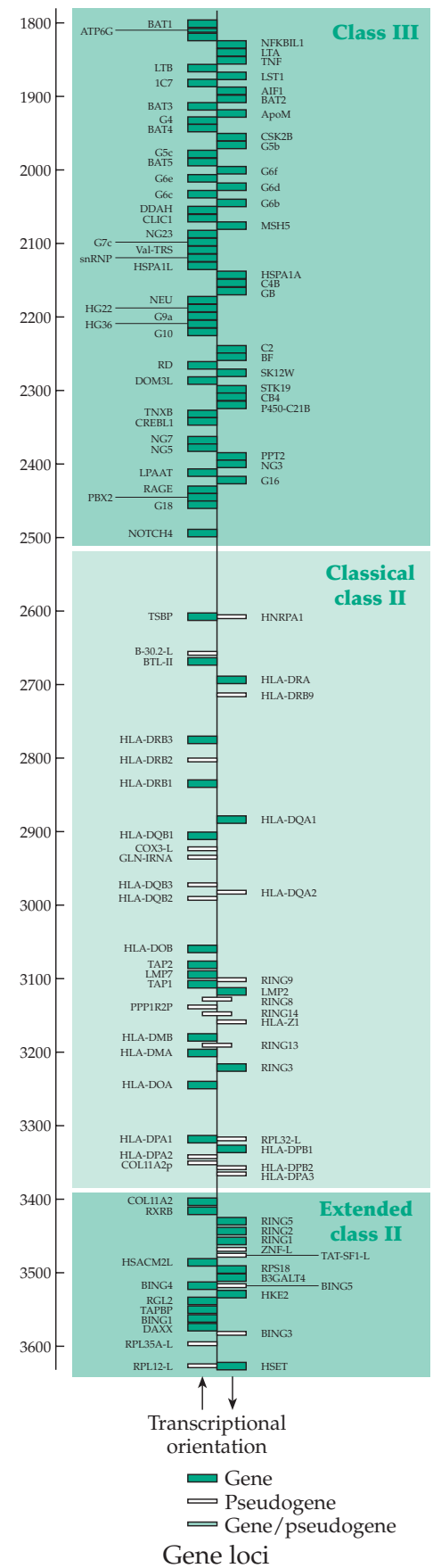
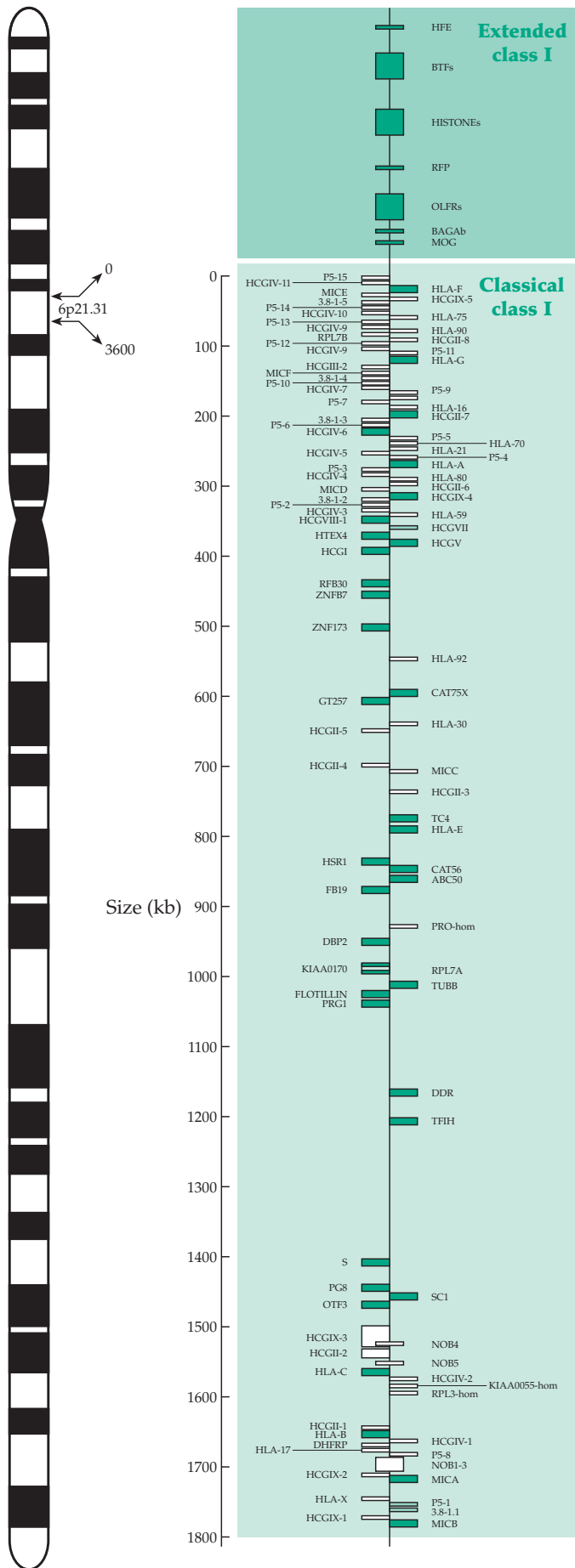
5. Identifying Self: The Major Histocompatibility Complex

Proteins encoded by a single cluster of genes are known as the **major histocompatibility complex** (MHC).²⁷⁹ These proteins, which are essential to T-cell function, were first recognized as the primary determinants of the compatibility of grafted tissues with the host's immune system.²⁸⁰ A lack of histocompatibility can be disastrous. Not only are grafted tissues rejected but T lymphocytes from the grafted tissues sometimes proliferate, attack, and kill the host. The MHC of mice is usually referred to as the **H-2 complex**^{281,282} and that of humans as **leukocyte locus A (HLA)**.^{283,284} Although the MHC is the most important determinant of histocompatibility, differences in other genes may also lead to a slow rejection of transplanted tissues. Since there are many different MHC genes, transplantation is successful only within inbred lines.

Some of the MHC genes have a large number (50–100) of alleles. So great is this genetic polymorphism that it is extremely unlikely that two individuals will have an identical set of histocompatibility genes. The MHC (HLA) genes are located in a 2-centimorgan (~3.6 kb) region of the short arm of human chromosome 6 (Fig. 31-12)^{284,285} and on chromosome 17 of mice. These genes are of at least three classes. **Class I genes** (called HLA-A, HLA-B, HLA-C, HLA-E, HLA-F, and HLA-G in humans; see Fig. 31-12) encode the major transplantation antigens, which are found on the surfaces of nearly all cells of the body. **Class II genes** encode proteins found largely on the membranes of B lymphocytes, macrophages, and dendritic cells. They are designated HLA-DP, HLA-DN, HLA-DM, HLA-DQ, HLA-DR, and HLA-DO.²⁸⁶ **Class III genes** encode several components of the complement system. Many other genes and pseudogenes are interspersed with those of the MHC.

All type I MHC molecules are integral membrane glycoproteins each of which is composed of a 45-kDa heavy chain of about 350 residues together with a noncovalently linked 17-kDa light chain. The genetic variation occurs in the heavy chain between residues 43 and 195 in the human proteins. This chain appears

Figure 31-12 (Opposite page) Arrangement of genes of the human major histocompatibility complex (MHC). Left: Banding pattern of a stained chromosome 6 with the MHC region marked. Center and right: locations of all genes and pseudogenes in this region. The MHC molecules can be divided into three classes on the basis of their structure and function. The class I antigens constitute a single class structurally but fall into two functional groups. The first of these contains the “classical” class I antigens, first discovered as the transplantation antigens and now known to function as target antigens in the recognition and destruction of virus-infected cells by cytotoxic T lymphocytes. They are expressed on virtually all somatic cells. The class II antigens are expressed largely on B lymphocytes and macrophages of the immune system and are essential for presenting antigen to the helper and suppressor T cells that regulate the immune response. Many class III products are components of the complement system. These maps are based on serological and biochemical data, together with complete sequences. From the MHC sequencing consortium.²⁸⁴



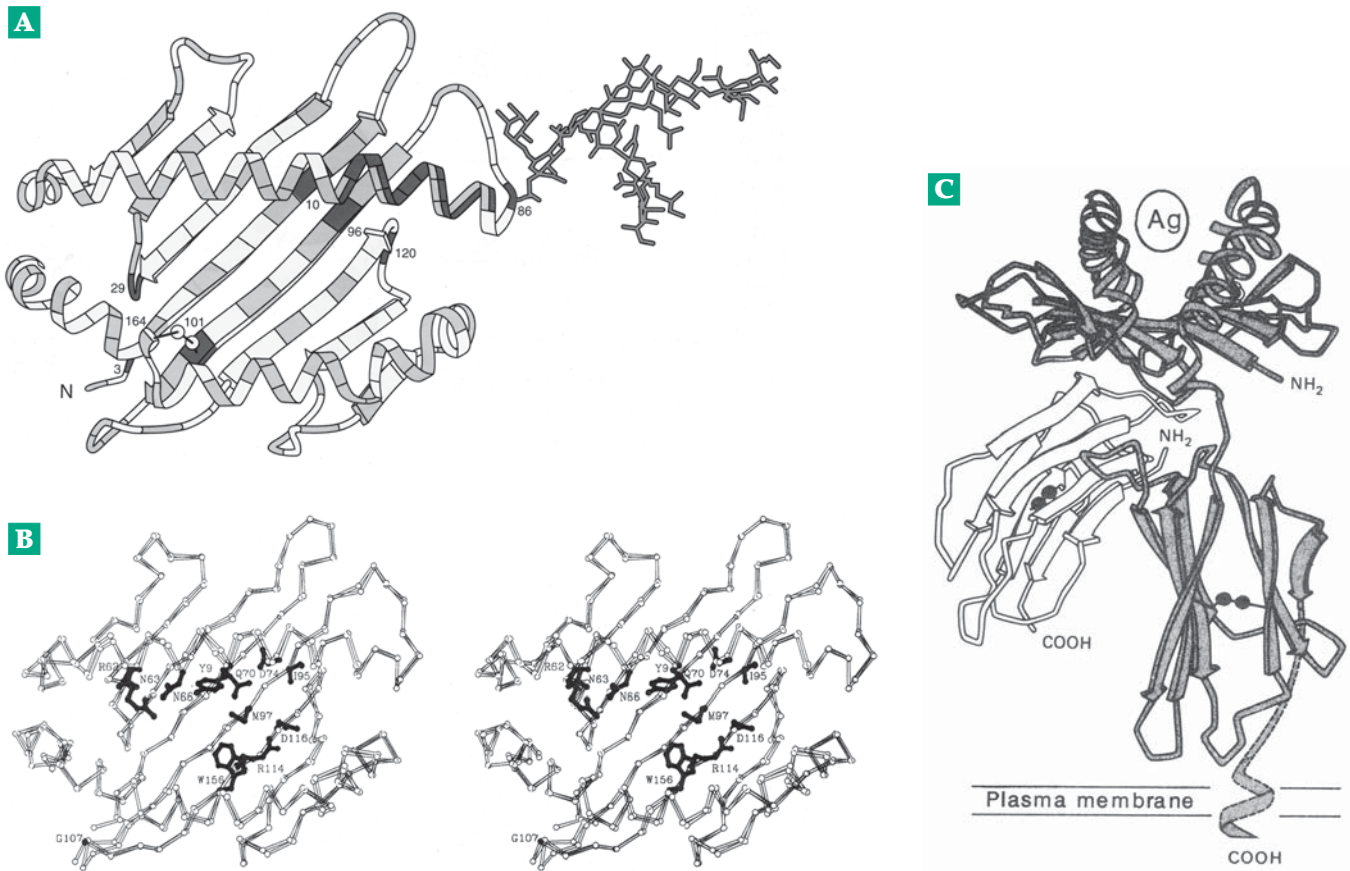


Figure 31-13 The structure of Class I MHC molecules. (A) The specificity pocket in the N-terminal part of the ~360-residue α chain. The numbered residues are invariant in all of the ~20 different Class I molecules. An oligosaccharide is shown on the invariant Asn 86. (B) A stereoscopic view of a similar MHC molecule showing some of the polar residues that protrude into the peptide-binding groove and may form hydrogen bonds with the peptide. From Garrett *et al.*²⁹¹ Courtesy of Don C. Wiley. (C) Side view of a complete MHC molecule with an antigenic peptide (Ag) bound into the peptide-binding groove. The C terminus of the long ~360-residue α chain is in the cytoplasm of the displaying cell. The small 99-residue β chain (unshaded) is a molecule of β microglobulin, which is also a constituent of blood plasma.²⁹² Courtesy of Peter Parham.

to consist of three ~90-residue domains protruding from the outside of the cell, about 25 residues embedded in the membrane, and a short C-terminal tail in the cytoplasm (Fig. 31-13).²⁸⁷ The light chain has an invariant composition and is identical to the plasma protein β_2 -microglobulin, whose gene is located on a different chromosome. Its structure closely resembles that of a single immunoglobulin domain.²⁸⁸ The MHC Class II antigens (Fig. 31-14) are also $\alpha\beta$ dimers, the α chains being 34-kDa glycoproteins and the 28-kDa β chains being larger than in the type I antigen.^{283,289} While the MHC of humans and mice have been studied the most, all vertebrates possess similar self-identification systems. Although both the sequences and the folding patterns of the MHC antigens (Figs. 31-13, 31-14) are somewhat similar to those of immunoglobulins, there are many differences. Furthermore, the cause and significance of the polymorphism is quite different in the two cases. Each individual has millions of antibodies with differ-

ent variable regions but only one set of HLA antigens, which are largely the same on germ, embryonic cells, and adult cells.

Serological tests allow tissue types to be defined by the HLA genes.²⁹⁰ Thus the commonest HLA type in Caucasian populations is HLA-A1 / B8 / Dw3, whereas A1 / B17 is common among Asian Indians. In every case subtypes can be defined, and this fact together with the polymorphism in other genes leads to a unique HLA type for nearly every individual. As is indicated in Fig. 31-12, complete nucleotide sequences are known for typical alleles of all of these genes.²⁸⁴ It is of medical interest that the susceptibility of an individual to many degenerative diseases is determined in part by the HLA type.^{289,295} Thus among patients with a kind of arthritis, **ankylosing spondylitis** that affects 1 or 2 per 1000 men of Caucasian origin, 96% have the HLA-B27 antigen. Of patients with **celiac disease**, a type of intolerance to gluten, 60% have the HLA-B8 antigen. Persons with

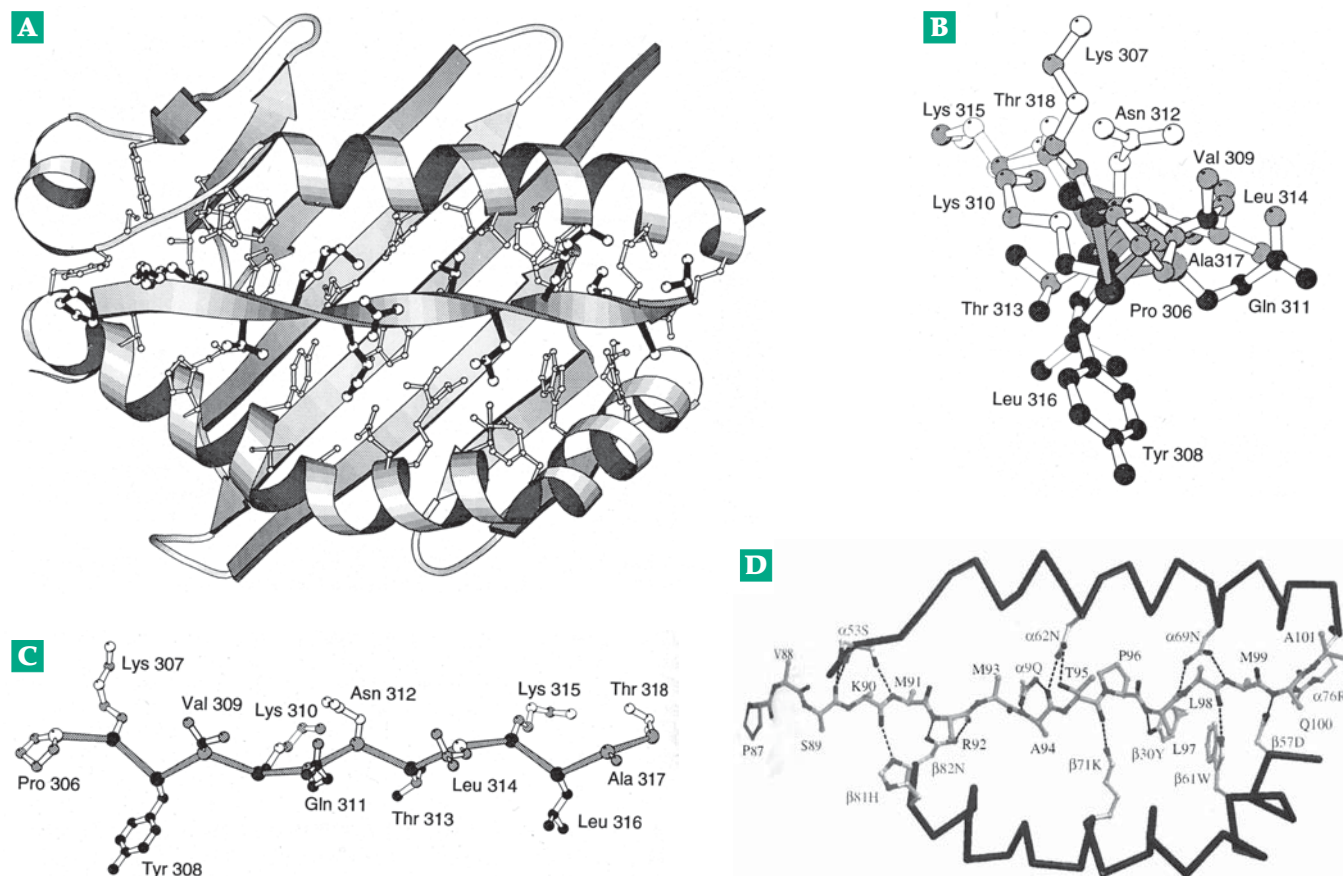


Figure 31-14 Illustration of the binding of a short peptide fragment (central ribbon) into the antigen-binding groove of an MHC type II molecule. (A) The binding groove of molecule HLA-DR1 with a bound peptide (HA) derived from an influenza virus. (B) End view of the bound peptide. (C) Side view of the same peptide. From Stern *et al.*²⁹³ (D) A similar HLA molecule (HLA-DR3) with the peptide CLIP (class II associated invariant chain peptide) bound into the antigen-binding groove. The binding is almost identical to that in (A). Notice the specific hydrogen bonding to side chains of the HLA molecule. From Ghosh *et al.*²⁹⁴ Courtesy of Don C. Wiley.

HLA-Bw17 and B13 have an increased susceptibility to **psoriasis** and those with HLA-DRw4 an increased tendency to develop **rheumatoid arthritis**. Susceptibility to **autoimmune insulin-dependent (Type I) diabetes** is strongly correlated with the presence of the neutral residues Ala, Val, or Ser at position 57 of the HLA-DQ β chain.^{296,297} However, aspartate in position 57 protects against the disease. It may prevent development of a dangerous autoantibody to this cell surface protein. The presence in populations of both humans and apes of a balanced polymorphism among the residues Ala, Val, Ser, and Asp at this position suggests an essential evolutionary origin to this disease susceptibility.²⁹⁷ HLA-B53 protects against severe malaria in Africa. Other diseases with a strong association with HLA type include multiple sclerosis, **Crohn disease** (inflammatory bowel disease),²⁹⁸ and several diseases induced by infections with viruses, bacteria, trypanosomes, etc. For example, arthritis can follow infection by *Salmonella*. This suggests that the killer T cells can be confused

when stimulated by foreign antigens, which are too closely related to the HLA antigens of the host.

6. Antigen Presentation and MHC Restriction

T cells usually do not respond to intact antigens on cell surfaces but only to partially degraded antigens. Antigen-presenting cells (APCs) of various types must process the antigen through endocytosis and partial digestion before the foreign antigen can bind as an MHC complex to a T-cell receptor. Apparently the processed foreign antigen must lie in the binding site of an MHC Class I or Class II chain (Figs. 31-13C, 31-14) before the T cell will recognize the complex and respond.

Antigen processing begins with cytosolic proteasomes that are present in all cells (Box 7-A; Chapter 29, Section D,8). They cleave proteins of the cell and of intracellular parasites into short peptide

fragments,^{55,299–301} which may need to be trimmed to shorter 8- to 11-residue peptides suitable for binding into the groove in an MHC Class I molecule.^{302,303} The peptides are carried into the ER with the aid of the **TAP** (transporter associated with antigen processing) complex,^{304–305a} which is discussed briefly in Chapter 29, Section D.3. Chaperones, such as hsp70, may also participate in the transport. In contrast, MHC Class II proteins receive their antigenic peptides via an endosomal–lysosomal pathway. Proteins from phagocytized pathogens are cleaved by proteases in an endosome or lysosome into fragments that tend to be longer (13–25 residues) than the 9- to 11-residue peptides generated by proteasomes.^{299,306}

Peptides bind into the groove in a Class I MHC molecule in a manner similar to that illustrated in Fig. 31-14 for a Class II MHC molecule. However, in the Class II complex the longer peptides extend from the two ends of the binding groove. The peptide, which assumes a polyproline II helical conformation, is held by hydrogen bonds from the Class II MHC molecule to the peptide backbone.^{307–311} A single peptide may shift and bind in a different register with the possibility of being recognized by a different T cell receptor, when it is displayed.³⁰⁸ In contrast, MHC Class I molecules bind best to 8- or 9-residue peptides, which are held by an array of hydrogen bonds to the $-\text{NH}_3^+$ and $-\text{COO}^-$ termini (not shown in Fig. 31-13, which displays the empty binding groove). The central part of this groove contains a deep pocket, which together with smaller pockets near the ends provides specificity.^{312–318} However, there is a puzzle. Because of the great genetic variability in the MHC genes there will be great differences (polymorphism) in the shapes of the binding pockets among different people. However, an individual has at most six different kinds of MHC molecules. Yet, a single MHC molecule has been estimated to be able to bind more than 10,000 different peptides.³¹⁴ Essential to this process is a final trimming of the peptides at their N termini to provide better fits.^{318a}

Peptides are loaded onto Class I MHC molecules, while they are together in the ER. They move as tightly bound complexes through the Golgi and into the external plasma membrane, where they remain tethered via the MHC molecule (Fig. 31-13C). In contrast, binding to Class II MHC molecules occurs in the endosomes or lysosomes. The process is somewhat complex. Class II MHC molecules are chaperoned from the cytosol into late endosomal/lysosomal organelles with their antigen-binding grooves occupied by a peptide fragment known as the Class II-associated invariant chain peptide (CLIP; Fig. 31-14D). This is cut from the end of an invariant chain known as Ii.^{319–321}

The MHC proteins HLA-DO and HLA-DM (Fig. 31-12) are resident in the lysosome-like organelles and chaperone the class II molecules, until they are ready for loading with peptides. HLA-DM assists in removal

of the CLIP peptide when loading occurs.³²² An asparaginyl endopeptidase may also be required³²³ as well as a disulfide reductase.³²⁴ After loading the Class II MHC•peptide complexes, like the Class I MHC•peptide complexes, are exported to the plasma membrane. There they may be recognized by a T cell, which utilizes its T-cell receptor to recognize an antigen and its CD4 or CD8 proteins to distinguish Class I from Class II MHC complexes. The binding of CD4 and CD8 to their cognate MHC molecules has also been described at the molecular level.^{310,325} As mentioned previously, some hydrophobic antigens are presented by CD1 molecules. They also have an MHC-like fold with a large hydrophobic binding groove.²⁷²

An interesting approach to the treatment of autoimmune diseases is design of peptide mimics that bind into the antigen-binding groove of specific MHC proteins. For example, a protease-resistant pyrrolinone–peptide hybrid has been designed to bind to the rheumatoid-arthritis-associated HLA-DR1.³²⁶

An important distinction between B- and T-cell responses is that T cells recognize a foreign antigen only when associated with an MHC antigen of the same type as is carried by the T cell. This “MHC restriction” limits the actions of T lymphocytes. The function of cytotoxic T cells appears to be primarily one of killing virus-infected cells and perhaps cancer cells. MHC restriction ensures that the T cell is attached by its receptor to a Class I MHC molecule belonging to self. The dual recognition ensures that the cell probably contains a foreign antigen and should be killed. The Class I MHC antigens are found on almost all body cells. Therefore, cytotoxic CD8⁺ T cells can attack infected cells of all types. They may kill their target cells by injecting their cell membranes with the complement C9-like cytolytins.

Regulatory CD4⁺ T cells recognize the Class II MHC molecules, which are found primarily on lymphocytes. Thus, the attention of regulatory T cells is directed towards other lymphocytes. In this case, too, MHC restriction enables helper T cells to recognize B lymphocytes as self. If foreign antigen is present so that B lymphocytes have been activated by the binding of a foreign antigen, they will be stimulated by the T_H cells to proliferate and make and release immunoglobulins. This is accomplished in part by secretion of the lymphokine interleukin-2 (Fig. 30-6) and B-cell growth factors. Some T cells become suppressor T cells.

7. T-Cell Receptors

T cells both mediate the recognition of self and also participate in the immunologic response to foreign antigens. Their surface receptors function much like the immunoglobulins that are attached to the surfaces of B cells. The T-cell receptors are $\alpha\beta$ disulfide-linked

heterodimeric glycoproteins (Fig. 31-15) consisting of 40- to 45-kDa α subunits and 42- to 44-kDa β subunits.^{267,327-330a} They are associated in the T-cell membrane with a larger complex called CD3, which contains additional 26- to 28-kDa γ , δ , ϵ , and ζ chains.^{267,331,332} The polypeptides of the CD3 complex have C-terminal cytoplasmic tails that contain tyrosine residues within several immune system tyrosine-based activation motifs (ITAMs; Fig. 31-15). As is illustrated in this figure, the activating antigen is cradled in the binding groove of an MHC molecule attached to the APC (top) with some side-chain and backbone atoms of the peptide available for bonding to the T-cell receptor.^{333,334} Notice that in the synapse the T-cell receptor also makes direct contact with the MHC molecule (Fig. 31-15).

Signaling by an activated T-cell receptor is quite complex. The ITAMs are sites of tyrosine phosphorylation by kinases of the Src family.^{335,336} Another tyrosine kinase, Zap-70 (Fig. 31-15), associates with the C-terminal tails of the disulfide-linked dimer of subunits ζ . It recognizes the phosphotyrosines groups via its SH2 domains (see Fig. 7-30). Zap-70 appears to act

synergistically with the Src kinases.^{332,337,338} The nature of the APC is also of importance. For example, some dendritic cells secrete Il-12, which favors formation of T_H1 helper cells. A second type of dendritic cell induces formation of T_H2 helper cells.^{338a} Dendritic cells may also control growth and proliferation of T cells by regulating the availability of cysteine, which is a nutritional essential for lymphocytes.^{338b} Other effects may result from endocytosis by T cells of occupied T-cell receptor•MHC complexes.^{338c}

A second type of T-cell receptor, the $\gamma\delta$ receptor, is carried by a small subgroup of T cells. It may have a distinct role in generating an immune response to certain microorganisms including *Mycobacterium tuberculosis*.^{339-342a} Like immunoglobulins, T-cell receptors have a great variety of amino acid sequences. They have C-terminal constant domains and N-terminal variable and hypervariable regions. Thus, T cells can bind to a variety of foreign antigens. However, except as a result of autoimmune diseases, they do not attack cells recognized as self unless these cells are infected with a virus or for some other reason carry foreign surface antigens.

Certain bacterial immunostimulatory molecules known as **superantigens** are able to stimulate MHC Class II molecules to activate large numbers of T cells without any assistance of an antigenic peptide, sometimes with disastrous results.^{343,344} Superantigens are a group of related proteins that includes enterotoxins from species of *Staphylococcus*³⁴⁴⁻³⁴⁶ and *Streptococcus*,³⁴⁷ a staphylococcal exfoliative toxin,³⁴⁸ and **toxic shock syndrome** toxins.³⁴⁹ Superantigen molecules don't occupy the peptide-binding groove in the MHC molecule but bind as intact proteins at an external site. They also bind to the variable region of the TCR in the MHC-TCR complex. Attempts are being made to design decoy molecules that prevent binding of a particular superantigen.³⁵⁰

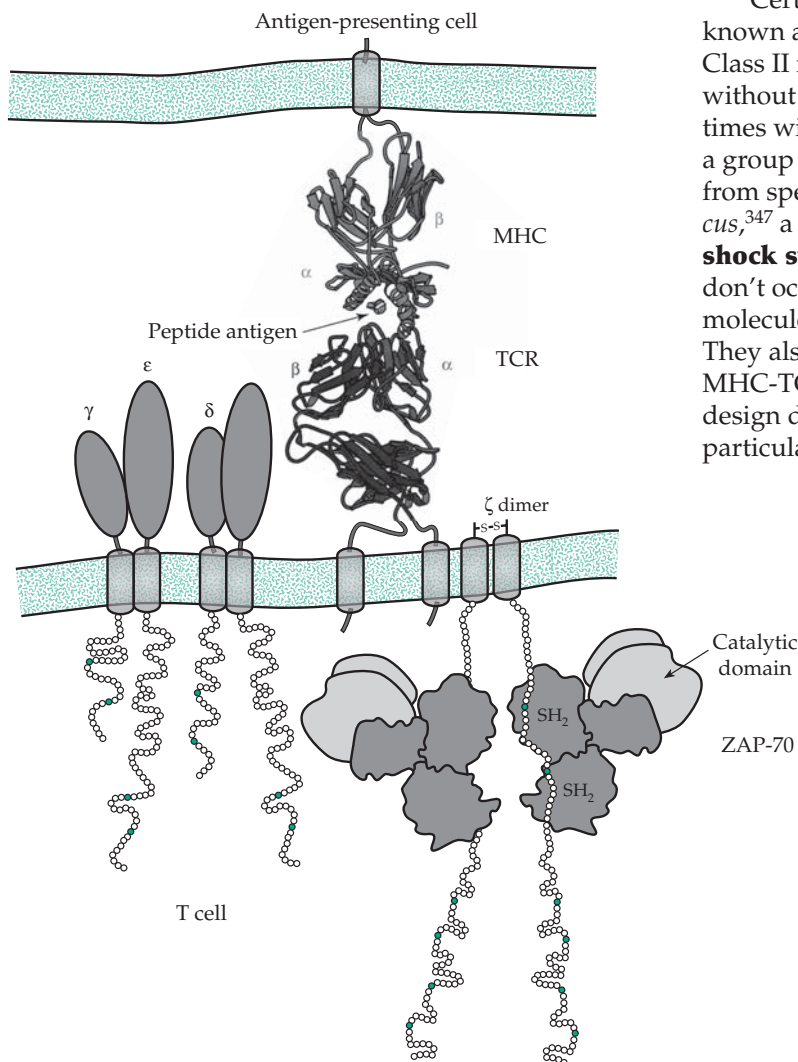


Figure 31-15 Interaction between an MHC•peptide complex on an antigen-presenting cell (APC) with a T-cell antigen receptor (TCR) that is attached to the plasma membrane of a T cell. Structures of the α - and β -subunits of the MHC molecule and of the T-cell receptor are based on crystallographic data. The detailed structures of the disulfide-linked $\sigma\epsilon$ and $\gamma\epsilon$ modules of the T-cell receptor are not shown. The dimeric ζ_2 subunit has large cytoplasmic domains that are thought to be involved in signaling the protein ZAP-70 (zeta-associated protein of M_r 70,000). The σ , ϵ , and γ subunits may also undergo phosphorylation of their tyrosyl groups (gray spheres), which are found in immune system tyrosine-based activation motifs (ITAMs). Drawing modified from those of Cochran *et al.*²⁶⁷ and Hatada *et al.*³³²

8. Self-Tolerance

The immune system is flexible enough and powerful enough to protect us from a great variety of dangers, even from viruses and organisms that may be entirely new. However, it is hard to understand how the immune system completely avoids fatal damage to our own bodies. The answer is complex. It has baffled generations of investigators^{350a,b} and is still not fully understood.

Since the discovery of vaccination in 1796 immunology has claimed the attention of many scientists. However, it was not until 1891 that the German bacteriologist Emil von Behring proposed the term antibody for the protective materials in blood.³⁵¹ By about 1900 Paul Ehrlich and Svante Tiselius, who wrote the first immunochemistry book,³⁵² initiated serious investigations. Ehrlich proposed that binding of an antigen to a surface receptor would induce the cell to make additional identical surface receptors, which would be released to become antibodies.^{351,353} The concept was correct, but it would be many years before knowledge of the structures and biosynthesis of antibodies became available.

Nevertheless, immunological tolerance interested Ehrlich and other immunologists.^{353a} One proposal, offered by Niels Jerne in 1974, was that self-tolerance depends upon **immunological networks**.^{354–356} Consider a lymphocyte bearing a bound immunoglobulin receptor or a bound T cell receptor. It will be specific for some epitope E. The receptors are shown in Fig. 31-16 as having V-shaped antigen-binding sites. Jerne pointed out that the variable region of this receptor will itself carry epitopes that can be recognized by other appropriate antibodies. These epitopes on the

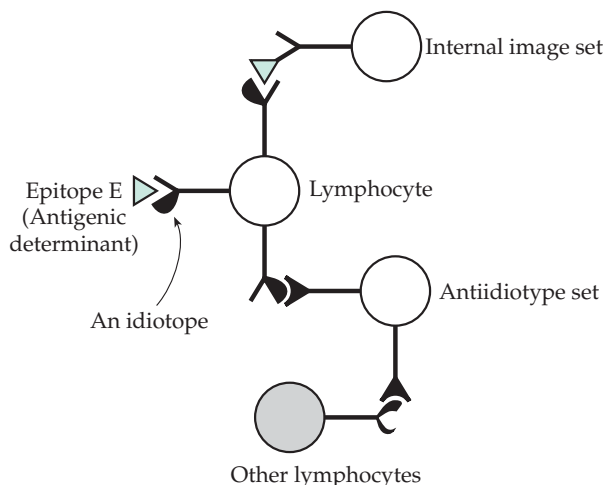


Figure 31-16 Schematic depiction of lymphocyte receptors forming anti-idiotypic and internal image sets as proposed by Jerne.

receptor are called **idiotopes** and as a group define the **idiotypic** of that receptor or immunoglobulin.^{356a} There will usually be other lymphocytes with receptors that recognize the idiotype of the first lymphocyte (see Fig. 31-16). These constitute an **antiidiotypic set**. In addition there will be lymphocytes, whose idiotopes resemble those of epitope E and which will therefore be recognized by the first lymphocyte as foreign. These lymphocytes constitute an **internal image set**. There will be other sets of lymphocytes that recognize the lymphocytes of the antiidiotypic set or of the internal image set. Thus, there will be an elaborate network of clones of interacting lymphocytes. When an immune response occurs many members of this network will respond. A B lymphocyte will recognize a particular antigen and gives rise to a clone of plasma cells making antibodies against that antigen. The body will then make new antibodies against the first antibodies formed, etc. A whole segment of the network will respond in this fashion. Jerne suggested that the overall effect would be to limit and suppress the immune response. Anti-idiotypic antibodies as well as anti-anti-idiotypic antibodies have been prepared³⁵⁶ and have been used in studies of receptors.³⁵⁷ However, Jerne's theory is generally regarded as incorrect or at least a great oversimplification.³⁵⁸

A process of **clonal selection**³⁵⁹ is now thought to be basic to self-tolerance. Credit for the theory, developed in the 1950s, is usually given to F. M. Burnet. However, Ehrlich, Jerne, and David Talmage were also prominent contributors.^{37,351,359,360} An essential postulate of the clonal selection theory is that each B lymphocyte is predetermined to make antibodies of only a single specificity. The mechanism of **allelic exclusion**,³⁵⁹ which makes this possible, is described in Section E. Clonal selection can occur because the B cells carry their antibodies as surface receptors. Binding of an antigen provides a signal for clonal expansion. However, during development in the thymus the progenitor B cells carrying self-reactive antibodies are killed by apoptosis. Later, peripheral B lymphocytes also undergo selection by a complex network of signaling and apoptosis.^{361–363} In a similar manner excess lymphocytes that build up during an immune response must be removed.³⁶⁴

The learning of self by T lymphocytes also happens in the thymus early in development, during the first three weeks of life in mice.^{365,366} This again involves selection against potentially autoreactive lymphocytes carrying idiotypes that are also present on the body's own tissues and which have a high affinity for self peptide•MHC complexes. However, T cells with a weaker affinity for a self MHC molecule but a high potential affinity for a nonself peptide are allowed to develop.^{366,367} Only about 1% of the lymphocytes that develop in the thymus emerge as mature T

cells.^{368,369} Others appear to be killed (clonal deletion) or to become unreactive toward antigen.^{353a} The latter enter a state referred to as **anergy**.^{370–373}

As with every aspect of metabolism, homeostasis is essential to the immune system, which must be able to both grow and shrink rapidly.^{373a–c} Antigens, cytokines, apoptosis-inducing signals, immune inhibitor receptors,^{229h} and receptor tyrosine kinases^{373d} all participate in preserving the delicate balance that is required.

9. Immunologic Memory and Vaccination

In 1796, Edward Jenner carried out the first human vaccination. Attempting to protect a teenaged boy from disfigurement and possible death from smallpox, he vaccinated him with material from a cowpox lesion on the hand of a milkmaid. (She had contracted the disease from a cow named Blossom, whose hide hangs in St. George's Hospital in London.)^{353,374,375} Six weeks later he inoculated the boy with virulent smallpox. Fortunately, the boy didn't contract the disease. Today vaccination is in use for more than 70 bacteria, viruses, parasites, and fungi, and the results have been impressive.³⁷⁶ Poliomyelitis has been almost eliminated.³⁷⁷ Smallpox has not been seen for many years, and the decreases in diphtheria, measles, mumps, whooping cough, and rubella have been impressive.

A nagging question is "How long will vaccination last?" One unplanned experiment resulted from two epidemics of measles in the remote Faroe Islands. The first outbreak was in 1781 after which the islands remained free of measles for 65 years. The second outbreak in 1846 affected 75–95 % of the population, but according to a physician who investigated the epidemic not a single one of the many aged people living who had had measles in 1781 contracted the disease a second time.³⁸ This bit of history confirms that immunological memory is sometimes very long-lived (although it doesn't prove that the smallpox vaccination older people received is still good!).^{377a}

Vaccines have been prepared traditionally by use of viruses or organisms killed by compounds such as formaldehyde or by attenuated viruses or live organisms. These are selected for a low degree of virulence after repeated passages through live animals or cell cultures.³⁷⁶ Newer methods utilize purified viral proteins, bacterial capsular polysaccharides, or DNA.^{378–379a} In the future edible vaccines may be produced in plants.³⁸⁰ Nevertheless, it is often very difficult to devise effective vaccines. In spite of 80 years of effort better vaccines against tuberculosis are needed.^{381,382} All efforts to produce an AIDS vaccine have failed.^{375,379} A satisfactory vaccine must activate both B cells and T cells. Activation of the latter may be especially difficult. Continuous development of new strains of bacteria is a problem for vaccination against

tuberculosis and has been an insurmountable barrier to vaccination against AIDS. However, in the latter case it may be possible to use vaccination to prevent an HIV infection from progressing to AIDS with complete destruction of CD4⁺ T cells.³⁷⁹ As the immune system and also diseases become better understood, it is possible that new strategies for induction of specific cytolytic T cells can be devised,³⁸³ e.g., for AIDS³⁸⁴ and even for cancer.³⁸⁵ A current obstacle to development of new vaccines is that pharmaceutical companies view vaccines as unprofitable.^{385a}

We need a better understanding of how memory B and T cells are formed and selected for long-term survival.³⁸⁶ After differentiation and selection in the bone marrow, B cells move to the spleen. These **transitional B cells** within germinal centers undergo further selection to become mature B cells and B memory cells.^{256,387,388} Interactions with cytokines and with coreceptors play important roles. A subset of B lymphocytes may remain in the germinal centers, serving as a kind of stem cell providing new memory B cells continuously.³⁹ **Naive T cells**, which have not yet encountered antigen, travel throughout the body but apparently don't enter nonlymphoid tissues. However, after being presented with antigen by dendritic cells within lymph nodes, some T cells move to the skin and other peripheral locations.^{389,390} After a pathogen is destroyed most of these T cells die, but a few remain as long-lived memory cells. These are able to respond to a second encounter with a pathogen.^{40,383,391} Apparently continuous new exposure to antigen is not needed for long-term immunity, but the slowly dividing CD8⁺ memory T cells may require continuous stimulation by IL-15 to counteract inhibition by IL-2.³⁹²

E. The Rearranging Genes for Immunoglobulins and T-Cell Receptors

An impressive example of the kind of permanent changes that can occur in the genome of specialized cells is provided by the genes of immunoglobulins and T-cell receptors. B lymphocytes make tens of millions of antibodies of differing sequence, and T lymphocytes make a similar number of different T-cell receptors. This diversity is established in major part in the DNA but also by alternative splicing and editing of RNA.

1. Rearrangements of Germline DNA

Each ~110-residue domain of an immunoglobulin is encoded by a single exon, but the exon for the amino-terminal or variable domain is assembled from two or three small genes (or segments) selected from a large family of such genes present in the germ cells and in the lymphoid progenitor cells.^{393–396} Within the

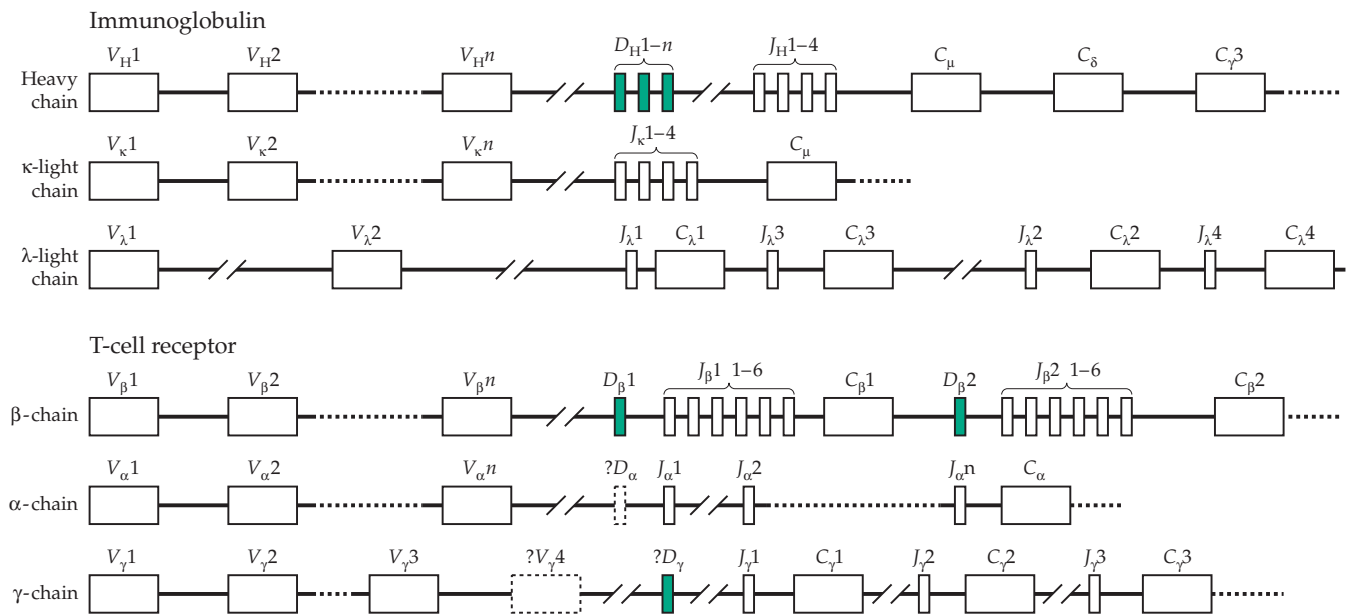


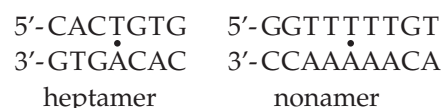
Figure 31-17 Organization of the immunoglobulin and T-cell receptor gene families of the mouse. The human γ -gene pool is larger as is the λ light-chain immunoglobulin gene pool. All six gene pools contain separate gene segments encoding the variable and constant regions of the antigen receptors of lymphocytes. In the course of lymphocyte development, one of the V segments is juxtaposed by chromosomal rearrangement with one of the J segments and, where applicable, a D segment to form a complete variable-region gene. Each V segment has two regions of hypervariability, which are known in the case of the immunoglobulins to contribute to the antigen-binding site in the folded molecule. A third hypervariable region, which also contributes to the antigen-binding site, is generated by the junction of the V segment with the J or the D and J segment(s) at recombination. There is more germline diversity in the T-cell receptor than in the immunoglobulin gene pools in the J segments. The β -gene pool is organized in a way that also allows more combinatorial diversity. The β -gene pool contains fewer J regions than the gene pool, but it has two distinctive features that allow for exceptional diversification during somatic rearrangement. First, the 'rules' for recombination allow in principle the joining of both D to D segments and V segments directly to J segments, neither of which is possible in the immunoglobulin heavy-chain pool. The D segments of the β genes can be read in any of the three possible reading frames, so that varying the site of the V–D junction alone can make a substantial contribution to the diversity of the third hypervariable region. From Robertson.³⁹⁷

V region the three short hypervariable segments alternate with four framework segments that have a more nearly constant structure. The V region of the light-chain genes (of either the κ or λ type) is put together in part from a V gene that encodes an approximately 95-residue sequence making up the first three framework regions plus two hypervariable regions and part of the third hypervariable segment. There are ~100 different V_κ genes and ~30 V_λ genes^{398–400} in the light-chain family. The similar arrangement of genes in the mouse is indicated in Fig. 31-17. These V genes are spaced at intervals of 14–30 kb within the DNA.

The rest of the third hypervariable region and the fourth framework region of the κ and λ chains are encoded by short **J (joining) genes** that specify ~15 residues. There are about five J_κ and three J_λ genes in the mouse. The first mechanism for creating antibody diversity lies in the large number of V genes, which are especially diverse in their hypervariable regions. The second mechanism is the joining of any one of these V genes with any one of the J genes, the joining taking place within the third hypervariable region.

There are also at least six different constant (C_λ) genes in the human genome.

The heavy-chain genes are more complex. There are over 200 V_H genes located at 14- to 16-kbp intervals, which are followed by ten ~15–17 bp **D (diversity) genes** and 4–6 J_H genes. During the differentiation of the lymphoid stem cells pre-pro-B cells can be identified in which the V, D, and J genes are still separate. Later there are pro-B cells containing joined DJ_H segments, then pre-B cells with joined VDJ_H segments, then B cells with a VJ_L (either κ or λ) segment also joined. The overall process of gene rearrangement, mRNA splicing, and immunoglobulin synthesis is outlined in Fig. 31-18. The joining of the gene segments occurs by recombinational mechanisms that involve 7-bp and 9-bp recognition sequences known as **recombination signal sequences (RSSs)**^{401–402b}:



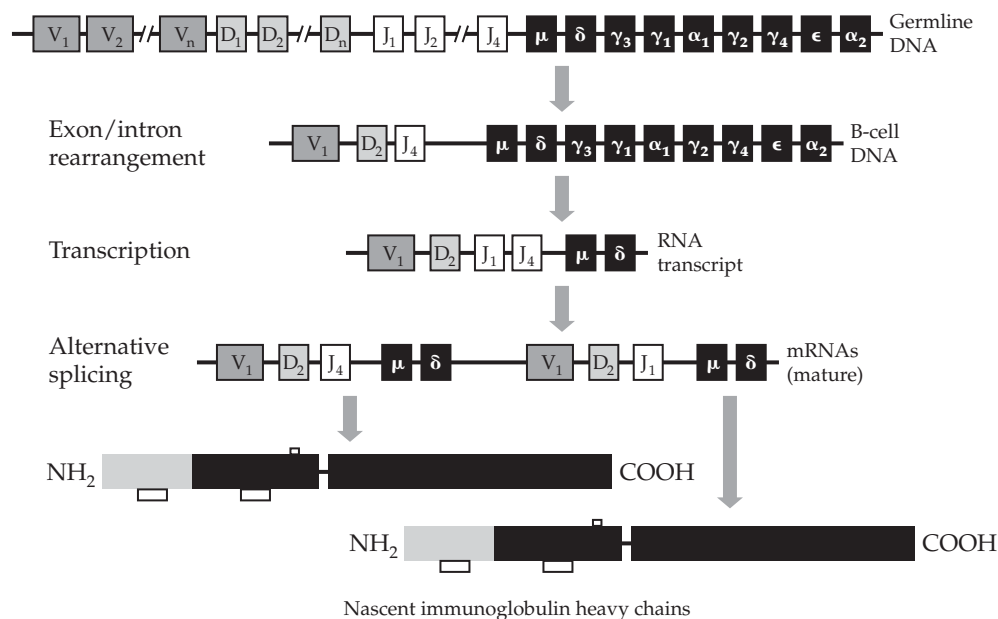


Figure 31-18 Human immunoglobulin heavy-chain gene structure and gene processing. Exons of the heavy-chain genes that encode the variable regions of the immunoglobulin molecule are labeled $V_1, V_2, \dots, V_n$. Selection from these V exons during embryonic development produces the unique sequences of each B-cell clone. The germline genes for the immunoglobulin heavy chains also contain diversity exons, labeled $D_1, D_2, \dots, D_n$. Recombination between the V and D regions occurs more frequently than that between the V and J exons in the light-chain exons. Introns between the V and D and between the D and J exons contain signal sequences that regulate synthesis of the Rag1/Rag2 recombinase. This enzyme is responsible for the efficient recombination that gives rise to the epitope-specific B-cell clones with their individual Ig genes. The heavy-chain genes contain exons that encode all of the isotype heavy chains. Class switching, i.e., the change in chain expression that occurs during antibody synthesis after B-cell activation, results from alternative splicing between the J exons and the exons for the various heavy-chain isotypes. Redrawn from Bhagavan.⁴⁰³ Courtesy of N. V. Bhagavan.

The heptamers have twofold rotational symmetry. The recognition sequences adjacent to one or both sides of the coding segments are separated by 12- or 23-bp spacers, and an empirical rule states that joining can occur only when one pair of recognition sequences are separated by 12 bp and the other by 23. Recombination involves cleavage and rejoining of the DNA, a process also used for reorganization of the T-cell receptor genes. The cleavage is catalyzed by a complex of two proteins RAG1 and RAG2, which are encoded by the adjacent **recombination activating genes** *RAG1* and *RAG2*.^{402–406} Cleavage is assisted by HMG chromatin proteins (Chapter 27, Section A.4) and by the level of histone acetylation in an associated enhancer.⁴⁰⁷ In the presence of a divalent metal such as Mg^{2+} or Mn^{2+} a single-stranded nick is made between the 3'-end of the DNA coding region and the heptamer sequence. The released 3'-OH group then attacks the phosphodiester bond on the opposite strand, cleaving it in a transesterification reaction. This leaves the coding sequences capped by a hairpin end and a blunt cut end on the RSS.⁴⁰⁸ The cut ends are apparently held until synapsis with a nonamer RSS further in the 3' direction and rejoining can occur.⁴⁰⁹ Rejoining requires DNA ligase IV.⁴¹⁰ The process resembles that

of transposition (Chapter 27, Section D.4).^{411,412} The details are still uncertain.^{408,413,413a}

The recombinational events are not entirely regular.^{401,404,414,415} There are excisions of pieces of DNA, and additional nucleotides may be inserted randomly through the action of **terminal deoxynucleotidyl transferase**.^{416,417} All of these mechanisms lead to additional diversity. After the rearrangements are completed, the fused VDJ and VJ gene segments are close enough to suitable constant-region (C) sequences that when the genes are transcribed the intervening sequences in the RNA are spliced out to yield the mature mRNAs for light κ and λ chains and H chains.

2. Somatic Hypermutation and Affinity Maturation

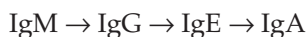
Yet another factor that introduces immunoglobulin diversity is the occurrence of somatic mutation at unusually high rates (**hypermutation**) in the hyper-variable regions. Hypermutation consists largely of point mutations in the V-region gene segment and occurs at a rate 10^5 - to 10^6 -fold higher than for the rest of the genome.^{51,418–423} It occurs after B cells have been

presented with antigen by dendritic cells in the germinal centers in a process called affinity maturation. The result is generation of an enlarged repertoire of B cells, some of which synthesize antibodies with an increased affinity for the antigen. These will be selected for clonal expansion.⁵¹ Further genetic alteration occurs by **gene conversion** (see p. 1566), which involves copying from a homologous segment of DNA, perhaps from a nearby pseudogene.^{423a,b} Some experiments suggested that RNA editing (p. 1642) may also contribute to antibody diversity.^{29,424,424a} However, it appears that the observed deamination of cytosine to uracil rings occurs at the DNA level and is initiated by an **activation-induced cytosine deaminase**.^{424b} The generation of uracil, which may be removed from the DNA by uracil-DNA glycosylase (p. 1579), apparently triggers both somatic hypermutation and gene conversion. Both processes also depend upon DNA repair via homologous recombination or nonhomologous end joining (p. 1581). The error-prone DNA polymerase τ is also needed.^{424c} Class-switching recombination, discussed in the next section, is apparently also initiated by the activation-induced deaminase.^{424d} These are affected by **RNA editing** (Chapter 28).^{29,424} **Receptor editing**, gene rearrangements that occur in the peripheral immune system, also contributes to affinity maturation.^{424,425}

It is important that the genes as finally assembled maintain correct reading frames so that a potentially useful antibody can be made.⁴²⁶ Although lymphocytes are diploid and therefore contain two sets of immunoglobulin genes, a single cell produces only one kind of immunoglobulin (**allelic exclusion**). When the genetic rearrangements produce a light chain able to combine with a heavy chain to form a functional immunoglobulin, a signal may be sent that stops the rearrangements in the other chromosome.⁴²⁷ However, recent findings suggest that one allele may be marked for inactivation early in development, just as one X chromosome becomes inactive in females (Chapter 32).⁴²⁸

3. Immunoglobulin Class (Isotype) Switching

A newly matured B cell produces initially IgM bound to its surfaces. The difference between the C_H domains of bound and secreted antibodies of a given type seems to lie in alternative splicing of the mRNA. More mysterious are the consecutive switches from IgM to other types in the following sequence:



This is the same sequence in which the C_H genes lie (Fig. 31-18). However, the C_δ gene, which codes for IgD, is not utilized in this sequence. A newly matured

B cell transcribes the C_μ gene to give IgM. Later, the class switch occurs, apparently by a looping-out recombination mechanism, allowing a C_γ gene (for IgG) to be expressed.^{429–433} The switches are mediated by tandemly repeated DNA sequences in 1- to 10-kbp switch regions and are controlled in part by cytokines. Other complexities are involved in synthesis of the J chain of IgM⁴³⁴ and in synthesis of IgD. Although the latter is a major surface immunoglobulin on B lymphocytes, its exact functions have been hard to understand. The δ exon of its heavy chain gene is joined to a J exon by alternative splicing of the mRNA (Fig. 31-18).¹¹² In a similar manner, the difference in the surface-bound and soluble forms of IgM arises by alternative mRNA splicing.⁴⁰³

The T-cell receptor gene families are also indicated in Fig. 31-17. Their development is remarkably similar to that of the immunoglobulin genes and involves most of the same mechanism of diversification^{365,397,435} with the exception of somatic hypermutation. The same recombinase may cut the DNA to initiate rearrangements of all of these gene families.⁴³⁶ A single T-cell precursor may give rise to 1000 or more clones with unique β -chain sequences.⁴³⁷ With a total of $\sim 10^6$ different β chains there are potentially $\sim 10^{15}$ unique T-cell receptor structures that could arise from the 42V and 61J segments of the α -chain gene and the 47V, 2D, and 13J segments of the human β -chain gene.⁴³⁸ Allelic exclusion is observed, as with the immunoglobulins.⁴³⁹

F. Disorders of the Immune System

Many things can go wrong with a system as complex as the human immune system. In immunodeficiency disease some component is missing or has been inactivated. In autoimmune diseases the immune system attacks some component of the body. Of the known problems none is more common than **allergy**,^{440–442} which may be described as the inappropriate activation of the immune system by environmental antigens (**allergens**).

1. Allergy

One in 10 persons, ~ 22 million people, in the United States have allergies. Ten million of these suffer from the nasal discomfort of “hay fever” and six million from the more serious **asthma**. Substantial numbers of people in the United States die of allergic reactions to insect stings (more than 30 per year) or to injections of penicillin (300 per year in 1970). Foods, drugs, pollens, mold spores, mites in house dust, and even heat or cold can evoke serious allergic reactions. Among these **eczema** (atopic dermatitis) is very common. A major cause of allergic reactions has been

traced to molecules of immunoglobulins IgE, which bind to the **basophils** in the blood and to the related **mast cells** of tissues. Binding of an antigen to these IgE molecules activates them. These activated antibodies bind (as in Fig. 31-6) to the α subunits of the $F_{c\epsilon}RI$, a transmembrane receptor on basophil or mast cell surfaces.^{108,109,443} If two or more IgE molecules bind to a mast cell, they may aggregate and activate the mast cell to release its histamine-containing granules.^{444,445} The granules also release cytokines and arachidonate, which is converted primarily into prostaglandin D_2 (Fig. 21-7) and into products of the 5-lipoxygenase pathway (Fig. 21-8). The products include the chemotactic leukotriene B_4 and leukotrienes C_4 and D_4 . The latter two constitute the slow-reacting substance of anaphylaxis (Fig. 21-8). The result is a rapid inflammatory response with dilation of blood vessels, increased vascular permeability, infiltration of leukocytes, and destruction of tissues.

What is the normal function of IgE and the mast cells? These cells are located at places where parasites might enter tissues. IgE is involved in killing of schistosomes, and elevated IgE levels are seen in patients infected with various parasites. The killing of schistosomes seems to be mediated by blood platelets as well as by neutrophils and eosinophils with the help of mast cells.^{446,447} Allergic persons often have an IgE level over ten times normal. This high level makes the individual especially sensitive to IgE-mediated reactions, a condition called **atopy** (meaning "strange disease").^{448,448a} Allergies may also be accompanied by increased B cell levels.⁴⁴² This can sometimes be responsible for the sudden and sometimes fatal systemic reaction of **anaphylaxis**. T-cell responses may also cause anaphylaxis.

Most allergy-inducing antigens are proteins, but proteins vary widely in their antigenicity. Only a few natural proteins are major allergens. Many of these are relatively small, with molecular masses of 5–50 kDa. Most are soluble, and some are glycoproteins.⁴⁴⁹ Mites are the closest animals to human life and carry allergens that are among the most important causes of asthma and allergic dermatitis. The allergens are 125- to 129-kDa proteins crosslinked by three disulfide bridges.^{450,451} Cockroaches^{449,452} and other insects also form many allergens. Among them are the hemoglobins of small flies of the chironomid family.⁴⁵³ Studies of the latter suggest that antigenicity may arise both from flexible regions and also from the presence of a preponderance of amino acids with polar side chains.

Proteins of cat saliva dry and flake off as dander, which contains major indoor allergens linked to asthma. Dogs, horses, cattle, and other animals also provide several allergens,⁴⁵⁴ some of which are lipocalins (Box 21-A).⁴⁵⁵ Other close-to-home allergens are provided by fungi that live on skin or nails.⁴⁵⁶

Plants provide a host of allergens. Major allergens

are found in pollen of rye-grass,⁴⁵⁷ of many other grasses, of ragweed,⁴⁵⁸ and of olive trees.⁴⁵⁹ Natural rubber latex would appear to be a harmless high polymer, but it contains antigenic proteins, which have been blamed for 1100 anaphylactic attacks with at least 15 deaths between 1988 and 1992.^{460,461}

Food is a major source of allergens, which are often overlooked. Food allergies may be hard to diagnose and symptoms such as headache, diarrhea, itching, and asthma may be attributed to other causes. However, the occasional rapid death from anaphylactic shock, e.g., from exposure to peanuts,^{461a} is a reminder that unrecognized food allergies exist. About 100–200 persons die annually of food allergies. About 90% of recognized food allergies involve milk, eggs, fish, crustacea, peanuts, tree nuts, soybeans, and wheat.⁴⁶² There are usually only a few allergenic proteins in any one food. Many of these are resistant to digestion in the stomach. Some but not all are compact proteins with multiple disulfide crosslinkages. However, no structural generalization can be applied to all food allergens.

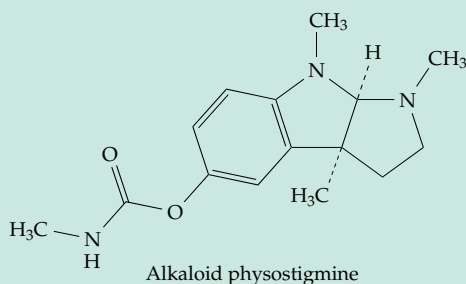
The increasing use of genetic engineering of foods poses both risks and hopes. The accidental incorporation of an allergenic protein into a plant or animal product can make a food previously safe for a person deadly. For this reason, attempts are now made to identify likely allergens and to avoid transferring their genes. However, this process can't be completely reliable, partly because we each have our own personal immune system. This is one reason for requiring accurate labeling of foods. On the positive side recognized major allergens can probably be eliminated by genetic engineering. The muscle protein tropomyosin (Fig. 19-9) is a well-known allergen whose allergenicity varies among different sources. Tropomyosin from beef, pork, and chicken is usually not highly allergenic, but that from shrimp often is.⁴⁶² Perhaps safer shrimp can be created.

Allergies are treated in various ways, often with antihistamines or corticosteroid ointments. Injection of adrenaline is an emergency treatment of anaphylaxis or asthma. Also important is **specific immunotherapy**, better known as "allergy shots."⁴⁴¹ Small amounts of allergen in increasing amounts are injected subcutaneously at intervals to desensitize the patient. Use of purer antigens, which may be engineered to decrease their antigenicity, may provide advances in this technique.⁴⁶³

Asthma is one of the most common chronic diseases in industrialized countries, affecting 10% or more of young children in some countries. An atopic disease with high IgE levels, asthma is induced by small particles of antigen, which are able to penetrate deep into the lungs.⁴⁶⁴ About 80% of asthmatic children are allergic to house mites.⁴⁵⁰ Animal danders are another major cause. The incidence of asthma appears to be increasing in many modern societies, but

BOX 31-D MYASTHENIA GRAVIS

One of the best-understood autoimmune diseases is myasthenia gravis, a condition associated with a decrease in the number of functional postsynaptic nicotinic acetylcholine receptors (Fig. 30-23) in neuromuscular junctions.^{a-e} The resulting extreme muscular weakness can be fatal. Myasthenia gravis is not rare and affects about one in 10,000 people.^c An interesting treatment consists of the administration of physostigmine, diisopropylphosphofluoridate (Chapter 12, Section C,1), or other acetylcholinesterase inhibitors (Box 12-E). These very toxic compounds, when administered in controlled amounts, permit accumulation of higher acetylcholine concentration with a resultant activation of muscular contraction. The same compounds



the reasons are unclear,^{465,466} and the increase doesn't appear to be linked to air pollution. A chronic atopic condition usually precedes an acute attack of asthma. In addition to IgE and eosinophils there are excessive numbers of neutrophils with high-affinity IgE receptors in the airway tissues.⁴⁶⁷ Prostaglandin D₂ released from mast cells may play a role in triggering an attack.⁴⁶⁸ Cytokines,⁴⁶⁹ nitric oxide,⁴⁴⁷ and nerve growth factor⁴⁷⁰ may also participate in the response. The presence of a high concentration of glutathione and of glutathione peroxidase, whose concentration increases in asthmatic lungs, may reflect the action of the antioxidant system in combatting inflammation.⁴⁷¹ The surfactant proteins SP-A and SP-D (Box 8-B) are Ca²⁺-dependent lectins, which serve as regulators of the innate immune response. Their concentrations also increase in asthma.^{472,473}

Treatment of asthma has depended upon inhaled glucocorticoids, quick-acting bronchodilators usually β -adrenergic antagonists, and long-acting beta agonists such as theophyllins and leukotriene antagonists.⁴⁶⁴

2. Autoimmune Diseases

There are numerous **autoimmune diseases** in which the body makes antibodies against its own cells

are widely used in the treatment of glaucoma.

More than 90% of patients with myasthenia gravis have circulating antibodies directed against a subunit of the acetylcholine receptor.^f Immunosuppressive drugs and steroids help to cut down on these autoantibodies, and many patients are benefited by removal of the thymus. Newer approaches involve specific immunotherapy aimed at increasing tolerance to either T cells or to B cells.^{c,d} For example, oral ingestion of purified acetylcholine receptors to desensitize the body's response or inhibition of production of IL-2.

A possible cause for the production of the damaging antibodies may be the sharing of common antigenic determinants between the receptor protein and surface proteins of bacteria such as *E. coli*.^f

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^b Tzartos, S. J. (1984) *Trends Biochem. Sci.* **9**, 63–67

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^d Steinman, L., and Mantegazza, R. (1990) *FASEB J.* **4**, 2726–2731

^e Barnes, D. M. (1986) *Science* **232**, 160–161

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(Table 31-3).^{363,474–476} In **myasthenia gravis** (Box 31-D) antibodies attack the acetylcholine receptors in postsynaptic membranes.⁴⁷⁷ In **Graves disease** aberrant antibodies are directed against receptors for thyrotropin. They have a stimulatory rather than an inhibitory effect and cause hyperthyroidism.⁴⁷⁸

Childhood onset (Type I) diabetes results from destruction of insulin-secreting cells by an autoimmune reaction triggered by environmental factors in genetically susceptible persons (Box 17-G).^{479,480} The principal autoantigen appears to be the 65-kDa form of glutamate decarboxylase (GAD).^{481–483} While GAD has an essential function in the formation of γ -aminobutyrate in the brain, its role in the pancreatic islets is not clear. What is established is the presence of specific GAD65-stimulated T cells in diabetic individuals. The stimulating autoantibodies, which may appear in the blood years before diabetes is evident, carry HLA-DR4 type surface MHC antigens.^{484,485} After the activated T cells kill enough of the pancreatic β cells, diabetes appears.⁴⁸⁶

Myasthenia gravis, Graves disease, and type I diabetes are organ-specific autoimmune diseases. Another group of autoimmune diseases are systemic, affecting many tissues. For example, in the severe **systemic lupus erythematosus** there are often antibodies against the victim's own DNA.^{487–488a} The

antibodies may then attack any tissue, e.g., the red blood cells. Antibodies against a variety of other nuclear constituents such as histones,²⁰⁷ ribosomal protein L7,⁴⁸⁹ ubiquitin,⁴⁹⁰ enzymes, cardiolipin,⁴⁹¹ and small nuclear RNAs⁴⁹² are also made. The primary defect appears to be an intolerance to chromatin and in particular to nucleosomes.⁴⁹³ Antibodies to nucleolar and other components of nuclei are also present in progressive systemic sclerosis (**scleroderma**).^{494,495} In **rheumatoid arthritis**, a chronic inflammation of joints, the serum and joint fluids contain abnormal complexes, which appear to consist entirely of immunoglobulins. They may be antibody-antiidiotype antibody complexes.^{496,496a} Immunization of animals with type II collagen induces a very similar arthritis,^{496,497} but this collagen probably doesn't supply the offending human antigen. Susceptibility to rheumatoid arthritis is linked to HLA-DR4 class II MHC genes. Molecules of the class II DR4 subtypes may associate with antigenic peptides of uncertain origin to induce a T-cell response. CD4⁺ T cells are thought to drive inflammation in the disease.⁴⁹⁷ Monocytes are attracted from the blood and become inflammatory macrophages.⁴⁹⁸

Another autoimmune disease in which antibodies

attack collagen is **Goodpasture disease**. It is mediated by B cells, which form antibodies directed at the N-terminal domain of the $\alpha 3$ chain of collagen IV (pp. 435–438).^{499–501} The antibodies attack the glomerular basement membranes causing a rapidly progressing glomerulonephritis and also lung hemorrhages. Primary **glomerulonephritis**, a major kidney disease, may be caused by a cross-reaction between the membrane of streptococci and the glomerular basement membranes.

In **Sjögren syndrome** autoantibodies are directed against α fodrin (p. 405).⁵⁰² In primary **biliary cirrhosis** they are directed at mitochondria and specifically to a pyruvate dehydrogenase subunit (Fig. 15-14).⁵⁰³ In the inflammatory muscle disease **polymyositis** autoantibodies are often directed against cytoplasmic proteins including aminoacyl-tRNA synthetases.⁵⁰⁴ In the rare **paroxysmal cold hemoglobinuria** autoantibodies attack red blood cell membranes only when the temperature of an extremity is lowered. **Paroxysmal nocturnal hemoglobinuria**, a serious complement-mediated condition, results from deficiency in the complement decay accelerating factor. This is a result of a defect in the PGI tail on this factor.⁵⁰⁵

Celiac disease (celiac sprue) is an allergic inflammatory condition caused by poorly digested proline-rich sequences of wheat gluten and related proteins (p. 74). The disease is usually not recognized, but it may occur in 3% or more of the United States population. A T-cell response that causes destruction of the smaller intestinal mucosa, celiac disease is characterized by malabsorption and diarrhea.^{505a–c} It can cause death by starvation. A primary target of the autoantibodies is a transglutaminase.^{505c,d}

Most cells of the immune system are ordinarily kept apart from those of the nervous system by means of the blood-brain barrier. However, allergic encephalomyelitis, in which T cells attack the myelin sheath of brain neurons, can easily be induced in mice.⁵⁰⁶ A similar autoimmune process is thought to be involved in human **multiple sclerosis** (see Chapter 30, pp. 1769, 1808, and Fig. 30-9).^{507,508} High levels of circulating IgM are found in some demyelinating diseases of peripheral neurons.⁵⁰⁸ In **Rasmussen's encephalitis**, which causes brain inflammation and epilepsy, serum antibodies attack a glutamate receptor subunit **GluR3**.⁵⁰⁹

The causes of autoimmune disease doubtless lie largely in the difficulty of developing a repertoire of immunoglobulin-forming B cells and of T-cell receptors that will always reliably distinguish self from a foreign antigen. The problem can lie either with B-cell recognition or with the T-cell receptors. Extensive medical use is made of **immunosuppressants** in treatment of persistent allergic reactions, autoimmune problems, and rejection of transplanted tissues. Among these compounds are the steroidal

TABLE 31-3
Some Autoimmune Diseases

Addison disease	Adrenal glands
Ankylosing spondylitis	
Celiac disease	Upper intestines
Crohn disease	Intestines
Diabetes, type I	Pancreatic islets
Glomerulonephritis	Kidney
Goodpasture disease	Kidney
Graves disease	Thyroid gland
Guillain – Barré syndrome	Gangliosides
Multiple sclerosis (MS)	Peripheral myelin
Myasthenia gravis	
Paroxysmal cold hemoglobinuria	Red blood cells
Primary biliary cirrhosis	
Psoriasis	Skin
Polymyositis	
Rasmussen encephalomyelitis	Cerebral cortex
Rheumatoid arthritis	Joints
Scleroderma	Skin
Sjögren syndrome	
Systemic lupus erythematosus	Many tissues
Thyroiditis	Thyroid gland
Ureitis	

BOX 31-E EVADING THE IMMUNE SYSTEM

Parasitic species always have a problem with the antibodies and killer T cells of their hosts, and the chemical makeup of the external coats of parasites tends to reflect this fact.^a An example is provided by **trypanosomes**, which cause sleeping sickness and which make much of Africa unsuitable for cattle grazing.^{b–d} Trypanosomes in the bloodstream evade the immune system by covering the outer surface of their plasma membrane, flagella and all, with a dense 12- to 15-mm thick monolayer of an ~60-kDa **variable surface glycoprotein**.^{e,f} The glycoprotein molecules are anchored in the cell membrane by C-terminal glycosylphosphatidylinositol (GPI) anchors (Fig. 8-13).^g The glycoprotein layer protects the parasite but is soon attacked by the host's immune system. However, the parasite has perhaps 1000 different genes for the variable surface protein, and every ten days or so new clones of trypanosomes appear with new coats that the immune system is not prepared to attack. To accomplish this cells occasionally copy one of the previously unused variable surface glycoprotein genes and place it into a new location in the genome, where it is expressed.^{h–j}

Parasitic **nematodes** shed the outer layers of their external cuticle and like trypanosomes reveal a new layer with different antigenic proteins.^k *Giardia* protects itself in a similar fashion.^l **Schistosomes**, tiny parasitic flatworms, evade a host's immune system by shedding complex glycoproteins from specialized double outer membranes.^m Antigenic determinants including MHC antigens characteristic of the mouse have been identified in the membrane of schistosomes from infected mice. Thus, one aspect of the parasite's defense may be to hide behind surface recognition markers stolen from its host.^{n,o} Schistosomes also secrete the peptide Thr-Lys-Pro, which inhibits macrophages, as well as a small molecule that inhibits T lymphocytes.^o

The malaria parasite *Plasmodium* has a complex life cycle with several forms and spends much of its life hiding within red blood cells.^p It may also suppress the immune system. The unicellular sporozoites, which are injected into the bloodstream by mosquitos, are protected by an external coat protein that is unusual in containing many short repeated sequences. For example, that of *P. falciparum*, which causes the most deadly form of malaria, contains the sequence Asn-Ala-Asn-Pro repeated 37 times.^q These coat proteins undergo unusually rapid evolution, which makes the preparation of vaccines difficult.^r

Trypanosomes, schistosomes, and malaria parasites still represent major health problems.

Malaria kills two to four million persons a year and endangers almost a third of the world's population. It has been impossible to produce suitable vaccines for any of these parasites. However, the cloning of genes for individual parasite proteins has given hope that effective vaccines can be devised.^{e,o–v} One problem is the lack of interest in financing the effort.^w

Many other protozoan parasites and bacteria invade cells and take up residence in macrophages.^x These include species of *Salmonella*, *Legionella*, and *Mycobacterium*.^y Bacteria often employ structural mimicry to gain access to a cell,^z e.g., by mimicking the type III secretion system (p. 520).^{aa} Some bacteria have developed defenses against reactive oxygen species, allowing them to evade the action of phagocytes.^{bb} *Borrelia burgdorferi*, the Lyme disease spirochete, synthesizes an unusual single-layer β -sheet outer surface protein,^{cc} which becomes coated with complement protein H. This may protect the bacteria and allow them to live for a long time within cells.^{dd}

Even the lowly **influenza virus** finds a way around our immunity so that it can strike us repeatedly. As this virus matures, it acquires a lipid membrane by budding from the host cell. Two virally encoded proteins are present in the membrane. One is a trimeric hemagglutinin, which forms small 7.6-nm spikes that protrude from the virus surface.^{ee,ff} The hemagglutinin monomer is a 550-residue peptide containing four antigenic regions. The RNA genome of the virus undergoes rapid mutation (Chapter 28, Section E,2). At least one amino acid substitution was found in each antigenic region, when hemagglutinins from influenza viruses causing epidemics in 1972 and 1975 were compared with the strain that caused a worldwide epidemic in 1968. Recently the type A influenza virus that caused the 1918–1919 pandemic, the greatest acute plague of the 20th century, has been “resurrected” and investigated using viral RNA from three victims.^{gg–ii} The globular part of the hemagglutinin appears to have come from a pig and the “stalk” from a human lineage. The virus takes advantage of the pool of virus in swine, humans, and birds to vary its structure and create new strains. The reason for the deadly nature of the 1918–1919 strain, which killed 20–40 million people, an unusually large number of whom were young, previously healthy adults, is not clear.

Viruses use a large variety of mechanisms to evade cellular defense mechanisms. Almost every aspect of the innate or adaptive immune systems provides some opportunity for evasion.^{jj} The rapid

BOX 31-E (continued)

mutation rate in a population of virus particles contributes greatly to this ability, allowing chronic infections such as those of hepatitis C^{kk} or delayed catastrophic infections such as those of HIV.

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anti-inflammatory agents such as prednisone and the folate antagonist methotrexate. A new vision of the possibilities for immunosuppression came, however, with the discovery of **cyclosporin A** (Box 9-F). This fungal metabolite inhibits lymphokine formation by helper T cells. It alleviates rejection of grafted tissues and prevents graft-versus-host disease. The use of cyclosporin and FK506 (Box 9-F) has permitted organ transplantation to the extent that by 1987 surgeons had transplanted in one year 1200 livers, 1500 hearts, and 9000 kidneys with one-year survival rates of 80% for hearts and over 90% for kidneys.⁵¹⁰ As mentioned in Box 9-F, cyclosporins bind very tightly to **cyclophilins**,⁵¹¹ which have peptidylprolyl *cis*–*trans* isomerase activity and perhaps other independent functions.⁵¹²

3. Immunodeficiencies

More than 95 different problems of impaired im-

munity have been identified. They affect about 1 in 10,000 persons born. The defects may involve T cells, B cells, NK cells, or phagocytic cells.^{513,514} There may be problems in lymphocyte development.⁵¹⁵ Complement proteins may be lacking,^{161,513,516} or their regulation may be faulty.⁵¹⁷ The immune system has specific “blind spots” and fails to recognize all dangerous foreign antigens.⁵¹⁸ In the fatal **X-linked immunoproliferative syndrome**,^{519,520} the immune system of susceptible males does not respond to the Epstein–Barr virus-induced mononucleosis by killing the persistently lymphoblastoid cells characteristic of that disease. Some individuals are born with **severe combined immunodeficiency** disease. This condition was made well known to the public by the plight of David, the “bubble boy,” who lived 12 years in protective sterile rooms and a plastic bubble-like “space suit.”⁵²¹ The condition is often caused by a defect in the interleukin-2 receptor, but there are a variety of other causes. About 15% of cases arise from a defect

BOX 31-F AN INSECTICIDAL PROTEIN

During sporulation the bacterium *Bacillus thuringiensis* forms within its own cells large protein crystals, which are highly toxic to some insect larvae. The crystals account for 20–30% of the dry weight of the bacterial spores and contain more than one toxin^a and, curiously, a 20-kbp piece of DNA.^b Dusting of garden plants with dried spores from these bacteria has become a popular and effective way of combating cabbage worms and other insects. The toxic protein from one strain of bacteria is encoded by a 4222-bp gene.^c The corresponding 133-kDa 1176-residue polypeptide protoxin undergoes glycosylation and perhaps other modifications, presumably prior to crystallization. After ingestion by susceptible insect larvae (largely Lepidoptera) the protein is cleaved to form a smaller ~65-kDa protease-resistant core, which is the active toxin. Other strains of bacteria produce toxins specific for Diptera or Coleoptera.^{d–f}

X-ray crystallography of the 65-kDa form reveal a three-domain structure. The central domain varies among different strains and is probably involved in recognition and in binding to cell surface receptors.^{e–h} The toxin binds to a receptor, apparently an aminopeptidase N,ⁱ after which the toxin is rapidly inserted into the membrane forming a 1- to 2-nm diameter pore. This leads to cell death.^j

Because the toxins appear to be harmless to human beings and higher animals the toxin genes have been transferred into various other bacteria, which are symbiotic with plants and into plants themselves. Toxin genes in suitably modified form (Chapter 27) were first transferred into bacteria that live naturally in association with roots of *Zea mays* and into tobacco and tomato plants. The new host organisms expressed the toxin genes and protected the plants from damage by caterpillars.^{k,l} Since then the toxin genes have been transferred into many crop plants, which are widely planted.

Two problems must be considered. Insects do develop resistance to the Bt toxin.^m This problem

can be combated by protein engineeringⁿ and by location of new sources of toxins.^{o,p} A second problem deals with the environmental impact.^q Will Bt toxin kill desirable insects? Will the gene be transferred in nature to other species and into the environment? The latter may seem unlikely, but as the toxins are applied to fight soil organisms such as nematodes, transfer into organisms of the largely unstudied soil ecosphere may pose problems.

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in adenosine deaminase, an enzyme of the purine salvage pathway (Fig. 25-17). As mentioned in Chapter 25, genetic therapy for this condition is being used. However, the most reliable treatment for these immunodeficiencies seems to be bone marrow transplantation. By 2000 more than 375 patients worldwide had received this treatment with up to 95% chance of survival.⁵¹³ Virus-induced immunodeficiency is the prime characteristic of HIV infection and **AIDS**.^{522,523} Both the amounts of autoantibodies and of amyloid

deposits increase with age, and immune complex disease is suspected of being a cause of aging.

4. Cancers of the Immune System

A major function of the immune system is thought to be destruction of cancer cells. In this case altered cell surface carbohydrates or proteins elicit an antibody response with destruction of the offending cells.

That this process works imperfectly may explain why the incidence of cancer increases with age and also why the concentration of autoantibodies increases. The immune system is also susceptible to cancers, which include multiple myeloma, leukemias, and lymphomas. Some of these, such as Burkitt's lymphoma, involve rearrangement of chromosome segments that carry immunoglobulin genes.^{524,525} These may result from errors in the gene rearrangements involved in the development of lymphocytes.

G. Defense Mechanisms of Plants

Plants make many compounds that repel or poison animals that eat them. Among such compounds are alkaloids, terpenes, calcium oxalate, fluoroacetate, cyanogenic glycosides, and phenolic compounds.⁵²⁶ The chewing of insects or other wounding of plant tissues releases phenolic glycosides and other reactive compounds from vesicles. Some of these compounds, which are often referred to as **phytoalexins** (see also Box 20-E),⁵²⁷ are repellent to predators, have antimicrobial activity, and /or participate in chemical cross-linking and strengthening of the plant cell wall.^{526,528} Some are protease inhibitors that interfere with a predator's nutrition.⁵²⁹ Some released compounds attract insects that may assist in defense by feeding on predator eggs or by attracting wasps that deposit eggs in predator larvae.⁵³⁰ These can all be regarded as part of an innate defense system that in some respects resembles our own innate immune system. For example, plant defensins (Fig. 31-7), most of which are directed against fungi,^{531,532} resemble those in our tissues.

A system of **receptor-mediated surveillance**, part of the innate system, triggers both immediate **local responses** and secondary immunity throughout the plant.^{530,533,534} Immediate responses include **programmed cell death** (called the **hypersensitive response**⁵³⁵), tissue reinforcement, and production of antimicrobial metabolites. Secondary responses, known as **systemic acquired resistance**, develop immunity throughout the plant. The surveillance system of some plants consists of a series of receptors known as **resistance (R) proteins**, which recognize signaling molecules produced by pathogens.^{535a,b} The R proteins are thought of as being paired with **avirulence (Avr) proteins** of the pathogen. If the resistance protein is missing, the plant will be susceptible

to attack by the pathogen. The pathogen's Avr protein is thought to be part of the chemical attack on the plant, apparently assisting in the invasion. However, if the Avr gene has been lost or is mutated, the R protein won't detect the invasion, and the pathogen may have increased virulence. What are the characteristics of the Avr proteins? They are often small and may be crosslinked by S-S bridges. They may be taken up by plant cells via receptors that resemble the type III translocation system of bacteria (p. 520).⁵³⁴ Similar small protein **elicitors** are released directly by wounding even in plants that do not have paired R-Avr genes.^{529,536}

The R proteins, which act as receptors for Avr, and other elicitor proteins, are usually leucine-rich-repeat proteins with a characteristic nucleotide binding site attached (NB-LRR proteins).^{534,537} Like other cell surface receptors they participate in signaling and utilize both ion channels and Ser/Thr protein kinases.⁵³⁸ The *Arabidopsis* genome contains ~150 sequences that may represent NB-LRR receptors.⁵³⁰

What do these receptors do? Like other cell membrane receptors they may induce both rapid and slower responses. The rapid responses may result from transmembrane flow of ions, just as in neurotransmitter action (Figs. 30-19, 30-20). The first response observed is an oxidative burst, which within minutes generates reduced oxygen intermediates (ROIs; pp. 1072–1074).^{533,539–543} These compounds may participate in crosslinking and lignification of cell walls. Together with nitric oxide (NO) and endogenous salicylic acid (Chapter 25, Section B,7),^{533,544,545} they promote transcription of defense-related genes and participate in the hypersensitive response. A second pathway, utilized against some pathogens especially those that kill plants to obtain nutrients,⁵³³ involves production of jasmonic acid (Eq. 21-18 and associated text)^{545a} and ethylene (Fig. 24-16).^{529,546}

Plants also have mechanisms for minimizing the damage from the over 500 known viruses. These don't often kill plants but can cause great damage. There are interferon-like responses⁵⁴⁷ and gene-silencing mechanisms.⁵⁴⁸ The latter often involve synthesis of dsRNA, cleavage by the enzyme Dicer, and interference with transcription as described on p. 1640.^{113a,b,549,550} This defensive reaction can spread between cells and throughout a plant, apparently by transport of RNA through plasmodesmata and the phloem.⁵⁵⁰

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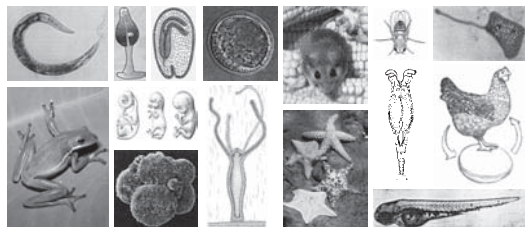
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Study Questions

1. Describe major aspects of the vertebrate innate and adaptive immune system. In what ways do they cooperate?
2. Describe briefly the functions of each of the following: antibodies, defensins, cytokines, complement, MHC proteins, B cells, T cells, dendritic cells, monocytes, macrophages, and neutrophils.
3. Why do antibodies produced using a native protein tend to bind only weakly to the corresponding denatured protein?
4. Discuss the topic of self-identity.
5. What are autoimmune diseases? How does the body avoid most autoimmune diseases?
6. List some methods by which viruses, bacteria, protozoa, and pathogenic fungi gain access to cells or to tissues.



Single cells develop into an astonishing variety of different species, all of which find their niches in the ecosystem. Whether a rectangular bacterium, a plant, a frog, or a human being, the size, shape, the body construction and metabolic pathways are established by the sequential expression of the organism's genes. Recent investigations have confirmed many similarities among major families of proteins from virtually all species. The same studies also emphasize the profound genetic differences between species. Understanding these differences, as well as the interrelationships among species, provides a continuing challenge to biochemists and biologists. Such understanding may even be essential to the survival of the human species in a changing environment.

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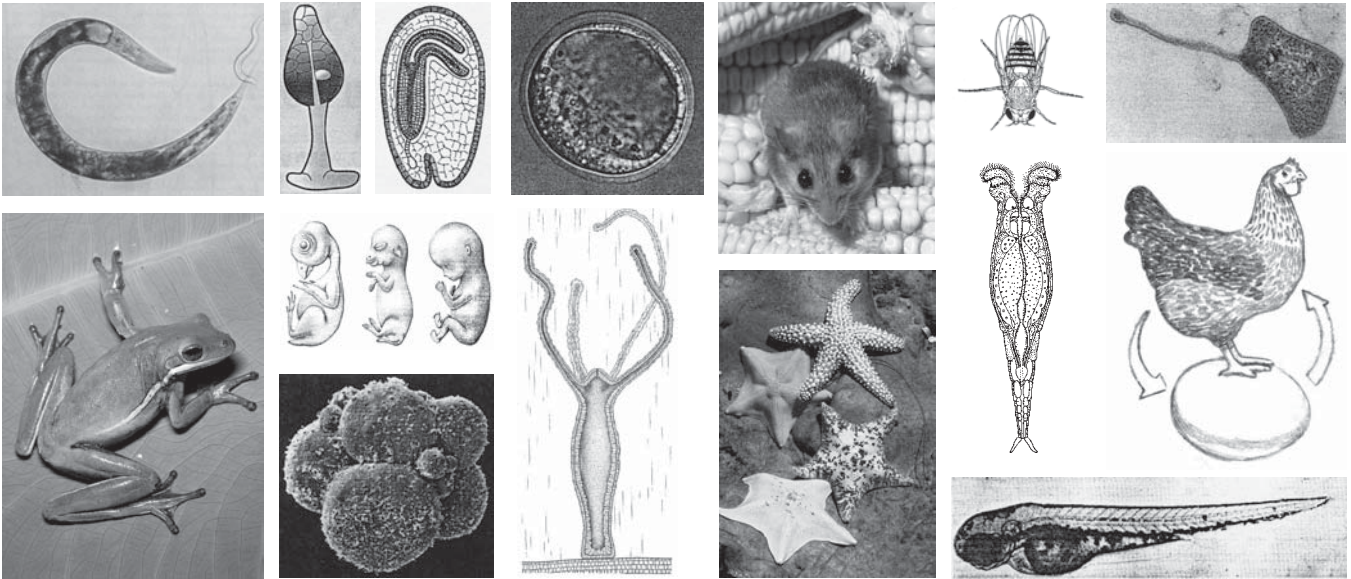
Tables

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1888 Table 32-2 Some Components of Apoptotic Systems

Growth and Development

32



One of the most fascinating of all biological phenomena is the development of an animal from a fertilized egg. From the early embryonic cells, which appear to be much alike, there arise during the course of a very few cell divisions differentiated organs and tissues such as liver, brain, kidney, muscle, skin, and red blood cells. The biochemical properties of differentiated cells are often highly specialized. Red blood cells make hemoglobin, while muscle cells make large amounts of myosin and actin. The endocrine cells of the pancreas make insulin or glucagon, while the exocrine cells form the digestive enzymes that are secreted into the intestinal tract.

Looking more broadly at the biological world we see many additional specialized features at every level of observation. Developmental patterns differ for every organism. Specialized organs abound. The structures of proteins, lipids, carbohydrates, and nucleic acids all vary, and every species has its own metabolic peculiarities. Even among bacteria we find extreme variation. Furthermore, many unicellular eukaryotic organisms undergo complex development within a single cell. The topics of this chapter are too complex for any detailed discussion. We will examine some aspects of growth and development for a few organisms and will ask whether there are simplifying generalizations. Comprehensive textbooks are available.^{1,2}

Enough has been learned about development to make it clear that the DNA contains genetically coded **developmental programs**, which are followed by cells.³ However, both transcription and translation are

controlled by many chemical signals, which influence the execution of the genetic program. Such signals may arise from within a single cell, from the external environment, and from adjacent cells. The tight control is reflected by the fact that in most human tissues at any stage of development no more than about 10% of the total genes are transcribed at any one time. Chemical analysis makes it clear that most specialized cells contain a full complement of DNA but 90% of the genes are turned off.

A. Basic Concepts and Molecular Essentials

Listed in Table 32-1 are some essential aspects of growth and differentiation. Some of them are obvious. **Cohesion** between molecules provides the basis of specificity. **Receptors**, whether they be enzymes, hormone receptors, or receptors for chemotaxis, are essential. They are usually activated by a **conformational change** that accompanies binding. A cell must have receptors that can respond to a variety of **signals**, which may come from within the cell, from the external medium, or from neighboring cells. The receptor-signal pairs are essential to **local control**, which provides the basis for all of development.⁴

To have any kind of spatial differentiation a cell must develop **directionality** (polarity).⁵ This permits **asymmetric cell division**⁵ and development of poles in a developing cell or embryo. **Adhesion** molecules hold cells together, allowing a cell to have a fixed

position relative to other cells. Development of directionality and of developmental **patterns**^{6–8} are dependent upon **gradients** of concentration, of foods, heat, light, gravitation, etc., that can be detected by receptors. Gradients of compounds called **morphogens**^{9,10} help to provide a **positional identity** to cells.^{10–12}

Movement of molecules of organelles and of intact cells is also essential. In multicellular organisms cells often migrate to new locations by following chemotactic signals.

Growth of individual cells enlarges them and often leads to **cell division**. The **cell cycle** describes this process with emphasis on replication of DNA.

Homeostasis encompasses adaptation to altered nutrient or other environmental variables and to all

processes which influence a cell to change. It provides for defensive reactions to many types of stress.^{12a,b}

1. DNA and Developmental Programs

The genetic developmental program of an organism is encoded in the DNA. The expression of the program is implemented initially at the transcriptional level by a host of transcription factors that act at appropriate times and in appropriate places. Their action usually requires the presence of many preformed compounds. **Alternative developmental programs** may be used by unicellular organisms to adapt to new environmental conditions or by multicellular organisms to differentiate tissues.⁶

Programmed alterations and rearrangements of DNA. Part of the developmental program may involve a temporary or permanent change in the DNA. One of the simplest of these changes makes use of the **transposable recombinational switch** in which a small piece of DNA is present in either of the two possible orientations. (See Chapter 27, Section D.3.) An example, illustrated by Eqs. 27-15 and 27-16, is the variation in “phase” of the flagella produced by *Salmonella*. A somewhat different example is provided by the unicellular yeast *Saccharomyces cerevisiae*, which changes the mating type of its haploid forms in a highly regulated pattern.^{13,14} The **a** mating type is expressed constitutively, but this is frequently switched to the **α** type, which produces a different mating pheromone (see Table 30-5) and responds to a pheromone from **a**-type cells. The change occurs through the transposition of different “cassettes” of DNA from “silent sites” into an expression site.^{14a,b} The cassettes contain several genes, which are copied into the expression site called MAT. This site always contains genes of either **a** or **α** type. However, both **a** and **α** genes are present in other storage locations. When the mating type is switched, a copy is made of one of the stored cassettes and is placed into the MAT locus replacing the cassette already present. The MAT **α** genes encode two regulatory proteins, **α1** and **α2**. Protein **α1** is a positive regulator of the **α**-cell-specific genes, while protein **α2** is a repressor of the **a**-cell-specific genes.^{15–17} A similar mechanism appears to be employed by trypanosomes in changing their variable cell surface proteins (Box 31-E).¹⁸

Inactivation of genes and imprinting. Under some circumstances a chromosome or part of a chromosome is permanently inactivated but remains within the cell as compactly folded heterochromatin. Heterochromatin often consists of reiterated sequences of unknown function, but it may also contain groups of inactivated genes. The most impressive case is the

TABLE 32-1
Some Essentials for Growth and Differentiation

Cohesion of molecules, utilizing specific, hydrogen-bonding, and complementary surface shapes

Recognition, and conformational changes

- **Receptors** and **signals**
- **Local controls**

Polarity (directionality)

- Asymmetric cell division
- Poles

Gradients that can be sensed by receptors

- Food, physical qualities, morphogens

Adhesion between cells is required to hold an organism together and also, together with morphogen gradients, to provide a **positional identity**

Movement of molecules, organelles, and cells

Growth to enlarge cell size and numbers of cells

- **A cell cycle** for **replication of DNA** and **cell division**

Homeostasis to permit adaptation to changes in nutrient concentrations, stress

A developmental program, which is encoded in the genome

- Implementing this program usually requires many preformed compounds
- **Alternative developmental programs** often provide flexibility to an organism

Stem cells of **totipotent**, **pluripotent**, or multipotent nature supply new germ cells and other cells for multicellular organisms when needed

Programmed cell death (apoptosis) is part of many developmental programs and provides for removal of unneeded cells without inflammation

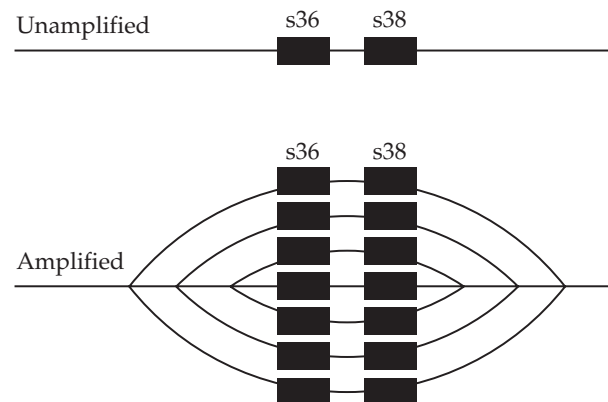
total inactivation of one of the two X chromosomes in cells of female mammals.^{19–21} The entire chromosome appears as heterochromatin. The inactivation occurs early in embryonic development and is random with respect to the two X chromosomes. In some cells the maternal X chromosome, in others the paternal X chromosome, becomes inactive. However, upon further cell divisions the same chromosome in each clone remains inactive. As a consequence of the random inactivation, the female body is a mosaic with respect to genes in the X chromosomes. Formation of heterochromatin in X chromosomes is described in Section C,1. It depends upon **epigenetic markers**, which consist, at least in part, of methylated cytosines in 5'-CpG-3' sequences. These CG pairs are palindromic and can be methylated on both strands (see Eq. 27-2 and Fig. 32-3).

Selective inactivation of genes by methylation accounts for **genomic imprinting**, which occurs in mammals and which marks a gene of either maternal or paternal origin for silencing.^{22–24} About 50 imprinted genes have been identified in mice and in humans.^{23,25} Imprinting, known as **epigenetic inheritance**, also appears to arise in part by methylation of cytosines in CpG sequences. Four m⁵C-methyltransferase genes have been identified in the human genome.²⁶ One of these presumably provides the initial methylation of the cytosine in one strand of the DNA. A second methyltransferase (apparently encoded by gene *Dnmt1*)²⁷ methylates the second strand and serves as the **maintenance methylase** that preserves the methylation pattern upon DNA replication (see Eq. 27-2). The faithful maintenance of imprinting has been demonstrated experimentally for genes of mice formed by nuclear transplantation, i.e., mice grown from somatic cell nuclei transplanted into ova.²⁸ Imprinted genes usually occur in clusters, which also contain imprinted genes for noncoding RNA molecules. The transcription of those genes often is correlated with repression (**silencing**) of protein-coding genes on the same chromosome. This observation is consistent with evidence that genes may be silenced by the binding of small RNA molecules to complementary sequences in the DNA or of mRNAs. The mechanism may be limited to **imprint control regions** or silencer units in the DNA, while the genes to be controlled are set off by **insulator elements** (see also pp. 1882, 1894 and Chapter 28, Section C,1). The significance of genomic imprinting is not clear.^{29,30} However, it is very important to nuclear transplantation because the methylation state of imprinted genes is normally reset (**epigenetic reprogramming**) before embryonic development begins.^{23,25,27,31–33} See Section 5. Methylation of DNA is not essential to the life of filamentous fungi. Nevertheless, *Neurospora* does have a DNA methyltransferase, which methylates about 1.5% of the cytosines present.

Loss of DNA. While some genes are selectively inactivated, others may be irreversibly lost during development. The extreme case is that of the human red blood cell from which the entire nucleus is expelled. Loss of DNA may result from recombinational events. For example, crossing-over between sister chromatids during mitosis has been demonstrated for some cells and chromosomes. This does not alter the genetics of the progeny cells if equal amounts of genetic material are exchanged. However, if two or more similar base sequences occur in tandem in a DNA molecule, unequal crossing-over (Chapter 27, Section D,2) can occur with the loss of genetic material from a chromosome of one of the progeny cells. This may be a deliberately programmed route of differentiation for some cells.

Alternatively, loss of DNA from a chromosome may occur through a **looping-out excision** mechanism. Like the excision of a prophage from the chromosome of *E. coli* (Chapter 27, Section D,3) this loss of genes occurs at specific sites in the DNA. The best known case is the joining of gene segments during the differentiation of lymphocytes (Fig. 31-18). The extent to which similar changes occur during terminal differentiation of other tissues or in nonmammalian species is uncertain.

Amplification of DNA of chromosomes. During formation of oocytes parts of the DNA are “amplified” by repeated replication. This provides a way for the ovum to accumulate ribosomal RNA and various proteins in large amounts. Similarly, genes for two abundant proteins of the egg shell or **chorion** of insects are amplified. Bidirectional replication initiated at discrete positions yields an “onion skin” structure containing many copies of an ~90-kb sequence containing the two genes. The polyploidy observed in some highly specialized cells such as the Purkinje cells



of the cerebellum and of many cells of Diptera larva (Chapter 26, Section F,3) represent another way of amplifying genes. Polyploid cells of animals generally represent a terminal stage of differentiation and do not divide. They tend to contain their full complement of genes in each copy of their DNA, but most genes are not expressed.

2. Receptors and Signals

Receptors and their ligands are numerous, varied, and essential to all forms of life. Cell-surface receptors on bacteria detect feeding attractants as well as dangerous molecules. From bacteria to humans seven-helix receptors function to detect light, odors, hormones, and other molecules. The numbers of different receptors are impressive. For example, the tiny nematode *C. elegans* has 650 seven-helix transmembrane receptors and 411 protein kinases, many of which may be associated with receptors.³⁴ Our bodies have thousands.

Every ligand that binds to a receptor is a signal of some kind. How many signaling molecules are there, and what are their structures? The number of proteins and small peptides affecting growth and development of cells may be enormous. Many of these have been considered in Chapter 30. In addition, the catecholamines, serotonin, histamine, and even bicarbonate ions may act as local hormones.³⁵ For example, HCO_3^- determines whether a thin-walled sporangium or a thick-walled heat-resistant sporangium will appear in the phycomycete water mold *Blastocladiella*. We know that a very large number of different proteins and small peptides are secreted from cells, many in very small amounts.³⁶ Some protein morphogens act at concentrations so low, e.g., 50 pM, that it has been hard to detect them.¹⁰ T-cell receptors respond to extremely small amounts of short peptides that stream outward from many cells bound to surface MHC molecules. Do some of these same peptides act as signals between other cells during development?

Small RNA molecules are now being found to function in many ways within cells,^{37–39} They may also act as messengers within cells and between cells. The small 22 nt RNAs transcribed from genes *lin-4* and *let-7* of the nematode *C. elegans* control key developmental decisions.^{38b,d} In green plants small RNAs move between cells and throughout the plants to trigger defensive responses (p. 1869).^{38c} Much of the RNA transcribed from genes of any eukaryotic cell lacks any known function. Since evolution tends to act on all of the molecular constituents of a cell, this vast amount of RNA may have acquired vital roles in the control of metabolism and growth.^{38e} A poorly understood intracellular structure, the **ribonucleoprotein vault** also remains a biochemical mystery. Differenti-

ated cells may contain 10^4 and embryonic cells over 10^7 of these $\sim 42 \times 75$ nm hollow objects. Vaults have internal cavities large enough to hold two intact ribosomes. Some vaults appear to be empty, but others contain materials, a fact that suggests some role in transport or storage.^{38f}

Homeostasis. A large fraction of the receptors and signaling system of cells is devoted to the maintenance of a constant internal environment. This homeostasis is essential if a cell is to respond to changes in external conditions without disastrous consequences.^{12a,b} Some special aspects of these processes are discussed in Chapters 11, 17, and 28. Within mammalian cells the hypothalamus, pituitary, and adrenals have a primary responsibility for maintaining homeostasis.⁴⁰

Transcriptional control in differentiation.

Development of an organism depends upon an orderly sequence of transcription of genes. Some genes are transcribed in germ cells, others within cells of an early embryo, and others later. As the embryo develops sequential waves of synthesis of the needed protein are observed.^{3,41–43} These are controlled by the actions of numerous transcription factors that act at a variety of **cis-regulatory modules** (CRMs) associated with promoter regions^{42,44–46} and also by controls on translation of mRNAs,⁴⁷ and by negative feedback loops.⁴⁸ The latter may involve hormones, morphogenic proteins, coregulatory proteins of various types, small RNAs, etc.⁴⁹ More than 2000 transcription factors are encoded in the human genome. Most are positive-acting, i.e., they *promote* transcription. Several families are specifically involved in regulation of development (see Table 28-2). These include the Zn^{2+} -containing GATA-1 (p. 1634), which regulates globin synthesis, embryonic factors of *Drosophila* (e.g., bicoid), vertebrate homeotic genes (Hox clusters), Pit1, the muscle-specific helix-loop-helix proteins MyoD and Myf5, and several forkhead (winged helix) proteins.³ A single CRM can bind many different regulatory molecules and single regulators can bind to a variety of CRMs. During development of the simple embryo of the sea urchin *Strongylocentrotus purpuratus* a network of 40 highly regulated genes is needed to coordinate growth and development with the production of the proteins needed at each step.^{41,43}

Part of the control of differentiation lies in the interaction of proteins that regulate transcription with metabolites and hormones. For example, substrate depletion not only decreases growth rate of bacteria but also alters gene transcription. This occurs in *E. coli* as a result of a rise in internal cAMP concentration. The presence of an alternative energy source such as lactose induces changes in gene transcription (Chapter 28, Section A,1). Such **physiological modulation** of a

developmental pattern can also be seen in higher organisms.

More striking is the fact that environmental signals can trigger a cell to switch to an alternative developmental program by which enough new genes are activated to rebuild the cell into a new form. An example is spore formation, a process that occurs with some bacteria when external conditions become unfavorable for vegetative growth (Section B,1). Alternative developmental programs are also evident in eukaryotic organisms that undergo metamorphosis, and they may be important to development. Perhaps persistent states of repression of groups of genes can be passed through several generations of cells until a specific chemical signal triggers the unwrapping of the appropriate nucleosomes and transcription of formerly inactive genes.

3. Adhesion, Cell–Cell Recognition, and Cell Migration

Development of multicellular organisms depends upon both adhesion and on recognition of a correct interaction. Like enzyme–substrate, receptor–hormone, and antibody–antigen binding these interactions of macromolecules on cell surfaces often show a high degree of specificity. They may be accompanied by conformational changes and may trigger signaling cascades. We have already discussed some of these interactions, for example, the binding of a molecule of IgG attached to a surface antigen to protein C1q of the complement system (Fig. 31-8) and the binding of an MHC–antigen complex to a T-cell receptor (Fig. 31-15).

There are many other cell–surface **adhesins**, several of which have been discussed on pp. 402–409. Among them are proteins that contain immunoglobulin-like domains and numerous **glycoproteins**. An example of the latter is the binding of a type of pili found in pathogenic strains of *E. coli* to epithelial cells of the urinary tract. The pilin subunits (Fig. 7-9), like lectins (Box 4-C), bind specifically to the disaccharide group Gal α 1 $\rightarrow$ 4Gal. A lectinlike protein specific for N-acetylglucosamine rings is involved in invasion of erythrocytes by the malaria parasite *Plasmodium*. The unicellular alga *Chlamydomonas* (Fig. 1-11) produces sexual gametes of two mating types. When mixed together, gametes of opposite mating types, prior to fusion, adhere to each other via **agglutinins** present on their flagella. The agglutinins are glycoproteins rich in hydroxyproline, serine, glycine, arabinose, and galactose.⁵⁰ As mentioned on p. 29, colored cells of different strains of the marine sponge *Microciona prolifera* find others of the same strain using highly specific proteoglycan-like aggregation factors.^{51–53} These compounds are highly polymorphic, and it has been suggested that they are part of a primitive immune

system. The aggregation reaction requires calcium ions. In our own bodies Ca²⁺-dependent lectins, the **selectins** (p. 187, 188), bind leukocytes and help to guide them to their sites of action.

Other adhesins include the **integrins**,^{53a} cellular adhesion molecules (**CAMs**), **cadherins**,^{53a–c} and **fibronectin** (Fig. 8-19). These are also discussed on pp. 402–409. The CAMs (Fig. 8-18A),^{54,55} which are members of the immunoglobulin-like protein family, are glycoproteins bearing large 2,8-linked sialic acid polymers.^{56–58} They promote Ca²⁺-dependent aggregation. However, the effect of NCAM, which is widely distributed in a developmentally regulated fashion, can be antiadhesive if long chains of sialic acid are present. NCAM appears to play a role in remodeling and repair of tissues. Adhesion of molecules within cell membranes and the binding of substances to membrane surfaces provides another driving force in development. Within membranes molecules spontaneously sort themselves into lipid **microdomains**, often called **lipid rafts**.^{58a} Related to lipid rafts are caveolae (p. 426). These little craters arise in cholesterol-rich microdomains. They often contain the protein **caveolin** as well as glycosphingolipids and GPI-tailed proteins (Fig. 8-13).^{58b} ATP-dependent linking reactions may also occur to provide more permanent bonding. Membrane-associated molecules, in turn, become centers for attachment of cytoskeletal proteins and other protein complexes. As with the cytosol and extracellular fluids homeostatic mechanisms act to provide a relatively constant membrane–lipid environment.^{58c}

Several types of cell junctions are associated with adhesion and participate in intercellular communication (Fig. 1-15).⁵⁹ The cadherins are transmembrane proteins with large extracellular domains (Fig. 8-18B). They are prominent components of adherens junctions^{59–61a} in which they join the exterior surfaces of pairs of cells in a zipper-like manner. Another protein, **β -catenin**, links the short C-terminal tails of cadherin through α -catenin subunits to the actin cytoskeleton.⁶⁰ In desmosomal junctions other specialized proteins including desmoglein have functions similar to that of cadherins.⁵⁹ Tight junctions, from zebrafish to humans, depend upon a complex of several proteins including those of the claudin family.⁶² Significantly, the cohesive powers of some adhesins, e.g., of cadherin, are altered during development. Cadherin E is nonadhesive in a four-cell mouse embryo but becomes adhesive after the eight-cell stage.⁵⁹ It is obvious that many other changes in intercellular adhesion must also occur during growth and development.

The integrins (see also p. 405) comprise a large family of adhesive receptors that are found in animals from sponges to humans.^{63–65} They have both adhesive and signaling functions. Both subunits of their $\alpha\beta$ heterodimeric structures⁶⁴ have single transmembrane

helices and short C-terminal cytoplasmic tails. The $\beta 1$ subunit tails interact with cytoplasmic proteins. The distribution of integrins varies among cell types. Human leukocytes contain alpha subunits of types αd , αl , αm , and αx with molecular masses of 150–180 kDa. Two ~95-kDa beta subunits ($\beta 1$ and $\beta 2$) are present. However, T lymphocytes express $\beta 1$, $\beta 2$, and $\beta 7$ integrins. Other patterns are observed for other leukocytes,⁶⁶ in skin,⁶⁷ and in other tissues.⁶⁸ Integrin molecules tend to aggregate into clusters, which are found together with other proteins, at the ends of actin stress fibers (p. 370).⁶³ The largest of these clusters are known as focal adhesions. Signals may be sent through integrins in either of the two directions.⁶³ The extracellular domains of integrins interact with a variety of proteins of the extracellular matrix. These include fibronectin, fibrinogen, vitronectin, collagen, and entactin.^{63,69} Other large cell surface adhesins include laminin and osteopontin (Chapter 8), thrombospondin, von Willebrand factor, and related proteins.⁷⁰ These adhesins appear to depend upon the sequence Arg-Gly-Asp (RGD), which binds noncovalently to integrins, which act as cell-surface receptors.^{71,71a} See also Chapter 12, Section C.9.

The functioning of the complex network of integrins, adhesins, and other components of the extracellular matrix is not understood in detail. One fundamental question is how the strength of adhesion can vary with time and stage of development. Roseman postulated an association of an oligosaccharide chain of a glycoprotein attached to one cell with a specific glycosyltransferase of another cell.⁷² The specific interaction would hold cells together, but addition of another glycosyl unit to the oligosaccharide by the transferase would alter the surface properties of the cell carrying the glycoprotein. This, in turn, could cause disaggregation of the cells. Glycosyltransferases can be found on the outer surfaces of cell membranes, and Roseman's proposal may correctly describe one aspect of cell adhesion.

Other molecules that are abundant on cell surfaces include heparan sulfate proteoglycans. Although they have often been regarded as providing a nonspecific "extracellular fly paper," recent evidence from studies of development in *Drosophila* suggest specific and important functions in signaling and in developmental patterning.⁷³ Both hyaluronan and chitin also have been proposed to play an important role in vertebrate development.^{74,75} Proteoglycans of plant cell surfaces, as well as the hydroxyproline-rich proteins of cell walls, may function in plant development.⁷⁶

Movement of cells from one location to another is essential to embryonic development as well as to wound repair and to the immune response. Many brain structures are composed exclusively of immigrant cells.⁷⁷ These **cell migrations** depend upon the cytoskeletal actin filaments, integrins, and focal

adhesions.^{78,79} Chemotactic signals are also required.⁸⁰ A great complexity of underlying chemistry is being elucidated.^{79,81} See also Chapter 19, Section C.

4. Polarity, Asymmetric Cell Division, and Morphogens

Cells of *E. coli* usually divide exactly in the center to form two seemingly identical cells.^{82–84a} However, under the right conditions some bacteria, e.g., *Caulobacter crescentus* and *Bacillus subtilis*, undergo asymmetric division to form two different types of cells (Fig. 32-1).⁸³ There is clearly an axial polarity. This polarity is evident even in *E. coli*, which has flagella streaming out at one end and its chemoreceptor-bearing "nose" at the other (Fig. 32-1).⁸⁵ Axial polarity is also obvious in other bacteria (Fig. 19-1).

Polarity is evident in eukaryotic cells from protists to higher organisms.^{85a,b} Cells of the yeast *S. cerevisiae*, whether haploid or diploid, divide in an asymmetric way by budding.^{5,7} Among body cells of higher animals those of the epithelium are among the most polarized (Fig. 32-2; see also Fig. 1-6, Box 8-F). Polarity is always present in ova of eukaryotes, but the ova may initially be radially symmetric.^{86–88} The **anterior-posterior** axis, which is formed first, establishes a head-to-tail direction.⁸⁹ In bacteria this major axis is determined as perpendicular to the division plane. In the tiny worm *C. elegans* the anterior-posterior axis of an ovum is determined by the position of entrance of the sperm. This marks the posterior end.⁹⁰ In higher animals the axis, which is also known as the **animal-vegetal axis**, is established by uneven distribution of materials that include mRNAs and proteins in the unfertilized ovum. During embryonic development of bilateral species two other axes, the **dorsal-ventral** and **right-left** axes, are also developed and help to establish the body plan. Throughout development polarized movements allow cells to intercalate between one another to help shape the body.^{90a}

Early studies of simple organisms such as *Hydra* (Fig. 1-13) and planaria (flatworms, Fig. 1-14A) showed that distinct chemical differences can be detected along the anterior-posterior axis. These organisms can be cut into pieces, many of which can regenerate complete individuals.^{91,92} Regions near the head regenerate most readily. These and other observations led to the concept of gradients of diffusible **morphogens**, or form-giving molecules.^{9,10} In *Drosophila* eggs an mRNA specified by the gene *bicoid* is localized at the anterior pole. The translation product, the bicoid protein, diffuses through the embryo, which in *Drosophila* lacks cell walls at this stage (see also Section C.4).^{9,93–95} Bicoid is a transcription factor and also one of a number of established morphogens. Many other morphogens are members of the **TGF- β**

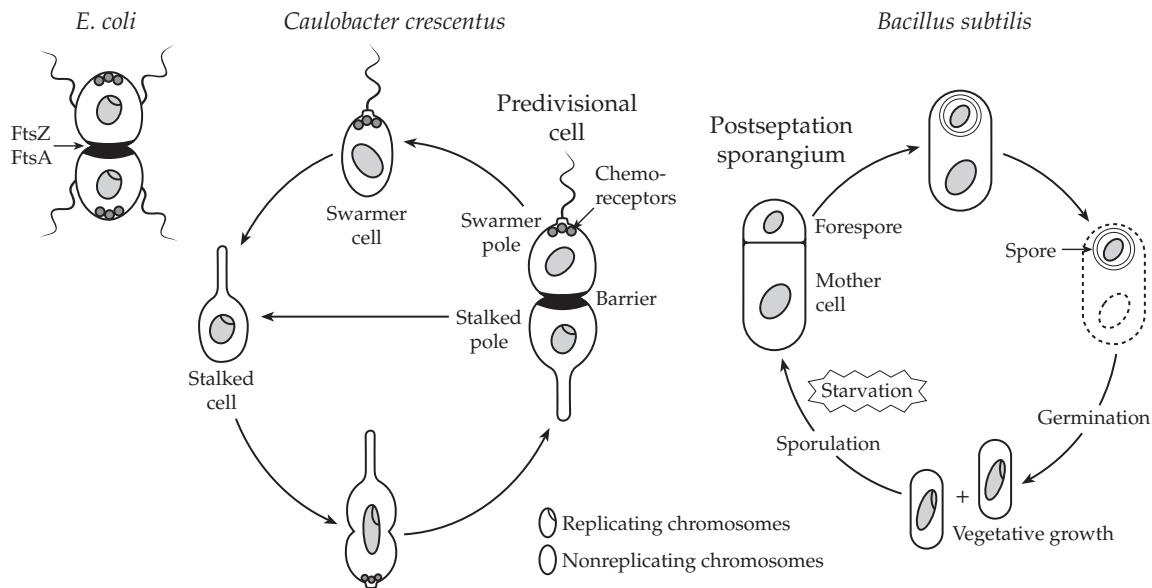


Figure 32-1 Comparison of cell division in three species of bacteria. *Escherichia coli* divides symmetrically after forming a septum in a plane marked by a ring of FtsZ (tubulin-like) and other cell division proteins. *Caulobacter crescentus* divides asymmetrically to give one flagellated swarmer cell and one stalked cell. *Bacillus subtilis*, under starvation conditions, divide to form a mother cell and a forespore. The latter is engulfed by the mother cell, which promotes its conversion to a resistant spore. From Shapiro and Losick.⁸³ Courtesy of L. Shapiro.

(transforming growth factor beta) family. Among them are proteins that establish the dorsal-ventral axis (Section C.4)^{86,96-98} and also bone morphogenic proteins (p. 443). Retinoids also appear to act as morphogens.

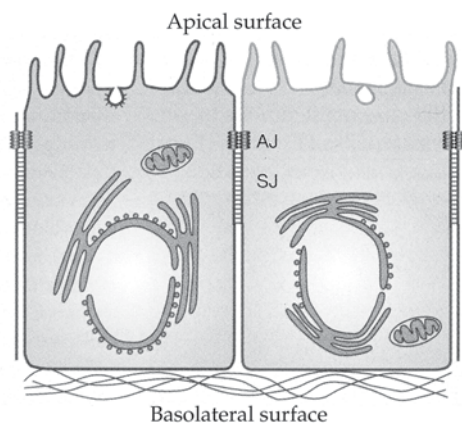


Figure 32-2 A pair of epithelial cells of *Drosophila*. The apical surface (top), e.g., of epithelial cells of the gut, faces the external surface, while the basolateral surface (bottom) binds to a basal membrane. Adheren junctions (AJ) and septate junctions (SJ) are shown between the cells. From Peifer and Tepass.⁸⁴ Drawing by S. Whitfield.

5. Totipotency and Stem Cells

The cambium layer of plant stems (Fig. 1-16) differentiates continuously to form phloem on the outside of the cambium and xylem on the inside. At the same time, cambium cells are retained. Thus, at each cell division one daughter cell becomes a differentiated cell, while another remains the less differentiated cambium. This pattern of continuous differentiation from a line of **stem cells** with constant properties is found in animals as well as in plants. In the differentiation of cambium it appears that chemical signals obtained from the surrounding cells on either the inside or the outside of the cambium layer determine whether the differentiated cell becomes phloem or xylem. Sucrose, auxin, and cytokinins are all involved.

Cloning. Asexual propagation (cloning) of plants ordinarily occurs by virtue of the ability of embryonic meristematic tissue to differentiate into roots and shoots. If isolated phloem cells or other more differentiated cells are cultured, the result is often the formation of a **callus**, a dedifferentiated mass of cells somewhat reminiscent of embryonic cells. Under proper conditions, e.g., in a coconut milk culture and in the presence of the correct auxin-to-cytokinin ratio, some carrot root phloem cells revert to embryonic cells and develop into intact plants.⁹⁹ This experiment provided proof that the differentiated carrot phloem cells

contained a complete genome for the plant. Its nucleus is **totipotent**, able to generate all cell types. However, the experiment cannot be done easily with most plants, and dedifferentiation is not always automatic. It does occur often enough to establish the totipotency of the nucleus of many differentiated cells.

Similar considerations apply to animal cells. In the earliest stages differentiation is readily reversible. Later it becomes difficult to convert a differentiated animal cell into one resembling an embryonic cell. However, Gurdon demonstrated that this is sometimes possible. Nuclei from cells of intestinal epithelia and other tissues were substituted, by transplantation, for the nuclei of egg cells. The process is called **nuclear transplantation**^{100–101a} or nuclear transfer cloning.¹⁰² The resulting eggs in some cases grew into adult toads. Thus, the full genetic information of the toad was present in the differentiated cells.^{103,104} However, it was not possible to accomplish this result with nuclei of neurons, which may have undergone irreversible differentiation. More recently mammalian nuclei have been utilized in the same way to create the famous sheep Dolly as well as mice, calves, pigs, and kittens.^{105–108} These animals are commonly said to have been **cloned**, a term that has long been used to denote asexual propagation, e.g., in a colony of dividing cells or in propagation of plants by grafting.

In nuclear transplantation it is the DNA that is hoped to be the same in every individual in a clone. However, the ovum used for the transplantation contains mitochondria. Some mitochondria may also accompany the nucleus during the transfer. If the donors of the ovum and of the nucleus are different individuals the offspring will be mitochondrial hybrids.¹⁰⁵ In addition, there are questions about the methylation state of DNA in the donated nucleus and about the age and health of the donated mitochondria. That these questions are significant is emphasized by a bit of 3000-year-old knowledge from mule breeders: a mare crossed with a donkey yields a mule but a stallion crossed with a donkey yields a hinny, which has shorter ears, a thicker mane and tail, and stronger legs than does a mule.¹⁰⁹ There are worries because Dolly and many other animals produced by nuclear transplantation have not been completely healthy.^{107b,110} Is something missing from the transplanted DNA or does it carry something extra, such as methyl groups? Recently it has been recognized that incorrect epigenetic marking of cytosine in CpG pairs that control maternally imprinted genes, especially those on chromosomes 11 or 15, may cause death of embryos or devastating human diseases.^{25,111} An important related question for those wishing to clone an animal by nuclear transplantation is “Should the cell that donates the nucleus be in the G_1 state of the cell cycle (Fig. 11-15) or the G_0 or paused state that precedes G_1 ?”¹⁰⁵ See also Chapter 27, Section B,6.

Stem cells. For many years it has been appreciated that, as shown in Fig. 31-2, both erythrocytes and other blood cells arise throughout life from self-renewing stem cells in the bone marrow.^{13,112} Stem cells are also needed for renewal of bones, muscle, skin, neurons, etc. Stem cells appear to be present only in small numbers and in well-protected special **niches** in the body.^{53c,113–115} They are able to live throughout an individual’s lifetime, dividing quite rarely and always producing one or more highly differentiated cells as well as a new stem cell.¹¹³ A fertilized egg (zygote) is totipotent, able to generate all the cells of an animal including those of the placenta and other tissues that are not part of the embryo. However, the most capable stem cells are **pluripotent**, able to form more than one type of specialized cell.^{113,116} Mammalian pluripotent stem cells include tumor cells, **embryonic stem cells**, derived from preimplantation embryos, and **embryonic germ cells**, derived from the primordial germ cells of the postimplantation embryo.¹¹⁷ These germ cells are not only totipotent but, with good luck, may be immortal.¹¹⁸

Recent results indicate that adult-derived somatic cell nuclei may still retain full pluripotency.^{107a,117a} Some confusion has arisen because of the discovery that stem cells may sometimes fuse with differentiated cells.^{118a,b} It is only recently that it has been possible to locate and to cultivate human stem cells. These cells, which may be recovered from both embryonic and mature tissues, include the blood-cell-forming **hematopoietic** stem cells, fetal **neuronal** stem cells, **melanocyte** stem cells, and **mesenchymal** stem cells (or marrow stromal cells). The last give rise to muscle, bone, cartilage, and tendons.^{119–121a} Most stem cells may arise late in development and function principally in tissue renewal.¹²² Among the most abundant are those of epithelial tissues, whose cells provide 60% of differentiated tissue types in the mammalian body.^{123,124} Epidermal stem cells must provide for regular replacement of the outer skin surface (Box 8-F) but must also provide cells for rapid repair of wounds.¹²⁵ The exact locations of epidermal skin cells have been difficult to find. The cells appear to be well-protected in areas deep in the skin. Some are located in hair follicles.^{120a,125} Stem cells of plants are present in specialized structures called **meristems**. A seedling typically has two meristems, at the tips of the shoot and root, respectively.¹²⁶ See Fig. 32-8B.

Cloning of human stem cells is of great medical interest because of the possibility of replacing defective cells or tissues. Tissue engineering may supply urgently needed differentiated cells for replacement purposes^{107,127–128a} and may eventually lead to replacement organs.^{129–131} These efforts must be pursued with caution, but most researchers see a bright future for cloning of tissue cells.^{128,132–134} At the same time there is nearly universal agreement that nuclear trans-

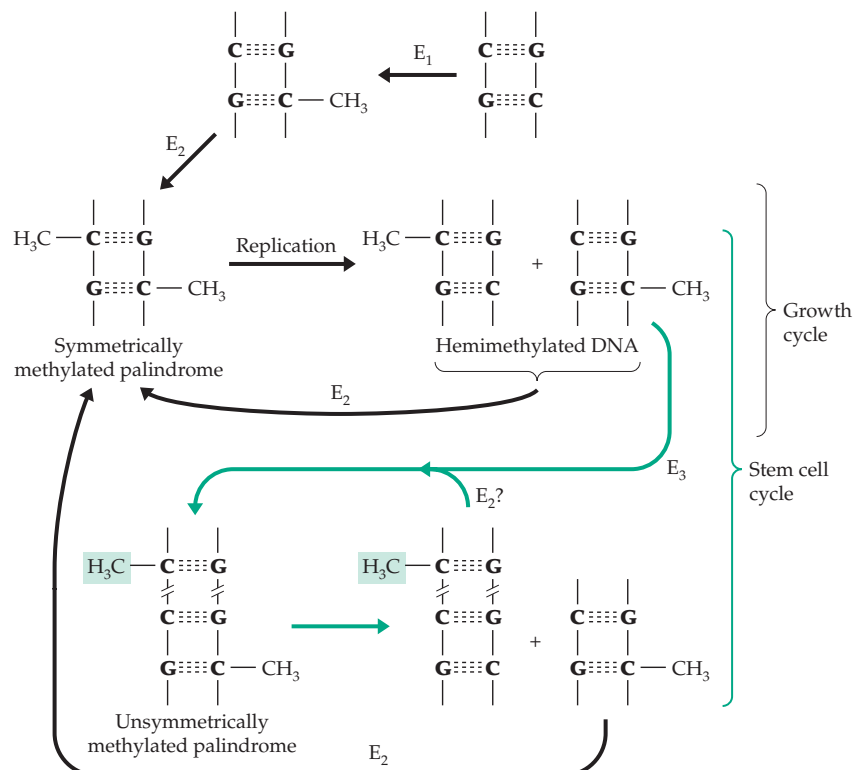
plantation cloning of human beings should not be attempted.^{132,135} One key objection is the near certainty that many seriously defective human embryos would be created.

How can we explain a pattern in which one daughter cell resulting from division of a stem cell undergoes differentiation while the other remains a stem cell? A hypothetical way in which this might happen is illustrated in Fig. 32-3. It depends upon methylation of CpG sequences in DNA (Eq. 27-2). A methyltransferase, E_1 in Fig. 32-3, would modify a site in the DNA that controls differentiation by methylating a base in one of the two strands of a palindromic sequence such as a CpG pair (Eq. 27-2). The maintenance methylase E_2 would further methylate the corresponding cytosine in the palindrome in the second strand. During tissue growth replication would produce a hemimethylated CpG in each daughter cell. These would also be methylated by E_2 (upper part of Fig. 32-3) to complete the replication process. In contrast, a stem cell would need to mark one DNA strand, e.g., by action of a third methyltransferase (E_3) or some other type of DNA-modifying enzyme. As depicted in Fig. 32-3, E_3 would add a methyl group to DNA on some location outside of the palindromic region. Replication in this case would yield one hemimethylated palindrome, which would allow one daughter cell to follow the normal growth and replication cycle. However, the other daughter cell would carry the mark

designating it as a stem cell. The presence of both E_1 and E_2 in cells would lead to the continuous differentiation of the modified cells from unmodified ones, the situation found in stem line cells. A “maintenance methylase” with the properties of E_2 has been identified (Section A,1).

One question that has been asked is whether the 200 cell types of the vertebrate body all arise as a result of chemical interactions between cells and hormones and other external signals? Alternatively, does a **developmental clock** count the number of cell divisions and at the appropriate time turn off one set of genes and turn on another?¹³⁶ Methylases as well as other enzymes might modify DNA at specific times during development. For example, a hydrolase might deaminate the adenine in an AT pair to inosine. Upon replication and cell division one daughter cell would receive an unaltered DNA molecule, but the other would contain in place of the AT base pair an IC pair. Following a second replication, a GC pair would be formed (Eq. 27-20) resulting in AT to GC mutation at a specific site in the DNA of some of the daughter cells. Such a simple change, occurring in response to an enzyme formed at a certain stage of development, could alter the expression of genes in a cell. Schemes involving palindromic sites and modification enzymes that could turn off specific genes after a given number of cell replications have been suggested.¹³⁶ In fact, there seems to be little support for such mechanisms.

Figure 32-3 Hypothetical way of controlling stem cell replication by methylation or other marking system. Methyltransferases E_1 and E_2 methylate the cytosine in a 5'-CpG-3' or other palindromic sequence. In freely replicating cells these two enzymes keep the CG sequences methylated on both DNA strands. In stem cells another enzyme, perhaps a third methylase (E_3), marks a location outside the palindromic DNA on one strand (■). Replication leaves the mark in the duplex, which is retained in a stem cell. The other strand will yield a hemimethylated duplex allowing the cell to follow the normal growth and replication pattern. Based on proposals of Holliday and Pugh.¹³⁶



Transcriptional controls (Section 2) and the sensing of a cell's location may provide adequate control.

If methylation and possibly other covalent modifications of DNA occur, how can one explain the totipotency observed for some nuclei of differentiated cells? During development of the ovum and the sperm there appears to be a “resetting” of the developmental clocks that led to differentiation. At this time all of the methyl groups on the CpG pairs of imprinted genes are removed (Section C,2).^{25,137} The mechanism is uncertain. Perhaps marking of newly replicated DNA stops. For example, in the case of the methylated DNA of Fig. 32-3 if E₁ and E₂ were absent in the cytoplasm of the ovum, no further methylation would occur during subsequent cell divisions. However, there may be active enzyme-catalyzed demethylation (see p.1541). Other mechanisms of gene silencing are also known (pp. 1881,1894).¹³⁸

6. Apoptosis or How the Tadpole Eats Its Tail

Observers have long been fascinated by the rapid resorption of a tadpole's tail as it turns into a frog or toad. The process, designated by the Greek word apoptosis (whose “pop” is pronounced),¹³⁹ or as **programmed cell death**, plays a major role in many aspects of development in nearly all organisms.^{140–142} For example, during human development about one-half of all the neurons generated die.^{143,144} Unneeded lymphocytes, some of which produce antibodies or T-cell receptors directed against a person's self, are also killed. Cells may die accidentally from injuries. In many cases the resulting death occurs by **necrosis** rather than apoptosis. Necrotic death is accompanied by swelling and bursting of the cell and a subsequent inflammatory response.^{145,146} In contrast, cells dying by apoptosis shrink, break into fragments, and are rapidly eaten by surrounding cells.^{146a,b} There is no inflammation. Because of this it has been difficult to determine the extent to which apoptosis contributes to normal development. Apoptosis is also distinguished from **autophagy**, which is intracellular turnover under starvation conditions. Cells may need to scavenge unneeded proteins and organelles, recycling them within the cell.¹⁴⁷

Cells damaged by disease, e.g., dopaminergic neurons in Parkinson disease, may die by apoptosis.^{147a,b} A second form of self-destruction occurs when an axon is cut.^{147b} Failure of the elaborate network of mechanism for repair of DNA and maintenance of the genome normally leads to apoptosis. In cancer essential steps in the apoptosis pathway are often inactivated.^{147c}

Our view of apoptosis (outlined in Fig. 32-4) changed with the tracing of the origins and fates of all of the ~1000 cells of the nematode *C. elegans*.¹³ During development of the adult worm just 131 specific cells

die by apoptosis. Studies of mutant worms revealed mutations in several cell-death (*ced*) genes. Three proteins, encoded by genes *ced-3*, *ced-4*, and *egl-1*, are essential for apoptosis.^{140,148} Somewhat surprisingly worms with defective Ced proteins are apparently healthy, even though they have 131 extra cells. On the other hand, in the fruit fly *Drosophila* mutations in similar death genes are sometimes fatal.¹⁴⁰

The nucleotide sequence of the *ced-3* gene revealed that the Ced3 protein is closely related to the **interleukin-converting enzyme ICE**,^{149–152} which is discussed on p. 619. ICE is a member of the **caspase family** of thiol proteases (p. 619). At least 14 different caspases are found in the human body. Some of them

TABLE 32-2
Some Components of Apoptotic Systems

Apaf1	Mammalian homolog of Ced4; component of apoptosome
Apoptosome	Cytosolic complex: Apaf1•caspase-9•cytochrome c
Bcl2	Mammalian homolog of Ced9 protein of <i>C. elegans</i> ; inhibitor of apoptosis
Bcl-2 family	Group of regulators of apoptosis, both inhibitory and stimulatory (Bad, Bax, Bik, etc.)
CARD	
CD95 (AP-1, Fas)	One of the most studied death receptors
Ced3	<i>C. elegans</i> thiol protease, related to mammalian caspase9
Ced4	Activator of Ced3, related to mammalian Apaf1
Ced9	Inhibitor of Ced3 and related caspases
DD	Death domain of a death receptor
DED	Death effector domain
DISC	Death-inducing signaling complex, formed in plasma membrane
FADD	Fas-associated death domain, an adapter protein
ICE	Interleukin-converting enzyme, structurally related to Ced 3
TNF	Tumor necrosis factor (a family of cytokines secreted by macrophages)
TNFR	Receptors for a TNF family member

function in apoptosis^{152a} and others in maturation of pro-inflammatory cytokines.^{152–155} Most exist as pro-enzymes, which must be activated by proteolysis.¹⁵⁶ The mammalian homology of Ced3 is caspase 9.^{153,157} The Ced4 protein of *C. elegans* is an activator for Ced3. Its mammalian counterpart is called **Apaf1** (apoptotic protease-activating factor 1).^{153,158} Protein **Ced9** is an inhibitor of apoptosis, which probably protects the worm from erroneous deaths.¹⁵⁹ Its mammalian equivalents are proteins of the *Bcl-2* gene family.^{142,152a}

It is well established that caspases participate in the final stages of apoptosis (Fig. 32-4), but what initiates the process? There appear to be many ways in which apoptosis can be triggered. If every cell has a proper location in the body, which is determined by signals from adjacent cells, what will happen if the cell becomes detached? There is evidence that such detachment with the loss of survival signals causes apoptosis.^{53a,152b} Cell damage is also a major trigger. In other cases the cell is “instructed” to die. An example is the death of unneeded lymphocytes, one of many cellular processes induced by cytokines of the tumor necrosis factor (TNF) family. To allow for this process cells have surface receptors of the TNF

receptor (TNFR) superfamily.^{159a} Some TNFRs are **death receptors**, which are called by many names.^{142,160–163} One of the best known is CD95^{164a} (also called Fas¹⁶⁴ or Apo1). CD95 is involved in death of mature T lymphocytes at the end of an immune response and also in the killing of virus-infected cells and cancer cells by cytotoxic T cells or NK cells.

Members of the TNF family that activate CD95 (CD95 ligands or CD95Ls) are trimers. They bind to the cysteine-rich external domains of the transmembrane CD95 molecules inducing them to aggregate (Fig. 32-4). The cytosolic portion of each of these death receptors contain a **death domain (DD)**. The bundle of aggregated receptors also bind to an adapter protein such as the Fas-associated death domain (FADD).¹⁵⁶ It is one of many proteins involved in apoptosis whose structures are known.^{157,163} The FADD molecule contains a **death effector domain (DED)**, which associates with a similar domain in the proenzyme procaspase 8. A rather large membrane-associated molecular complex, the **death-inducing signaling complex (DISC; Fig. 32-4)**, is assembled in this way.^{161,165} Oligomerization of the procaspase domain causes activation via self-cleavage to give active

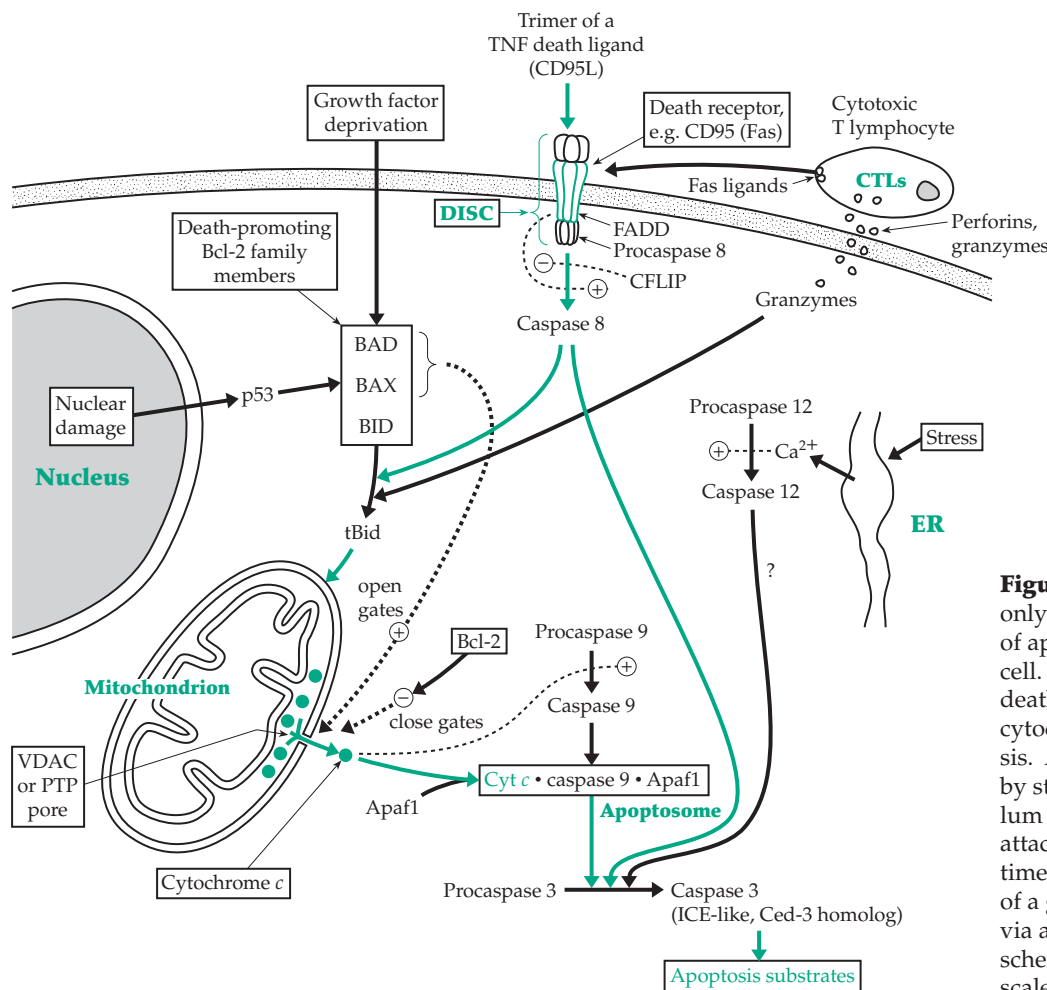


Figure 32-4 Sketch illustrating only a few of the many aspects of apoptosis in a mammalian cell. Emphasis here is on the death receptor pathways and cytochrome *c*-activated apoptosis. A third pathway is initiated by stress in endoplasmic reticulum membranes. In addition, attack by cytolytic T cells sometimes causes apoptosis by action of a granzyme on protein Bid or via a death receptor. Objects in scheme are not drawn to a single scale.

caspase 8, which initiates the apoptotic response. Other “upstream” caspases (caspases 2, 9, and 10) also participate in initiation of apoptosis. In contrast, caspases 3, 6, and 7, the “executioner caspases,” are thought to participate directly in demolition of the cell.¹⁵⁴ A caspase-activated DNase also participates by degrading DNA.^{166,167}

A second major pathway of activation of apoptosis depends upon mitochondria. Various stresses such as lack of needed growth factors, exposure to ultraviolet light, or other apoptosis-inducing signals apparently open pores or gates in mitochondrial membranes allowing materials that would promote apoptosis to flow out into the cytoplasm and stimulate the effector caspases.^{167a} This possibility was supported by the discovery that cytochrome *c* stimulates apoptosis.¹⁶⁸ Cytochrome *c* is a small protein, which is present in inner mitochondrial membranes in a 1:1 ratio with other electron carriers. It is more mobile and less tightly bound than the other components. It carries electrons from complex III to cytochrome *c* oxidase within the intermembrane space (Fig. 18-5). As a result of apoptotic stimuli cytochrome *c* rapidly flows out of the intermembrane space into the cytoplasm both interfering with respiration and triggering other changes in the cell.^{169–172} The outflow of cytochrome *c* may occur via the mitochondrial porin VDAC (p. 1047) or under some circumstances via the mitochondrial permeability transition pore (PTP; p. 1049).¹⁷³ Within the cytosol the escaped cytochrome *c*, together with caspase 9 and Apaf1, forms a large multimeric complex (cyt *c* • caspase 9 • Apaf1) called an **apoptosome**. The apoptosome catalyzes activation of caspase 3, initiating the caspase cascade.

Control of the gates or pores by which cytochrome *c* escapes from mitochondria is poorly understood.^{171,171a,173} Whereas in *C. elegans* a single protein Ced9 has been identified as an inhibitor of apoptosis, vertebrate animals have a large family of proteins that are related to the Ced9 homolog Bcl-2.^{157,174} Of these Bcl-2 and Bcl-*x_L* inhibit the flow of cytochrome *c* out of mitochondria, but several other members of the family, e.g., Bad, Bid, Bik, and Bax, promote apoptosis.^{152a,175,176} Bad carries a signal that indicates a lack of growth factor stimulation. Bid carries a death message from QD95R and other death receptors. Bax carries a signal from p53 (Fig. 11-15) indicating unacceptable DNA damage. However, a truncated form of Bax may prevent apoptosis of neurons.¹⁷⁷ In every one of these pathways there are many complexities. In one of the best known pathways Bid is cleaved by caspase 8 to form a 15-kDa fragment t-Bid that becomes an integral membrane protein in the outer membrane of mitochondria. There it promotes the release of cytochrome *c* (Fig. 32-4).¹⁷⁸

A quite different source of apoptotic signals are ER membranes, which respond to stress by releasing Ca²⁺

ions that activate caspase 12 (Fig. 32-4).¹⁷⁹ Yet another type of apoptosis is sometimes induced by granzyme B (p. 610), which is released from cytolytic T cells.^{180–182}

B. Differentiation in Prokaryotic Cells and in Simple Eukaryotes

Every species undergoes developmental changes. Only a few of these will be considered here briefly.

1. Bacteria

Although they are usually regarded as unicellular, some bacteria develop more than one type of cell,⁸³ and some even form colonies with filamentous growth^{183,184} or other distinct morphology.¹⁸⁵ Many bacteria alter their development in response to changes in environment. For example, unfavorable conditions lead bacteria such as *Bacillus subtilis* to form compact endospores inside the vegetative cells.^{83,184} Many other bacteria including *E. coli* divide symmetrically. This fact also poses a question. How does a cell locate its center and divide? The answer is only partially understood. In all kinds of bacteria a protein known as **FtsZ** (filamentation temperature-sensitive protein Z), a GTP-binding protein homologous to eukaryotic tubulins (Fig. 7-33), is essential. Prior to division FtsZ accumulates as a **septal ring** at the center of the *E. coli* cell. Contraction of the ring is thought to be an essential step in cell division.^{186–188} The FtsZ ring nucleates a growing complex of eight additional proteins known as FtsA, T, K, L, N, Q, W, and ZipA. While ZipA is not highly conserved among bacteria, in *E. coli* it is the first protein to add to the FtsZ ring.^{187,189} ZipA is somewhat related to eukaryotic actin. Another group of proteins is also needed for location of the midcell plane. These are known as MinC, MinD, and MinE. A MinC•MinD complex inhibits potential binding sites for FtsZ. MinD is an ATPase, which structurally resembles the Fe protein of nitrogenase (Fig. 24-2) and appears to propel the MinC•MinD complex in an oscillatory fashion from pole-to-pole.^{190–193} This behavior is not understood, but in some manner the 10-kDa MinE is able to overcome the inhibition and bind to FtsZ initiating division. Division in *E. coli* follows DNA replication by a constant time period (20 min at 37°C). The timing apparently depends upon diadenosine 5'-tetraphosphate (Ap₄A), which acts as a signal to couple division to replication.¹⁹⁴

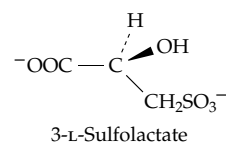
Bacteria with stalks. *Caulobacter crescentis* spreads to new areas while retaining a presence at home. As is illustrated in Fig. 32-1, asymmetric division produces two distinctly different cells. One, like

the maternal cell, has a stalk, while the other has a rotary flagellum with which it travels. In time the motile “swarmer” cell sheds its flagellum and undergoes metamorphoses into a stalked cell. What controls this process? There is apparently a two-component system similar to the one that controls flagellar movement (Fig. 19-5). Two sensor histidine kinases phosphorylate a central response regulator **CtrA**.^{195,195a,b} This represses initiation of replication in the swarmer cells as well as transcription of the cell division gene *ftsZ*. As mentioned on p.1094 changes in DNA methylation may also occur. Flagellar biosynthesis and other steps of differentiation then occur in the swarmer cells but not in the stalked cells in which normal growth and replication take place.^{196–201} Action of proteases is also essential.²⁰²

Sporulation. Bacteria of the genera *Bacillus* and *Clostridium* form metabolically inert spores when deprived of adequate nutrients (Fig. 32-1).^{83,203,204} Bacterial spores are remarkably resistant to heat and can survive boiling water for prolonged periods. Their metabolic rate is essentially zero, but they can be revived and can grow even after many years. For example, bacteria have been grown from a 118-year old can of meat. Some data suggest that spores can survive for no more than ~1000 years, but recent reports, not yet fully verified, say that spores have survived when embedded in salt crystals for 250 million years.^{203,205} At the onset of sporulation the synthesis of ribosomal RNA is turned off completely, and new classes of mRNA are made. More than 50 genetic loci are affected by mutations that cause spore formation. As was mentioned in Chapter 28, one or more specific forms of the 70-kDa σ subunit of RNA polymerase (Chapter 28, Section A,2) are produced and direct the initiation of the new mRNA molecules encoding new proteins.^{204,206–209} Prior to asymmetric cell division the first of the new σ factors, σ_F , is formed together with two regulatory proteins, Spo0A and Spo0B. Spo0B, a protein kinase, phosphorylates Spo0A, inactivating it.²¹⁰ SpoAB also forms an inactive complex with σ_F . After asymmetric cell division σ_F remains inactive in the mother cell but is released in the prespore by action of Spo0E. This is a membrane-bound phosphatase, which dephosphorylates Spo0A-P, allowing it to form a complex with Spo0B with release of σ_F . Another protein, SpoIIIE, appears to direct one copy of the replicated DNA into the forespore.²¹¹ The σ_F factor then directs the transcription of genes in the forespore. In contrast, σ_E is produced only in the mother cell.

One of the most striking metabolic changes in metabolism during sporulation is the accumulation of large amounts of dipicolinic acid (Fig. 24-14). This requires the appearance of at least one new enzyme. In addition, as the spores develop the bacteria take up

large amounts of Ca^{2+} and substantial concentrations of Mn^{2+} and other metal ions. In many bacteria 3-L-sulfolactic acid is also formed.



These components account for the following percentages of the total dry weight of spores of *B. subtilis*: dipicolinic acid, 10%; sulfolactic acid, 3–6%; Ca^{2+} , 3%; and Mn^{2+} , 0.3%. It is often suggested that the dipicolinic acid and other ions protect the proteins from denaturation. However, the heat resistance may arise from the maintenance of the core of the spore in a highly dehydrated state.²¹² When conditions become appropriate for growth again, the spore germinates, and the bacterium again follows the cell growth and division program.

More complex alternative developmental programs are followed by colonial forms of bacteria such as the myxobacteria. The life cycle involves aggregation of cells and formation of fruiting bodies as well as sporulation.¹⁸⁵

Signaling among bacteria. Even bacteria respond to signals from other bacteria. Individuals of a single species often react by secreting pheromones called **autoinducers** using a process called **quorum sensing**. Among the responses are swarming of cells, emission of light by luminous bacteria, synthesis of antibiotics, and formation of biofilms. As mentioned on p.1758, autoinducers used by gram-negative bacteria are often *N*-acetylhomoserine lactones.^{213–215} A furanoyl borate diester (see Box 11-F) may be a more nearly universal autoinducer.²¹⁶ Programmed cell death can also be observed among bacterial populations.²¹⁷

2. Yeasts

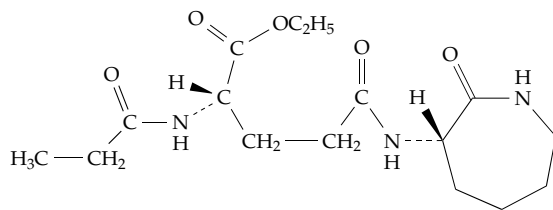
The budding yeast *Saccharomyces cerevisiae* and the fission yeast *Schizosaccharomyces pombe* are the best known fungi. Although they usually grow as individual cells, they can grow with a filamentous form under some conditions.²¹⁸ Other yeast, notably *Candida albicans*, are important pathogens and can also grow in either yeast or pseudohyphal filamentous forms.^{219,220} Like *E. coli*, *S. pombe* undergoes symmetric cell division. However, the strong $\beta(1\rightarrow3)$ linked glycans with their $\beta(1\rightarrow6)$ crosslinkages, mannose polysaccharides, and chitin provide a cell wall very different from those of bacteria.²²¹ Cells of *S. pombe* grow mainly at their tips and begin early in mitosis to form a ring of actomyosin and other proteins at the center. This

corresponds to the mammalian contractile ring. At the end of anaphase the ring contracts, and the septum that separates the two cells develops.^{222–223a} In both types of yeast the septum is rich in chitin, which is secreted from the cell membrane or from vesicles known as chitosomes. After the septum is fully formed and thickened, a chitinase partially hydrolyzes the chitin releasing the cells. In *S. cerevisiae* the cell division is asymmetric.^{5,221} The position of the bud seems to be directed by the actin cytoskeleton.

All fungi form spores (gametes) during their haploid stage, which follows meiosis (Fig. 1-10). The transcriptional program for *S. cerevisiae* involves at least four sets of genes, which are transcribed consecutively. During spore formation the mRNA levels of more than 1000 of the ~6200 protein-encoding genes are changed. About 50% are elevated and ~50% are depressed.²²⁴ The mating type changes in the haploid state have been mentioned on p. 1574. Similar mating and sporulation pathways are observed for *S. pombe*.²²⁵

3. The Cellular Slime Molds

The life cycle of *Dictyostelium discoideum* is described briefly in Box 11-C. About 10^5 individual amebas aggregate to form a moving “slug” in response to the chemoattractant cAMP. Some other species of *Dictyostelium* are attracted to a folic acid derivative or to the ethyl ester of *N*-propionyl- γ -L-glutamyl-L-ornithine- δ -lactam.²²⁶



N-Propionyl- γ -L-glutamyl-L-ornithine- δ -lactam

In all cases the cells also utilize cAMP as an internal second messenger. For *D. discoideum* the components of the chemotactic-aggregation system include a 41-kDa cAMP receptor on the outside, adenylate cyclase, an extracellular diesterase that specifically hydrolyzes the cAMP to AMP, and a diesterase inhibitor protein.^{35,227–230} The inhibitor keeps the phosphodiesterase largely inactive initially, but when cAMP concentrations build up synthesis of the inhibitor is repressed and the cAMP is hydrolyzed, a necessary condition for retaining sensitivity of the receptors for the arriving pulses of cAMP.

The slug of aggregated amebas continues to move and to undergo differentiation into two cell types: about 80% of the cells become pre-spores and the remaining 20%, which are at the “head” of the slug,

become pre-stalk cells. The front-to-back gradient of cAMP within the moving aggregate seems to be involved in differentiation. However, another “differentiation-inducing factor” as well as NH_3 may be involved in the formation of stalk.²³⁰ As the aggregate forms, the cells become cohesive, an 80-kDa surface glycoprotein being involved.²³¹ Later other adhesive **discoidins**, 24- to 27-kDa RGD-containing galactose-binding lectins,²³² also participate in holding the colony together. Some cells begin to produce cellulose. Trehalose is also formed and is stored in the spores. New enzymes have to be made to synthesize these materials. An alternative developmental pattern for some strains of *Dictyostelium* is formation of macrocysts between cells of two different mating types. A diffusible 12-kDa inducing factor appears to be released by cells of one strain.²³³

4. The Hydra

A well-fed hydra (Fig. 1-13) appears immortal. Its body cells are sloughed off and replaced at a steady rate so that within a month or so its body has been completely renewed.³⁵ The hydra contains only ten cell types. These include two kinds of stem cells that give rise to the ectodermal and endodermal cells of the body wall as well as small **interstitial stem cells** (Fig. 1-13) that differentiate nerve cells, germ cells, and the nematocytes or stinging cells. Of the $\sim 10^5$ cells in a hydra about 3600 are interstitial stem cells. Each day they generate 400 nerve cells and 1800 nematocyte precursor cells as well as 3500 new interstitial cells. The nematocyte precursors move up the body of the hydra and take up residence in the tentacles. Their movement is thought to be guided by chemotaxis. The head activator peptide (Table 30-5), which was identified following isolation from 3×10^6 hydras (3 kg),²³⁴ diffuses from the foot end of the animal forming a gradient. A foot activator may diffuse from the opposite end. The interstitial stem cells of hydra also give rise to clones that develop into the gametes. Female hydra always develop female gametes, but stem cells of male hydra give rise to both male and female gametes.²³⁵ This sex switching is reminiscent of the mating type variation of yeast.

5. Cell-Constant Animals

While the hydra is almost immortal as a result of the continuous differentiation of its stem cell lines, other small invertebrates follow a very different course of development. Both the rotifers and the annelid worms (Fig. 1-14) tend to have a constant number of cells in the adult body. The entire developmental program is specified genetically in strict detail.

The one millimeter long adult nematode *Caenorhabditis elegans* contains only 959 somatic cells. The lineal descent of all of these has been traced.^{236–240} The development follows an almost exactly defined pathway with 113 programmed cell deaths during formation of the 558-cell newly hatched larva. In addition, each adult worm contains 302 neurons that make about 8000 synapses. This little nematode also has an alternative developmental pathway. The larvae shed their cuticles in four consecutive molts. If the food supply is inadequate, they enter a persistent nonfeeding state in which they may survive for months and are able to resume development when conditions are appropriate.²⁴¹

C. Development of Animal Embryos

The shapes and body plans of animals vary enormously. Consequently, the study of embryonic development of sea urchins, insects, frogs, chickens, mice, and humans might appear to lead to quite unrelated conclusions. However, there are many similarities as well as variety.

1. Germ Cells and Gametes

Throughout the animal kingdom from protozoa to human beings sexual reproduction predominates. It is true that there are about 1000 species that reproduce asexually.^{242,243} Among them are ~350 species of all-female rotifers²⁴² and even a species of tiny mites, all of which are haploid females.²⁴⁴ Nevertheless, sex seems to have conferred some advantage on most species. There are two theories that may explain this: (1) Sex brings different combinations of genes together, allowing especially favorable combinations to survive, when changing conditions make life difficult.²⁴² (2) Sex helps to remove deleterious mutations from a population.²⁴³ A large fraction of human fetuses (at least 10–25%) contain an “incorrect” number of chromosomes and as many as 20% of oocytes are defective. In contrast only 3–4% of sperm are chromosomally abnormal. Female meiosis I appears to be highly error-prone.^{243a} Abnormal fertilized eggs or embryos are eliminated later in development.

Sex determination. The sex of an individual is determined by the chromosomes. In humans and other mammals presence or absence of a Y chromosome determines the sex. However, in many organisms including *C. elegans* and *Drosophila* this is not true. Although *Drosophila* males like human males are XY, it is the ratio of the number of X chromosomes to the number of sets of autosome (A) that determines the sex. This is also true for *C. elegans*, which has no

Y chromosomes.^{244,245} Apparently because of the differing ratios of X:A in the two sexes, organisms utilize a variety of **dosage compensation** methods. In cells of human females only one X chromosome is active. In *Drosophila* the rate of expression of genes from the X chromosome is roughly doubled in males.^{245–248} In *C. elegans* the expression from both X chromosomes of the (hermaphroditic) female is roughly halved.^{245,246,249,250} The biochemistry underlying these processes is quite complex.

The mammalian Y chromosome. The basic plan of the gonads prior to differentiation is female. However, if a Y chromosome is present (or if genes from a Y chromosome have been translocated to other locations) testes develop and begin to secrete androgen as early as the 60th day of gestation. A male-specific DNA sequence, **SRY** (sex determining region Y), constituting the gene for the **testes-determining factor**, is located in the small arm of the Y chromosome (Fig. 32-5).^{251–253} A small pseudoautosomal segment at the end of the short arm of the Y chromosome carries other genes and undergoes crossing-over during meiosis.^{254,255} The SRY gene lies between this and the centromere. The SRY protein is a member of the HMGA subgroup of **HMG** DNA-binding proteins (p. 1535).^{256,257} It binds tightly to the sequence AACAA(A/T)(G/C) broadening the minor groove of the B-DNA and bending the DNA by more than 70°.^{252,258,259}

Both SRY and the related SOX proteins are critical developmental regulators.²⁶⁰ In early fetal life the mammalian embryo contains an indifferent gonad, able to differentiate into either a testis or ovary. Adjacent to the gonad are two simple ducts, the Müllerian (female) and Wolffian (male).²⁶¹ In the male SRY acts in the developing gonads to induce differentiation into the Sertoli cells of the testis. In the mouse the *Sry* gene is active for only a brief period about ten days after fertilization. During that period cells of the genital ridge start to differentiate. In the absence of protein SRY they develop into the female follicle (granulosa) cells but in the presence of SRY into Sertoli cells.^{252,262} This is, in part, a result of production of the **Müllerian inhibitory substance** (MIS), which induces regression of the Müllerian duct, and later production of testosterone. MIS is a glycoprotein of the TGF- β family. Binding of SRY to a site in the *Mis* gene promoter appears to be involved in activation of the *Mis* gene.²⁵² Recent evidence points to a role for both SRY and SOX proteins in pre-mRNA splicing.²⁶⁰ At least 25 other genes are also involved in spermatogenesis in the mouse.²⁶³ Many of these testis-specific genes have completely unmethylated CpG sequences.^{264,265} For example, a cAMP-responsive element present in a promoter sequence for a testis-specific subunit of pyruvate dehydrogenase must be demethylated for

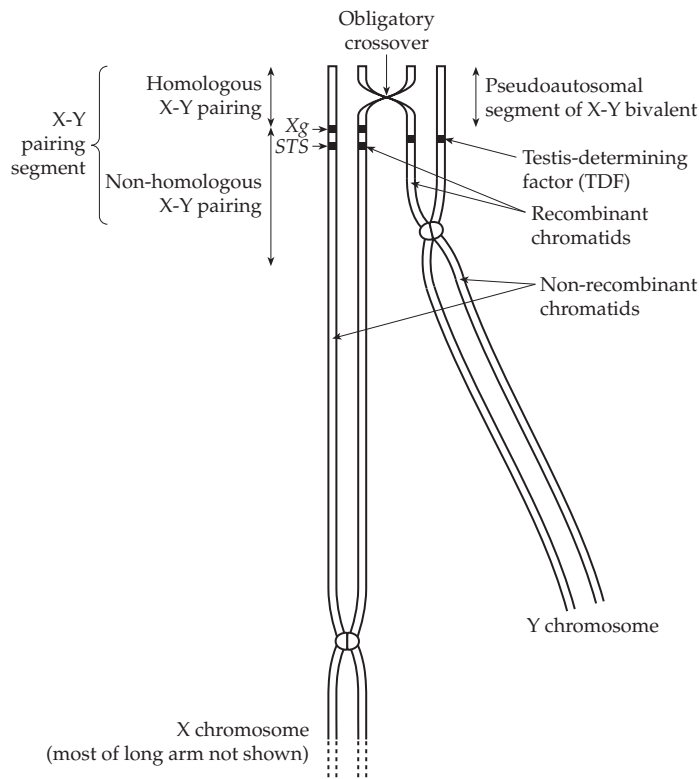


Figure 32-5 Schematic diagram showing crossing-over between the human X and Y chromosome. The pseudoautosomal segment is that part of the X-Y bivalent where there can be X-Y exchange by crossing over. X-Y homology in this segment is maintained by, and may be necessary for, this crossing over. There is always one “obligatory” X-Y crossover, whose position varies. The length of the X-Y pairing segment varies with meiotic stage and can extend well beyond the pseudoautosomal segment into the Y long arm. Much (perhaps all) of the synaptonemal complex formed outside the pseudoautosomal segment represents non-homologous pairing. From Burgoyne.²⁵⁴

transcription to occur. The developing germ cells interact with the surrounding Sertoli cells at every stage both through direct cell-cell contacts and via secreted signals.²⁶⁶ Interstitial cells of the testis differentiate into Leydig cells, which secrete testosterone, promoting development of the Wolffian duct.²⁶¹ A small population of germ-line stem cells provide for continuing spermatogenesis. In *Drosophila* their self-renewal depends upon signals from special hub cells.^{115,267} Other proteins needed for normal male development include the X-linked androgen receptor, whose absence causes testicular feminization, and dihydrotestosterone reductase (Chapter 22).

The development of spermatozoa is unlike that of somatic cells. Extensive reorganization of chromatin occurs under the direction of cis-regulatory elements that are controlled by cells of the testis.^{267a} Among specialized proteins that are synthesized is a testis-specific polyadenylate polymerase.^{267b} During the remodeling histones are replaced by arginine- and cysteine-rich protamines.^{267a-c} In mammals this occurs in two stages. Small intermediate proteins (TP1 and TP2) replace histones in the first stage and are displaced by protamines in the second.^{267a,d} Sulfolipids, which are also present in myelin, are essential to spermatogenesis.^{267e} Both the sphingolipid 3-sulfogalactosylceramide and **seminolipid**, a sulfate ester of monogalactosylalkylacylglycerol (structure on p. 387) are present in large amounts. However, their functions are not clear.

Selenium plays a special role in development and protection of spermatozoa (Chapter 15). The selenoprotein **phospholipid hydroperoxide glutathione peroxidase** (PHGPx; Eq. 15-58, Table 15-4) has a high activity in the testis and in spermatids. However, in mature spermatozoa it forms an enzymatically inactive oxidatively crosslinked capsular material around the midpiece of the cell perhaps providing mechanical stability.²⁶⁸ A similar 34-kDa selenoprotein is present in sperm nuclei and may be essential for condensation of DNA.²⁶⁹ Sperm tails contain specialized cytoskeletal proteins which form “outer dense fibers.”²⁷⁰ In contrast to mammalian spermatozoa, nematode sperm move by ameboid motility that depends upon a specialized actin-like molecule.²⁷¹ Sperm cells are unusually rich in polyamines, most of which are bound to RNA and DNA (Chapter 24).

The X chromosomes. The phenomenon of X chromosome inactivation in mammalian female cells is closely related to imprinting, which has been discussed in Section 1. The inactivation process is quite complex. It involves methylation of 5'-CpG-3' sequences of DNA, as is described in Chapter 27, Section B.6. It also depends upon an **inactivation center**, the *Xist* gene, which is expressed only from the inactivated X_i chromosome, whose *Xist* DNA is unmethylated. On the X_a chromosome this DNA is methylated, and the gene is silent.²⁷² The *Xist* transcript is a long RNA that may bind to and coat much of the X_i

DNA.^{19,21,273–275} The associated chromatin is enriched in a variant of histone H2A and is underacetylated on the tails of histones H2A, H3, and H4 (Fig. 27-4). Also noteworthy is the fact that not all genes on X_i are inactivated. As many as 19% escape this control.²⁷⁶ Another gene *Tsix*, which is adjacent to *Xist* in the DNA, is also involved. *Tsix* encodes an RNA that is *antisense* to the *Xist* transcript^{21,275,277} but is transcribed from the active X chromosome X_a . One hypothesis is that the *Xist* transcript causes X inactivation and that the *Tsix* transcript acts in an opposite way to favor activation of the chromosome. A transcription factor known as **CTCF** has been identified as a possible regulator of the inactivation process.^{21,277} This is a trans-acting factor that is encoded on a chromosome other than X or Y. The process also depends upon methylation of histone H3.²⁷⁸ Methylation of H3 may also be a factor in gene silencing in other organisms.²⁷⁹ CTCF also regulates a number of other genes, e.g., those of the globin gene cluster (Fig. 27-10). It binds to 60-bp sequences, perhaps in enhancer elements.

2. Development of the Ovum

In the early mammalian female embryo the absence of the Müllerian inhibitory substance MIS permits continuing development of the Müllerian duct, while the absence of testosterone permits the Wolffian duct to degenerate. However, positive developmental signals are also required. Among these is the protein **Wnt-4**, a member of a large family of locally acting signal molecules (Section 4). Wnt-4 may be needed both for oocyte development and for further suppression of male development.²⁶¹

The earliest studies of oocyte development were done with sea urchins (often *S. purpuratus*) and with amphibians (often the South African clawed toad *Xenopus laevis*) whose eggs are as much as 1000 times larger than those of mammals.²⁸⁰ However, despite the differences in size, modes of fertilization, and ovary development, oocytes of nematodes, sea urchins, frogs, insects, and mammals have much in common. Oocytes of *C. elegans* and of most other animals undergo a temporary arrest in development at the prophase stage of the first meiotic division (Fig. 26-12).^{13,281–283a} At this stage oocytes transcribe many genes. In some species chromosomes may develop a “lampbrush” appearance (Fig. 27-6) as a result of the transcriptional activity. Many mRNA molecules are stored in the expanding cytoplasm. Proteins are also synthesized and stored.²⁸⁴ Among these are specialized proteins of yolk granules and proteins used to construct an outer egg coat. Surrounding **follicle cells** also contribute nutrients to the oocyte.^{284a} In insects, whose early embryonic development has some special characteristics, 16 surrounding **nurse cells** are connected to the

oocyte by cytoplasmic bridges.^{285,286}

Oocytes may remain in arrest at the beginning of meiosis for prolonged periods before continuing through the **maturation** stage to form an ovum (egg). Women and other female mammals are born with thousands of oocytes, but only a few at a time develop into eggs. Maturation is often delayed until sexual maturity, when it is stimulated by hormones.¹³ In *X. laevis* progesterone stimulates maturation.^{282,287} In *C. elegans* and many other animals a signal from a sperm cell is needed to induce maturation.^{281,283} Maturation of the oocyte is often arrested again, this time at metaphase of the second meiotic division. Transcription is halted, and protein synthesis is slowed. Fertilization then induces rapid completion of meiosis. Penetration of the sperm leads to “activation” of the egg and completion of meiosis. In lower organisms activation can often be carried out by chemical or physical treatment in the absence of a sperm cell, with formation of parthenogenetic offspring.

3. Fertilization

Fertilization of the egg is a biochemically complex process.^{35,288} It involves recognition of sperm and egg, often in a species-specific manner.²⁸⁹ The jelly layer around sea urchin eggs contains peptides such as the **sperm-activating peptide** (speract; Table 30-4), which stimulates increased respiration and motility of the sperm cells.²⁹⁰ Additional chemotactic peptides may also be released from the jelly layer of invertebrate eggs. Chemoattractants for vertebrate eggs are less well known, but a 21-kDa sperm attractant protein from *X. laevis* egg jelly has been characterized.²⁹¹ Both in sea urchins and in mammals the jelly layer, which is called the **zona pellucida**, contains sperm cell receptors.^{288,292–295} These are glycoproteins that interact with proteins (spermadhesins)²⁹⁶ of the sperm cell membrane. One of these is the integrin-associated CD9, an integral membrane protein.^{297,298} Penetration of the sperm through the zona pellucida often involves a large specialized secretory vesicle, the **acrosome**, as well as the enzyme hyaluronidase.^{298a} In some species the acrosome releases a large amount of monomeric G actin, which polymerizes suddenly into a tube of polymeric F actin, which in some way assists the penetration of sperm.³⁵ In the horseshoe crab *Limulus polyphemus* the acrosome in an unactivated sperm cell contains a twisted bundle of as many as 120 cross-linked actin filaments. When the sperm is activated by contact with the jelly coat of the egg, the acrosome straightens into a 50- μ m-long crystalline bundle, which is driven into the egg coat.²⁹⁹ Of importance to all types of sperm cells are proteases and other materials that are also released from the acrosome and which help to etch a hole that allows the sperm to enter the

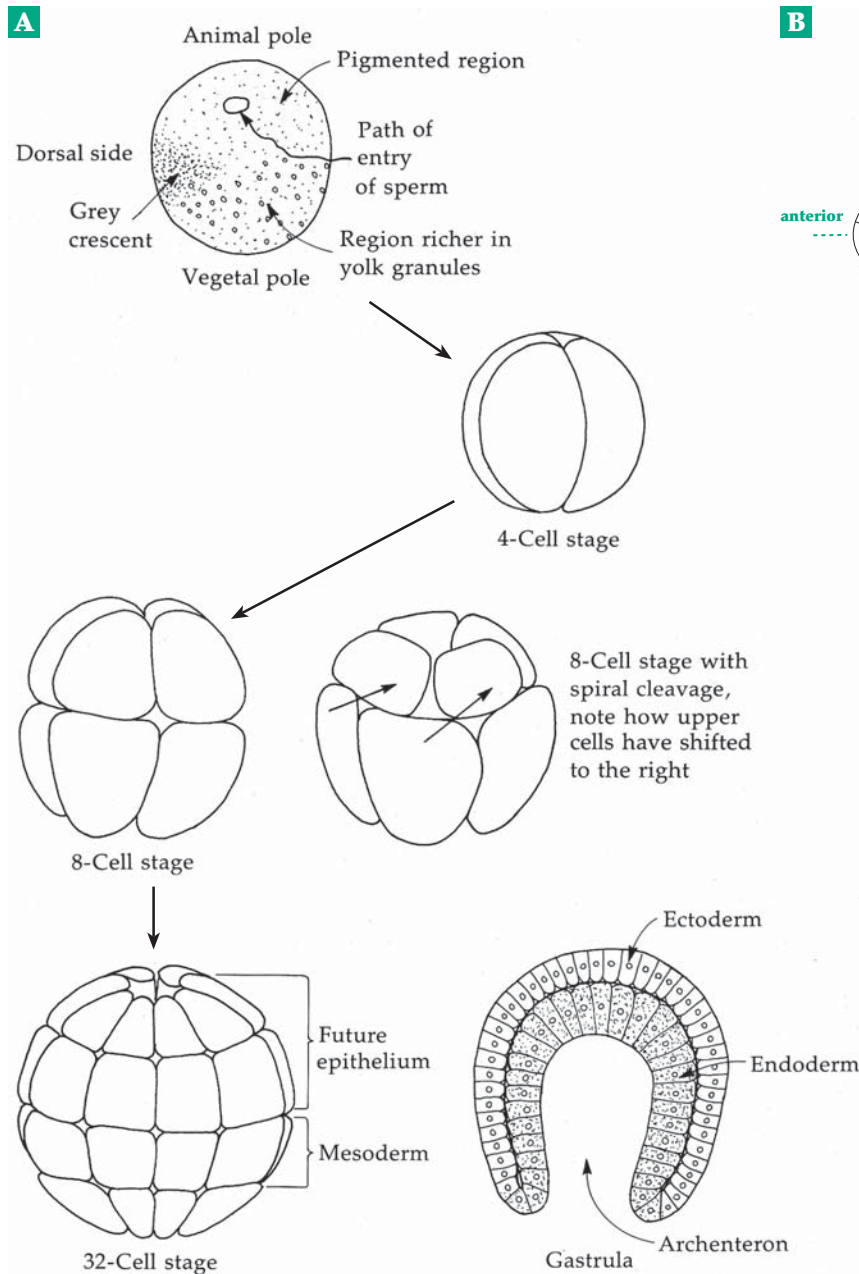


Figure 32-6 Cleavage and gastrula formation. (A) Fertilization and cleavage of an amphibian egg is illustrated at the left. Spiral cleavage, usual among molluscs and other annelids,³⁰⁹ is also shown. The gastrula might be from an echinoderm or some lower invertebrate. (B) An egg of the fruitfly *Drosophila melanogaster*, showing early development. Anterior–posterior and dorsal–ventral axes are labeled. After Nüsslein-Volhard (after a drawing by Laurie Grace).⁹

ovum.^{300,301} The acrosome reaction also activates the egg, a process that may depend upon NO.³⁰² In the sea urchin the acrosome contains a large amount of the protein **bindin**, which mediates a species-specific adhesion of the gametes and presumably fusion with the egg membrane.^{302a}

Fusion of the sperm membrane with that of the ovum causes a rapid depolarization of the membrane of the ovum and an influx of calcium ions.^{303,304} This causes an immediate block to the entrance of any other sperm cells in most species. It also causes the fusion

of the membranes of **cortical granules** (several thousand in a mouse ovum) with the cell membrane and release of their contents.³⁰⁵ The material released to the outside of the ovum includes various proteins and enzymes such as a peroxidase that catalyzes crosslinking of tyrosine side chains (Chapter 25) and hardens the material immediately around the ovum into a tough **fertilization membrane**. Within the ovum a respiratory burst resembling that of activated neutrophils (p. 1074) provides H₂O₂ to the peroxidase.

Fertilization also induces completion of meiosis

and formation of a one-cell embryo containing a maternal pronucleus contributed by the egg and a paternal pronucleus derived from the sperm. Each pronucleus undergoes DNA replication and then enters the first mitosis, which yields a two-cell embryo containing one diploid zygotic nucleus per cell.^{283,306} Under the influence of the cyclin-dependent kinase *cdc2* (see Eq. 26-3) and a hyperphosphorylated form of the protein **nucleoplasmin**, the chromatin of the compact sperm nucleus undergoes decondensation. The sperm basic proteins that coated the DNA are replaced by histones H₂A and H₂B.³⁰⁷

Although an amphibian egg is nearly spherical, there is already a strong polarity. The nucleus lies nearer to the **animal pole** than to the **vegetal pole**, which in many eggs is rich in yolk granules. In eggs of amphibians the animal pole is highly pigmented, but the vegetal pole is less so. On one side above the equator, there is a gray crescent. In some animals this marking appears on the opposite side of the egg from the point of sperm cell entry (Fig. 32-6). The gray crescent marks the future back (dorsal) side of the organism and the opposite part of the cell, the future ventral side.²⁸⁰ The point of sperm entry also marks the ventral side for the mouse, a fact that suggests that the plan of development of mammalian embryos may be basically the same as that of frogs.³⁰⁸ However, it is the internal components of the cell that actually determine the cell's axes. The cytoplasm of the mature ovum contains an unequal distribution of many materials with a well-developed bilateral symmetry. That this distribution is important is seen from the fact that centrifugation of eggs prior to fertilization often leads to formation of abnormal embryos because of

displacement of preformed ribosomes and other materials. It is probably gradients in the concentrations of dormant mRNAs²⁸⁴ and other metabolites that lead to uneven growth of cells and to the indentation of cells at the vegetal pole, a process that initiates the formation of the endodermal layer of the gastrula (Fig. 32-6). In insects the polarity of the developing ovum (the oocyte) is established by the cytoplasmic bridges from surrounding nurse cells, which are asymmetrically arranged (Fig. 32-6B).²⁸⁵

4. Embryonic Development

The fertilized (activated) ovum rapidly undergoes several mitotic divisions, known as **cleavage**, during which no overall growth occurs. The number of cells increases and the DNA replicates at each division, but the overall size of the resulting cell cluster is the same as that of the original ovum (Fig. 32-6). Further development leads quickly to a stage in which a layer of cells (called blastomeres at this stage) surrounds an internal cavity forming a **blastula**. In the sea urchin the blastula, which is released from its protective fertilization membrane ~11h after fertilization,^{310a} consists of a single layer of cells. In frogs and many other organisms there are two or more layers. In *X. laevis* about 4000 cells are formed in eight hours.^{307,310} In mammals a solid cell mass (**morula**) forms first and is later transformed to a **blastocyst**, a hollow ball with an internal cavity.

Early mammalian development has been hard to study because of two facts: the ovum is very small, and a first priority is development of the placenta and of the layers of tissues that surround the embryo.^{308,311,312} This occurs in humans within the first week after fertilization. Both the trophoblast and cells of the inner cell mass (Fig. 32-7) contribute to the extra embryonic tissues.

Development of a mouse beyond the one-cell stage is dependent upon a regulatory gene of the *Oct* family (see p. 1631). The **Oct-3/4** protein, which binds specifically to the DNA motif 5'-ATTTGGAT, consists of two domains, both of which are essential for tight binding to this sequence. One domain is a 75-residue **POU domain** that consists of a helix-turn-helix motif with an amino acid sequence that is highly conserved among mammalian **Pit** and **Oct** regulatory proteins as well as some **Unc** proteins of *C. elegans*.³¹³ The second domain is a 60-residue homeodomain (see p. 1900).^{313,314} Oct-3/4 appears to be essential not only for cleavage of a one-cell egg but also for progression from a two-cell to a four-cell egg (Fig. 32-7) and also in other embryonic cells.³¹⁵ Up to the two-cell stage very little transcription of zygotic genes is observed but further development requires zygotic genes and by the 8-cell stage protein synthesis from maternal mRNAs

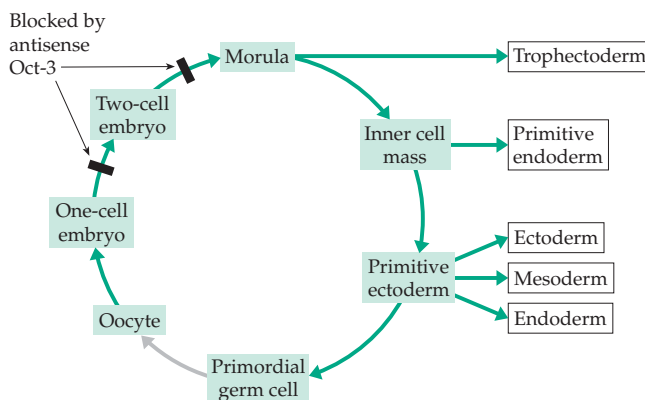


Figure 32-7 Expression pattern of Oct-3 mRNA during mouse development. The green boxes indicate those stages in which Oct-3 is expressed, while the white boxes at the right indicate all types in which little or no Oct-3 mRNA is formed. After Rosner *et al.*³¹⁴

ends.²⁸³ Oct-3/4 continues to be synthesized in the embryo and is necessary for establishment of pluripotent stem cells in the embryo.³¹⁵

Development in *Drosophila* and other insects follows a somewhat different pathway, as is indicated in Fig. 32-6B. The egg, which is surrounded by follicle cells and 16 nurse cells, does not divide. However, its nucleus divides repeatedly, about once every nine minutes, to form ~6000 nuclei. Only then do separating membranes form to give individual cells.^{9,285,316,317} During the first two hours the nuclei form a **syncytium**, in which they are embedded in a common cytoplasm, that allows free diffusion of signaling compounds. At first the nuclei are in the center, but later most migrate to the periphery and form a single layer of cells comparable to the blastoderm of amphibian cells. A few nuclei remain in the central cavity to become yolk cells, and some at the posterior pole become separated into pole cells.

The next stage in embryo formation, which occurs universally, is the invagination of the blastula at the vegetal pole to form a **gastrula**. At this stage the embryo has distinct ectoderm and endoderm cell layers. The cavity, formed in the gastrulation process and connecting to the outside, is referred to as the **archenteron** and is the forerunner of the gastrointestinal tract or enteron. Gastrula formation is more complex in the frog embryo and still more so in the human embryo. In all but the most primitive of animals a third layer of **mesosomal** cells is formed between the endoderm and ectoderm. These three **germ layers** differentiate further as follows. The **ectoderm** yields the skin and nervous system; the **mesoderm** the skeleton, muscles, connective tissues, and circulatory system; and the **endoderm** the digestive tract, lungs, and other internal organs and germ cells.

Organ development occurs largely by infoldings of cells from the endoderm and ectoderm. These infoldings appear to be induced by chemical substances secreted by cells of an adjacent germ layer. Thus, ectodermal cells form the **neural plate**, the prospective brain, and spinal cord in response to induction by mesodermal cells lying beneath the neural plate area. The mammary glands also arise from interactions of mesodermal and ectodermal cells, while the formation of the pancreas, liver, and lungs depends upon interactions of groups of cells from endoderm and mesoderm. Because of their transparency zebrafish are especially useful for study of organ formation.³¹⁸⁻³²¹

During the early stages of embryo growth, development seems to be directed largely by the polarity and gradients of the large amounts of mRNA, yolk and other constituents, which form a prepattern in the ovum.^{321a} However, even at very early stages signaling from the nucleus of the ovum to surrounding cells is a necessary part of establishing the cell axes and developmental pattern.

The anterior-posterior axis. Only recently has it become possible to identify some of the specific mRNAs, the signaling molecules, and receptors that are involved in establishing the principal axes of the ovum (Fig. 32-6). Even for this aspect of development the genetic program is very complex, there are many uncertainties, and the details go far beyond the scope of this book. Much of the most important work has been done with *Drosophila* for which numerous mutants have been identified and characterized. Many names of genes and of proteins are derived from a description of the phenotype of a mutant fly. These same names are often used for the corresponding genes or proteins in other organisms. However, a protein may be known by more than one name, depending upon the species.

A *Drosophila* mother deposits mRNA for ~80% of all of her genes in the egg, but not all of the encoded information is used. Because genetic experiments can be done so readily with *Drosophila*, it is possible to ask what **maternal effects** do come from this mRNA and what effects come from the genes of the zygote.^{9,88,316} For example, the *bicoid* mRNA that accumulates at the anterior end of the *Drosophila* egg is produced by nurse cells and is transported into the ovum. If the mother has defective *bicoid* genes the eggs die. A normal *bicoid* gene present in the father does not prevent death. Another maternal effect protein is encoded by *nanos*, whose mRNA accumulates at the posterior pole of the ovum. The maternal *torso* gene, which acts on follicle cells, is also needed for the anterior-posterior axis formation.³¹⁶ About 30 of these maternal effect genes are active in organizing the pattern of the embryo.

The *Drosophila bicoid* mRNA forms an anterior-posterior (A-P) concentration gradient, which controls early development along the axis. Another protein called **Staufen**, which forms a complex with the *bicoid* mRNA, assists in moving the RNA along on microtubules to form the gradient.^{321b} The microtubules, together with associated transport proteins, are also essential to the prepatterning of the ovum.^{9,88,285,322,323} Bicoid protein is a transcription factor that is synthesized at the sites of its mRNA accumulation. It is absent from the eggs when they are laid but soon appears.³¹⁶ Bicoid binds to the CRMs of target genes and activates them. Cooperative binding of multiple copies of the transcription factor may be necessary to provide the observed sensitivity to concentration. A transcription factor may bind at many sites on the DNA of the zygote or of nurse cells. For example, a gene called *even-skipped* (*eve*) is expressed in seven stripes in the blastoderm embryo. The enhancer that controls *eve*'s second stripe contains at least five binding sites for Bicoid as well as others for the Hunchback, Giant, and Krüppel proteins.⁴⁴

A different gradient along the A-P axis is formed by the **Nanos** protein, whose mRNA localizes in the

cytoplasm of the posterior part of the ovum. Like the *bicoid* mRNA, *nanos* mRNA forms a complex with the Staufen protein. Messenger RNA transcribed from a gene called *oskar* is also necessary for development of the posterior region of the ovum. An additional gene, called *gurken*, is also involved in establishing the A-P axis. The nucleus of the ovum secretes *gurken* mRNA (Fig. 32-6), which is translated to the protein **Gurken**, a TGF- α -like protein that carries signals to follicle cells. They, in turn, help to organize the microtubules in the ovum. The nucleus of the ovum moves, sending the Gurken signal first to the posterior pole cells, then to a position that marks the dorsal side of the ovum. Thus, it participates in establishing both the A-P and dorsal-ventral axes.³²⁴

The animal-vegetal axis of sea urchin eggs is established during oogenesis, but the mechanisms are still unclear. After fertilization a distinct pattern along the axis is established by the 60-cell stage. Signaling from the vegetal pole appears to play an important role.⁸⁹ In the presence of lithium chloride, which is known to affect inositol triphosphate (InsP₃) metabolism (Fig. 11-9), embryos develop an excessive fraction of endoderm and mesoderm (tissues of the vegetal half) at the expense of ectoderm (tissues of the animal half).^{89,325} Some data suggest that this effect implicates the **Wnt signaling pathway** and the protein **β -catenin**. This is the same protein that links C-terminal tails of cadherin to the actin cytoskeleton.⁶⁰ Its cytoskeletal and signaling functions seem to be controlled separately.

In the frog *Xenopus* the ovum accumulates an RNA (Vg1 mRNA) that encodes a growth factor of the TGF- β family (Chapter 30, Section A,6).²⁸⁰ Initially present throughout the ovum, it moves and forms a gradient of concentration that is highest at the vegetal pole.

The dorsal-ventral (D-V) and right-left axes.

Establishment of the D-V axis in *Drosophila* requires the participation of several genes. As mentioned above, *gurken* affects follicle cells. These cells cooperate with the ovum to set up a concentration gradient of the maternal gene **Dorsal**, the protein encoded by the *dorsal* gene.^{9,280,326–328} Dorsal is a transcription factor related to NF- κ B (Fig. 5-40). In the early embryo of the fly it forms a gradient in which it remains largely in the cytoplasm on the dorsal side but is mostly taken up into the nuclei on the ventral side. There it binds to a series of complex enhancers, each 300–1000 kbp in length, which act along the D-V axis. The enhancers interpret the concentration of Dorsal at five different threshold levels.³²⁸

The establishment of a second axis in vertebrate embryos is a complex process, which depends upon prior formation of a mesoderm layer. Development of mesodermal cells (not shown in the more primitive gastrula illustrated in Fig. 32-6) is induced by diffusion

of a growth factor from the vegetal pole.^{329–331}

Activin, a member of the TGF- β family, has been proposed as the natural inducer of mesoderm. More recently **nodal**, a different TGF- β -like protein, has been suggested.³³² The fact that induction can occur through thin (20 μ m) filters without any cell-cell contact indicates that specific chemical agents are responsible. Induction of the mesodermal layer in *X. laevis* appears to be an effect of an epidermal growth-factor-like protein. Additional factors are also needed to establish the D-V axis in vertebrates³²⁶ in which the dorsal side is homologous to the ventral region in *Drosophila*.

Expression of another set of genes establishes the **right-left** axis and characteristic asymmetries of the body.^{333–336} In the chick activin 2 β , also a member of the TGF- β family, as well as Nodal and **Sonic hedgehog** (Shh) participate in control. The gene *Pitx2* is a downstream transcription target for this signaling cascade in vertebrates.

Spemann's organizer. In 1924, the German physiologist Hans Spemann with Hilde Mangold transplanted a small piece of tissue from the dorsal lip of a newt blastula to a site on the ventral side of an early blastula of a differently colored species. The embryo developed a small secondary embryo, most of the tissues of which came from the host, not from the transplanted piece. It was concluded from this famous experiment that the transplanted vegetal tissues had supplied a diffusible inducer.^{13,14,337–339} This morphogen caused the cells of the ventral surface of the blastula to secrete other morphogens. The signaling center in this part of the blastula surface is known as **Spemann's organizer**, or simply the organizer. It utilizes more than one inducer and a complex set of signaling interactions.³⁴⁰ The cascade that induces formation of the organizer involves the Wnt- β -catenin and TGF- β pathways and the transcription factor **Smad4**.^{41,341,342} The organizer secretes protein such as **noggin**, **folistatin**, and **chordin**, as well as nodal and other members of the TGF- β family.^{332,343,344} They establish the D-V axis and also direct the development of the head and the initial patterning of the central nervous system. Noggin seems to be a neural inducer.³³⁸

Patterns, signaling pathways, and homeotic genes. While gradients in *Drosophila* eggs establish the anterior-posterior axis, products of other genes specify the developmental fates of cells of specific lineages and of cells found in particular spatial domains. Many *Drosophila* genes are needed to establish patterns, e.g., spacings of similar elements such as hairs, components of compound eyes, and whorls of plants.³⁴⁵ Insects are organized into a pattern of consecutive segments from head to tail along the A-P

axis.^{13,14} In *Drosophila* there are typically 17 of these segments, some carrying appendages such as antenna, legs, and wings. Each segment develops under a different set of influences from neighboring cells. Development is controlled by ~30 **segmentation genes**, which determine the number of segments and their internal organization. Of these genes one set of at least six **gap genes**, among them *hunchback* (Hb) and *Krüppel*, are expressed first. Both of these genes encode zinc-finger transcription factors (see Fig. 5-38). The Hunchback protein acts mainly on head parts and upper thorax, while Krüppel influences development of the thorax. Also among the segmentation genes are eight “pair-rule” genes and at least 16 segment polarity genes. The interactions of the products of these genes creates a prepattern that provides positional information and guides further development.³⁴⁶ Somewhat similar to the segmentation of the insect body is the development of skeletal muscle from a series of mammalian embryonic blocks known as somites.^{13,347,348} In *Drosophila* signaling pathways involving the secreted proteins EGF, Decapentaplegic (Dpp),³⁴⁹ Wingless (Wg), Hedgehog (Hh),^{349,350} and Notch are used repeatedly to provide positional information. The names of these proteins describe effects of mutations on the limbs of the insect, but the proteins have a much broader significance. They represent an evolutionarily conserved set of intercellular signaling pathways.³⁴⁶ Wingless is the first member of the previously mentioned Wnt family. Proteins of the Hh family, including the vertebrate Shh, control a large variety of processes that include development both of limbs and of the nervous system.^{351–355} The Hh and Shh proteins all carry a molecule of cholesterol covalently bound in ester linkage to the C termini of the biologically active N-terminal domains of these proteins.^{350,353} Defects in Shh signaling causes some human birth defects, and it is possible that drugs that inhibit cholesterol synthesis may have adverse effects on embryonic development.^{352,355}

The *Drosophila* Notch 1 is a 300-kDa integral membrane protein that contains 36 EGF-like repeats. Its activation by proteolysis produces a 200-kDa N-terminal portion and a 120-kDa C-terminal fragment, which contains a transmembrane domain. The small intracellular domain of this fragment is then released by protease action and travels to the nucleus where it activates several target genes.^{356–358} The Notch signaling pathway is conserved in all metazoans and influences many interactions that control cell fate during development.³⁵⁹ The proteolytic cleavages of the Notch protein parallel those of both the ErbB-4 growth factor receptor (see Table 11-3) and the amyloid precursor protein APP that is pictured in Fig. 30-34. The same type of protease (γ -secretase or presenilin in the human brain) cleaves all three of these proteins.^{359–361} Actions of Notch are modulated by posttranslational

alteration, e.g., glycosylation by a fucosyltransferase encoded by gene *fringe*.^{362,363} A homolog of *fringe*, called *lunatic fringe*, encodes an essential component of somite formation in the mouse.^{348,364}

Functioning together with the signaling pathways are **selector genes** that determine which specific pathway is to be influenced. For example, eyes, antennae, legs, or wings of a fly may be selected.³⁴⁶ *Antennapedia* (*Antp*) is one of these genes. Several of the *Drosophila* selector genes as well as some pair-rule genes³⁶⁵ are also known as **homeotic genes** (see also Chapter 28, Section C,6). Homeotic genes were first recognized by the fact that mutation causes conversion of one segment of an insect’s body into the homologous tissues of another segment.^{366,367} For example, a mutation in *antennapedia* (*ante*) changes the antenna into a leg. Similar genes are also active in vertebrates, e.g., in the development of the chick embryo limb bud, the very tip (the last 20 cell diameters of length) contains cells that differentiate into the various elements of the limb in a relatively autonomous manner. If this *progress zone* from one limb bud is grafted onto the end of another limb bud, the bones and cartilaginous elements of the limb are repeated. Both the number and morphology of fingers and toes are determined by homeotic genes^{367a} as is the formation of sphincters in the developing gut.^{367b} Homeotic genes (also known as *Hox* genes) contain a conserved sequence of 180 bp that specifies a 60-residue protein **homeodomain** (also known as a homeobox). The homeodomain folds into a helix–turn–helix motif characteristic of many transcriptional regulators (Figs. 5-35, 28-3, p. 1631).^{365,367–371} *Hox* genes are found among all forms of life. Hundreds have been described.³⁷² They include genes for the previously mentioned transcription factors of the Oct and Pit families as well as for the yeast mating type proteins MATa1 and MAT α 2 (p. 1880).¹⁷ The *Drosophila* genome contains eight *Hox* genes, while the human genome has ~40, which are organized into *Hox* clusters.³⁷³ *Hox* genes are also abundant in plants.^{345,372,374}

Despite intense interest the role of homeodomain proteins in development is not well understood. The highly conserved motif binds to DNA at many places in the genome. Current thinking is that homeodomain proteins interact with other regulatory proteins, and that various combinations of these proteins provide the information needed to direct development.³⁷⁵

D. Specialized Tissues and Organs

Here are a few details about development of mammalian tissues. We’ll begin with the blood and connective tissues, which arise from embryonic mesodermal cells.

1. Blood Cells and the Circulatory System

Every second of life a human must produce about 2.5 million red blood cells, about 2 million granulocytes, and many lymphocytes as well as other less numerous leukocytes. All of these arise from **multi-potential stem cells** found in the bone marrow.³⁷⁶⁻³⁷⁹ Each of these stem cells divides to form one daughter stem cell and one **progenitor cell**.³⁸⁰ The progenitor cells are also stem cells but have differentiated into **myeloid**,³⁸¹ **erythroid**, and **lymphoid**^{382,383} cells. These differentiate further as is indicated in Fig. 31-2. Mature blood cells of most types have short lifespans and must be regenerated from stem cells continuously.

At all stages the differentiation process is regulated by the microenvironment which is rich in specific protein growth factors, several of which have been discussed in Chapter 30. Among the 20 interleukins, three stimulate growth of both multipotential progenitor cells and erythroid progenitor cells.^{380,384} The acetylated tetrapeptide Ac-Ser-Asp-Lys-Pro inhibits stem cell proliferation. Granulocyte-macrophage colony-stimulating factors stimulate proliferation of both granulocytes and macrophages. The kidney cytokine **erythropoietin**, a 30.4-kDa glycoprotein,³⁸⁵ is a primary regulator of red blood cell formation. Its action on a differentiated stem cell initiates massive hemoglobin synthesis and terminal differentiation of the erythrocyte. **Thrombopoietin** promotes formation of megakaryocytes and also their maturation and release of platelets to the blood.³⁸⁶ **Thymopoietin** promotes early T-cell differentiation. Activated macrophages secrete interleukin-1, which stimulates maturation and proliferation of B lymphocytes. Interleukin-2 (T-cell growth factor; Fig. 30-8) is produced by activated T lymphocytes and is needed by T lymphocytes for long-term helper and cytotoxic functions. Differentiation of the stem cells into the erythroid lineage requires transcription factor GATA-1, development into erythrocytes requires GATA-2, while development into T lymphocytes requires GATA-3.³⁸⁷

Globin genes. The genes that encode the human globins from which hemoglobin is formed are found in two clusters, the α -like genes on chromosome 16 and the β -like on chromosome 11. They are developmentally regulated, different genes in the clusters being active at different stages of development. Mammalian hemoglobins (Chapter 7) each contain two α chains or two related ζ chains and two other chains, β , γ , δ , or ϵ . Adult hemoglobin is mainly $\alpha_2\beta_2$ but contains small amounts of $\alpha_2\delta_2$. In early embryos the hemoglobin is $\zeta_2\epsilon_2$, but during the second to sixth weeks of embryonic life the two fetal hemoglobin chains γ^G and γ^A replace the ϵ chains. The switch from fetal to adult hemoglobin begins a few weeks before birth and is complete by about ten weeks after birth. The β -like

gene cluster contains five genes encoding globins ϵ , γ^G , γ^A , δ and β (Fig. 27-10). Each gene consists of three exons separated by two introns and has rather similar control signals. These include CACCC at $\sim$ -100, CACA at $\sim$ -92, CCAAT at $\sim$ -75, and ATAAAA (TATA sequence) at $\sim$ -30, as well as AATAAA (cleavage and polyadenylation). The CCAAT sequence appears twice in the γ^A promoter. A variety of transcription factors and chromatin modifiers influence the expression of these genes.^{388,389}

Only a small amount of fetal hemoglobin ($\alpha_2\gamma_2$) is produced after infancy. There are two genes, γ^G and γ^A , for the β -like chains of fetal hemoglobin. A few adults make large amounts of fetal hemoglobin and this hereditary **persistence of fetal hemoglobin** has survival value for persons carrying thalassemia genes. This condition may result from a single base change in the CCAAT sequence found upstream ($\sim$ 75 bp) of the globin genes. Many other genes are also preceded by the same sequence, which in extended form is often TTGGPyCAAT. In one individual with persistence of fetal hemoglobin the first G in this sequence was replaced by A in one of the two CCAAT sequences present in the γ^A gene.

Thalassemias. In these important hemoglobin diseases the α or β chain either is absent or is present in far less than stoichiometric amounts.^{14,390} About 40 point mutations in the β globin gene have been described among patients with β thalassemia, in which β chains are missing (β^0 thalassemia) or are present in reduced amounts (β^+ thalassemia). These mutations sometimes occur in control regions at the 5' end of the gene. For example, a change of the TATA sequence from ATAAAA to ATACAA causes decreased transcriptional efficiency and β^+ thalassemia. Other mutations result in abnormal splicing or in instability of the β globin. Deletions may result in the complete absence of the gene or in a frameshift that results in nonfunctional globin and β^0 thalassemia.

How do embryonic cells choose to transcribe only the embryonic globin genes? The decisions to switch from embryonic to fetal and from fetal to adult at appropriate times appear to be controlled by interactions with appropriately expressed transcription factors. Especially impressive is the total and permanent cessation of transcription of the embryonic globin ϵ gene at about the seventh week of gestation. Transcription of the β globin gene cluster is controlled by a powerful enhancer, the **locus control region** $\sim$ 6-22 kbp upstream of the promoter.³⁹¹ In addition, $\sim$ 270 bp upstream of the ϵ gene promoter is a **silencer**, a DNA sequence to which inhibitory proteins may bind and, in cooperation with the enhancer, may completely silence the ϵ gene while allowing transcription of other genes in the cluster.^{392,393} The ζ gene, in the α globin cluster, is silenced by the binding of an NF- κ B transcription factor to a 108-bp

segment of DNA located 1.2 kbp to the 3' end of the gene.³⁹⁴ The globin genes are shielded from action of nearby enhancers by **insulators**, DNA sequences that often contain CpG islands.³⁹⁵

Blood vessels: vasculogenesis and angiogenesis. Early in development of an embryo a network of blood vessels is formed from mesenchymal progenitor cells (vasculogenesis).^{396–398} Later, in either the embryo or the adult, new capillaries are formed (angiogenesis). As the organism develops these new vessels are “pruned,” and the vascular bed is remodeled to a tree-like form with vessels of both large and small diameter.^{399,400} A key activator of angiogenesis is the vascular endothelial growth factor **VEGF**.^{401,401a} However, a number of other proteins including **endoglin**, a TGF- β binding protein,⁴⁰² the clotting factor thrombin,^{400,403} and the ribonuclease **angiogenin** (p. 648) exert their influences.⁴⁰⁴ There are diseases that result from defective angiogenesis. However, a major interest in this process arises because of the essential role of angiogenesis in the growth of cancer⁴⁰⁵ and as a complication of inflammatory processes. Angiogenesis in the retina is a major cause of blindness resulting from diabetes mellitus or from macular degeneration.^{405,406} There are natural antagonists of angiogenesis,^{407–409} and efforts are being made to utilize them in therapy. A related project is development of completely tissue-engineered blood vessels for surgical use.⁴¹⁰

2. Cartilage, Tendons, Bone, Muscle, and Fat

Mesenchymal cells differentiate into cartilage, bone, muscle, adipose, and other connective tissues.⁴¹¹ **Chondrocytes** synthesize the variety of collagens (pp. 431–426) that are needed for synthesis of cartilage and other connective tissues. The 32 or more genes encoding the polypeptides needed for synthesis of the 19 types of vertebrate collagens (Table 8-4) are developmentally regulated in a complex manner.⁴¹¹ Their promoters contain TATA and CCAAT sequences as well as other presumed regulatory codes. The first intron of several collagen genes has also been identified as a control region containing enhancers.⁴¹² The elastic fibers (p. 436) owe their properties in part to elastin. The control region of elastin genes lacks the TATA sequence but has SP1 binding sites. As with many other mammalian genes, a diversity of protein products, many in small amounts, are made by alternative splicing.

Collagen fibrils provide the scaffolding for formation of bone, whose composition is considered on pp. 440–443. Bone develops under the influence of **bone morphogenic proteins** BMP-2 to BMP-7.^{413–415} Most of these are cytokines of the TGF- β family. Noggin

(p. 1899) antagonizes the action of the BMPs.³⁴⁴ A characteristic of bone is rapid remodeling (p. 441) by which ten per cent of skeletal bone is replaced every year. The balance between action of the bone-forming osteoblasts and the bone-resorbing osteoclasts is regulated by surface proteins responding to **c-Fos** and to **interferon- β** .⁴¹⁶

Muscle, whose structure and function are discussed in Chapter 19, develops in response to four members of the myoD family. These include myoD, **myogenin**, **myf5**, and **MRF4**.^{417–419} All are muscle-specific transcription factors of the basic helix-loop-helix class. An unusual aspect of muscle development is formation of multinucleate **myotubes** (muscle fibers; p. 1096).⁴²⁰ Apoptosis plays an important role in muscle development and can present significant complications in damaged cardiac muscle.⁴²¹ Defects in several developmental control genes are responsible for congenital heart diseases.⁴²²

3. Epithelia

Epithelial tissues, which line both internal and external surfaces, arise from all three cell layers of the blastula. The epidermis (Box 8-F) arises from ectoderm, while the lining of the digestive tract is formed by endodermal cells. Mesoderm provides the linings of blood vessels. About 60% of differentiated tissue types in the mammalian body are epithelia.¹²³ Stem cells or progenitor cells are present and provide for renewal.^{123,124} While epidermal stem cells are located in deep layers of the skin, **keratinocytes** are readily cultured *in vitro* and can give rise to fully differentiated multilayered skin.⁴²³ Development appears to require transcription factors related to **Oct-2**⁴²⁴ as well as **p63**, a homolog of the tumor suppressor p53.^{425,426} Mice deficient in the aspartyl protease **presenilin 1** (which is defective in some forms of Alzheimer disease, Chapter 30) develop characteristic epidermal skin tumors. The β -catenin–Wnt signaling pathway (p. 1899) seems to be involved.⁴²⁷ The gastrointestinal endoderm develops its highly convoluted surface under some control by the Notch signaling pathway.^{428,428a} Endothelial progenitor cells (**angioblasts**) are responsive to many signaling molecules⁴²⁹ including thrombin.⁴³⁰ The *Drosophila* eye develops from the epithelium, again through signaling via Notch and other morphogens.^{430a} Of outstanding importance to epithelial cells in general is their ability to form complex communicating junctions.^{430b}

4. The Nervous System

Development of the vertebrate central nervous systems is initiated during gastrulation through an

interaction between the dorsal ectoderm and an infolding of the dorsal mesoderm. Several different diffusible inducers are involved. These include noggin,³⁴⁴ follistatin, and other members of the TGF- β family as well as thyroid hormones, basic fibroblast growth factor (bFGF), and sonic hedgehog.³²⁹ The nervous system develops over a period of a few days with differentiation of precise numbers of neurons, astrocytes, and oligodendrites in successive waves. The order in which various cell types arise is determined by the order in which transcription factors such as Hunchback, Krüppel, and others are expressed.⁴³¹ Multipotential neural stem cells provide the new cells that are required.^{432,433} Neural tissue from a region called the **neural plate** develops into a neural fold. The latter is closed to form the **neural tube**,⁴³⁴ within which the notochord, the precursor to the spine, as well as the neurons, glia, and other cells grow. The **neural crest** forms as an outgrowth from the dorsal surface of the neural tube under the influence of inducers of the Wnt and BMP families.^{434a} Cells migrate from this crest to form the peripheral nervous system, melanocytes, and cranial cartilage.^{434b} The pituitary, a central component of the neuroendocrine system, develops from tissues from the midline part of the anterior neural ridge.⁴³⁵

The **floor plate**, which develops along the midline of the ventral surface of the neural tube, is the source of Sonic hedgehog,^{436,437} **netrin-1**, and other secreted molecules.⁴³⁶ Some of these participate in **axon guidance** by which the growing tips (**growth cones**) of axons are able to connect to the correct "targets."^{437a} For example, every visual receptor cell in the retina must send its signal to the correct locations in the visual cortex of the brain.⁴³⁸ How can this be accomplished? Over a century ago Ramon y Cajal proposed that chemoattraction, analogous to chemotaxis of bacteria, might be involved.^{439,440} A hundred years later Tessier-Lavigne and coworkers isolated the first of these attractants, **netrin-1** and **netrin-2**, from 25,000 pulverized chick brains.^{441,442} Of these two closely related 75- and 78-kDa molecules netrin-1 is produced only in the floor-plate cells. Like the less well understood nerve growth factor (Fig. 30-7), netrins induce outgrowths of neurites and also are chemoattractants for nerve growth cones.⁴⁴³ The netrin receptor is known as **DCC** (Deleted in Colorectal Cancer).⁴⁴⁴ A nematode protein UNC-6 is related to the netrins.⁴⁴⁵

Growth cones are subject to both chemoattractant and chemorepellent effects of guidance molecules and also to attraction or repulsion resulting from cell-cell contacts. To complicate the picture further, the netrins and also the brain-derived neurotrophic factor BDNF (see Fig. 30-6D) may first attract, and then after a period of adaptation or desensitization, repel a growth cone.^{443,444,446-448} Consecutive phases of desensitization and resensitization may result in a zig-zag path of

growth. The netrins were recognized first by observing growth of **commissural neurons**. These neurons originate within the spinal cord on one side or the other or the midline. They grow down toward the floor plate attracted by the netrin-1 or **BDNF** produced there. The neurons then cross the midline before turning and growing toward the brain. After crossing the midline the growth cones become insensitive to netrin but are repelled by a molecule (first recognized in *Drosophila*) called **slit**. Its receptor is appropriately named roundabout (**Robo**).⁴⁴⁹ A similar receptor in zebrafish (called **astray**) is required for retinal axon guidance.⁴⁴⁹ An important aspect of neuron guidance is apoptosis induced by misdirected growth.⁴⁵⁰ The Notch receptor apparently participates in this decision in the mammalian CNS.³⁵⁷ Positive signals for axon growth often involve the MAP kinase pathways, while inhibition may involve INS-2P-Ca²⁺ signaling.⁴⁵¹

Chemoattractants that function in development of the cerebral cortex include several **semaphorins**.^{445,452,453} A separate family of attractant and repellent compounds, the semaphorins have been identified in insects, chickens, and mammals. They play a role in regulation of communication between neurons. Because of the complexity of the brain, the study of growth of neurons with the brain is difficult. The >10¹² neurons each contact, on the average, 100 different cells. Some insight comes from mutant mice with names such as *reeler*, *scrambler*, *stargazer*, and *Yotari* (Japanese for tottering).⁴⁵⁴ The single defective gene in these mice can be identified and studied. For example, *reeler* mice are defective in **reelin**, a large glycoprotein of the extracellular matrix (ECM).^{366,379,455,456} The *reeler* phenotype can also result from mutation of the gene *disabled-1*, which encodes a cytosolic tyrosine kinase. Other mutations in mice implicate **VLDL** and **apoE** receptors (Chapter 21) in these developmental abnormalities.^{379,455}

The *stargazer* mutant mouse is ataxic and epileptic. It lacks functional **AMPA receptors** (Fig. 30-1), which apparently are not delivered successfully to the synapses in the cerebellum in which they function.^{380,386} Mutation of a transmembrane protein **stargazin**, which may interact with the AMP receptor, causes the symptoms.^{457,458} **NMDA receptors** (Fig. 30-20) are involved in synapse formation in the brain. Filopodial extensions on dendrites, triggered by electrical activity, are essential for synapse formation,⁴⁵⁹ which occurs rapidly.^{459a} Activation of NMDA receptors is apparently also necessary.^{379,460} Without this stimulation the excitatory glutamatergic neurons of the developing brain undergo apoptosis.

Why do neurons grow in the embryo but not in most parts of the adult CNS? Two proteins called **Nogo** were isolated from bovine brain. Their sequences were utilized in identifying the *Nogo* gene and three human

isoforms of the protein.^{389,461} The large 250-kDa Nogo-A is present both in myelin and in the endoplasmic reticulum. Both Nogo-A and the diffusible 35-kDa Nogo-B are inhibitory of neurite outgrowth.⁴⁶² This effect, as well as the crowding of regenerating neurons by the chondroitin and other matrix components,³⁹⁶ may provide obstacles to nerve regeneration.⁴⁶³

E. Development of Green Plants

Green plants may have diverged from a common ancestor with animals ~1.6 billion (1.6×10^9) years ago. How do the genomes of present-day plants and animals compare? There are many similarities in basic metabolism. These arise from the intrinsic chemical properties and reactivities of cellular components and from the coevolution of plants and animals. Plants and animals also utilize similar structures and similar control of chromatin. However, in the control of development there are great differences.⁴⁶⁴ For example, the *Arabidopsis* genome contains no relative of the *Drosophila* Gurken, no receptor tyrosine kinases, no relatives of transcription factor NF- κ B. However, there are similarities in parallel pathways utilized by plants and animals.

The structures and life cycle of angiosperms⁴⁶⁵ are described briefly on pp. 29–30. The alternating haploid (n , **gametophytic**) and diploid ($2n$, **sporophytic**) phases of the life cycle^{466,467} are diagrammed in Fig. 32-8A. Following flowering a diploid **mother cell** within the **ovule** undergoes meiosis to form four haploid **megaspores**. After mitosis a single egg cell is formed. Within pollen sacs in the anthers of each mother undergoes meiosis to yield four haploid **microspores**. Following mitosis these develop into pollen grains each of which contains two sperm cells as well as a vegetative nucleus. After falling upon the stigma surface and growth of the pollen tube, one of the sperm cells fuses with the egg to give the diploid zygote. The other sperm unites with the specialized diploid **central cell** in the ovule to form a triploid ($3n$) **endosperm nucleus**, which develops into the **endosperm**, the food storage tissue of the seed. Endosperm contains two tissues, a starchy inner layer and a protein- and oil-rich outer layer.⁴⁶⁸

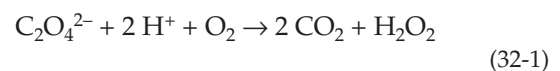
Variations of the life cycle occur. For example, a process called **apomixis** leads to asexual formation of seed.⁴⁶⁹ In many plants, including maize, separate flowers form the ovule and the pollen. This is one mechanism for avoiding inbreeding.⁴⁷⁰ In many plants systems of **self-incompatibility** have evolved.⁴⁷¹ In some, e.g., *Arabidopsis* and other crucifers, pollen germination is disrupted unless it falls on a stigma possessing a different allele-specific receptor. In other cases development of the pollen tube is disrupted at a later stage. In maize and in more than 150 other

species a mitochondrial and therefore maternally inherited trait prevents formation of a functional pollen.^{472,473} Male-sterile plants, which carry this trait, are very useful in plant breeding. However, a near disaster occurred in the United States in 1970 when the fungal disease Southern corn leaf blight attacked the male-sterile maize that had been used for production of 85% of commercial hybrid maize. The mitochondrial defect in pollen formation also resulted in an increased sensitivity to the fungal toxin. The target of the toxin is a 115-residue pore-forming polypeptide in the inner mitochondrial membrane.⁴⁷⁴ The male sterility can be reversed if the plant carries two nuclear **restorer genes**.⁴⁷² One of these encodes an aldehyde dehydrogenase, but its mechanism of action is unclear.

The plant embryo is a juvenile form, the seedling. In *Arabidopsis* the zygote, which is surrounded by maternal diploid tissue, divides asymmetrically. The resulting apical and basal cells (Fig. 32-8B) differ in several ways. The small cytoplasm-rich apical cell is partitioned into eight proembryo cells by two rounds of vertical division and one horizontal division. The larger basal cell contains a vacuole and divides repeatedly horizontally to give 7–9 aligned cells. Only the uppermost of these becomes a part of the embryo. The others form an extra-embryonic suspensor (Fig. 32-8B).⁴⁷⁵ The apical part of the embryo develops the shoot meristem and the central part the radial pattern of tissue layers characteristic of plants. The root meristem develops from the basal portion of the embryo. Movements of proteins that provide positional cues are involved in the development of the embryo.^{476,477} Early embryonic and endosperm development is largely under maternal control. Most paternal genes may be initially silent.⁴⁷⁸

Many angiosperms develop **fruit** from tissues of the ovary (Fig. 32-8A). The development and ripening of fruit is also complex and highly regulated.^{479,480}

Formation of seeds is a slow process. For example, in wheat the mature embryo, which consists of $\sim 10^5$ cells, develops over a seven week period. Seeds may live from a few years to 1000 years or more.⁴⁸¹ Subsequent germination of the seed into a seedling requires only two days.⁴⁸² The very dry embryo is converted into a highly hydrated plant whose further growth requires uptake of very large amounts of water. Many plants also synthesize large amounts of oxalic acid. This may arise from ascorbate (p. 1135) or via oxidation of glycine (Fig. 24-20). One of the earliest mRNAs to appear during seed germination encodes a 125-kDa glycoprotein called **germin**. This protein, which exists as multiple isoforms, is a copper-dependent oxalate oxidase (Eq. 32-1) which generates hydrogen peroxide. The latter is probably needed to



crosslink cell-wall polymers. Germin may also be useful to plants in defense against oxalate-forming fungi.⁴⁸²

The rapid vegetative growth, which includes development of shoots, leaves, and flowers, is controlled by a variety of transcription factors.⁴⁸³ Among these are homeodomain proteins that control differentiation of meristem cells.^{484–486} The induction of flowering is

especially complex, involved day length, light quality, and effects of gibberellins.⁴⁸⁶ At the ends of their lives plant cells die slowly from **senescence**. In this process many materials are recycled for use by new cells. Other plant cells die via the **hypersensitive response**, a form of programmed cell death.^{486a}

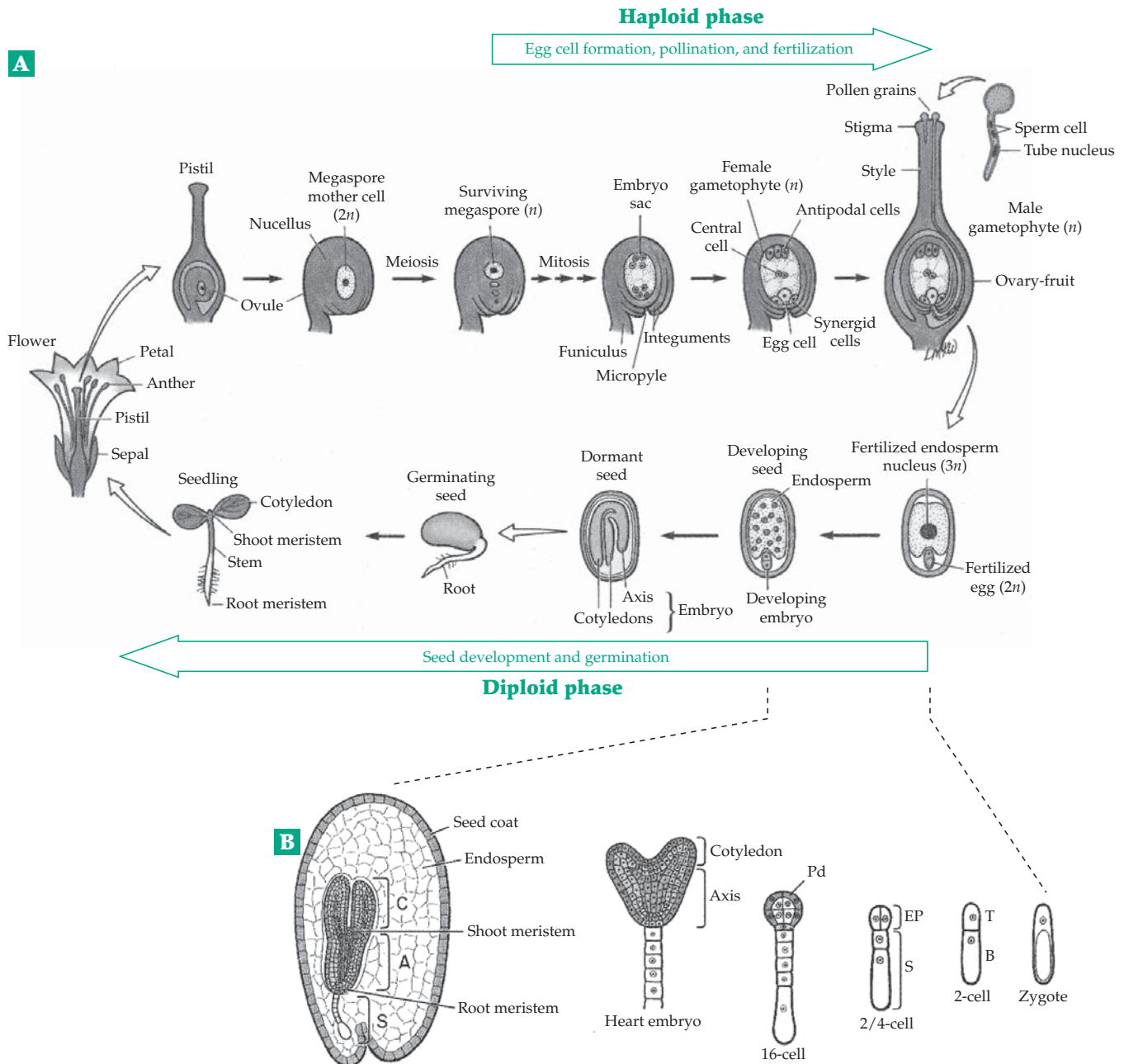


Figure 32-8 (A) The life cycle of a flowering plant with emphasis on egg-cell formation and seed development. (B) Some further details of embryo development. T, terminal cell; B, basal cell; C, cotyledon; A, axis; SC, seed coat; En, endosperm; EP, embryo proper; S, suspensor; SM, shoot meristem; Pd, protoderm; RM, root meristem. From Goldberg *et al.*⁴⁶⁶ with modification.

F. Aging

Why do we age? This question is often asked but the answers are not simple. Do our tissues deteriorate with age as a result of damage to proteins? From an accumulation of mutations in our DNA? From attacks by free radicals? From loss of hormone receptors? From misregulation of mitosis?⁴⁸⁷ From loss of telomeres on the ends of replicating DNA (Chapter 27, Section C,8)?^{488,489} From an internal genetic program that dictates our life span? All of these possibilities may be partially correct. The simple answer is that “we just wear out.” However, different parts wear out at different rates and in different ways.

Perhaps we should be amazed that the human body can live for an average of 75–80 years.^{490,491} If we all avoided accidents and could cure all recognized diseases we might live an average of ~90 years.⁴⁹² A few very healthy people live for 100 years or more, ~20 years longer than average.⁴⁹³ The maximum lifespan at present seems to be ~114 years. Long life tends to run in families, indicating a genetic component that can be identified.⁴⁹³ However, this component is relatively small.⁴⁹⁴

Why do small rodents live only 2–3 years while we often live nearly 100 years?^{494,495} Is it because their rate of metabolism is high? But bats have a comparable metabolic rate to mice, yet live ten times longer.⁴⁹⁶ Nematodes live only ~20 days and fruitflies ~10 days. At the other extreme fish and some reptiles continue to grow throughout their lifetimes, surviving even longer than mammals.⁴⁹⁵ Except for their germ cells nematodes, rotifers, and many insects have no dividing cells in their adult bodies. Their lifespan is presumably determined by the loss of cells through injury or death. In contrast, some simple animals, such as *Hydra*, other coelenterates, and flatworms maintain a pool of pluripotent stem cells that, except for accidental death, seems to make them immortal.⁴⁹⁵

Considerations such as these have helped move contemporary thinking toward an evolutionary view.^{488,489,495–499} If the mortality of an animal in the wild (**extrinsic mortality**) is high it will evolve to have rapid development, good reproductive ability, and a short lifetime. If the extrinsic mortality is low the lifetime will be long. Such animals will require development of good protective functions including a highly developed brain.

Many factors must affect aging. It is generally agreed that one of these is the deleterious effects of free radicals derived from oxygen^{500–503} (see pp. 1074, 1075). The lowered turnover rate of aging tissues may allow the damage to become lethal. According to this theory we might anticipate that free radical scavengers such as vitamin E could prolong life as might a restriction in food consumption.⁵⁰⁴ For example, decreased fat intake might cut down on production of

malondialdehyde (p. 1205) and lipid peroxides that may be especially damaging to cell membranes. The life of rodents can be prolonged substantially by a semi-starvation diet. Although there are uncertainties, a convincing case can be made for humans to keep food intake to a minimum and to eat foods rich in antioxidants and other nutrients.^{491,504a}

For many years after techniques of cell culture had been developed it was commonly believed that cells in tissue culture were potentially immortal.⁵⁰⁵ Challenging this idea, Medvedev⁵⁰⁶ and others proposed that cells are internally programmed for a certain lifetime. This might explain why we have short-lived mammals as well as long-lived mammals. Support for the idea was supplied by Hayflick,^{507,508} who observed that animal cells in culture have a limited potential for doubling. For example, normal human diploid embryonic fibroblasts grow in culture and double their number approximately 50 ± 10 times. Regardless of cultural conditions, the cells die after this number of doublings. Cells taken from older humans undergo a smaller number of doublings before dying, as do those taken from shorter lived animals such as the mouse (14–28 doublings).⁵⁰⁸ These experiments suggested that there is an internal program by which cells are scheduled to die from **replicative senescence**.^{509–511} Malignant transformation overrides this program and transformed cells appear “immortal.”⁵¹² However, unlike fibroblasts some glial stem cells have been identified as possibly having unlimited proliferative capacity. These include cultured rat oligodendrocyte⁵⁰⁹ and Schwann cells.⁵¹³ These results suggest that replicative senescence may not be inevitable.

As mentioned in Chapter 27 (p. 1568), “erosion” of the telomere ends on chromosomes is thought to be a major cause of cell senescence. Old cells have little or no telomerase. However, most human cancer cells, as well as those from immortalized cell cultures, do synthesize telomerase and maintain telomeres of adequate length.^{514,515} Inhibition of telomerase activity in immortalized cells causes telomere shortening and cell death.⁵¹⁶ A second pathway for telomere maintenance is based upon homologous recombination.⁵¹⁷ Experimental elongation of telomeres extends the lifespan of cells in culture.⁵¹⁸ Furthermore, apparently healthy calves have been produced by nuclear transfer cloning from senescent fibroblast cells for which four or fewer cell doublings were expected to be possible. The cells of the cloned animals had the capacity for 90 or more cell doublings.⁵¹⁹

What information about aging can we obtain by study of the “model” organisms *S. cerevisiae*, *C. elegans*, *D. melanogaster*, and the mouse? In every case a variety of mutations may shorten or lengthen the lifespan. In every case dietary energy restriction can lengthen life. Yeast cells grown on 0.5% glucose instead of 2%

glucose may undergo ~25% more cell doublings before a culture becomes senescent. However, mutant strains with a defect in the *SIR2* gene have a shortened life-span, which is not increased by caloric restriction.^{520–522} *SIR2* encodes Sir2p, an NAD⁺-dependent histone deacetylase (p. 1626).^{522,523} It is likely that caloric restriction causes the yeast to switch from anaerobic fermentation to oxidative metabolism. The resulting increase in the [NAD⁺] / [NADH] ratio activates Sir2p, thereby altering chromatin and silencing a group of genes. Mechanisms by which this shift in metabolism decreases deleterious mutations, even though respiration is increased, are probably complex.⁵²³

Nematodes (*C. elegans*) usually live about three weeks but the simultaneous presence of life-extending mutations in two different groups of genes lengthen the lifespan as much as fivefold.⁵²⁴ One of the genes is the maternal-effect Clock gene *Clk-1*. It has been found to code for a hydroxylase involved in the final step of synthesis of ubiquinone-9 (p. 1429, Fig. 25-4).⁵²⁵ The lifetime of wild-type nematodes is lengthened by ~60% by elimination of ubiquinone from the diet.⁵²⁶ The second group of genes that affect longevity regulate an insulin-like signaling system.^{527–529} In every case metabolism is slowed, an effect which may reduce the rate of harmful mutations. However, mutant animals may not be healthy. Some of these mutations induce formation of long-lived “dauer” larvae, providing a means for the larvae to survive for up to two months during periods of starvation.^{527,530} Others affect sensory cilia.⁵³¹ Mutation of a cytosolic catalase gene *reduces* the lifespan.⁵³⁰ The nematode’s lifetime is also affected negatively by its own germ cells, perhaps via a steroid hormone.⁵³² The heat shock proteins (p. 1630), by chaperoning newly synthesized proteins and preventing aggregation, also increase life span both in *C. elegans* and *Drosophila*.^{532a,b}

Some mutants of *D. melanogaster* with extended lifespans have a defective insulin / IGF signaling pathway.^{526,533} The **methuselah** mutant, whose lifespan is 35% greater than average, appears to involve a G-protein-coupled transmembrane receptor.⁵³⁴ Mutation of an insulin receptor homolog extends lifespan, apparently by causing a juvenile hormone deficiency.⁵³⁵ *Drosophila* lifespan is also lengthened by mutation of a transmembrane dicarboxylate transporter⁵³⁶ or by overexpression of a protein repair carboxyl methyltransferase (p. 594).⁵³⁷

Some mutant mice have extended lifespans. The Ames dwarf mouse has a mutation in *p66^{shc}*, a cell-surface protein that contains both Src-homology and collagen-homology domains. It lives almost one-third longer than do wild-type mice.⁵³⁸ Mice deficient in methionine sulfoxide reductase have a reduced lifespan⁵³⁹ but fruit flies with overexpressed activity of the enzyme are more resistant than wild-type flies to oxidative damage.⁵⁴⁰

In humans 100 or more years in age some mitochondrial mutations are associated with good health and longevity.⁵⁴¹ Dietary factors doubtless play a role. For example, supplementation of rats’ diet with lipoic acid improved mitochondrial function and increased the metabolic rate of old animals.⁵⁴²

A number of genetic **progeroid diseases** result in premature aging.^{543,544} Several of these arise from deficiencies in repair of DNA (Box 27-A). Among them are some types of cancer, and **Werner syndrome**, which arises from a defect in a 1432-residue protein with a central domain homologous to the RecQ family of DNA helicases (p. 1550).⁵⁴⁵ Defects in other RecQ homologs cause **Hutchinson-Gilford progeria** as well as the Bloom syndrome (see Box 27-A).⁵⁴⁶ Yet another DNA helicase, a subunit of transcription factor II (TFIIh, p. 1628), is defective in trichothiodystrophy (TTD, see Box 27-A).^{547,548} Another gene which helps to prevent aging is *KLOTHO*. First identified in mice, it encodes a transmembrane protein that has sequence similarities to β -glucosidases.^{549,550} Some mice with mutations in the tumor suppressor **p53** (Box 11-D) have enhanced resistance to tumors but age rapidly.^{551,552}

Aging seems to be inevitably linked to an increase in the incidence of cancer. This uncontrolled growth of cells appears to be allowed by the stepwise accumulation of mutations that affect growth, differentiation, and survival.⁵⁵³ Several aspects of cancer are discussed in other chapters of this book (see Box 11-D). However, the topic is so complex and research so active that it is hard to give even a thumbnail sketch of more recent discoveries.

Much effort is being dedicated to identifying the many signaling pathways that control growth, the mechanisms that cells employ to recognize problems in the control of growth, and the means by which cells can correct the problems or undergo apoptosis and avoid cancer.^{553–558} Some of the complexity arises because of the large number of signaling pathways in which mutations may produce activated proto-onco-genes or faulty tumor suppressors. A large network of these suppressors is present in human cells.^{553,559} Among the relevant signaling pathways are the following:

RAS – RAF – ERK (Fig. 11-12)^{553–555}
 p53^{556–558}
 the PtdIns 3-kinase – PKB/Akt pathway (Fig 11-9), which is opposed by PTEN^{560,561}
 EFG receptor (EGFR) signaling
 Wnt-Catenin signaling^{559,562}
 E. Cadherin^{559,563}

The importance of oncogenes and tumor suppressors has been demonstrated by conversion of human cells in culture into tumor cells in vitro.⁵⁶⁴ Introduction of

an activated *ras* gene, an SV40 viral protein that inhibits formation of both p53 and the Rb gene (Fig. 11-15), and an active telomerase gene sufficed. However, there is some doubt about the relevance of this work to human cancer.

Most cancerous cells have extra chromosomes. The karyotype (p. 1472) is rarely normal.⁵⁶⁵ This and other evidence suggest that **genomic instability** may be the major cause of cancer.^{566,567} In healthy cells stalled RNA polymerase is removed by transcription-coupled repair and lesions in DNA are either repaired (Chapter 27) or the cell undergoes apoptosis. Telomere dysfunction is also a factor.⁵⁶⁷ The two breast cancer susceptibility genes *BRCA1* and *BRCA2* are apparently responsible for about half of all hereditary breast and ovarian cancers.^{568,569} Protein *BRCA1* is an 1863-residue nuclear protein, which is thought to function in transcription. However, recent evidence indicates that *BRCA2* is directly involved in repair of double-strand breaks in DNA by homologous recombination.^{569,570} Other data implicate the Neu-Ras pathway, proto-oncogenes *c-myc* and *Wnt-1*, and cyclin D1 in breast cancer.⁵⁷¹

Yet another aspect of cancer is the **aberrant glycosylation** observed for many proteins.⁵⁷² The state of glycosylation of cell-surface proteins is one of many factors that affect metastasis, which is critical to growth of tumors.⁵⁷³⁻⁵⁷⁵ The recognition that causes of cancer are numerous has led to a new large-scale project to identify as many cancer-associated mutations as possible within the entire human genome. One early success from this effort is identification of mutations in the gene *BRAF*, one of the three human *RAF* genes. These mutations are present in 15% of human malignant melanomas.^{555,576}

G. Ecological Matters (Author's Personal Postscript)

The final section of this chapter deals with interactions among different species. As humans, beset by problems arising from our inability to communicate with other humans, we may feel that ecological relationships are relatively unimportant. However, any careful look at what can be regarded as an extension of metabolic cycles into the biosphere should convince us of the significance of this aspect of biochemistry.

Recall that the original development of eukaryotic creatures may have started with a symbiotic relationship between two prokaryotes and that symbiosis between algae and nonphotosynthetic organisms may have led to development of higher plants. Associations between species are still important today. For example, the bacteria in the protozoa of the digestive tract of ruminant animals are essential to production of meat. Our own bodies play host to bacteria, fungi,

and other organisms with whom we have to try to maintain friendly relations. We depend upon antibiotics produced by bacteria or by fungi to fight our bacterial infections. Plants provide both essential nutrients and oxygen. Our environment has been created in large part by other living forms that coexist with us and which are subject to ecological checks and balances. It is therefore important that we learn more about the effects of one group of organisms on another and also about the effects of human activities on plants and animals of all degrees of complexity. This includes the poorly understood world of soil microorganisms. The consequences of environmental pollution, of depletion of atmospheric ozone or other alterations that affect the radiant energy reaching us, and of the availability to humans of excessive amounts of energy must all be considered. Just as a steady state within cells is often essential to the life of organisms, maintenance of a steady state in the chemical cycles of the biosphere may also be a necessity.

Biochemists and molecular biologists are being called upon to play an increasing role in medicine, agriculture, and industry. As such, they must be prepared to help in the making of decisions that may affect the future of life on earth. Biochemical approaches will be required to cope with many important problems. Among these are the long-term effects of the growing number of synthetic compounds in the environment, problems of antibiotic resistance, and effects of bioengineering of plants, fishes, and other organisms in the biosphere. Some of these scientific and ethical questions have been discussed in Chapter 26, and more are considered in the Study Questions that follow in this chapter.

Despite attempts to ignore it, we cannot avoid facing the war problem. The possibility of virtually total destruction of the more complex forms of life by genetic damage from radiation is real. That we have lived with nuclear weapons as long as we have is encouraging but continuing threats to use them as a last resort may bring eventual catastrophe. A race to put weapons into space might result in having computers decide to fight a war in which all people could be destroyed, but one computer might win! Perhaps biochemists, who understand the technical problems of radiation damage and mutation, have a special obligation to point out the hazard to others.

Just as threatening is the possibility of biochemical warfare, e.g., the use of artificial viruses. Biological weapons have been little used because of their lack of discrimination between friend and foe. However, our increasing knowledge of molecular biology makes possible insidious attacks on a population of unvaccinated persons. Since biochemical work does not require elaborate facilities, the development of biological weapons can be carried on by small groups in a clandestine manner. The recent assembly of a viable

polio virus from oligonucleotides purchased from a commercial supplier emphasizes the ease with which virus warfare might be launched. Finding a way to protect ourselves may be more difficult.

Should we really worry about such matters? Since biochemistry is unable to ascribe any purpose to life, shouldn't we scientists stick to science? Science is amoral, isn't it? And besides, won't society do just what it wants to regardless of our opinions? Questions like these will always be with us, but most of the best scientists in the world seem to act with a great deal of responsibility. Not only do they want the pleasure and excitement of discovery and recognition for their work, but also they want a world for their children and grandchildren. They tend to feel compassion for other human beings. Many of them will give as a principal motivation for becoming biochemists the desire to contribute to the understanding of living things for the purpose of improving health, medical care, nutrition, etc. Most of them would not like to see the evolution of human beings ended through a disaster with nuclear or biological weapons or by irreversible pollution of land and sea. It will be a strange irony if we use our marvelous inquisitive, ingenious, inventive, and compassionate brains, the pinnacle of biological evolution, to destroy our environment and ourselves.

At a conference in Berkeley in 1971,⁵⁷⁷ Joshua Lederberg, discoverer of genetic recombination in

bacteria, talked about these matters. Lederberg asked if fairness and objectivity are possible outside the laboratory. He thought so. He pointed out that the nations of the world agreed to stop production of biological weapons and that genuine steps had been taken to decrease some of the hazards facing us. Nevertheless, progress is slow. Some insist on inspection for violation of agreements. But how can one inspect thoroughly enough? Lederberg suggested that the only possible form of control is now evolving. It must come from scientists themselves who must step out of their roles as "pure" scientists and accept the responsibility of preventing foolish uses of new biological discoveries. It may seem impossible that there could be a scientific community which could be counted on always to act in a responsible way, but it may be the only way that the human beings can survive for long on this planet. Lederberg believes it possible (and so do I).

If this book has helped to bring to the reader some awareness of the knowledge and power of molecular biology, I hope that these final words may lead the reader to heed the advice of Professor Lederberg. I sincerely hope that all the young people now studying biochemistry and modern biology will commit themselves to using the fantastic new knowledge available to us for the betterment of mankind and to proceeding with caution and responsibility as they move into positions of influence in the scientific community.

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Study Questions

1. Discuss the roles of the following proteins in development: receptors, transcription factors, protein kinases, histones, DNA methylases, adhesion molecules, ubiquitin. How do small RNA molecules participate in development?
2. Are all body cells totipotent?
3. Discuss the roles of apoptosis in various groups of organisms.
4. Compare signaling between bacteria and other unicellular organisms with signaling in higher eukaryotes.
5. Are human beings the most highly developed organisms? If so, in what ways? Has evolution of humans stopped or will it continue? Will it be upward?
6. Is it important for the world to achieve a sustainable state in which the population is constant and the environment stable?⁵⁷⁸ How will the world support a projected increase in population from the present 6 billion to 9 billion in 50 years?^{579,580}
7. How seriously is the earth's ecosystem dominated by human activity?^{581,582} Human activities have greatly reduced the amount of area available to wild species. Will the ensuing extinction of many organisms impoverish future diversity?^{583–585} Can the world's fisheries become sustainable?^{586,587}
8. Is science losing its objectivity because of an emphasis on monetary gain rather than on meeting social needs?⁵⁸⁸

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Page numbers set in **boldface** refer to major discussions.
The symbol *s* after a page number refers to a chemical structure.

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